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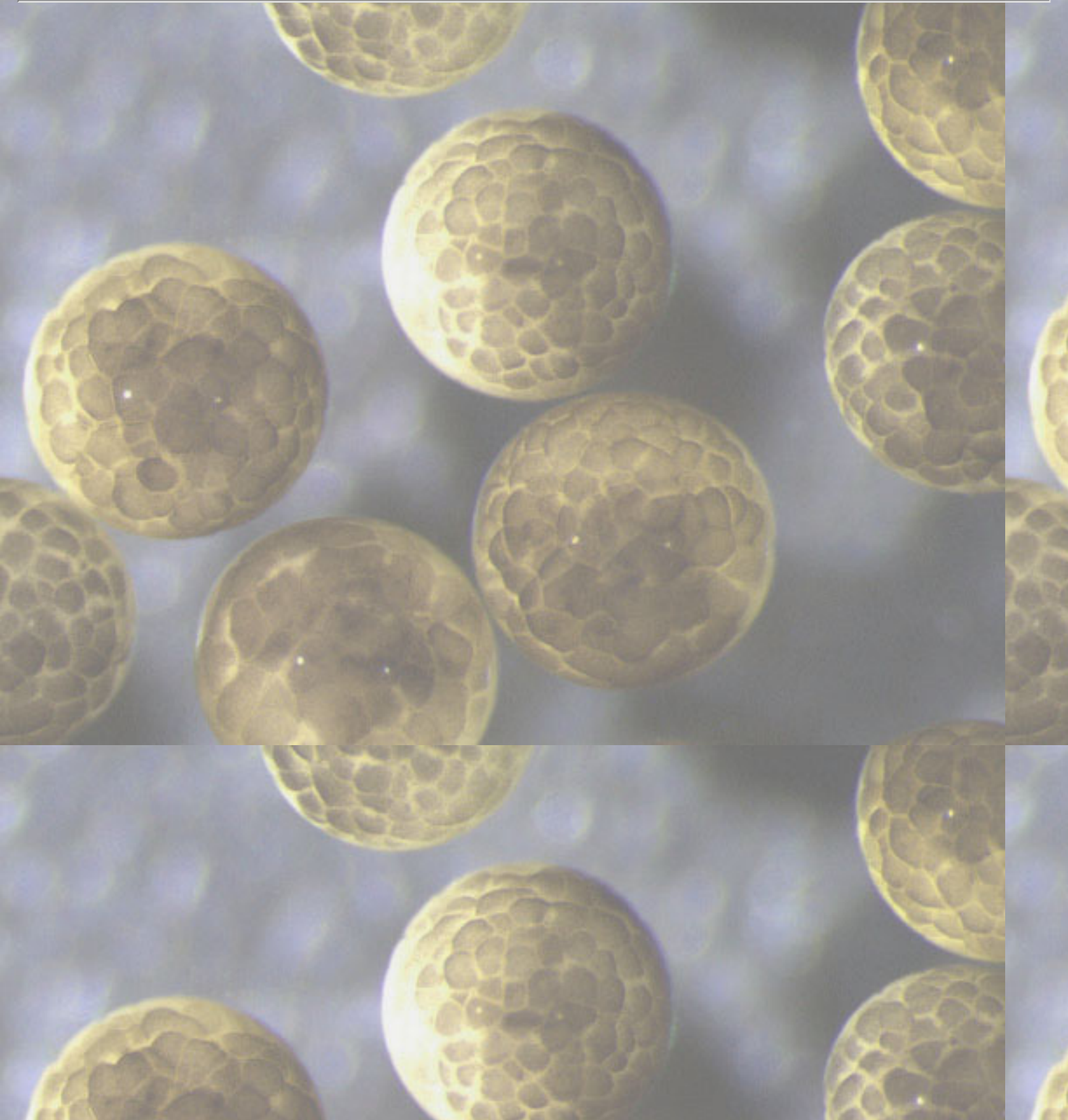
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## $K_{eq}$ , $K_w$ and pH

As  $H_2O$  is the medium of biological systems one must consider the role of this molecule in the dissociation of ions from biological molecules. Water is essentially a neutral molecule but will ionize to a small degree. This can be described by a simple equilibrium equation:



This equilibrium can be calculated as for any reaction:

$$K_{eq} = [H^+][OH^-]/[H_2O] \text{ Eqn. 2}$$

Since the concentration of  $H_2O$  is very high (55.5M) relative to that of the  $[H^+]$  and  $[OH^-]$ , consideration of it is generally removed from the equation by multiplying both sides by 55.5 yielding a new term,  $K_w$ :

$$K_w = [H^+][OH^-] \text{ Eqn. 3}$$

This term is referred to as the ion product. In pure water, to which no acids or bases have been added:

$$K_w = 1 \times 10^{-14} \text{ M}^2 \text{ Eqn. 4}$$

As  $K_w$  is constant, if one considers the case of pure water to which no acids or bases have been added:

$$[H^+] = [OH^-] = 1 \times 10^{-7} \text{ M Eqn. 5}$$

This term can be reduced to reflect the hydrogen ion concentration of any solution. This is termed the pH, where:

$$\text{pH} = -\log[H^+] \text{ Eqn. 6}$$

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## $\text{pK}_a$

Acids and bases can be classified as proton donors and proton acceptors, respectively. This means that the conjugate base of a given acid will carry a net charge that is more negative than the corresponding acid. In biologically relevant compounds various weak acids and bases are encountered, e.g. the acidic and basic amino acids, nucleotides, phospholipids etc.

Weak acids and bases in solution do not fully dissociate and, therefore, there is an equilibrium between the acid and its conjugate base. This equilibrium can be calculated and is termed the **equilibrium constant =  $K_a$** . This is also sometimes referred to as the dissociation constant as it pertains to the dissociation of protons from acids and bases.

In the reaction of a weak acid:



the equilibrium constant can be calculated from the following equation:

$$K_a = [H^+][A^-]/[HA] \text{ Eqn. 8}$$

As in the case of the ion product:

$$\text{pK}_a = -\log K_a \text{ Eqn. 9}$$

Therefore, in obtaining the  $-\log$  of both sides of the equation describing the dissociation of a weak acid we arrive at the following equation:

$$-\log K_a = -\log[\text{H}^+][\text{A}^-]/[\text{HA}] \text{ Eqn. 10}$$

Since as indicated above  $-\log K_a = \text{pK}_a$  and taking into account the laws of logarithms:

$$\text{pK}_a = -\log[\text{H}^+] - \log[\text{A}^-]/[\text{HA}] \text{ Eqn. 11}$$

$$\text{pK}_a = \text{pH} - \log[\text{A}^-]/[\text{HA}] \text{ Eqn. 12}$$

From this equation it can be seen that the smaller the  $\text{pK}_a$  value the stronger is the acid. This is due to the fact that the stronger an acid the more readily it will give up  $\text{H}^+$  and, therefore, the value of  $[\text{HA}]$  in the above equation will be relatively small.

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## The Henderson-Hasselbalch Equation

By rearranging the above equation we arrive at the **Henderson-Hasselbalch equation**:

$$\text{pH} = \text{pK}_a + \log[\text{A}^-]/[\text{HA}] \text{ Eqn. 13}$$

It should be obvious now that the pH of a solution of any acid (for which the equilibrium constant is known, and there are numerous tables with this information) can be calculated knowing the concentration of the acid, HA, and its conjugate base  $[\text{A}^-]$ .

At the point of the dissociation where the concentration of the conjugate base  $[\text{A}^-] =$  to that of the acid  $[\text{HA}]$ :

$$\text{pH} = \text{pK}_a + \log[1] \text{ Eqn. 14}$$

The  $\log$  of 1 = 0. Thus, at the mid-point of a titration of a weak acid:

$$\text{pK}_a = \text{pH} \text{ Eqn. 15}$$

In other words, the term  $\text{pK}_a$  is that pH at which an equivalent distribution of acid and conjugate base (or base and conjugate acid) exists in solution.



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## Buffering

It should be noted that around the  $pK_a$  the pH of a solution does not change appreciably even when large amounts of acid or base are added. This phenomenon is known as **buffering**. In most biochemical studies it is important to perform experiments, that will consume  $H^+$  or  $OH^-$  equivalents, in a solution of a buffering agent that has a  $pK_a$  near the pH optimum for the experiment.

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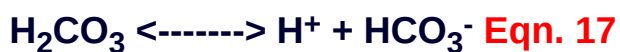
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## Clinical Significance of Blood Buffering

The pH of blood is maintained in a narrow range around 7.4. Even relatively small changes in this value of blood pH can lead to severe metabolic consequences. Therefore, **blood buffering** is extremely important in order to maintain homeostasis. Although the blood contains numerous cations (e.g.,  $Na^+$ ,  $K^+$ ,  $Ca^{2+}$  and  $Mg^{2+}$ ) and anions (e.g.,  $Cl^-$ ,  $PO_4^{3-}$  and  $SO_4^{2-}$ ) that can, as a whole, play a role in buffering, the **primary buffers in blood are hemoglobin in erythrocytes and bicarbonate ion ( $HCO_3^-$ ) in the plasma**. Buffering by hemoglobin is accomplished by ionization of the imidazole ring of histidines in the protein.

The formation of bicarbonate ion in blood from  $CO_2$  and  $H_2O$  allows the transfer of relatively insoluble  $CO_2$  from the tissues to the lungs, where it is expelled. The major source of  $CO_2$  in the tissues comes from the oxidation of ingested carbon compounds.

Carbonic acid is formed from the reaction of dissolved  $CO_2$  with  $H_2O$ . The relationship between carbonic acid and bicarbonate ion formation is shown in equations 16 and 17.



The reactions shown in equations 16 and 17 occur predominately in the erythrocytes, since nearly all of the  $CO_2$  leaving tissues via the capillary endothelium is taken up by these cells. This reaction is catalyzed by **carbonic anhydrase**. Ionization of carbonic acid then occurs spontaneously (as shown in equation 17), yielding bicarbonate ion.

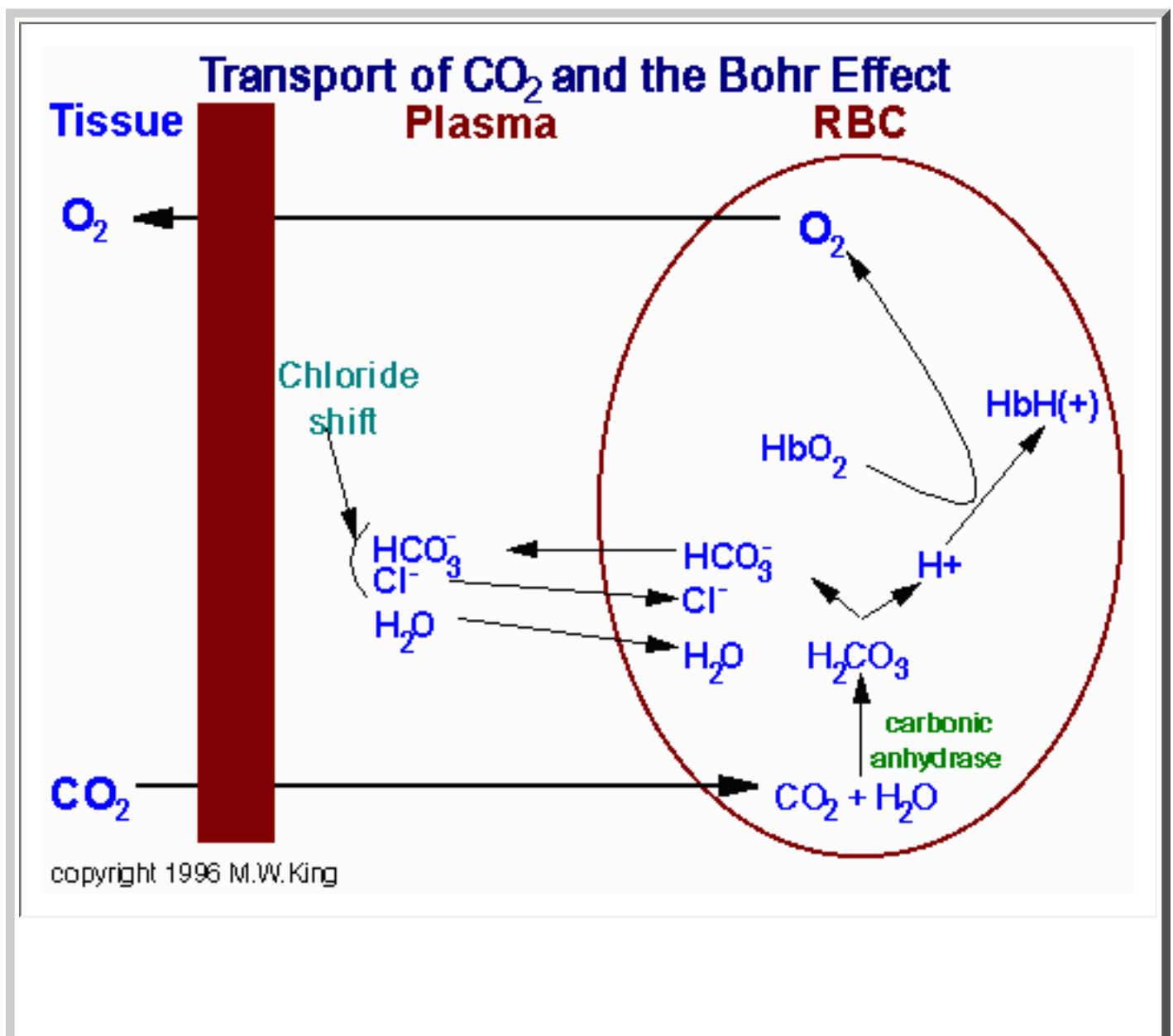
Carbonic acid is a relatively strong acid with a  $pK_a$  of 3.8. However, carbonic acid is in equilibrium with dissolved  $CO_2$ . Therefore, the equilibrium equation for the sum of equations 16 and 17 requires a conversion factor, since  $CO_2$  is a dissolved gas. This factor has been shown to be approximately 0.03 times the partial pressure of  $CO_2$  ( $PCO_2$ ). When this is

entered into the Henderson-Hasselbalch equation:

$$\text{pH} = 6.1 + \log \left[ \frac{[\text{HCO}_3^-]}{(0.03)(\text{PCO}_2)} \right] \quad \text{Eqn. 18}$$

where the apparent pKa for bicarbonate formation, 6.1, has been introduced into equation 18.

The  $\text{PCO}_2$  in the peripheral tissues is approximately 50mm Hg, whereas in the blood entering the peripheral tissues it is approximately 40mm Hg. This difference results in the diffusion of  $\text{CO}_2$  from the tissues into the blood in the capillaries of the periphery. When the  $\text{CO}_2$  is converted to  $\text{H}_2\text{CO}_3$  within the erythrocytes and then ionizes, the hydrogen ions ( $\text{H}^+$ ) are buffered by hemoglobin. The production of  $\text{H}^+$  ions within erythrocytes and their subsequent buffering by hemoglobin results in a reduced affinity of hemoglobin for oxygen. This leads to a release of  $\text{O}_2$  to the peripheral tissues, a phenomenon is termed the **Bohr effect**.



Representation of the transport of  $\text{CO}_2$  from the tissues to the blood with delivery of  $\text{O}_2$  to the tissues. The opposite process occurs when  $\text{O}_2$  is taken up from the alveoli of the lungs and the  $\text{CO}_2$  is expelled. All of the processes of the transport of  $\text{CO}_2$  and  $\text{O}_2$  are not shown such as the formation and ionization of carbonic acid in the plasma. The latter is a major mechanism for the transport of  $\text{CO}_2$  to the lungs, i.e. in the plasma as  $\text{HCO}_3^-$ . The  $\text{H}^+$  produced in the plasma by the ionization of carbonic acid is buffered by phosphate ( $\text{HPO}_4^{2-}$ ) and by proteins. Additionally, some 15% of the  $\text{CO}_2$  is transported from the tissues to the lungs as **hemoglobin carbamate** as shown in Eqn. 19.

As  $\text{CO}_2$  passes from the tissues to the plasma a minor amount of carbonic acid takes form and ionizes. The  $\text{H}^+$  ions are then buffered predominantly by proteins and phosphate ions in the plasma. As the concentration of bicarbonate ions rises in erythrocytes, an osmotic imbalance occurs. The imbalance is relieved as bicarbonate ion leaves the erythrocytes in exchange for chloride ions from the plasma. This phenomenon is known as the **chloride shift** which is also shown in the diagram above. Therefore, the majority of the bicarbonate ion formed as  $\text{CO}_2$  leaves the peripheral tissues is transported by the plasma to the lungs.

Around 15% of  $\text{CO}_2$  transport from the tissues to the lungs occurs through a reversible combination with non-ionized amino groups ( $-\text{NH}_2$ ) of hemoglobin forming what is termed **hemoglobin carbamate**.



The formation of hemoglobin carbamate results in a reduced affinity of hemoglobin for  $\text{O}_2$  thus favoring dissociation of bound oxygen in the tissues where the concentration of  $\text{CO}_2$  is high. The process is reversed when the erythrocytes enter the lungs and the partial pressure of  $\text{O}_2$  is elevated.

The partial pressure of  $\text{O}_2$  ( $PO_2$ ) in the pulmonary alveoli is higher than the  $PO_2$  of the entering erythrocytes that contain predominantly deoxygenated hemoglobin. This increased  $PO_2$  leads to oxygenation of hemoglobin and release of  $\text{H}^+$  ions from the hemoglobin. The released  $\text{H}^+$  ions combine with the bicarbonate ions to form  $\text{H}_2\text{CO}_3$ . Cellular **carbonic anhydrase** then catalyzes the reverse of reaction 17, leading to release of  $\text{CO}_2$  from erythrocytes. Owing to the  $PCO_2$  gradient (described above), the  $\text{CO}_2$  diffuses from the blood to the alveoli where it is expelled.

The great utility of bicarbonate as a physiological buffer stems from the fact that if excess acid is added to the blood the concentration of bicarbonate ion declines and the level of



CO<sub>2</sub> increases. The CO<sub>2</sub> then passes from capillaries in the pulmonary alveoli and is expelled. As a consequence, the H<sup>+</sup> ion concentration drives reaction 17 to the left and bicarbonate ion acts as a buffer until all of the hydrogen ion is consumed. Conversely, when excess base is added to the blood, CO<sub>2</sub> is consumed by carbonic acid and replaced by metabolic reactions within the body.

If blood is not adequately buffered, the result may be **metabolic acidosis** or **metabolic alkalosis**. These physiological states can be reached if a metabolic defect results in the inappropriate accumulation or loss of acidic or basic compounds. These compounds may be ingested, or they may accumulate as metabolic by-products such as acetoacetic acid and lactic acid. Both of these will ionize, thereby increasing the level of H<sup>+</sup> ions that will in turn remove bicarbonate ions from the blood and alter blood pH. The predominant defect in acid or base elimination arises when the excretory system of the kidneys is impaired. Alternatively, if the lungs fail to expel accumulated CO<sub>2</sub> adequately and CO<sub>2</sub> accumulates in the body, the result will be **respiratory acidosis**. If a decrease in *P*CO<sub>2</sub> within the lungs occurs, as during hyperventilation, the result will be **respiratory alkalosis**.

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## Ampholytes, Polyampholytes, pI and Zwitterion

Many substances in nature contain both acidic and basic groups as well as many different types of these groups in the same molecule. (e.g. proteins). These are called **ampholytes** (one acidic and one basic group) or **polyampholytes** (many acidic and basic groups). Proteins contains many different amino acids some of which contain ionizable side groups, both acidic and basic. Therefore, a useful term for dealing with the titration of ampholytes and polyampholytes (e.g. proteins) is the **isoelectric point, pI**. This is described as the pH at which the effective net charge on a molecule is zero.

For the case of a simple ampholyte like the amino acid glycine the pI, when calculated from the Henderson-Hasselbalch equation, is shown to be the average of the pK for the α-COOH group and the pK for the α-NH<sub>2</sub> group:

$$pI = [pK_{a-(COOH)} + pK_{a-(NH_3^+)}]/2 \text{ Eqn. 20}$$

For more complex molecules such as polyampholytes the pI is the average of the pK<sub>a</sub> values that represent the boundaries of the zwitterionic form of the molecule. The pI value, like that of pK, is very informative as to the nature of different molecules. A molecule with a low pI would contain a predominance of acidic groups, whereas a high pI indicates predominance of basic groups.

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## Solvation and Hydration shells

Depending on the pH of a solution, macromolecules such as proteins which contain many charged groups, will carry substantial net charge, either positive or negative. Cells of the body and blood contain many **polyelectrolytes** (molecules that contain multiple same charges, e.g. DNA and RNA) and **polyampholytes** that are in close proximity. The close association allows these molecules to interact through opposing charged groups. The presence, in cells and blood, of numerous small charged ions (e.g.  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Mg}^{2+}$ ,  $\text{Mn}^{2+}$ ,  $\text{K}^+$ ) leads to the interaction of many small ions with the larger macroions. This interaction can result in a **shielding** of the electrostatic charges of like-charged molecules. This electrostatic shielding allows macroions to become more closely associated than predicted based upon their expected charge repulsion from one another. The net effect of the presence of small ions is to maintain the solubility of macromolecules at pH ranges near their pI. This interaction between solute (e.g. proteins, DNA, RNA, etc.) and solvent (e.g. blood) is termed **solvation** or **hydration**. The opposite effect to solvation occurs when the salt (small ion) concentration increases to such a level as to interfere with the solvation of proteins by  $\text{H}_2\text{O}$ . This results from the  $\text{H}_2\text{O}$  forming **hydration shells** around the small ions.

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## Role of the Kidneys in Acid-Base Balance

The kidneys function to filter the plasma that passes through the nephrons. Filtration of the plasmas occurs in the glomerular capillaries of the nephron. These capillaries allow the passage of water and low molecular weight solutes (less than 70 kDa) into the capsular space. The filtrate then passes through the proximal and distal convoluted tubules where reabsorption of water and many solutes takes place. In the course of glomerular filtration and tubule reabsorption the composition of the plasma changes generating the typical composition of urine. From a biochemical standpoint the kidneys serve important roles in the regulation of plasma acid-base balance and the elimination of nitrogenous wastes.

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## Sodium Bicarbonate Reabsorption

Regulation of plasma acid-base balance is primarily effected within the kidney through control over  $\text{HCO}_3^-$  reabsorption and secretion of  $\text{H}^+$ . Secretion of  $\text{H}^+$ , in excess of its capacity to react with  $\text{HCO}_3^-$  in the tubular luminal fluid, requires the presence of other buffers (see below). The generation of  $\text{HCO}_3^-$  and  $\text{H}^+$  occurs by dissociation of **carbonic acid** ( $\text{H}_2\text{CO}_3$ ), formed in the tubule cells from  $\text{H}_2\text{O}$  and  $\text{CO}_2$ , through the action of **carbonic anhydrase**. Secretion of  $\text{H}^+$  into the lumen of the tubule is accompanied by an exchange for  $\text{Na}^+$ . This reabsorption of  $\text{Na}^+$  occurs by an antiport mechanism during the exchange for  $\text{H}^+$ . Reduction in the intracellular concentration of  $\text{Na}^+$  occurs by an active transport process involving a  $\text{Na}^+/\text{K}^+$

$^+$ -ATPase pump which pumps the excess  $\text{Na}^+$  into the interstitial fluid. The intracellular  $\text{HCO}_3^-$  then diffuses from the tubule cell into the interstitial fluid.

The capacity of the kidney to secrete  $\text{H}^+$  is regulated by the maximal  $\text{H}^+$  gradient that can form between the tubule and lumen and still allow transport mechanisms to operate. This gradient is determined by the pH of the urine which in humans is near 4.5. The capacity to secrete  $\text{H}^+$  would be rapidly reached if it were not for the presence of buffers within the interstitial fluid. The  $\text{H}^+$  secreted into the tubular lumen can undergo three different fates depending upon the concentration of the three primary buffers of the interstitial fluid. These buffers are  $\text{HCO}_3^-$ ,  $\text{HPO}_4^{2-}$  and  $\text{NH}_3$ . Reaction of  $\text{H}^+$  with  $\text{HCO}_3^-$  forms  $\text{H}_2\text{O}$  and  $\text{CO}_2$  which diffuse back into the tubule cell. The net result of this process is the regeneration of  $\text{HCO}_3^-$  within the tubule cell. This process is termed reabsorption of sodium bicarbonate. The **reabsorption of sodium bicarbonate** takes place primarily within the proximal convoluted tubules.

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## Excretion of Acid

As the concentration of  $\text{HCO}_3^-$  in the tubular lumen drops the pH of the fluid drops due to an increasing concentration of  $\text{H}^+$ . The pH of the tubular fluid gradually approaches the  $\text{pK}_a$  for the dibasic/monobasic phosphate buffering system ( $\text{pK}_a = 6.8$ ). The excess  $\text{H}^+$  reacts with dibasic phosphate ( $\text{HPO}_4^{2-}$ ) forming monobasic phosphate ( $\text{H}_2\text{PO}_4^-$ ). The  $\text{H}_2\text{PO}_4^-$  so formed is not reabsorbed and its excretion results in the net excretion of  $\text{H}^+$ . The greatest extent of  $\text{H}_2\text{PO}_4^-$  formation occurs within the distal convoluted tubules and the collecting ducts.

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## Ammonia Secretion

Buffering of  $\text{H}^+$  is also accomplished by reaction with ammonia,  $\text{NH}_3$ , to form ammonium ion,  $\text{NH}_4^+$ . Elimination of  $\text{NH}_4^+$  is the major contributory factor in the ability of the body to excrete acid. Because the  $\text{pK}_a$  of  $\text{NH}_4^+$  is 9.3, excretion of acid in this form can be accomplished without lowering the pH of the urine. Additionally important is the fact that excretion of acid in the form of  $\text{NH}_4^+$  occurs without depleting  $\text{Na}^+$  nor  $\text{K}^+$ .

Two principal reactions within tubule cells result in the generation of  $\text{NH}_3$ , conversion of glutamine to glutamate and conversion of glutamate to  $\alpha$ -ketoglutarate. These reactions are catalyzed by **glutaminase** and **glutamate dehydrogenase**, respectively (Equations 21 and 22).



Both of these enzymes are abundant in tubule cells. Ammonia is lipid soluble and will diffuse down its concentration gradient out of the tubule cell into the tubular fluid. There it reacts with  $\text{H}^+$  to yield  $\text{NH}_4^+$  which is excreted in the urine.

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## Neurotoxicity of Ammonia

Excess ammonia is severely neurotoxic. Marked brain damage is seen in cases of failure to make urea via the [urea cycle](#) or to eliminate urea through the kidneys. The result of either of these events is a buildup of circulating levels of ammonium ion. Aside from its effect on blood pH, ammonia readily traverses the brain blood barrier and in the brain is converted to glutamate via **glutamate dehydrogenase**, depleting the brain of  $\alpha$ -ketoglutarate. As the  $\alpha$ -ketoglutarate is depleted, oxaloacetate falls correspondingly, and ultimately TCA cycle activity comes to a halt. In the absence of aerobic oxidative phosphorylation and TCA cycle activity, irreparable cell damage and neural cell death ensue.

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## Acidosis and Alkalosis

The kidneys play an important role in the control of acidosis by responding with an increase in the excretion of  $\text{H}^+$ . When  $\text{H}^+$  is excreted as a titratable acid such as  $\text{H}_2\text{PO}_4^-$  or when the anions of strong acids such as acetoacetate are excreted there is a requirement for simultaneous excretion of cations to maintain electrical neutrality. The principal cation excreted is  $\text{Na}^+$ . As the level of excretable  $\text{Na}^+$  is depleted excretion of  $\text{K}^+$  increases. In conditions of acidosis the kidney will increase the production of  $\text{NH}_3$  from tubular amino acids or amino acids absorbed from the plasma. As indicated the  $\text{NH}_3$  can diffuse across the tubule cell membrane where it will react with  $\text{H}^+$  to form the excretable ammonium ion without a concomitant requirement for cation excretion. This demonstrates that an inability of the kidney to generate  $\text{NH}_3$  would rapidly lead to fatal acidosis.

When the kidneys fail to modulate  $\text{HCO}_3^-$  excretion, metabolic alkalosis will develop. Alkalosis is normally countered quite effectively by the kidney allowing  $\text{HCO}_3^-$  to freely escape. Alkalosis generally only becomes problematic if the kidneys are restricted in their ability to secrete  $\text{HCO}_3^-$ . This situation can occur in patients taking diuretics since several of this class of drug cause a reduction in the ability of the kidney to reabsorb an anion (e.g.  $\text{Cl}^-$ )

concomitant with the reabsorption of  $\text{Na}^+$ .

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## Introduction

Most, if not all, cancer cells contain genetic damage that appears to be the responsible event leading to tumorigenesis. The genetic damage present in a parental tumorigenic cell is maintained (i.e. not correctable) such that it is a heritable trait of all cells of subsequent generations. Genetic damage found in cancer cells is of two types:

- **1.** Dominant and the genes have been termed **proto-oncogenes**. The distinction between the terms proto-oncogene and oncogene relates to the activity of the protein product of the gene. A proto-oncogene is a gene whose protein product has the capacity to induce cellular transformation given it sustains some genetic insult. An oncogene is a gene that has sustained some genetic damage and, therefore, produces a protein capable of cellular transformation.

The process of activation of proto-oncogenes to oncogenes can include retroviral transduction or retroviral integration (see below), point mutations, insertion mutations, gene amplification, chromosomal translocation and/or protein-protein interactions.

Proto-oncogenes can be classified into many different groups based upon their normal function within cells or based upon sequence homology to other known proteins. As predicted, proto-oncogenes have been identified at all levels of the various signal transduction cascades that control cell growth, proliferation and differentiation. The list of proto-oncogenes identified to date is too lengthy to include here, therefore, only those genes that have been highly characterized are described. Proto-oncogenes that were originally identified as resident in transforming retroviruses are designated as **c-** indicative of the cellular origin as opposed to **v-** to signify original identification in retroviruses.

- **2. Recessive and the genes variously termed tumor suppressors, growth suppressors, recessive oncogenes or anti-oncogenes.**

Given the complexity of inducing and regulating cellular growth, proliferation and differentiation, it was suspected for many years that genetic damage to genes encoding growth factors, growth factor receptors and/or the proteins of the various signal transduction cascades would lead to cellular transformation. This suspicion has proven true with the identification of numerous genes, whose products function in cellular signaling, that are involved in some way in the genesis of the tumorigenic state. The majority of these proto-oncogenes were identified by either of two means: as the transforming genes (oncogenes) of transforming retroviruses or through transfection of DNA from tumor cell lines into non-transformed cell lines and screening for resultant tumorigenesis.

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## Viruses and Cancer

Tumor cells also can arise by non-genetic means through the actions of specific tumor viruses. Tumor viruses are of two distinct types. There are viruses with DNA genomes (e.g. papilloma and adenoviruses) and those with RNA genomes (termed **retroviruses**).

RNA tumor viruses are common in chickens, mice and cats but rare in humans. The only currently known human retroviruses are the **human T-cell leukemia viruses (HTLVs)** and the related retrovirus, **human immunodeficiency virus (HIV)**.

Retroviruses can induce the transformed state within the cells they infect by two mechanisms. Both of these mechanisms are related to the life cycle of these viruses. When a retrovirus infects a cell its RNA genome is converted into DNA by the viral encoded RNA-dependent DNA polymerase (**reverse transcriptase**). The DNA then integrates into the genome of the host cell where it can remain being copied as the host genome is duplicated during the process of cellular division. Contained within the sequences at the ends of the retroviral genome are powerful transcriptional promoter sequences termed **long terminal repeats (LTRs)**. The LTRs promote the transcription of the viral DNA leading to the production of new virus particles.

At some frequency the integration process leads to rearrangement of the viral genome and the consequent incorporation of a portion of the host genome into the viral genome. This process is termed transduction. Occasionally this transduction process leads to the virus acquiring a gene from the host that is normally involved in cellular growth control. Because of the alteration of the host gene during the transduction process as well as the gene being transcribed at a higher rate due to its association with the retroviral LTRs the transduced gene confers a growth advantage to the infected cell. The end result of this process is unrestricted cellular proliferation leading to tumorigenesis. The transduced genes are termed oncogenes. The normal cellular gene in its unmodified, non-transduced form is termed a proto-oncogene since it has the capacity to transform cells if altered in some way or expressed in an uncontrolled manner. Numerous oncogenes have been discovered in the genomes of transforming retroviruses.

The second mechanism by which retroviruses can transform cells relates to the powerful transcription promoting effect of the LTRs. When a retrovirus genome integrates into a host genome it does so randomly. At some frequency this integration process leads to the placement of the LTRs close to a gene that encodes a growth regulating protein. If the protein is expressed at an abnormally elevated level it can result in cellular transformation. This is termed retroviral integration induced transformation. It has recently been shown that HIV induces certain forms of cancers in infected



individuals by this integration induced transformation process.

Cellular transformation by DNA tumor viruses, in most cases, has been shown to be the result of protein-protein interaction. Proteins encoded by the DNA tumor viruses, termed **tumor antigens** or T antigens, can interact with cellular proteins. This interaction effectively sequesters the cellular proteins away from their normal functional locations within the cell. The predominant types of proteins that are sequestered by viral T antigens have been shown to be of the tumor suppressor type. It is the loss of their normal suppressor functions that results in cellular transformation.

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## Classifications of Proto-Oncogenes

Although there are numerous examples of each classification of proto-oncogene, the lists here are by no means exhaustive. New genes with tumor causing capabilities are being isolated continuously.

### Growth Factors

The c-Sis gene (the v-sis gene is the oncogene in simian sarcoma virus) encodes the PDGF B chain. The v-sis gene was the first oncogene to be identified as having homology to a known cellular gene. The int-2 gene (named for the fact that it is a common site of **integration** of mouse mammary tumor virus) encodes an FGF-related growth factor. The KGF (also called Hst) gene also encodes an FGF-related growth factor and was identified in gastric carcinoma and Kaposi's sarcoma cells.

### Receptor Tyrosine Kinases

The c-Fms (*fms*) gene encodes the colony stimulating factor-1 (CSF-1) receptor. and was first identified as a retroviral oncogene. The Flg (*flag*) gene (named because it has homology to the Fms [*fms*] gene, hence **fms-like g**ene) encodes a form of the FGF receptor. The Neu (*new*) gene was identified as an EGF receptor-related gene in an ethylnitrosourea-induced **neuroblastoma**. The conversion of proto-oncogenic to oncogenic Neu requires only a single amino acid change in the transmembrane domain. The Trk (*track*) genes encodes the NGF receptor-like proteins. The first Trk gene was found in a pancreatic cancer. Subsequently, two additional Trk-related genes were identified. These three are now identified as TrkA, TrkB and TrkC. The Met gene encodes the hepatocyte growth factor(HGF)/scatter factor (SF) receptor. The c-Kit gene encodes the mast cell growth factor receptor.

### Membrane Associated Non-Receptor Tyrosine Kinases

The v-src gene was the first identified oncogene. The c-Src gene is the archetypal protein tyrosine kinase. The Lck gene was isolated from a T cell tumor line (**L**YSTRA **c**ell **k**inase) and has been shown to be associated with the CD4 and CD8 antigens of T cells.

### G-Protein Coupled Receptors

The Mas gene was identified in a mammary carcinoma and has been shown to be the

angiotensin receptor.

## Membrane Associated G-Proteins

There are three different homologs of the c-Ras gene, each of which was identified in a different type of tumor cell. The Ras gene is one of the most frequently disrupted genes in colorectal carcinomas.

## Serine/Threonine Kinases

The Raf gene is involved in the signaling pathway of most RTKs. It is likely responsible for threonine phosphorylation of MAP kinase following receptor activation.

## Nuclear DNA-Binding/Transcription Factors

The Myc gene was originally identified in the avian **myelocytomatosis** virus. A disrupted human c-Myc gene has been found to be involved in numerous hematopoietic neoplasias. Disruption of c-Myc has been shown to be the result of retroviral integration and transduction as well as chromosomal rearrangements. The Fos gene was identified in the **feline osteosarcoma** virus. The protein interacts with a second proto-oncogenic protein, Jun to form a transcriptional regulatory complex. The p53 gene was originally identified as a major nuclear antigen in transformed cells. The p53 gene is the single most identified mutant protein in human tumors. Mutant forms of the p53 protein interfere with cell growth suppressor effects of wild-type p53 indicating that the p53 gene product is actually a **tumor suppressor**.

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## Hereditary Cancer Syndromes

| Syndrome                                                                 | Cloned Gene             | Function                         | Chromosomal Location | Tumor Types                                     |
|--------------------------------------------------------------------------|-------------------------|----------------------------------|----------------------|-------------------------------------------------|
| <a href="#">Li-Fraumeni Syndrome</a><br><br><a href="#">OMIM data</a>    | P53<br>tumor suppressor | cell cycle regulation, apoptosis | 17p13                | brain tumors, sarcomas, leukemia, breast cancer |
| <a href="#">Familial Retinoblastoma</a><br><br><a href="#">OMIM data</a> | RB1<br>tumor suppressor | cell cycle regulation            | 13q14                | retinoblastoma, osteogenic sarcoma              |
| <a href="#">Wilms Tumor</a><br><br><a href="#">OMIM data</a>             | WT1<br>tumor suppressor | transcriptional regulation       | 11p13                | pediatric kidney cancer                         |

|                                                                           |                                                                       |                                                                  |         |                                                                              |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|---------|------------------------------------------------------------------------------|
| <b>Neurofibromatosis Type 1</b><br><br><a href="#">OMIM data</a>          | <b>NF1</b><br>protein=neurofibromin 1<br>tumor suppressor             | catalysis of RAS inactivation                                    | 17q11.2 | neurofibromas, sarcomas, gliomas                                             |
| <b>Neurofibromatosis Type 2</b><br><br><a href="#">OMIM data</a>          | <b>NF2</b><br>protein = merlin or neurofibromin 2<br>tumor suppressor | linkage of cell membrane to cytoskeleton                         | 22q12.2 | Schwann cell tumors, astrocytomas, meningiomas, ependynomas                  |
| <b>Familial Adenomatous Polyposis</b><br><br><a href="#">OMIM data</a>    | <b>APC</b><br>tumor suppressor                                        | signaling through adhesion molecules to nucleus                  | 5q21    | colon cancer                                                                 |
| <b>Tuberous sclerosis 1</b><br><br><a href="#">OMIM data</a>              | <b>TSC1</b><br>protein = hamartin<br>tumor suppressor                 |                                                                  | 9q34    | facial angiofibromas                                                         |
| <b>Tuberous sclerosis 2</b><br><br><a href="#">OMIM data</a>              | <b>TSC2</b><br>protein = tuberlin<br>tumor suppressor                 | GTPase activation                                                | 16      | benign growths (hamartomas) in many tissues, astrocytomas, rhabdomyosarcomas |
| <b>Deleted in Pancreatic Carcinoma 4</b><br><br><a href="#">OMIM data</a> | <b>DPC4</b><br>also known as Smad4<br>tumor suppressor                | regulation of TGF- $\beta$ /BMP signal transduction              | 18q21.1 | pancreatic carcinoma, colon cancer                                           |
| <b>Deleted in Colorectal Carcinoma</b><br><br><a href="#">OMIM data</a>   | <b>DCC</b><br>tumor suppressor                                        | transmembrane receptor involved in axonal guidance via netrins   | 18q21.3 | colorectal cancer                                                            |
| <b>Familial Breast Cancer</b><br><br><a href="#">OMIM data</a>            | <b>BRCA1</b><br>tumor suppressor                                      | repair of double strand breaks by association with Rad51 protein | 17q21   | breast and ovarian cancer                                                    |

|                                                                                                                       |                                                                                     |                                                                                          |          |                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------|
| <p>Familial Breast Cancer</p> <p><a href="#">OMIM data</a></p>                                                        | BRCA2<br>tumor suppressor                                                           | similar to<br>BRCA1?                                                                     | 13q12.3  | breast and ovarian cancer                                                                               |
| <p>Peutz-Jeghers Syndrome</p> <p><a href="#">OMIM data</a></p>                                                        | STK11<br>tumor suppressor<br>protein = serine-<br>threonine kinase 11               | potential<br>regulation of<br>vascular<br>endothelial<br>growth factor<br>(VEGF) pathway | 19p13.3  | hyperpigmentation,<br>multiple<br>hamartomatous<br>polyps, colorectal,<br>breast and ovarian<br>cancers |
| <p>Hereditary<br/>Nonpolyposis<br/>Colorectal Cancer<br/>type 1<br/>HNPCC1</p> <p><a href="#">OMIM data</a></p>       | MSH2<br>tumor suppressor                                                            | DNA mismatch<br>repair                                                                   | 2p22-p21 | colorectal cancer                                                                                       |
| <p>Hereditary<br/>Nonpolyposis<br/>Colorectal Cancer<br/>type 2<br/>HNPCC2</p> <p><a href="#">OMIM data</a></p>       | MLH1<br>tumor suppressor                                                            | DNA mismatch<br>repair                                                                   | 3p21.3   | colorectal cancer                                                                                       |
| <p>von Hippel-Lindau<br/>Syndrome</p> <p><a href="#">OMIM data</a></p>                                                | VHL<br>tumor suppressor                                                             | regulation of<br>transcription<br>elongation                                             | 3p26-p25 | renal cancers,<br>hemangioblastomas,<br>pheochromocytoma                                                |
| <p>Familial Melanoma</p> <p><a href="#">OMIM data</a></p>                                                             | CDKN2A<br>protein = cyclin-<br>dependent kinase<br>inhibitor 2A<br>tumor suppressor | inhibits cell-cycle<br>kinases CDK4<br>and CDK6                                          | 9p21     | melanoma,<br>pancreatic cancer,<br>others                                                               |
| <p>Gorlin Syndrome:<br/>Nevoid basal cell<br/>carcinoma<br/>syndrome<br/>(NBCCS)</p> <p><a href="#">OMIM data</a></p> | PTCH<br>protein = patched<br>tumor suppressor                                       | transmembrane<br>receptor for<br><i>hedgehog</i><br>signaling protein                    | 9q22.3   | basal cell skin cancer                                                                                  |

|                                                                                                            |                                                                                |                                                                                     |          |                                                                     |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------|
| Multiple Endocrine Neoplasia Type 1<br><a href="#">OMIM data</a>                                           | MEN1<br>tumor suppressor                                                       | unknown                                                                             | 11q13    | parathyroid and pituitary adenomas, islet cell tumors, carcinoid    |
| Multiple Endocrine Neoplasia Type 2<br><a href="#">OMIM data</a>                                           | RET, MEN2                                                                      | transmembrane receptor tyrosine kinase for glial-derived neurotrophic factor (GDNF) | 10q11.2  | medullary thyroid cancer, type 2A pheochromocytoma, mucosal hartoma |
| Beckwith-Wiedmann Syndrome<br><a href="#">OMIM data</a>                                                    | p57, KIP2                                                                      | cell cycle regulator                                                                | 11p15.5  | Wilms tumor, adrenocortical cancer, hepatoblastoma                  |
| Hereditary papillary renal cancer (HPRC)<br><a href="#">OMIM data</a>                                      | MET                                                                            | transmembrane receptor for hepatocyte growth factor (HGF)                           | 7q31     | renal papillary cancer                                              |
| Cowden syndrome<br><a href="#">OMIM data</a>                                                               | PTEN<br>tumor suppressor                                                       | phosphoinositide 3-phosphatase protein tyrosine phosphatase                         | 10q23.3  | breast cancer, thyroid cancer, head & neck squamous carcinomas      |
| Hereditary prostate cancer numerous loci: HPC1(PRCA1), HPCX, MXI1, KAI1, PCAP<br><a href="#">OMIM data</a> | HPC1 and PRCA1 are same designation ribonuclease L (RNaseL) maps to this locus | RNaseL involved in mRNA degradation                                                 | 1q24-q25 | prostate cancer                                                     |
| Ataxia telangiectasia (AT)<br><a href="#">OMIM data</a>                                                    | ATM                                                                            | DNA repair                                                                          | 11q22.3  | lymphoma, cerebellar ataxia, immunodeficiency                       |
| Bloom syndrome<br><a href="#">OMIM data</a>                                                                | BLM                                                                            | DNA helicase?                                                                       | 15q26.1  | solid tumors, immunodeficiency                                      |



|                                                                                                                                                     |                                                                    |                                                           |                                                                                                   |                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <p>Xeroderma pigmentosum (XP)<br/>7 complementation groups</p> <p>OMIM data for<br/><a href="#">XPA</a> <a href="#">XPC</a> <a href="#">XPD</a></p> | XPA - XPG                                                          | DNA repair<br>helicases,<br>nucleotide<br>excision repair | <p>XPA = 9q22.3<br/>XPC = 3p25<br/>XPD=19q13.2-q13.3<br/>XPE=11p12-p11<br/>XPF=16p13.3-p13.13</p> | skin cancer                                                                  |
| <p>Fanconi's anemia<br/>11 complementation groups</p> <p>OMIM data for<br/><a href="#">FANCA</a><br/><a href="#">FANCC</a></p>                      | <p>FANCA, B, C, D1, D2, E, F, G, I, J, L</p> <p>FANCD1 = BRCA2</p> | components of<br>DNA repair<br>machinery                  | <p>FANCA=16q24.3<br/>FANCC = 9q22.3<br/>FANCD2=3p25.3<br/>FANCE=11p15</p>                         | acute myeloid leukemia (AML),<br>pancytopenia,<br>chromosomal<br>instability |

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# Tumor Suppressors and Cancer

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## Introduction

Tumor suppressor genes were first identified by making cell hybrids between tumor and normal cells. On some occasions a chromosome from the normal cell reverted the transformed phenotype.

Several familial cancers have been shown to be associated with the loss of function of a tumor suppressor gene. The table below lists several of these syndromes.

A few of these tumor suppressor genes are described in more detail below. They include the **retinoblastoma susceptibility gene (RB)**, **Wilms' tumors (WT1)**, **neurofibromatosis type-1 (NF1)**, **familial adenomatosis polyposis coli (FAP)**, **von Hippel-Lindau syndrome**, and those identified through loss of heterozygosity such as in colorectal carcinomas (called **DCC** for **deleted in colon carcinoma**) and **p53** which was originally thought to be a proto-oncogene. However, the wild-type p53 protein suppresses the activity of mutant alleles of p53 which are the oncogenic forms of p53.

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| Familial Cancer Syndrome                                              | Tumor Suppressor Gene | Function                         | Chromosomal Location | Tumor Types Observed                            |
|-----------------------------------------------------------------------|-----------------------|----------------------------------|----------------------|-------------------------------------------------|
| <a href="#">Li-Fraumeni Syndrome</a><br><br><a href="#">OMIM data</a> | P53                   | cell cycle regulation, apoptosis | 17p13.1              | brain tumors, sarcomas, leukemia, breast cancer |

|                                                                                                                                            |                                          |                                                     |               |                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------|---------------|------------------------------------------------------------------------------|
| <a href="#">Familial Retinoblastoma</a><br><br><a href="#">OMIM data</a>                                                                   | RB1                                      | cell cycle regulation                               | 13q14.1-q14.2 | retinoblastoma, osteogenic sarcoma                                           |
| <a href="#">Wilms Tumor</a><br><br><a href="#">OMIM data</a>                                                                               | WT1                                      | transcriptional regulation                          | 11p13         | pediatric kidney cancer                                                      |
| <a href="#">Neurofibromatosis Type 1</a><br><br><a href="#">OMIM data</a>                                                                  | NF1<br>protein = neurofibromin 1         | catalysis of RAS inactivation                       | 17q11.2       | neurofibromas, sarcomas, gliomas                                             |
| <a href="#">Neurofibromatosis Type 2</a><br><br><a href="#">OMIM data</a>                                                                  | NF2<br>protein=merlin or neurofibromin 2 | linkage of cell membrane to actin cytoskeleton      | 22q12.2       | Schwann cell tumors, astrocytomas, meningiomas, ependymomas                  |
| <a href="#">Familial Adenomatous Polyposis</a><br><br><a href="#">OMIM data</a>                                                            | APC                                      | signaling through adhesion molecules to nucleus     | 5q21-q22      | colon cancer                                                                 |
| <a href="#">Tuberous sclerosis 1</a><br><br><a href="#">OMIM data</a>                                                                      | TSC1<br>protein = hamartin               | interacts with tuberin, exact function unknown      | 9q34          | facial angiofibromas                                                         |
| <a href="#">Tuberous sclerosis 2</a><br><br><a href="#">OMIM data</a>                                                                      | TSC2<br>protein = tuberin                | GTPase activation of RAP1 and RAB5                  | 16p13.3       | benign growths (hamartomas) in many tissues, astrocytomas, rhabdomyosarcomas |
| <a href="#">Deleted in Pancreatic Carcinoma 4</a><br><a href="#">Familial juvenile polyposis syndrome</a><br><br><a href="#">OMIM data</a> | DPC4<br>also known as Smad4              | regulation of TGF- $\beta$ /BMP signal transduction | 18q21.1       | pancreatic carcinoma, colon cancer                                           |

|                                                                                 |                                                                                   |                                                                                                                           |          |                                                                                          |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------|
| Deleted in Colorectal Carcinoma<br><a href="#">OMIM data</a>                    | DCC                                                                               | transmembrane receptor involved in axonal guidance via netrins                                                            | 18q21.3  | colorectal cancer                                                                        |
| Familial Breast Cancer<br><a href="#">OMIM data</a>                             | BRCA1                                                                             | repair of double strand breaks by interaction with Rad51 protein                                                          | 17q21    | breast and ovarian cancer                                                                |
| Familial Breast Cancer<br><a href="#">OMIM data</a>                             | BRCA2                                                                             | similar to BRCA1 activity?                                                                                                | 13q12.3  | breast and ovarian cancer                                                                |
| Cowden syndrome<br><a href="#">OMIM data</a>                                    | PTEN                                                                              | phosphoinositide 3-phosphatase protein tyrosine phosphatase                                                               | 10q23.3  | gliomas, breast cancer, thyroid cancer, head & neck squamous carcinoma                   |
| Peutz-Jeghers Syndrome<br><a href="#">OMIM data</a>                             | LKB1<br>a nuclear localized kinase also called STK11 (serine-threonine kinase 11) | phosphorylates and activates AMP-activated kinase (AMPK), AMPK involved in stress responses, lipid and glucose metabolism | 19p13.3  | hyperpigmentation, multiple hamartomatous polyps, colorectal, breast and ovarian cancers |
| Hereditary Nonpolyposis Colon Cancer type 1 HNPCC1<br><a href="#">OMIM data</a> | MSH2                                                                              | DNA mismatch repair                                                                                                       | 2p22-p21 | colon cancer                                                                             |
| Hereditary Nonpolyposis Colon Cancer type 2 HNPCC2<br><a href="#">OMIM data</a> | MLH1                                                                              | DNA mismatch repair                                                                                                       | 3p21.3   | colon cancer                                                                             |

|                                                                             |                                                                                            |                                                                                                                           |          |                                                                     |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------|
| Familial diffuse-type gastric cancer<br><br><a href="#">OMIM data</a>       | CDH1<br>protein = E-cadherin                                                               | cell-cell adhesion protein                                                                                                | 16q22.1  | gastric cancer, lobular breast cancer                               |
| <a href="#">von Hippel-Lindau Syndrome</a><br><br><a href="#">OMIM data</a> | VHL                                                                                        | regulation of transcription elongation through activation of a ubiquitin ligase complex                                   | 3p26-p25 | renal cancers, hemangioblastomas, pheochromocytoma, retinal angioma |
| Familial Melanoma<br><br><a href="#">OMIM data</a>                          | p16 <sup>INK4a</sup><br>also called CDKN2A<br>protein=cyclin-dependent kinase inhibitor 2A | cell-cycle regulation                                                                                                     | 9p21     | melanoma, pancreatic cancer, others                                 |
| Gorlin Syndrome<br><br><a href="#">OMIM data</a>                            | PTCH<br>protein = patched                                                                  | transmembrane receptor for sonic hedgehog (shh), involved in early development through repression of action of smoothened | 9q22.3   | basal cell skin carcinoma                                           |
| Multiple Endocrine Neoplasia Type 1<br><br><a href="#">OMIM data</a>        | MEN1                                                                                       | unknown                                                                                                                   | 11q13    | parathyroid and pituitary adenomas, islet cell tumors, carcinoid    |

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## p53

Loss of heterozygosity at the short arm of chromosome 17 has been associated with tumors of the lung, colon and breast. This region of chromosome 17 includes the p53 gene.

The P53 gene was originally discovered because the protein product complexes with the SV40 large T antigen. It was first thought that P53 was a dominant oncogene since cDNA clones isolated from tumor lines were able to cooperate with the RAS oncogene in transformation assays. This proved to be a misleading since the cDNA clones used in all these studies were mutated forms of wild-type p53 and cDNAs from normal tissue were later shown to be incapable of RAS co-transformation. The mutant p53



proteins were shown to be altered in stability and conformation as well as binding to hsp70.

Subsequent analysis of several murine leukemia cell lines showed that the P53 locus was lost by either insertions or deletions on both alleles. This suggested that wild-type p53 may be a tumor suppressor not a dominant proto-oncogene. Direct confirmation came when it was shown that wild-type p53 could suppress transformation in oncogene cooperation assays with mutant p53 and ras.

It has now been demonstrated that mutation at the P53 locus occurs in cancers of the colon, breast, liver and lung. Indeed, p53 involvement in neoplasia is more frequent than any other known tumor suppressor or dominant proto-oncogene!!!!

The protein encoded by P53 is a nuclear localized phosphoprotein. A domain near the N-terminus of the p53 protein is highly acidic like similar domains found in various transcription factors. When this domain is fused to the DNA-binding domain of the yeast GAL4 protein, the resulting chimera is able to activate transcription from genes containing GAL4 response elements. This suggests that p53 may be involved in transcriptional regulation. p53 has been shown to bind DNA, *in vitro*, that contains at least 2 copies of the motif 5'-PuPuC(A/T)(A/T)GPyPyPy-3'. This sequence motif suggests that p53 may bind to DNA as a tetramer. Binding as a tetrameric complex explains the fact that mutant p53 proteins act in a dominant manner. They are present in complexes with wild type p53 and alter the function of the normal tetramer.

Like pRB, p53 forms a complex with SV40 large T antigen, as well as the E1B transforming protein of adenovirus and E6 protein of human papilloma viruses. Complexing with these tumor antigens increases the stability of the p53 protein. This increased stability of p53 is characteristic of mutant forms found in tumor lines. The complexes of T antigens and p53 renders p53 incapable of binding to DNA and inducing transcription. A cellular protein, originally identified in a spontaneous transformed mouse cell line and termed MDM2, has been shown to bind to p53. Complexing of p53 and MDM2 results in loss of p53 mediated trans-activation of gene expression. Significantly, amplification of the MDM2 gene is observed in a significant fraction of most common human sarcomas.

Phosphorylation also regulates the activity of p53. The level of p53 is low after mitosis but increases during G<sub>1</sub>. During S phase the protein becomes phosphorylated by the M-phase cyclin-CDK complex of the [cell cycle](#) and also by casein kinase II (CKII). Sequences at the N-terminus of the p53 protein function as a transcription activator indicating the role of p53 in the transcription of genes involved in suppression of cell growth. One major cell-cycle regulating gene that is a target for p53 is the CDK inhibitory protein (CIP), p21<sup>CIP</sup>. Activation of p53 results in increased expression of p21<sup>CIP</sup> with a resultant arrest in the G<sub>1</sub> and G<sub>2</sub> phase of the cell cycle.

Additionally, p53 protein has been shown to block the binding of DNA polymerase- $\alpha$  to SV40 large T, blocking replication of SV40 DNA. It is suggested that p53 may also regulate the initiation of DNA synthesis. Due to the involvement of p53 in both transcription and DNA replication, the various mutants of p53 may affect these properties in different ways. This may account for why some mutants lose tumor suppressor activity while others behave as dominant oncogenes.

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[More detailed information on p53](#)

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## Retinoblastoma (RB)

In the familial form of this disease individuals inherit a mutant, loss of function allele from an affected parent. A subsequent later somatic mutational event inactivates the normal allele resulting in

retinoblastoma development. This leads to an apparently dominant mode of inheritance. The requirement for an additional somatic mutational event at the unaffected allele means that penetration of the defect is not always complete.

In sporadic forms of tumors involving the RB locus 2 somatic mutational events must occur, the second of which must occur in the descendants of the cell receiving the first mutation. This combination of mutational events is extremely rare.

The locus of the RB gene, identified cytogenetically, is chromosome 13q14.1. A 4.7 kb RB transcript has been identified (by chromosomal walking and subsequent Northern blotting with genomic DNA probes) and subsequently cloned. The RB gene encompasses 27 exons that span 180 kb of chromosome 13. Two of the introns in this gene are extremely large, 35 kb and 70 kb. The RB RNA encodes a p110 kDa protein (pRB) of 928 amino acids. pRB is a nuclear localized phosphoprotein. pRB is not detectable in any retinoblastoma cells. However, surprisingly detectable levels of pRB can be found in most proliferating cells even though there is a restricted number of tissues affected by mutations in the RB gene (i.e. retina, bone and connective tissue).

Many different types of mutations occur to result in loss of RB function. The largest percentage (30%) of retinoblastomas contain large scale deletions. Splicing errors, point mutations and small deletions in the promoter region have also been observed in some retinoblastomas.

The germ line mutations at RB occur predominantly during spermatogenesis as opposed to oogenesis. However, the somatic mutations occur with equal frequency at the paternal or maternal locus. In contrast, somatic mutations at RB in sporadic osteosarcomas occur preferentially at the paternal locus. This may be the result of genomic imprinting.

The major function of pRB is in the regulation of [cell cycle](#) progression. Its ability to regulate the cell cycle correlates to the state of phosphorylation of pRB. Phosphorylation is maximal at the start of S phase and lowest after mitosis and entry into G<sub>1</sub>. Stimulation of quiescent cells with mitogen induces phosphorylation of pRB, while in contrast, differentiation induces hypophosphorylation of pRB. It is, therefore, the hypophosphorylated form of pRB that suppresses cell proliferation. One of the most significant substrates for phosphorylation by the G<sub>1</sub> cyclin-CDK complexes that regulate progression through the cell cycle is pRB. pRB forms a complex with the E2F family of transcription factors, a result of which renders E2F inactive. When pRB is phosphorylated by G<sub>1</sub> cyclin-CDK complexes it is released from E2F allowing E2F to transcriptionally activate genes. In the context of the cell cycle, E2F increases the transcription of the S-phase cyclins as well as leads to increases in its' own transcription.

One element in the growth suppressive pathway of pRB involves the c-MYC gene. Proliferation of keratinocytes by TGF- $\beta$  is accompanied by suppression of c-MYC expression. The inhibition of c-MYC expression can be abrogated by introducing vectors that express the SV40 and adenovirus large T antigens which bind pRB. Therefore, a link exists between TGF- $\beta$ , pRB and c-MYC expression in keratinocytes.

Transformation by the DNA tumor viruses, SV40, adeno, polyoma, human papilloma and BK is accomplished by binding of the transforming proteins of these viruses to pRB when pRB is in the hypophosphorylated (and thus the proliferation inhibitory) state.

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## Wilms Tumor (WT1)

Genetic evidence indicates that at least 3 distinct loci may be involved in the development of Wilms tumors, a kidney cancer found in children. Either one (unilateral) or both (bilateral) kidneys can be involved. Sporadic evolution of Wilms tumors is associated with chromosomal deletions, identified cytogenetically, at both 11p13 and 11p15. The 11p15 deletions may involve the IGF-2 or c-Ha-RAS loci. There are also familial forms of Wilms tumors that do not involve either locus. These studies suggest that lesions at 3 different loci could lead to Wilms tumors.

Only a single candidate locus has thus far been characterized. The potential Wilms tumor gene at 11p13 is found in a deleted region of about 345 kb. This region contains a single transcription unit identified as WT1 that spans 50-60 kb, contains 10 exons and specifies a 3-kb mRNA. The WT1 gene codes for a protein that is highly conserved evolutionarily suggesting an important role for this gene. Several functional domains have been identified in the WT1 protein. The first hint of the potential function for WT1 came from the identification of 4 zinc finger domains suggesting it to be a transcription factor. There is a potential leucine zipper motif in the center of the protein indicating that WT1 may associate other leucine zipper containing proteins. The NH<sub>2</sub>-terminal 180 amino acids are involved in self-association. Separate domains have also been characterized that are involved in autonomous repression and activation of transcription.

Expression of WT1 is very restrictive, unlike the RB gene which is expressed everywhere. Wilms tumors that are homozygous for 11p13 deletions do not contain WT1 mRNA. Most other Wilms tumors show high expression of WT1 mRNA but these are likely to be produced from mutated WT1 genes.

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## Neurofibromatosis Type 1 (NF1)

All cases of neurofibromatosis arise by inheritance of a mutant allele. Roughly 50% of all affected individuals carry new mutations which appear to arise paternally, possibly reflecting genomic imprinting. Germ line mutations at the NF1 locus result in multiple abnormal melanocytes (café-au-lait spots) and benign neurofibromas. Some patients also develop benign pheochromocytomas and CNS tumors. A small percentage of patients develop neurofibrosarcomas which are likely to be Schwann cell derived.

Assignment of the NF1 locus to chromosome 17q11.2 was done by linkage studies of affected pedigrees. The NF1 locus is extremely large as is the transcript encoded by the locus. The mRNA is 11-13 kb and contains a 7.5 kb coding region. The protein encoded is 2485 amino acids and shares striking homology to rasGAP. The NF1 protein has been given the name **neurofibromin**. Expression of NF1 is observed in all tissues thus far examined.

Development of benign neurofibromas versus malignant neurofibrosarcomas may be the difference between inactivation of one NF1 allele versus both alleles, respectively. However, changes other than at the NF1 locus are clearly indicated in the genesis of neurofibrosarcomas. A consistent loss of genetic material on the short arm of chromosome 17 is seen in neurofibrosarcomas but not neurofibromas. The losses at 17p affect the wild type p53 locus and may be associated with a mutant p53 allele on the other chromosome.

Characterization of the NF1 protein was carried out by generating antibodies against both fusion

proteins and synthetic peptides. These antibodies specifically recognize a 220kDa protein, called **neurofibromin**, in both human and rat spinal cord. Neurofibromin is most abundant in the nervous system. Immunostaining of tissue sections indicates that neurons, oligodendrocytes, and nonmyelinating Schwann cells contained neurofibromin, whereas astrocytes and myelinating Schwann cells do not. In schwannoma cell lines from patients with neurofibromatosis, loss of neurofibromin is associated with impaired regulation of the GTP-bound form of the proto-oncogene RAS (GTP-RAS). Analysis of other neural crest-derived tumor cell lines showed that some melanoma and neuroblastoma cell lines established from tumors occurring in patients without neurofibromatosis also contained reduced or undetectable levels of neurofibromin, with concomitant genetic abnormalities of the NF1 locus. In contrast to the schwannoma cell lines, however, GTP-RAS was appropriately regulated in the melanoma and neuroblastoma lines that were deficient in neurofibromin. These results demonstrate that some neural crest tumors not associated with neurofibromatosis have acquired somatically inactivated NF1 genes and suggested a tumor-suppressor function for neurofibromin that is independent of RAS GTPase activation.

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## Familial Adenomatosis Polyposis (FAP)

Somatic mutations in the adenomatous polyposis coli (APC) gene appear to initiate colorectal cancer development in the general population, whereas it is germ line mutations that are responsible for familial adenomatous polyposis (FAP). The APC gene exhibits a dominant pattern of inheritance. Multiple colonic polyp development characterizes the disease. These polyps arise during the second and third decades of life and become malignant carcinomas and adenomas later in life. Genetic linkage analysis assigned the APC locus to 5q21. This region of the chromosome is also involved in nonfamilial forms of colon cancer. FAP adenomas appear as a result of loss-of function mutations to the APC gene. This is characteristic of tumor suppressors.

Identification of the APC gene was aided by the observation that 2 patients contained deletions at the locus spanning 100 kb of DNA. Three candidate genes in this region, DP1, SRP19 and DP2.5 were examined for mutations that could be involved in APC. The DP2.5 gene has sustained 4 distinct mutations specific to APC patients indicating this to be the APC gene. To date, more than 120 different germ line and somatic mutations have been identified in the APC gene. The vast majority of these mutations lead to COOH-terminal truncation of the APC protein.

The APC gene contains 15 exons spanning approximately 125 kb of DNA encoding an 8.5 kb coding region in the mRNA. Northern blotting detects an RNA of around 10 kb. An alternative form of exon 9 (9A) was also found that splices into the interior of exon 9 removing 101 amino acids from the full length APC transcript. The protein coding region of the APC gene is also extremely large encompassing 2844 amino acids. No similarities to known proteins was found except for several stretches of sequence related to intermediate filament proteins.

Using antibodies specific for the NH<sub>2</sub>-terminus of APC, it is possible to co-precipitate additional APC-associated proteins. One of these APC-associated proteins is  $\beta$ -catenin. The catenins are a family of proteins that interact with the cytoplasmic portion of the cadherins (cell-cell adhesion family of proteins), thus linking the cadherins to the actin cytoskeleton.

Catenins are equally important in the signaling cascade initiated by the [Wnt family of proteins](#) that

are involved in embryonic patterning, development of the nervous system. The Wnt proteins are secreted factors that interact with cell-surface receptors. Wnt-receptor interaction induces the activity of the cytoplasmic phosphoprotein dishevelled. Activated dishevelled inhibits the serine/threonine kinase glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ). When GSK-3 $\beta$  is inhibited,  $\beta$ -catenin becomes hypophosphorylated. The hypophosphorylated form of  $\beta$ -catenin migrate to the nucleus and interacts with transcription factors (in particular with T-cell factor/lymphoid enhancer-binding factor-1 (Tcf/LEF-1), thereby, inducing expression of various genes. The suspected role of APC in this pathway is to bind phosphorylated  $\beta$ -catenin. The APC- $\beta$ -catenin complex stimulates the breakdown of  $\beta$ -catenin. Therefore, mutations which lead to a loss of APC, or to a loss of the portion of the APC protein that interacts with  $\beta$ -catenin, would lead to constitutive activation of Tcf/LEF-1 and unrestricted growth.

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## Deleted in Colon Carcinoma (DCC)

Loss of heterozygosity (LOH) on chromosome 18 is frequently observed in colorectal carcinomas (73%) and in advanced adenomas (47%), but only occasionally in earlier-stage adenomas (11 to 13%). The area of chromosome 18 which is observed to be lost resides between 18q21.3 and the telomere. A 370 kbp stretch of DNA from the region of 18q suspected to contain the tumor suppressor gene was cloned. Expressed exons were used as probes for screening cDNA libraries to obtain clones that encoded a gene which was given the name **DCC ('deleted in colorectal carcinomas')**. A YAC contig, containing the entire DCC coding region, has been characterized showing that the DCC gene spans approximately 1.4 Mbp and contains 29 exons

The expression of the DCC gene has been detected in most normal tissues, including colonic mucosa. Somatic mutations have been observed within the DCC gene in colorectal cancers. The types of mutations seen included a homozygous deletion of the 5' end, a point mutation within one of the introns, and 10 examples of DNA insertions within a 170 bp fragment immediately downstream of one of the exons.

Evaluation of sporadic colon cancers for allelic deletions defined an area of chromosome 18 that included two candidate tumor suppressor. One was **DPC4** (see Table above) and the other was DCC. DPC4 is deleted in up to one-third of cases assayed and DCC, or a closely linked gene, was deleted in the remaining tumors. Tumor suppressor genes located on chromosome 17p and 18q are critically involved in the development of most gastric cancers. Involvement of DCC may be rather selective for gastrointestinal cancers. Loss of DCC gene expression is also an important factor in the development or progress of pancreatic adenocarcinoma.

The DCC protein is a transmembrane protein of the immunoglobulin superfamily and has structural features in common with certain types of cell-adhesion molecules, including **neural-cell adhesion molecule (N-CAM)**. It is known that the establishment of neuronal connections requires the accurate guidance of developing axons to their targets. This guidance process involves both attractive and repulsive cues in the extracellular environment. The **netrins** and **semaphorins** are proteins that can function as diffusible attractants or repellents for developing neurons. However, the receptors and signal transduction mechanisms through which they produce their effects are poorly understood. Netrins are chemoattractants for commissural axons in the vertebral spinal cord. Recent work has shown that DCC is



expressed on spinal commissural axons and possesses netrin-1-binding activity. This suggests that DCC is a receptor or a component of a receptor that mediates the effects of netrin-1 on commissural axons. Genetic evidence showing an interaction between DCC and netrin homologs in *C. elegans* (the **UNC-40** protein) and *Drosophila melanogaster* (the **frazzled** protein) supports the role of DCC as a receptor in the axonal guidance pathway. Mice carrying a null allele of DCC harbor defects in axonal projections that are similar to those seen in netrin-1-deficient mice, further supporting the interaction between DCC and axon development. However, the DCC-deficient mice exhibited no effects on intestinal growth, differentiation or morphogenesis which fails to demonstrate a tumor-suppressor role for DCC.

DCC has been shown to induce apoptosis in the absence of ligand binding, but blocks apoptosis when engaged by netrin-1. Furthermore, DCC is a caspase substrate, and mutation of the site at which caspase-3 cleaves DCC suppresses the pro-apoptotic effect of DCC completely. DCC may function as a tumor-suppressor protein by inducing apoptosis during metastasis or tumor growth beyond the local blood supply, both of which are conditions that lack the DCC ligand. This would likely occur through functional caspase cascades leading to cleavage of DCC.

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## von Hippel-Lindau Syndrome (VHL)

Like many diseases caused by loss of tumor suppressor gene activity, individuals afflicted with von Hippel-Lindau syndrome (VHL) inherit one normal copy and one mutant copy of the VHL gene. As a consequence of somatic mutation or loss of the normal VHL gene, individuals are predisposed to a wide array of tumors that include renal cell carcinomas, retinal angiomas, cerebellar hemangioblastomas and pheochromocytomas. In addition, some individuals with VHL sustain somatic alterations to both wild-type genes. This latter phenomenon is evident in the majority of sporadic clear cell renal carcinomas.

One characteristic feature of tumors from VHL patients is the high degree of vascularization, primarily as a result of the constitutive expression of the **vascular endothelial growth factor (VEGF)** gene. Many of the **hypoxia-inducible factor (HIF)** controlled genes are also constitutively expressed in these tumors whether or not high oxygen levels are present. In addition, the VHL gene has been shown to be required for cells to exit out of the **cell cycle** during nutrient depletion.

The role of the protein encoded by the VHL locus (pVHL) has been deduced from studies on the alterations in HIF control of hypoxia-inducible genes in VHL tumors. HIF is composed of an  $\alpha$ -subunit and a  $\beta$ -subunit. The  $\alpha$ -subunit is encoded by one of three genes (HIF1 $\alpha$ , HIF2 $\alpha$ , and HIF3 $\alpha$ ). The  $\beta$ -subunit (HIF1 $\beta$ ) is also known as the **aryl hydrocarbon receptor nuclear translocator, ARNT**. Normally the HIF $\alpha$  subunits are degraded in the presence of oxygen due to the regulated addition of multiple ubiquitin molecules. Polyubiquitination is a key modification directing proteins for rapid degradation by the proteasome machinery. Cells lacking pVHL do not ubiquitinate HIF1 $\alpha$ . These observations led to the identification that pVHL was a component of a ubiquitin ligase complex that is responsible for polyubiquitination of HIF1 $\alpha$  subunits.

pVHL has also been implicated in additional functions. With respect to cell cycle control, pVHL has been implicated in the downregulation of cyclin D1 which would explain why VHL tumor cells fail to exit the cell cycle on nutrient deprivation.

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## Introduction to the Wnt Family of Proteins

The Wnt genes constitute a large family of cysteine-rich, secreted glycoproteins that are involved in critical aspects of early embryonic development. The term **wnt** is an amalgam of **wingless** (Wg) and **int**. The Wg gene was identified as a locus in *Drosophila* required in each segment for the establishment of normal pattern. In the absence of Wg, posterior pattern elements (i.e. naked cuticle) are replaced by mirror image duplications of anterior denticle bands. The **int** loci (there were 2 termed **int-1** and **int-2**) were originally identified in mice as sites of frequent integration (hence **int** loci) of the Moloney murine leukemia virus. Subsequently it was shown that the murine int-1 locus was homologous to *Drosophila* Wg and the family name was changed to Wnt to reflect both origins. To date there have been at least 18 mammalian members of the Wnt family (visit the [Roeland Nusse lab](#) for an extensive listing of Wnt genes). Several of the mouse Wnt genes are described in the Table below.

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## Wnt Expression Patterns in Mice

| Gene  | Embryonic Expression | Adult Expression |
|-------|----------------------|------------------|
| Wnt-1 | brain, spinal cord   | testes           |

|        |                                                  |                           |
|--------|--------------------------------------------------|---------------------------|
| Wnt-2  | ventral-lateral<br>mesoderm, heart,<br>allantois | lung, brain, heart        |
| Wnt-3  | brain, spinal cord, limbs                        | thalamus, Purkinje cells  |
| Wnt-3a | brain, primitive streak                          | lung                      |
| Wnt-4  | not determined                                   | brain, lung               |
| Wnt-5a | ventral brain, spinal cord                       | heart, lung               |
| Wnt-5b | not restricted                                   | heart, brain, liver, lung |
| Wnt-6  | not determined                                   | testes                    |
| Wnt-7a | not determined                                   | brain, lung               |
| Wnt-8  | not determined                                   | brain                     |

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## Classification of Wnt Activities

Results from experiments designed to ectopically express Wnt proteins in mammary epithelial cells and in *Xenopus* embryos allowed for the loose division of the Wnts into two classes. One class are the axis inducing and transforming **Wnt-1 class**. All the Wnts of this class are capable of transforming mammalian cells in culture and of inducing a secondary neural axis when ectopically expressed in the future ventral side of early frog embryos. The Wnts in this first class are the mouse Wnt-1, -3a and -7a genes and the *Xenopus* Xwnt-1, -3a, -8 and -8b genes.

The second class of Wnts are the non-transforming **Wnt-5a class**. This class of Wnts cannot induce the transformed state in cells in culture and do not induce a secondary neural axis in frog embryos. The Wnts of this class include mouse the Wnt-4 and -5a genes and the *Xenopus* Xwnt-5a and -4 genes. Chimeric proteins generated by combining elements of Xwnt-8 (a Wnt-1 class protein) and Xwnt-5A (a Wnt-5a class protein) and then used for axis induction experiments in frog embryos, demonstrated that the different activities of the two Wnt classes is determined by the C-terminal portion of the proteins.

The Wnt proteins are capable of inducing cells to proliferate, to differentiate and to survive by signaling through both autocrine and paracrine pathways. The Wnt proteins exert these effects on cells and during early development by interacting with specific cell-surface receptors which in turn initiate a signaling cascade culminating with changes in gene expression. It is this Wnt-induced alteration in the expression of specific genes that leads to the multitude of effects observed for Wnt proteins.

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## Wnt Genes and Central Nervous System Development

By using the technique of *in situ* hybridization, several studies have suggested that Wnt signaling plays an important role during the early patterning of the neural tube. In mouse embryos, the timing of Wnt gene expression in the central nervous system (CNS) indicates that these genes likely assist in the regional specification of the neural tube, in particular in the forebrain region and the spinal cord. Wnt gene expression within the developing spinal cord exhibits particular patterns of restriction along the dorsal-ventral axis prior to the differentiation of neurons. Wnt-1, Wnt-3 and Wnt-3a are the earliest markers observed to be expressed along the dorsal midline of the developing spinal cord. Wnt-4 expression correlates with the region of the presumptive alar plate that is destined to produce sensory interneurons. Wnt-7a and Wnt-7b are expressed in the ventral spinal cord. Additionally, Wnt-4 is expressed in the ventral region of the floor plate. Since it has been shown that signals emanate from the notochord to induce the formation of the floor plate within the CNS (these are in the form of the inducing protein, **sonic hedgehog, shh**), the expression patterns of these Wnts suggest that they may play important roles in reinforcing the dorsal-ventral polarities of the CNS established by inducers such as shh. Overexpression of Wnt-1 within the spinal cord severely disrupts normal spinal cord morphology, most probably as a result of overproliferation. However, this effect does not interfere with the primary dorsal-ventral patterning of the spinal cord indicating that Wnts likely regulate the patterns of cell proliferation within the spinal cord not the actual patterns induced by other genes.

Several Wnts have been shown to be expressed in the developing forebrain. Wnt-5a and Wnt-7a are expressed in overlapping domains in the ventral and lateral diencephalon. Wnt-1, Wnt-3, Wnt-3a and Wnt-4 are expressed in the dorsal diencephalon. Wnt-7b is expressed throughout much of the diencephalon and the optic stalks. Wnt-3b and Wnt-7b expression extends into the telencephalon along the dorsal regions. These patterns of expression indicate a role for the Wnts as important regulators of the developing forebrain. The pattern of segmentation in the vertebrate hindbrain is divided into regions identified as rhombomeres. Evidence on the expression of numerous genes suggests the the forebrain may be similarly identifiable by specific patterns of segmentation. Wnt gene expression patterns generally coincide with these predicted segments of the forebrain. As an example, an early patch of Wnt-3a expression in the diencephalon appears to correspond to the presumptive pretectum (synencephalon). A domain of Wnt-3

expression correlates with the future dorsal thalamus. Expression of Wnts in the developing embryonic CNS clearly demarcates dorsal-ventral zones whose subdivisions are comparable to the roof, alar and basal plates of the more posterior neural tube.

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## Wnt Genes and Axis Specification

As indicated above the Wnts are divided into two classes determined in part on an ability, or lack thereof, to induce a secondary neural axis when ectopically expressed in the future dorsal side of frog embryos. The earliest experiments with Wnts and axis duplication were performed with Wnt-1. Subsequently, numerous *Xenopus* Wnt genes (identified as Xwnts) have been isolated and characterized. Three possible functions for Wnts in the establishment of axial pattern have been suggested.

One mechanism suggests that Wnts could assist in the initial establishment of dorsal versus ventral sides of the embryo very early following fertilization and be components of the dorsalizing or ventralizing centers of mesoderm induction. Induction of mesoderm has been established as being critically important in future neuralization of the embryo. A second mechanism has the Wnts acting as parts of the dorsal or ventral signals that modify the inherent character of pre-existing mesoderm. Lastly, Wnts could function by assisting in the production of specified mesoderm, as the primary axis information is established during gastrulation. Strong evidence more closely links Wnts to the latter two models.

Both Xwnt-8 and Xwnt-11 have the capacity to modify the character of mesoderm that has been induced by other signals. Xwnt-8 has the capability to dorsalize regions of the embryo when ectopically expressed. However, it is normally expressed on the ventral side of the embryo during the blastula and gastrula stages. When Xwnt-8 is expressed dorsally it has the capability to ventralize the embryo indicating that it is a good candidate for a ventralizing signal. Conversely, Xwnt-11 is capable of dorsalizing previously induced mesoderm and the RNA is localized to the marginal zone (future mesodermal tissue is derived from this region of the embryo) with highest levels being found dorsally. When embryos are exposed to *uv*-irradiation on the vegetal hemisphere they are completely ventralized and form no dorsal structures. Xwnt-11 is capable of partially rescuing the dorsal axis in *uv*-ventralized embryos. Both Xwnt-8 and Xwnt-11 are incapable of inducing mesoderm on their own, therefore, the above described results indicate that both Wnts must play a role in the modification of pre-existing mesoderm. Wnt proteins are important mediators of axis establishment when the primary patterns of the embryo are determined during gastrulation. Direct evidence for this has been shown in mice in which Wnt genes have been knocked out by targeted



disruption. Expression of Wnt-3a is first detected throughout the primitive streak of the mouse embryo in regions fated to generate dorsal mesoderm. Targeted disruption of the mouse Wnt-3a gene results in a severe truncation of the body axis posterior to the forelimbs. The most affected tissue being dorsal mesoderm; the notochord is disrupted and the tailbud fails to form. These data demonstrate that Wnt-3a is required for the production of somitic mesoderm and for the generation of all new embryonic mesoderm by the late primitive streak stage.

The Wnt-5a and -5b genes are also expressed in the primitive streak during gastrulation in the mouse. Overexpression of Xwnt-5a in frog embryos leads to complex head and tail malformations. Xwnt-5a does not ectopically induce mesoderm or change the type of pre-existing mesoderm but can inhibit the normal morphogenetic movements that occur during gastrulation. The chicken Cwnt-8c gene is expressed in the posterior marginal zone during gastrulation and later in the primitive streak and Hensen's node. When Cwnt-8c is ectopically expressed in frog embryos it can dorsalize mesoderm and induce axis duplication.

Although no Wnts are candidates for endogenous mesoderm induction or initial patterning, it is clear that this family of proteins is capable of modifying pre-existing mesoderm (Xwnt-8, Xwnt-11 and Cwnt-8c) and affecting gastrulation (Wnt-3a, Wnt-5a, Wnt-5b and Cwnt-8c).

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## Wnt Signaling Pathways

Wnts are soluble secreted factors that exert the effects described above through interaction with cell surface receptors. Wnts bind to receptors of the **Frizzled** family, named after the *Drosophila* tissue polarity gene **frizzled**. There are currently 8 mammalian frizzled proteins. Structurally the Frizzled receptors have an extracellular Wnt-binding domain, seven transmembrane-spanning regions and an intracellular C-terminal tail. The presence of the seven transmembrane-spanning domains places the Frizzled proteins in the class of receptors known as the **serpentine receptors** (see the [Signal Transduction](#) page). This class includes the receptors for serotonin (5-HT), angiotensin and glucagon for example. The serpentine receptors couple ligand binding to activation of tripartite G-proteins. Additional components of the Frizzled receptor-Wnt interaction pathway are the soluble Wnt inhibitors which consist of secreted proteins containing a cysteine-rich domain (CRD) similar to that in the ligand binding domain of the Frizzled receptors. This class of Wnt inhibitory proteins are termed **frizzled receptor-like proteins (FRPs)**. By binding Wnts, FRP proteins can sequester Wnts away from the cell-surface Frizzled receptors and thereby, reduce the effective concentration of available Wnt protein.

Direct evidence for the interaction of Wnts and Frizzled receptors came from

several studies. When *Drosophila* frizzled-2 (Dfz2) was expressed in fly cells in culture, the cells attained the ability to bind Wg protein. When rat Frizzled 1 (RFz1) or human Frizzled 5 (Hfz5) were expressed in frog embryos, Xwnt-5a and Xwnt-8 activated axis duplication through interaction with these exogenous receptors. The presence of a large family of Frizzled receptors suggests that there may exist the ability to differentially bind various Wnt proteins. Indeed, evidence shows that Dfz1, Dfz2, MFz4 (mouse Frizzled 4), MFz7, MFz8 and HFz5 but not MFz3 were able to bind Wg in fly cell in culture. Since Wg can bind to at least 6 different Frizzled receptors it is likely that multiple Wnts can be bound by any given Frizzled receptor and that many Wnts will be able to bind to multiple Frizzled receptors. Since endogenous Wnt expression patterns overlap (particularly in the mouse hindbrain) there may be functional redundancy in Wnt expression.

It is also probable that overlapping patterns of Wnt expression may allow for different Wnts to antagonize each other during development. Evidence for this latter possibility has been most clearly demonstrated in studies of axis duplication in *Xenopus*. Prior expression of Xwnt-5a blocks the axis inducing activity of Xwnt-8. The precise mechanism for this interference is not clear. It could be that Xwnt-5a blocked Xwnt-8 action by receptor competition or by blocking the action of intracellular signaling components.

Since the Frizzled receptors have no enzymic motifs on their intracellular domains, the Wnt signals are likely transmitted through receptor coupling to G-proteins as for the other members of the serpentine receptor family. One candidate family of G-proteins, shown by genetic studies in *Drosophila* to be downstream of Wg, is that characterized by **Dishevelled (dsh)**. Members of the Dishevelled family encode cytoplasmic proteins with no known enzymic functions. Three homologues have been identified in mice (Dvl-1, Dvl-2 and Dvl-3) and shown to have overlapping patterns of expression. This overlapping pattern of expression suggests redundancy of function for Dishevelled proteins. This supposition is supported by the fact that targeted deletion of Dvl-1 results in mice that are structurally normal. There are three regions of sequence homology between the different Dishevelled family members. The N-terminally conserved domain is termed the Dsh homology domain. This domain is similar to the C-terminus of **Axin** (encoded by the *Drosophila* **fused** gene) a protein shown to interfere with the formation of the endogenous dorsal axis and to prevent Xwnt-8 from inducing an ectopic axis. The central portion of Dishevelled proteins contains a PDZ domain which is present in many other proteins and has been shown to bind C-terminal peptides of the sequence motif X-(S/T)-X-V (where X is any amino acid). The *Drosophila* Dsh PDZ domains have been shown to be essential for downstream signaling in frog embryos. The third homology domain in Dishevelled proteins is the DEP domain. This domain has similarity to **GTPase activating proteins (GAPs)** and **gunaine**

**nucleotide exchange factors (GEFs)**. The DEP domain may, therefore, be important for recruiting downstream targets in G-protein-coupled pathways. The next step in the Wnt signaling cascade is the inhibition of the serine/threonine kinase, **glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ )** by Dishevelled. GSK-3 $\beta$  is a cytoplasmic kinase originally identified for its role in **glycogen metabolism**. GSK-3 $\beta$  has been shown to be the homolog of the *Drosophila* gene zeste-white-3 (zw3). Genetic studies showed that zw3 functioned downstream of dsh and probably inactivated Wg signaling. The mammalian GSK-3 $\beta$  protein can partially compensate for loss of zw3 during *Drosophila* embryogenesis. The fact that activation of Dishevelled leads to an inactivation of GSK-3 $\beta$  activity was demonstrated by the failure of GSK-3 $\beta$  to phosphorylate the microtubule-associated protein tau in Dvl-1 overexpression studies.

When a dominant negative form of *Xenopus* GSK-3 $\beta$  (dnGSK-3 $\beta$ ), which lacks the kinase domain, is expressed in *Xenopus* embryos an ectopic axis forms which closely resembles the axis induced by ectopic expression of Xwnt-8 and Xdsh. In contrast, wild-type XGSK-3 $\beta$  is able to suppress the ectopic axis inducing activities of Xwnt-8 and Xdsh. This suggests that the normal function of GSK-3 $\beta$  is to suppress Wnt signaling from Dsh.

The consequences of Wnt-induced inhibition of GSK-3 $\beta$  activity is that the next protein identified in the cascade,  $\beta$ -catenin, is stabilized.  $\beta$ -Catenin is a member of a multigene family of proteins characterized by the presence of a domain found in the *Drosophila* Armadillo (Arm) gene. Genetic studies showed that Arm functions downstream of zw3/GSK-3 $\beta$ . Evidence indicates that Wnt inhibition of GSK-3 $\beta$  activity prevents the turnover of  $\beta$ -catenin leading to its accumulation. The loss of zw3 function or the use of dnGSK-3 $\beta$  stabilizes  $\beta$ -catenin and Arm levels in both *Drosophila* and *Xenopus* indicating that GSK-3 $\beta$  normally functions to lower  $\beta$ -catenin levels. Additionally, the loss of zw3 or the use of dnGSK-3 $\beta$  leads to a reduction in the phosphorylation of Arm and  $\beta$ -catenin. These observations indicate that phosphorylation of  $\beta$ -catenin leads to its degradation, by an as yet unidentified mechanism.

There are several structural domains in the  $\beta$ -catenins. The N-terminus is the site of GSK-3 $\beta$ -mediated phosphorylation. Another region binds to  $\alpha$ -catenin, a cytoplasmic protein with similarity to vinculin which links actin filaments to the adherens junctions. The central region of the protein contains a positively charged groove that is suspected to be involved in interactions with acidic regions in the adenomatous polyposis coli (APC) locus encoded protein, the transcription factor TCF/LEF-1 (T-cell factor/Lymphocyte enhancer factor-1) and the cadherin family of calcium-dependent cell-adhesion molecules. The APC gene is a tumor-suppressor

linked to inherited and sporadic forms of colon cancer (see the [Tumor Suppressor](#) page for more information on the connection between APC and Wnts).

Mutational analyses have shown that regions of  $\beta$ -catenin that are required for Wnt signaling are separable from those required for cell adhesion suggesting that cadherins are not required for Wnt signaling. However, changes in cadherin expression can affect Wnt signaling and conversely, Wnt signaling can affect cell adhesion. This could possibly lead to altered levels of free  $\beta$ -catenin and thus indirectly alter the level of  $\beta$ -catenin-TCF complexes.

It is not clear whether APC is directly involved in the Wnt signal cascade. However,  $\beta$ -catenin has been shown to directly interact with APC and mutations in APC that led to colon cancer result in a protein that does not bind to  $\beta$ -catenin. Complexes of APC and  $\beta$ -catenin bind to GSK-3 $\beta$ . These results suggest that APC functions as a molecular scaffold coordinating the activities of  $\beta$ -catenin and GSK-3 $\beta$  such that APC may assist in the ability of GSK-3 $\beta$  to phosphorylate  $\beta$ -catenin leading to its degradation. Lending confusion to the role of APC in Wnt signaling is the observation that overexpression of APC in frog embryos positively induced axis duplication instead of suppressing it as expected from the supposition that APC enhances GSK-3 $\beta$  activity.

The final link in the Wnt signal cascade is the interaction between  $\beta$ -catenin and the transcription factors of the TCF/LEF-1 family. Proteins of the TCF family bind to DNA and contain a high-mobility group (HMG) domain. Murine family members include LEF-1 and TCF-1 which function in lymphopoiesis and epithelial-mesenchymal interaction, respectively. There are three isoforms of the *Xenopus* XTCTF3 gene. The pangolin gene in *Drosophila* and the pop-1 gene of *C. elegans* are family members. The N-terminus of TCF mediates the interaction with  $\beta$ -catenin. TCF: $\beta$ -catenin complexes then enter the nucleus and interact with TCF DNA elements. Several candidate TCF: $\beta$ -catenin target genes have thus far been identified. A few are the *Xenopus* homeobox genes ***siamois*** and ***Twin***, the secreted protein ***nodal related 3*** and ***fibronectin***. The promoters of these genes all contain TCF/Lef binding sites. For a more complete updated listing, visit the [Wnt Web site](#) maintained by Roel Nusse which has a [table of Wnt/ \$\beta\$ -catenin targets](#).

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## TGFs- $\beta$ /BMP Family Member Signaling and the Smads

TGF- $\beta$  was originally characterized as a protein (secreted from a tumor cell line) that was capable of inducing a transformed phenotype in non-neoplastic cells in culture. This effect was reversible, as demonstrated by the reversion of the cells to



a normal phenotype following removal of the TGF- $\beta$ . Subsequently, many proteins homologous to TGF- $\beta$  have been identified. The four closest relatives are TGF- $\beta$ 1 (the original TGF- $\beta$ ) through TGF- $\beta$ 5 (TGF- $\beta$ 1 = TGF- $\beta$ 4). All four of these proteins share extensive regions of similarity in their amino acids. Many other proteins, possessing distinct biological functions, have stretches of amino-acid homology to the TGF- $\beta$  family of proteins, particularly the C-terminal region of these proteins where a conserved series of six cysteine residues is found. There are at least 42 members in humans.

The TGF- $\beta$ -related family of proteins includes the **activin** and **inhibin** proteins. There are activin A, B and AB proteins, as well as an inhibin A and inhibin B protein. The **Mullerian inhibiting substance** (MIS) is also a TGF- $\beta$ -related protein, as are members of the **bone morphogenetic protein** (BMP) family of growth-regulatory factors first characterized as having effects on bone but known to exert numerous effects in a variety of different systems. Two additional groups of related proteins belonging to the TGF- $\beta$  family are the **nodal** proteins and the **growth and differentiation factor** (GDF) proteins. Based upon sequence similarities of the various members of the TGF- $\beta$  family, the family can be divided into two subfamilies: the TGF- $\beta$ /activin/nodal subfamily and the BMP/MIS/GDF subfamily.

TGFs- $\beta$  have proliferative effects on many mesenchymal and epithelial cell types. Under certain conditions TGFs- $\beta$  will demonstrate anti-proliferative effects on endothelial cells, macrophages, and T- and B-lymphocytes. Such effects include decreasing the secretion of immunoglobulin and suppressing hematopoiesis, myogenesis, adipogenesis and adrenal steroidogenesis. Several members of the TGF- $\beta$  family are potent inducers of mesodermal differentiation in early embryos, in particular TGF- $\beta$  and activin A.

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## TGF- $\beta$ Family Receptors

The TGF- $\beta$  family members interact with their cell-surface receptors in a dimeric form. There are several classes of cell-surface receptors that bind different TGFs- $\beta$  with differing affinities. There are also cell-type specific differences in receptor subtypes. Unlike the EGF, PDGF and FGF receptors, the TGF- $\beta$  family of receptors all have intrinsic serine/threonine kinase activity and, therefore, induce distinct cascades of signal transduction. The TGF- $\beta$  family receptors consist of a type I and a type II protein component. Currently there are 7 type I receptors and 5 type II receptors known to exist in the human genome. Ligand binding brings together the type I and type II components of the receptor. This interaction induces the type II



receptor to phosphorylate the type I receptor kinase domain. Activation of the type I kinase domain propagates the signal through phosphorylation of the Smad proteins (discussed below). The type I receptors contain a sequence motif: SGSGSG (termed the GS domain), that is the target for type II receptor phosphorylation. Adding to the potential complexity is the fact that specific ligand traps have been identified. These traps are secreted proteins that bind TGF- $\beta$  ligands in the circulation and thus interfere with the ability of ligand to dock with receptor.

| <b>Ligand</b>        | <b>Receptors</b>                       | <b>Ligand Traps</b>                                                         | <b>Accessory Receptors</b> | <b>Smads</b>                     |
|----------------------|----------------------------------------|-----------------------------------------------------------------------------|----------------------------|----------------------------------|
| TGF- $\beta$         | T $\beta$ R-II<br>T $\beta$ R-I (ALK5) | Latency-associated polypeptide (LAP)<br>Decorin<br>$\alpha$ 2-macroglobulin | Betaglycan                 | Smad2<br>Smad3<br>Smad4          |
| Inhibin              | ActR-IIB<br>ALK4                       |                                                                             | betaglycan                 | Smad2<br>Smad3<br>Smad4          |
| Activin              | ActR-IIB<br>ALK4                       | folistatin                                                                  |                            | Smad2<br>Smad3<br>Smad4          |
| Nodal<br>GDF1<br>Vg1 | ActR-IIB<br>ALK4, ALK7                 | DAN<br>Cerberus                                                             | Cripto                     | Smad2<br>Smad3<br>Smad4          |
| BMP2<br>BMP4<br>BMP7 | BMPR-II<br>ALK3, ALK6                  | DAN<br>Cerberus<br>Chordin/SOG<br>Follistatin<br>Noggin                     |                            | Smad1<br>Smad5<br>Smad8<br>Smad4 |
| BMP4<br>GDF5         | ActR-IIB<br>ALK3, ALK6                 | Chordin/SOG<br>Follistatin<br>Noggin                                        |                            | Smad1<br>Smad5<br>Smad8<br>Smad4 |
| BMP7                 | ActR-IIB<br>ALK2                       | Chordin/SOG<br>Follistatin<br>Noggin                                        |                            | Smad1<br>Smad5<br>Smad8<br>Smad4 |
| MIS                  | AMHR-II<br>ALK2, ALK3, ALK6            |                                                                             |                            | Smad1<br>Smad5<br>Smad8<br>Smad4 |

|              |                        |  |          |                                  |
|--------------|------------------------|--|----------|----------------------------------|
| TGF- $\beta$ | T $\beta$ R-II<br>ALK1 |  | Endoglin | Smad1<br>Smad5<br>Smad8<br>Smad4 |
|--------------|------------------------|--|----------|----------------------------------|

Ligand-receptor interactions within the TGF- $\beta$  family are characterized by two distinct modes. The BMP subfamily of ligands has highest affinity for the type I receptor and low affinity for the type II receptor. The TGF- $\beta$  subfamily has highest affinity for the type II receptor and low affinity for the type I receptor.

The ability of TGF- $\beta$  ligands to bind and activate receptors is tightly regulated. A group of soluble secreted proteins serves as ligand binding traps that sequester ligand preventing interaction with receptor. One ligand trap comprises the latency-associated polypeptide (LAP) which is actually the pro-region of the TGF- $\beta$  protein that remains noncovalently bound to the active domain. Other traps include the glycoprotein decorin and  $\alpha$ 2-macroglobulin which both bind to free TGF- $\beta$ . Follistatin traps activins and BMPs. Then there are the Noggin, Chordin/SOG (short gastrulation), and DAN/Cerberus families of traps that bind to the BMPs. Another member of the inhibitory class of proteins is BAMBI (**B**MP and **A**ctivin receptors **M**embrane **B**ound **I**nhibitor). BAMBI functions as a decoy receptor whose extracellular domain has sequence similarities to the type I receptors.

In addition to ligand traps that serve to regulate the action of the TGF-b proteins, there exist membrane anchored proteins that serve as accessory receptors or co-receptors. The membrane anchored proteoglycan, betaglycan, has long been known as the TGF- $\beta$  type III receptor. Betaglycan does not bind to the activins or the BMPs but does interact with inhibins facilitating interaction with the activin receptors. The betaglycan-related protein endoglin facilitates binding of TGF-b family members to the type I receptor ALK1 on endothelial cells. Cripto is a member of the CFC-EGF family of both secreted and membrane anchored proteins. Cripto mediates the binding of Nodal, Vg1, and GDF1 to the activin receptors.

Regulation of TGF- $\beta$  signaling at the level of the receptors also occurs intracellularly. The immunophilin, FKBP12 (a protein that binds to the immunosuppressive drug FK506) binds to the unphosphorylated GS domain of type I receptors preventing ligand-independent phosphorylation and activation of the type I receptor. Following activation the receptors can be negatively regulated by the inhibitory Smad (I-Smad) Smad7. Smad7 binding leads to the ubiquitination and degradation of the receptors. This involves the **Smad ubiquitination regulatory factors** (Smurfs) which are ubiquitin ligases.

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# The Role of Smads in TGF- $\beta$ Signaling

The first intracellular mediator of TGF- $\beta$  signaling was identified in the fruitfly *Drosophila melanogaster* and the locus was called **mothers against decapentaplegic** (MAD), decapentaplegic (Dpp) being the *D.m.* TGF- $\beta$  protein. Subsequently related proteins were found in other invertebrates (the worm genes were called **sma**) and in vertebrates. Using an amalgam of the worm and fly nomenclature the proteins were named **Smad** proteins.

There are three classes of Smads, the R-Smads (those that are activated as a consequence of phosphorylation by the kinase domain of the type I receptors), the I-Smads (inhibitory) and the Co-Smads. The R-Smads are Smads1, 2, 3, 5 and 8. Smad2 and Smad3 respond to signaling via the TGF- $\beta$  subfamily and Smad1, 5, and 8 to the BMP subfamily. The Co-Smad is Smad4. The R-Smads and Co-Smads contain two conserved structural domains termed MH1 (**MAD Homology** domain) and MH2. Once phosphorylated the R-Smads associate with the Co-Smad, Smad4, and the complex migrates to the nucleus where target gene transcription is affected. It is the MH1 domain of the R-Smads that exhibits DNA-binding activity. The MH2 domain, which is highly conserved among all Smad proteins, is responsible for receptor interaction.

Stimulation of cells by TGF- $\beta$  leads to both positive and negative regulation changes in the expression of several hundred different genes. Both the activation and repression events use the same set of activated Smad proteins. Smad access to target genes along with the recruitment of the various transcriptional co-activators and co-repressors necessary to effect the “proper” signals in response to TGF- $\beta$  is dependent upon cell-type specific interactions. Smad proteins thus cooperate with each other as well as with other DNA binding proteins to ultimately elicit the correct responses. Many different families of DNA-binding proteins have been shown to function as Smad partners. These include homeobox proteins, forkhead proteins, E-box proteins, CREBP (cAMP response element binding protein), Jun/Fos, and E2F.

Equally important are inhibitors of Smad activity. Members of the Ski family of protooncogenes, c-Ski and SnoN, have been shown to antagonize TGF- $\beta$  signaling via direct interaction with the Co-Smad, Smad4, and the R-Smads. Smads also bind to the transcriptional corepressor TGIF. TGIF recruits **histone deacetylase** (HDAC) and other repressors to transcriptional complexes resulting in transcriptional repression.

Termination of Smad signaling occurs via dephosphorylation of the R-Smads as well as ubiquitination and proteasome-mediated degradation. As discussed above, Smurfs target Smads for ubiquitination and thus destruction. As yet the phosphatase(s) responsible for Smad dephosphorylation has not been

identified.

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## Introduction

Stores of readily available glucose to supply the tissues with an oxidizable energy source are found principally in the liver, as glycogen. A second major source of stored glucose is the glycogen of skeletal muscle. However, muscle glycogen is not generally available to other tissues, because muscle lacks the enzyme glucose-6-phosphatase.

The major site of daily glucose consumption (75%) is the brain via aerobic pathways. Most of the remainder of is utilized by erythrocytes, skeletal muscle, and heart muscle. The body obtains glucose either directly from the diet or from amino acids and lactate via gluconeogenesis. Glucose obtained from these two primary sources either remains soluble in the body fluids or is stored in a polymeric form, **glycogen**. Glycogen is considered the principal storage form of glucose and is found mainly in liver and muscle, with kidney and intestines adding minor storage sites. With up to 10% of its weight as glycogen, the liver has the highest specific content of any body tissue. Muscle has a much lower amount of glycogen per unit mass of tissue, but since the total mass of muscle is so much greater than that of liver, total glycogen stored in muscle is about twice that of liver. Stores of glycogen in the liver are considered the main buffer of blood glucose levels.

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## Glycogenolysis

Degradation of stored glycogen (**glycogenolysis**) occurs through the action of glycogen phosphorylase. The action of phosphorylase is to phosphorolytically remove single glucose residues from  $\alpha$ -(1,4)-linkages within the glycogen molecules. The product of this reaction is glucose-1-phosphate. The advantage of the reaction proceeding through a phosphorolytic step is that:

- **1.** The glucose is removed from glycogen is an activated state, i.e. phosphorylated and this occurs without ATP hydrolysis.
- **2.** The concentration of  $P_i$  in the cell is high enough to drive the equilibrium of the reaction the favorable direction since the free energy change of the standard state reaction is positive.

The glucose-1-phosphate produced by the action of phosphorylase is converted to glucose-6-phosphate by phosphoglucomutase: this enzyme, like phosphoglycerate mutase (of [glycolysis](#)), contains a



phosphorylated amino acid in the active site (in the case of phosphoglucomutase it is a Ser residue). The enzyme phosphate is transferred to C-6 of glucose-1-phosphate generating glucose-1,6-phosphate as an intermediate. The phosphate on C-1 is then transferred to the enzyme regenerating it and glucose-6-phosphate is the released product.

As mentioned above the phosphorylase mediated release of glucose from glycogen yields a charged glucose residue without the need for hydrolysis of ATP. An additional necessity of releasing phosphorylated glucose from glycogen ensures that the glucose residues do not freely diffuse from the cell. In the case of muscle cells this is acutely apparent since the purpose in glycogenolysis in muscle cells is to generate substrate for glycolysis.

The conversion of glucose-6-phosphate to glucose, which occurs in the liver, kidney and intestine, by the action of glucose-6-phosphatase does not occur in skeletal muscle as these cells lack this enzyme. Therefore, any glucose released from glycogen stores of muscle will be oxidized in the glycolytic pathway. In the liver the action of glucose-6-phosphatase allows glycogenolysis to generate free glucose for maintaining blood glucose levels.

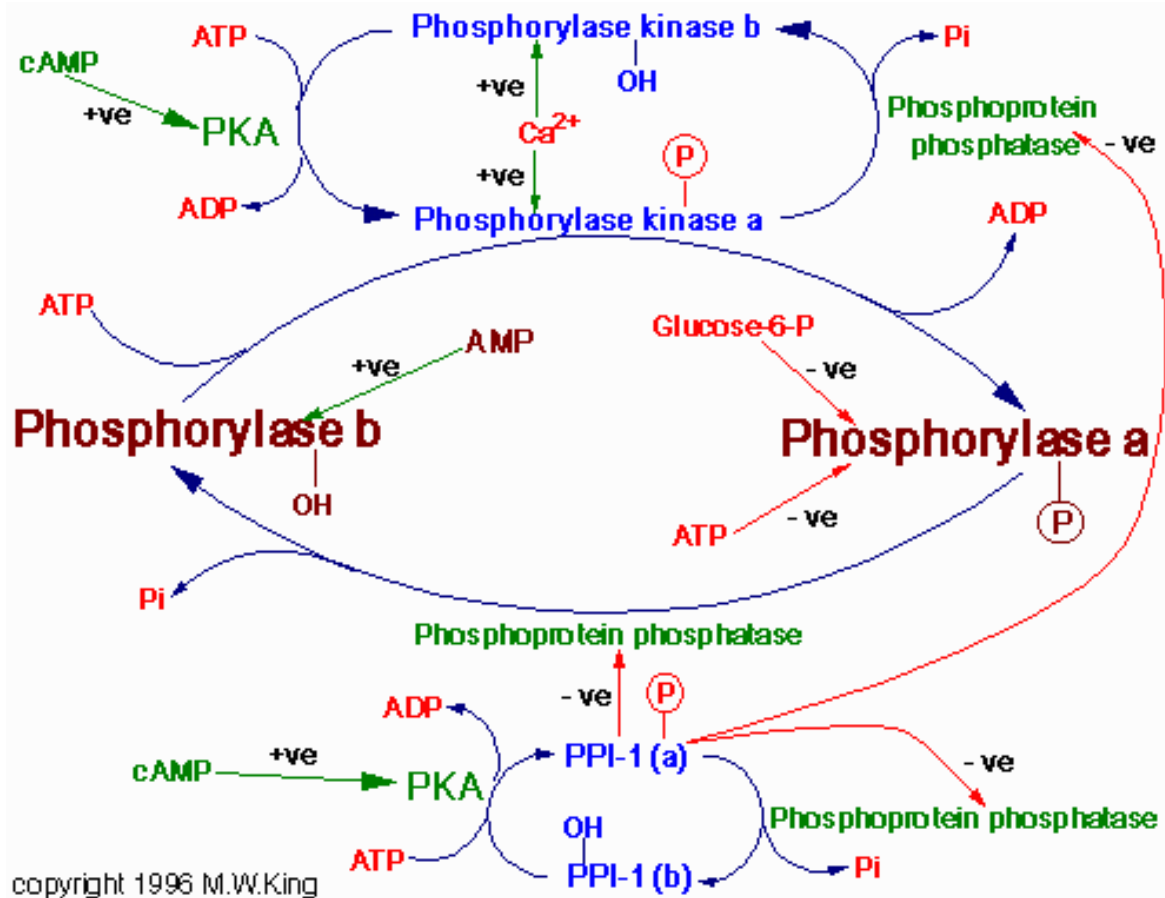
Glycogen phosphorylase cannot remove glucose residues from the branch points ( $\alpha$ -1,6 linkages) in glycogen. The activity of phosphorylase ceases 4 glucose residues from the branch point. The removal of these branch point glucose residues requires the action of debranching enzyme (also called glucan transferase) which contains 2 activities: glucotransferase and glucosidase. The transferase activity removes the terminal 3 glucose residues of one branch and attaches them to a free C-4 end of a second branch. The glucose in  $\alpha$ -(1,6)-linkage at the branch is then removed by the action of glucosidase. This glucose residue is uncharged since the glucosidase-catalyzed reaction is not phosphorylytic. This means that theoretically glycogenolysis occurring in skeletal muscle could generate free glucose which could enter the blood stream. However, the activity of hexokinase in muscle is so high that any free glucose is immediately phosphorylated and enters the glycolytic pathway. Indeed, the precise reason for the temporary appearance of the free glucose from glycogen is the need of the skeletal muscle cell to generate energy from glucose oxidation, thereby, precluding any chance of the glucose entering the blood.

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## Regulation of Glycogenolysis

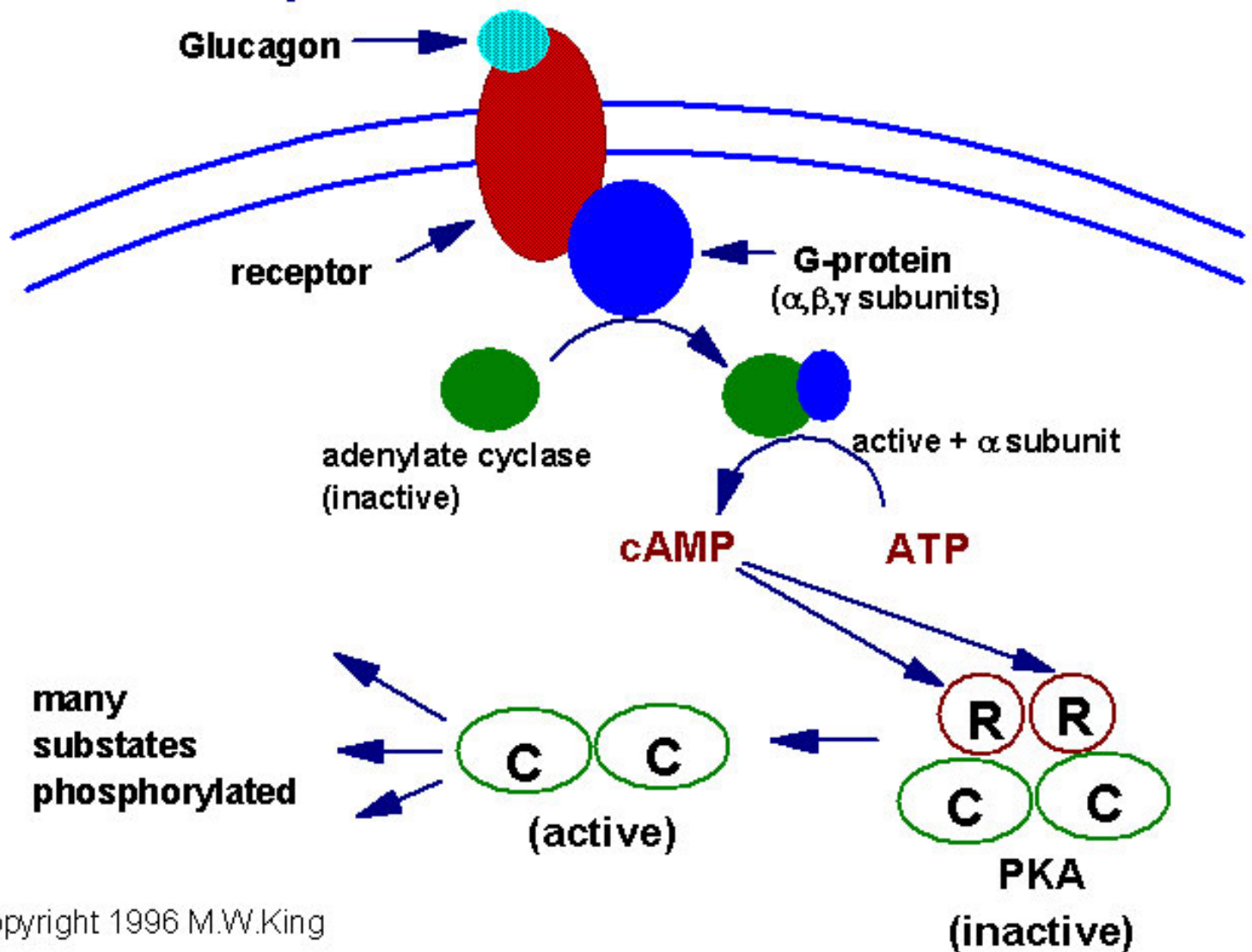
Glycogen phosphorylase is a homodimeric enzyme that exist in two distinct conformational states: a T (for tense, less active) and R (for relaxed, more active) state. Phosphorylase is capable of binding to glycogen when the enzyme is in the R state. This conformation is enhanced by binding of AMP and inhibited by binding ATP or glucose-6-phosphate. The enzyme is also subject to covalent modification by phosphorylation as a means of regulating its activity. The relative activity of the un-modified phosphorylase enzyme (given the name phosphorylase-b) is sufficient to generate enough glucose-1-phosphate for entry into glycolysis for the production of sufficient ATP to maintain the normal resting activity of the cell. This is true in both liver and muscle cells.



Pathways involved in the regulation of glycogen phosphorylase. See the text for details of the regulatory mechanisms. PKA is cAMP-dependent protein kinase. PPI-1 is phosphoprotein phosphatase-1 inhibitor. Whether a factor has positive (+ve) or negative (-ve) effects on any enzyme is indicated. Briefly, phosphorylase b is phosphorylated, and rendered highly active, by phosphorylase kinase. Phosphorylase kinase is itself phosphorylated, leading to increased activity, by PKA (itself activated through receptor-mediated mechanisms). PKA also phosphorylates PPI-1 leading to an inhibition of phosphate removal allowing the activated enzymes to remain so longer. Calcium ions can activate phosphorylase kinase even in the absence of the enzyme being phosphorylated. This allows neuromuscular stimulation by **acetylcholine** to lead to increased glycogenolysis in the absence of receptor stimulation.

In response to lowered blood glucose the  $\alpha$  cells of the pancreas secrete **glucagon** which binds to cell surface receptors on liver and several other cells. Liver cells are the primary target for the action of this peptide hormone. The response of cells to the binding of glucagon to its cell surface receptor is the activation of the enzyme adenylate cyclase which is associated with the receptor. Activation of adenylate cyclase leads to a large increase in the formation of cAMP. cAMP binds to an enzyme called cAMP-dependent protein kinase, **PKA** (see Figure below). Binding of cAMP to the regulatory subunits of PKA leads to the release and subsequent activation of the catalytic subunits. The catalytic subunits then phosphorylate a number of proteins on serine and threonine residues.

# Receptor-Mediated Activation of PKA



Representative pathway for the activation of **cAMP-dependent protein kinase (PKA)**. In this example glucagon binds to its' cell-surface receptor, thereby activating the receptor. Activation of the receptor is coupled to the activation of a receptor-coupled G-protein (GTP-binding and hydrolyzing protein) composed of 3 subunits. Upon activation the alpha subunit dissociates and binds to and activates adenylate cyclase. Adenylate cyclase then converts ATP to cyclic-AMP (cAMP). The cAMP thus produced then binds to the regulatory subunits of PKA leading to dissociation of the associated catalytic subunits. The catalytic subunits are inactive until dissociated from the regulatory subunits. Once released the catalytic subunits of PKA phosphorylate numerous substrate using ATP as the phosphate donor.

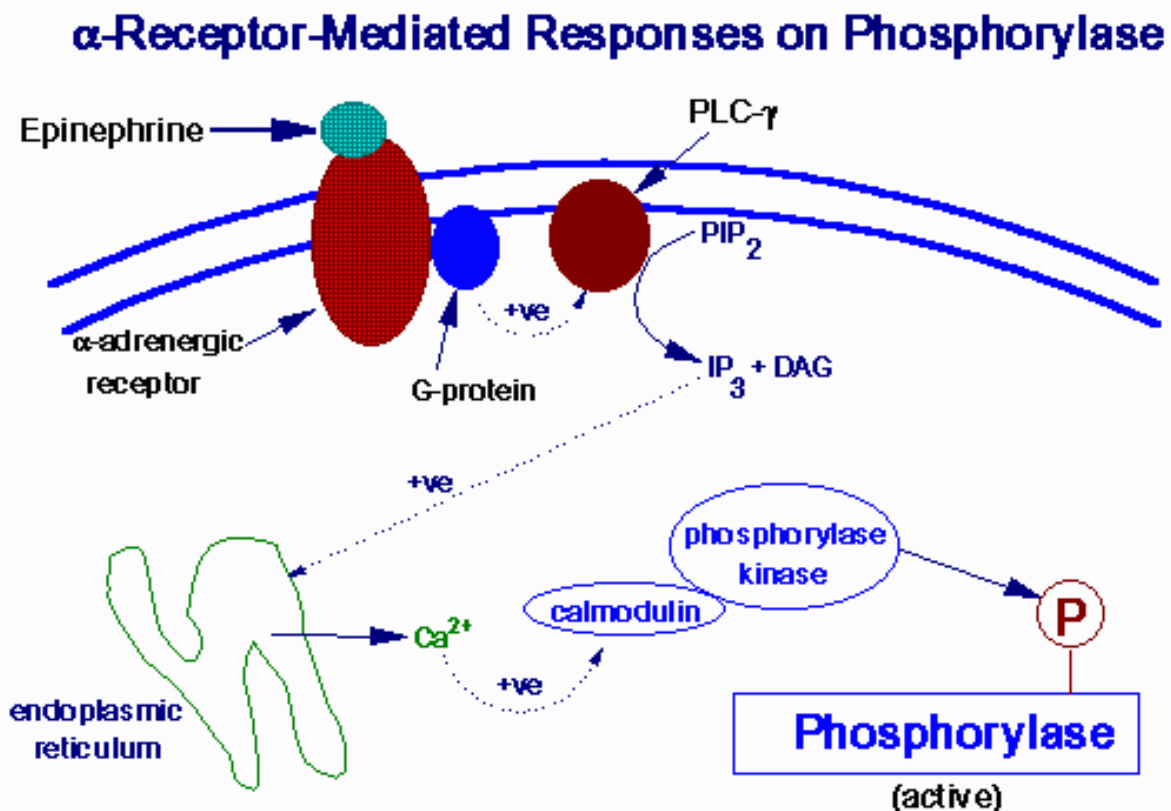
Of significance to this discussion is the PKA-mediated phosphorylation of phosphorylase kinase as shown in the diagram above. Phosphorylase kinase is a multi-subunit enzyme composed of  $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\delta$  subunits. The  $\alpha$  and  $\beta$  subunits are the regulatory subunits that are phosphorylated. The  $\gamma$  subunit is the catalytic subunit and the  $\delta$  subunit is calmodulin (as described below). Phosphorylation of phosphorylase kinase activates the enzyme which in turn phosphorylates the b form of phosphorylase. Phosphorylation of

phosphorylase-b greatly enhances its activity towards glycogen breakdown. The modified enzyme is called phosphorylase-a. The net result is an extremely large induction of glycogen breakdown in response to glucagon binding to cell surface receptors.

This identical cascade of events occurs in skeletal muscle cells as well. However, in these cells the induction of the cascade is the result of epinephrine binding to receptors on the surface of muscle cells. Epinephrine is released from the adrenal glands in response to neural signals indicating an immediate need for enhanced glucose utilization in muscle, the so called **fight or flight response**. Muscle cells lack glucagon receptors. The presence of glucagon receptors on muscle cells would be futile anyway since the role of glucagon release is to increase blood glucose concentrations and muscle glycogen stores cannot contribute to blood glucose levels.

Regulation of phosphorylase kinase activity is also affected by two distinct mechanisms involving  $\text{Ca}^{2+}$  ions. The ability of  $\text{Ca}^{2+}$  ions to regulate phosphorylase kinase is through the function of one of the subunits of this enzyme. One of the subunits of this enzyme is the ubiquitous protein, **calmodulin**. Calmodulin is a calcium binding protein. Binding induces a conformational change in calmodulin which in turn enhances the catalytic activity of the phosphorylase kinase towards its substrate, phosphorylase-b. This activity is crucial to the enhancement of glycogenolysis in muscle cells where muscle contraction is induced via acetylcholine stimulation at the neuromuscular junction. The effect of acetylcholine release from nerve terminals at a neuromuscular junction is to depolarize the muscle cell leading to increased release of sarcoplasmic reticulum stored  $\text{Ca}^{2+}$ , thereby activating phosphorylase kinase. Thus, not only does the increased intracellular calcium increase the rate of muscle contraction it increases glycogenolysis which provides the muscle cell with the increased ATP it also needs for contraction.

The second  $\text{Ca}^{2+}$  ion-mediated pathway to phosphorylase kinase activation is through activation of  $\alpha$ -adrenergic receptors by epinephrine.



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Pathways involved in the regulation of glycogen phosphorylase by epinephrine activation of  $\alpha$ -adrenergic receptors. See the text for details of the regulatory mechanisms. PLC- $\gamma$  is phospholipase C- $\gamma$ . The substrate for PLC- $\gamma$  is **phosphatidylinositol-4,5-bisphosphate (PIP<sub>2</sub>)** and the products are **IP<sub>3</sub>, inositol trisphosphate** and **DAG, diacylglycerol**.

Unlike  $\beta$ -adrenergic receptors which are coupled to activation of adenylate cyclase,  $\alpha$ -adrenergic receptors are coupled through G-proteins that activate phospholipase-C- $\gamma$  (PLC- $\gamma$ ). Activation of PLC- $\gamma$  leads to increased hydrolysis of membrane **phosphatidylinositol-4,5-bisphosphate (PIP<sub>2</sub>)**, the products of which are **inositol trisphosphate (IP<sub>3</sub>)** and **diacylglycerol (DAG)**. DAG binds to and activates protein kinase C (PKC) an enzyme that phosphorylates numerous substrate, one of which is glycogen synthase (see below). IP<sub>3</sub> binds to receptors on the surface of the endoplasmic reticulum leading to release of Ca<sup>2+</sup> ions. The Ca<sup>2+</sup> ions then interact the calmodulin subunits of phosphorylase kinase resulting in its' activation. Additionally, the Ca<sup>2+</sup> ions activate PKC in conjunction with DAG.

In order to terminate the activity of the enzymes of the glycogen phosphorylase activation cascade, once the needs of the body are met, the modified enzymes need to be un-modified. In the case of Ca<sup>2+</sup> induced activation, the level of Ca<sup>2+</sup> ion release from muscle stores will terminate when the incoming nerve impulses cease. The removal of the phosphates on phosphorylase kinase and phosphorylase-a is carried out by phosphoprotein phosphatase-1 (**PP-1**). In order that the phosphate residues placed on these enzymes by PKA and phosphorylase kinase are not immediately removed, the activity of PP-1 must also be regulated. This is accomplished by the binding of PP-1 to **phosphoprotein phosphatase inhibitor (PPI-1)**. This protein also is phosphorylated by PKA and dephosphorylated by PP-1 (see diagram above). The phosphorylation of PPI allows it to bind to PP-1, an activity it is incapable of carrying out when not phosphorylated. When PPI binds PP-1 its phosphorylations are removed by PP-1 but at a much reduced rate than by free PP-1 thus temporarily trapping PP-1 from other substrates.

The effects of the activation of this regulatory phosphorylation cascade on the rate of glycogen synthesis is described below.

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## Glycogen Synthesis

Synthesis of glycogen from glucose is carried out the enzyme glycogen synthase. This enzyme utilizes UDP-glucose as one substrate and the non-reducing end of glycogen as another. The activation of glucose to be used for glycogen synthesis is carried out by the enzyme UDP-glucose pyrophosphorylase. This enzyme exchanges the phosphate on C-1 of glucose-1-phosphate for UDP. The energy of the phosphoglycosyl bond of UDP-glucose is utilized by glycogen synthase to catalyze the incorporation of glucose into glycogen. UDP is subsequently released from the enzyme. The  $\alpha$ -1,6 branches in glucose are produced by amylo-(1,4 - 1,6)-transglycosylase, also termed the branching enzyme. This enzyme transfers a terminal fragment of 6-7 glucose residues (from a polymer at least 11 glucose residues long) to an internal glucose residue at the C-6 hydroxyl position.

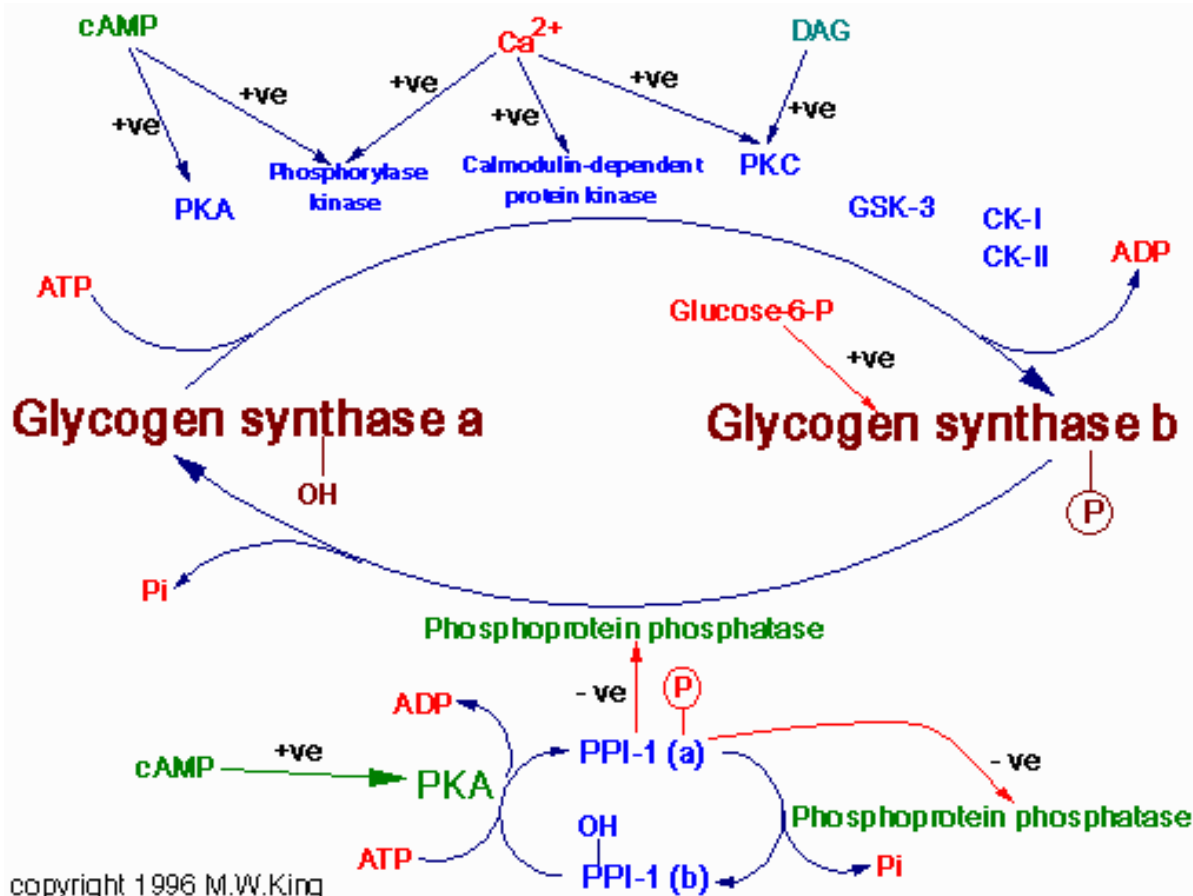
Until recently, the source of the first glycogen molecule that might act as a primer in glycogen synthesis was unknown. Recently it has been discovered that a protein known as **glycogenin** is located at the core of glycogen molecules. Glycogenin has the unusual property of catalyzing its own glycosylation, attaching C-1 of a UDP-glucose to a tyrosine residue on the enzyme. The attached glucose is believed to serve as the primer required by glycogen synthase.

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## Regulation of Glycogen Synthesis

Glycogen synthase is a tetrameric enzyme consisting of 4 identical subunits. The activity of glycogen synthase is regulated by phosphorylation of serine residues in the subunit proteins. Phosphorylation of glycogen synthase reduces its activity towards UDP-glucose. When in the non-phosphorylated state, glycogen synthase does not require glucose-6-phosphate as an allosteric activator---when phosphorylated it does. The two forms of glycogen synthase are identified by the same nomenclature as used for glycogen phosphorylase. The unphosphorylated and most active form is synthase-a and the phosphorylated glucose-6-phosphate-dependent form is synthase-b shift.

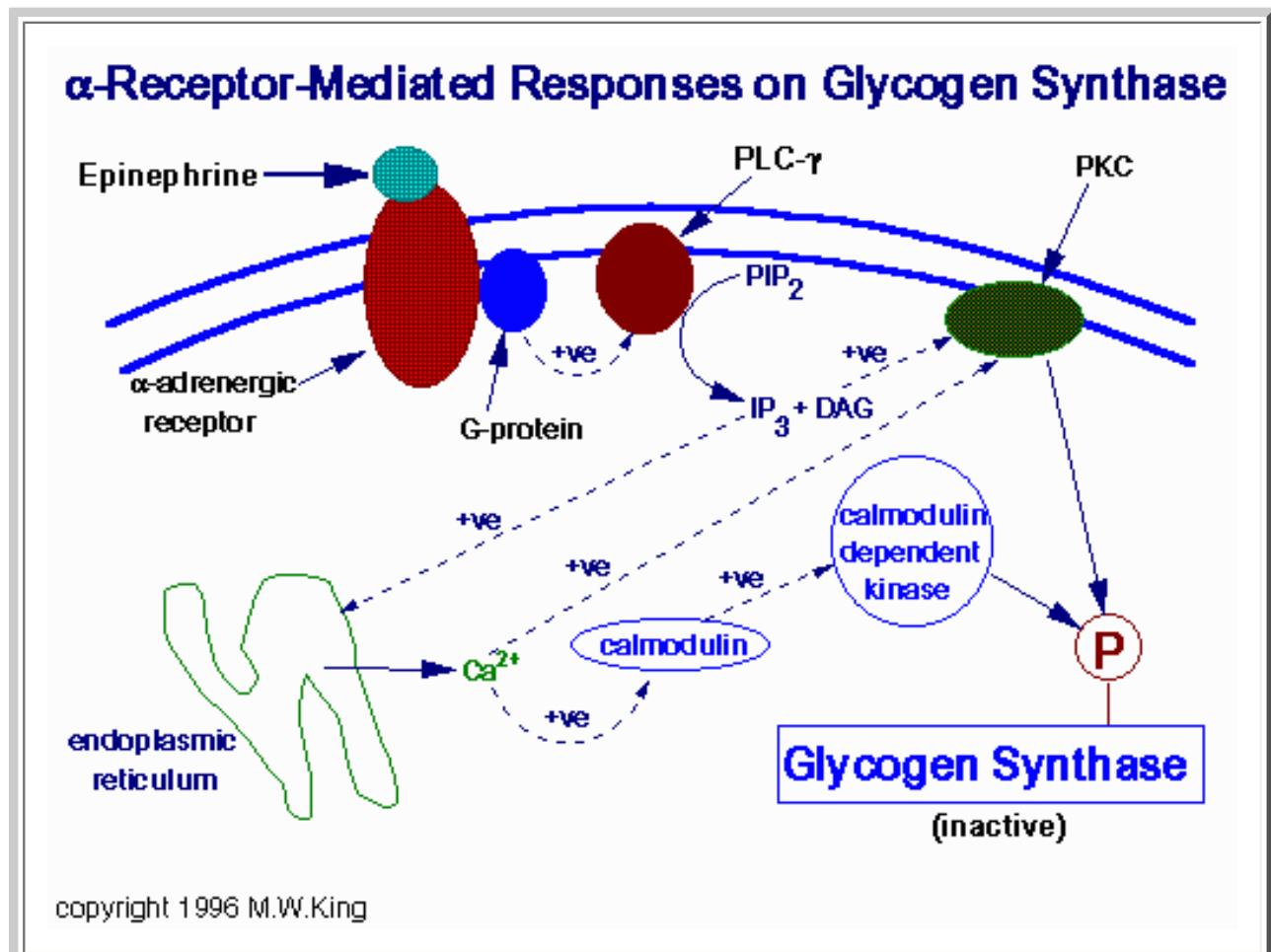


Pathways involved in the regulation of glycogen synthase. See the text for details of the regulatory mechanisms. PKA is cAMP-dependent protein kinase. PPI-1 is phosphoprotein phosphatase-1 inhibitor. Whether a factor has positive (+ve) or negative (-ve) effects on any enzyme is indicated. Briefly, glycogen synthase a is phosphorylated, and rendered much less active and requires glucose-6-phosphate to have any activity at all. Phosphorylation of glycogen synthase is accomplished by several different enzymes. The most important is synthase-phosphorylase kinase the same enzyme responsible for phosphorylation (and activation) of glycogen phosphorylase. PKA (itself activated through receptor mediated mechanisms) also phosphorylates glycogen synthase directly. The effects of PKA on PPI-1 are the same as those described above for the regulation of glycogen phosphorylase. The other enzymes shown to directly phosphorylate glycogen synthase are protein kinase C (PKC), calmodulin-dependent protein kinase, glycogen synthase kinase-3 (GSK-3) and two

forms of casein kinase (CK-I and CK-II). The enzyme PKC is activated by  $\text{Ca}^{2+}$  ions and phospholipids, primarily **diacylglycerol, DAG**. DAG is formed by receptor-mediated hydrolysis of membrane **phosphatidylinositol bisphosphate ( $\text{PIP}_2$ )**.

Phosphorylation of synthase occurs primarily in response to hormonal activation of PKA (see Figure above). One of the major kinases active on synthase is synthase-phosphorylase kinase; the same enzyme that phosphorylates glycogen phosphorylase. However, at least 5 additional enzymes have been identified that phosphorylate glycogen synthase directly. One of these glycogen synthase phosphorylating enzymes is PKA itself. One important glycogen synthase phosphorylating enzyme is active independently of increases in cAMP levels. This enzyme is glycogen synthase kinase 3 (**GSK-3**). Each phosphorylation event occurs at distinct serine residues which can result in a progressively increased state of synthase phosphorylation.

Glycogen synthase activity can also be affected by epinephrine binding to  $\alpha$ -adrenergic receptors through a pathway like that described above for regulation of glycogen phosphorylase.



Pathways involved in the regulation of glycogen synthase by epinephrine activation of  $\alpha$ -adrenergic receptors. See the text for details of the regulatory mechanisms. PKC is protein kinase C. PLC- $\gamma$  is phospholipase C- $\gamma$ . The substrate for PLC- $\gamma$  is **phosphatidylinositol-4,5-bisphosphate ( $\text{PIP}_2$ )** and the products are  **$\text{IP}_3$ , inositol trisphosphate** and **DAG, diacylglycerol**.

When  $\alpha$ -adrenergic receptors are stimulated there is an increase in the activity of PLC- $\gamma$  with a resultant increase in PIP<sub>2</sub> hydrolysis. The products of PIP<sub>2</sub> hydrolysis are DAG and IP<sub>3</sub>. As described above for glycogen phosphorylase, DAG and the Ca<sup>2+</sup> ions released by IP<sub>3</sub> activate PKC which phosphorylates and inactivates glycogen synthase. Additional responses of calcium are the activation of calmodulin-dependent protein kinase (calmodulin is a component of many enzymes that are responsive to Ca<sup>2+</sup>) which also phosphorylates glycogen synthase.

The effects of these phosphorylations leads to:

- **1.** Decreased affinity of synthase for UDP-glucose.
- **2.** Decreased affinity of synthase for glucose-6-phosphate.
- **3.** Increased affinity of synthase for ATP and P<sub>i</sub>.

Reconversion of synthase-b to synthase-a requires dephosphorylation. This is carried out predominately by protein phosphatase-1 (PP-1) the same phosphatase involved in dephosphorylation of phosphorylase.

The activity of PP-1 is also affected by insulin. The pancreatic hormone exerts an opposing effect to that of glucagon and epinephrine. This should appear obvious since the role of insulin is to increase the uptake of glucose from the blood.

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## Glycogen Storage Diseases

Since glycogen molecules can become enormously large, an inability to degrade glycogen can cause cells to become pathologically engorged; it can also lead to the functional loss of glycogen as a source of cell energy and as a blood glucose buffer. Although [glycogen storage diseases](#) are quite rare, their effects can be most dramatic. The debilitating effect of many glycogen storage diseases depends on the severity of the mutation causing the deficiency. In addition, although the glycogen storage diseases are attributed to specific enzyme deficiencies, other events can cause the same characteristic symptoms. For example, Type I glycogen storage disease ([von Gierke's disease](#)) is attributed to lack of glucose-6-phosphatase. However, this enzyme is localized on the cisternal surface of the endoplasmic reticulum (ER); in order to gain access to the phosphatase, glucose-6-phosphate must pass through a specific translocase in the ER membrane. Mutation of either the phosphatase or the translocase makes transfer of liver glycogen to the blood a very limited process. Thus, mutation of either gene leads to symptoms associated with von Gierke's disease, which occurs at a rate of about 1 in 200,000 people.

Several glycogenoses are the result of deficiencies in enzymes of glycolysis whose symptoms and signs are similar to those seen in type V glycogen storage disease. These include deficiencies in muscle phosphoglycerate kinase and muscle pyruvate kinase as well as deficiencies in fructose 1,6-bisphosphatase, lactate dehydrogenase and phosphoglycerate mutase.

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## Table of Glycogen Storage Diseases

| Type: Name | Enzyme Affected | Primary Organ | Manifestations |
|------------|-----------------|---------------|----------------|
|            |                 |               |                |

|                                    |                                                                                              |                                    |                                                                                                   |
|------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------|
| Type 0                             | glycogen synthase                                                                            | liver                              | hypoglycemia, early death, hyperketonia                                                           |
| Type Ia:<br>von Gierke's           | glucose-6-phosphatase                                                                        | liver                              | hepatomegaly, kidney failure, thrombocyte dysfunction                                             |
| Type Ib                            | microsomal glucose-6-phosphate translocase                                                   | liver                              | like Ia, also neutropenia, bacterial infections                                                   |
| Type Ic                            | microsomal P <sub>i</sub> transporter                                                        | liver                              | like Ia                                                                                           |
| Type II:<br>Pompe's                | lysosomal $\alpha$ -1,4-glucosidase,<br>lysosomal acid $\alpha$ -glucosidase<br>acid maltase | skeletal and cardiac muscle        | infantile form = death by 2;<br>juvenile form = myopathy;<br>adult form = muscular dystrophy-like |
| Type IIIa:<br>Cori's or<br>Forbe's | liver and muscle<br>debranching enzyme                                                       | liver, skeletal and cardiac muscle | infant hepatomegaly, myopathy                                                                     |
| Type IIIb                          | liver debranching enzyme<br>normal muscle enzyme                                             | liver, skeletal and cardiac muscle | liver symptoms same as type IIIa                                                                  |
| Type IV:<br>Anderson's             | branching enzyme                                                                             | liver, muscle                      | hepatosplenomegaly, cirrhosis                                                                     |
| Type V:<br>McArdle's               | muscle phosphorylase                                                                         | skeletal muscle                    | exercise-induced cramps and pain, myoglobinuria                                                   |
| Type VI:<br>Her's                  | liver phosphorylase                                                                          | liver                              | hepatomegaly, mild hypoglycemia, hyperlipidemia and ketosis, improvement with age                 |
| Type VII:<br>Tarui's               | muscle PFK-1                                                                                 | muscle, RBC's                      | like V, also hemolytic anemia                                                                     |

|                              |                                    |                                 |                                                                                       |
|------------------------------|------------------------------------|---------------------------------|---------------------------------------------------------------------------------------|
| Type VIa, VIII<br>or Type IX | phosphorylase kinase               | liver,<br>leukocytes,<br>muscle | like VI                                                                               |
| Type XI:<br>Fanconi-Bickel   | glucose transporter-<br>2 (GLUT-2) | liver                           | failure to thrive,<br>hepatomegaly, rickets,<br>proximal renal tubular<br>dysfunction |

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## Digestion of Dietary Carbohydrates

Dietary carbohydrate from which humans gain energy enter the body in complex forms, such as disaccharides and the polymers **starch** (amylose and amylopectin) and **glycogen**. The polymer **cellulose** is also consumed but not digested. The first step in the metabolism of digestible carbohydrate is the conversion of the higher polymers to simpler, soluble forms that can be transported across the intestinal wall and delivered to the tissues. The breakdown of polymeric sugars begins in the mouth. Saliva has a slightly acidic pH of 6.8 and contains lingual amylase that begins the digestion of carbohydrates. The action of lingual amylase is limited to the area of the mouth and the esophagus; it is virtually inactivated by the much stronger acid pH of the stomach. Once the food has arrived in the stomach, acid hydrolysis contributes to its degradation; specific gastric proteases and lipases aid this process for proteins and fats, respectively. The mixture of gastric secretions, saliva, and food, known collectively as **chyme**, moves to the small intestine.

The main polymeric-carbohydrate digesting enzyme of the small intestine is  $\alpha$ -amylase. This enzyme is secreted by the pancreas and has the same activity as salivary amylase, producing disaccharides and trisaccharides. The latter are converted to monosaccharides by intestinal saccharidases, including maltases that hydrolyze di- and trisaccharides, and the more specific disaccharidases, sucrase, lactase, and trehalase. The net result is the almost complete conversion of digestible carbohydrate to its constituent monosaccharides.

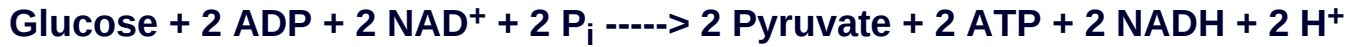
The resultant glucose and other simple carbohydrates are transported across the intestinal wall to the hepatic portal vein and then to liver parenchymal cells and other tissues. There they are converted to fatty acids, amino acids, and glycogen, or else oxidized by the various catabolic pathways of cells.

Oxidation of glucose is known as **glycolysis**. Glucose is oxidized to either lactate or pyruvate. Under aerobic conditions, the dominant product in most tissues is **pyruvate** and the pathway is known as **aerobic glycolysis**. When oxygen is depleted, as for instance during prolonged vigorous exercise, the dominant glycolytic product in many tissues is **lactate** and the process is known as **anaerobic glycolysis**.

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## The Energy Derived from Glucose Oxidation

Aerobic glycolysis of glucose to pyruvate, requires two equivalents of ATP to activate the process, with the subsequent production of four equivalents of ATP and two equivalents of NADH. Thus, conversion of one mole of glucose to two moles of pyruvate is accompanied by the net production of two moles each of ATP and NADH.



The NADH generated during glycolysis is used to fuel mitochondrial ATP synthesis via [oxidative phosphorylation](#), producing either two or three equivalents of ATP depending upon whether the [glycerol phosphate shuttle](#) or the [malate-aspartate shuttle](#) is used to transport the electrons from cytoplasmic NADH into the mitochondria. The net yield from the oxidation of 1 mole of glucose to 2 moles of pyruvate is, therefore, either 6 or 8 moles of ATP. Complete oxidation of the 2 moles of pyruvate, through the [TCA cycle](#), yeilds an additional 30 moles of ATP; the total yield, therefore being either 36 or 38 moles of ATP from the complete oxidation of 1 mole of glucose to CO<sub>2</sub> and H<sub>2</sub>O.

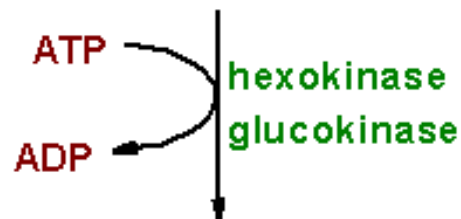
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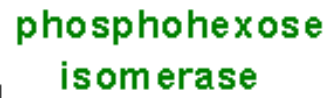
## The Individual Reactions of Glycolysis

The pathway of glycolysis can be seen as consisting of 2 separate phases. The first is the chemical priming phase requiring energy in the form of ATP, and the second is considered the energy-yielding phase. In the first phase, 2 equivalents of ATP are used to convert glucose to fructose 1,6-bisphosphate (F1,6BP). In the second phase F1,6BP is degraded to pyruvate, with the production of 4 equivalents of ATP and 2 equivalents of NADH.

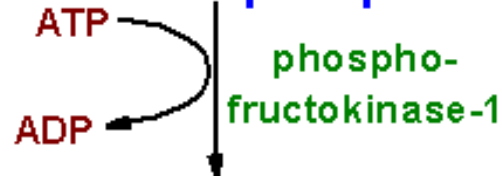
# Glucose



## Glucose-6-phosphate



## Fructose-6-phosphate



## Fructose-1,6-bisphosphate

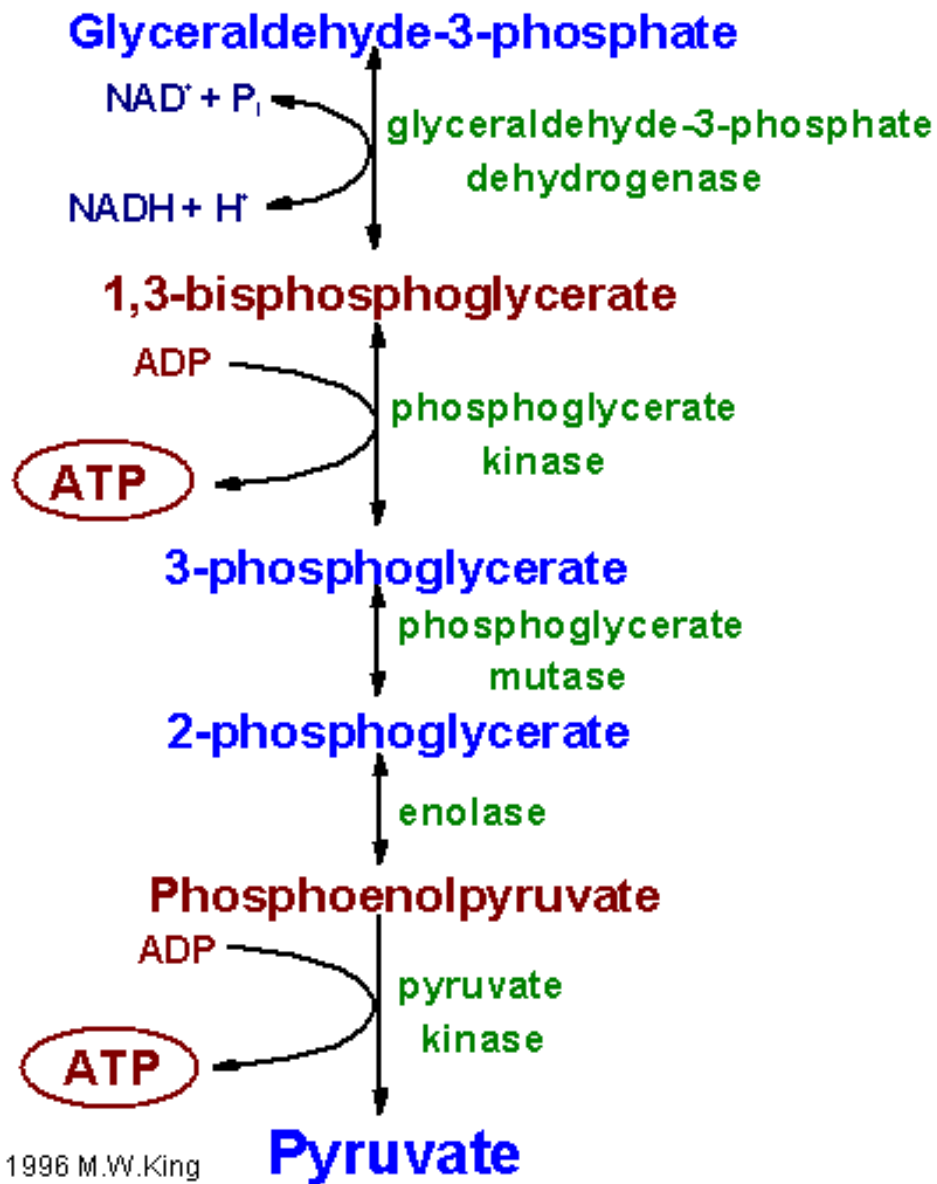


## Glyceraldehyde-3-phosphate



## Dihydroxyacetone phosphate

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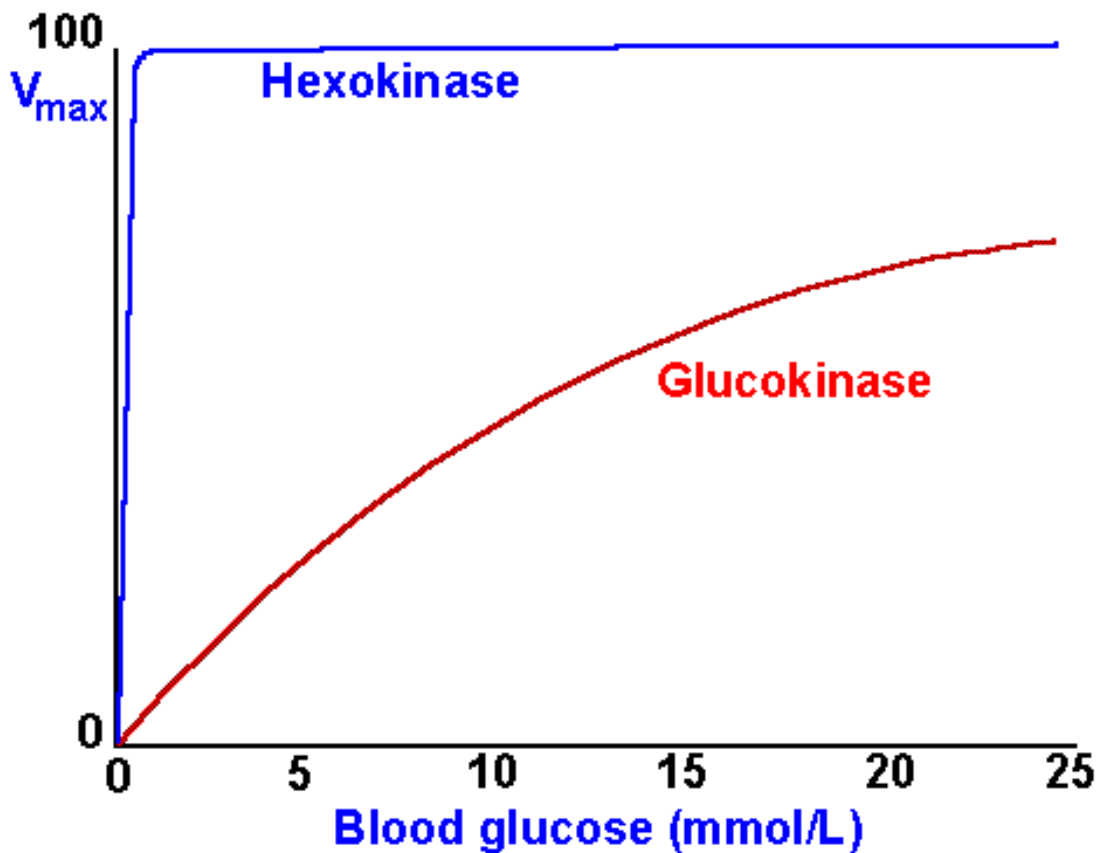


Pathway of glycolysis from glucose to pyruvate. Substrates and products are in blue, enzymes are in green. The two high energy intermediates whose oxidations are coupled to ATP synthesis are shown in red (1,3-bisphosphoglycerate and phosphoenolpyruvate).

## The Hexokinase Reaction:

The ATP-dependent phosphorylation of glucose to form glucose 6-phosphate (G6P) is the first reaction of glycolysis, and is catalyzed by tissue-specific isoenzymes known as hexokinases. The phosphorylation accomplishes two goals: First, the hexokinase reaction converts nonionic glucose into an anion that is trapped in the cell, since cells lack transport systems for phosphorylated sugars. Second, the otherwise biologically inert glucose becomes activated into a labile form capable of being further metabolized.

Four mammalian isozymes of hexokinase are known (Types I - IV), with the Type IV isozyme often referred to as glucokinase. Glucokinase is the form of the enzyme found in hepatocytes. The high  $K_m$  of glucokinase for glucose means that this enzyme is saturated only at very high concentrations of substrate.



Comparison of the activities of hexokinase and glucokinase. The  $K_m$  for hexokinase is significantly lower (0.1mM) than that of glucokinase (10mM). This difference ensures that non-hepatic tissues (which contain hexokinase) rapidly and efficiently trap blood glucose within their cells by converting it to glucose-6-phosphate. One major function of the liver is to deliver glucose to the blood and this is ensured by having a glucose phosphorylating enzyme (glucokinase) whose  $K_m$  for glucose is sufficiently higher than the normal circulating concentration of glucose (5mM).

This feature of hepatic glucokinase allows the liver to **buffer** blood glucose. After meals, when postprandial blood glucose levels are high, liver glucokinase is significantly active, which causes the liver preferentially to trap and to store circulating glucose. When blood glucose falls to very low levels, tissues such as liver and kidney---which contain glucokinases but are not highly dependent on glucose---do not continue to use the meager glucose supplies that remain available. At the same time, tissues such as the brain, which are critically dependent on glucose, continue to scavenge blood glucose using their low  $K_m$  hexokinases, and as a consequence their viability is protected. Under various conditions of glucose deficiency, such as long periods between meals, the liver is stimulated to supply the blood with glucose through the pathway of **gluconeogenesis**. The levels of glucose produced during gluconeogenesis are insufficient to activate glucokinase, allowing the glucose to pass out of hepatocytes and into the blood.

The regulation of hexokinase and glucokinase activities is also different. Hexokinases I, II, and III are allosterically inhibited by product (G6P) accumulation, whereas glucokinases are not. The latter further insures liver accumulation of glucose stores during times of glucose excess, while favoring peripheral



glucose utilization when glucose is required to supply energy to peripheral tissues.

## Phosphohexose Isomerase:

The second reaction of glycolysis is an isomerization, in which G6P is converted to fructose 6-phosphate (F6P). The enzyme catalyzing this reaction is phosphohexose isomerase (also known as phosphoglucose isomerase). The reaction is freely reversible at normal cellular concentrations of the two hexose phosphates and thus catalyzes this interconversion during glycolytic carbon flow and during gluconeogenesis.

## 6-Phosphofructo-1-Kinase (Phosphofructokinase-1, PFK-1):

The next reaction of glycolysis involves the utilization of a second ATP to convert F6P to fructose 1,6-bisphosphate (F1,6BP). This reaction is catalyzed by 6-phosphofructo-1-kinase, better known as phosphofructokinase-1 or **PFK-1**. This reaction is not readily reversible because of its large positive free energy ( $\Delta G^0 = +5.4$  kcal/mol) in the reverse direction. Nevertheless, fructose units readily flow in the reverse (gluconeogenic) direction because of the ubiquitous presence of the hydrolytic enzyme, fructose-1,6-bisphosphatase (F-1,6-BPase).

The presence of these two enzymes in the same cell compartment provides an example of a metabolic futile cycle, which if unregulated would rapidly deplete cell energy stores. However, the activity of these two enzymes is so highly regulated that PFK-1 is considered to be the **rate-limiting enzyme** of glycolysis and F-1,6-BPase is considered to be the **rate-limiting enzyme** in gluconeogenesis.

## Aldolase:

Aldolase catalyses the hydrolysis of F1,6BP into two 3-carbon products: dihydroxyacetone phosphate (DHAP) and glyceraldehyde 3-phosphate (G3P). The aldolase reaction proceeds readily in the reverse direction, being utilized for both glycolysis and gluconeogenesis.

**Triose Phosphate Isomerase:** The two products of the aldolase reaction equilibrate readily in a reaction catalyzed by triose phosphate isomerase. Succeeding reactions of glycolysis utilize G3P as a substrate; thus, the aldolase reaction is pulled in the glycolytic direction by mass action principals.

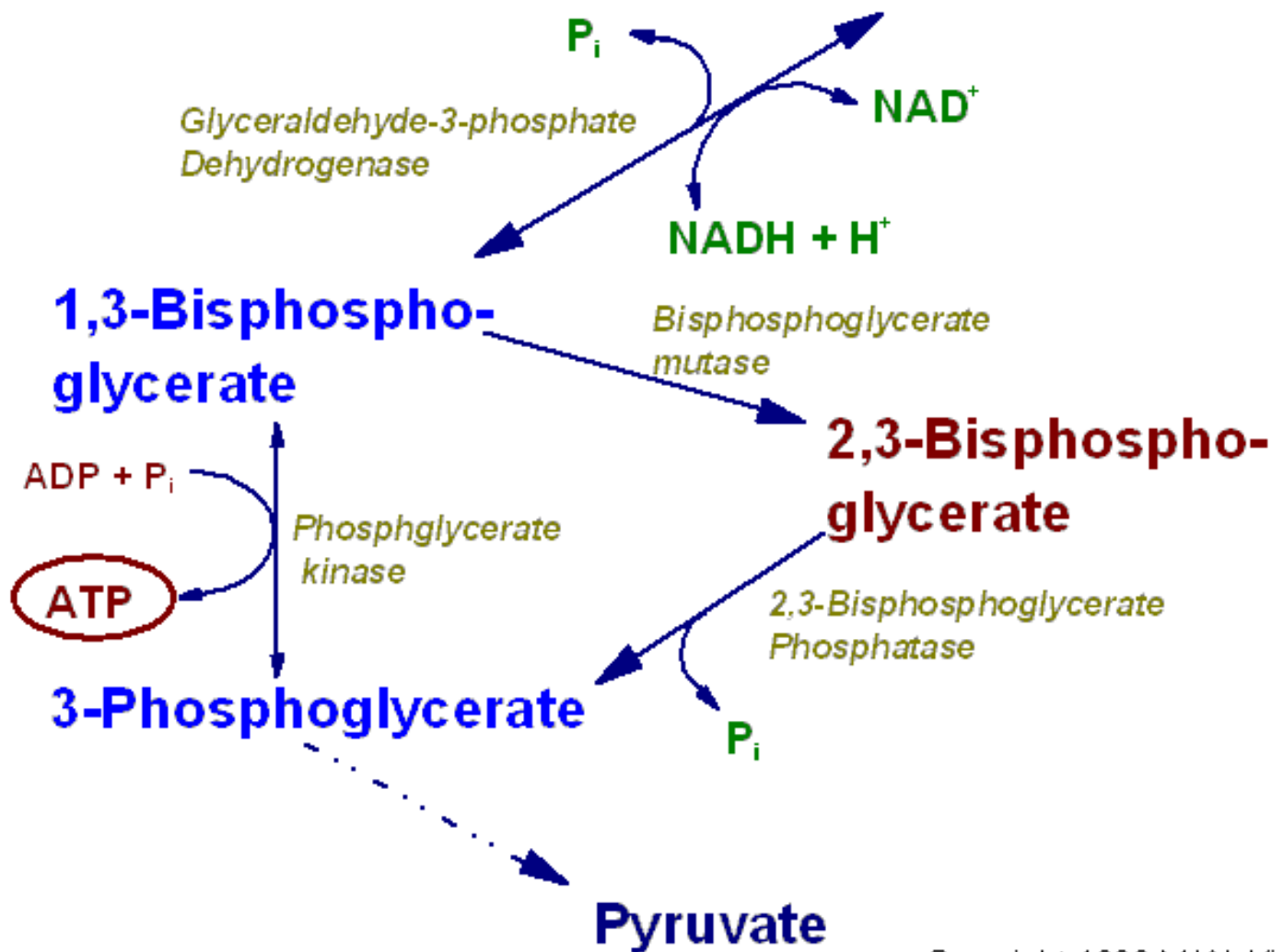
## Glyceraldehyde-3-Phosphate Dehydrogenase:

The second phase of glucose catabolism features the energy-yielding glycolytic reactions that produce ATP and NADH. In the first of these reactions, glyceraldehyde-3-P dehydrogenase (G3PDH) catalyzes the  $\text{NAD}^+$ -dependent oxidation of G3P to 1,3-bisphosphoglycerate (1,3BPG) and NADH. The G3PDH reaction is reversible, and the same enzyme catalyzes the reverse reaction during gluconeogenesis.

## Phosphoglycerate Kinase:

The high-energy phosphate of 1,3-BPG is used to form ATP and 3-phosphoglycerate (3PG) by the enzyme phosphoglycerate kinase. Note that this is the only reaction of glycolysis or gluconeogenesis that involves ATP and yet is reversible under normal cell conditions. Associated with the phosphoglycerate kinase pathway is an important reaction of erythrocytes, the formation of 2,3-bisphosphoglycerate, 2,3BPG (see Figure below) by the enzyme bisphosphoglycerate mutase. 2,3BPG is an important regulator of [hemoglobin's](#) affinity for oxygen. Note that 2,3-bisphosphoglycerate phosphatase degrades 2,3BPG to 3-phosphoglycerate, a normal intermediate of glycolysis. The 2,3BPG shunt thus operates with the expenditure of 1 equivalent of ATP per triose passed through the shunt. The process is not reversible under physiological conditions.

**1/2 Glucose**  $\xrightarrow{\quad}$  **Glyceraldehyde-3-phosphate**



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The pathway for 2,3-bisphosphoglycerate (**2,3-BPG**) synthesis within erythrocytes. Synthesis of 2,3-BPG represents a major reaction pathway for the consumption of glucose in erythrocytes. The synthesis of 2,3-BPG in erythrocytes is critical for controlling hemoglobin affinity for oxygen. Note that when glucose is oxidized by this pathway the erythrocyte loses the ability to gain 2 moles of ATP from glycolytic oxidation of 1,3-BPG to 3-phosphoglycerate via the phosphoglycerate kinase reaction.

## Phosphoglycerate Mutase and Enolase:

The remaining reactions of glycolysis are aimed at converting the relatively low energy phosphoacyl-ester of 3PG to a high-energy form and harvesting the phosphate as ATP. The 3PG is first converted to 2PG by phosphoglycerate mutase and the 2PG conversion to phosphoenolpyruvate (PEP) is catalyzed by enolase

## Pyruvate Kinase:

The final reaction of aerobic glycolysis is catalyzed by the highly regulated enzyme pyruvate kinase

(PK). In this strongly exergonic reaction, the high-energy phosphate of PEP is conserved as ATP. The loss of phosphate by PEP leads to the production of pyruvate in an unstable enol form, which spontaneously tautomerizes to the more stable, keto form of pyruvate. This reaction contributes a large proportion of the free energy of hydrolysis of PEP.

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## Anaerobic Glycolysis

Under aerobic conditions, pyruvate in most cells is further metabolized via the [TCA cycle](#). Under anaerobic conditions and in erythrocytes under aerobic conditions, pyruvate is converted to lactate by the enzyme lactate dehydrogenase (LDH), and the lactate is transported out of the cell into the circulation. The conversion of pyruvate to lactate, under anaerobic conditions, provides the cell with a mechanism for the oxidation of NADH (produced during the G3PDH reaction) to  $\text{NAD}^+$ ; which occurs during the LDH catalyzed reaction. This reduction is required since  $\text{NAD}^+$  is a necessary substrate for G3PDH, without which glycolysis will cease. Normally, during aerobic glycolysis the electrons of cytoplasmic NADH are transferred to mitochondrial carriers of the [oxidative phosphorylation](#) pathway generating a continuous pool of cytoplasmic  $\text{NAD}^+$ .

Aerobic glycolysis generates substantially more ATP per mole of glucose oxidized than does anaerobic glycolysis. The utility of anaerobic glycolysis to a muscle cell when it needs large amounts of energy stems from the fact that the rate of ATP production from glycolysis is approximately 100X faster than from oxidative phosphorylation. During exertion muscle cells do not need to energize anabolic reaction pathways. The requirement is to generate the maximum amount of ATP, for muscle contraction, in the shortest time frame. This is why muscle cells derive almost all of the ATP consumed during exertion from anaerobic glycolysis.

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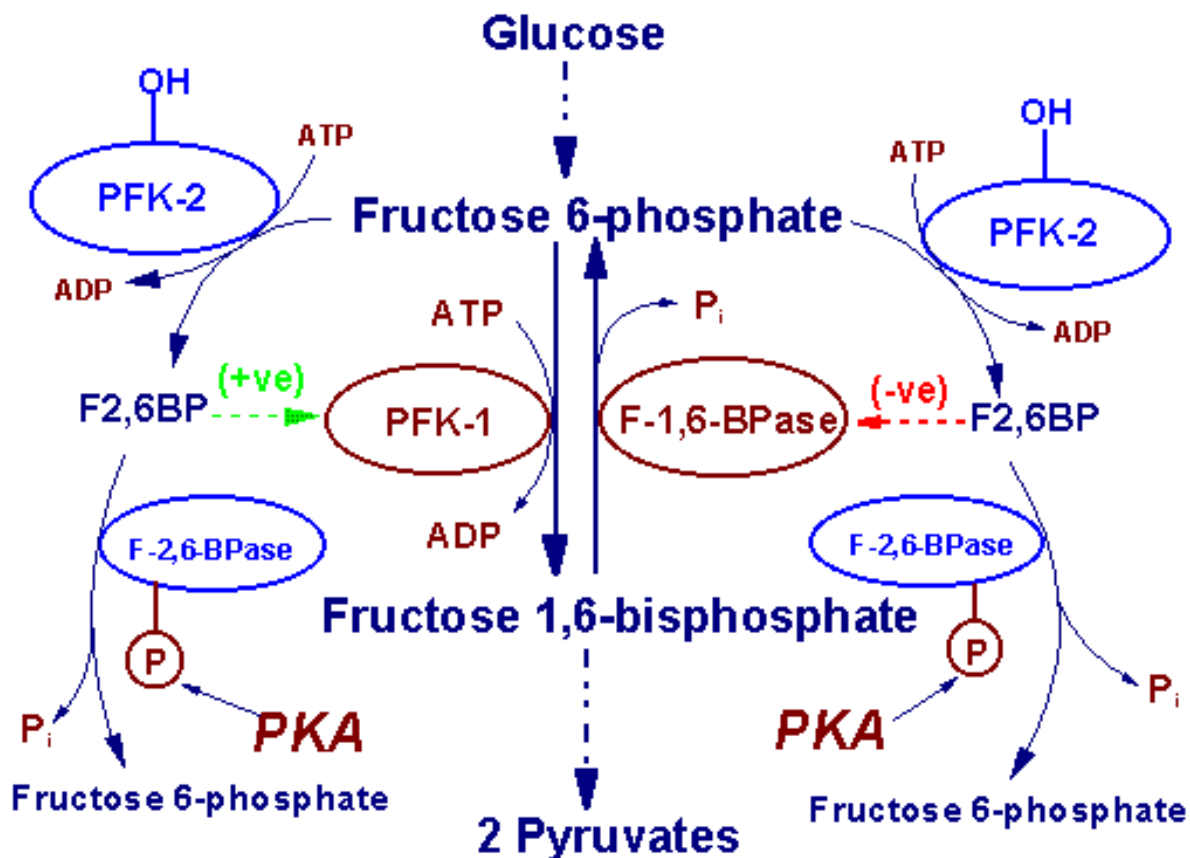
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## Regulation of Glycolysis

The reactions catalyzed by hexokinase, PFK-1 and PK all proceed with a relatively large free energy decrease. These nonequilibrium reactions of glycolysis would be ideal candidates for regulation of the flux through glycolysis. Indeed, *in vitro* studies have shown all three enzymes to be allosterically controlled.

Regulation of hexokinase, however, is not the major control point in glycolysis. This is due to the fact that large amounts of G6P are derived from the breakdown of glycogen (the predominant mechanism of carbohydrate entry into glycolysis in skeletal muscle) and, therefore, the hexokinase reaction is not necessary. Regulation of PK is important for reversing glycolysis when ATP is high in order to activate gluconeogenesis. As such this enzyme catalyzed reaction is not a major control point in glycolysis. The rate limiting step in glycolysis is the reaction catalyzed by PFK-1.

PFK-1 is a tetrameric enzyme that exist in two conformational states termed R and T that are in equilibrium. ATP is both a substrate and an allosteric inhibitor of PFK-1. Each subunit has two ATP binding sites, a substrate site and an inhibitor site. The substrate site binds ATP equally well when the tetramer is in either conformation. The inhibitor site binds ATP essentially only when the enzyme is in the T state. F6P is the other substrate for PFK-1 and it also binds preferentially to the R state enzyme. At high concentrations of ATP, the inhibitor site becomes occupied and shifting the equilibrium of PFK-1 conformation to that of the T state decreasing PFK-1's ability to bind F6P. The inhibition of PFK-1 by ATP is overcome by AMP which binds to the R state of the enzyme and, therefore, stabilizes the conformation of the enzyme capable of binding F6P. The most important allosteric regulator of both glycolysis and gluconeogenesis is **fructose 2,6-bisphosphate, F2,6BP**, which is not an intermediate in glycolysis or in gluconeogenesis.



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Regulation of glycolysis and gluconeogenesis by **fructose 2,6-bisphosphate (F2,6BP)**. The major sites for regulation of glycolysis and gluconeogenesis are the phosphofructokinase-1 (PFK-1) and fructose-1,6-bisphosphatase (F-1,6-BPase) catalyzed reactions. PFK-2 is the kinase activity and F-2,6-BPase is the phosphatase activity of the bi-functional regulatory enzyme, phosphofructokinase-2/fructose-2,6-bisphosphatase. PKA is cAMP-dependent protein kinase which phosphorylates PFK-2/F-2,6-BPase turning on the phosphatase activity. (+ve) and (-ve) refer to positive and negative activities, respectively.

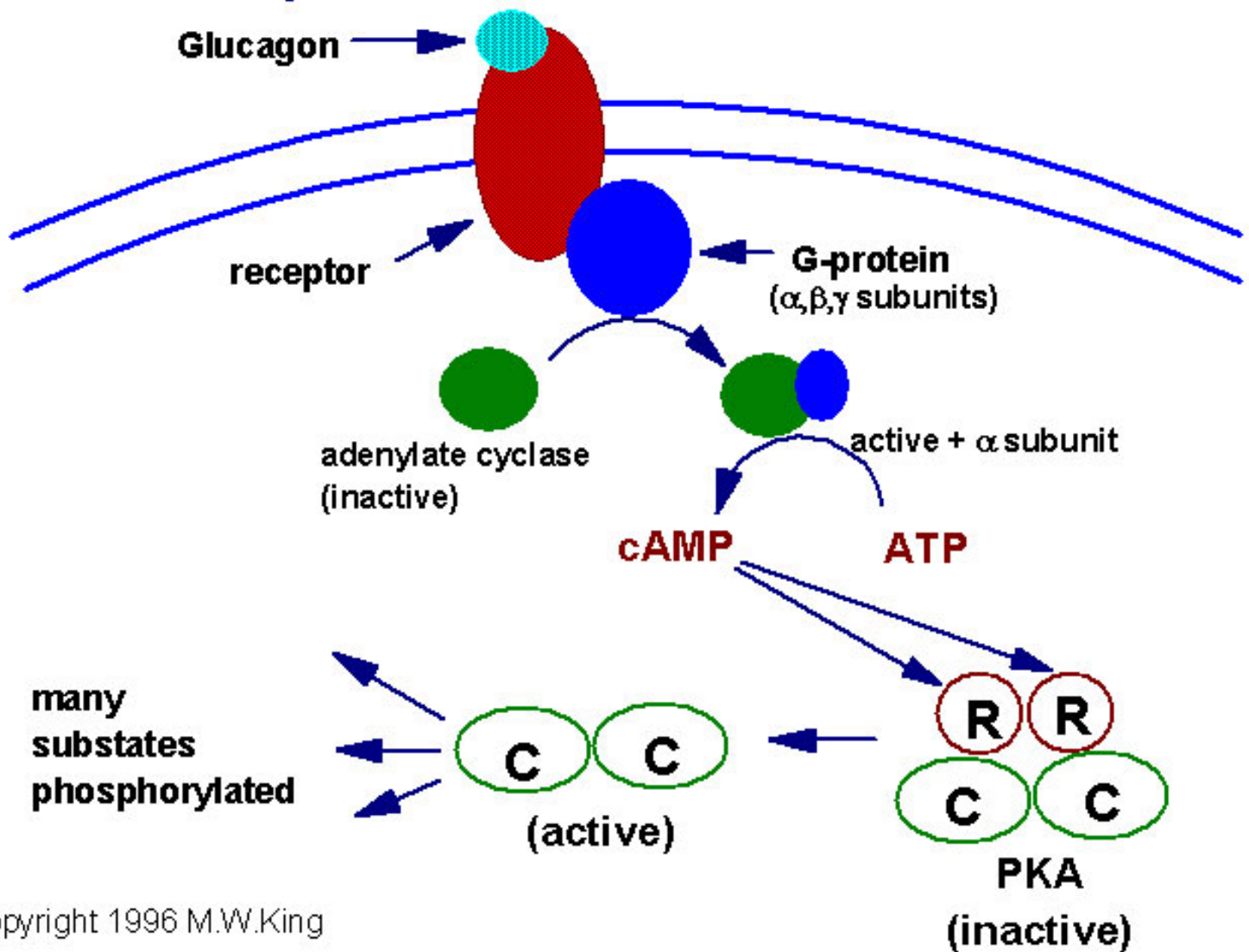
The synthesis of F2,6BP is catalyzed by the bifunctional enzyme phosphofructokinase-2/fructose-2,6-bisphosphatase (PFK-2/F-2,6-BPase). In the nonphosphorylated form the enzyme is known as PFK-2 and serves to catalyze the synthesis of F2,6BP by phosphorylating fructose 6-phosphate. The result is that the activity of PFK-1 is greatly stimulated and the activity of F-1,6-BPase is greatly inhibited.

Under conditions where PFK-2 is active, fructose flow through the PFK-1/F-1,6-BPase reactions takes place in the glycolytic direction, with a net production of F1,6BP. When the bifunctional enzyme is phosphorylated it no longer exhibits kinase activity, but a new active site hydrolyzes F2,6BP to F6P and inorganic phosphate. The metabolic result of the phosphorylation of the bifunctional enzyme is that allosteric stimulation of PFK-1 ceases, allosteric inhibition of F-1,6-BPase is eliminated, and net flow of fructose through these two enzymes is gluconeogenic, producing F6P and eventually glucose.

The interconversion of the bifunctional enzyme is catalyzed by cAMP-dependent protein kinase (PKA), which in turn is regulated by circulating peptide hormones. When blood glucose levels drop, pancreatic

insulin production falls, glucagon secretion is stimulated, and circulating glucagon is highly increased. Hormones such as glucagon bind to plasma membrane receptors on liver cells, activating membrane-localized adenylate cyclase leading to an increase in the conversion of ATP to cAMP (see diagram below). cAMP binds to the regulatory subunits of PKA, leading to release and activation of the catalytic subunits. PKA phosphorylates numerous enzymes, including the bifunctional PFK-2/F-2,6-BPase. Under these conditions the liver stops consuming glucose and becomes metabolically gluconeogenic, producing glucose to reestablish normoglycemia.

## Receptor-Mediated Activation of PKA



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Representative pathway for the activation of **cAMP-dependent protein kinase (PKA)**. In this example glucagon binds to its' cell-surface receptor, thereby activating the receptor. Activation of the receptor is coupled to the activation of a receptor-coupled G-protein (GTP-binding and hydrolyzing protein) composed of 3 subunits. Upon activation the alpha subunit dissociates and binds to and activates adenylate cyclase. Adenylate cyclase then converts ATP to cyclic-AMP (cAMP). The cAMP thus produced then binds to the regulatory subunits of PKA leading to dissociation of the associated catalytic subunits. The catalytic subunits are inactive until dissociated from the regulatory subunits. Once released the catalytic subunits of PKA phosphorylate numerous substrate using ATP as the phosphate donor.

Regulation of glycolysis also occurs at the step catalyzed by pyruvate kinase, (PK). The liver enzyme has been most studied *in vitro*. This enzyme is inhibited by ATP and acetyl-CoA and is activated by F1,6BP. The inhibition of PK by ATP is similar to the effect of ATP on PFK-1. The binding of ATP to the inhibitor site reduces its affinity for PEP. The liver enzyme is also controlled at the level of synthesis. Increased carbohydrate ingestion induces the synthesis of PK resulting in elevated cellular levels of the enzyme.

A number of PK isozymes have been described. The liver isozyme (L-type), characteristic of a gluconeogenic tissue, is regulated via phosphorylation by PKA, whereas the M-type isozyme found in brain, muscle, and other glucose requiring tissue is unaffected by PKA. As a consequence of these differences, blood glucose levels and associated hormones can regulate the balance of liver gluconeogenesis and glycolysis while muscle metabolism remains unaffected.

In erythrocytes, the fetal PK isozyme has much greater activity than the adult isozyme; as a result, fetal erythrocytes have comparatively low concentrations of glycolytic intermediates. Because of the low steady-state concentration of fetal 1,3BPG, the 2,3BPG shunt (see diagram above) is greatly reduced in fetal cells and little 2,3BPG is formed. Since 2,3BPG is a negative effector of hemoglobin affinity for oxygen, fetal erythrocytes have a higher oxygen affinity than maternal erythrocytes. Therefore, transfer of oxygen from maternal hemoglobin to fetal hemoglobin is favored, assuring the fetal oxygen supply. In the newborn, an erythrocyte isozyme of the M-type with comparatively low PK activity displaces the fetal type, resulting in an accumulation of glycolytic intermediates. The increased 1,3BPG levels activate the 2,3BPG shunt, producing 2,3BPG needed to regulate oxygen binding to hemoglobin.

Genetic diseases of adult erythrocyte PK are known in which the kinase is virtually inactive. The erythrocytes of affected individuals have a greatly reduced capacity to make ATP and thus do not have sufficient ATP to perform activities such as ion pumping and maintaining osmotic balance. These erythrocytes have a short half-life, lyse readily, and are responsible for some cases of **hereditary hemolytic anemia**.

The liver PK isozyme is regulated by phosphorylation, allosteric effectors, and modulation of gene expression. The major allosteric effectors are F1,6BP, which stimulates PK activity by decreasing its  $K_{m(app)}$  for PEP, and for the negative effector, ATP. Expression of the liver PK gene is strongly influenced by the quantity of carbohydrate in the diet, with high-carbohydrate diets inducing up to 10-fold increases in PK concentration as compared to low carbohydrate diets. Liver PK is phosphorylated and inhibited by PKA, and thus it is under hormonal control similar to that described earlier for PFK-2.

Muscle PK (M-type) is not regulated by the same mechanisms as the liver enzyme. Extracellular conditions that lead to the phosphorylation and inhibition of liver PK, such as low blood glucose and high levels of circulating glucagon, do not inhibit the muscle enzyme. The result of this differential regulation is that hormones such as glucagon and epinephrine favor liver gluconeogenesis by inhibiting liver glycolysis, while at the same time, muscle glycolysis can proceed in accord with needs directed by intracellular conditions.

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## Metabolic Fates of Pyruvate

Pyruvate is the branch point molecule of glycolysis. The ultimate fate of pyruvate depends on the oxidation state of the cell. In the reaction catalyzed by G3PDH a molecule of  $\text{NAD}^+$  is reduced to NADH. In order to maintain the re-dox state of the cell, this NADH must be re-oxidized to  $\text{NAD}^+$ . During aerobic glycolysis this occurs in the mitochondrial electron transport chain generating ATP. Thus, during aerobic glycolysis ATP is generated from oxidation of glucose directly at the PGK and PK reactions as well as indirectly by re-oxidation of NADH in the [oxidative phosphorylation](#) pathway. Additional NADH molecules are generated during the complete aerobic oxidation of pyruvate in the [TCA cycle](#). Pyruvate enters the TCA cycle in the form of **acetyl-CoA** which is the product of the pyruvate dehydrogenase reaction. The fate of pyruvate during anaerobic glycolysis is reduction to lactate.

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## Lactate Metabolism

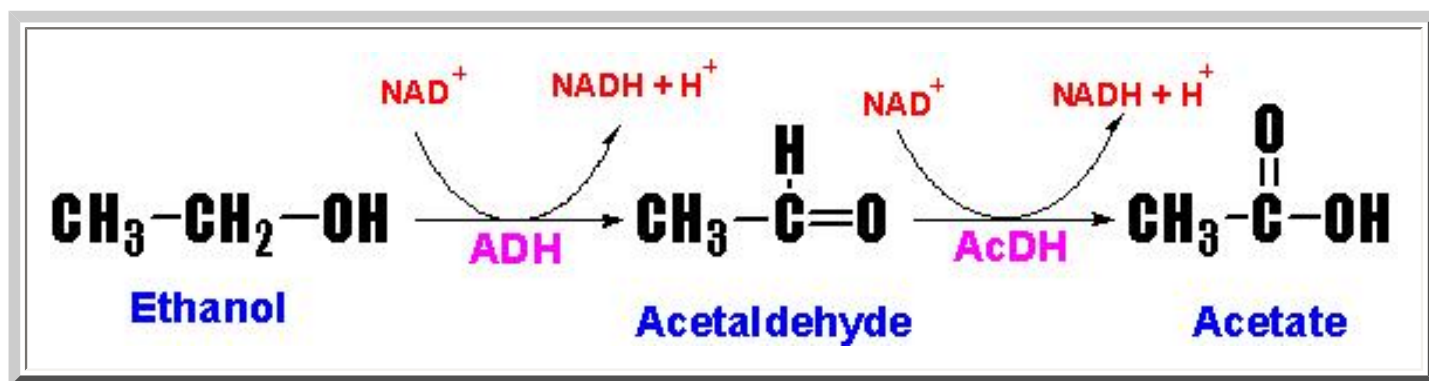
During anaerobic glycolysis, that period of time when glycolysis is proceeding at a high rate (or in anaerobic organisms), the oxidation of NADH occurs through the reduction of an organic substrate. Erythrocytes and skeletal muscle (under conditions of exertion) derive all of their ATP needs through anaerobic glycolysis. The large quantity of NADH produced is oxidized by reducing pyruvate to lactate. This reaction is carried out by lactate dehydrogenase, (LDH). The lactate produced during anaerobic glycolysis diffuses from the tissues and is transported to highly aerobic tissues such as cardiac muscle and liver. The lactate is then oxidized to pyruvate in these cells by LDH and the pyruvate is further oxidized in the TCA cycle. If the energy level in these cells is high the carbons of pyruvate will be diverted back to glucose via the gluconeogenesis pathway.

Mammalian cells contain two distinct types of LDH subunits, termed M and H. Combinations of these different subunits generates LDH isozymes with different characteristics. The H type subunit predominates in aerobic tissues such as heart muscle (as the H<sub>4</sub> tetramer) while the M subunit predominates in anaerobic tissues such as skeletal muscle as the M<sub>4</sub> tetramer). H<sub>4</sub> LDH has a low  $K_m$  for pyruvate and also is inhibited by high levels of pyruvate. The M<sub>4</sub> LDH enzyme has a high  $K_m$  for pyruvate and is not inhibited by pyruvate. This suggests that the H-type LDH is utilized for oxidizing lactate to pyruvate and the M-type the reverse.

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## Ethanol Metabolism

Animal cells (primarily hepatocytes) contain the cytosolic enzyme alcohol dehydrogenase (ADH) which oxidizes ethanol to acetaldehyde. Acetaldehyde then enters the mitochondria where it is oxidized to acetate by acetaldehyde dehydrogenase (AcDH).



Acetaldehyde forms adducts with proteins, nucleic acids and other compounds, the results of which are the toxic side effects (the **hangover**) that are associated with alcohol consumption. The ADH and AcDH catalyzed reactions also leads to the reduction of  $\text{NAD}^+$  to NADH. The metabolic effects of ethanol intoxication stem from the actions of ADH and AcDH and the resultant cellular imbalance in the  $\text{NADH}/\text{NAD}^+$ . The NADH produced in the cytosol by ADH must be reduced back to  $\text{NAD}^+$  via either the [malate-aspartate shuttle](#) or the [glycerol-phosphate shuttle](#). Thus, the ability of an individual to metabolize ethanol is dependent upon the capacity of hepatocytes to carry out either of these 2 shuttles, which in turn is affected by the rate of the [TCA cycle](#) in the mitochondria whose rate of function is being impacted by the NADH produced by the AcDH reaction. The reduction in  $\text{NAD}^+$  impairs the flux of glucose through glycolysis at the glyceraldehyde-3-phosphate dehydrogenase reaction, thereby limiting energy production. Additionally, there is an increased rate of hepatic lactate production due to the effect of increased NADH on direction of the hepatic lactate dehydrogenase (LDH) reaction. This reversal of the LDH reaction in hepatocytes diverts pyruvate from gluconeogenesis leading to a reduction in the capacity of the liver to deliver glucose to the blood.

In addition to the negative effects of the altered  $\text{NADH}/\text{NAD}^+$  ratio on hepatic gluconeogenesis, fatty acid oxidation is also reduced as this process requires  $\text{NAD}^+$  as a cofactor. In fact the opposite is true, fatty acid synthesis is increased and there is an increase in triacylglyceride production by the liver. In the mitochondria, the production of acetate from acetaldehyde leads to increased levels of acetyl-CoA. Since the increased generation of NADH also reduces the activity of the TCA cycle, the acetyl-CoA is diverted to fatty acid synthesis. The reduction in cytosolic  $\text{NAD}^+$  leads to reduced activity of glycerol-3-phosphate dehydrogenase (in the glycerol 3-phosphate to DHAP direction) resulting in increased levels of glycerol 3-phosphate which is the backbone for the synthesis of the triacylglycerides. Both of these two events lead to fatty acid deposition in the liver leading to **fatty liver syndrome**.

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## Regulation of Blood Glucose Levels

If for no other reason, it is because of the demands of the brain for oxidizable glucose that the human body exquisitely regulates the level of glucose circulating in the blood. This level is maintained in the range of 5mM.

Nearly all carbohydrates ingested in the diet are converted to glucose following transport to the liver. Catabolism of dietary or cellular proteins generates carbon atoms that can be utilized for glucose synthesis via [gluconeogenesis](#). Additionally, other tissues besides the liver that incompletely oxidize glucose (predominantly skeletal muscle and erythrocytes) provide lactate that can be converted to glucose via gluconeogenesis.

Maintenance of blood glucose homeostasis is of paramount importance to the survival of the human organism. The predominant tissue responding to signals that indicate reduced or elevated blood glucose levels is the liver. Indeed, one of the most important functions of the liver is to produce glucose for the circulation. Both elevated and reduced levels of blood glucose trigger hormonal responses to initiate pathways designed to restore glucose homeostasis. Low blood glucose triggers release of **glucagon** from pancreatic  $\alpha$ -cells. High blood glucose triggers release of **insulin** from pancreatic  $\beta$ -cells. Additional signals, **ACTH and growth hormone**, released from the pituitary act to increase blood glucose by inhibiting uptake by extrahepatic tissues. **Glucocorticoids** also act to increase blood glucose levels by inhibiting glucose uptake. **Cortisol**, the major glucocorticoid released from the adrenal cortex, is secreted in response to the increase in circulating ACTH. The adrenal medullary hormone, **epinephrine**, stimulates production of glucose by activating glycogenolysis in response to stressful stimuli.

Glucagon binding to its' receptors on the surface of liver cells triggers an increase in cAMP production

leading to an increased rate of [glycogenolysis](#) by activating glycogen phosphorylase via the PKA-mediated cascade. This is the same response hepatocytes have to epinephrine release. The resultant increased levels of G6P in hepatocytes is hydrolyzed to free glucose, by glucose-6-phosphatase, which then diffuses to the blood. The glucose enters extrahepatic cells where it is re-phosphorylated by hexokinase. Since muscle and brain cells lack glucose-6-phosphatase, the glucose-6-phosphate product of hexokinase is retained and oxidized by these tissues.

In opposition to the cellular responses to glucagon (and epinephrine on hepatocytes), insulin stimulates extrahepatic uptake of glucose from the blood and inhibits glycogenolysis in extrahepatic cells and conversely stimulates glycogen synthesis. As the glucose enters hepatocytes it binds to and inhibits glycogen phosphorylase activity. The binding of free glucose stimulates the de-phosphorylation of phosphorylase thereby, inactivating it. Why is it that the glucose that enters hepatocytes is not immediately phosphorylated and oxidized? Liver cells contain an isoform of hexokinase called glucokinase. Glucokinase has a much lower affinity for glucose than does hexokinase. Therefore, it is not fully active at the physiological ranges of blood glucose. Additionally, glucokinase is not inhibited by its product G6P, whereas, hexokinase is inhibited by G6P.

One major response of non-hepatic tissues to insulin is the recruitment, to the cell surface, of glucose transporter complexes. Glucose transporters comprise a family of five members, **GLUT-1 to GLUT-5**. GLUT-1 is ubiquitously distributed in various tissues. GLUT-2 is found primarily in intestine, kidney and liver. GLUT-3 is also found in the intestine and GLUT-5 in the brain and testis. GLUT-5 is also the major glucose transporter present in the membrane of the endoplasmic reticulum (ER) and serves the function of transporting glucose to the cytosol following its' dephosphorylation by the ER enzyme glucose 6-phosphatase. Insulin-sensitive tissues such as skeletal muscle and adipose tissue contain GLUT-4. When the concentration of blood glucose increases in response to food intake, pancreatic GLUT-2 molecules mediate an increase in glucose uptake which leads to increased insulin secretion. Recent evidence has shown that the cell surface receptor for the human T cell leukemia virus (HTLV) is the ubiquitous GLUT-1.

Hepatocytes, unlike most other cells, are freely permeable to glucose and are, therefore, essentially unaffected by the action of insulin at the level of increased glucose uptake. When blood glucose levels are low the liver does not compete with other tissues for glucose since the extrahepatic uptake of glucose is stimulated in response to insulin. Conversely, when blood glucose levels are high extrahepatic needs are satisfied and the liver takes up glucose for conversion into [glycogen](#) for future needs. Under conditions of high blood glucose, liver glucose levels will be high and the activity of glucokinase will be elevated. The G6P produced by glucokinase is rapidly converted to G1P by phosphoglucomutase, where it can then be incorporated into glycogen.

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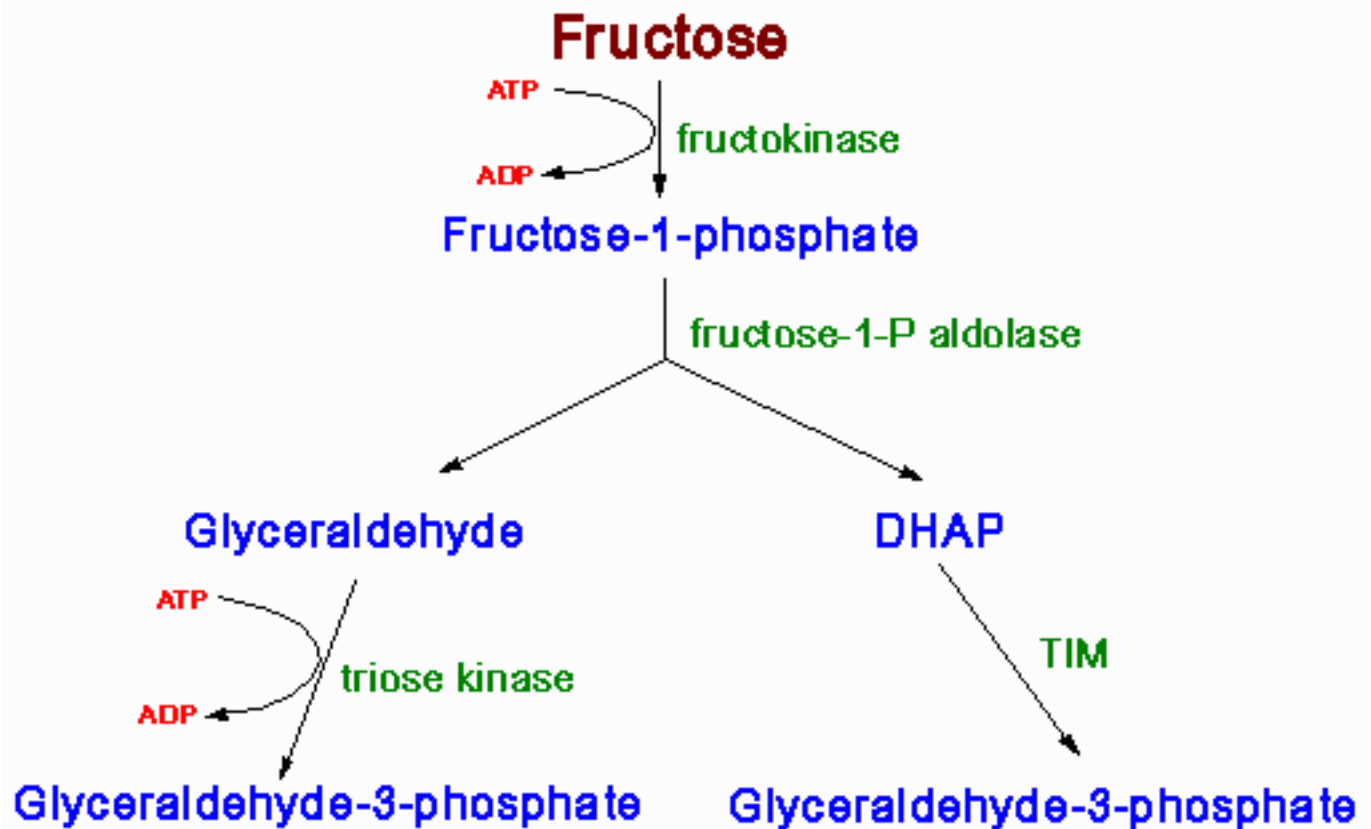
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## Fructose Metabolism

Diets containing large amounts of sucrose (a disaccharide of glucose and fructose) can utilize the fructose as a major source of energy. The pathway to utilization of fructose differs in muscle and liver. Muscle which contains only hexokinase can phosphorylate fructose to F6P which is a direct glycolytic intermediate.

In the liver which contains mostly glucokinase, which is specific for glucose as its substrate, requires the function of additional enzymes to utilize fructose in glycolysis. Hepatic fructose is phosphorylated on C-1 by fructokinase yielding fructose-1-phosphate (F1P). In liver the form of aldolase that predominates (aldolase B) can utilize both F-1,6-BP and F1P as substrates. Therefore, when presented with F1P the enzyme generates DHAP and glyceraldehyde. The DHAP is converted, by triose phosphate isomerase, to G3P and enters glycolysis. The glyceraldehyde can be phosphorylated to G3P by glyceraldehyde kinase or converted to DHAP through the concerted actions of alcohol dehydrogenase, glycerol kinase and glycerol phosphate dehydrogenase.



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Entry of fructose carbon atoms into the glycolytic pathway in hepatocytes.

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## Clinical Significances of Fructose Metabolism

Three inherited abnormalities in fructose metabolism have been identified. **Essential fructosuria** is a benign metabolic disorder caused by the lack of fructokinase which is normally present in the liver, pancreatic islets and kidney cortex. The fructosuria of this disease depends on the time and amount of fructose and sucrose intake. Since the disorder is asymptomatic and harmless it may go undiagnosed.

**Hereditary fructose intolerance** is a potentially lethal disorder resulting from a lack of aldolase B which is normally present in the liver, small intestine and kidney cortex. The disorder is characterized by severe hypoglycemia and vomiting following fructose intake. Prolonged intake of fructose by infants with this defect leads to vomiting, poor feeding, jaundice, hepatomegaly, hemorrhage and eventually hepatic failure and death. The hypoglycemia that result following fructose uptake is caused by fructose-1-phosphate inhibition

of glycogenolysis, by interfering with the phosphorylase reaction, and inhibition of gluconeogenesis at the deficient aldolase step. Patients remain symptom free on a diet devoid of fructose and sucrose.

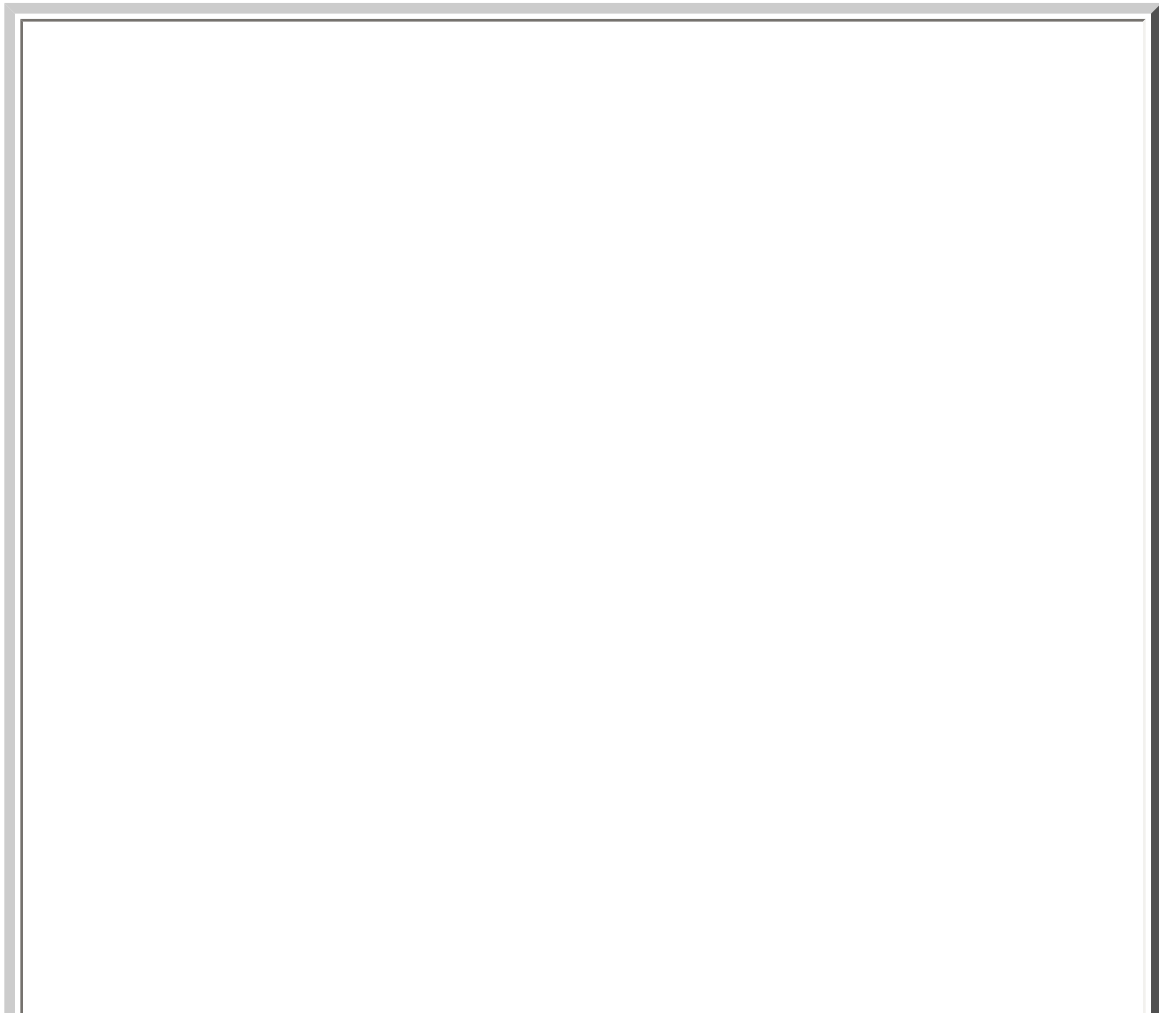
[Hereditary fructose-1,6-bisphosphatase deficiency](#) results in severely impaired hepatic gluconeogenesis and leads to episodes of hypoglycemia, apnea, hyperventilation, ketosis and lactic acidosis. These symptoms can take on a lethal course in neonates. Later in life episodes are triggered by fasting and febrile infections.

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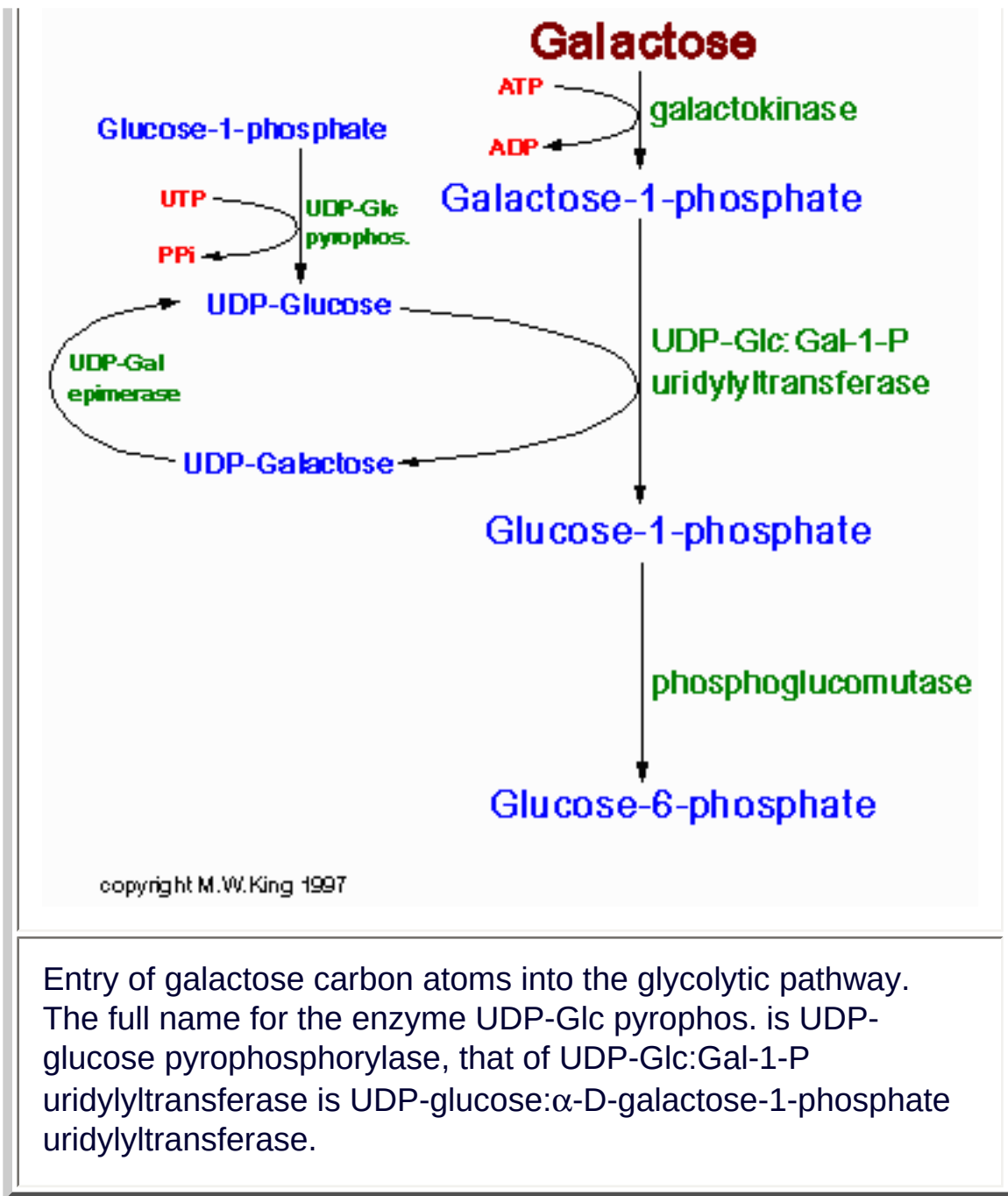
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## Galactose Metabolism

Galactose, which is metabolized from the milk sugar, lactose (a disaccharide of glucose and galactose), enters glycolysis by its conversion to glucose-1-phosphate (G1P). This occurs through a series of steps. First the galactose is phosphorylated by galactokinase to yield galactose-1-phosphate. Epimerization of galactose-1-phosphate to G1P requires the transfer of UDP from uridine diphosphoglucose (UDP-glucose) catalyzed by galactose-1-phosphate uridyl transferase. This generates UDP-galactose and G1P. The UDP-galactose is epimerized to UDP-glucose by UDP-galactose-4 epimerase. The UDP portion is exchanged for phosphate generating glucose-1-phosphate which then is converted to G6P by phosphoglucose mutase.







Entry of galactose carbon atoms into the glycolytic pathway. The full name for the enzyme UDP-Glc pyrophos. is UDP-glucose pyrophosphorylase, that of UDP-Glc:Gal-1-P uridylyltransferase is UDP-glucose:α-D-galactose-1-phosphate uridylyltransferase.

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## Clinical Significances of Galactose Metabolism

Three inherited disorders of galactose metabolism have been delineated. **Classic galactosemia** is a major symptom of two enzyme defects. One results from loss of the enzyme **galactose-1-phosphate uridyl transferase**. The second form of galactosemia results from a loss of **galactokinase**. These two defects are manifest by a failure of neonates to thrive. Vomiting and diarrhea occur following ingestion of milk, hence individuals are termed **lactose intolerant**. Clinical findings of these disorders include impaired liver function (which if left untreated leads to severe cirrhosis), elevated blood galactose, hypergalactosemia,

hyperchloremic metabolic acidosis, urinary galactitol excretion and hyperaminoaciduria. Unless controlled by exclusion of galactose from the diet, these galactosemias can go on to produce blindness and fatal liver damage. Even on a galactose-restricted diet, transferase-deficient individuals exhibit urinary galactitol excretion and persistently elevated erythrocyte galactose-1-phosphate levels. Blindness is due to the conversion of circulating galactose to the sugar alcohol **galacitol**, by an NADPH-dependent galactose reductase that is present in neural tissue and in the lens of the eye. At normal circulating levels of galactose this enzyme activity causes no pathological effects. However, a high concentration of galacitol in the lens causes osmotic swelling, with the resultant formation of cataracts and other symptoms. The principal treatment of these disorders is to eliminate lactose from the diet.

The third disorder of galactose metabolism result from a deficiency of **UDP-galactose-4-epimerase**. Two different forms of this deficiency have been found. One is benign affecting only red and white blood cells. The other affects multiple tissues and manifests symptoms similar to the transferase deficiency. Treatment involves restriction of dietary galactose.  
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## Mannose Metabolism

The digestion of many polysaccharides and glycoproteins yields mannose which is phosphorylated by hexokinase to generate mannose-6-phosphate. Mannose-6-phosphate is converted to fructose-6-phosphate, by the enzyme phosphomannose isomerase, and then enters the glycolytic pathway or is converted to glucose-6-phosphate by the gluconeogenic pathway of hepatocytes.  
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## Glycerol Metabolism

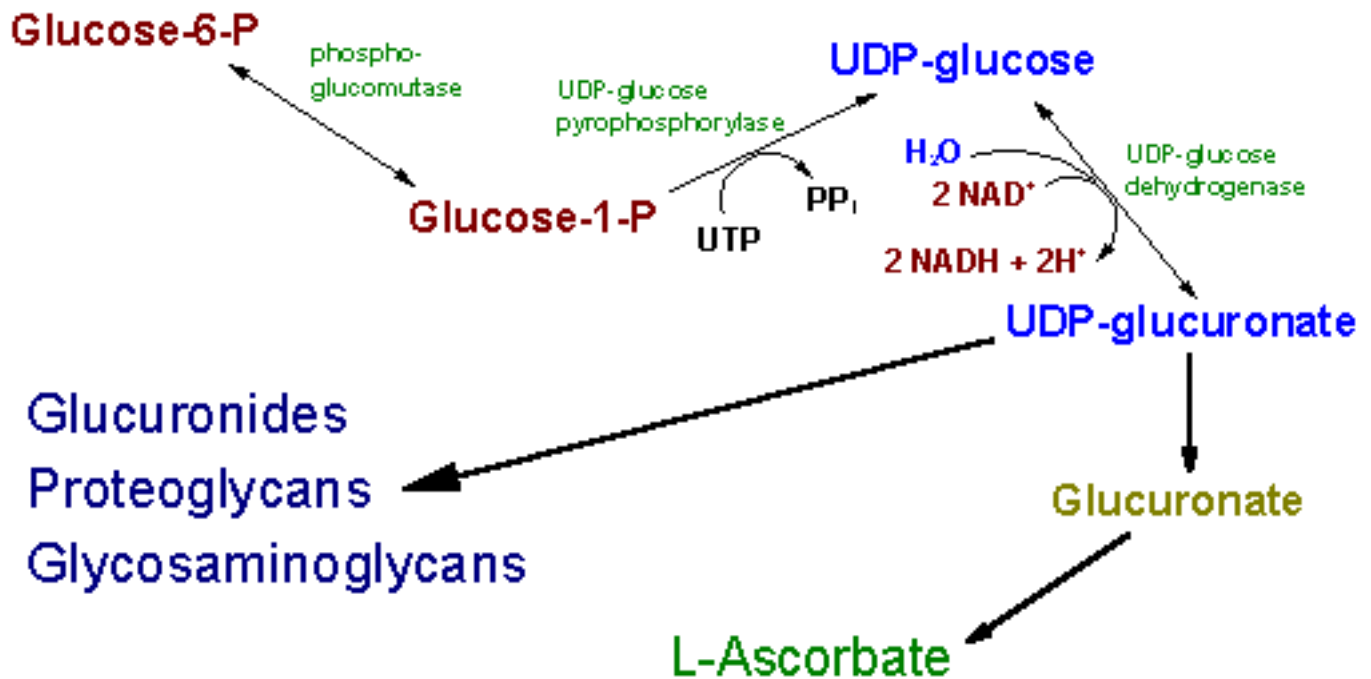
The predominant source of glycerol is adipose tissue. This molecule is the backbone for the triacylglycerols. Following release of the fatty acid portions of triacylglycerols the glycerol backbone is transported to the liver where it is phosphorylated by glycerol kinase yielding glycerol-3-phosphate. Glycerol-3-phosphate is oxidized to DHAP by glycerol-3-phosphate dehydrogenase. DHAP then enters the glycolytic if the liver cell needs energy. However, the more likely fate of glycerol is to enter the gluconeogenesis pathway in order for the liver to produce glucose for use by the rest of the body.  
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## Glucuronate Metabolism

Glucuronate is a highly polar molecule which is incorporated into proteoglycans as well as combining with **bilirubin** and **steroid hormones**; it can also be combined with certain drugs to increase their solubility. Glucuronate is derived from glucose in the **uronic acid pathway**.

# Glucuronic Acid Synthesis



The uronic acid pathway is utilized to synthesize UDP-glucuronate, glucuronate and L-ascorbate. The pathway involves the oxidation of glucosae-6-phosphate to UDP-glucuronate. The oxidation is uncoupled from energy production. UDP-glucuronate is used in the synthesis of **glycosaminoglycan and proteoglycans** as well as forming complexes with bilirubin, steroids and certain drugs. The glucuronate complexes form to solubilize compounds for excretion. The synthesis of ascorbate (**vitamin C**) does not occur in primates.

The uronic acid pathway is an alternative pathway for the oxidation of glucose that does not provide a means of producing ATP, but is utilized for the generation of the activated form of glucuronate, **UDP-glucuronate**. The uronic acid pathway of glucose conversion to glucuronate begins by conversion of glucose-6-phosphate to glucose-1-phosphate by phosphoglucomutase, and then activated to UDP-glucose by UDP-glucose pyrophosphorylase. UDP-glucose is oxidized to UDP-glucuronate by the NAD<sup>+</sup>-requiring enzyme, UDP-glucose dehydrogenase. UDP-glucuronate then serves as a precursor for the synthesis of **iduronic acid** and **UDP-xylose** and is incorporated into proteoglycans and glycoproteins or forms conjugates with bilirubin, steroids, xenobiotics, drugs and many

compounds containing hydroxyl (-OH) groups.

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## Clinical Significance of Glucuronate

In the adult human, a significant number of erythrocytes die each day. This turnover releases significant amounts of the iron-free portion of heme, **porphyrin**, which is subsequently degraded. The primary sites of porphyrin degradation are found in the reticuloendothelial cells of the liver, spleen and bone marrow. The breakdown of porphyrin yields **bilirubin**, a product that is non-polar and therefore, insoluble. In the liver, to which is transported in the plasma bound to albumin, bilirubin is solubilized by conjugation to glucuronate. The soluble conjugated **bilirubin diglucuronide** is then secreted into the bile. An inability to conjugate bilirubin, for instance in hepatic disease or when the level of bilirubin production exceeds the capacity of the liver, is a contributory cause of **jaundice**.

The conjugation of glucuronate to certain non-polar drugs is important for their solubilization in the liver. Glucuronate conjugated drugs are more easily cleared from the blood by the kidneys for excretion in the urine. The glucuronate-drug conjugation system can, however, lead to drug resistance; chronic exposure to certain drugs, such as barbiturates and AZT, leads to an increase in the synthesis of the UDP-glucuronyltransferases in the liver that are involved in glucuronate-drug conjugation. The increased levels of these hepatic enzymes result in a higher rate of drug clearance leading to a reduction in the effective dose of glucuronate cleared drugs.

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## Introduction

Iron serves numerous important functions in the body relating to the metabolism of oxygen, not the least of which is its role in hemoglobin transport of oxygen. Within the body iron exist in two oxidation states: ferrous ( $\text{Fe}^{2+}$ ) or ferric ( $\text{Fe}^{3+}$ ). Because iron has an affinity electronegative atoms such as oxygen, nitrogen and sulfur, these atoms are found at the heart of the iron-binding centers of macromolecules.

Under conditions of neutral or alkaline pH, iron is found in the  $\text{Fe}^{3+}$  state and at acidic pH the  $\text{Fe}^{2+}$  state is favored. When in the  $\text{Fe}^{3+}$  state, iron will form large complexes with anions, water and peroxides. These large complexes have poor solubility and upon their aggregation lead to pathological consequences. In addition, iron can bind to and interfere with the structure and function of various macromolecules. For this reason the body must protect itself against the deleterious effects of iron. This is the role served by numerous iron-binding proteins (see below).

Aside from its importance as the prosthetic group of hemoglobin and a small number of enzymes (e.g., redox cytochromes and the  $\text{P}_{450}$  class of detoxifying cytochromes), **heme** is important because a number of genetic disease states are associated with deficiencies of the enzymes used in its biosynthesis. Some of these disorders are readily diagnosed because they cause  **$\delta$ -aminolevulinic acid**, (ALA) and other abnormally colored heme intermediates to appear in the circulation, the urine, and in other tissues such as teeth and bones. Some disorders of heme biosynthesis are more insidious such as the various **porphyrias**, a list of which can be found [below](#) and in the [Inborn Errors Page](#).

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## Iron Metabolism

Iron is associated with proteins either by incorporation into protoporphyrin IX or by binding to other ligands. When the ferrous form of iron and protoporphyrin IX are complexed

the structure is referred to as heme. There are a number of heme containing proteins involved in the transport of oxygen (hemoglobin), oxygen storage (myoglobin) and enzyme catalysis such as [nitric oxide synthase \(NOS\)](#) and [prostaglandin synthase \(cyclooxygenase\)](#). A number of non-heme iron containing proteins are also known such as the iron-sulfur proteins of [oxidative phosphorylation](#) and the iron transport and storage proteins, transferrin and ferritin, respectively.

Transferrin, made in the liver, is the serum protein responsible for the transport of iron. Although several metals can bind to transferrin, the highest affinity is for the ferric ( $\text{Fe}^{3+}$ ) form of iron. The ferrous form of iron does not bind to transferrin. Transferrin can bind two moles of iron. Cells take up the transported iron through interaction of transferrin with cell-surface receptors. Internalization of the iron-transferrin-receptor complexes is initiated following receptor phosphorylation by PKC. Following internalization, the iron is released due to the acidic nature of the lysosomes. The transferrin receptor is then recycled back to the cell surface.

Ferritin is the major protein used for intracellular storage of iron. The majority of intracellularly stored iron is found in the liver, skeletal muscle and reticuloendothelial cells. If the storage capacity of the ferritin is exceeded, iron will deposit adjacent to the ferritin-iron complexes in the cell. Histologically these amorphous iron deposits are referred to as **hemosiderin**.

The regulation of iron utilization in the body is primarily controlled via iron-mediated regulation of mRNA translation. The description of this process can be found in the [Protein Synthesis](#) page. Both the transferrin receptor and the ferritin mRNAs contain stem-loop structures termed **iron responsive elements, IREs**. These IREs are recognized by an iron-binding protein containing an iron-sulfur center similar to that of the [TCA cycle](#) enzyme aconitase. Other IRE containing mRNAs include those encoding the erythrocyte protoporphyrin synthesis enzyme, ALA synthase (see below), mitochondrial aconitase and ferroportin 1. [back to the top](#)

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## Clinical Aspects of Iron Metabolism

Iron can bind to and form complexes with numerous macromolecules, the consequences of which can be a disruption in normal activities of the affected complexes. Excess intracellular iron results in formation and deposition of hemosiderin which can lead to cellular dysfunction and damage. Thus, the consequences of excess iron intake and storage can have profound consequences. However, one must also consider that a reduction in iron intake can also lead to untoward consequences. Most notably, a reduced iron level negatively affects the function of oxygen transport in red blood cells. Defects in iron metabolism can result from impaired intestinal absorption, excess loss of heme iron due to bleeding as well as to mutations in the iron response elements of iron regulated mRNAs.

**Hemochromatosis** is defined as a disorder in iron metabolism that is characterized by excess iron absorption, saturation of iron-binding proteins and deposition of hemosiderin in the tissues. The primary affected tissues are the liver pancreas and skin. Iron deposition in the liver leads to cirrhosis and in the pancreas causes diabetes. The excess iron deposition leads



to bronze pigmentation of the organs and skin. In fact, the bronze skin pigmentation seen in hemochromatosis, coupled with the resultant diabetes lead to the designation of this condition as **bronze diabetes**.

The primary cause of hemochromatosis is the inheritance of an autosomal recessive allele. The locus causing hemochromatosis has been designated the HFE and is a **major histocompatibility complex (MHC) class-1** gene. The gene encodes an  $\alpha$  chain protein with three immunoglobulin-like domains. This  $\alpha$  chain protein associates with  $\beta_2$ -microglobulin. Normal HFE has been shown to form a complex with the transferrin receptor and in so doing is thought to regulate the rate of iron transfer into cells. A mutation in HFE will therefore, lead to increased iron uptake and storage.

The majority of hereditary hemochromatosis patients have inherited a mutation in HFE that results in the substitution of Cys 282 for a Tyr. This mutation causes loss of conformation of one of the immunoglobulin domains in HFE. Another mutation found in HFE causes a change of His 68 to Asp.

Secondary hemochromatosis (not caused by mutation in HFE and thus not inherited) can result from excess oral intake of iron or in patients receiving blood transfusions.

**Iron deficiency anemia** is characterized by microcytic (small) and hypochromic (low pigment) red blood cells. Reduced iron intake and/or excess iron excretion results in a decreased globin protein content in red blood cells as a consequence of the heme control of globin synthesis (see the [Protein Synthesis](#) page for details).

The most common causes of iron deficient anemia are excess menstrual flow or gastrointestinal (GI) bleeding. Causes of GI bleeding can include the use of medications that lead to ulceration or erosion of the gastric mucosa, peptic ulcer disease, gastric tumors, hiatal hernia or the gastritis associated with chronic alcohol consumption.

Treatment of iron deficiency anemia is to first determine the cause and source of the excess bleeding. Oral administration of ferrous sulfate is commonly used to supplement the iron loss, however, intravenous iron therapy may be called for in some cases. Severe iron deficiency anemia may necessitate transfusion with packed red blood cells.

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## Synthesis of Porphobilinogen and Heme

The first reaction in heme biosynthesis takes place in the mitochondrion and involves the condensation of 1 glycine and 1 succinylCoA by the pyridoxal phosphate-containing enzyme,  $\delta$ -aminolevulinic acid synthase (ALA synthase). This reaction is both the rate-limiting reaction of heme biosynthesis, and the most highly regulated reaction (see [Regulation](#) below). Mitochondrial  $\delta$ -aminolevulinic acid (ALA) is transported to the cytosol, where ALA dehydratase (also called porphobilinogen synthase or hydroxymethylbilane synthase) dimerizes 2 molecules of ALA to produce the pyrrole ring compound **porphobilinogen**.

The next step in the pathway involves the head-to-tail condensation of 4 molecules of porphobilinogen to produce the linear tetrapyrrole intermediate, **hydroxymethylbilane**. The enzyme for this condensation is porphobilinogen deaminase (PBG deaminase). This enzyme

is also called uroporphyrinogen I synthase. Hydroxymethylbilane has two main fates. The most important is regulated, enzymatic conversion to **uroporphyrinogen III**, the next intermediate on the path to heme. This step is mediated by a holoenzyme comprised of uroporphyrinogen synthase plus a protein known as uroporphyrinogen III cosynthase.

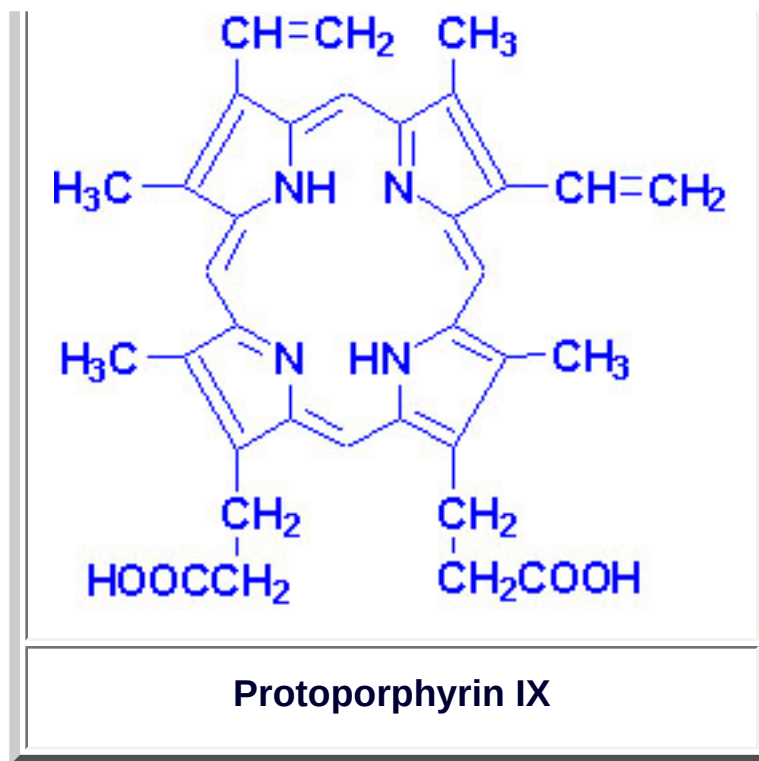
Genetic defects that cause increased ALA synthase activity or decreased uroporphyrinogen I synthase activity lead to the disease known as **acute intermittent porphyria**, which is diagnosed by the excretion of excess porphobilinogen (a condition that is not obvious from the color of the urine).

Genetic defects in enzymes beyond hydroxymethylbilane in the biosynthetic pathway result in the accumulation of excess hydroxymethylbilane, leading to an alternative non-enzymatic cyclization. In **erythropoietic protoporphyria**, bone marrow uroporphyrinogen III cosynthase is only present at about 30% of the normal level; the result is that massive amounts of Type I uroporphyrinogen and its highly colored oxidation products are found in the urine and deposited in a variety of tissues, including teeth and bones.

In the cytosol, the acetate substituents of uroporphyrinogen (normal uroporphyrinogen III or abnormal uroporphyrinogen I) are all decarboxylated by the enzyme uroporphyrinogen decarboxylase. The resultant products have methyl groups in place of acetate and are known as coproporphyrinogens, with **coproporphyrinogen III** being the , important, normal intermediate in heme synthesis.

Coproporphyrinogen III is transported to the interior of the mitochondrion, where 2 propionate residues are decarboxylated, yielding vinyl substituents on the 2 pyrrole rings. The colorless product is **protoporphyrinogen IX**. In the mitochondrion, protoporphyrinogen IX is converted to **protoporphyrin IX** (structure shown below) by protoporphyrinogen IX oxidase. The oxidase reaction requires molecular oxygen and results in the loss of 6 protons and 6 electrons, yielding a completely conjugated ring system, which is responsible for the characteristic red color to hemes. The final reaction in heme synthesis also takes place in the mitochondrion and involves the insertion of the iron atom into the ring system generating **heme b**. The enzyme catalyzing this reaction is known as ferrochelatase.





The enzymes ferrochelatase, ALA synthase and ALA dehydratase (a sulfhydryl containing enzyme) are highly sensitive to inhibition by heavy metal poisoning. Indeed, a characteristic of lead poisoning is an increase in ALA in the circulation in the absence of an increase in porphobilinogen.

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## Regulation of Heme Biosynthesis

Although heme is synthesized in virtually all tissues, the principal sites of synthesis are erythroid cells (~85%) and hepatocytes (accounting for nearly all the rest of heme synthesis). The differences in these two tissues and their needs for heme result in quite different mechanisms for regulation of heme biosynthesis.

In hepatocytes, heme is required for incorporation into the cytochromes, in particular, the P<sub>450</sub> class of cytochromes that are important for detoxification. In addition numerous cytochromes of the [oxidative-phosphorylation](#) pathway contain heme. The rate-limiting step in hepatic heme biosynthesis occurs at the ALA synthase catalyzed step, which is the committed step in heme synthesis. The Fe<sup>3+</sup> oxidation product of heme is termed **hemin**. Hemin acts as a feed-back inhibitor on ALA synthase. Hemin also inhibits transport of ALA synthase from the cytosol (its' site of synthesis) into the mitochondria (its' site of action) as well as represses synthesis of the enzyme.

In erythroid cells all of the heme is synthesized for incorporation into hemoglobin and occurs only upon differentiation when synthesis of hemoglobin proceeds. When red cells mature both heme and hemoglobin synthesis ceases. The heme (and hemoglobin) must, therefore, survive for the life of the erythrocyte (normally this is 120 days). In reticulocytes

(immature erythrocytes) heme stimulates protein synthesis. The mechanism of this mode of heme-mediate regulation of protein synthesis is described in the [Protein Synthesis](#) section. Additionally, control of heme biosynthesis in erythrocytes occurs at numerous sites other than at the level of ALA synthase. Control has been shown to be exerted on ferrochelatase, the enzyme responsible for iron insertion into protoporphyrin IX, and on uroporphobilinogen deaminase.

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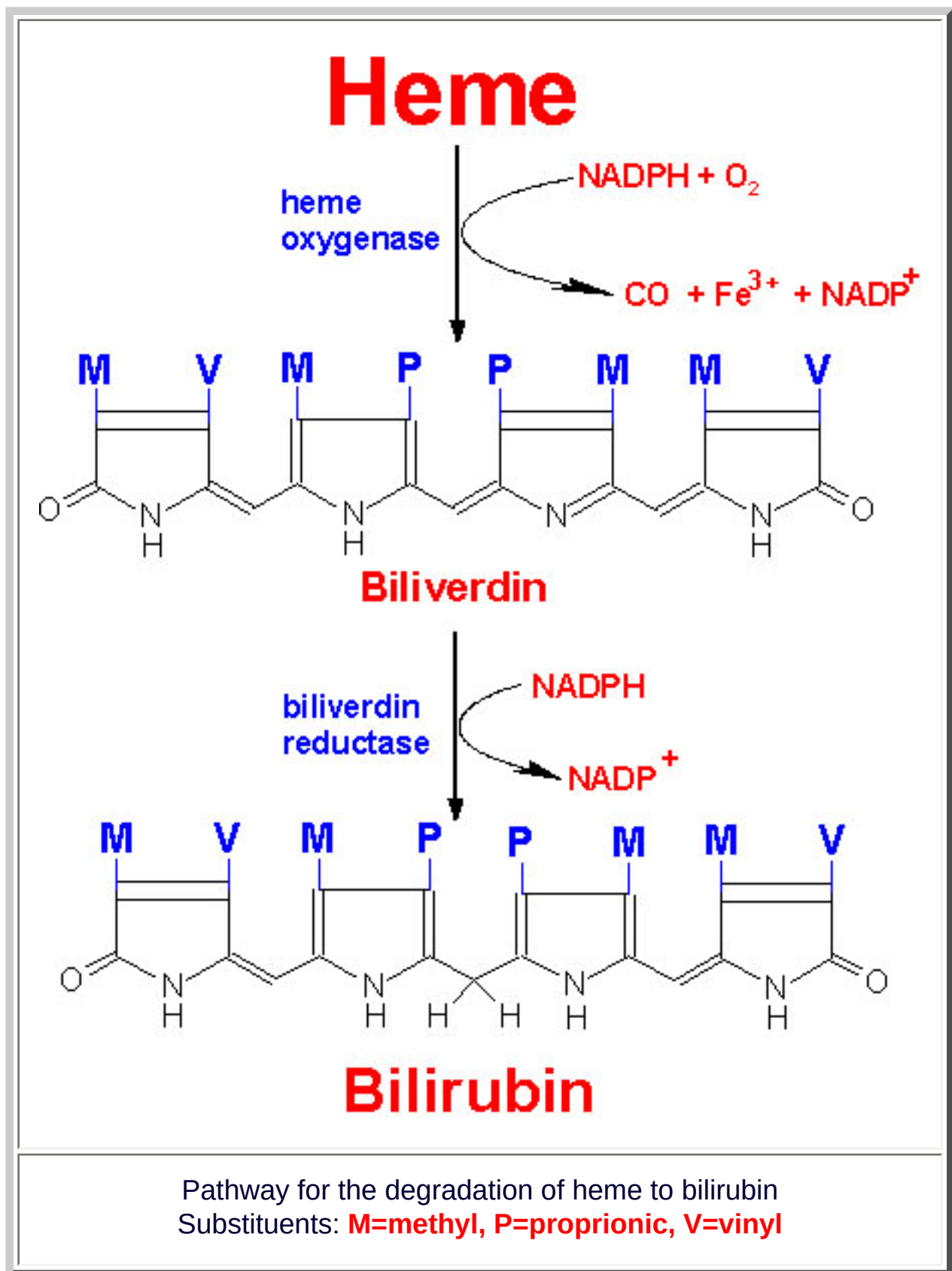
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## Heme Metabolism

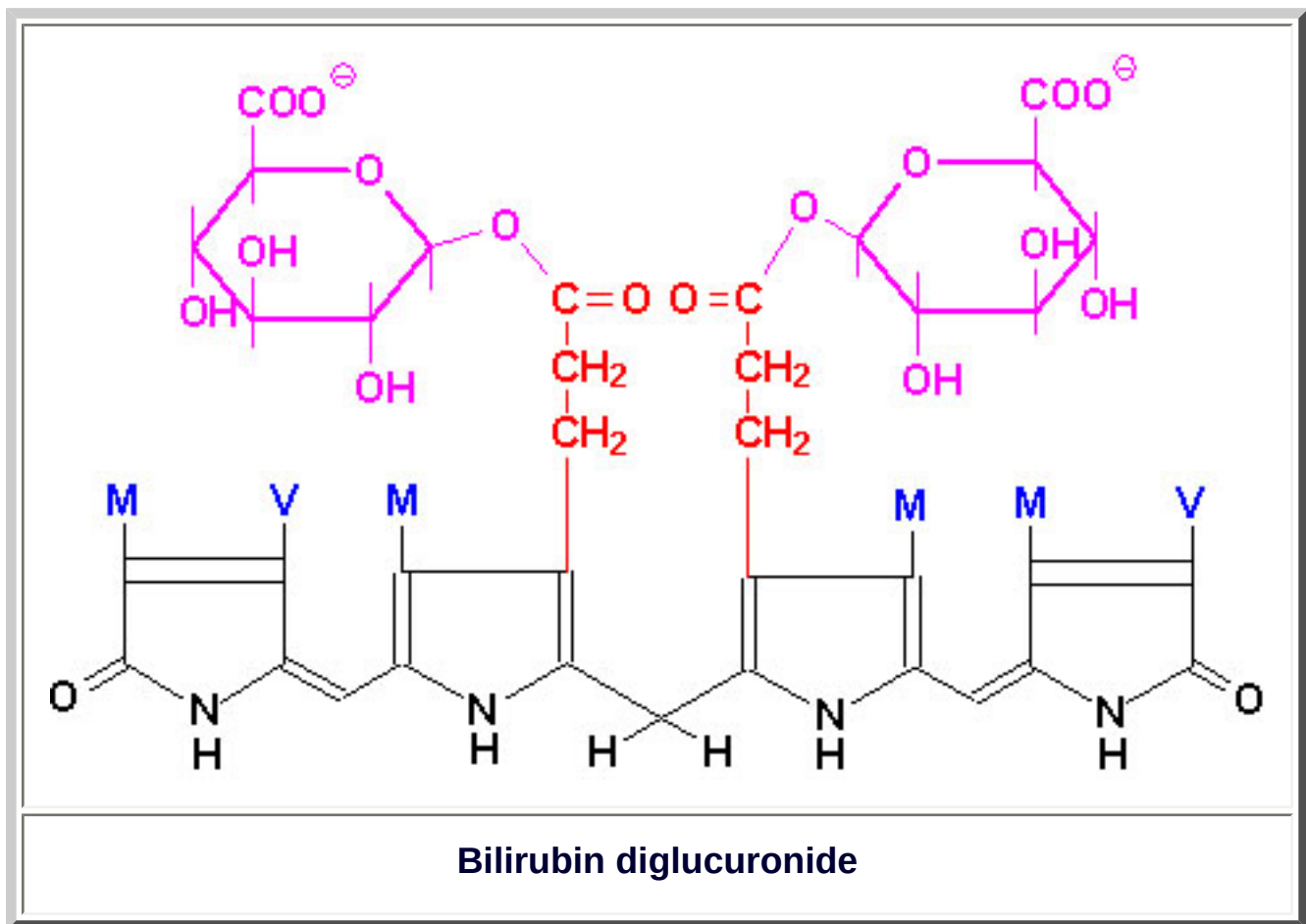
The largest repository of heme in the human body is in red blood cells, which have a life span of about 120 days. There is thus a turnover of about 6 g/day of hemoglobin, which presents 2 problems. First, the porphyrin ring is hydrophobic and must be solubilized to be excreted. Second, iron must be conserved for new heme synthesis.

Normally, senescent red blood cells and heme from other sources are engulfed by cells of the reticuloendothelial system. The globin is recycled or converted into amino acids, which in turn are recycled or catabolized as required. Heme is oxidized, with the heme ring being opened by the endoplasmic reticulum enzyme, heme oxygenase. The oxidation step requires heme as a substrate, and any hemin ( $\text{Fe}^{3+}$ ) is reduced to heme ( $\text{Fe}^{2+}$ ) prior to oxidation by heme oxygenase. The oxidation occurs on a specific carbon producing the linear tetrapyrrole **biliverdin**, ferric iron ( $\text{Fe}^{3+}$ ), and carbon monoxide (CO). This is the only reaction in the body that is known to produce CO. Most of the CO is excreted through the lungs, with the result that the CO content of expired air is a direct measure of the activity of heme oxygenase in an individual.

In the next reaction a second bridging methylene (between rings III and IV) is reduced by biliverdin reductase, producing **bilirubin**. Bilirubin is significantly less extensively conjugated than biliverdin causing a change in the color of the molecule from blue-green (biliverdin) to yellow-red (bilirubin). The latter catabolic changes in the structure of tetrapyrroles are responsible for the progressive changes in color of a hematoma, or bruise, in which the damaged tissue changes its color from an initial dark blue to a red-yellow and finally to a yellow color before all the pigment is transported out of the affected tissue. Peripherally arising bilirubin is transported to the liver in association with albumin, where the remaining catabolic reactions take place.



In hepatocytes, UDP glucuronyl transferase adds 2 equivalents of glucuronic acid to bilirubin to produce the more water soluble, **bilirubin diglucuronide** derivative. The increased water solubility of the tetrapyrrole facilitates its excretion with the remainder of the bile as the bile pigments.



In individuals with abnormally high red cell lysis, or liver damage with obstruction of the bile duct, the bilirubin and its precursors accumulate in the circulation; the result is **hyperbilirubinemia**, the cause of the abnormal body pigmentation known as **jaundice**. In normal individuals, intestinal bilirubin is acted on by bacteria to produce the final porphyrin products, **urobilinogens** and **urobilins**, that are found in the feces. Bilirubin and its catabolic products are collectively known as the **bile pigments**.

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## Clinical Aspect of Heme Metabolism

Clinical problems associated with heme metabolism are of two types. Disorders that arise from defects in the enzymes of heme biosynthesis are termed the **porphyrias** and cause elevations in the serum and urine content of intermediates in heme synthesis. Inherited disorders in bilirubin metabolism lead to **hyperbilirubinemia**.

Bilirubin is potentially toxic waste product of heme catabolism. The body eliminates bilirubin by transporting it to the liver bound to albumin in the serum. In the liver it is conjugated with glucuronate which renders it water soluble. The glucuronide conjugate is then excreted in the bile. Persons with extreme elevation in unconjugated bilirubin are susceptible to **bilirubin encephalopathy**, also referred to as **kernicterus**. Accumulation of bilirubin in the plasma and tissues results in **jaundice**.



Several inherited disorders in bilirubin metabolism have been identified. [Gilbert syndrome](#) and [Crigler-Najjar syndrome](#) result from predominantly unconjugated hyperbilirubinemia. [Dubin-Johnson syndrome](#) and [Rotor syndrome](#) result from conjugated hyperbilirubinemia.

The porphyrias are both inherited and acquired disorders in heme synthesis. These disorders are classified as either erythroid or hepatic, depending upon the principal site of expression of the enzyme defect. Eight different porphyrias have been classified. With the exception of the reaction catalyzed by ALA synthase, defects in each of the other enzymes of heme synthesis have been identified. The most commonly occurring porphyria is **acute intermittent porphyria, AIP** which is caused by a defect in porphobilinogen deaminase, (PBG deaminase). This enzyme is also called hydroxymethylbilane synthase or uroporphyrinogen synthase.

All of the porphyrias lead to excretion of heme biosynthetic byproducts that turn the urine red and when deposited in the teeth turn them reddish brown. Accumulation of these byproducts in the skin renders it extremely sensitive to sunlight causing ulceration and disfiguring scars. Increased hair growth is also a symptom of the porphyrias leading to appearance of fine hairs over the entire face and on the extremities.

| Porphyria                                                 | Enzyme Defect                   | Primary Symptom  |
|-----------------------------------------------------------|---------------------------------|------------------|
| <b>Erythropoietic Class</b>                               |                                 |                  |
| <a href="#">Congenital erythropoietic porphyria, CEP</a>  | Uroporphyrinogen III cosynthase | Photosensitivity |
| <a href="#">Erythropoietic protoporphyria, EPP</a>        | Ferrochelatase                  | Photosensitivity |
| <b>Hepatic Class</b>                                      |                                 |                  |
| <a href="#">ALA dehydratase deficiency porphyria, ADP</a> | ALA dehydratase                 | Neurovisceral    |
| <a href="#">Acute intermittent porphyria, AIP</a>         | PBG deaminase                   | Neurovisceral    |

|                                                     |                                |                                      |
|-----------------------------------------------------|--------------------------------|--------------------------------------|
| <a href="#">Hereditary coproporphyria, HCP</a>      | Coproporphyrinogen oxidase     | Neurovisceral, some photosensitivity |
| <a href="#">Variegate porphyria, VP</a>             | Protoporphyrinogen oxidase     | Neurovisceral, some photosensitivity |
| <a href="#">Porphyria cutanea tarda, PCT</a>        | Uroporphyrinogen decarboxylase | Photosensitivity                     |
| <a href="#">Hepatoerythropoietic porphyria, HEP</a> | Uroporphyrinogen decarboxylase | Photosensitivity, some neurovisceral |

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# Information on Inborn Errors in Metabolism

Choose from the following categories:

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- Defects in Carbohydrate Metabolism
- Defects in Cholesterol and Lipoprotein Metabolism
- Mucopolysaccharide and Glycolipid Disorders
- Defects in Amino and Organic Acid Metabolism
- Porphyrrias and Bilirubinemias
- Errors in Fatty Acid Metabolism
- Defects in Nucleotide Metabolism
- Disorders in Metal Metabolism and Transport
- Defects in Peroxisomes
- Diseases Associated with Defective DNA Repair

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- [Mucolipidoses](#)
  - [Canadian Society for MPS and related Diseases](#) additional information
  - [National MPS Society](#) a USA-based support group

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# Diseases Associated with Oxygen Carrying Proteins

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- [Sickle-Cell Anemia](#)
  - [β-Thalassemias](#)
  - [α-Thalassemias](#)
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# Diseases Associated with Connective Tissue Proteins

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- [Marfan's Syndrome](#)
- [Ehlers-Danlos](#) Type I
- [Ehlers-Danlos](#) Type II
- [Ehlers-Danlos](#) Type III
- [Ehlers-Danlos](#) Type IV autosomal dominant
- [Ehlers-Danlos](#) Type IV autosomal recessive
- [Ehlers-Danlos](#) Type IV-D
- [Ehlers-Danlos](#) Type V
- [Ehlers-Danlos](#) Type VI
- [Ehlers-Danlos](#) Type VII autosomal dominant
- [Ehlers-Danlos](#) Type VII autosomal recessive
- [Ehlers-Danlos](#) Type VIII
- [Ehlers-Danlos](#) with Platelet Dysfunction
- [Cutis Laxa](#)
- [Cutis Laxa](#) recessive Type I
- [Occipital Horn Syndrome](#) Cutis Laxa, X-linked
- [Osteogenesis Imperfecta](#) Type I
- [Osteogenesis Imperfecta](#) Type I-C
- [Osteogenesis Imperfecta](#) Silent Type II/III
- [Osteogenesis Imperfecta](#) Type IV
- [Osteogenesis Imperfecta](#) Neonatal Lethal form
- [Osteogenesis Imperfecta](#) progressively deforming

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# Diseases Associated with Clotting Factor Dysfunction

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- [Afibrinogenemia](#) complete loss of fibrinogen, Factor I
- [Dysfibrinogenemia](#) dysfunctional fibrinogen, Factor I
- [Factor II Disorders](#)
- Factor III (tissue factor) is the only coagulation factor for which a congenital defect has not been identified
- [Factor V Deficiency](#) Labile Factor deficiency
- [Factor VII Deficiency](#)
- [Hemophilia A](#) Factor VIII deficiency
- [Hemophilia B](#) Factor IX deficiency
- [Factor X Deficiency](#)
- [Factor XI Deficiency](#) Rosenthal Syndrome, Plasma Thromboplastin Antecedent (PTA) deficiency
- [Factor XII Deficiency](#) Hageman factor deficiency
- [Factor XIII Deficiency](#)
  
- [Factor V & VIII Combined Deficiency](#)
- [Factor VIII & IX combined Deficiency](#)
- [Factor IX & XI Combined Deficiency](#)
  
- [Protein C Deficiency](#)
- [Protein S Deficiency](#)
  
- [Thrombophilia](#) Antithrombin III deficiency
- [Giant Platelet Syndrome](#) platelet glycoprotein Ib deficiency
- [von Willebrand Disease](#)
- [Fletcher Factor Deficiency](#) Prekallikrein deficiency

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# Glycogen Storage Diseases

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- [Type 0](#)
- [Type I](#) - von Gierke's disease
- [Type Ib](#)
- [Type Ic](#)
- [Type II](#) - Pompe disease
- [Type IIb](#) - Danon disease
- [Type III](#) - Cori disease or Forbes disease
- [Type IV](#) - Andersen disease
- [Type V](#) - McArdle disease
- [Type VI](#) - Hers disease
- [Type VII](#) - Tarui disease
- [Type VIII](#)
- [Type IX](#)
- [Type XI](#) - Fanconi-Bickel syndrome

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## Defects in Fructose, Galactose and Glycerol Metabolism

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- [Hereditary Fructose Intolerance](#) - Aldolase B Deficiency
  - [Fructosuria](#) - Hepatic Fructokinase Deficiency
  - [Classic Galactosemia](#)
  - [Galactose Epimerase Deficiency](#)
  - [Galactokinase Deficiency](#)
  - [Hyperglycerolemia](#) - Glycerol Kinase Deficiency
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# Hyperlipoproteinemias

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- Apolipoprotein(a) - Lp(a)
- Hyperlipoproteinemia Type I
- Hyperlipoproteinemia Type Ib - Apolipoprotein C-II Deficiency
- Hyperlipoproteinemia Type Ic - Chylomicronemia
- Familial Hypercholesterolemia - Type II Hyperlipoproteinemia
- Hyperlipoproteinemia Type II - Familial Hyperbetalipoproteinemia
- Hyperlipoproteinemia Type III - Apolipoprotein E Deficiency
- Hyperlipoproteinemia Type IV
- Hyperlipoproteinemia Type V
- Familial LCAT Deficiency
- Wolman Disease
- Lipoprotein Lipase Deficiency
- Familial Hypertriglyceridemia
- Hyperlipidemia Type V
- Hyperlipidemia Type VI
- Familial ligand-defective apo-B - 2 different mutations affecting LDL-receptor interaction
- Familial hyperalphalipoproteinemia

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## Intestinal Uptake of Lipids

In order for the body to make use of dietary lipids, they must first be absorbed from the small intestine. Since these molecules are oils, they are essentially insoluble in the aqueous environment of the intestine. The solubilization (or emulsification) of dietary lipids is therefore accomplished by means of bile salts, which are synthesized from cholesterol in the liver and then stored in the gallbladder; they are secreted following the ingestion of fat.

The emulsification of dietary fats renders them accessible to pancreatic lipases (primarily lipase and phospholipase A<sub>2</sub>). These enzymes, secreted into the intestine from the pancreas, generate free fatty acids and a mixture of mono- and diacylglycerols from dietary triacylglycerols. Pancreatic lipase degrades triacylglycerols at the 1 and 3 positions sequentially to generate 1,2-diacylglycerols and 2-acylglycerols. Phospholipids are degraded at the 2 position by pancreatic phospholipase A<sub>2</sub> releasing a free fatty acid and the lysophospholipid. The products of pancreatic lipases then diffuse into the intestinal epithelial cells, where the re-synthesis of triacylglycerols occurs.

Dietary triacylglycerols and cholesterol, as well as triacylglycerols and cholesterol

synthesized by the liver, are solubilized in lipid-protein complexes. These complexes contain triacylglycerol lipid droplets and cholesteryl esters surrounded by the polar phospholipids and proteins identified as apolipoproteins. These lipid-protein complexes vary in their content of lipid and protein.

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## Composition of the Major Lipoprotein Complexes

| Complex           | Source                                    | Density (g/ml) | % Protein | % TG <sup>a</sup> | % PL <sup>b</sup> | % CE <sup>c</sup> | % C <sup>d</sup> | % FFA <sup>e</sup> |
|-------------------|-------------------------------------------|----------------|-----------|-------------------|-------------------|-------------------|------------------|--------------------|
| Chylomicron       | Intestine                                 | <0.95          | 1-2       | 85-88             | 8                 | 3                 | 1                | 0                  |
| VLDL              | Liver                                     | 0.95-1.006     | 7-10      | 50-55             | 18-20             | 12-15             | 8-10             | 1                  |
| IDL               | VLDL                                      | 1.006-1.019    | 10-12     | 25-30             | 25-27             | 32-35             | 8-10             | 1                  |
| LDL               | VLDL                                      | 1.019-1.063    | 20-22     | 10-15             | 20-28             | 37-48             | 8-10             | 1                  |
| *HDL <sub>2</sub> | Intestine, liver (chylomicrons and VLDLs) | 1.063-1.125    | 33-35     | 5-15              | 32-43             | 20-30             | 5-10             | 0                  |
| *HDL <sub>3</sub> | Intestine, liver (chylomicrons and VLDLs) | 1.125-1.21     | 55-57     | 3-13              | 26-46             | 15-30             | 2-6              | 6                  |
| Albumin-FFA       | Adipose tissue                            | >1.281         | 99        | 0                 | 0                 | 0                 | 0                | 100                |

<sup>a</sup>Triacylglycerols, <sup>b</sup>Phospholipids, <sup>c</sup>Cholesteryl esters, <sup>d</sup>Free cholesterol, <sup>e</sup>Free fatty acids

\*HDL<sub>2</sub> and HDL<sub>3</sub> derived from nascent HDL as a result of the acquisition of cholesteryl esters

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## Apoprotein Classifications

| <b>Apoprotein -<br/>MW (Da)</b> | <b>Lipoprotein<br/>Association</b> | <b>Function and<br/>Comments</b>                                                                                                                                           |
|---------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apoA-I - 29,016                 | Chylomicrons, HDL                  | major protein of HDL,<br>activates lecithin:cholesterol<br>acyltransferase, LCAT                                                                                           |
| apoA-II - 17,400                | Chylomicrons, HDL                  | primarily in HDL, enhances<br>hepatic lipase activity                                                                                                                      |
| apoA-IV - 46,000                | Chylomicrons and HDL               | present in triacylglycerol rich<br>lipoproteins                                                                                                                            |
| apoB-48 - 241,000               | Chylomicrons                       | exclusively found in<br>chylomicrons, derived from<br>apoB-100 gene by RNA<br>editing in intestinal<br>epithelium; lacks the LDL<br>receptor-binding domain of<br>apoB-100 |
| apoB-100 - 513,000              | VLDL, IDL and LDL                  | major protein of LDL, binds<br>to LDL receptor; one of the<br>longest known proteins in<br>humans                                                                          |
| apoC-I - 7,600                  | Chylomicrons, VLDL, IDL<br>and HDL | may also activate LCAT                                                                                                                                                     |
| apoC-II - 8, 916                | Chylomicrons, VLDL, IDL<br>and HDL | activates lipoprotein lipase                                                                                                                                               |
| apoC-III - 8,750                | Chylomicrons, VLDL, IDL<br>and HDL | inhibits lipoprotein lipase                                                                                                                                                |
| apoD, 33,000                    | HDL                                | closely associated with<br>LCAT                                                                                                                                            |



|                                                                                                                             |                                         |                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cholesterol ester transfer protein, CETP                                                                                    | HDL                                     | exclusively associated with HDL, cholesteryl ester transfer                                                                                                                                                                                                               |
| apoE - 34,000 (at least 3 alleles [E <sub>2</sub> , E <sub>3</sub> , E <sub>4</sub> ] each of which have multiple isoforms) | Chylomicron remnants, VLDL, IDL and HDL | binds to LDL receptor, apoE <sub>ε</sub> -4 allele amplification associated with late-onset Alzheimer's disease                                                                                                                                                           |
| apoH - 50,000 (also known as β-2-glycoprotein I)                                                                            | Chylomicrons                            | triacylglycerol metabolism                                                                                                                                                                                                                                                |
| apo(a) - at least 19 different alleles; protein ranges in size from 300,000 - 800,000                                       | LDL                                     | disulfide bonded to apoB-100, forms a complex with LDL identified as <b>lipoprotein (a)</b> , <b>Lp(a)</b> ; strongly resembles plasminogen; may deliver cholesterol to sites of vascular injury, high risk association with premature coronary artery disease and stroke |

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## Chylomicrons

Chylomicrons are assembled in the intestinal mucosa as a means to transport dietary cholesterol and triacylglycerols to the rest of the body. Chylomicrons are, therefore, the molecules formed to mobilize dietary (exogenous) lipids. The predominant lipids of chylomicrons are triacylglycerols (see Table above). The apolipoproteins that predominate before the chylomicrons enter the circulation include apoB-48 and apoA-I, -A-II and IV. ApoB-48 combines only with chylomicrons.

Chylomicrons leave the intestine via the lymphatic system and enter the circulation at the left subclavian vein. In the bloodstream, chylomicrons acquire apoC-II and apoE from plasma HDLs. In the capillaries of adipose tissue and muscle, the fatty acids of chylomicrons are removed from the triacylglycerols by the action of lipoprotein lipase (**LPL**), which is found on the surface of the endothelial cells of the capillaries. The apoC-II in the chylomicrons

activates LPL in the presence of phospholipid. The free fatty acids are then absorbed by the tissues and the glycerol backbone of the triacylglycerols is returned, via the blood, to the liver and kidneys. Glycerol is converted to the glycolytic intermediate dihydroxyacetone phosphate (DHAP). During the removal of fatty acids, a substantial portion of phospholipid, apoA and apoC is transferred to HDLs. The loss of apoC-II prevents LPL from further degrading the **chylomicron remnants**.

Chylomicron remnants--- containing primarily cholesterol, apoE and apoB-48--- are then delivered to, and taken up by, the liver through interaction with the chylomicron remnant receptor. The recognition of chylomicron remnants by the hepatic remnant receptor requires apoE. Chylomicrons function to deliver dietary triacylglycerols to adipose tissue and muscle and dietary cholesterol to the liver.

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## Very Low Density Lipoproteins, LDLs

The dietary intake of both fat and carbohydrate, in excess of the needs of the body, leads to their conversion into triacylglycerols in the liver. These triacylglycerols are packaged into VLDLs and released into the circulation for delivery to the various tissues (primarily muscle and adipose tissue) for storage or production of energy through oxidation. VLDLs are, therefore, the molecules formed to transport endogenously derived triacylglycerols to extra-hepatic tissues. In addition to triacylglycerols, VLDLs contain some cholesterol and cholesteryl esters and the apoproteins, apoB-100, apoC-I, apoC-II, apoC-III and apoE. Like nascent chylomicrons, newly released VLDLs acquire apoCs and apoE from circulating HDLs.

The fatty acid portion of VLDLs is released to adipose tissue and muscle in the same way as for chylomicrons, through the action of lipoprotein lipase. The action of lipoprotein lipase coupled to a loss of certain apoproteins (the apoCs) converts VLDLs to intermediate density lipoproteins (IDLs), also termed VLDL remnants. The apoCs are transferred to HDLs. The predominant remaining proteins are apoB-100 and apoE. Further loss of triacylglycerols converts IDLs to LDLs.

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## Intermediate Density Lipoproteins, IDLs

IDLs are formed as triacylglycerols are removed from VLDLs. The fate of IDLs is either conversion to LDLs or direct uptake by the liver. Conversion of IDLs to LDLs occurs as more triacylglycerols are removed. The liver takes up IDLs after they have interacted with the LDL receptor to form a complex, which is endocytosed by the cell. For LDL receptors in the liver to recognize IDLs requires the presence of both apoB-100 and apoE (the LDL receptor is also called the apoB-100/apoE receptor). The importance of apoE in cholesterol uptake by LDL receptors has been demonstrated in transgenic mice lacking functional apoE genes. These mice develop severe atherosclerotic lesions at 10 weeks of age.

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## Low Density Lipoproteins, LDLs

The cellular requirement for cholesterol as a membrane component is satisfied in one of two ways: either it is synthesized *de novo* within the cell, or it is supplied from extra-cellular sources, namely, chylomicrons and LDLs. As indicated above, the dietary cholesterol that goes into chylomicrons is supplied to the liver by the interaction of chylomicron remnants with the remnant receptor. In addition, cholesterol synthesized by the liver can be transported to extra-hepatic tissues if packaged in VLDLs. In the circulation VLDLs are converted to LDLs through the action of lipoprotein lipase. LDLs are the primary plasma carriers of cholesterol for delivery to all tissues.

The exclusive apolipoprotein of LDLs is apoB-100. LDLs are taken up by cells via LDL receptor-mediated endocytosis, as described above for IDL uptake. The uptake of LDLs occurs predominantly in liver (75%), adrenals and adipose tissue. As with IDLs, the interaction of LDLs with LDL receptors requires the presence of apoB-100. The endocytosed membrane vesicles (endosomes) fuse with lysosomes, in which the apoproteins are degraded and the cholesterol esters are hydrolyzed to yield free cholesterol. The cholesterol is then incorporated into the plasma membranes as necessary. Excess intracellular cholesterol is re-esterified by acyl-CoA-cholesterol acyltransferase (**ACAT**), for intracellular storage. The activity of ACAT is enhanced by the presence of intracellular cholesterol.

Insulin and tri-iodothyronine (T3) increase the binding of LDLs to liver cells, whereas glucocorticoids (e.g., dexamethasone) have the opposite effect. The precise mechanism for these effects is unclear but may be mediated through the regulation of apoB degradation. The effects of insulin and T3 on hepatic LDL binding may explain the hypercholesterolemia and increased risk of atherosclerosis that have been shown to be associated with uncontrolled diabetes or hypothyroidism.

An abnormal form of LDL, identified as **lipoprotein-X (Lp-X)**, predominates in the circulation of patients suffering from lecithin-cholesterol acyl transferase (**LCAT**, see HDL discussion for LCAT function) deficiency or cholestatic liver disease. In both cases there is an elevation in the level of circulating free cholesterol and phospholipids.

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## High Density Lipoproteins, HDLs

HDLs are synthesized *de novo* in the liver and small intestine, as primarily protein-rich disc-shaped particles. These newly formed HDLs are nearly devoid of any cholesterol and cholesteryl esters. The primary apoproteins of HDLs are apoA-I, apoC-I, apoC-II and apoE. In fact, a major function of HDLs is to act as circulating stores of apoC-I, apoC-II and apoE.

HDLs are converted into spherical lipoprotein particles through the accumulation of cholesteryl esters. This accumulation converts nascent HDLs to HDL<sub>2</sub> and HDL<sub>3</sub>. Any free cholesterol present in chylomicron remnants and VLDL remnants (IDLs) can be esterified through the action of the HDL-associated enzyme, lecithin:cholesterol acyltransferase, **LCAT**.

LCAT is synthesized in the liver and so named because it transfers a fatty acid from the C-2 position of lecithin to the C-3-OH of cholesterol, generating a cholesteryl ester and lysolecithin. The activity of LCAT requires interaction with apoA-I, which is found on the surface of HDLs.

Cholesterol-rich HDLs return to the liver, where they are endocytosed. Hepatic uptake of HDLs, or **reverse cholesterol transport**, may be mediated through an HDL-specific apoA-I receptor or through lipid-lipid interactions. Macrophages also take up HDLs through apoA-I receptor interaction. HDLs can then acquire cholesterol and apoE from the macrophages; cholesterol-enriched HDLs are then secreted from the macrophages. The added apoE in these HDLs leads to an increase in their uptake and catabolism by the liver.

HDLs also acquire cholesterol by extracting it from cell surface membranes. This process has the effect of lowering the level of intracellular cholesterol, since the cholesterol stored within cells as cholesteryl esters will be mobilized to replace the cholesterol removed from the plasma membrane.

The cholesterol esters of HDLs can also be transferred to VLDLs and LDLs through the action of the HDL-associated enzyme, cholesterol ester transfer protein (**CETP**). This has the added effect of allowing the excess cellular cholesterol to be returned to the liver through the LDL-receptor pathway as well as the HDL-receptor pathway.

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## LDL Receptors

LDLs are the principal plasma carriers of cholesterol delivering cholesterol from the liver (via hepatic synthesis of VLDLs) to peripheral tissues, primarily the adrenals and adipose tissue. LDLs also return cholesterol to the liver. The cellular uptake of cholesterol from LDLs occurs following the interaction of LDLs with the LDL receptor (also called the apoB-100/apoE receptor). The sole apoprotein present in LDLs is apoB-100, which is required for interaction with the LDL receptor.

The LDL receptor is a polypeptide of 839 amino acids that spans the plasma membrane. An extracellular domain is responsible for apoB-100/apoE binding. The intracellular domain is responsible for the clustering of LDL receptors into regions of the plasma membrane termed **coated pits**. Once LDL binds the receptor, the complexes are rapidly internalized (**endocytosed**). ATP-dependent proton pumps lower the pH in the endosomes, which results in dissociation of the LDL from the receptor. The portion of the endosomal membranes harboring the receptor are then recycled to the plasma membrane and the LDL-containing endosomes fuse with lysosomes. Acid hydrolases of the lysosomes degrade the apoproteins and release free fatty acids and cholesterol. As indicated above, the free cholesterol is either incorporated into plasma membranes or esterified (by ACAT) and stored within the cell.

The level of intracellular cholesterol is regulated through cholesterol-induced suppression of LDL receptor synthesis and cholesterol-induced inhibition of cholesterol synthesis. The increased level of intracellular cholesterol that results from LDL uptake has the additional effect of activating ACAT, thereby allowing the storage of excess cholesterol within cells. However, the effect of cholesterol-induced suppression of LDL receptor synthesis is a

decrease in the rate at which LDLs and IDLs are removed from the serum. This can lead to excess circulating levels of cholesterol and cholesteryl esters when the dietary intake of fat and cholesterol exceeds the needs of the body. The excess cholesterol tends to be deposited in the skin, tendons and (more gravely) within the arteries, leading to **atherosclerosis**.  
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## Clinical Significances of Lipoprotein Metabolism

Fortunately, few individuals carry the inherited defects in lipoprotein metabolism that lead to [hyper-](#) or [hypolipoproteinemias](#) (see Tables below for brief descriptions). Persons suffering from [diabetes mellitus](#), hypothyroidism and kidney disease often exhibit abnormal lipoprotein metabolism as a result of secondary effects of their disorders. For example, because lipoprotein lipase (LPL) synthesis is regulated by insulin, LPL deficiencies leading to [Type I hyperlipoproteinemia](#) may occur as a secondary outcome of diabetes mellitus. Additionally, insulin and thyroid hormones positively affect hepatic LDL-receptor interactions; therefore, the hypercholesterolemia and increased risk of atherosclerosis associated with uncontrolled diabetes or hypothyroidism is likely due to decreased hepatic LDL uptake and metabolism.

Of the many disorders of lipoprotein metabolism, [familial hypercholesterolemia \(FH\)](#) may be the most prevalent in the general population. Heterozygosity at the FH locus occurs in 1:500 individuals, whereas, homozygosity is observed in 1:1,000,000 individuals. FH is an inherited disorder comprising four different classes of mutation in the LDL receptor gene. The class 1 defect (the most common) results in a complete loss of receptor synthesis. The class 2 defect results in the synthesis of a receptor protein that is not properly processed in the Golgi apparatus and therefore is not transported to the plasma membrane. The class 3 defect results in an LDL receptor that is incapable of binding LDLs. The class 4 defect results in receptors that bind LDLs but do not cluster in coated pits and are, therefore, not internalized.

FH sufferers may be either heterozygous or homologous for a particular mutation in the receptor gene. Homozygotes exhibit grossly elevated serum cholesterol (primarily in LDLs). The elevated levels of LDLs result in their phagocytosis by macrophages. These lipid-laden phagocytic cells tend to deposit within the skin and tendons, leading to **xanthomas**. A greater complication results from cholesterol deposition within the arteries, leading to **atherosclerosis**, the major contributing factor of nearly all cardiovascular diseases.

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## Hyperlipoproteinemias

| Disorder | Defect | Comments |
|----------|--------|----------|
|----------|--------|----------|



|                                                                                                                             |                                                                                                                                                              |                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type I</b> (familial LPL deficiency, familial hyperchylomicronemia)                                                      | (a) deficiency of LPL;<br>(b) production of abnormal LPL;<br>(c) apoC-II deficiency                                                                          | slow chylomicron clearance, reduced LDL and HDL levels; treated by low fat/complex carbohydrate diet; no increased risk of coronary artery disease                                                |
| <b>Type II</b> (familial hypercholesterolemia, FH)                                                                          | 4 classes of LDL receptor defect                                                                                                                             | reduced LDL clearance leads to hypercholesterolemia, resulting in atherosclerosis and coronary artery disease                                                                                     |
| <b>Type III</b> (familial dysbetalipoproteinemia, remnant removal disease, broad beta disease, apolipoprotein E deficiency) | hepatic remnant clearance impaired due to apoE abnormality; patients only express the apoE <sub>2</sub> isoform that interacts poorly with the apoE receptor | causes xanthomas, hypercholesterolemia and atherosclerosis in peripheral and coronary arteries due to elevated levels of chylomicrons and VLDLs                                                   |
| <b>Type IV</b> (familial hypertriacylglycerolemia)                                                                          | elevated production of VLDL associated with glucose intolerance and hyperinsulinemia                                                                         | frequently associated with type-II non-insulin dependent diabetes mellitus, obesity, alcoholism or administration of progestational hormones; elevated cholesterol as a result of increased VLDLs |
| <b>Type V</b> familial                                                                                                      | elevated chylomicrons and VLDLs due to unknown cause                                                                                                         | hypertriacylglycerolemia and hypercholesterolemia with decreased LDLs and HDLs                                                                                                                    |



|                                                              |                                                                                                                                                                      |                                                                                                                                                                                          |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Familial<br>hyperalphalipoproteinemia                        | increased level of<br>HDLs                                                                                                                                           | a rare condition that is<br>beneficial for health and<br>longevity                                                                                                                       |
| Type II<br>Familial<br>hyperbetalipoproteinemia              | increased LDL<br>production and<br>delayed clearance<br>of triacylglycerols<br>and fatty acids                                                                       | strongly associated with<br>increased risk of coronary<br>artery disease                                                                                                                 |
| Familial ligand-defective apoB                               | 2 different<br>mutations: Gln for<br>Arg (amino acid<br>3500) or Cys for<br>Arg (amino acid<br>3531); both lead to<br>reduced affinity of<br>LDL for LDL<br>receptor | dramatic increase in LDL<br>levels; no affect on HDL,<br>VLDL or plasma triglyceride<br>levels; significant cause of<br>hypercholesterolemia and<br>premature coronary artery<br>disease |
| Familial LCAT deficiency                                     | absence of LCAT<br>leads to inability of<br>HDLs to take up<br>cholesterol<br>(reverse cholesterol<br>transport)                                                     | decreased levels of plasma<br>cholesteryl esters and<br>lysolecithin; abnormal LDLs<br>(Lp-X) and VLDLs;<br>symptoms also found<br>associated with cholestasis                           |
| Wolman's disease<br>(cholesteryl ester storage<br>disease)   | defect in lysosomal<br>cholesteryl ester<br>hydrolase; affects<br>metabolism of LDLs                                                                                 | reduced LDL clearance<br>leads to<br>hypercholesterolemia,<br>resulting in atherosclerosis<br>and coronary artery disease                                                                |
| heparin-releasable hepatic<br>triglyceride lipase deficiency | deficiency of the<br>lipase leads to<br>accumulation of<br>triacylglycerol-rich<br>HDLs and VLDL<br>remnants (IDLs)                                                  | causes xanthomas and<br>coronary artery disease                                                                                                                                          |

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## Hypolipoproteinemias

| Disorder                                                                                                                  | Defect                                                                                                                     | Comments                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Abetalipoproteinemia</a><br>(acanthocytosis, Bassen-Kornzweig syndrome)                                       | no chylomicrons, VLDLs or LDLs due to defect in apoB expression                                                            | rare defect; intestine and liver accumulate, malabsorption of fat, retinitis pigmentosa, ataxic neuropathic disease, erythrocytes have thorny appearance |
| <a href="#">Familial hypobetalipoproteinemia</a>                                                                          | at least 20 different apoB gene mutations identified, LDL concentrations 10-20% of normal, VLDL slightly lower, HDL normal | mild or no pathological changes                                                                                                                          |
| <a href="#">Familial alpha-lipoprotein deficiency</a> (Tangier disease, Fish-eye disease, apoA-I and -C-III deficiencies) | all of these related syndromes have reduced HDL concentrations, no effect on chylomicron or VLDL production                | tendency to hypertriacylglycerolemia; some elevation in VLDLs; Fish-eye disease characterized by severe corneal opacity                                  |

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## Pharmacologic Intervention

Drug treatment to lower plasma lipoproteins and/or cholesterol is primarily aimed at reducing the risk of atherosclerosis and subsequent coronary artery disease that exists in patients with elevated circulating lipids. Drug therapy usually is considered as an option only if non-pharmacologic interventions (altered diet and exercise) have failed to lower plasma lipids

- **Mevinolin, Mevastatin, Lovastatin:** These drugs are fungal HMG-CoA reductase inhibitors. The net result of treatment is an increased cellular uptake of LDLs, since the intracellular synthesis of cholesterol is inhibited and cells are therefore dependent on extracellular sources of cholesterol. However, since mevalonate (the product of the HMG-CoA reductase reaction) is required for the synthesis of other important isoprenoid compounds besides cholesterol, long-term treatments carry some risk of toxicity.
- **Nicotinic acid:** Nicotinic acid reduces the plasma levels of both VLDLs and LDLs by inhibiting hepatic VLDL secretion, as well as suppressing the flux of FFA release from adipose tissue by inhibiting lipolysis. Because of its ability to cause large reductions in circulating levels of cholesterol, nicotinic acid is used to treat Type II, III, IV and V hyperlipoproteinemias.
- **Clofibrate, Gemfibrozil, Fenofibrate:** These compounds are derivatives of fibric acid and promote rapid VLDL turnover by activating lipoprotein lipase. They also induce the diversion of hepatic free fatty acids from esterification reactions to those of oxidation, thereby decreasing the liver's secretion of triacylglycerol- and cholesterol-rich VLDLs.
- **Probucol:** Probucol increases the rate of LDL metabolism and may block the intestinal transport of cholesterol. The net result is a significant reduction in plasma cholesterol levels.
- **Cholestyramine or colestipol (resins):** These compounds are nonabsorbable resins that bind bile acids which are then not reabsorbed by the liver but excreted. The drop in hepatic reabsorption of bile acids releases a feedback inhibitory mechanism that had been inhibiting bile acid synthesis. As a result, a greater amount of cholesterol is converted to bile acids to maintain a steady level in circulation. Additionally, the synthesis of LDL receptors increases to allow increased cholesterol uptake for bile acid synthesis, and the overall effect is a reduction in plasma cholesterol. (This treatment is ineffective in homozygous FH patients, since they are completely deficient in LDL receptors.

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Last modified: **Thursday, 07-Oct-2004 15:19:47 EST**

# Hypolipoproteinemias

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- [Hypobetalipoproteinemia](#) - Apolipoprotein B-100 Deficiency
  - [Abetalipoproteinemia](#) - Kornzweig Syndrome
  - [Tangier Disease](#) - Familial High-Density Lipoprotein Deficiency
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Last modified: **Thursday, 24-Jun-2004 15:46:04 EST**

- [Actions of Insulin](#)
  - [Nutrient Intake and Hormonal Control of Insulin Action](#)
  - [Definition of Diabetes](#)
  - [Types of Diabetes Mellitus](#)
  - [Insulin-Dependent Diabetes Mellitus \(IDDM\)](#)
  - [Non-Insulin-Dependent Diabetes Mellitus \(NIDDM\)](#)
  - [Diabetes and The "Metabolic Syndrome"](#)
  - [Therapeutic Intervention for Hyperglycemia](#)
- [Diabetes-guide.org](#)  
An excellent site with diabetes information  
focusing on preventing and treating symptoms of diabetes

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## Actions of Insulin

The major function of insulin is to counter the concerted action of a number of hyperglycemia-generating hormones and to maintain low blood glucose levels. Because there are numerous hyperglycemic hormones, untreated disorders associated with insulin generally lead to severe hyperglycemia and shortened life span.

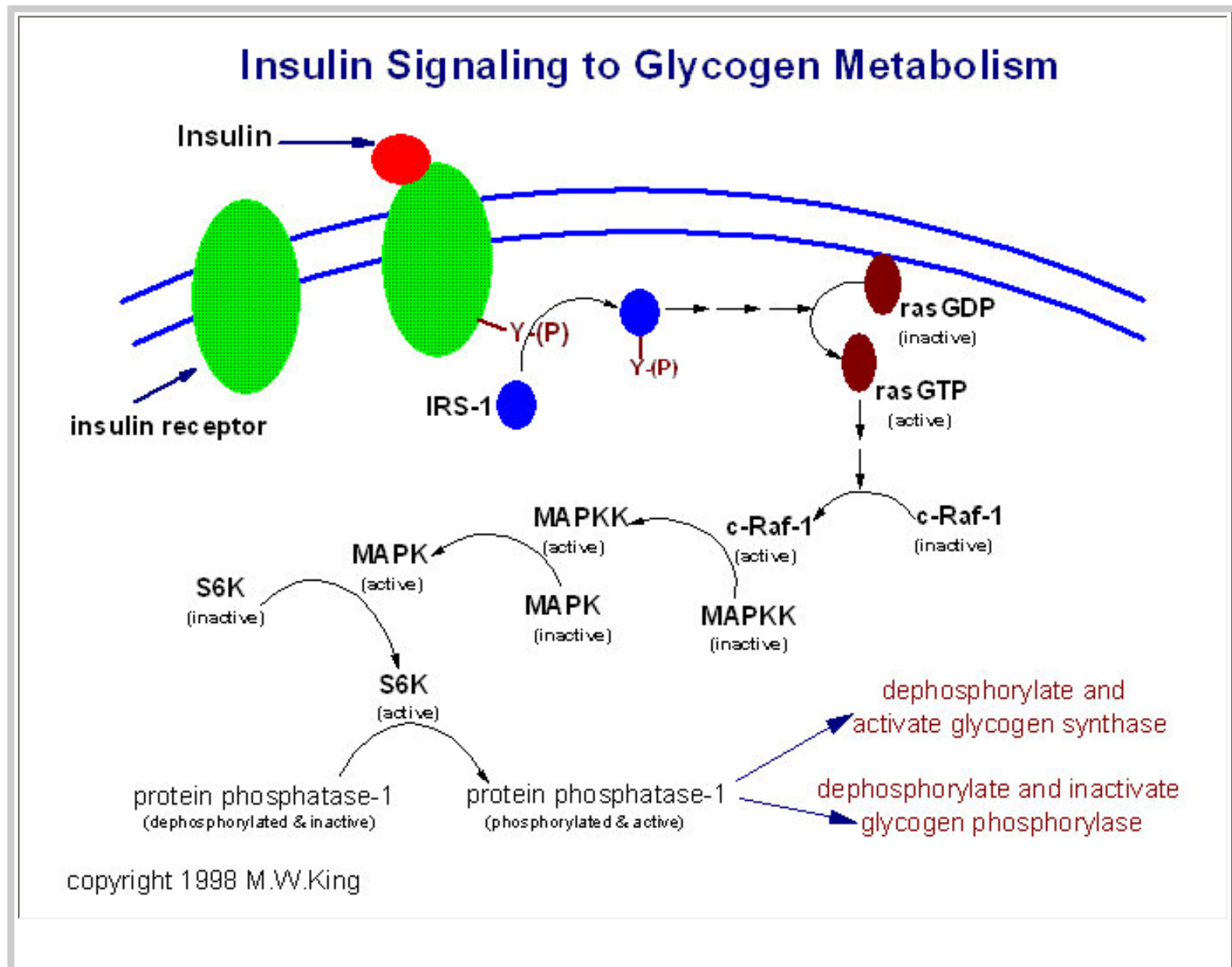
In addition to its role in regulating glucose metabolism, insulin stimulates lipogenesis, diminishes lipolysis, and increases amino acid transport into cells. Insulin also modulates transcription, altering the cell content of numerous mRNAs. It stimulates growth, DNA synthesis, and cell replication, effects that it holds in common with the insulin-like growth factors (IGFs) and relaxin.

Insulin is synthesized as a preprohormone in the  $\beta$  cells of the islets of Langerhans. Its signal peptide is removed in the cisternae of the endoplasmic reticulum and it is packaged into secretory vesicles in the Golgi, folded to its native structure, and locked in this conformation by the formation of 2 disulfide bonds. Specific protease activity cleaves the center third of the molecule, which dissociates as **C peptide**, leaving the amino terminal **B peptide** disulfide bonded to the carboxy terminal **A peptide**.

Insulin secretion from  $\beta$  cells is principally regulated by plasma glucose levels. Increased uptake of glucose by pancreatic  $\beta$ -cells leads to a concomitant increase in metabolism. The increase in metabolism leads to an elevation in the ATP/ADP ratio. This in turn leads to an inhibition of an ATP-sensitive  $K^+$  channel. The net result is a depolarization of the cell leading to  $Ca^{2+}$  influx and insulin secretion. In fact, the role of  $K^+$  channels in insulin secretion presents a viable [therapeutic target](#) for treating hyperglycemia due to insulin insufficiency.

Chronic increases in numerous other hormones---including **growth hormone, placental lactogen, estrogens, and progestins** --- up-regulate insulin secretion, probably by increasing the preproinsulin mRNA and enzymes involved in processing the increased preprohormone.

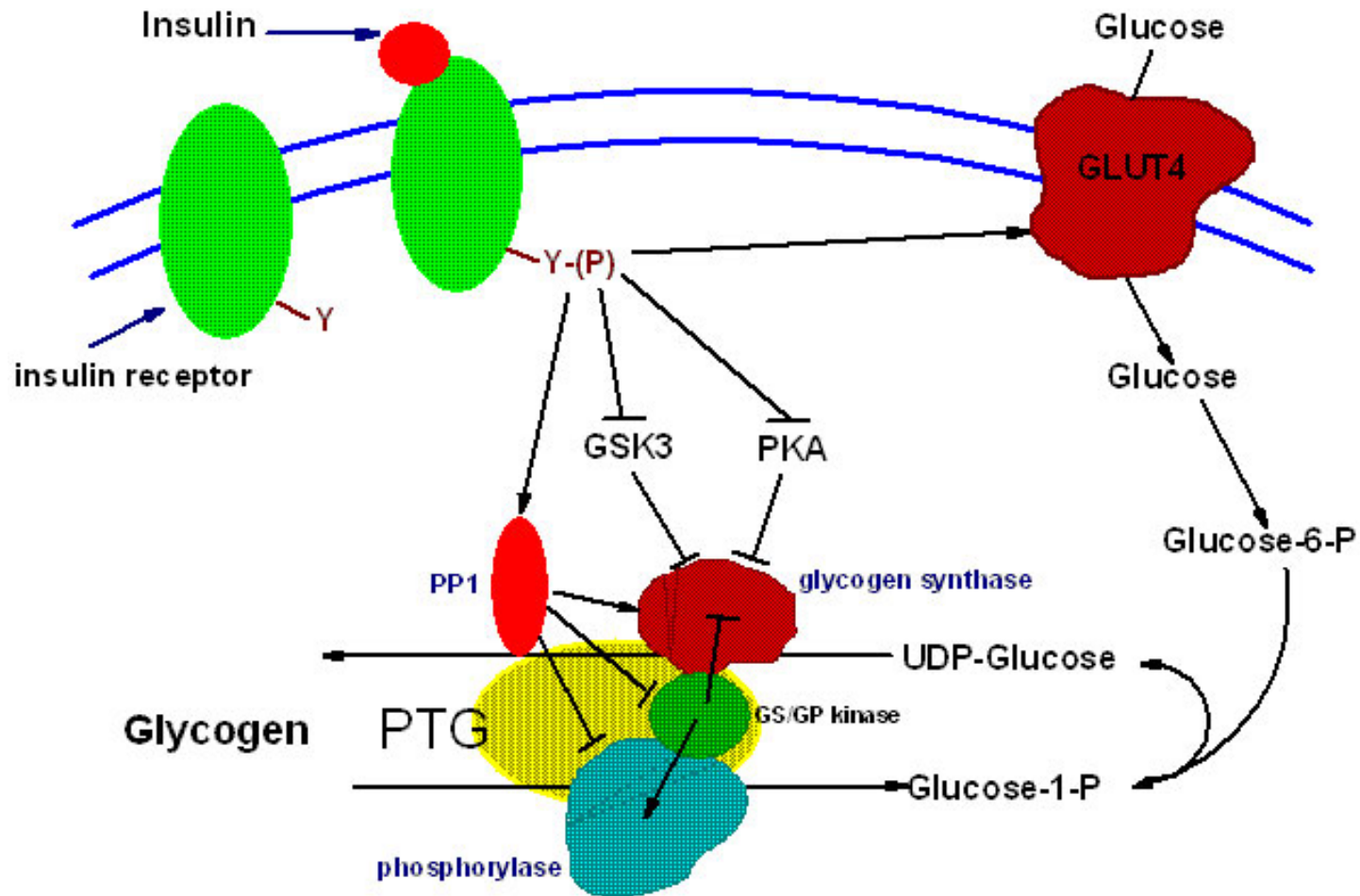
Insulin, secreted by the  $\beta$ -cells of the pancreas, is directly infused via the portal vein to the liver, where it exerts profound metabolic effects. These effects are the response of the activation of the insulin receptor which belongs to the class of cell surface receptors that exhibit intrinsic tyrosine kinase activity (see [Signal Transduction](#)). The insulin receptor is a heterotetramer of 2 extracellular  $\alpha$ -subunits disulfide bonded to 2 transmembrane  $\beta$ -subunits. With respect to hepatic glucose homeostasis, the effects of insulin receptor activation are specific phosphorylation events that lead to an increase in the storage of glucose with a concomitant decrease in hepatic glucose release to the circulation as diagrammed below (only those responses at the level of glycogen synthase and glycogen phosphorylase are represented).





Actions of insulin-insulin receptor interactions at the level of IRS1 and activation of the kinase cascade leading to altered activities of glycogen phosphorylase and glycogen synthase.

## Insulin-Mediated Regulation of Glycogen metabolism



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Insulin-insulin receptor actions on glycogen homeostasis showing the role of **protein targeting glycogen, PTG** in complexing many of the enzymes and substrates together. Also diagrammed is response of insulin at the level of glucose transport into cells via GLUT4 translocation to the plasma membrane. GS/GP kinase = glycogen synthase: glycogen phosphorylase kinase. PPI = protein phosphatase inhibitor. Arrows denote either direction of flow or positive effects, T lines represent inhibitory effects.

In most nonhepatic tissues, insulin increases glucose uptake by increasing the number of plasma membrane **glucose transporters: GLUTs**. Glucose transporters are in a continuous state of turnover. Increases in the plasma membrane content of transporters stem from an increase in the rate of recruitment of new transporters into the plasma membrane, deriving from a special pool of preformed transporters localized in the cytoplasm. GLUT1 is present in most tissues, GLUT2 is found in liver and pancreatic  $\beta$ -cells, GLUT3 is in the brain and GLUT4 is found in heart, adipose tissue and skeletal muscle.

In liver glucose uptake is dramatically increased because of increased activity of the enzymes glucokinase, phosphofructokinase-1 (PFK-1), and pyruvate kinase (PK), the key regulatory enzymes of **glycolysis**. The latter effects are induced by insulin-dependent activation of phosphodiesterase, with decreased PKA activity and diminished phosphorylation of pyruvate kinase and phosphofructokinase-2, PFK-2. Dephosphorylation of pyruvate kinase increases its' activity while dephosphorylation of PFK-2 renders it active as a kinase. The kinase activity of PFK-2 converts fructose-6-phosphate into **fructose-2,6-bisphosphate** (F2,6BP). F2,6BP is a potent allosteric activator of the rate limiting enzyme of glycolysis, PFK-1, and an inhibitor of the gluconeogenic enzyme, fructose-1,6-bisphosphatase. In addition, phosphatases specific for the phosphorylated forms of the glycolytic enzymes increase in activity under the influence of insulin. All these events lead to conversion of the glycolytic enzymes to their active forms and consequently a significant increase in glycolysis. In addition, glucose-6-phosphatase activity is down-regulated. The net effect is an increase in the content of hepatocyte glucose and its phosphorylated derivatives, with diminished blood glucose.

In addition to the above described events, diminished cAMP and elevated phosphatase activity combine to convert glycogen phosphorylase to its inactive form and glycogen synthase to its active form, with the result that not only is glucose funneled to glycolytic products, but glycogen content is increased as well.

All of the post-receptor responses initiated by insulin binding to its receptor are mediated as a consequence of the activation of several signal transduction pathways. These include receptor activation of **phosphatidylinositol-3-kinase, (PI3K)**. Activation of PI3K involves a linkage to receptor activation of **insulin receptor substrates** (of which there are four: IRS1, IRS2, IRS3 and IRS4). Activated PI3K phosphorylates membrane phospholipids, the major product being **phosphatidylinositol 3,4,5 trisphosphate, PIP<sub>3</sub>**. PIP<sub>3</sub> in turn activates the enzymes **protein kinase B, PKB** (also called Akt), **PIP<sub>3</sub>-dependent kinase (PDK)**, some isoforms of **protein kinase C (PKC)**, principally PKC- $\lambda$ ) and **small ribosomal subunit protein 6 kinase, S6K**. The MAP kinase pathway is also activated either through receptor activation of the protein tyrosine phosphatase (**Shp-2**) or growth factor receptor binding protein-2 (**Grb2**).

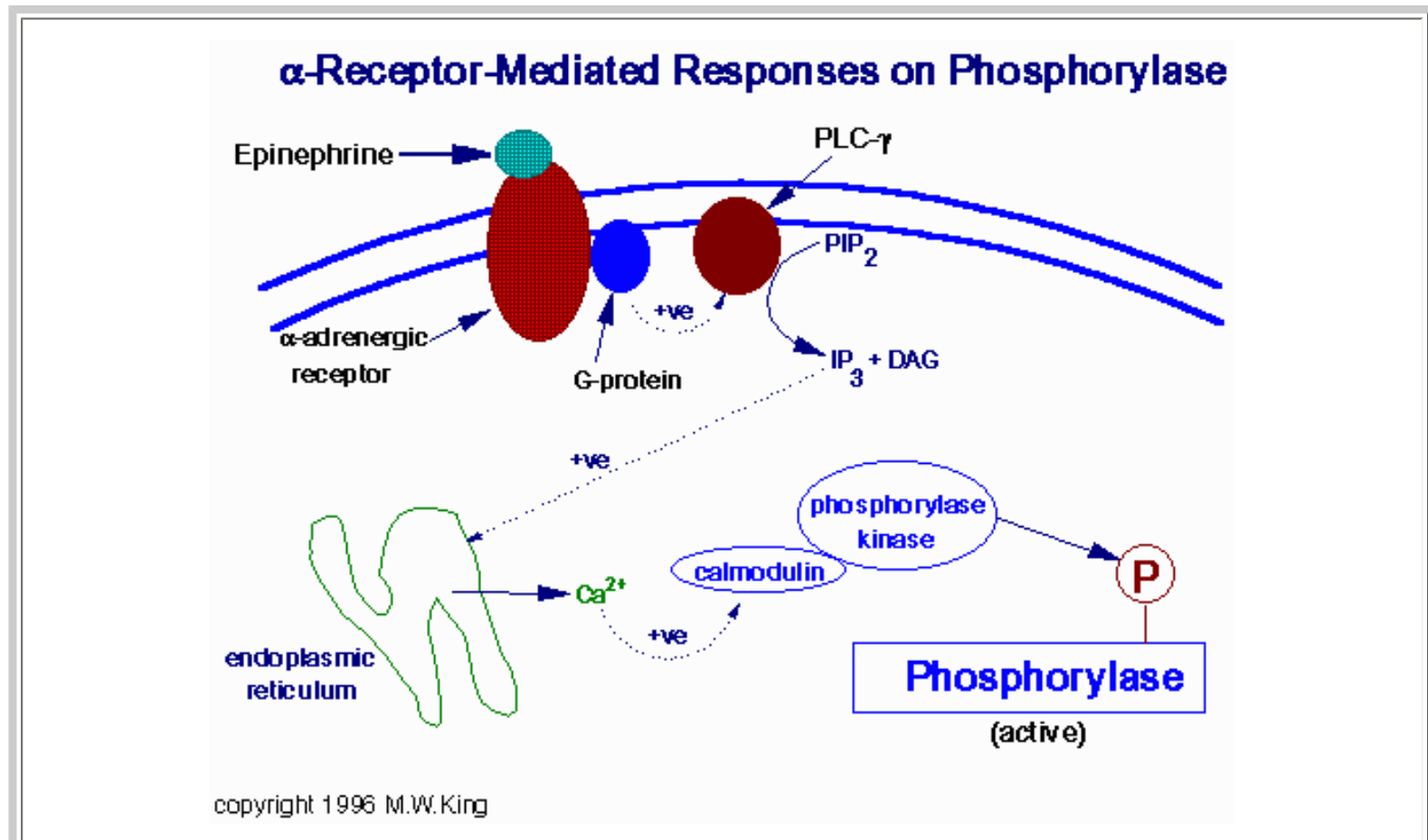
With respect to insulin responses, activation of PKB and PKC- $\lambda$  lead to translocation of GLUT4 molecules to the cell surface resulting in increased glucose uptake which is significant in skeletal muscle. Activation of PKB also leads to the phosphorylation and inhibition of **glycogen synthase kinase-3 (GSK3)**, which is a major regulatory kinase of **glycogen homeostasis**. In addition, PKB phosphorylates and inhibits the activity of a transcription factor (**FKHRL1**, now called **FoxO3a**) that has pro-apoptotic activity. This results in reduced apoptosis in response to insulin action.

Activation of S6K leads to the phosphorylation of the regulator of translation **eIF-4E binding protein, 4E-BP**. Phosphorylation of 4E-BP

prevents it from binding to eIF-4E the consequences of which would normally lead to a reduction in translation rate.

Insulin also has profound effects on the transcription of numerous genes, effects that are primarily mediated by regulated function of **sterol-regulated element binding protein, SREBP**. These transcriptional effects include (but are not limited to) increases in glucokinase, pyruvate kinase, lipoprotein lipase (LPL), fatty acid synthase (FAS) and acetylCoA carboxylase (ACC) and decreases in glucose 6-phosphatase, fructose 1,6-bisphosphatase and phosphoenolpyruvate carboxykinase (PEPCK)

In contrast, epinephrine diminishes insulin secretion by a cAMP-coupled regulatory path. In addition, epinephrine counters the effect of insulin in liver and peripheral tissue, where it binds to  $\beta$ -adrenergic receptors, induces adenylate cycles activity, increases cAMP, and **activates PKA** similarly to that of glucagon. The latter events induce glycogenolysis and gluconeogenesis, both of which are hyperglycemic and which thus counter insulin's effect on blood glucose levels. In addition, epinephrine influences glucose homeostasis through interaction with  $\alpha$ -adrenergic receptors.



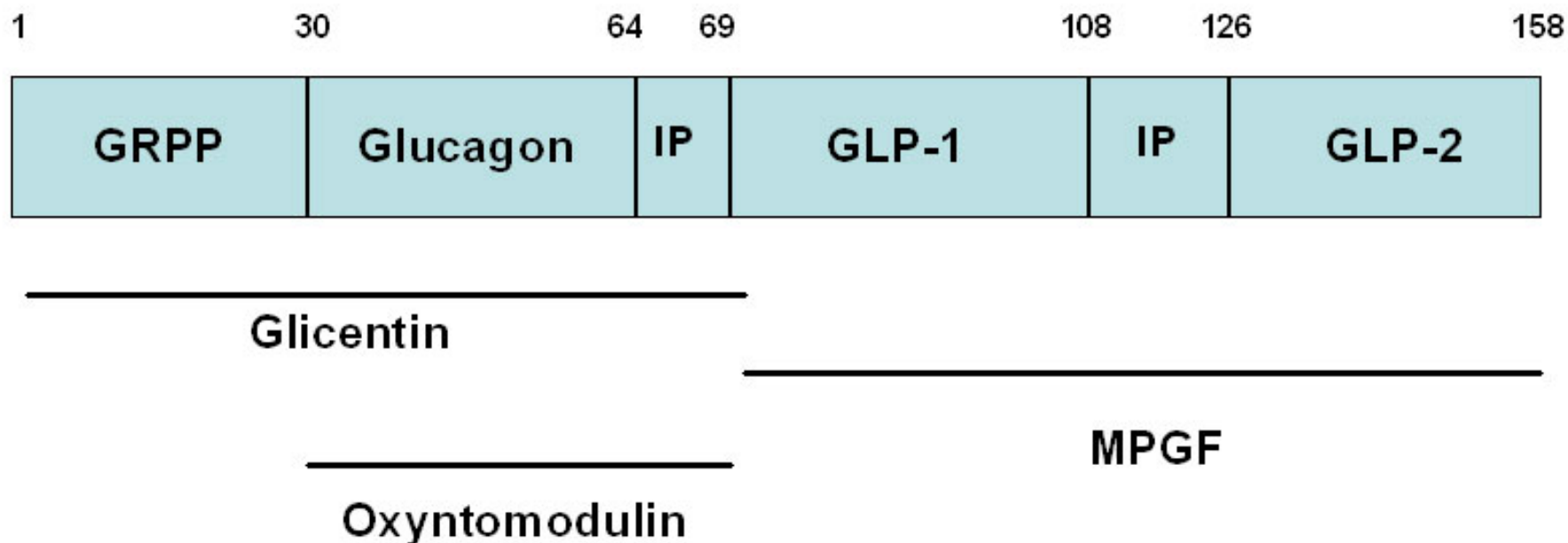
Pathways involved in the regulation of glycogen phosphorylase by epinephrine activation of  $\alpha$ -adrenergic receptors. See [Glycogen Metabolism](#) for details of the epinephrine action. PLC- $\gamma$  is phospholipase C- $\gamma$ . The substrate for PLC- $\gamma$  is **phosphatidylinositol-4,5-bisphosphate (PIP<sub>2</sub>)** and the products are **IP<sub>3</sub>**, **inositol trisphosphate** and **DAG, diacylglycerol**. Similar calmodulin-mediated phosphorylations lead to inhibition of glycogen synthase.

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## Nutrient Intake and Hormonal Control of Insulin Action

Two of the many [gastrointestinal hormones](#) have significant effects on insulin secretion and glucose regulation. These hormones are the glucagon-like peptides (principally glucagon-like peptide-1, GLP-1) and glucose-dependent insulintropic peptide (GIP). Both of these gut hormones constitute the class of molecules referred to as the **incretins**. Incretins are molecules associated with food intake-stimulation of insulin secretion from the pancreas.

GLP-1 is derived from the product of the proglucagon gene. This gene encodes a preproprotein that is differentially cleaved dependent upon the tissue in which it is synthesized. For example, in pancreatic  $\alpha$ -cells prohormone convertase 2 action leads to the release of glucagon. In the gut prohormone convertase 1/3 action leads to release of several peptides including GLP-1. Upon nutrient ingestion GLP-1 is secreted from intestinal enteroendocrine L-cells that are found predominantly in the ileum and colon with some production from these cell types in the duodenum and jejunum. Bioactive GLP-1 consists of 2 forms; GLP-1(7-37) and GLP-1(7-36)amide, where the latter form constitutes the majority (80%) of the circulating hormone.



Structure of the mammalian preproglucagon product. GRPP=glucagon-related pancreatic peptide. IP=intervening peptide. GLP-2=glucagon-related peptide-2. Additional peptides are derived from the preproprotein including glicentin which is composed of amino acids 1-69, oxyntomodulin is composed of amino acids 30-69 and the major proglucagon fragment (MPGF) comprises amino acids 72-158.

The primary physiological responses to GLP-1 are glucose-dependent insulin secretion, inhibition of glucagon secretion and inhibition of gastric acid secretion and gastric emptying. The latter effect will lead to increased satiety with reduced food intake along with a reduced desire to ingest food. The action of GLP-1 at the level of insulin and glucagon secretion results in significant reduction in circulating levels of glucose following nutrient intake. This activity has obvious significance in the context of this discussion of diabetes, in particular the hyperglycemia associated with poorly controlled type 2 diabetes. The glucose lowering activity of GLP-1 is highly transient as the half-life of this hormone in the circulation is less than 2 minutes. Removal of bioactive GLP-1 is a consequence of N-terminal proteolysis catalyzed by dipeptidyl peptidase IV (DPP IV). DPP IV is also known as the lymphocyte surface antigen CD26 and has numerous activities unrelated to incretin inactivation (see [therapeutic intervention](#) below for more information on DPP IV activity).

All of the effects of GLP-1 are mediated following activation of the GLP-1 receptor (GLP-1R). The GLP-1R is a typical seven-transmembrane spanning receptor coupled to G-protein activation, increased cAMP production and activation of PKA. However, there are also PKA-independent responses initiated through the GLP-1R. Other major responses to the actions of GLP-1 include pancreatic  $\beta$ -cell

proliferation and expansion concomitant with a reduction of  $\beta$ -cell apoptosis (death). In addition, GLP-1 activity results in increased expression of the glucose transporter-2 (GLUT-2) and glucokinase genes in pancreatic cells.

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## Definition of Diabetes

Diabetes is any disorder characterized by excessive urine excretion. The most common form of diabetes is **diabetes mellitus**, a metabolic disorder in which there is an inability to oxidize carbohydrate due to disturbances in insulin function. Diabetes mellitus is characterized by elevated glucose in the plasma and episodic ketoacidosis. Additional symptoms of diabetes mellitus include excessive thirst, glucosuria, polyuria, lipemia and hunger. If left untreated the disease can lead to fatal ketoacidosis. Other forms of diabetes include **diabetes insipidus** and **brittle diabetes**. Diabetes insipidus is the result of a deficiency of **antidiuretic hormone**. The major symptom of diabetes insipidus (excessive urine output) results from an inability of the kidneys to resorb water. Brittle diabetes is a form that is very difficult to control. It is characterized by unexplained oscillations between hypoglycemia and acidosis.

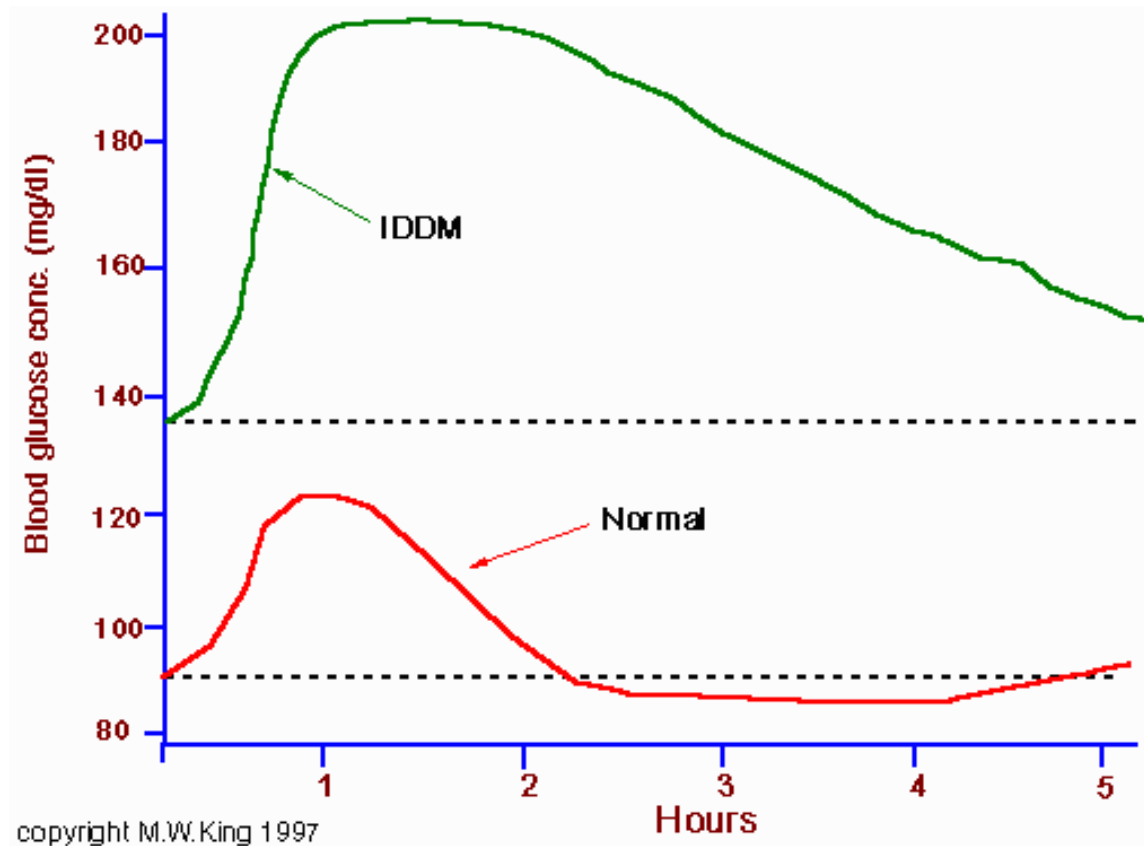
Criteria, which clinically establish an individual as suffering from diabetes mellitus, include:

- **1.** Having a fasting plasma glucose level in excess of 126mg/dL (7mmol/L). Normal levels should be less than 100mg/dL (<5.6mmol/L).
- **2.** or having plasma glucose levels in excess of 200mg/dL (11mmol/L) at two times points during a **glucose tolerance test (GTT)**, one of which must be within 2 hrs of ingestion of glucose.

The earlier a person is diagnosed with diabetes (principally type 2) the better chance the person has of staving off the primary negative consequences which are renal failure, blindness and limb amputations due to circulatory problems. The American Diabetes Association is planning to recommend that physicians consider patients to be pre-diabetic if their fasting blood glucose level is above 100mg/dL but less than 125mg/dL and whose glucose levels are at least 140mg/dL but less than 200mg/dL following an oral glucose tolerance test (OGTT).







Glucose tolerance curve for a normal person and one with insulin-dependent diabetes mellitus (IDDM, Type 1 diabetes). The dotted lines indicate the range of glucose concentration expected in a normal individual.

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## Types of Diabetes Mellitus

Diabetes mellitus is a heterogeneous clinical disorder with numerous causes. Two main classifications of diabetes mellitus exist, **idiopathic** and **secondary**.

Idiopathic diabetes is divided into two main types; insulin dependent and non-insulin-dependent. **Insulin-dependent diabetes mellitus (IDDM)**, also called **type 1 diabetes** is defined by the development of ketoacidosis in the absence of insulin therapy. Type 1 diabetes most often manifests in childhood (hence also called **juvenile onset diabetes**) and is the result of an autoimmune destruction of the  $\beta$ -cells of the pancreas. **Non-insulin-dependent diabetes mellitus (NIDDM or type 2 diabetes)** is characterized by persistent hyperglycemia but rarely leads to ketoacidosis. Type 2 diabetes generally manifests after age 40 and therefore has the obsolete name of

adult onset-type diabetes. Type 2 diabetes can result from genetics defects that cause both insulin resistance and insulin deficiency. There are two main forms of type 2 diabetes:

- **1.** Late onset associated with obesity.
- **2.** Late onset not associated with obesity.

There is a strong correlation between obesity and the onset of type 2 diabetes with its associated insulin resistance. It should be pointed out that in the United States the proportion of the population under 40 that can be clinically defined as obese now exceeds 25%. Many children are obese and are developing type 2 diabetes at an alarming epidemic rate. The dramatic rise in obesity in the US has lead to an equally alarming increase in the percentage of the population who suffer from the **metabolic syndrome**. The **metabolic syndrome** is a clustering of atherosclerotic cardiovascular disease risk factors, one of which involves insulin resistance characteristic in type 2 diabetes. It should be pointed out that obesity alone does not always lead to insulin resistance as some individuals who are obese do not experience insulin resistance and conversely, some individuals who manifest insulin resistance are not obese. These latter observations point to the added role of genetics in the acquisition of insulin resistance.

Secondary, or other specific types of diabetes mellitus are the result of many causes including:

- **1.** Maturity onset type diabetes of the young (MODY) was previously considered to be a third form of type 2 diabetes. However, with the discovery of specific mutations leading to MODY, it is now classified under secondary or other specific types of diabetes. MODY is characterized by onset prior to age 25. All cases to date have shown impaired  $\beta$ -cell function. Patients may also exhibit insulin resistance and late  $\beta$ -cell failure.

Evidence indicates that mutations in 10-12 different genes have been correlated with the development of MODY. Mutations in the 6 genes described here are all clearly correlated to MODY:

- MODY1: the transcription factor identified as hepatic nuclear factor-4 $\alpha$  (HNF-4 $\alpha$ ).
  - MODY2: pancreatic glucokinase
  - MODY3: the transcription factor HNF-1 $\alpha$ . This gene is also called hepatocyte transcription factor-1 (TCF1).
  - MODY4: the homeodomain transcription factor insulin promoter factor-1 (IPF-1). This gene is more commonly called PDX1 derived from pancreas duodenum homeobox-1.
  - MODY5: the transcription factor HNF-1 $\beta$ . This gene is also called hepatocyte transcription factor-2 (TCF2).
  - MODY6: the bHLH transcription factor NeuroD1. NeuroD1 was first identified as a neural fate-inducing gene. The hamster beta-2 gene, shown to regulate insulin transcription is identical to NeuroD1 so the gene is often called NeuroD/ $\beta$ 2.
- **2.** Pancreatic disease: Pancreatectomy leads to the clearest example of secondary diabetes. Cystic fibrosis and pancreatitis can also lead to destruction of the pancreas.
- **3.** Endocrine disease: Some tumors can produce counter-regulatory hormones that oppose the action of insulin or inhibit insulin secretion. These counter-regulatory hormones are glucagon, epinephrine, growth hormone and cortisol.
  - **a.** Glucagonomas are pancreatic cancers that secrete glucagon.
  - **b.** Pheochromocytomas secrete epinephrine.
  - **c.** **Cushing syndrome** results in excess cortisol secretion.

- **d. Acromegaly** results in excess growth hormone production.
- **4.** Drug-induced diabetes; treatment with glucocorticoids and diuretics can interfere with insulin function.
- **5.** Anti-insulin receptor autoantibodies (Type B insulin resistance)
- **6.** Mutations in the insulin gene
- **7.** [Mutations in insulin receptor gene](#) which lead to the syndromes listed below. Two clinical features are common in all syndromes that result from mutations in the insulin receptor gene: acanthosis nigricans and hyperandrogenism (the latter being observed only in females).
  - **a.** Leprachaunism
  - **b.** Rabson-Mendenhall syndrome
  - **c.** Type A insulin resistance
- **8.** Gestational diabetes; this syndrome sets in during pregnancy and usually resolves itself following childbirth.
- **9.** Many other genetic syndromes have either diabetes or impaired glucose tolerance associated with them; [lipoatrophic diabetes](#), [Wolfram syndrome](#), [Down syndrome](#), Klinefelter syndrome (XXY males), [Turner syndrome](#), [myotonic dystrophy](#), [muscular dystrophy](#), [Huntington disease](#), [Friedrich ataxia](#) (associated with deficiency in purine nucleotide phosphorylase), [Prader-Willi syndrome](#), [Werner syndrome](#), [Cockayne syndrome](#), and others such as those indicated above in 2 and 6.

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## Insulin-Dependent Diabetes Mellitus (IDDM)

### Etiology of IDDM

Type 1 diabetes has been shown to be the result of an autoimmune reaction to antigens of the islet cells of the pancreas. There is a strong association between IDDM and other endocrine autoimmunities (e.g. [Addison disease](#)). Additionally, there is an increased prevalence of autoimmune disease in family members of IDDM patients.

### Types of Autoantibodies:

- **1.** Islet cell cytoplasmic antibodies: The primary antibodies found in 90% of type 1 diabetics are against islet cell cytoplasmic proteins (termed ICCA, islet cell cytoplasmic antibodies). In non-diabetics ICCA frequency is only 0.5% - 4%. The presence of ICCA is a highly accurate predictor of future development of IDDM.
 

ICCA are not specific for the  $\beta$ -cells and recognize antigens in other cell types in the islet. However, the autoimmune attack appears to selectively destroy  $\beta$ -cells. Therefore, the antibodies may play a primary role in the destruction of islet cells. It is an equally likely possibility that the production of anti-islet antibodies occurs as a result of the destruction of  $\beta$ -cells. Whether a direct cause or an effect of islet cell destruction, the titer of the ICCA tends to decline over time.
- **2.** Islet cell surface antibodies: Autoantibodies directed against cell-surface antigens (ICSA) have also been described in as many as 80% of type 1 diabetics. Similar to ICCA, the titer of ICSA declines over time. Some patients with type 2 diabetes have been identified that are ICSA positive.
- **3.** Specific antigenic targets of islet cells: Antibodies to glutamic acid decarboxylase (GAD) have been identified in over 80% of patients newly diagnosed with IDDM. Like ICCA, anti-GAD antibodies decline over time in type 1 diabetics. The presence of anti-GAD

antibodies is a strong predictor of the future development of IDDM in high-risk populations. Anti-insulin antibodies (IAA) have been identified in IDDM patients and in relatives at risk to develop IDDM. These IAA are detectable even before the onset of insulin therapy in type 1 diabetics. IAA are detectable in around 40% of young children with IDDM.

### Pathophysiology of IDDM

The autoimmune destruction of pancreatic  $\beta$ -cells leads to a deficiency of insulin secretion. It is this loss of insulin secretion that leads to the metabolic derangements associated with IDDM. In addition to the loss of insulin secretion, the function of pancreatic  $\alpha$ -cells is also abnormal. There is excessive secretion of glucagon in IDDM patients. Normally, hyperglycemia leads to reduced glucagon secretion. However, in patients with IDDM, glucagon secretion is not suppressed by hyperglycemia. The resultant inappropriately elevated glucagon levels exacerbates the metabolic defects due to insulin deficiency (see below). The most pronounced example of this metabolic disruption is that patients with IDDM rapidly develop diabetic ketoacidosis in the absence of insulin administration. If somatostatin is administered to suppress glucagon secretion, there is a concomitant suppression in the rise of glucose and ketone bodies. Particularly problematic for long term IDDM patients is an impaired ability to secrete glucagon in response to hypoglycemia. This leads to potentially fatal hypoglycemia in response to insulin treatment in these patients.

Although insulin deficiency is the primary defect in IDDM, in patients with poorly controlled IDDM there is also a defect in the ability of target tissues to respond to the administration of insulin. There are multiple biochemical mechanisms that account for this impairment of tissues to respond to insulin. Deficiency in insulin leads to elevated levels of free fatty acids in the plasma as a result of uncontrolled lipolysis in adipose tissue. Free fatty acids suppress glucose metabolism in peripheral tissues such as skeletal muscle. This impairs the action of insulin in these tissues, i.e. the promotion of glucose utilization. Additionally, insulin deficiency decreases the expression of a number of genes necessary for target tissues to respond normally to insulin -- such as glucokinase in liver and the GLUT 4 class of glucose transporters in adipose tissue.

The major metabolic derangements which result from insulin deficiency in IDDM are impaired glucose, lipid and protein metabolism.

**Glucose Metabolism:** Uncontrolled IDDM leads to increased hepatic glucose output. First, liver glycogen stores are mobilized then hepatic **gluconeogenesis** is used to produce glucose. Insulin deficiency also impairs non-hepatic tissue utilization of glucose. In particular in adipose tissue and skeletal muscle, insulin stimulates glucose uptake. This is accomplished by insulin-mediated movement of glucose transporter proteins to the plasma membrane of these tissues. Reduced glucose uptake by peripheral tissues in turn leads to a reduced rate of glucose metabolism. In addition, the level of hepatic glucokinase is regulated by insulin. Therefore, a reduced rate of glucose phosphorylation in hepatocytes leads to increased delivery to the blood. Other enzymes involved in anabolic metabolism of glucose are affected by insulin (primarily through covalent modifications). The combination of increased hepatic glucose production and reduced peripheral tissues metabolism leads to elevated plasma glucose levels. When the capacity of the kidneys to absorb glucose is surpassed, **glucosuria** ensues. Glucose is an osmotic diuretic and an increase in renal loss of glucose is accompanied by loss of water and electrolytes, termed **polyuria**. The result of the loss of water (and overall volume) leads to the activation of the thirst mechanism (**polydipsia**). The negative caloric balance which results from the glucosuria and tissue catabolism leads to an increase in appetite and food intake (**polyphagia**).

**Lipid Metabolism:** One major role of insulin is to stimulate the storage of food energy following the consumption of a meal. This energy storage is in the form of glycogen in hepatocytes and skeletal muscle. Additionally, insulin stimulates hepatocytes to synthesize

triglycerides and storage of triglycerides in adipose tissue. In opposition to increased adipocyte storage of triglycerides is insulin-mediated inhibition of lipolysis. In uncontrolled IDDM there is a rapid mobilization of triglycerides leading to increased levels of plasma free fatty acids. The free fatty acids are taken up by numerous tissues (however, not the brain) and metabolized to provide energy. Free fatty acids are also taken up by the liver.

Normally, the levels of malonyl-CoA are high in the presence of insulin. These high levels of malonyl-CoA inhibit carnitine palmitoyltransferase I, the enzyme required for the transport of fatty acyl-CoA's into the mitochondria where they are subject to oxidation for energy production. Thus, in the absence of insulin, malonyl-CoA levels fall and transport of fatty acyl-CoA's into the mitochondria increases. Mitochondrial oxidation of fatty acids generates acetyl-CoA which can be further oxidized in the [TCA cycle](#). However, in hepatocytes the majority of the acetyl-CoA is not oxidized by the TCA cycle but is metabolized into the ketone bodies, acetoacetate and  $\beta$ -hydroxybutyrate. These ketone bodies leave the liver and are used for energy production by the brain, heart and skeletal muscle. In IDDM, the increased availability of free fatty acids and ketone bodies exacerbates the reduced utilization of glucose furthering the ensuing hyperglycemia. Production of ketone bodies in excess of the body's ability to utilize them leads to ketoacidosis. In diabetics, this can be easily diagnosed by smelling the breath. A spontaneous breakdown product of acetoacetate is acetone which is volatilized by the lungs producing a distinctive odor.

Normally, plasma triglycerides are acted upon by lipoprotein lipase (LPL), an enzyme on the surface of the endothelial cells lining the vessels. In particular, LPL activity allows fatty acids to be taken from circulating triglycerides for storage in adipocytes. The activity of LPL requires insulin and in its absence a hypertriglyceridemia results.

**Protein Metabolism:** Insulin regulates the synthesis of many genes, either positively or negatively that then affect overall metabolism. Insulin has a global effect on protein metabolism -- increasing the rate of protein synthesis and decreasing the rate of protein degradation. Thus, insulin deficiency will lead to increased catabolism of protein. The increased rate of proteolysis leads to elevated concentrations in plasma amino acids. These amino acids serve as precursors for hepatic and renal gluconeogenesis. In liver, the increased gluconeogenesis further contributes to the hyperglycemia seen in IDDM.

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## Non-Insulin-Dependent Diabetes Mellitus (NIDDM)

### Etiology of NIDDM

NIDDM is characterized by a lack of the need for insulin to prevent ketoacidosis. Type 2 diabetes refers to the common form of idiopathic NIDDM. NIDDM is not an autoimmune disorder, however, there is a strong genetic correlation to the susceptibility to NIDDM. The susceptibility genes that predispose one to NIDDM have not been identified in most patients. This is due in part to the heterogeneity of the genes responsible for the susceptibility to NIDDM. Obesity is a major risk factor that predisposes one to NIDDM. Genetic studies in mice and rats have demonstrated a link between genes responsible for obesity and those that cause diabetes mellitus.

### Pathophysiology of NIDDM

Unlike patients with IDDM, those with NIDDM have detectable levels of circulating insulin. On the basis of oral glucose tolerance testing the essential elements of NIDDM can be divided into 4 distinct groups; those with normal glucose tolerance, chemical diabetes (called impaired glucose tolerance), diabetes with minimal fasting hyperglycemia (fasting plasma glucose  $<140$  mg/dL), and diabetes mellitus in association with overt fasting hyperglycemia (fasting plasma glucose  $>140$  mg/dL). In patients with the highest levels of plasma insulin (impaired glucose tolerance group) there was also elevated plasma glucose. This indicates that these individuals are resistant to the

action of insulin. In the progression from impaired glucose tolerance to diabetes mellitus the level of insulin declines indicating that patients with NIDDM have decreased insulin secretion.

Additional studies have subsequently demonstrated that both insulin resistance and insulin deficiency is common in the average NIDDM patient. Many experts conclude that insulin resistance is the **primary** cause of NIDDM, however, others contend that insulin deficiency is the primary cause because a moderate degree of insulin resistance is not sufficient to cause NIDDM. As indicated above, most patients with the common form of NIDDM have both defects.

A number of candidate genes have been screened for having causative roles in type 2 diabetes. Although several monogenic loci are associated with type 2 diabetes (see MODY descriptions above) none has been shown to be a significant cause of the disease (i.e. >50% in all cases). Several of the genes having roles in progression to type 2 diabetes include pancreatic glucokinase (MODY 2), GLUT-2 (glucose transporter), glucagon receptor, glucagon-like protein-1 (GLIP-1), glucokinase regulatory protein and hexokinase-1.

Recent evidence has demonstrated a role for a member of the nuclear hormone receptor superfamily of proteins in the etiology of type 2 diabetes. A relatively new class of drugs used to increase the sensitivity of the body to insulin are the **thiazolidinedione** drugs (see [below](#)). These compounds bind to and alter the function of the **peroxisome proliferator-activated receptor  $\gamma$  (PPAR $\gamma$ )**. PPAR $\gamma$  is also a transcription factor and, when activated, binds to another transcription factor known as the **retinoid X receptor (RXR)**. When these two proteins are complexed a specific set of genes becomes activated. PPAR $\gamma$  is a key regulator of adipocyte differentiation, it can induce the differentiation of fibroblasts or other undifferentiated cells into mature fat cells. PPAR $\gamma$  is also involved in the synthesis of biologically active compounds from vascular endothelial cells and immune cells.

Mutations in the gene for PPAR $\gamma$  have been correlated with insulin resistance. It is still not completely clear how impaired PPAR $\gamma$  signaling can affect the sensitivity of the body to insulin or indeed if the observed mutations are a direct or indirect cause of the symptoms of insulin resistance.

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## Diabetes and The "Metabolic Syndrome"

Although the metabolic syndrome (also called **syndrome X**) is not exclusively associated with type 2 diabetes and the associated insulin resistance, the increasing prevalence of obesity and associated development of type 2 diabetes places insulin resistance as a major contributor to the syndrome. The metabolic syndrome is defined as a clustering of atherosclerotic cardiovascular disease risk factors that include visceral adiposity (obesity), insulin resistance, low levels of HDLs and a systemic proinflammatory state. There are key components to the metabolic syndrome which include hypertension, dyslipidemia, insulin resistance, chronic inflammation, impaired fibrinolysis, procoagulation and most telling central obesity.

The role of adipose tissue (fat) stems from the fact that the organ is active at secretion of cytokines, termed **adipocytokines**. These include tumor necrosis factor- $\alpha$  (TNF $\alpha$ ), interleukin-6 (IL-6), leptin, adiponectin and resistin. Leptin has received particular attention of late due to its role in obesity in addition to the fact that recent data indicates that plasma leptin levels are found to be predictive of the potential for cardiovascular pathology.

Many clinicians and researchers believe that insulin resistance underlies the cardiovascular pathologies of the metabolic syndrome. One primary reason for this is the role of insulin in fat homeostasis. As indicated above, the major role of insulin is to induce the storage of fuel. This can be as fat (triacylglycerides, TGs) in adipose tissue or as carbohydrate in the form of glycogen in liver and skeletal muscle. The effect of insulin resistance at the level of fat homeostasis is an increase in circulating TGs, referred to as dyslipidemia. Due to insulin resistance there is an increase in the delivery of peripheral fatty acids to the liver which in turn drives hepatic TG synthesis. These TGs are then packaged into **lipoprotein particles** termed VLDLs (very low density lipoproteins) which are returned to the circulation.



Taken together, the insulin resistance and its associated negative effects on metabolism, the increased levels of circulating TGs, the reduced levels of HDLs and hypertension, all contribute to the progression of atherosclerosis. With associated coagulation and fibrinolysis pathologies, the cardiovascular events of the metabolic syndrome can be devastating.

Since many of these pathologies can be reversed with proper diet and exercise, it is in a person's best interest to take responsibility for the role their life style choices play in the development of the metabolic syndrome.

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## Therapeutic Intervention for Hyperglycemia

Many, if not all, of the vascular consequences of insulin resistance are due to the persistent hyperglycemia seen in type 2 diabetes. For this reason a major goal of therapeutic intervention in type 2 diabetes is to reduce circulating glucose levels. There are many pharmacologic strategies to accomplish these goals.

**1. The  $\alpha$ -Glucosidase inhibitors:** Alpha-glucosidase inhibitors such as acarbose (**Precose**) and miglitol (**Glyset**) function by interfering with the action of the  $\alpha$ -glucosidases present in the small intestinal brush border. The consequence of this inhibition is a reduction in digestion and the consequent absorption of glucose into the systemic circulation. The reduction in glucose uptake allows the pancreatic  $\beta$ -cells to more effectively regulate insulin secretion. The advantage to the use of the  $\alpha$ -glucosidase inhibitors is that they function locally in the intestine and have no major systemic action. Hypoglycemia does not usually occur with the use of  $\alpha$ -glucosidase inhibitors but they are effective at reducing fasting plasma glucose (FPG) levels and levels of glycosylated hemoglobin ( $HbA_{1c}$ ). The common adverse side effects of these inhibitors are abdominal bloating and discomfort, diarrhea and flatulence.

**2. The Sulfonylureas:** The sulfonylurea and meglitinide classes of oral hypoglycemic drugs are referred to as **endogenous insulin secretagogues** because they induce the pancreatic release of endogenous insulin.

The sulfonylureas have been used in the US for nearly 50 years. The first generation sulfonylureas (**tolbutamide**, **acetohexamide**, **chlorpropamide** and **tolazamide**) are not routinely prescribed any longer in the US. The second generation sulfonylureas include glipizide (**Glucotrol**), glimepiride (**Amaryl**) and glyburide (**DiaBeta**, **Micronase**, **Glynase**). Because all of these drugs can induce pronounced hypoglycemia, treatment is initiated with the lowest possible dose and carefully monitored until the dose is found that results in a FPG of 110-140mg/dL. Sulfonylureas function by binding to and inhibiting the pancreatic ATP-dependent potassium channel that is normally involved in glucose-mediated insulin secretion (see above under insulin function). Sulfonylureas have no significant effects on circulating triglycerides, lipoproteins or cholesterol.

**3. The Meglitinides:** As indicated, the meglitinides repaglinide (**Prandin**) and nateglinide (**Starlix**) are non-sulfonylurea insulin secretagogues that are both fast acting and of short duration. Like the sulfonylureas, meglitinides therapy results in significant reduction in FPG as well as  $HbA_{1c}$ . The mechanism of action of the meglitinides is initiated by binding to a receptor on the pancreatic  $\beta$ -cell that is distinct from the receptors for the sulfonylureas. However, meglitinides do exert effects on potassium conductance. Like the sulfonylureas, the meglitinides have no direct effects on the circulating levels of plasma lipids.

**4. The Biguanides:** The biguanides are a class of drugs that function to lower serum glucose levels by enhancing insulin-mediated suppression of hepatic glucose production and enhancing insulin-stimulated glucose uptake by skeletal muscle. **Metformin (Glucophage)** is a member of this class and is currently the most widely prescribed insulin-sensitizing drug in current clinical use. Metformin administration does not lead to increased insulin release from the pancreas and as such the risk of hypoglycemia is minimal. Because the major site of action for metformin is the liver its use can be contraindicated in patients with liver dysfunction. The drug is ideal for obese patients and for younger type 2 diabetics.

Evidence on the mode of action of metformin shows that it improves insulin sensitivity by increasing insulin receptor tyrosine kinase

activity, enhancing glycogen synthesis and increasing recruitment and transport of GLUT4 transporters to the plasma membrane. Additionally, it has been shown that metformin affects mitochondrial activities dependent upon the model system studied. Metformin has a mild inhibitory effect on complex I of [oxidative phosphorylation](#), has antioxidant properties, and activates both glucose 6-phosphate dehydrogenase (G6PDH) and AMP-activated protein kinase, AMPK. The importance of AMPK metformin action stems from the role of AMPK in the regulation of both lipid and carbohydrate metabolism (see [AMPK: Master Metabolic Regulator](#) for more details). In adipose tissue, metformin inhibits lipolysis while enhancing re-esterification of fatty acids.

In adolescent females with type 2 diabetes, the use of metformin is highly recommended to reduce the incidence as well as the potential for **polycystic ovarian syndrome, PCOS**. PCOS is brought on by the hyperinsulinemia of type 2 diabetes. Insulin effects on the ovary drive conversion of progesterone to testosterone and a reduction in serum hormone binding globulin (SHBG). Taken together, the effects of hyperinsulinemia lead to a hyperandrogenic state in the ovary resulting in follicular atresia and ovulatory dysfunction.

**5. The Thiazolidinediones (TZDs):** The TZDs, such as troglitazone (**Rezulin**: Warner Lambert Co. but this drug was voluntarily removed from the market in 2000), rosiglitazone (**Avandia**: GlaxoSmithKline) and pioglitazone (**Actos**: Eli Lilly and Co.) have been proven useful in treating the hyperglycemia associated with insulin-resistance in both type 2 diabetes and non-diabetic conditions. The TZDs function as agonists for the nuclear receptor **peroxisome proliferator-activated receptor- $\gamma$  (PPAR $\gamma$ )**. PPAR $\gamma$  is a member of a superfamily of transcription factors that also include the closely related members, PPAR $\alpha$  and PPAR $\delta$ . PPAR $\gamma$  exists as a heterodimer with the nuclear **retinoid X receptor (RXR)**. The heterodimer binds to PPAR response elements (PPREs) in a number of target genes. Without ligand bound the heterodimer is associated with a co-repressor complex that includes a histone deacetylase. Deacetylated histone keeps DNA in a transcriptionally repressed state. When ligand binds to PPAR $\gamma$  the co-repressor complex dissociates and a co-activator complex containing histone acetylase associates resulting chromatin structural changes and transcriptional activation. The net effect of the TZDs is a potentiation of the actions of insulin in liver, adipose tissue and skeletal muscle, increased peripheral glucose disposal and a decrease in glucose output by the liver. Genes shown to be affected by PPAR $\gamma$  action include those encoding glucokinase, GLUT4, malic enzyme, lipoprotein lipase, fatty acyl-CoA synthase and adipocyte fatty acid binding protein. Given that PPAR $\gamma$  is predominantly expressed in adipose tissue, the effects of PPAR $\gamma$  agonists seen in the liver and skeletal muscle may be exerted via endocrine signaling from adipocytes. Recently it was shown that mutations in the PPAR $\gamma$  gene were correlated to familial insulin resistance.

**6. Targeting glucagon-like peptide-1 (GLP-1):** The synthesis and activities associated with GLP-1 were discussed above. As review, the primary metabolic responses to GLP-1 release from the enteroendocrine L-cells of the gut are inhibition of glucagon secretion and enhancement of glucose-dependent insulin release from the pancreas, both effects lead to decreased glycemic excursion. As indicated above, the hormonal action of GLP-1 is rapidly terminated as a consequence of enzymatic cleavage by DPP IV. Recent clinical evidence has shown that either infusion of GLP-1 or inhibition of DPP IV can result in dramatic reductions in plasma glucose concentrations, reductions in HbA<sub>1c</sub> and improvement in pancreatic  $\beta$ -cell function. Thus, both represent potential targets for the prevention of the hyperglycemia associated with diabetes and impaired insulin function (see the [DPP IV review site](#) maintained by Dr. Daniel J. Drucker).

There are advantages and disadvantages with the current therapeutic approaches to targeting GLP-1 action in diabetic patients. Current use of GLP-1 mimetics and/or GLP-1 receptor (GLP-1R) agonists focus on peptides or modified peptides and these must be injected. The need for chronic injection as a means of therapy always runs into the problem of patient compliance. One of the most promising GLP-1R agonists that has recently been approved for use is **BYETTA** (developed by Amylin Pharmaceuticals and Eli Lilly and Co.). Byetta is composed of **exenatide** which is the lizard salivary peptide called exendin-4. Exenatide is 53% identical to GLP-1 at the level of amino acids and binds to and activates the GLP-1R. The advantage of exenatide as a therapeutic is that it is resistant to cleavage and inactivation by DPP IV. In a recent trial in patients with type 2 diabetes, Byetta was shown not only to lower blood glucose levels and HbA<sub>1c</sub>, but patients also had an associated weight loss. Another promising GLP-1R agonist is **liraglutide**. This compound is a fatty acid-linked

derivative of GLP-1 that is resistant to DPP IV cleavage.

Although targeting compounds that can inhibit the enzymatic action of DPP IV would seem like ideal candidates for treating the hyperglycemia of uncontrolled diabetes, there are several unknowns associated with DPP IV inhibition. One of these issues is the fact that GLP-1 and GIP are only two of the many known substrates for DPP IV cleavage. Thus, prolonged inhibition of DPP IV enzymatic activity may have unexpected consequences unrelated to control of hyperglycemia.

DPP IV was originally identified as the lymphocyte cell surface antigen CD26. In humans CD26 functions in many pathways that are not directly related to its peptidase activity. It harbors adenosine deaminase-binding (ADA) properties and is involved in extracellular matrix binding. Of importance to the immune system, CD26 expression and activity are enhanced upon T-cell activation. CD26 interacts with other lymphocyte cell surface antigens including ADA, CD45 and the chemokine receptor CXCR4 (notable is the fact that CXCR4 is a T-cell attachment site for HIV). As yet it has not been clearly delineated as to whether the enzymatic activity of DPP IV is essential for the T-cell activating and co-stimulatory functions assigned to CD26. This issue must be resolved before chronic administration of DPP IV inhibitors can be applied in the clinic. Of significance, however, is that in gene knock-out mice lacking CD26 there is enhanced insulin secretion and improved glucose tolerance.

The major clinical advantages to the use of DPP IV inhibitors is that the ones in current trials are orally delivered. Compliance in patients is much higher with orally delivered drugs than with those that require injection. The drug NVPDPP728 was shown to have significant DPP IV inhibitory action and led to significant lowering of blood glucose and HbA<sub>1c</sub> levels in type 2 diabetics. A second generation DPP IV inhibitor LAF237 is currently in clinical trials.

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## Mechanisms of Signal Transduction

Signal transduction at the cellular level refers to the movement of signals from outside the cell to inside. The movement of signals can be simple, like that associated with receptor molecules of the acetylcholine class: receptors that constitute channels which, upon ligand interaction, allow signals to be passed in the form of small ion movement, either into or out of the cell. These ion movements result in changes in the electrical potential of the cells that, in turn, propagates the signal along the cell. More complex signal transduction involves the coupling of ligand-receptor interactions to many intracellular events. These events include phosphorylations by tyrosine kinases and/or serine/threonine kinases. Protein phosphorylations change enzyme activities and protein conformations. The eventual outcome is an alteration in cellular activity and changes in the program of genes expressed within the responding cells.

Please refer to the page on [Growth Factors](#) for descriptions of the growth factors described in this page and the explanation of their abbreviations.  
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## Classifications of Signal Transducing Receptors

Signal transducing receptors are of three general classes:

- **1.** Receptors that penetrate the plasma membrane and have intrinsic enzymatic activity. Receptors that have intrinsic enzymatic activities include those that are **tyrosine kinases** (e.g. PDGF, insulin, EGF and FGF receptors), **tyrosine phosphatases** (e.g. CD45 [*cluster determinant-45*] protein of T cells and macrophages), **guanylate cyclases** (e.g. natriuretic peptide receptors) and **serine/threonine kinases** (e.g. activin and TGF- $\beta$  receptors). Receptors with intrinsic tyrosine kinase activity are capable of autophosphorylation as well as phosphorylation of other substrates.

Additionally, several families of receptors lack intrinsic enzyme activity, yet are coupled to intracellular tyrosine kinases by direct protein-protein interactions (see below).

- **2.** Receptors that are coupled, inside the cell, to GTP-binding and hydrolyzing proteins (termed **G-proteins**). Receptors of the class that interact with G-proteins all have a structure that is characterized by 7 transmembrane spanning domains. These receptors are termed *serpentine* receptors. Examples of this class are the adrenergic receptors, odorant receptors, and certain hormone receptors (e.g. glucagon, angiotensin, vasopressin and bradykinin).
- **3.** Receptors that are found intracellularly and upon ligand binding migrate to the nucleus where the ligand-receptor complex directly affects gene transcription.

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## Receptor Tyrosine Kinases (RTKs)

The proteins encoding RTKs contain four major domains:

- An extracellular ligand binding domain.
- An intracellular tyrosine kinase domain.
- An intracellular regulatory domain.
- A transmembrane domain.

The amino acid sequences of the tyrosine kinase domains of RTKs are highly conserved with those of cAMP-dependent protein kinase (**PKA**) within the ATP binding and substrate binding regions. Some RTKs have an insertion of non-kinase domain amino acids into the kinase domain termed the **kinase insert**. RTK proteins are classified into families based upon structural features in their extracellular portions (as well as the presence or absence of a kinase insert) which include the cysteine rich domains, immunoglobulin-like domains, leucine-rich domains, Kringle domains, cadherin domains, fibronectin type III repeats, discoidin I-like domains, acidic domains, and EGF-like domains. Based upon the presence of these various extracellular domains the RTKs have been sub-divided into at least 14 different families.

## Characteristics of the Common Classes of RTKs



| Class | Examples                                                         | Structural Features of Class                                                                                                                                                                               |
|-------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I     | EGF receptor, NEU/HER2, HER3                                     | cysteine-rich sequences                                                                                                                                                                                    |
| II    | insulin receptor, IGF-1 receptor                                 | cysteine-rich sequences; characterized by disulfide-linked heterotetramers                                                                                                                                 |
| III   | PDGF receptors, c-Kit                                            | contain 5 immunoglobulin-like domains; contain the kinase insert                                                                                                                                           |
| IV    | FGF receptors                                                    | contain 3 immunoglobulin-like domains as well as the kinase insert; acidic domain                                                                                                                          |
| V     | vascular endothelial cell growth factor (VEGF) receptor          | contain 7 immunoglobulin-like domains as well as the kinase insert domain                                                                                                                                  |
| VI    | hepatocyte growth factor (HGF) and scatter factor (SC) receptors | heterodimeric like the class II receptors except that one of the two protein subunits is completely extracellular. The HGF receptor is a proto-oncogene that was originally identified as the Met oncogene |
| VII   | neurotrophin receptor family (trkA, trkB, trkC) and NGF receptor | contain no or few cysteine-rich domains; NGFR has leucine rich domain                                                                                                                                      |

Many receptors that have intrinsic tyrosine kinase activity as well as the tyrosine kinases that are associated with cell surface receptors contain tyrosines residues, that upon phosphorylation, interact with other proteins of the signaling cascade. These other proteins contain a domain of amino acid sequences that are homologous to a domain first identified in the c-Src proto-oncogene (**c** - designates the cellular form of proto-oncogenes that were first identified in transforming retrovirus). These domains are termed **SH2 domains** (Src homology domain 2). Another conserved protein-protein interaction domain identified in many signal transduction proteins is related to a third domain in c-Src identified as the **SH3 domain**.

The interactions of SH2 domain containing proteins with RTKs or receptor associated tyrosine kinases leads to tyrosine phosphorylation of the SH2 containing proteins. The result of the phosphorylation of SH2 containing proteins that have enzymatic activity is an alteration (either positively or negatively) in that activity. Several SH2 containing proteins that have intrinsic enzymatic activity include **phospholipase C- $\gamma$  (PLC- $\gamma$ )**, the proto-oncogene c-Ras associated GTPase activating protein (**rasGAP**), **phosphatidylinositol-3-kinase (PI-3K)**, **protein phosphatase-1C (PTP1C)**, as well as members of the Src family of **protein tyrosine kinases (PTKs)**.

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## Non-Receptor Protein Tyrosine Kinases (PTKs)

There are numerous intracellular PTKs that are responsible for phosphorylating a variety of intracellular proteins on tyrosine residues following activation of cellular growth and proliferation signals. There is now recognized two distinct families of non-receptor PTKs. The archetypal PTK family is related to the Src protein. The Src protein is a tyrosine kinase first identified as the transforming protein in Rous sarcoma virus. Subsequently, a cellular homolog was identified as **c-Src**. Numerous **proto-oncogenes** were identified as the transforming proteins carried by retroviruses. The second family is related to the **Janus kinase (Jak)**.

Most of the proteins of both families of non-receptor PTKs couple to cellular receptors that lack enzymatic activity themselves. This class of receptors includes all of the cytokine receptors (eg the interleukin-2 (IL-2) receptor) as well as the CD4 and CD8 cell surface glycoproteins of T cells and the T cell antigen receptor (TCR). This mode of coupling receptors to intracellular PTKs suggests a split form of RTK.

Another example of receptor-signaling through protein interaction involves the insulin receptor (IR). This receptor has intrinsic tyrosine kinase activity but does not directly interact, following autophosphorylation, with enzymatically active proteins containing SH2 domains (e.g. PI-3K or PLC- $\gamma$ ). Instead, the principal IR substrate is a protein termed IRS-1. IRS-1 contains several motifs that resemble SH2 binding consensus sites for the catalytically active subunit of PI-3K. These domains allow complexes to form between IRS-1 and PI-3K. This model suggests that IRS-1 acts as a **docking** or **adapter protein** to couple the IR to SH2 containing signaling proteins.

Additional adapter proteins have been identified, the most commonly occurring being a protein termed **growth factor receptor-binding protein 2, Grb2**.

An example of an alteration in receptor activity in response to association with an intracellular PTK is the nicotinic acetylcholine receptor (AChR). These receptors comprise an ion channel consisting of four distinct subunits ( $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\delta$ ). The  $\beta$ ,  $\gamma$ , and  $\delta$  subunits are tyrosine phosphorylated in response to acetylcholine binding which leads to an increase in the rate of desensitization to acetylcholine.

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## Receptor Serine/Threonine Kinases (RSTKs)

The receptors for the TGF- $\beta$  superfamily of ligands have intrinsic serine/threonine kinase activity. A more complete description of the TGF- $\beta$  signaling cascade can be found in the [Signaling by Wnts and the TGFs- \$\beta\$ /BMP Families](#) page.

There are more than 30 multifunctional proteins of the TGF- $\beta$  superfamily which also includes the activins, inhibins and the bone morphogenetic proteins (BMPs). This superfamily of proteins can induce and/or inhibit cellular proliferation or differentiation and regulate migration and adhesion of various cell types. The signaling pathways utilized by the TGF- $\beta$ ,

activin and BMP receptors are different than those for receptors with intrinsic tyrosine kinase activity or that associate with intracellular tyrosine kinases.

At least 17 RSTKs have been isolated and can be divided into 2 subfamilies identified as the type I and type II receptors. Ligands first bind to the type II receptors which then leads to interaction with the type I receptors. When the complex between ligand and the 2 receptor subtypes forms, the type II receptor phosphorylates the type I receptor leading to initiation of the signaling cascade. One predominant effect of TGF- $\beta$  is regulation of progression through the [cell cycle](#). One nuclear protein involved in the responses of cells to TGF- $\beta$  is the proto-oncogene, c-Myc ("mick") which directly affects the expression of genes harboring Myc-binding elements.

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## Non-Receptor Serine/Threonine Kinases

There are several serine/threonine kinases that function in signal transduction pathways. The two more commonly known are **cAMP-dependent protein kinase (PKA)** and **protein kinase C (PKC)**. Additional serine/threonine kinases important for signal transduction are the **mitogen activated protein kinases (MAP kinases)**.

### Protein Kinase C (PKC)

PKC was originally identified as a serine/threonine kinase that was maximally active in the presence of diacylglycerols (DAG) and calcium ion. It is now known that there are at least ten proteins of the PKC family. Each of these enzymes exhibits specific patterns of tissue expression and activation by lipid and calcium. PKCs are involved in the signal transduction pathways initiated by certain hormones, growth factors and neurotransmitters. The phosphorylation of various proteins, by PKC, can lead to either increased or decreased activity. Of particular importance is the phosphorylation of the EGF receptor by PKC which down-regulates the tyrosine kinase activity of the receptor. This effectively limits the length of the cellular responses initiated through the EGF receptor.

### MAP Kinases

MAP kinases were identified by virtue of their activation in response to growth factor stimulation of cells in culture, hence the name **mitogen activated protein kinases**. MAP kinases are also called ERKs for **extracellular-signal regulated kinases**. On the basis of *in vitro* substrates the MAP kinases have been variously called microtubule associated protein-2 kinase (MAP-2 kinase), myelin basic protein kinase (MBP kinase), ribosomal S6 protein kinase (RSK-kinase: i.e. a kinase that phosphorylates a kinase) and EGF receptor threonine kinase (ERT kinase). All of these proteins have similar biochemical properties, immuno-crossreactivities, amino acid sequence and ability to *in vitro* phosphorylate similar substrates.

Maximal MAP kinase activity requires that both tyrosine and threonine residues are phosphorylated. This indicates that MAP kinases act as switch kinases that transmits

information of increased intracellular tyrosine phosphorylation to that of serine/threonine phosphorylation. Although MAP kinase activation was first observed in response to activation of the EGF, PDGF, NGF and insulin receptors, other cellular stimuli such as T cell activation (which signals through the Lck [**lick**] tyrosine kinase), phorbol esters (that function through activation of PKC), thrombin, bombesin and bradykinin (that function through G-proteins) as well as N-methyl-D-aspartate (NMDA) receptor activation and electrical stimulation rapidly induce tyrosine phosphorylation of MAP kinases.

MAP kinases are, however, not the direct substrates for RTKs nor receptor associated tyrosine kinases but are in fact activated by an additional class of kinases termed MAP kinase kinases (MAPK kinases) and MAPK kinase kinases (MAPKK kinases). One of the MAPK kinases has been identified as the proto-oncogenic serine/threonine kinase, Raf. Ultimate targets of the MAP kinases are several transcriptional regulators e.g. serum response factor (SRF), and the proto-oncogenes Fos, Myc and Jun as well as members of the steroid/thyroid hormone receptor super family of proteins.

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## Phospholipases and Phospholipids in Signal Transduction

Phospholipases and phospholipids are involved in the processes of transmitting ligand-receptor induced signals from the plasma membrane to intracellular proteins. The primary protein affected by the activation of phospholipases is PKC which is maximally active in the presence of calcium ion and DAG. The generation of DAG occurs in response to agonist activation of various phospholipases. The principal mediators of PKC activity are receptors coupled to activation of phospholipase C- $\gamma$  (PLC- $\gamma$ ). PLC- $\gamma$  contains SH2 domains that allow it to interact with tyrosine phosphorylated RTKs. This allows PLC- $\gamma$  to be intimately associated with the signal transduction complexes of the membrane as well as membrane phospholipids that are its substrates. Activation of PLC- $\gamma$  leads primarily to the hydrolysis of membrane phosphatidylinositol bisphosphate (PIP<sub>2</sub>) leading to an increase in intracellular DAG and inositol trisphosphate (IP<sub>3</sub>). The released IP<sub>3</sub> interacts with intracellular membrane receptors leading to an increased release of stored calcium ions. Together, the increased DAG and intracellular free calcium ion concentrations lead to increased activity of PKC.

Recent evidence indicates that phospholipases D and A<sub>2</sub> (PLD and PLA<sub>2</sub>) also are involved in the sustained activation of PKC through their hydrolysis of membrane phosphatidylcholine (PC). PLD action on PC leads to the release of phosphatidic acid which in turn is converted to DAG by a specific phosphatidic acid phosphomonoesterase. PLA<sub>2</sub> hydrolyzes PC to yield free fatty acids and lysoPC both of which have been shown to potentiate the DAG mediated activation of PKC. Of medical significance is the ability of **phorbol ester tumor promoters** to activate PKC directly. This leads to elevated and unregulated activation of PKC and the consequent disruption in normal cellular growth and proliferation control leading ultimately to neoplasia.

## Phosphatidylinositol-3-Kinase (PI-3K)

PI-3K is tyrosine phosphorylated, and subsequently activated, by various RTKs and receptor-associated PTKs. PI-3K is a heterodimeric protein containing an 85 kDa and 110 kDa subunits. The p85 subunit contains SH2 domains that interact with activated receptors or other receptor-associated PTKs and is itself subsequently tyrosine phosphorylated and activated. The 85 kDa subunit is non-catalytic, however, it does contain a domain homologous to GTPase activating (GAP) proteins. It is the 110 kDa subunit that is enzymatically active. PI-3K, associates with and is activated by, the PDGF, EGF, insulin, IGF-1, HGF and NGF receptors. PI-3K phosphorylates various phosphatidylinositols at the 3 position of the inositol ring. This activity generates additional substrates for PLC- $\gamma$  allowing a cascade of DAG and IP<sub>3</sub> to be generated by a single activated RTK or other protein tyrosine kinases.

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## G-Protein Coupled Receptors

There are several different classifications of receptors that couple signal transduction to G-proteins. These classes of receptor are termed **G-protein coupled receptors, GPCRs**. Well over 1000 different GPCRs have been cloned, most being orphan receptors having no as yet identified ligand. Three different classes of GPCR are reviewed:

- **1.** GPCRs that modulate adenylate cyclase activity. One class of adenylate cyclase modulating receptors activate the enzyme leading to the production of cAMP as the second messenger. Receptors of this class include the  $\beta$ -adrenergic, glucagon and odorant molecule receptors. Increases in the production of cAMP leads to an increase in the activity of PKA in the case of  $\beta$ -adrenergic and glucagon receptors. In the case of odorant molecule receptors the increase in cAMP leads to the activation of ion channels. In contrast to increased adenylate cyclase activity, the  $\alpha$ -type adrenergic receptors are coupled to inhibitory G-proteins that repress adenylate cyclase activity upon receptor activation.
- **2.** GPCRs that activate PLC- $\gamma$  leading to hydrolysis of polyphosphoinositides (e.g. PIP<sub>2</sub>) generating the second messengers, diacylglycerol (DAG) and inositoltrisphosphate (IP<sub>3</sub>). This class of receptors includes the angiotensin, bradykinin and vasopressin receptors.
- **3.** A novel class of GPCRs are the photoreceptors. This class is coupled to a G-protein termed **transducin** that activates a phosphodiesterase which leads to a decrease in the level of cGMP. The drop in cGMP then results in the closing of a Na<sup>+</sup>/Ca<sup>2+</sup> channel leading to hyperpolarization of the cell. See the [Role of Vitamin A in Vision](#) for more details.

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## G-Protein Regulators

The activity of G-proteins with respect to GTP hydrolysis is regulated by a family of proteins termed **GTPase activating proteins, GAPs**. The proto-oncogenic protein, Ras, is a G-protein involved in the genesis of numerous forms of cancer (when the protein sustains specific mutations). Of particular clinical significance is the fact that oncogenic activation of Ras occurs with higher frequency than any other gene in the development of colo-rectal cancers. Regulation of Ras GTPase activity is controlled by **rasGAP**.

There are several other GAP proteins besides rasGAP that are important in signal transduction. There are two clinically important proteins of the GAP family of proteins. One is the gene product of the **neurofibromatosis type-1 (NF1) susceptibility locus**. The NF1 gene is a tumor suppressor gene and the protein encoded is called **neurofibromin**. The second is the protein encoded by the BCR locus (**break point cluster region** gene). The BCR locus is rearranged in the Philadelphia+ chromosome (Ph<sup>+</sup>) observed with high frequency in chronic myelogenous leukemias (CMLs) and acute lymphocytic leukemias (ALLs).

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## Intracellular Hormone Receptors

Hormone receptors are proteins that effectively bypass all of the signal transduction pathways described thus far by residing within the cytoplasm. Additionally, all of the hormone receptors are bi-functional. They are capable of binding hormone as well as directly activating gene transcription.

The **steroid/thyroid hormone receptor** superfamily (e.g. glucocorticoid, vitamin D, retinoic acid and thyroid hormone receptors) is a class of proteins that reside in the cytoplasm and bind the lipophilic steroid/thyroid hormones. These hormones are capable of freely penetrating the hydrophobic plasma membrane. Upon binding ligand the hormone-receptor complex translocates to the nucleus and bind to specific DNA sequences termed **hormone response elements (HREs)**. The binding of the complex to an HRE results in altered transcription rates of the associated gene.

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## Phosphatases in Signal Transduction

Substantial evidence links both tyrosine and serine/threonine phosphorylation with increased cellular growth, proliferation and differentiation. Removal of the incorporated phosphates must be a necessary event in order to turn off the proliferative signals. This suggests that phosphatases may function as anti-oncogenes or growth suppressor genes. The loss of a functional phosphatase involved in regulating growth promoting signals could lead to neoplasia. However, examples are known where dephosphorylation is required for promotion of cell growth. This is particularly true of specialized kinases that are directly involved in



regulating cell cycle progression. Therefore, it is difficult to envision all phosphatases as being tumor suppressor genes.

There are two broad classes of **protein tyrosine phosphatases (PTPs)**. One class are transmembrane enzymes which contain the phosphatase activity domain in the intracellular portion of the protein. The other are intracellularly localized enzymes. The first transmembrane PTP characterized was the **leukocyte common antigen protein, CD45**. This protein was shown to have homology to the intracellular PTP, PTP1B. There are at least six sub-classes of the transmembrane PTPs.

The clearest studies of a role for transmembrane PTPs in signal transduction have involved the CD45 protein. These studies have shown that CD45 is involved in the regulation of the tyrosine kinase activity of Lck in T cells. As indicated above Lck is associated with T cell antigens CD4 and CD8 generating a split-RTK involved in T cell activation. It is suspected that CD45 dephosphorylates a regulatory tyrosine phosphorylation site in the C-terminus of Lck, thereby, increasing the activity of Lck towards its substrate(s).

The second class of PTPs are the intracellular proteins. The C-terminal residues of most if not all intracellular PTPs are very hydrophobic and suggest these sites are membrane attachment domains of these proteins. One role of intracellular PTPs is in the maturation of *Xenopus* oocytes in response to hormone. Over expression of PTP-1B in oocytes resulted in a marked retardation in the rate of insulin- and progesterone-induced maturation. These results suggest a role for PTP-1B in countering the signals leading to cellular activation.

The above observation as well as several others have demonstrated a link between insulin function and PTP-1B. PTP-1B directly interacts with the insulin receptor and removes the tyrosine phosphates incorporated by autophosphorylation in response to insulin binding, thereby, negatively affecting the activity of the insulin receptor. Recently the PTP-1B gene was disrupted in mice by targeted deletion. Mice lacking a functional PTP-1B gene exhibit increased insulin sensitivity as well as resistance to obesity induced by a high fat diet.

As with the transmembrane PTPs little is known about the regulation of the activity of the intracellular PTPs.

Two intracellular PTPs (PTP-1C and PTP-1D) have been shown to contain SH2 domains. These SH2 domains allow these PTPs to directly interact with tyrosine phosphorylated RTKs and PTKs, thereby, dephosphorylating tyrosines in these proteins. Following receptor stimulation of signal transduction events, the SH2 containing PTPs are directed to several of the RTKs and/or PTKs with the net effect being a termination of the signaling events by tyrosine dephosphorylation.

Other phosphatases that recognize serine and/or threonine phosphorylated proteins also exist in cells. These are referred to as **protein serine phosphatases (PSPs)**. At least 15 distinct PSPs have been identified. The type 2A PSPs exhibit selective substrate specificity towards PKC phosphorylated proteins; in particular serine and threonine phosphorylated receptors. Type 2A PSPs are more effective than other PSPs in dephosphorylating RSKs, proteins that are involved in signaling cascades by phosphorylating ribosomal S6 protein (see above). However, a type 1 PSP is required to dephosphorylate S6 itself.

The type 2A PSPs have 2 subunits (a regulatory and a catalytic) both of which can associate with one of the tumor antigens of the DNA tumor virus, polyoma. Transformation by



DNA tumor viruses such as polyoma appears to be mediated by the formation of a signal transduction unit consisting of a virally encoded T antigen and several host encoded proteins. Several host proteins are tyrosine kinases of the src family. Polyoma middle T antigen also can bind to PI-3K. The association of type 2A PSPs in these complexes may lead to dephosphorylation of regulatory serine/threonine phosphorylated sites resulting in increased signal transduction and subsequent cellular proliferation.

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the list is not intended to be comprehensive and the information supplied is only a brief overview

**For a more comprehensive listing see: [The C.O.P.E. Site](#)**

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## Introduction

Growth factors are proteins that bind to receptors on the cell surface, with the primary result of activating cellular proliferation and/or differentiation. Many growth factors are quite versatile, stimulating cellular division in numerous different cell types; while others are specific to a particular cell-type. The lists in the following Tables as well as the descriptions of several factors are not intended to be comprehensive nor complete but a look at some of the more

commonly known factors and their principal activities.

| Factor        | Principal Source                                        | Primary Activity                                                                                          | Comments                                                                                         |
|---------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| PDGF          | platelets, endothelial cells, placenta                  | promotes proliferation of connective tissue, glial and smooth muscle cells                                | two different protein chains form 3 distinct dimer forms; AA, AB and BB                          |
| EGF           | submaxillary gland, Brunners gland                      | promotes proliferation of mesenchymal, glial and epithelial cells                                         |                                                                                                  |
| TGF- $\alpha$ | common in transformed cells                             | may be important for normal wound healing                                                                 | related to EGF                                                                                   |
| FGF           | wide range of cells; protein is associated with the ECM | promotes proliferation of many cells; inhibits some stem cells; induces mesoderm to form in early embryos | at least 19 family members, 4 distinct receptors                                                 |
| NGF           |                                                         | promotes neurite outgrowth and neural cell survival                                                       | several related proteins first identified as proto-oncogenes; trkA ( <i>trackA</i> ), trkB, trkC |

|                |                                                                          |                                                                                                                                                          |                                                             |
|----------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Erythropoietin | kidney                                                                   | promotes proliferation and differentiation of erythrocytes                                                                                               |                                                             |
| TGF- $\beta$   | activated TH <sub>1</sub> cells (T-helper) and natural killer (NK) cells | anti-inflammatory (suppresses cytokine production and class II MHC expression), promotes wound healing, inhibits macrophage and lymphocyte proliferation | at least 100 different family members                       |
| IGF-I          | primarily liver                                                          | promotes proliferation of many cell types                                                                                                                | related to IGF-II and proinsulin, also called Somatomedin C |
| IGF-II         | variety of cells                                                         | promotes proliferation of many cell types primarily of fetal origin                                                                                      | related to IGF-I and proinsulin                             |

Cytokines are a unique family of growth factors. Secreted primarily from leukocytes, cytokines stimulate both the humoral and cellular immune responses, as well as the activation of phagocytic cells. Cytokines that are secreted from lymphocytes are termed **lymphokines**, whereas those secreted by monocytes or macrophages are termed **monokines**. A large family of cytokines are produced by various cells of the body. Many of the lymphokines are also known as **interleukins (ILs)**, since they are not only secreted by leukocytes but also able to affect the cellular responses of leukocytes. Specifically, interleukins are growth factors targeted to cells of hematopoietic origin. The list of identified interleukins grows continuously with the total number of individual activities now at 22 (13 are listed in the Table below).

| Interleukins                | Principal Source                                           | Primary Activity                                                                                                                                  |
|-----------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| IL1- $\alpha$ and - $\beta$ | macrophages and other antigen presenting cells (APCs)      | costimulation of APCs and T cells, inflammation and fever, acute phase response, hematopoiesis                                                    |
| IL-2                        | activated TH <sub>1</sub> cells, NK cells                  | proliferation of B cells and activated T cells, NK functions                                                                                      |
| IL-3                        | activated T cells                                          | growth of hematopoietic progenitor cells                                                                                                          |
| IL-4                        | TH <sub>2</sub> and mast cells                             | B cell proliferation, eosinophil and mast cell growth and function, IgE and class II MHC expression on B cells, inhibition of monokine production |
| IL-5                        | TH <sub>2</sub> and mast cells                             | eosinophil growth and function                                                                                                                    |
| IL-6                        | activated TH <sub>2</sub> cells, APCs, other somatic cells | acute phase response, B cell proliferation, thrombopoiesis, synergistic with IL-1 and TNF on T cells                                              |
| IL-7                        | thymic and marrow stromal cells                            | T and B lymphopoiesis                                                                                                                             |
| IL-8                        | macrophages, other somatic cells                           | chemoattractant for neutrophils and T cells                                                                                                       |
| IL-9                        | T cells                                                    | hematopoietic and thymopoietic effects                                                                                                            |

|                             |                                                                              |                                                                                                                                                                                            |
|-----------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IL-10                       | activated TH <sub>2</sub> cells, CD8 <sup>+</sup> T and B cells, macrophages | inhibits cytokine production, promotes B cell proliferation and antibody production, suppresses cellular immunity, mast cell growth                                                        |
| IL-11                       | stromal cells                                                                | synergistic hematopoietic and thrombopoietic effects                                                                                                                                       |
| IL-12                       | B cells, macrophages                                                         | proliferation of NK cells, INF- $\gamma$ production, promotes cell-mediated immune functions                                                                                               |
| IL-13                       | TH <sub>2</sub> cells                                                        | IL-4-like activities                                                                                                                                                                       |
| <b>Interferons</b>          | <b>Principal Source</b>                                                      | <b>Primary Activity</b>                                                                                                                                                                    |
| INF- $\alpha$ and - $\beta$ | macrophages, neutrophils and some somatic cells                              | antiviral effects, induction of class I MHC on all somatic cells, activation of NK cells and macrophages                                                                                   |
| INF- $\gamma$               | activated TH <sub>1</sub> and NK cells                                       | induces class I MHC on all somatic cells, induces class II MHC on APCs and somatic cells, activates macrophages, neutrophils, NK cells, promotes cell-mediated immunity, antiviral effects |

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## Epidermal Growth Factor (EGF)

EGF, like all growth factors, binds to specific high-affinity, low-capacity receptors on the surface of responsive cells. Intrinsic to the EGF receptor is tyrosine kinase activity, which is activated in response to EGF binding. The kinase domain of the EGF receptor phosphorylates the EGF receptor itself (**autophosphorylation**) as well as other proteins, in [signal transduction](#)



cascades, that associate with the receptor following activation. Experimental evidence has shown that the **Neu** proto-oncogene is a homologue of the EGF receptor.

EGF has proliferative effects on cells of both mesodermal and ectodermal origin, particularly keratinocytes and fibroblasts. EGF exhibits negative growth effects on certain carcinomas as well as hair follicle cells. Growth-related responses to EGF include the induction of nuclear proto-oncogene expression, such as **Fos**, **Jun** and **Myc**. EGF also has the effect of decreasing gastric acid secretion.

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## Platelet-Derived Growth Factor (PDGF)

PDGF is composed of two distinct polypeptide chains, A and B, that form homodimers (AA or BB) or heterodimers (AB). The **c-Sis** proto-oncogene has been shown to be homologous to the PDGF A chain. Only the dimeric forms of PDGF interact with the PDGF receptor. Two distinct classes of PDGF receptor have been cloned, one specific for AA homodimers and another that binds BB and AB type dimers. Like the EGF receptor, the PDGF receptors have intrinsic tyrosine kinase activity. Following autophosphorylation of the PDGF receptor, numerous signal-transducing proteins associate with the receptor and are subsequently tyrosine phosphorylated.

Proliferative responses to PDGF action are exerted on many mesenchymal cell types. Other growth-related responses to PDGF include cytoskeletal rearrangement and increased polyphosphoinositol turnover. Again, like EGF, PDGF induces the expression of a number of nuclear localized proto-oncogenes, such as **Fos**, **Myc** and **Jun**. The primary effects of TGF- $\beta$  are due to the induction, by TGF- $\beta$ , of PDGF expression.

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## Fibroblast Growth Factors (FGFs)

There are at least 19 distinct members of the FGF family of growth factors. The two originally characterized FGFs were identified by biological assay and are termed **FGF1 (acidic-FGF, aFGF)** and **FGF2 (basic-FGF, bFGF)**. Kaposi's sarcoma cells (prevalent in patients with AIDS) secrete a homologue of FGF called the K-FGF proto-oncogene. In mice the mammary tumor virus integrates at two predominant sites in the mouse genome identified as Int-1 and Int-2. The protein encoded by the Int-2 locus is a homologue of the FGF family of growth factors.

Studies of human disorders as well as gene knock-out studies in mice show the prominent role for FGFs is in the development of the skeletal system and nervous system in mammals. FGFs also are neurotrophic for cells of both the peripheral and central nervous system. Additionally, several members of the FGF family are potent inducers of mesodermal differentiation in early embryos. Non-proliferative effects include regulation of pituitary and ovarian cell function.

The FGFs interact with specific cell-surface receptors. There have been identified 4 distinct receptor types identified as FGFR1 - FGFR4. Each of these receptors has intrinsic

tyrosine kinase activity like both the EGF and PDGF receptors. As with all transmembrane receptors that have tyrosine kinase activity, autophosphorylation of the receptor is the immediate response to FGF binding. Following activation of FGF receptors, numerous signal-transducing proteins associate with the receptor and become tyrosine-phosphorylated. The **Flg** proto-oncogene is a homologue of the FGF receptor family. The FGFR1 receptor also has been shown to be the portal of entry into cells for herpes viruses. FGFs also bind to cell-surface heparan-sulfated proteoglycans with low affinity relative to that of the specific receptors. The purpose in binding of FGFs to these proteoglycans is not completely understood but may allow the growth factor to remain associated with the extracellular surface of cells that they are intended to stimulate under various conditions.

The FGF receptors are widely expressed in developing bone and several common autosomal dominant disorders of bone growth have been shown to result from mutations in the FGFR genes. The most prevalent is **achondroplasia, ACH**. ACH is characterized by disproportionate short stature, where the limbs are shorter than the trunk, and macrocephaly (excessive head size). Almost all persons with ACH exhibit a glycine to arginine substitution in the transmembrane domain of FGFR3. This mutation results in ligand-independent activation of the receptor. FGFR3 is predominantly expressed in quiescent chondrocytes where it is responsible for restricting chondrocyte proliferation and differentiation. In mice with inactivating mutations in FGFR3 there is an expansion of long bone growth and zones of proliferating cartilage further demonstrating that FGFR3 is necessary to control the rate and amount of chondrocyte growth.

Several other disorders of bone growth collectively identified as **craniosynostosis syndromes** have been shown to result from mutations in FGFR1, FGFR2 and FGFR3. Sometimes the same mutation can cause two or more different craniosynostosis syndromes. A cysteine to tyrosine substitution in FGFR2 can cause either Pfeiffer or Crouzon syndrome. This phenomenon indicates that additional factors are likely responsible for the different phenotypes.

| Affected Receptor | Syndrome        | Phenotypes                            |
|-------------------|-----------------|---------------------------------------|
| FGFR1             | Pfeiffer        | broad first digits, hypertelorism     |
| FGFR2             | Apert           | mid-face hypoplasia, fusion of digits |
| FGFR2             | Beare-Stevenson | mid-face hypoplasia, corrugated skin  |

|       |                                     |                                                                   |
|-------|-------------------------------------|-------------------------------------------------------------------|
| FGFR2 | Crouzon                             | mid-face hypoplasia,<br>ocular proptosis                          |
| FGFR2 | Jackson-Weiss                       | mid-face hypoplasia, foot<br>anomalies                            |
| FGFR2 | Pfeiffer                            | same as for FGFR1<br>mutations                                    |
| FGFR3 | Crouzon                             | mid-face hypoplasia,<br>acanthosis nigricans,<br>ocular proptosis |
| FGFR3 | Non-syndromatic<br>craniosynostosis | digit defects, hearing loss                                       |

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## Transforming Growth Factors- $\beta$ (TGFs- $\beta$ )

A more detailed description of the TGF- $\beta$  family of growth factors and associated signaling pathways can be found on the [Signaling by Wnts and TGFs- \$\beta\$ /BMP](#) page.

TGF- $\beta$  was originally characterized as a protein (secreted from a tumor cell line) that was capable of inducing a transformed phenotype in non-neoplastic cells in culture. This effect was reversible, as demonstrated by the reversion of the cells to a normal phenotype following removal of the TGF- $\beta$ . Subsequently, many proteins homologous to TGF- $\beta$  have been identified. The four closest relatives are TGF- $\beta$ -1 (the original TGF- $\beta$ ) through TGF- $\beta$ -5 (TGF- $\beta$ -1 = TGF- $\beta$ -4). All four of these proteins share extensive regions of similarity in their amino acids. Many other proteins, possessing distinct biological functions, have stretches of amino-acid homology to the TGF- $\beta$  family of proteins, particularly the C-terminal region of these proteins.

The TGF- $\beta$ -related family of proteins includes the activin and inhibin proteins. There are activin A, B and AB proteins, as well as an inhibin A and inhibin B protein. The Mullerian inhibiting substance (MIS) is also a TGF- $\beta$ -related protein, as are members of the bone morphogenetic protein (BMP) family of bone growth-regulatory factors. Indeed, the TGF- $\beta$  family may comprise as many as 100 distinct proteins, all with at least one region of amino-acid sequence homology.

There are several classes of cell-surface receptors that bind different TGFs- $\beta$  with

differing affinities. There also are cell-type specific differences in receptor sub-types. Unlike the EGF, PDGF and FGF receptors, the TGF- $\beta$  family of receptors all have intrinsic serine/threonine kinase activity and, therefore, induce distinct cascades of signal transduction.

TGFs- $\beta$  have proliferative effects on many mesenchymal and epithelial cell types. Under certain conditions TGFs- $\beta$  will demonstrate anti-proliferative effects on endothelial cells, macrophages, and T- and B-lymphocytes. Such effects include decreasing the secretion of immunoglobulin and suppressing hematopoiesis, myogenesis, adipogenesis and adrenal steroidogenesis. Several members of the TGF- $\beta$  family are potent inducers of mesodermal differentiation in early embryos, in particular TGF- $\beta$  and activin A.

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## Transforming Growth Factor- $\alpha$ (TGF- $\alpha$ )

TGF- $\alpha$ , like the  $\beta$  form, was first identified as a substance secreted from certain tumor cells that, in conjunction with TGF- $\beta$ -1, could reversibly transform certain types of normal cells in culture. TGF- $\alpha$  binds to the EGF receptor, as well as its own distinct receptor, and it is this interaction that is thought to be responsible for the growth factor's effect. The predominant sources of TGF- $\alpha$  are carcinomas, but activated macrophages and keratinocytes (and possibly other epithelial cells) also secrete TGF- $\alpha$ . In normal cell populations, TGF- $\alpha$  is a potent keratinocyte growth factor; forming an autocrine growth loop by virtue of the protein activating the very cells that produce it.

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## Erythropoietin (Epo)

Epo is synthesized by the kidney and is the primary regulator of erythropoiesis. Epo stimulates the proliferation and differentiation of immature erythrocytes; it also stimulates the growth of erythroid progenitor cells (e.g. erythrocyte burst-forming and colony-forming units) and induces the differentiation of erythrocyte colony-forming units into proerythroblasts. When patients suffering from anemia due to kidney failure are given Epo, the result is a rapid and significant increase in red blood cell count.

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## Insulin-Like Growth Factor-I (IGF-I)

IGF-I (originally called **somatomedin C**) is a growth factor structurally related to insulin. IGF-I is the primary protein involved in responses of cells to growth hormone (GH): that is, IGF-I is produced in response to GH and then induces subsequent cellular activities, particularly on bone growth. It is the activity of IGF-I in response to GH that gave rise to the term **somatomedin**. Subsequent studies have demonstrated, however, that IGF-I has autocrine

and paracrine activities in addition to the initially observed endocrine activities on bone. The IGF-I receptor, like the insulin receptor, has intrinsic tyrosine kinase activity. Owing to their structural similarities IGF-I can bind to the insulin receptor but does so at a much lower affinity than does insulin itself.

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## Insulin-Like Growth Factor-II (IGF-II)

IGF-II is almost exclusively expressed in embryonic and neonatal tissues. Following birth, the level of detectable IGF-II protein falls significantly. For this reason IGF-II is thought to be a fetal growth factor. The IGF-II receptor is identical to the mannose-6-phosphate receptor that is responsible for the integration of lysosomal enzymes (which contain mannose-6-phosphate residues) to the lysosomes.

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## Interleukin-1 (IL-1)

IL-1 is one of the most important immune response-- modifying interleukins. The predominant function of IL-1 is to enhance the activation of T-cells in response to antigen. The activation of T-cells, by IL-1, leads to increased T-cell production of IL-2 and of the IL-2 receptor, which in turn augments the activation of the T-cells in an autocrine loop. IL-1 also induces expression of interferon- $\gamma$  (IFN- $\gamma$ ) by T-cells. This effect of T-cell activation by IL-1 is mimicked by TNF- $\alpha$  which is another cytokine secreted by activated macrophages. There are 2 distinct IL-1 proteins, termed IL-1- $\alpha$  and -1- $\beta$ , that are 26% homologous at the amino acid level. The IL-1s are secreted primarily by macrophages but also from neutrophils, endothelial cells, smooth muscle cells, glial cells, astrocytes, B- and T-cells, fibroblasts and keratinocytes. Production of IL-1 by these different cell types occurs only in response to cellular stimulation. In addition to its effects on T-cells, IL-1 can induce proliferation in non-lymphoid cells.

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## Interleukin-2 (IL-2)

IL-2, produced and secreted by activated T-cells, is the major interleukin responsible for clonal T-cell proliferation. IL-2 also exerts effects on B-cells, macrophages, and natural killer (NK) cells. The production of IL-2 occurs primarily by CD4+ T-helper cells. As indicated above, the expression of both IL-2 and the IL-2 receptor by T-cells is induced by IL-1. Indeed, the IL-2 receptor is not expressed on the surface of resting T-cells and is present only transiently on the surface of T-cells, disappearing within 6-10 days of antigen presentation. In contrast to T-helper cells, NK cells constitutively express IL-2 receptors and will secrete TNF- $\alpha$ , IFN- $\gamma$  and GM-CSF in response to IL-2, which in turn activate macrophages.

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## Interleukin-6 (IL-6)

IL-6 is produced by macrophages, fibroblasts, endothelial cells and activated T-helper cells. IL-6 acts in synergy with IL-1 and TNF- $\alpha$  in many immune responses, including T-cell activation. In particular, IL-6 is the primary inducer of the **acute-phase response** in liver. IL-6 also enhances the differentiation of B-cells and their consequent production of immunoglobulin. Glucocorticoid synthesis is also enhanced by IL-6. Unlike IL-1, IL-2 and TNF- $\alpha$ , IL-6 does not induce cytokine expression; its main effects, therefore, are to augment the responses of immune cells to other cytokines.

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## Interleukin-8 (IL-8)

IL-8 is an interleukin that belongs to an ever-expanding family of proteins that exert chemoattractant activity to leukocytes and fibroblasts. This family of proteins is termed the **chemokines**. IL-8 is produced by monocytes, neutrophils, and NK cells and is chemoattractant for neutrophils, basophils and T-cells. In addition, IL-8 activates neutrophils to degranulate.

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## Tumor Necrosis Factor- $\alpha$ (TNF- $\alpha$ )

TNF- $\alpha$  (also called **cachectin**), like IL-1 is a major immune response-- modifying cytokine produced primarily by activated macrophages. Like IL-1, TNF- $\alpha$  induces the expression of other autocrine growth factors, increases cellular responsiveness to growth factors and induces signaling pathways that lead to proliferation. TNF- $\alpha$  acts synergistically with EGF and PDGF on some cell types. Like other growth factors, TNF- $\alpha$  induces expression of a number of nuclear proto-oncogenes as well as of several interleukins.

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## Tumor Necrosis Factor- $\beta$ (TNF- $\beta$ )

TNF- $\beta$  (also called **lymphotoxin**) is characterized by its ability to kill a number of different cell types, as well as the ability to induce terminal differentiation in others. One significant non-proliferative response to TNF- $\beta$  is an inhibition of lipoprotein lipase present on the surface of vascular endothelial cells. The predominant site of TNF- $\beta$  synthesis is T-lymphocytes, in particular the special class of T-cells called cytotoxic T-lymphocytes (CTL cells). The induction of TNF- $\beta$  expression results from elevations in IL-2 as well as the interaction of antigen with T-cell receptors.

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## Interferon- $\gamma$ (INF- $\gamma$ )

IFN- $\alpha$ , IFN- $\beta$  and IFN- $\omega$  are known as **type I interferons**: they are predominantly responsible for the antiviral activities of the interferons. In contrast, IFN- $\gamma$  is a **type II** or **immune interferon**. Although IFN- $\gamma$  has antiviral activity it is significantly less active at this function than the type I IFNs. Unlike the type I IFNs, IFN- $\gamma$  is not induced by infection nor by double-stranded RNAs. IFN- $\gamma$  is secreted primarily by CD8<sup>+</sup> T-cells. Nearly all cells express receptors for IFN- $\gamma$  and respond to IFN- $\gamma$  binding by increasing the surface expression of class I MHC proteins, thereby promoting the presentation of antigen to T-helper (CD4<sup>+</sup>) cells. IFN- $\gamma$  also increases the presentation of class II MHC proteins on class II cells further enhancing the ability of cells to present antigen to T-cells.

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## Colony Stimulating Factors (CSFs)

CSFs are cytokines that stimulate the proliferation of specific pluripotent stem cells of the bone marrow in adults. Granulocyte-CSF (**G-CSF**) is specific for proliferative effects on cells of the granulocyte lineage. Macrophage-CSF (**M-CSF**) is specific for cells of the macrophage lineage. Granulocyte-macrophage-CSF (**GM-CSF**) has proliferative effects on both classes of lymphoid cells. Epo is also considered a CSF as well as a growth factor, since it stimulates the proliferation of erythrocyte colony-forming units. **IL-3** (secreted primarily from T-cells) is also known as **multi-CSF**, since it stimulates stem cells to produce all forms of hematopoietic cells.

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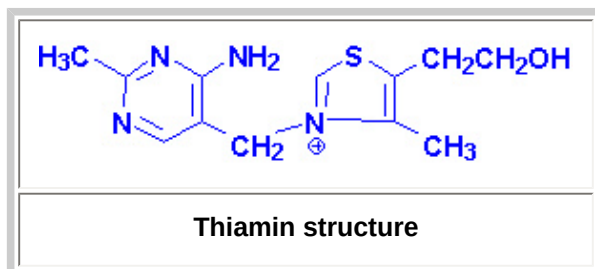
## Introduction to Vitamins

Vitamins are organic molecules that function in a wide variety of capacities within the body. The most prominent function is as **cofactors** for enzymatic reactions. The distinguishing feature of the vitamins is that they generally cannot be synthesized by mammalian cells and, therefore, must be supplied in the diet. The vitamins are of two distinct types:

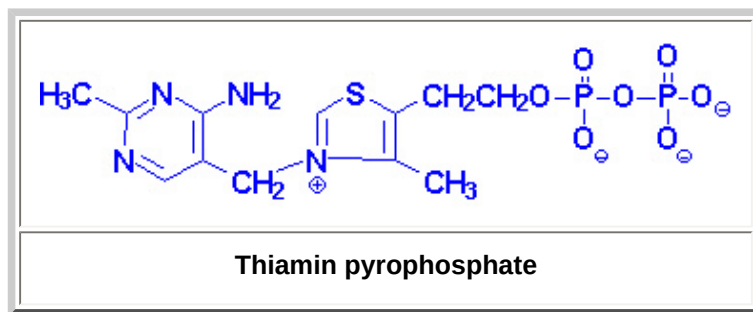
| Water Soluble Vitamins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fat Soluble Vitamins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <ul style="list-style-type: none"> <li>• <a href="#">Thiamin (B<sub>1</sub>)</a> <ul style="list-style-type: none"> <li>◦ <a href="#">B<sub>1</sub> Deficiency and Disease</a></li> </ul> </li> <li>• <a href="#">Riboflavin (B<sub>2</sub>)</a> <ul style="list-style-type: none"> <li>◦ <a href="#">B<sub>2</sub> Deficiency and Disease</a></li> </ul> </li> <li>• <a href="#">Niacin (B<sub>3</sub>)</a> <ul style="list-style-type: none"> <li>◦ <a href="#">B<sub>3</sub> Deficiency and Disease</a></li> </ul> </li> <li>• <a href="#">Pantothenic Acid (B<sub>5</sub>)</a></li> <li>• <a href="#">Pyridoxal, Pyridoxamine, Pyridoxine (B<sub>6</sub>)</a></li> <li>• <a href="#">Biotin</a></li> <li>• <a href="#">Cobalamin (B<sub>12</sub>)</a> <ul style="list-style-type: none"> <li>◦ <a href="#">B<sub>12</sub> Deficiency and Disease</a></li> </ul> </li> <li>• <a href="#">Folic Acid</a> <ul style="list-style-type: none"> <li>◦ <a href="#">Folate Deficiency and Disease</a></li> </ul> </li> <li>• <a href="#">Ascorbic Acid</a></li> </ul> | <ul style="list-style-type: none"> <li>• <a href="#">Vitamin A</a> <ul style="list-style-type: none"> <li>◦ <a href="#">Gene Control by Vitamin A</a></li> <li>◦ <a href="#">Role of Vitamin A in Vision</a></li> <li>◦ <a href="#">Additional Roles of Vitamin A</a></li> <li>◦ <a href="#">Clinical Significances of Vitamin A</a></li> </ul> </li> <li>• <a href="#">Vitamin D</a> <ul style="list-style-type: none"> <li>◦ <a href="#">Clinical Significances of Vitamin D</a></li> </ul> </li> <li>• <a href="#">Vitamin E</a> <ul style="list-style-type: none"> <li>◦ <a href="#">Clinical Significances of Vitamin E</a></li> </ul> </li> <li>• <a href="#">Vitamin K</a> <ul style="list-style-type: none"> <li>◦ <a href="#">Clinical Significance of Vitamin K</a></li> </ul> </li> </ul> |

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## Thiamin



Thiamin is also known as **vitamin B<sub>1</sub>**. Thiamin is derived from a substituted pyrimidine and a thiazole which are coupled by a methylene bridge. Thiamin is rapidly converted to its active form, **thiamin pyrophosphate, TPP**, in the brain and liver by a specific enzymes, thiamin diphosphotransferase.



TPP is necessary as a cofactor for the pyruvate and  $\alpha$ -ketoglutarate dehydrogenase catalyzed reactions as well as the transketolase catalyzed reactions of the pentose phosphate pathway. A deficiency in thiamin intake leads to a severely reduced capacity of cells to generate energy as a result of its role in these reactions.

The dietary requirement for thiamin is proportional to the caloric intake of the diet and ranges from 1.0 - 1.5 mg/day for normal adults. If the carbohydrate content of the diet is excessive then an in thiamin intake will be required.

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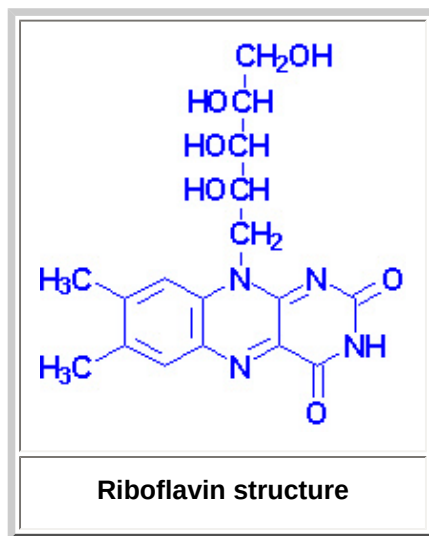
## Clinical Significances of Thiamin Deficiency

The earliest symptoms of thiamin deficiency include constipation, appetite suppression, nausea as well as mental depression, peripheral neuropathy and fatigue. Chronic thiamin deficiency leads to more severe neurological symptoms including ataxia, mental confusion and loss of eye coordination. Other clinical symptoms of prolonged thiamin deficiency are related to cardiovascular and musculature defects.

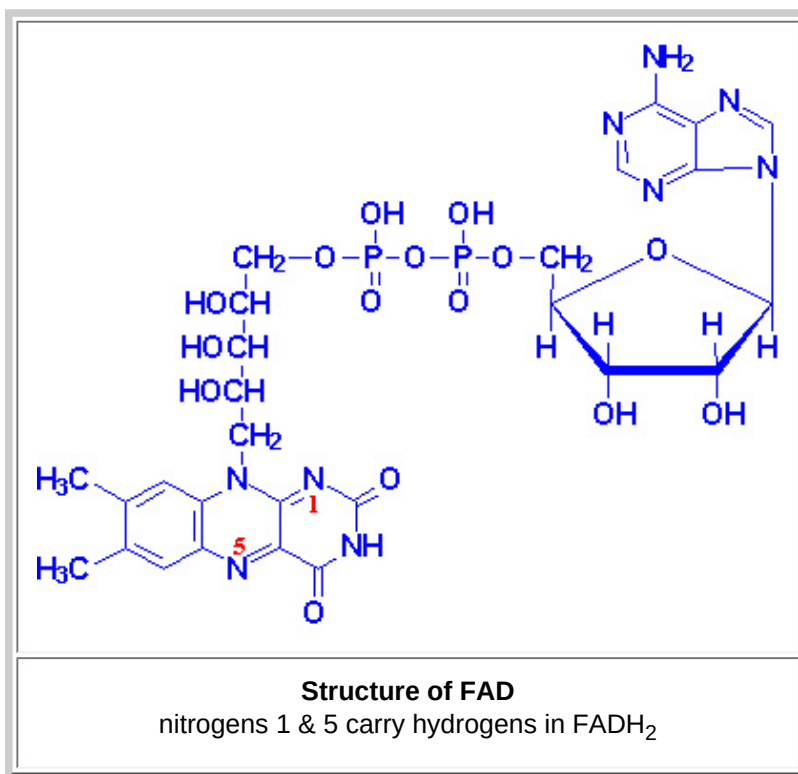
The severe thiamin deficiency disease known as **Beriberi**, is the result of a diet that is carbohydrate rich and thiamin deficient. An additional thiamin deficiency related disease is known as **Wernicke-Korsakoff syndrome**. This disease is most commonly found in chronic alcoholics due to their poor dietetic lifestyles.

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## Riboflavin



Riboflavin is also known as **vitamin B<sub>2</sub>**. Riboflavin is the precursor for the coenzymes, **flavin mononucleotide (FMN)** and **flavin adenine dinucleotide (FAD)**. The enzymes that require FMN or FAD as cofactors are termed flavoproteins. Several flavoproteins also contain metal ions and are termed metalloflavoproteins. Both classes of enzymes are involved in a wide range of redox reactions, e.g. succinate dehydrogenase and xanthine oxidase. During the course of the enzymatic reactions involving the flavoproteins the reduced forms of FMN and FAD are formed, FMNH<sub>2</sub> and FADH<sub>2</sub>, respectively.



The normal daily requirement for riboflavin is 1.2 - 1.7 mg/day for normal adults.

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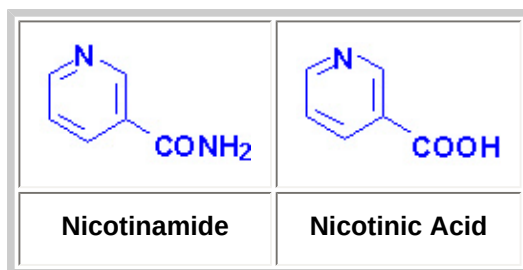
## Clinical Significances of Flavin Deficiency

Riboflavin deficiencies are rare in the United States due to the presence of adequate amounts of the vitamin in eggs, milk, meat and cereals. Riboflavin deficiency is often seen in chronic alcoholics due to their poor dietetic habits.

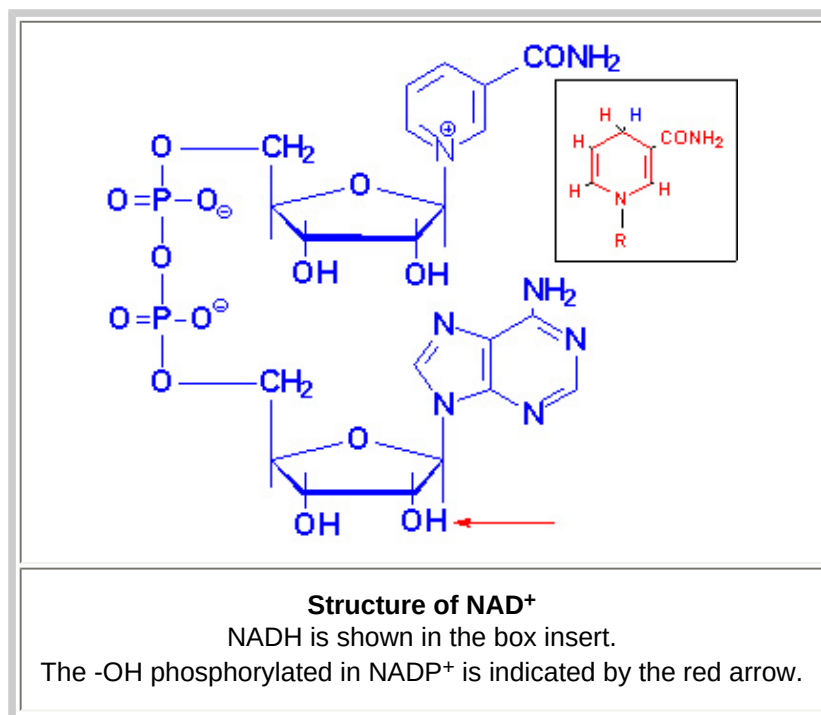
Symptoms associated with riboflavin deficiency include, glossitis, seborrhea, angular stomatitis, cheilosis and photophobia. Riboflavin decomposes when exposed to visible light. This characteristic can lead to riboflavin deficiencies in newborns treated for hyperbilirubinemia by phototherapy.

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## Niacin



Niacin (nicotinic acid and nicotinamide) is also known as **vitamin B<sub>3</sub>**. Both nicotinic acid and nicotinamide can serve as the dietary source of vitamin B<sub>3</sub>. Niacin is required for the synthesis of the active forms of vitamin B<sub>3</sub>, **nicotinamide adenine dinucleotide (NAD<sup>+</sup>)** and **nicotinamide adenine dinucleotide phosphate (NADP<sup>+</sup>)**. Both NAD<sup>+</sup> and NADP<sup>+</sup> function as cofactors for numerous dehydrogenase, e.g., lactate and malate dehydrogenases.



Niacin is not a true vitamin in the strictest definition since it can be derived from the amino acid tryptophan. However, the ability to utilize tryptophan for niacin synthesis is inefficient (60 mg of tryptophan are required to synthesize 1 mg of niacin). Also, synthesis of niacin from tryptophan requires vitamins B<sub>1</sub>, B<sub>2</sub> and B<sub>6</sub> which would be limiting in themselves on a marginal diet.

The recommended daily requirement for niacin is 13 - 19 niacin equivalents (NE) per day for a normal adult. One NE is equivalent to 1 mg of free niacin).

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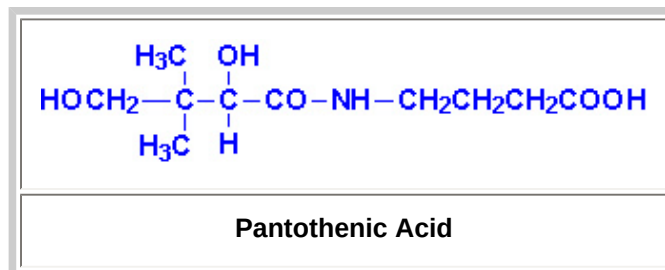
## Clinical Significances of Niacin and Nicotinic Acid

A diet deficient in niacin (as well as tryptophan) leads to glossitis of the tongue, dermatitis, weight loss, diarrhea, depression and dementia. The severe symptoms, depression, dermatitis and diarrhea, are associated with the condition known as **pellagra**. Several physiological conditions (e.g. **Hartnup disease** and **malignant carcinoid syndrome**) as well as certain drug therapies (e.g. isoniazid) can lead to niacin deficiency. In Hartnup disease tryptophan absorption is impaired and in malignant carcinoid syndrome tryptophan metabolism is altered resulting in excess serotonin synthesis. Isoniazid (the hydrazide derivative of isonicotinic acid) is the primary drug for chemotherapy of tuberculosis.

Nicotinic acid (but not nicotinamide) when administered in pharmacological doses of 2 - 4 g/day lowers plasma cholesterol levels and has been shown to be a useful therapeutic for **hypercholesterolemia**. The major action of nicotinic acid in this capacity is a reduction in fatty acid mobilization from adipose tissue. Although nicotinic acid therapy lowers blood cholesterol it also causes a depletion of glycogen stores and fat reserves in skeletal and cardiac muscle. Additionally, there is an elevation in blood glucose and uric acid production. For these reasons nicotinic acid therapy is not recommended for diabetics or persons who suffer from **gout**.

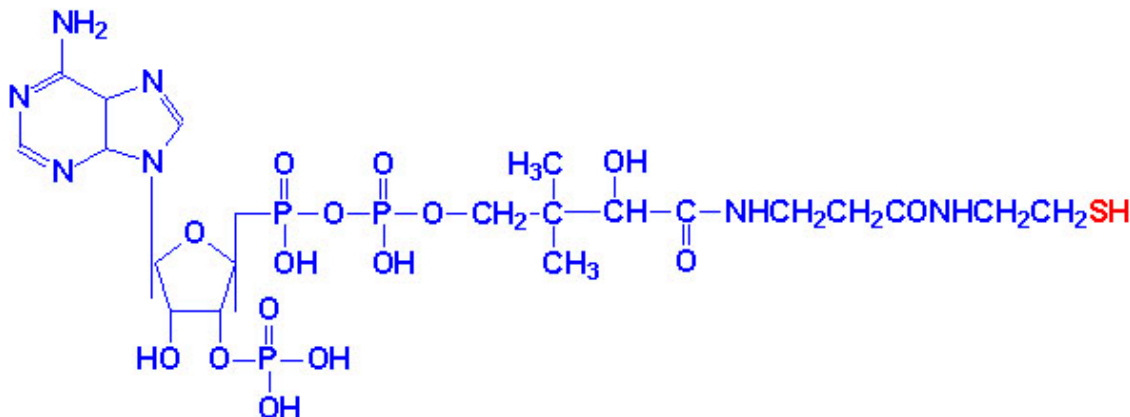
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## Pantothenic Acid



Pantothenic acid is also known as **vitamin B<sub>5</sub>**. Pantothenic acid is formed from β-alanine and pantoic acid. Pantothenate is required for synthesis of coenzyme A, CoA and is a component of the acyl carrier protein (ACP) domain of fatty acid synthase. Pantothenate is, therefore, required for the metabolism of carbohydrate via the TCA cycle and all fats and proteins. At least 70 enzymes have been identified as requiring CoA or ACP derivatives for their function.

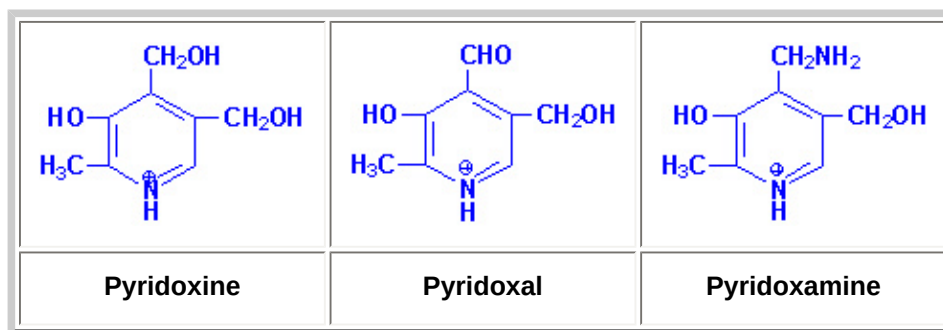
Deficiency of pantothenic acid is extremely rare due to its widespread distribution in whole grain cereals, legumes and meat. Symptoms of pantothenate deficiency are difficult to assess since they are subtle and resemble those of other B vitamin deficiencies.



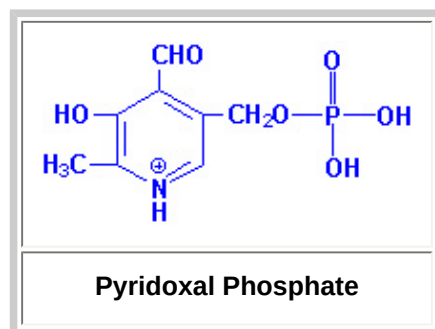
Coenzyme A

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## Vitamin B<sub>6</sub>



**Pyridoxal, pyridoxamine** and **pyridoxine** are collectively known as **vitamin B<sub>6</sub>**. All three compounds are efficiently converted to the biologically active form of vitamin B<sub>6</sub>, **pyridoxal phosphate**. This conversion is catalyzed by the ATP requiring enzyme, pyridoxal kinase.



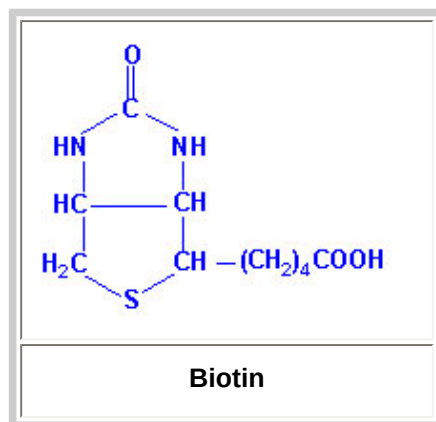
Pyridoxal phosphate functions as a cofactor in enzymes involved in transamination reactions required for the synthesis and catabolism of the amino acids as well as in glycogenolysis as a cofactor for glycogen phosphorylase. The requirement for vitamin B<sub>6</sub> in the diet is proportional to the level of protein consumption ranging from 1.4 - 2.0 mg/day for a normal adult. During pregnancy and lactation the requirement for vitamin B<sub>6</sub> increases approximately 0.6 mg/day.



Deficiencies of vitamin B<sub>6</sub> are rare and usually are related to an overall deficiency of all the B-complex vitamins. Isoniazid (see niacin deficiencies above) and penicillamine (used to treat rheumatoid arthritis and cystinurias) are two drugs that complex with pyridoxal and pyridoxal phosphate resulting in a deficiency in this vitamin.

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## Biotin



Biotin is the cofactor required of enzymes that are involved in carboxylation reactions, e.g. acetyl-CoA carboxylase and pyruvate carboxylase. Biotin is found in numerous foods and also is synthesized by intestinal bacteria and as such deficiencies of the vitamin are rare. Deficiencies are generally seen only after long antibiotic therapies which deplete the intestinal fauna or following excessive consumption of raw eggs. The latter is due to the affinity of the egg white protein, **avidin**, for biotin preventing intestinal absorption of the biotin.

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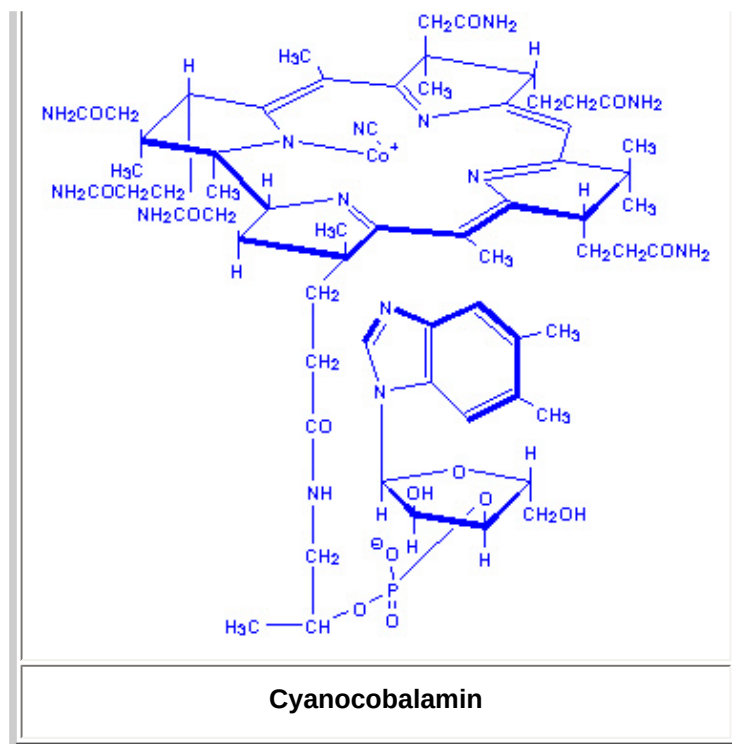
## Cobalamin

Cobalamin is more commonly known as **vitamin B<sub>12</sub>**. Vitamin B<sub>12</sub> is composed of a complex tetrapyrrol ring structure (corrin ring) and a cobalt ion in the center. Vitamin B<sub>12</sub> is synthesized exclusively by microorganisms and is found in the liver of animals bound to protein as methylcobalamin or 5'-deoxyadenosylcobalamin. The vitamin must be hydrolyzed from protein in order to be active. Hydrolysis occurs in the stomach by gastric acids or the intestines by trypsin digestion following consumption of animal meat. The vitamin is then bound by **intrinsic factor**, a protein secreted by parietal cells of the stomach, and carried to the ileum where it is absorbed. Following absorption the vitamin is transported to the liver in the blood bound to **transcobalamin II**.

There are only two clinically significant reactions in the body that require vitamin B<sub>12</sub> as a cofactor. During the catabolism of fatty acids with an odd number of carbon atoms and the amino acids valine, isoleucine and threonine the resultant propionyl-CoA is converted to succinyl-CoA for oxidation in the TCA cycle. One of the enzymes in this pathway, methylmalonyl-CoA mutase, requires vitamin B<sub>12</sub> as a cofactor in the conversion of methylmalonyl-CoA to succinyl-CoA. The 5'-deoxyadenosine derivative of cobalamin is required for this reaction.

The second reaction requiring vitamin B<sub>12</sub> catalyzes the conversion of homocysteine to methionine and is catalyzed by methionine synthase. This reaction results in the transfer of the methyl group from N<sup>5</sup>-methyltetrahydrofolate to hydroxycobalamin generating tetrahydrofolate (THF) and methylcobalamin during the process of the conversion.





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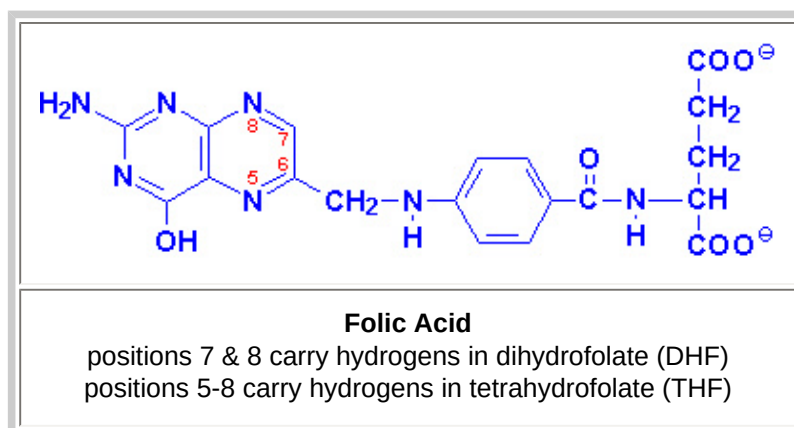
## Clinical Significances of B<sub>12</sub> Deficiency

The liver can store up to six years worth of vitamin B<sub>12</sub>, hence deficiencies in this vitamin are rare. [Pernicious anemia](#) is a megaloblastic anemia resulting from vitamin B<sub>12</sub> deficiency that develops as a result a lack of intrinsic factor in the stomach leading to malabsorption of the vitamin. The anemia results from impaired DNA synthesis due to a block in [purine and thymidine biosynthesis](#). The block in nucleotide biosynthesis is a consequence of the effect of vitamin B<sub>12</sub> on folate metabolism. When vitamin B<sub>12</sub> is deficient essentially all of the folate becomes trapped as the N<sup>5</sup>-methylTHF derivative as a result of the loss of functional methionine synthase. This trapping prevents the synthesis of other THF derivatives required for the purine and thymidine nucleotide biosynthesis pathways.

Neurological complications also are associated with vitamin B<sub>12</sub> deficiency and result from a progressive demyelination of nerve cells. The demyelination is thought to result from the increase in methylmalonyl-CoA that result from vitamin B<sub>12</sub> deficiency. Methylmalonyl-CoA is a competitive inhibitor of malonyl-CoA in fatty acid biosynthesis as well as being able to substitute for malonyl-CoA in any fatty acid biosynthesis that may occur. Since the myelin sheath is in continual flux the methylmalonyl-CoA-induced inhibition of fatty acid synthesis results in the eventual destruction of the sheath. The incorporation methylmalonyl-CoA into fatty acid biosynthesis results in branched-chain fatty acids being produced that may severely alter the architecture of the normal membrane structure of nerve cells

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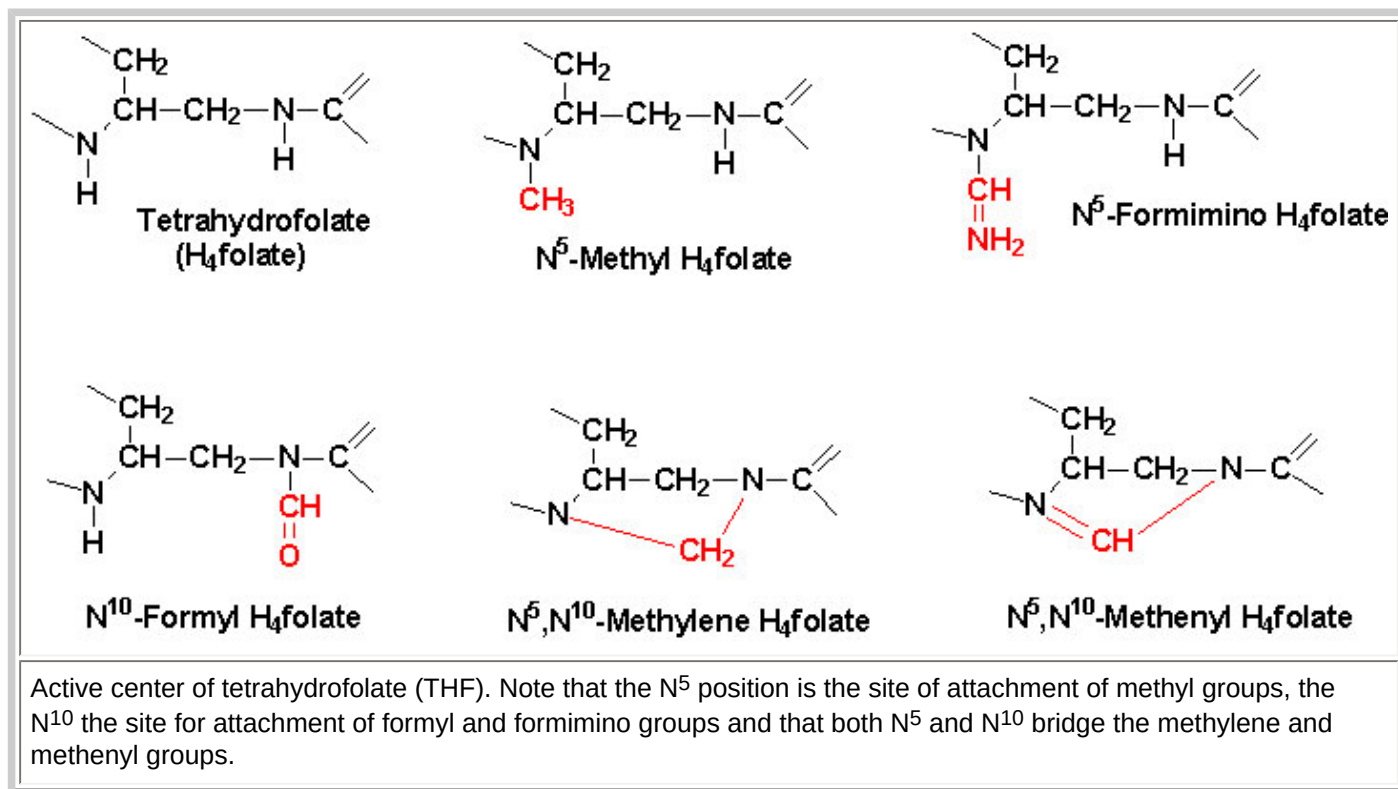
## Folic Acid



Folic acid is a conjugated molecule consisting of a pteridine ring structure linked to para-aminobenzoic acid (**PABA**) that forms pteric acid. Folic acid itself is then generated through the conjugation of glutamic acid residues to pteric acid. Folic acid is obtained primarily from yeasts and leafy vegetables as well as animal liver. Animal cannot synthesize PABA nor attach glutamate residues to pteric acid, thus, requiring folate intake in the diet.

When stored in the liver or ingested folic acid exists in a polyglutamate form. Intestinal mucosal cells remove some of the glutamate residues through the action of the lysosomal enzyme, conjugase. The removal of glutamate residues makes folate less negatively charged (from the polyglutamic acids) and therefore more capable of passing through the basal laminal membrane of the epithelial cells of the intestine and into the bloodstream. Folic acid is reduced within cells (principally the liver where it is stored) to tetrahydrofolate (THF also H<sub>4</sub>folate) through the action of dihydrofolate reductase (DHFR), an NADPH-requiring enzyme.

The function of THF derivatives is to carry and transfer various forms of one carbon units during biosynthetic reactions. The one carbon units are either methyl, methylene, methenyl, formyl or formimino groups.



These one carbon transfer reactions are required in the biosynthesis of serine, methionine, glycine, choline and the purine nucleotides and dTMP.

The ability to acquire choline and amino acids from the diet and to salvage the purine nucleotides makes the role of N<sup>5</sup>,N<sup>10</sup>-methylene-THF in dTMP synthesis the most metabolically significant function for this vitamin. The role of vitamin B<sub>12</sub> and N<sup>5</sup>-methyl-THF in the conversion of homocysteine to methionine also can have a significant impact on the ability of cells to regenerate needed THF.

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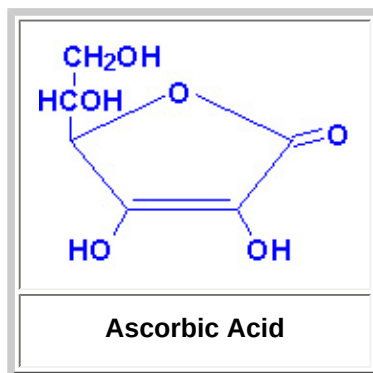
## Clinical Significance of Folate Deficiency

Folate deficiency results in complications nearly identical to those described for vitamin B<sub>12</sub> deficiency. The most pronounced effect of folate deficiency on cellular processes is upon DNA synthesis. This is due to an impairment in dTMP synthesis which leads to cell cycle arrest in S-phase of rapidly proliferating cells, in particular hematopoietic cells. The result is **megaloblastic anemia** as for vitamin B<sub>12</sub> deficiency. The inability to synthesize DNA during erythrocyte maturation leads to abnormally large erythrocytes termed **macrocytic anemia**.

Folate deficiencies are rare due to the adequate presence of folate in food. Poor dietary habits as those of chronic alcoholics can lead to folate deficiency. The predominant causes of folate deficiency in non-alcoholics are impaired absorption or metabolism or an increased demand for the vitamin. The predominant condition requiring an increase in the daily intake of folate is pregnancy. This is due to an increased number of rapidly proliferating cells present in the blood. The need for folate will nearly double by the third trimester of pregnancy. Certain drugs such as anticonvulsants and oral contraceptives can impair the absorption of folate. Anticonvulsants also increase the rate of folate metabolism.

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## Ascorbic Acid



Ascorbic acid is more commonly known as **vitamin C**. Ascorbic acid is derived from glucose via the uronic acid pathway. The enzyme L-gulonolactone oxidase responsible for the conversion of gulonolactone to ascorbic acid is absent in primates making ascorbic acid required in the diet.

The active form of vitamin C is ascorbate acid itself. The main function of ascorbate is as a reducing agent in a number of different reactions. Vitamin C has the potential to reduce cytochromes a and c of the respiratory chain as well as molecular oxygen. The most important reaction requiring ascorbate as a cofactor is the hydroxylation of proline residues in collagen. Vitamin C is, therefore, required for the maintenance of normal connective tissue as well as for wound healing since synthesis of connective tissue is the first event in wound tissue remodeling. Vitamin C also is necessary for bone remodeling due to the presence of collagen in the organic matrix of bones.

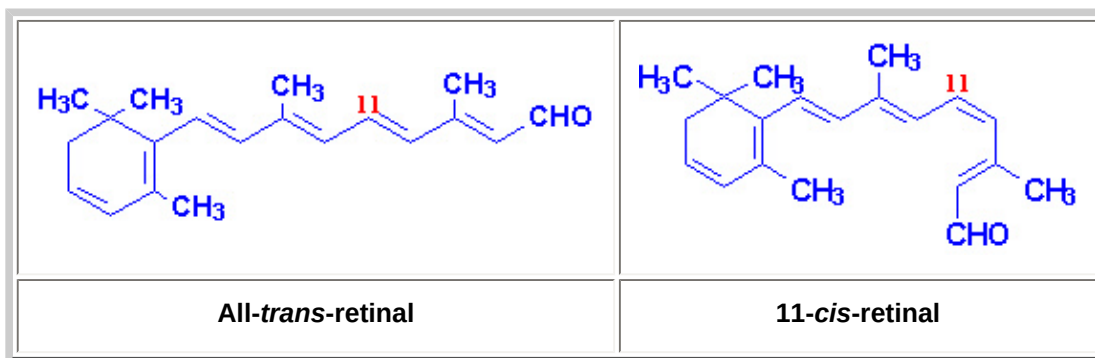
Several other metabolic reactions require vitamin C as a cofactor. These include the catabolism of tyrosine and the synthesis of epinephrine from tyrosine and the synthesis of the bile acids. It is also believed that vitamin C is involved in the process of steroidogenesis since the adrenal cortex contains high levels of vitamin C which are depleted upon adrenocorticotrophic hormone (ACTH) stimulation of the gland.

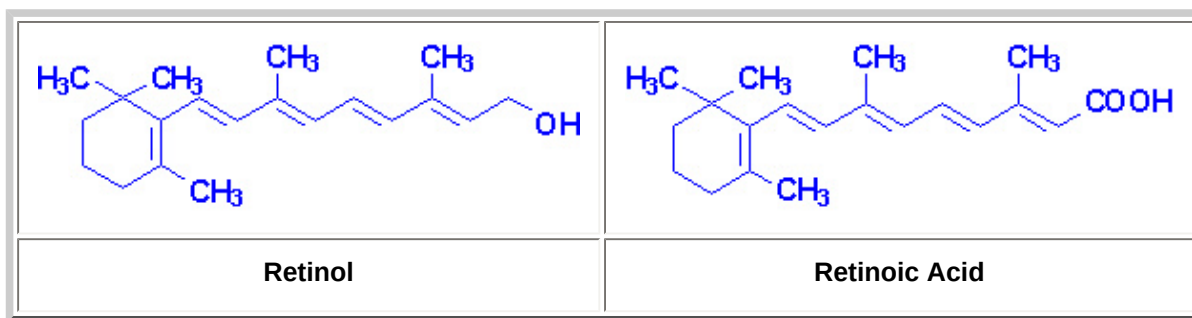
Deficiency in vitamin C leads to the disease **scurvy** due to the role of the vitamin in the post-translational modification of collagens. Scurvy is characterized by easily bruised skin, muscle fatigue, soft swollen gums, decreased wound healing and hemorrhaging, osteoporosis, and anemia. Vitamin C is readily absorbed and so the primary cause of vitamin C deficiency is poor diet and/or an increased requirement. The primary physiological state leading to an increased requirement for vitamin C is severe stress (or trauma). This is due to a rapid depletion in the adrenal stores of the vitamin. The reason for the decrease in adrenal vitamin C levels is unclear but may be due either to redistribution of the vitamin to areas that need it or an overall increased utilization.

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## Vitamin A

Vitamin A consists of three biologically active molecules, **retinol**, **retinal** (retinaldehyde) and **retinoic acid**.





Each of these compounds are derived from the plant precursor molecule,  **$\beta$ -carotene** (a member of a family of molecules known as **carotenoids**). Beta-carotene, which consists of two molecules of retinal linked at their aldehyde ends, is also referred to as the provitamin form of vitamin A.

Ingested  $\beta$ -carotene is cleaved in the lumen of the intestine by  $\beta$ -carotene dioxygenase to yield retinal. Retinal is reduced to retinol by retinaldehyde reductase, an NADPH requiring enzyme within the intestines. Retinol is esterified to palmitic acid and delivered to the blood via chylomicrons. The uptake of chylomicron remnants by the liver results in delivery of retinol to this organ for storage as a lipid ester within lipocytes. Transport of retinol from the liver to extrahepatic tissues occurs by binding of hydrolyzed retinol to **aporetinol binding protein (RBP)**. the retinol-RBP complex is then transported to the cell surface within the Golgi and secreted. Within extrahepatic tissues retinol is bound to **cellular retinol binding protein (CRBP)**. Plasma transport of retinoic acid is accomplished by binding to albumin.

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## Gene Control Exerted by Retinol and Retinoic Acid

Within cells both retinol and retinoic acid bind to specific receptor proteins. Following binding, the receptor-vitamin complex interacts with specific sequences in several genes involved in growth and differentiation and affects expression of these genes. In this capacity retinol and retinoic acid are considered hormones of the steroid/thyroid hormone superfamily of proteins. Vitamin D also acts in a similar capacity. Several genes whose patterns of expression are altered by retinoic acid are involved in the earliest processes of embryogenesis including the differentiation of the three germ layers, organogenesis and limb development.

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## Vision and the Role of Vitamin A

Photoreception in the eye is the function of two specialized cell types located in the retina; the rod and cone cells. Both rod and cone cells contain a photoreceptor pigment in their membranes. The photosensitive compound of most mammalian eyes is a protein called **opsin** to which is covalently coupled an aldehyde of vitamin A. The opsin of rod cells is called **scotopsin**. The photoreceptor of rod cells is specifically called **rhodopsin** or **visual purple**. This compound is a complex between scotopsin and the 11-*cis*-retinal (also called 11-*cis*-retinene) form of vitamin A. Rhodopsin is a serpentine receptor imbedded in the membrane of the rod cell. Coupling of 11-*cis*-retinal occurs at three of the transmembrane domains of rhodopsin. Intracellularly, rhodopsin is coupled to a specific G-protein called **transducin**.

When the rhodopsin is exposed to light it is **bleached** releasing the 11-*cis*-retinal from opsin. Absorption of photons by 11-*cis*-retinal triggers a series of conformational changes on the way to conversion **all-trans-retinal**. One important conformational intermediate is **metarhodopsin II**. The release of opsin results in a conformational change in the photoreceptor. This conformational change activates transducin, leading to an increased GTP-binding by the  $\alpha$ -subunit of transducin. Binding of GTP releases the  $\alpha$ -subunit from the inhibitory  $\beta$ - and  $\gamma$ -subunits. The GTP-activated  $\alpha$ -subunit in turn activates an associated phosphodiesterase; an enzyme that hydrolyzes cyclic-GMP (cGMP) to GMP. Cyclic GMP is required to maintain the  $\text{Na}^+$  channels of the rod cell in the open conformation. The drop in cGMP concentration results in complete closure of the  $\text{Na}^+$  channels. Metarhodopsin II appears to be responsible for initiating the closure of the channels. The closing of the channels leads to hyperpolarization of the rod cell with concomitant propagation of nerve impulses to the brain.

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## Additional Role of Retinol

Retinol also functions in the synthesis of certain glycoproteins and mucopolysaccharides necessary for mucous production and normal growth regulation. This is accomplished by phosphorylation of retinol to **retinyl phosphate** which then functions similarly to dolichol phosphate.

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## Clinical Significances of Vitamin A Deficiency

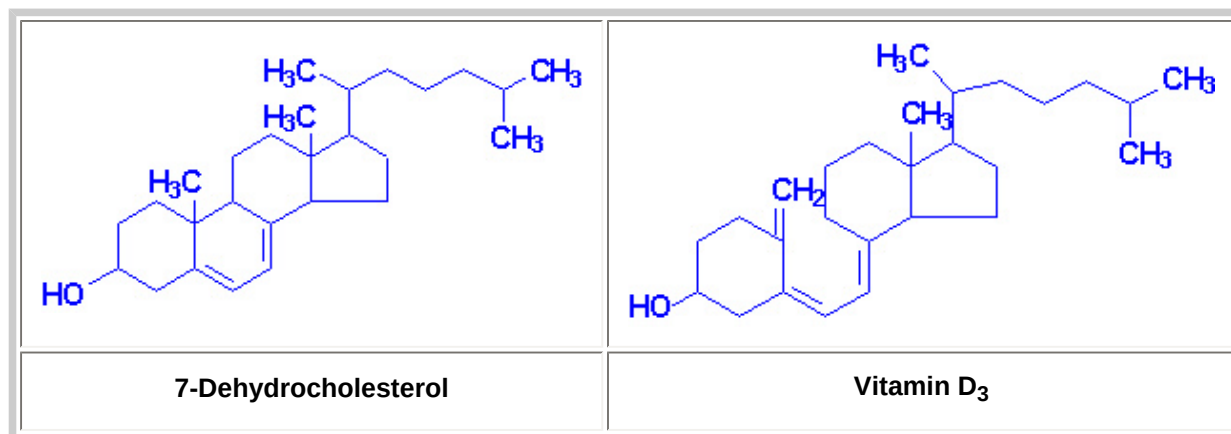
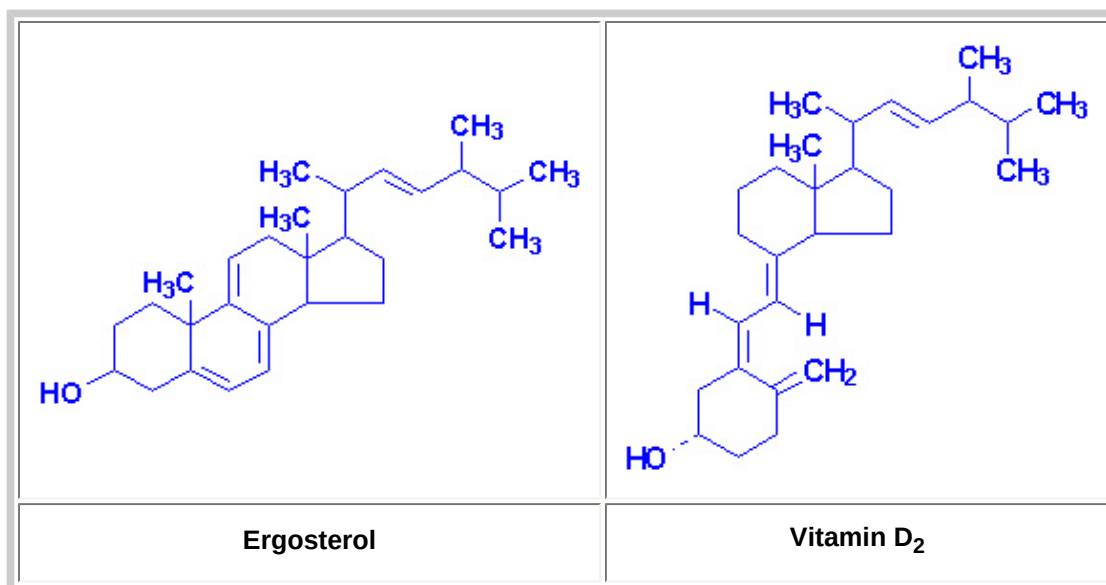
Vitamin A is stored in the liver and deficiency of the vitamin occurs only after prolonged lack of dietary intake. The earliest symptoms of vitamin A deficiency are **night blindness**. Additional early symptoms include follicular hyperkeratinosis, increased susceptibility to infection and cancer and anemia equivalent to iron deficient anemia. Prolonged lack of vitamin A leads to deterioration of the eye tissue through progressive keratinization of the cornea, a condition known as xerophthalmia.

The increased risk of cancer in vitamin deficiency is thought to be the result of a depletion in  $\beta$ -carotene. Beta-carotene is a very effective antioxidant and is suspected to reduce the risk of cancers known to be initiated by the production of free radicals. Of particular interest is the potential benefit of increased  $\beta$ -carotene intake to reduce the risk of lung cancer in smokers. However, caution needs to be taken when increasing the intake of any of the lipid soluble vitamins. Excess accumulation of vitamin A in the liver can lead to toxicity which manifests as bone pain, hepatosplenomegaly, nausea and diarrhea.

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## Vitamin D

Vitamin D is a steroid hormone that functions to regulate specific gene expression following interaction with its intracellular receptor. The biologically active form of the hormone is **1,25-dihydroxy vitamin D<sub>3</sub>** (1,25-(OH)<sub>2</sub>D<sub>3</sub>, also termed **calcitriol**). Calcitriol functions primarily to regulate calcium and phosphorous homeostasis.

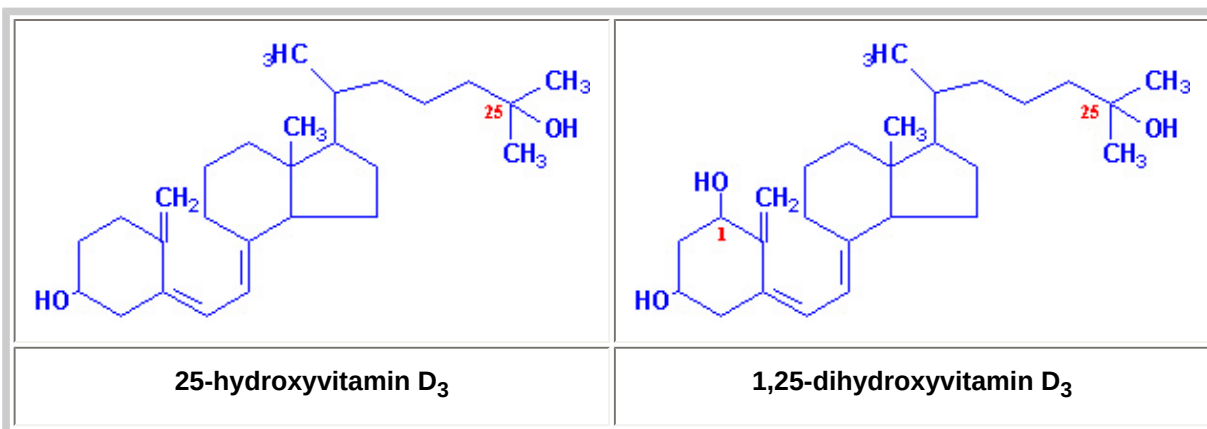


Active calcitriol is derived from **ergosterol** (produced in plants) and from **7-dehydrocholesterol** (produced in the skin). **Ergocalciferol** (vitamin D<sub>2</sub>) is formed by uv irradiation of ergosterol. In the skin 7-dehydrocholesterol is converted to **cholecalciferol** (vitamin D<sub>3</sub>) following uv irradiation.

Vitamin D<sub>2</sub> and D<sub>3</sub> are processed to D<sub>2</sub>-calcitriol and D<sub>3</sub>-calcitriol, respectively, by the same enzymatic pathways in the body. Cholecalciferol (or ergocalciferol) are absorbed from the intestine and transported to the liver bound to a specific **vitamin D-binding**



**protein.** In the liver cholecalciferol is hydroxylated at the 25 position by a specific  $D_3$ -25-hydroxylase generating 25-hydroxy- $D_3$  [ $25-(OH)D_3$ ] which is the major circulating form of vitamin D. Conversion of  $25-(OH)D_3$  to its biologically active form, calcitriol, occurs through the activity of a specific  $D_3$ -1-hydroxylase present in the proximal convoluted tubules of the kidneys, and in bone and placenta.  $25-(OH)D_3$  can also be hydroxylated at the 24 position by a specific  $D_3$ -24-hydroxylase in the kidneys, intestine, placenta and cartilage.



Calcitriol functions in concert with **parathyroid hormone (PTH)** and **calcitonin** to regulate serum calcium and phosphorous levels. PTH is released in response to low serum calcium and induces the production of calcitriol. In contrast, reduced levels of PTH stimulate synthesis of the inactive  $24,25-(OH)_2D_3$ . In the intestinal epithelium, calcitriol functions as a steroid hormone in inducing the expression of **calbindin $D_{28K}$** , a protein involved in intestinal calcium absorption. The increased absorption of calcium ions requires concomitant absorption of a negatively charged counter ion to maintain electrical neutrality. The predominant counter ion is  $Pi$ . When plasma calcium levels fall the major sites of action of calcitriol and PTH are bone where they stimulate bone resorption and the kidneys where they inhibit calcium excretion by stimulating reabsorption by the distal tubules. The role of calcitonin in calcium homeostasis is to decrease elevated serum calcium levels by inhibiting bone resorption.

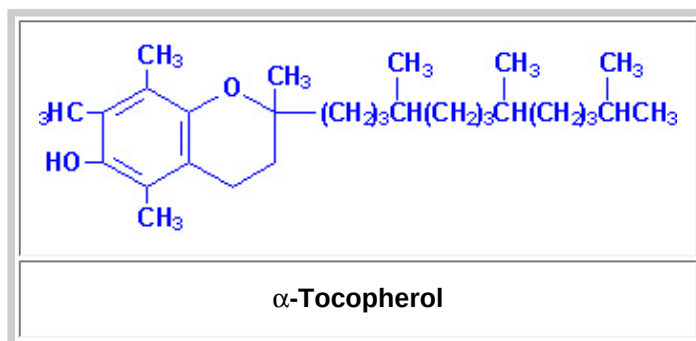
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## Clinical Significance of Vitamin D Deficiency

As a result of the addition of vitamin D to milk, deficiencies in this vitamin are rare in this country. The main symptom of vitamin D deficiency in children is **rickets** and in adults is **osteomalacia**. Rickets is characterized improper mineralization during the development of the bones resulting in soft bones. Osteomalacia is characterized by demineralization of previously formed bone leading to increased softness and susceptibility to fracture.

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## Vitamin E



Vitamin E is a mixture of several related compounds known as **tocopherols**. The  $\alpha$ -tocopherol molecule is the most potent of the tocopherols. Vitamin E is absorbed from the intestines packaged in chylomicrons. It is delivered to the tissues via chylomicron transport and then to the liver through chylomicron remnant uptake. The liver can export vitamin E in VLDLs. Due to its lipophilic nature, vitamin E accumulates in cellular membranes, fat deposits and other circulating lipoproteins. The major site of vitamin E storage is in adipose tissue.

The major function of vitamin E is to act as a natural **antioxidant** by scavenging free radicals and molecular oxygen. In particular vitamin E is important for preventing peroxidation of polyunsaturated membrane fatty acids. The vitamins E and C are interrelated in their antioxidant capabilities. Active  $\alpha$ -tocopherol can be regenerated by interaction with vitamin C following scavenge of

a peroxy free radical. Alternatively,  $\alpha$ -tocopherol can scavenge two peroxy free radicals and then be conjugated to glucuronate for excretion in the bile.

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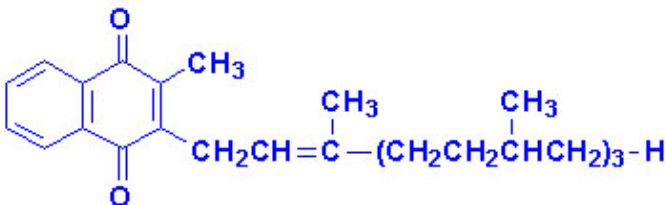
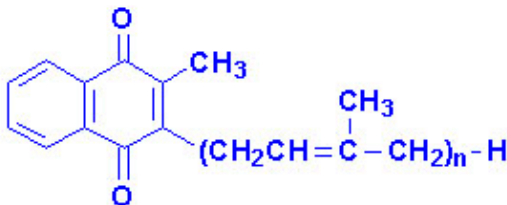
## Clinical significances of Vitamin E Deficiency

No major disease states have been found to be associated with vitamin E deficiency due to adequate levels in the average American diet. The major symptom of vitamin E deficiency in humans is an increase in red blood cell fragility. Since vitamin E is absorbed from the intestines in chylomicrons, any fat malabsorption diseases can lead to deficiencies in vitamin E intake. Neurological disorders have been associated with vitamin E deficiencies associated with fat malabsorptive disorders. Increased intake of vitamin E is recommended in premature infants fed formulas that are low in the vitamin as well as in persons consuming a diet high in polyunsaturated fatty acids. Polyunsaturated fatty acids tend to form free radicals upon exposure to oxygen and this may lead to an increased risk of certain cancers.

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## Vitamin K

The K vitamins exist naturally as  $K_1$  (phyloquinone) in green vegetables and  $K_2$  (menaquinone) produced by intestinal bacteria and  $K_3$  is synthetic menadione. When administered, vitamin  $K_3$  is alkylated to one of the vitamin  $K_2$  forms of menaquinone.

|                                                                                   |                                                                                     |               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
|  |  |               |
| Vitamin $K_1$                                                                     | Vitamin $K_2$<br>"n" can be 6, 7 or 9 isoprenoid groups                             | Vitamin $K_3$ |

The major function of the K vitamins is in the maintenance of normal levels of the **blood clotting** proteins, **factors II, VII, IX, X** and **protein C** and **protein S**, which are synthesized in the liver as inactive precursor proteins. Conversion from inactive to active clotting factor requires a **posttranslational modification** of specific glutamate (E) residues. This modification is a carboxylation and the enzyme responsible requires vitamin K as a cofactor. The resultant modified E residues are  **$\gamma$ -carboxyglutamate (gla)**. This process is most clearly understood for factor II, also called **preprothrombin**. Prothrombin is modified preprothrombin. The gla residues are effective calcium ion chelators. Upon chelation of calcium, prothrombin interacts with phospholipids in membranes and is proteolysed to thrombin through the action of activated factor X (Xa).

During the carboxylation reaction reduced hydroquinone form of vitamin K is converted to a 2,3-epoxide form. The regeneration of the hydroquinone form requires an uncharacterized reductase. This latter reaction is the site of action of the **dicumarol** based anticoagulants such as **warfarin**.

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## Clinical significance of Vitamin K Deficiency

Naturally occurring vitamin K is absorbed from the intestines only in the presence of bile salts and other lipids through interaction with chylomicrons. Therefore, fat malabsorptive diseases can result in vitamin K deficiency. The synthetic vitamin  $K_3$  is water soluble and absorbed irrespective of the presence of intestinal lipids and bile. Since the vitamin  $K_2$  form is synthesized by intestinal bacteria, deficiency of the vitamin in adults is rare. However, long term antibiotic treatment can lead to deficiency in adults. The intestine of newborn infants is sterile, therefore, vitamin K deficiency in infants is possible if lacking from the early diet. The primary symptom of a deficiency in infants is a **hemorrhagic syndrome**.

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## Introduction

The metabolic requirements for the nucleotides and their cognate bases can be met by both dietary intake or synthesis *de novo* from low molecular weight precursors. Indeed, the ability to salvage nucleotides from sources within the body alleviates any nutritional requirement for nucleotides, thus the purine and pyrimidine bases are not required in the diet. The salvage pathways are a major source of nucleotides for synthesis of DNA, RNA and enzyme co-factors.

Extracellular hydrolysis of ingested nucleic acids occurs through the concerted actions of endonucleases, phosphodiesterases and nucleoside phosphorylases. Endonucleases degrade DNA and RNA at internal sites leading to the production of oligonucleotides. Oligonucleotides are further digested by phosphodiesterases that act from the ends inward yielding free nucleosides. The bases are hydrolyzed from nucleosides by the action of phosphorylases that yield ribose-1-phosphate and free bases. If the nucleosides and/or bases are not re-utilized the purine bases are further degraded to **uric acid** and the pyrimidines to  $\beta$ -aminoisobutyrate,  $\text{NH}_3$  and  $\text{CO}_2$ .

Both the salvage and *de novo* synthesis pathways of purine and pyrimidine biosynthesis lead to production of nucleoside-5'-phosphates through the utilization of an activated sugar intermediate and a class of enzymes called phosphoribosyltransferases. The activated sugar used is **5-phosphoribosyl-1-pyrophosphate, PRPP**. PRPP is generated by the action of PRPP synthetase and requires energy in the form of ATP as shown:

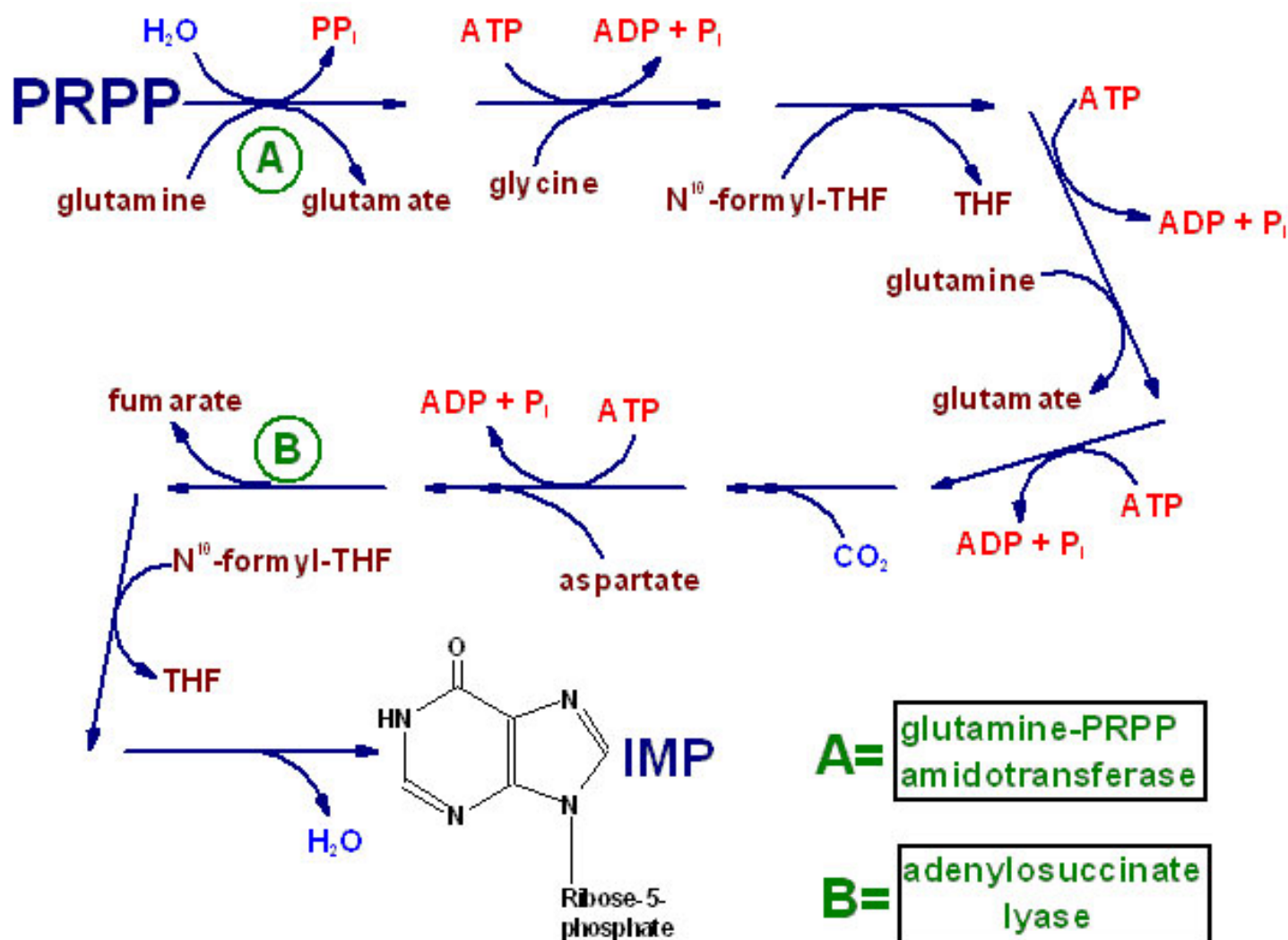


Note that this reaction releases AMP. Therefore, 2 high energy phosphate equivalents are consumed during the reaction.

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## Purine Nucleotide Biosynthesis

The major site of purine synthesis is in the liver. Synthesis of the purine nucleotides begins with PRPP and leads to the first fully formed nucleotide, **inosine 5'-monophosphate (IMP)**. This pathway is diagrammed below. The purine base without the attached ribose moiety is **hypoxanthine**. The purine base is built upon the ribose by several amidotransferase and transformylation reactions. The synthesis of IMP requires five moles of ATP, two moles of glutamine, one mole of glycine, one mole of CO<sub>2</sub>, one mole of aspartate and two moles of formate. The formyl moieties are carried on **tetrahydrofolate (THF)** in the form of N<sup>5</sup>,N<sup>10</sup>-methenyl-THF and N<sup>10</sup>-formyl-THF. IMP represents a branch point for purine biosynthesis, because it can be converted into either AMP or GMP through two distinct reaction pathways. The pathway leading to AMP requires energy in the form of GTP; that leading to GMP requires energy in the form of ATP. The utilization of GTP in the pathway to AMP synthesis allows the cell to control the proportions of AMP and GMP to near equivalence. The accumulation of excess GTP will lead to accelerated AMP synthesis from IMP instead, at the expense of GMP synthesis. Conversely, since the conversion of IMP to GMP requires ATP, the accumulation of excess ATP leads to accelerated synthesis of GMP over that of AMP.



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Synthesis of the first fully formed purine nucleotide, **inosine monophosphate, IMP** begins with **5-phospho- $\alpha$ -ribosyl-1-pyrophosphate, PRPP**. Through a series of reactions utilizing ATP, tetrahydrofolate (THF) derivatives, glutamine, glycine and aspartate this pathway yields IMP. The two indicated enzymes (A and B) are those catalyzing the rate limiting step and the reaction necessary for the **purine nucleotide cycle**, respectively. The structure of the nucleobase of IMP (hypoxanthine) is shown.

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## Regulation of Purine Nucleotide Synthesis

The essential rate limiting steps in purine biosynthesis occur at the first two steps of the pathway. The synthesis of PRPP by PRPP synthetase is feed-back inhibited by purine-5'-nucleotides (predominantly AMP and GMP). Combinatorial effects of those two nucleotides are greatest, e.g., inhibition is maximal when the correct concentration of both adenine and guanine nucleotides is achieved.

The amidotransferase reaction catalyzed by PRPP amidotransferase is also feed-back inhibited allosterically by binding ATP, ADP and AMP at one inhibitory site and GTP, GDP and GMP at another. Conversely the activity of the enzyme is stimulated by PRPP.

Additionally, purine biosynthesis is regulated in the branch pathways from IMP to AMP and GMP. The accumulation of excess ATP leads to accelerated synthesis of GMP, and excess GTP leads to accelerated synthesis of AMP.

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## Catabolism and Salvage of Purine Nucleotides

Catabolism of the purine nucleotides leads ultimately to the production of **uric acid** which is insoluble and is excreted in the urine as sodium urate crystals.

The synthesis of nucleotides from the purine bases and purine nucleosides takes place in a series of steps known as the **salvage pathways**. The free purine bases---adenine, guanine, and hypoxanthine---can be reconverted to their corresponding nucleotides by phosphoribosylation. Two key transferase enzymes are involved in the salvage of purines: adenosine phosphoribosyltransferase (APRT), which catalyzes the following reaction:



and hypoxanthine-guanine phosphoribosyltransferase (HGPRT), which catalyzes the following reactions:

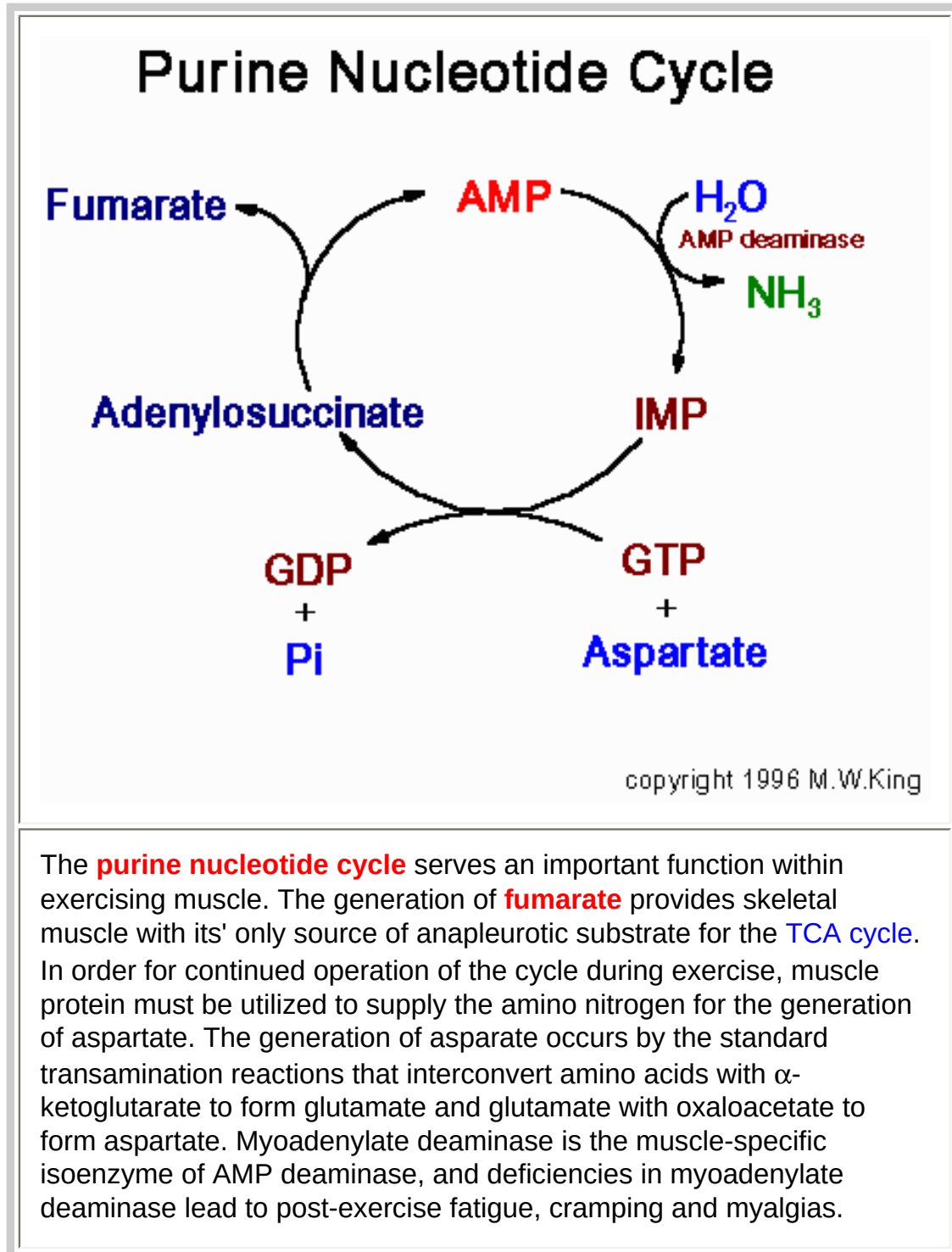


Purine nucleotide phosphorylases can also contribute to the salvage of the bases through a reversal of the catabolism pathways. However, this pathway is less significant than those catalyzed by the



phosphoribosyltransferases.

The synthesis of AMP from IMP and the salvage of IMP via AMP catabolism have the net effect of deaminating aspartate to fumarate. This process has been termed the **purine nucleotide cycle** (see diagram below). This cycle is very important in muscle cells. Increases in muscle activity create a demand for an increase in the **TCA cycle**, in order to generate more NADH for the production of ATP. However, muscle lacks most of the enzymes of the major anapleurotic reactions. Muscle replenishes TCA-cycle intermediates in the form of fumarate generated by the purine nucleotide cycle.



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## Clinical Significances of Purine Metabolism

Clinical problems associated with nucleotide metabolism in humans are predominantly the result of abnormal catabolism of the purines. The clinical consequences of abnormal purine metabolism range from mild to severe and even fatal disorders. Clinical manifestations of abnormal purine catabolism arise from the insolubility of the degradation byproduct, uric acid. Excess accumulation of uric acid leads to **hyperuricemia**, more commonly known as **gout**. This condition results from the precipitation of sodium urate crystals in the synovial fluid of the joints, leading to severe inflammation and arthritis.

Most forms of gout are the result of excess purine or of a partial deficiency in the salvage enzyme, HGPRT. Most forms of gout can be treated by administering the **antimetabolite, allopurinol**. This compound is a structural analog of hypoxanthine that strongly inhibits xanthine oxidase.

Two severe disorders, both quite well described, are associated with defects in purine metabolism: **Lesch-Nyhan syndrome** and **severe combined immunodeficiency disease (SCID)**. **Lesch-Nyhan syndrome** results from the loss of a functional HGPRT gene. The disorder is inherited as a sex-linked trait, with the HGPRT gene on the X chromosome (Xq26-q27.2). Patients with this defect exhibit not only severe symptoms of gout but also a severe malfunction of the nervous system. In the most serious cases, patients resort to self-mutilation. Death usually occurs before patients reach their 20th year.

**SCID** is caused by a deficiency in the enzyme adenosine deaminase (ADA). This is the enzyme responsible for converting adenosine to inosine in the catabolism of the purines. This deficiency selectively leads to a destruction of B and T lymphocytes, the cells that mount immune responses. In the absence of ADA, deoxyadenosine is phosphorylated to yield levels of dATP that are 50-fold higher than normal. The levels are especially high in lymphocytes, which have abundant amounts of the salvage enzymes, including nucleoside kinases. High concentrations of dATP inhibit ribonucleotide reductase (see below), thereby preventing other dNTPs from being produced. The net effect is to inhibit DNA synthesis. Since lymphocytes must be able to proliferate dramatically in response to antigenic challenge, the inability to synthesize DNA seriously impairs the immune responses, and the disease is usually fatal in infancy unless special protective measures are taken. A less severe immunodeficiency results when there is a lack of purine nucleoside phosphorylase (PNP), another purine-degradative enzyme.

One of the many glycogen storage diseases **von Gierke's disease** also leads to excessive uric acid production. This disorder results from a deficiency in glucose 6-phosphatase activity. The increased availability of glucose-6-phosphate increases the rate of flux through the pentose phosphate pathway, yielding an elevation in the level of ribose-5-phosphate and consequently PRPP. The increases in PRPP then result in excess purine biosynthesis.

## Disorders of Purine Metabolism

| Disorder    | Defect          | Nature of Defect                                           | Comments      |
|-------------|-----------------|------------------------------------------------------------|---------------|
| <b>Gout</b> | PRPP synthetase | increased enzyme activity due to elevated V <sub>max</sub> | hyperuricemia |
| Gout        | PRPP synthetase | enzyme is resistant to feed-back inhibition                | hyperuricemia |

|                                      |                       |                                                                       |                                           |
|--------------------------------------|-----------------------|-----------------------------------------------------------------------|-------------------------------------------|
| Gout                                 | PRPP synthetase       | enzyme has increased affinity for ribose-5-phosphate (lowered $K_m$ ) | hyperuricemia                             |
| Gout                                 | PRPP amidotransferase | loss of feed-back inhibition of enzyme                                | hyperuricemia                             |
| Gout                                 | HGPRT <sup>a</sup>    | partially defective enzyme                                            | hyperuricemia                             |
| <a href="#">Lesch-Nyhan syndrome</a> | HGPRT                 | lack of enzyme                                                        | see above                                 |
| <a href="#">SCID</a>                 | ADA <sup>b</sup>      | lack of enzyme                                                        | see above                                 |
| Immunodeficiency                     | PNP <sup>c</sup>      | lack of enzyme                                                        | see above                                 |
| <a href="#">Renal lithiasis</a>      | APRT <sup>d</sup>     | lack of enzyme                                                        | 2,8-dihydroxyadenine renal lithiasis      |
| <a href="#">Xanthinuria</a>          | Xanthine oxidase      | lack of enzyme                                                        | hypouricemia and xanthine renal lithiasis |
| <a href="#">von Gierke's disease</a> | Glucose-6-phosphatase | enzyme deficiency                                                     | see above                                 |

<sup>a</sup>Hypoxanthine-guanine phosphoribosyltransferase; <sup>b</sup>adenosine deaminase;

<sup>c</sup>purine nucleotide phosphorylase; <sup>d</sup>adenosine phosphoribosyltransferase

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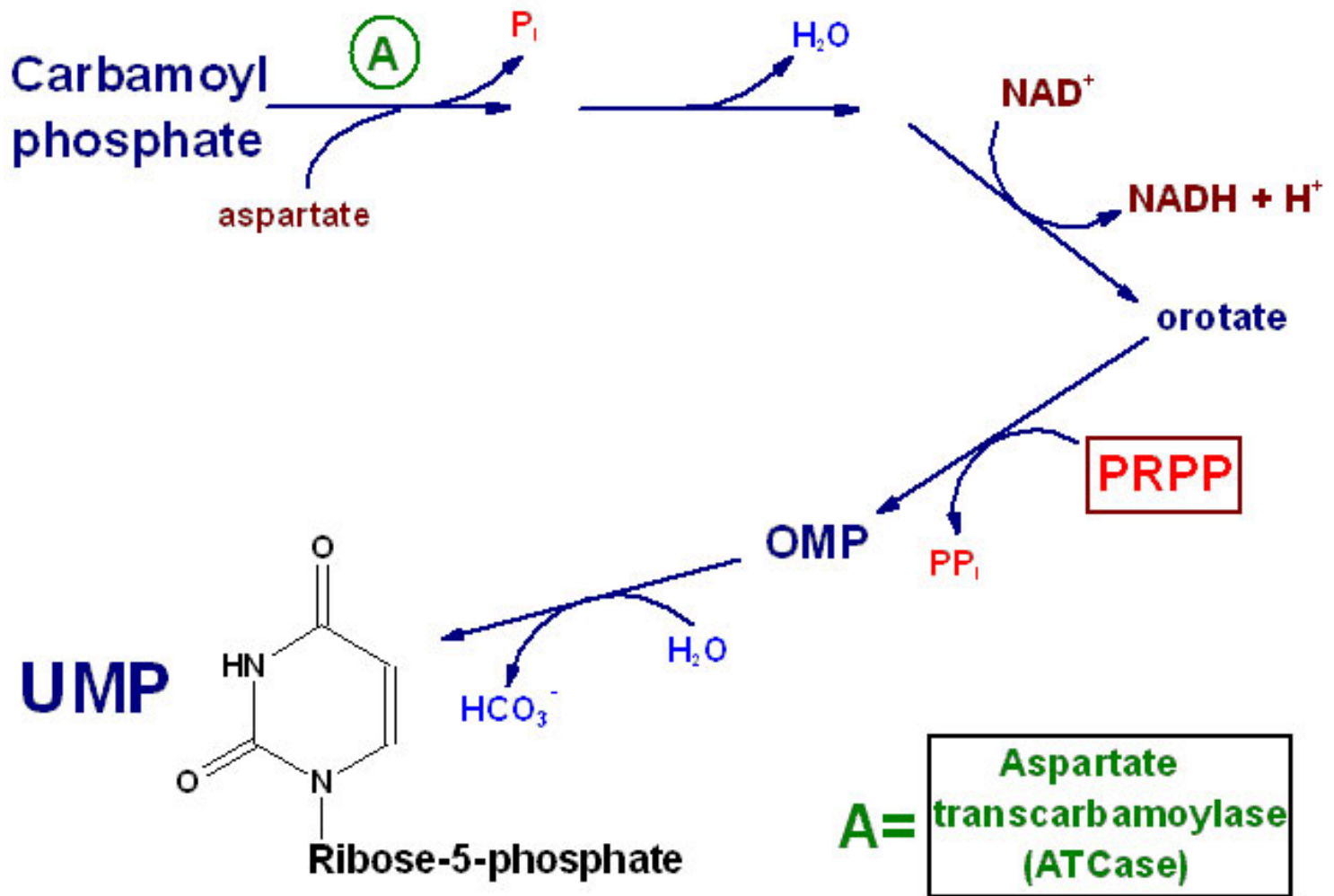
## Pyrimidine Nucleotide Biosynthesis

Synthesis of the pyrimidines is less complex than that of the purines, since the base is much simpler. The first completed base is derived from 1 mole of glutamine, one mole of ATP and one mole of CO<sub>2</sub> (which form **carbamoyl phosphate**) and one mole of aspartate. An additional mole of glutamine and ATP are required in the conversion of UTP to CTP. The pathway of pyrimidine biosynthesis is diagrammed below.

The carbamoyl phosphate used for pyrimidine nucleotide synthesis is derived from glutamine and bicarbonate, within the cytosol, as opposed to the urea cycle carbamoyl phosphate derived from ammonia and bicarbonate in the mitochondrion. The urea cycle reaction is catalyzed by carbamoyl phosphate synthetase I (CPS-I) whereas the pyrimidine nucleotide precursor is synthesized by **CPS-II**. Carbamoyl phosphate is then condensed with aspartate in a reaction catalyzed by the rate limiting enzyme of pyrimidine nucleotide biosynthesis, aspartate transcarbamoylase (ATCase).

The synthesis of pyrimidines differs in two significant ways from that of purines. First, the ring structure is assembled as a free base, not built upon PRPP. PRPP is added to the first fully formed pyrimidine base (**orotic acid**), forming **orotate monophosphate (OMP)**, which is subsequently decarboxylated to UMP. Second, there is no branch in the pyrimidine synthesis pathway. UMP is phosphorylated twice to yield UTP (ATP is the phosphate donor). The first phosphorylation is catalyzed by

uridylate kinase and the second by ubiquitous nucleoside diphosphate kinase. Finally UTP is aminated by the action of CTP synthase, generating CTP. The thymine nucleotides are in turn derived by *de novo* synthesis from dUMP or by salvage pathways from deoxyuridine or deoxythymidine.



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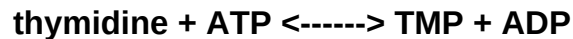
Synthesis of UMP from carbamoyl phosphate. Carbamoyl phosphate utilized in pyrimidine nucleotide synthesis differs from that synthesized in the urea cycle; it is synthesized from glutamine instead of ammonia and is synthesized in the cytosol. The reaction is catalyzed by carbamoyl phosphate synthetase II (CPS-II). Subsequently carbamoyl phosphate is incorporated into the pyrimidine nucleotide biosynthesis pathway through the action of aspartate transcarbamoylase (ATCase) which is the rate limiting step in pyrimidine biosynthesis. Following completion of UMP synthesis it can be phosphorylated to UTP and utilized as a substrate for CTP synthase for the synthesis of CTP. Uridine nucleotides are also the precursors for *de novo* synthesis of the thymine nucleotides.

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## Synthesis of the Thymine Nucleotides

The *de novo* pathway to dTTP synthesis first requires the use of dUMP from the metabolism of either UDP or CDP. The dUMP is converted to dTMP by the action of thymidylate synthase. The methyl group (recall that thymine is 5-methyl uracil) is donated by tetrahydrofolate, similarly to the donation of methyl groups during the biosynthesis of the purines.

The salvage pathway to dTTP synthesis involves the enzyme thymidine kinase which can use either thymidine or deoxyuridine as substrate:



The activity of thymidine kinase (one of the various deoxyribonucleotide kinases) is unique in that it fluctuates with the cell cycle, rising to peak activity during the phase of DNA synthesis; it is inhibited by dTTP.

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## Clinical Relevance of Tetrahydrofolate

Tetrahydrofolate (THF) is regenerated from the dihydrofolate (DHF) product of the thymidylate synthase reaction by the action of dihydrofolate reductase (DHFR), an enzyme that requires NADPH. Cells that are unable to regenerate THF suffer defective DNA synthesis and eventual death. For this reason, as well as the fact that dTTP is utilized only in DNA, it is possible therapeutically to target rapidly proliferating cells over non-proliferative cells through the inhibition of thymidylate synthase. Many anti-cancer drugs act directly to inhibit thymidylate synthase, or indirectly, by inhibiting DHFR.

The class of molecules used to inhibit thymidylate synthase is called the **suicide substrates**, because they irreversibly inhibit the enzyme. Molecules of this class include **5-fluorouracil** and **5-fluorodeoxyuridine**. Both are converted within cells to **5-fluorodeoxyuridylate, FdUMP**. It is this drug metabolite that inhibits thymidylate synthase. Many DHFR inhibitors have been synthesized, including **methotrexate**, **aminopterin**, and **trimethoprim**. Each of these is an analog of folic acid.

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## Regulation of Pyrimidine Biosynthesis

The regulation of pyrimidine synthesis occurs mainly at the first step which is catalyzed by aspartate transcarbamoylase, ATCase. Inhibited by CTP and activated by ATP, ATCase is a multifunctional protein in mammalian cells. It is capable of catalyzing the formation of **carbamoyl phosphate**, **carbamoyl aspartate**, and **dihydroorotate**. The carbamoyl synthetase activity of this complex is termed carbamoyl phosphate synthetase II (CPS-II) as opposed to CPS-I, which is involved in the **urea cycle**. ATCase, and therefore the activity of CPS-II, is localized to the cytoplasm and prefers glutamine as a substrate. CPS-I of the urea cycle is localized in the mitochondria and utilizes ammonia. The CPS-II domain is activated by ATP and inhibited by UDP, UTP, dUTP, and CTP.

The role of glycine in ATCase regulation is to act as a competitive inhibitor of the glutamine binding site. As in the regulation of purine synthesis, ATP levels also regulate pyrimidine biosynthesis at the level of PRPP formation. An increase in the level of PRPP results in an activation of pyrimidine synthesis.

There is also regulation of OMP decarboxylase: this enzyme is competitively inhibited by UMP and, to a lesser degree, by CMP. Finally, CTP synthase is feedback-inhibited by CTP and activated by GTP.

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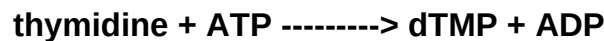
## Catabolism and Salvage of Pyrimidine Nucleotides

Catabolism of the pyrimidine nucleotides leads ultimately to  $\beta$ -alanine (when CMP and UMP are degraded) or  $\beta$ -aminoisobutyrate (when dTMP is degraded) and  $\text{NH}_3$  and  $\text{CO}_2$ . The  $\beta$ -alanine and  $\beta$ -aminoisobutyrate serve as  $-\text{NH}_2$  donors in transamination of  $\alpha$ -ketoglutarate to glutamate. A subsequent reaction converts the products to malonyl-CoA (which can be diverted to fatty acid synthesis) or methylmalonyl-CoA (which is converted to succinyl-CoA and can be shunted to the TCA cycle).

The salvage of pyrimidine bases has less clinical significance than that of the purines, owing to the solubility of the by-products of pyrimidine catabolism. However, as indicated above, the salvage pathway to thymidine nucleotide synthesis is especially important in the preparation for cell division. Uracil can be salvaged to form UMP through the concerted action of uridine phosphorylase and uridine kinase, as indicated:



Deoxyuridine is also a substrate for uridine phosphorylase. Formation of dTMP, by salvage of dTMP requires thymine phosphorylase and the previously encountered thymidine kinase:



The salvage of deoxycytidine is catalyzed by deoxycytidine kinase:



Deoxyadenosine and deoxyguanosine are also substrates for deoxycytidine kinase, although the  $K_m$  for these substrates is much higher than for deoxycytidine.

The major function of the pyrimidine nucleoside kinases is to maintain a cellular balance between the level of pyrimidine nucleosides and pyrimidine nucleoside monophosphates. However, since the overall cellular and plasma concentrations of the pyrimidine nucleosides, as well as those of ribose-1-phosphate, are low, the salvage of pyrimidines by these kinases is relatively inefficient.

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## Clinical Significances of Pyrimidine Metabolism

Because the products of pyrimidine catabolism are soluble, few disorders result from excess levels of their synthesis or catabolism. Two inherited disorders affecting pyrimidine biosynthesis are the result of deficiencies in the bifunctional enzyme catalyzing the last two steps of UMP synthesis, orotate phosphoribosyl transferase and OMP decarboxylase. These deficiencies result in [orotic aciduria](#) that causes



retarded growth, and severe anemia caused by hypochromic erythrocytes and megaloblastic bone marrow. Leukopenia is also common in orotic acidurias. The disorders can be treated with uridine and/or cytidine, which leads to increased UMP production via the action of nucleoside kinases. The UMP then inhibits CPS-II, thus attenuating orotic acid production.

## Disorders of Pyrimidine Metabolism

| Disorder                                         | Defective Enzyme                                                                                                 | Comments                                                                                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orotic aciduria, Type I                          | orotate phosphoribosyl transferase and OMP decarboxylase                                                         | see above                                                                                                                                             |
| Orotic aciduria, Type II                         | OMP decarboxylase                                                                                                | see above                                                                                                                                             |
| Orotic aciduria (mild, no hematologic component) | the urea cycle enzyme, ornithine transcarbamoylase, is deficient                                                 | increased mitochondrial carbamoyl phosphate exits and augments pyrimidine biosynthesis; hepatic encephalopathy                                        |
| $\beta$ -aminoisobutyric aciduria                | Transaminase, affects urea cycle function during deamination of $\alpha$ -amino acids to of $\alpha$ -keto acids | benign, frequent in Orientals                                                                                                                         |
| Drug induced orotic aciduria                     | OMP decarboxylase                                                                                                | Allopurinol and 6-azauridine treatments cause orotic acidurias without a hematologic component; their catabolic by-products inhibit OMP decarboxylase |

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## Formation of Deoxyribonucleotides

The typical cell contains 5--10 times as much RNA (mRNAs, rRNAs and tRNAs) as DNA. Therefore, the majority of nucleotide biosynthesis has as its purpose the production of rNTPs. However, because proliferating cells need to replicate their genomes, the production of dNTPs is also necessary. This process begins with the reduction of rNDPs, followed by phosphorylation to yield the dNTPs. The phosphorylation of dNDPs to dNTPs is catalyzed by the same nucleoside diphosphate kinases that phosphorylates rNDPs to rNTPs, using ATP as the phosphate donor.

Ribonucleotide reductase (RR) is a multifunctional enzyme that contains redox-active thiol groups for

the transfer of electrons during the reduction reactions. In the process of reducing the rNDP to a dNDP, RR becomes oxidized. RR is reduced in turn, by either **thioredoxin** or **glutaredoxin**. The ultimate source of the electrons is NADPH. The electrons are shuttled through a complex series of steps involving enzymes that regenerate the reduced forms of thioredoxin or glutaredoxin. These enzymes are thioredoxin reductase and glutathione reductase respectively.

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## Regulation of dNTP Formation

Ribonucleotide reductase is the only enzyme used in the generation of all the deoxyribonucleotides. Therefore, its activity and substrate specificity must be tightly regulated to ensure balanced production of all four of the dNTPs required for DNA replication. Such regulation occurs by binding of nucleoside triphosphate effectors to either the activity sites or the specificity sites of the enzyme complex. The activity sites bind either ATP or dATP with low affinity, whereas the specificity sites bind ATP, dATP, dGTP, or dTTP with high affinity. The binding of ATP at activity sites leads to increased enzyme activity, while the binding of dATP inhibits the enzyme. The binding of nucleotides at specificity sites effectively allows the enzyme to detect the relative abundance of the four dNTPs and to adjust its affinity for the less abundant dNTPs, in order to achieve a balance of production. thioredoxin reductase and glutathione reductase respectively.

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## Interconversion of the Nucleotides

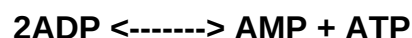
During the catabolism of nucleic acids, nucleoside mono- and diphosphates are released. The nucleosides do not accumulate to any significant degree, owing to the action of nucleoside kinases. These include both nucleoside monophosphate (NMP) kinases and nucleoside diphosphate (NDP) kinases. The NMP kinases catalyze ATP-dependent reactions of the type:



There are four classes of NMP kinases that catalyze, respectively, the phosphorylation of:

- 1. AMP and dAMP; this kinase is known as adenylate kinase.
- 2. GMP and dGMP.
- 3. CMP, UMP and dCMP.
- 4. dTMP.

The enzyme adenylate kinase is important for ensuring adequate levels of energy in cells such as liver and muscle. The predominant reaction catalyzed by adenylate kinase is:



The NDP kinases catalyze reaction of the type:



$N_1$  can represent a purine ribo- or deoxyribonucleotide;  $N_2$  a pyrimidine ribo- or deoxyribonucleotide. The activity of the NDP kinases can range from 10 to 100 times higher than that of the NMP kinases. This difference in activity maintains a relatively high intracellular level of (d)NTPs relative to that of (d)NDPs. Unlike the substrate specificity seen for the NMP kinases, the NDP kinases recognize a wide spectrum of (d)NDPs and (d)NTPs.

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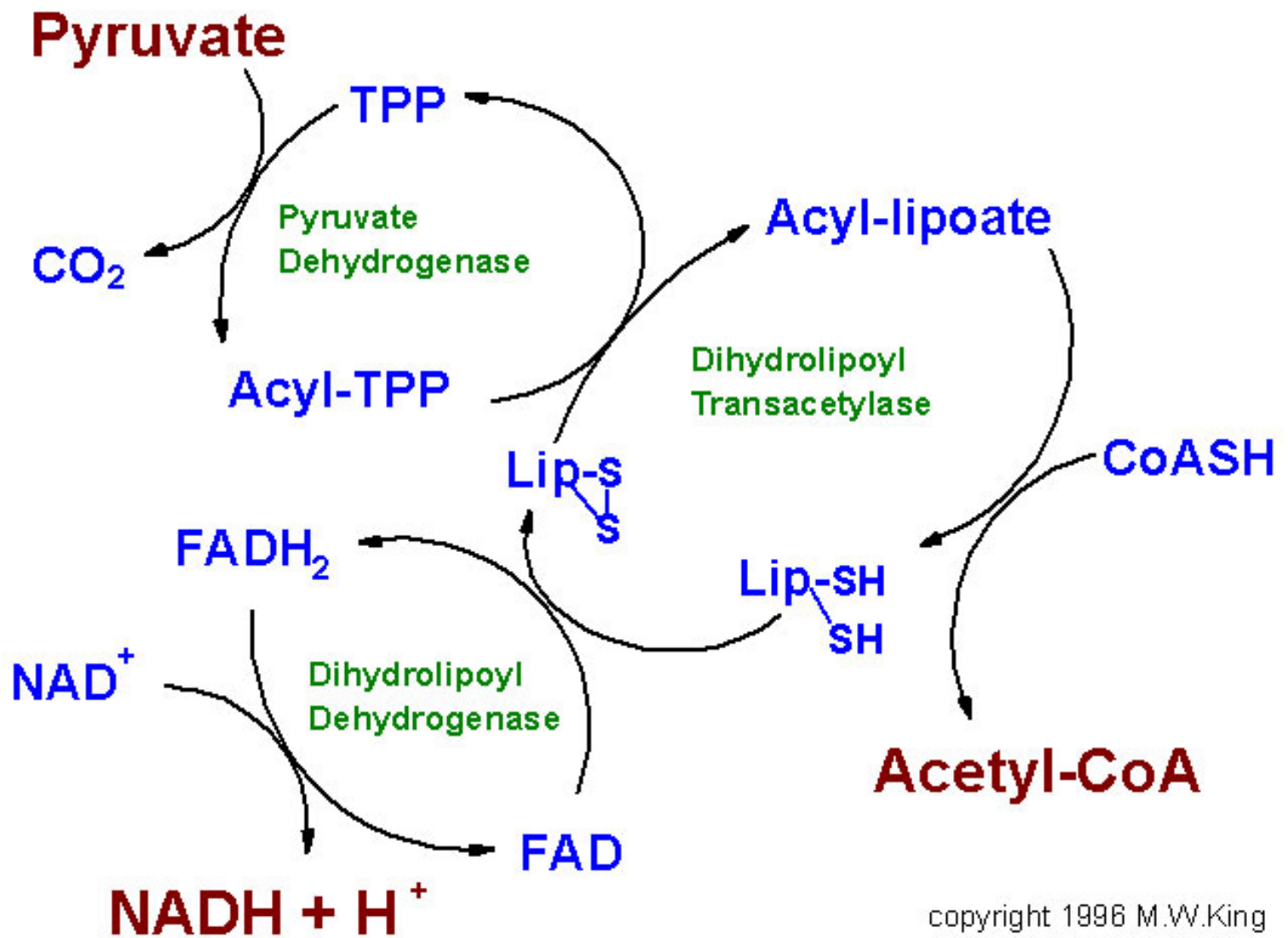
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## The Pyruvate Dehydrogenase (PDH) Complex

The bulk of ATP used by many cells to maintain homeostasis is produced by the oxidation of pyruvate in the TCA cycle. During this oxidation process, reduced **nicotinamide adenine dinucleotide (NADH)** and reduced **flavin adenine dinucleotide (FADH<sub>2</sub>)** are generated. The NADH and FADH<sub>2</sub> are principally used to drive the processes of [oxidative phosphorylation](#), which are responsible for converting the reducing potential of NADH and FADH<sub>2</sub> to the high energy phosphate in ATP.

The fate of pyruvate depends on the cell energy charge. In cells or tissues with a high energy charge pyruvate is directed toward [gluconeogenesis](#), but when the energy charge is low pyruvate is preferentially oxidized to CO<sub>2</sub> and H<sub>2</sub>O in the TCA cycle, with generation of 15 equivalents of ATP per pyruvate. The enzymatic activities of the TCA cycle (and of oxidative phosphorylation) are located in the mitochondrion. When transported into the mitochondrion, pyruvate encounters two principal metabolizing enzymes: pyruvate carboxylase (a gluconeogenic enzyme) and pyruvate dehydrogenase (PDH), the first enzyme of the PDH complex. With a high cell-energy charge coenzyme A (CoA) is highly acylated, principally as acetyl-CoA, and able allosterically to activate pyruvate carboxylase, directing pyruvate toward gluconeogenesis. When the energy charge is low CoA is not acylated, pyruvate carboxylase is inactive, and pyruvate is preferentially metabolized via the PDH complex and the enzymes of the TCA cycle to CO<sub>2</sub> and H<sub>2</sub>O. Reduced NADH and FADH<sub>2</sub> generated during the oxidative reactions can then be used to drive ATP synthesis via oxidative phosphorylation.

The PDH complex is comprised of multiple copies of 3 separate enzymes: pyruvate dehydrogenase (20-30 copies), dihydrolipoyl transacetylase (60 copies) and dihydrolipoyl dehydrogenase (6 copies). The complex also requires 5 different coenzymes: **CoA**, **NAD<sup>+</sup>**, **FAD<sup>+</sup>**, **lipoic acid** and **thiamine pyrophosphate (TPP)**. Three of the coenzymes of the complex are tightly bound to enzymes of the complex (TPP, lipoic acid and FAD<sup>+</sup>) and two are employed as carriers of the products of PDH complex activity (CoA and NAD<sup>+</sup>). The pathway of PDH oxidation of pyruvate to acetyl-CoA is diagrammed below.



Flow diagram depicting the overall activity of the **pyruvate dehydrogenase complex**. During the oxidation of pyruvate to CO<sub>2</sub> by pyruvate dehydrogenase the electrons flow from pyruvate to the lipoamide moiety of dihydrolipoyl transacetylase then to the FAD cofactor of dihydrolipoyl dehydrogenase and finally to reduction of NAD<sup>+</sup> to NADH. The acetyl group is linked to **coenzyme A (CoASH)** in a high energy thioester bond. The acetyl-CoA then enters the TCA cycle for complete oxidation to CO<sub>2</sub> and H<sub>2</sub>O.

The first enzyme of the complex is PDH itself which oxidatively decarboxylates pyruvate. During the course of the reaction the acetyl group derived from decarboxylation of pyruvate is bound to TPP. The next reaction of the complex is the transfer of the 2--carbon acetyl group from acetyl-TPP to lipoic acid, the covalently bound coenzyme of lipoyl transacetylase. The transfer of the acetyl group from acyl-lipoamide to CoA results in the formation of 2 sulfhydryl (SH) groups in lipoate requiring reoxidation to the disulfide (S-S) form to regenerate lipoate as a competent acyl acceptor. The enzyme dihydrolipoyl dehydrogenase, with FAD<sup>+</sup> as a cofactor, catalyzes that oxidation reaction. The final activity of the PDH complex is the transfer of reducing equivalents from the FADH<sub>2</sub> of dihydrolipoyl dehydrogenase to NAD<sup>+</sup>. The fate of the NADH is oxidation via mitochondrial electron transport, to produce 3 equivalents of ATP:

The net result of the reactions of the PDH complex are:



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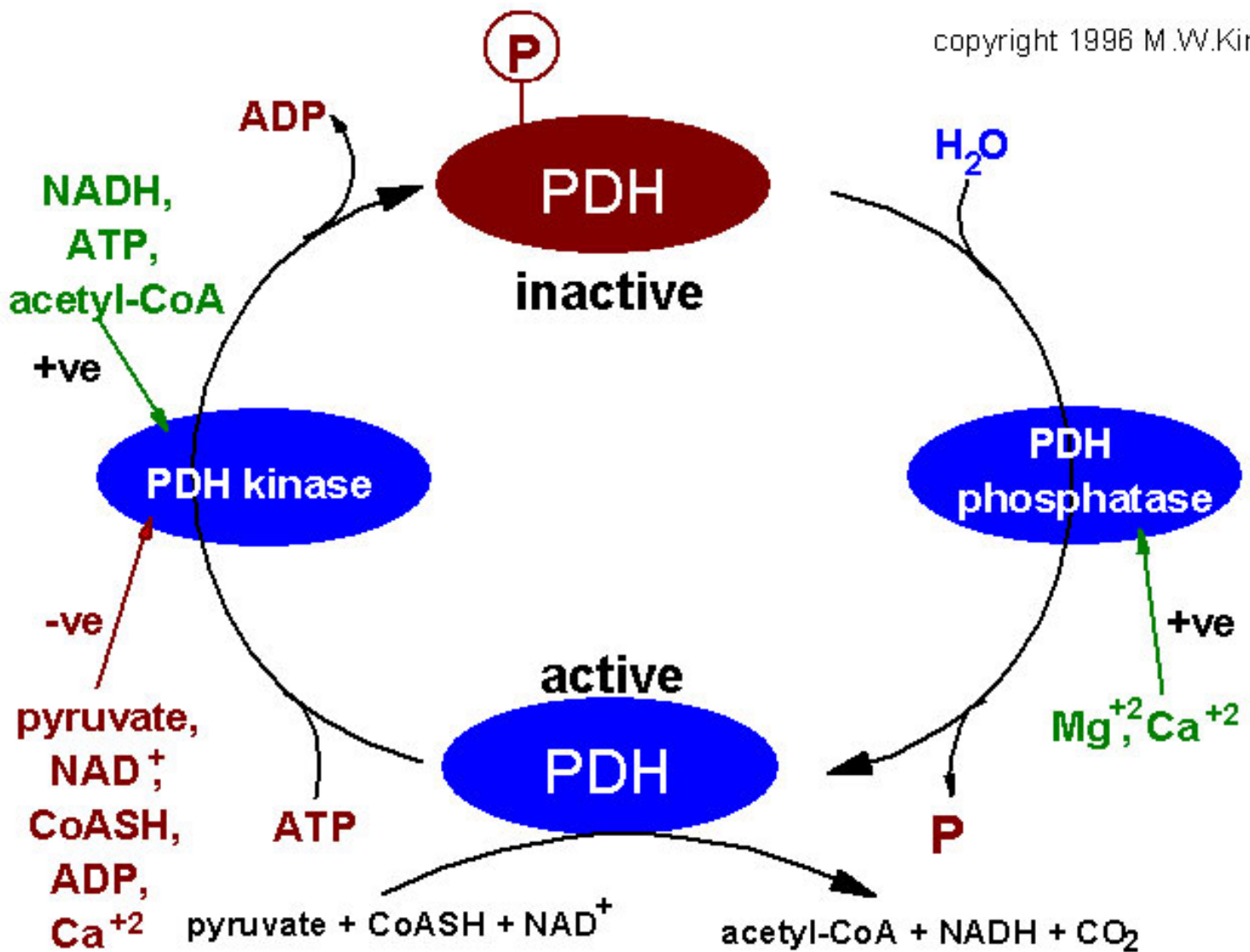
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## Regulation of the PDH Complex

The reactions of the PDH complex serves to interconnect the metabolic pathways of glycolysis, gluconeogenesis and fatty acid synthesis to the TCA cycle. As a consequence, the activity of the PDH complex is highly regulated by a variety of allosteric effectors and by covalent modification. The importance of the PDH complex to the maintenance of homeostasis is evident from the fact that although diseases associated with deficiencies of the PDH complex have been observed, affected individuals often do not survive to maturity. Since the energy metabolism of highly aerobic tissues such as the brain is dependent on normal conversion of pyruvate to acetyl-CoA, aerobic tissues are most sensitive to deficiencies in components of the PDH complex. Most genetic diseases associated with PDH complex deficiency are due to mutations in PDH. The main pathologic result of such mutations is moderate to severe **cerebral lactic acidosis** and **encephalopathies**.

The main regulatory features of the PDH complex are diagrammed below.





Factors regulating the activity of pyruvate dehydrogenase, (PDH). PDH activity is regulated by its' state of phosphorylation, being most active in the dephosphorylated state. Phosphorylation of PDH is catalyzed by a specific PDH kinase. The activity of the kinase is enhanced when cellular energy charge is high which is reflected by an increase in the level of ATP, NADH and acetyl-CoA. Conversely, an increase in pyruvate strongly inhibits PDH kinase. Additional negative effectors of PDH kinase are ADP, NAD<sup>+</sup> and CoASH, the levels of which increase when energy levels fall. The regulation of PDH phosphatase is not completely understood but it is known that  $Mg^{2+}$  and  $Ca^{2+}$  activate the enzyme. In adipose tissue **insulin** increases PDH activity and in cardiac muscle PDH activity is increased by **catecholamines**.

Two products of the complex, NADH and acetyl-CoA, are negative allosteric effectors on PDH-a, the non-phosphorylated, active form of PDH. These effectors reduce the affinity of the enzyme for pyruvate, thus limiting the flow of carbon through the PDH complex. In addition, NADH and acetyl-CoA are powerful positive effectors on PDH kinase, the enzyme that inactivates PDH by converting it to the phosphorylated PDH-b form. Since NADH and acetyl-CoA accumulate when the cell energy charge is high, it is not surprising that high ATP levels also up-regulate PDH kinase activity, reinforcing down-regulation of PDH activity in energy-rich cells. Note, however, that pyruvate is a potent negative effector on PDH kinase, with the result that when pyruvate levels rise, PDH-a will be favored even with high levels of NADH and acetyl-

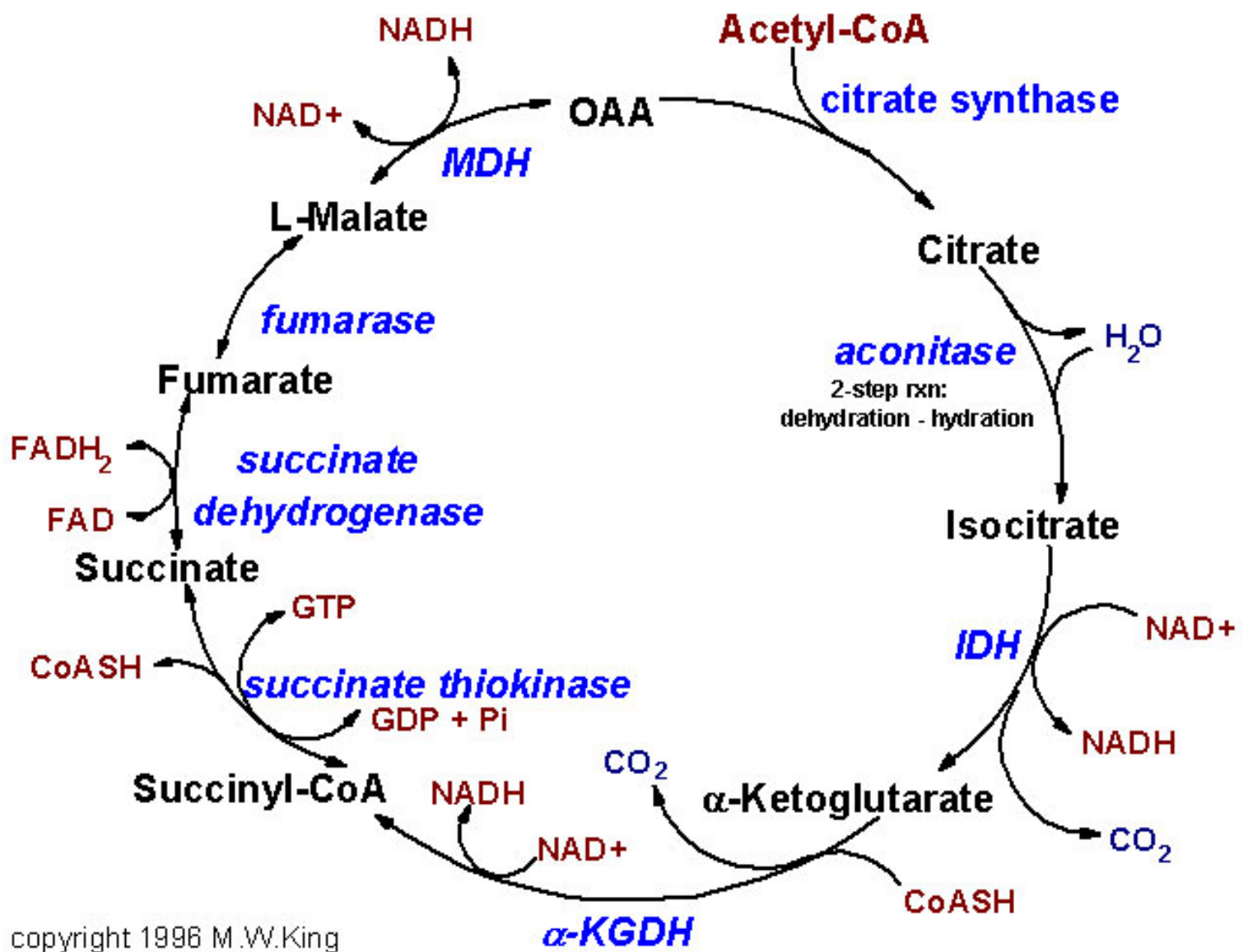
CoA.

Concentrations of pyruvate which maintain PDH in the active form (PDH-a) are sufficiently high so that, in energy-rich cells, the allosterically down-regulated, high  $K_m$  form of PDH is nonetheless capable of converting pyruvate to acetyl-CoA. With large amounts of pyruvate in cells having high energy charge and high NADH, pyruvate carbon will be directed to the 2 main storage forms of carbon---glycogen via gluconeogenesis and fat production via fatty acid synthesis---where acetyl-CoA is the principal carbon donor.

Although the regulation of PDH-b phosphatase is not well understood, it is quite likely regulated to maximize pyruvate oxidation under energy-poor conditions and to minimize PDH activity under energy-rich conditions.

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## Reactions of the TCA Cycle



The TCA cycle showing enzymes, substrates and products. The abbreviated enzymes are: IDH = isocitrate dehydrogenase and  $\alpha$ -KGDH =  $\alpha$ -ketoglutarate dehydrogenase. The GTP generated during the succinate thiokinase (succinyl-CoA synthetase) reaction is equivalent to a mole of ATP by virtue of the presence of nucleoside diphosphokinase. The 3 moles of NADH and 1 mole of FADH<sub>2</sub> generated during each round of the cycle feed into the [oxidative phosphorylation](#) pathway. Each mole of NADH leads to 3 moles of ATP and each mole of FADH<sub>2</sub> leads to 2 moles of ATP. Therefore, for each mole of pyruvate which enters the TCA cycle, 12 moles of ATP can be generated.

## Citrate Synthase (Condensing enzyme)

The first reaction of the cycle is condensation of the methyl carbon of acetyl-CoA with the keto carbon (C-2) of oxaloacetate (OAA). The standard free energy of the reaction, -8.0 kcal/mol, drives it strongly in the forward direction. Since the formation of OAA from its precursor is thermodynamically unfavorable, the highly exergonic nature of the citrate synthase reaction is of central importance in keeping the entire cycle going in the forward direction, since it drives oxaloacetate formation by mass action principals.

When the cellular energy charge increases the rate of flux through the TCA cycle will decline leading to a build-up of citrate. Excess citrate is used to transport acetyl-CoA carbons from the mitochondrion to the cytoplasm where they can be used for [fatty acid](#) and [cholesterol biosynthesis](#). Additionally, the increased levels of citrate in the cytoplasm activate the key regulatory enzyme of fatty acid biosynthesis, acetyl-CoA carboxylase (ACC) and inhibit PFK-1. In non-hepatic tissues citrate is also required for [ketone body synthesis](#).

## Aconitase

The isomerization of citrate to isocitrate by aconitase is stereospecific, with the migration of the -OH from the central carbon of citrate (formerly the keto carbon of OAA) being always to the adjacent carbon which is derived from the methylene (-CH<sub>2</sub>-) of OAA. The stereospecific nature of the isomerization determines that the CO<sub>2</sub> lost, as isocitrate is oxidized to succinyl-CoA, is derived from the oxaloacetate used in citrate synthesis.

Aconitase is one of several mitochondrial enzymes known as **non-heme-iron proteins**. These proteins contain inorganic iron and sulfur, known as **iron sulfur centers**, in a coordination complex with cysteine sulfurs of the protein. There are two prominent classes of non-heme-iron complexes, those containing two equivalents each of inorganic iron and sulfur Fe<sub>2</sub>S<sub>2</sub>, and those containing 4 equivalents of each Fe<sub>4</sub>S<sub>4</sub>. Aconitase is a member of the Fe<sub>4</sub>S<sub>4</sub> class. Its iron sulfur centers are often designated as Fe<sub>4</sub>S<sub>4</sub>Cys<sub>4</sub>, indicating that 4 cystine sulfur atoms are involved in the complete structure of the complex. In iron sulfur compounds the iron is generally involved in oxidation-reduction events.

## Isocitrate Dehydrogenase

Isocitrate is oxidatively decarboxylated to  $\alpha$ -ketoglutarate by isocitrate dehydrogenase, (IDH). There are two different IDH enzymes. The IDH of the TCA cycle uses NAD<sup>+</sup> as a cofactor, whereas the other IDH uses NADP<sup>+</sup> as a cofactor. Unlike the NAD<sup>+</sup>-requiring enzyme, which is located only in the mitochondrial matrix, the NADP<sup>+</sup>-requiring enzyme is found in both the mitochondrial matrix and the cytosol. IDH catalyzes the **rate-limiting step**, as well as the first NADH-yielding reaction of the TCA cycle. The CO<sub>2</sub> produced by the IDH reaction is the original C-1 of the oxaloacetate used in the citrate synthase reaction.

It is generally considered that control of carbon flow through the cycle is regulated at IDH by the powerful negative allosteric effectors NADH and ATP and by the potent positive effectors; isocitrate, ADP and AMP. From the latter it is clear that cell energy charge is a key factor in regulating carbon flow through the TCA cycle.

## $\alpha$ -Ketoglutarate Dehydrogenase Complex

$\alpha$ -ketoglutarate is oxidatively decarboxylated to succinyl-CoA by the  $\alpha$ -ketoglutarate dehydrogenase ( $\alpha$ -KGDH) complex. This reaction generates the second TCA cycle equivalent of CO<sub>2</sub> and NADH. This multienzyme complex is very similar to the PDH complex in the intricacy of its protein makeup, cofactors, and its mechanism of action. Also, as with the PDH complex, the reactions of the  $\alpha$ -KGDH complex proceed with a large negative standard free energy change. Although the  $\alpha$ -KGDH of the complex is not subject to covalent modification, allosteric regulation is quite complex, with activity being regulated by energy charge, the NAD<sup>+</sup>/NADH ratio, and effector activity of substrates and products.

Succinyl-CoA and  $\alpha$ -ketoglutarate are also important metabolites outside the TCA cycle. In particular,  $\alpha$ -ketoglutarate represents a key anapleurotic metabolite linking the entry and exit of carbon atoms from the TCA cycle to pathways involved in [amino acid metabolism](#).  $\alpha$ -ketoglutarate is also important for driving the [malate-aspartate shuttle](#). Succinyl-CoA, along with glycine, contributes all the carbon and nitrogen atoms required for the synthesis of [protoporphyrin heme biosynthesis](#) and for non-hepatic tissue utilization of [ketone bodies](#).

## Succinyl CoA Synthetase (Succinyl Thiokinase )

The conversion of succinyl-CoA to succinate by succinyl CoA synthetase involves use of the high-energy thioester of succinyl-CoA to drive synthesis of a high-energy nucleotide phosphate, by a process known as **substrate-level phosphorylation**. In this process a high energy enzyme--phosphate intermediate is formed, with the phosphate subsequently being transferred to GDP. Mitochondrial GTP is used in a trans-phosphorylation reaction catalyzed by the mitochondrial enzyme nucleoside diphosphate kinase to phosphorylate ADP, producing ATP and regenerating GDP for the continued operation of succinyl CoA synthetase.

## Succinate Dehydrogenase (SDH)

Succinate dehydrogenase catalyzes the oxidation of succinate to fumarate with the sequential reduction of enzyme-bound FAD and non-heme-iron. In mammalian cells the final electron acceptor is coenzyme Q<sub>10</sub> (CoQ<sub>10</sub>), a mobile carrier of reducing equivalents that is restricted by its lipophilic nature to the lipid phase of the mitochondrial membrane.

## Fumarase (fumarate hydratase)

The fumarase-catalyzed reactions specific for the *trans* form of fumarate. The result is that the hydration of fumarate proceeds stereospecifically with the production of L-malate.

## Malate Dehydrogenase (MDH)

L-malate is the specific substrate for MDH, the final enzyme of the TCA cycle. The forward reaction of the cycle, the oxidation of malate yields oxaloacetate (OAA). In the forward direction the reaction has a standard free energy of about +7 kcal/mol, indicating the very unfavorable nature of the forward direction. As noted earlier, the citrate synthase reaction that condenses oxaloacetate with acetyl-CoA has a standard free energy of about -8 kcal/mol and is responsible for pulling the MDH reaction in the forward direction. The

overall change in standard free energy change is about -1 kcal/mol for the conversion of malate to oxaloacetate and on to succinate.

The overall stoichiometry of the TCA cycle is:



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## Regulation of the TCA Cycle

Regulation of the TCA cycle, like that of glycolysis, occurs at both the level of entry of substrates into the cycle as well as at the key reactions of the cycle. Fuel enters the TCA cycle primarily as acetyl-CoA. The generation of acetyl-CoA from carbohydrates is, therefore, a major control point of the cycle. This is the reaction catalyzed by the PDH complex.

By way of review, the PDH complex is inhibited by acetyl-CoA and NADH and activated by non-acetylated CoA (CoASH) and  $\text{NAD}^+$ . The pyruvate dehydrogenase activities of the PDH complex are regulated by their state of phosphorylation. This modification is carried out by a specific kinase (PDH kinase) and the phosphates are removed by a specific phosphatase (PDH phosphatase). The phosphorylation of PDH inhibits its activity and, therefore, leads to decreased oxidation of pyruvate. PDH kinase is activated by NADH and acetyl-CoA and inhibited by pyruvate, ADP, CoASH,  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ . The PDH phosphatase, in contrast, is activated by  $\text{Mg}^{2+}$  and  $\text{Ca}^{2+}$ .

Since three reactions of the TCA cycle as well as PDH utilize  $\text{NAD}^+$  as co-factor it is not difficult to understand why the cellular ratio of  $\text{NAD}^+/\text{NADH}$  has a major impact on the flux of carbon through the TCA cycle.

Substrate availability can also regulate TCA flux. This occurs at the citrate synthase reaction as a result of reduced availability of oxaloacetate. Product inhibition also controls the TCA flux, e.g. citrate inhibits citrate synthase,  $\alpha$ -KGDH is inhibited by NADH and succinyl-CoA. The key enzymes of the TCA cycle are also regulated allosterically by  $\text{Ca}^{2+}$ , ATP and ADP.

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## Introduction

While the large quantity of NADH resulting from TCA cycle activity can be used for reductive biosynthesis, the reducing potential of mitochondrial NADH is most often used to supply the energy for ATP synthesis via **oxidative phosphorylation**. Oxidation of NADH with phosphorylation of ADP to form ATP are processes supported by the mitochondrial electron transport assembly and ATP synthase, which are integral protein complexes of the inner mitochondrial membrane. The electron transport assembly is comprised of a series of protein complexes that catalyze sequential oxidation reduction reactions; some of these reactions are thermodynamically competent to support ATP production via ATP synthase provided a coupling mechanism, such as a common intermediate, is available. Proton translocation and the development of a transmembrane proton gradient provides the required coupling mechanism.

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## Principals of Reduction/Oxidation (Redox) Reactions

Redox reactions involve the transfer of electrons from one chemical species to another. The oxidized plus the reduced form of each chemical species is referred to as an **electrochemical half cell**. Two half cells having at least one common intermediate comprise a complete, coupled, redox reaction. Coupled electrochemical half cells have the thermodynamic properties of other coupled chemical reactions. If one half cell is far from electrochemical equilibrium, its tendency to achieve equilibrium (i.e., to gain or lose electrons) can be used to alter the equilibrium position of a coupled half cell. An example of a coupled redox reaction is the oxidation of NADH by the electron transport chain:



The thermodynamic potential of a chemical reaction is calculated from equilibrium constants and concentrations of reactants and products. Because it is not practical to measure electron concentrations directly, the electron energy potential of a redox system is determined from the electrical potential or voltage of the individual half cells, relative to a standard half cell. When the reactants and products of a half cell are in their standard state and the voltage is determined relative to a standard hydrogen half cell (whose voltage, by



convention, is zero), the potential observed is defined as the **standard electrode potential,  $E_0$** . If the pH of a standard cell is in the biological range, pH 7, its potential is defined as the **standard biological electrode potential** and designated  **$E_0'$** . By convention, standard electrode potentials are written as potentials for reduction reactions of half cells. The free energy of a typical reaction is calculated directly from its  $E_0'$  by the **Nernst equation** as shown below, where  $n$  is the number of electrons involved in the reaction and  $F$  is the Faraday constant (23.06 kcal/volt/mol or 94.4 kJ/volt/mol):

$$\Delta G^{0'} = -nF\Delta E_0'$$

For the oxidation of NADH, the standard biological reduction potential is -52.6 kcal/mol. With a free energy change of -52.6 kcal/mol, it is clear that NADH oxidation has the potential for driving the synthesis of a number of ATPs since the standard free energy for the reaction:



is +7.3 kcal/mole. Classically, the description of ATP synthesis through oxidation of reduced electron carriers indicated 3 moles of ATP could be generated for every mole of NADH and 2 moles for every mole of  $\text{FADH}_2$ . However, direct chemical analysis has shown that for every 2 electrons transferred from NADH to oxygen, 2.5 equivalents of ATP are synthesized and 1.5 for  $\text{FADH}_2$ .

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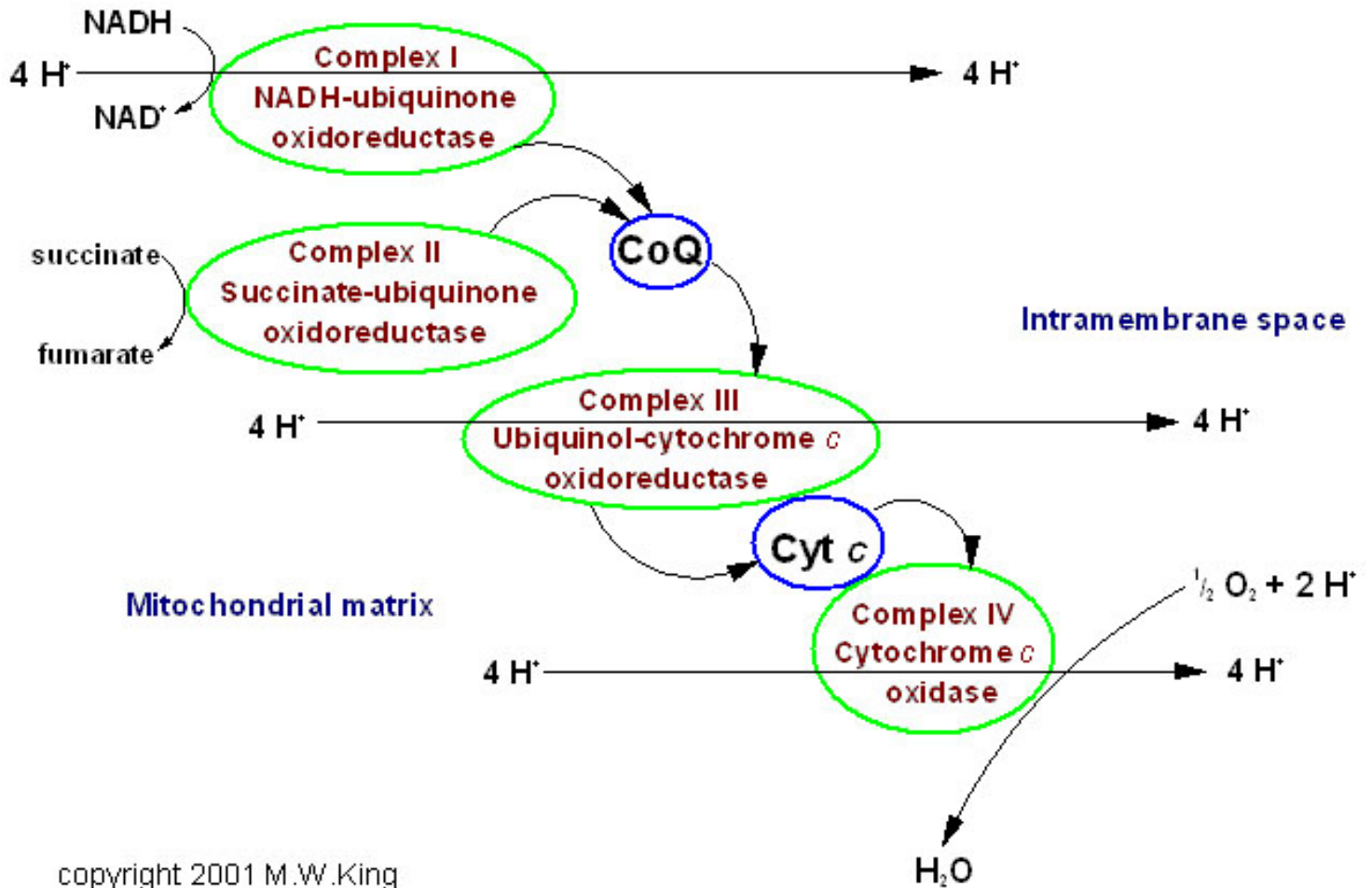
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## Complexes of the Electron Transport Chain

NADH is oxidized by a series of catalytic redox carriers that are integral proteins of the inner mitochondrial membrane. The free energy change in several of these steps is very exergonic. Coupled to these oxidation reduction steps is a transport process in which protons ( $\text{H}^+$ ) from the mitochondrial matrix are translocated to the space between the inner and outer mitochondrial membranes. The redistribution of protons leads to formation of a proton gradient across the mitochondrial membrane. The size of the gradient is proportional to the free energy change of the electron transfer reactions. The result of these reactions is that the redox energy of NADH is converted to the energy of the proton gradient. In the presence of ADP, protons flow down their thermodynamic gradient from outside the mitochondrion back into the mitochondrial matrix. This process is facilitated by a proton carrier in the inner mitochondrial membrane known as ATP synthase. As its name implies, this carrier is coupled to ATP synthesis.

Electron flow through the mitochondrial electron transport assembly is carried out through several enzyme complexes. Electrons enter the transport chain primarily from cytosolic NADH to mitochondrial NADH but can also be supplied by succinate (to mitochondrial  $\text{FADH}_2$ ) or by the [glycerol phosphate shuttle](#) via mitochondrial  $\text{FADH}_2$ .

# Flow of Electrons During Oxidative Phosphorylation



Diagrammatic representation of the flow of electrons from either NADH or succinate to oxygen (O<sub>2</sub>) in the electron transport chain of oxidative phosphorylation. Complex I contains FMN and 22-24 iron-sulfur (Fe-S) proteins in 5-7 clusters. Complex II contains FAD and 7-8 Fe-S proteins in 3 clusters and cytochrome *b*<sub>560</sub>. Complex III contains cytochrome *b*, cytochrome *c*<sub>1</sub> and one Fe-S protein. Associated with complex III by electrostatic interaction is cytochrome *c*, the ultimate electron acceptor in complex III. Complex IV contains cytochrome *a*, cytochrome *a*<sub>3</sub> and 2 copper ions. As the two electrons pass through the proteins of complex I, four protons (H<sup>+</sup>) are pumped into the intramembrane space of the mitochondrion. Similarly, four protons are pumped into the intramembrane space as each electron pair flows through complexes III and as four electrons are used to reduce O<sub>2</sub> to H<sub>2</sub>O in complex IV. The free energy released as electrons flow through complex II is insufficient to be coupled to proton pumping. These protons are returned to the matrix of the mitochondrion, down their concentration gradient, by passing through ATP synthase coupling electron flow and proton pumping to ATP synthesis.

With the exception of NADH, succinate, and CoQ, all of the components of the pathway are integral proteins of the inner mitochondrial membrane whose cofactors undergo redox reactions. NADH and succinate are soluble in the mitochondrial matrix, while CoQ is a small, mobile carrier that transfers electrons between the primary dehydrogenases and cytochrome *b*. CoQ is also restricted to the membrane phase because of its hydrophobic character.

The mitochondrial electron transport proteins are clustered into complexes (as shown above) known as **Complexes I, II, III, and IV**. Complex I, also known as **NADH:CoQ oxidoreductase**, is composed of NADH dehydrogenase with FMN as cofactor, plus non-heme-iron proteins having at least 1 iron sulfur center. Complex I is responsible for transferring electrons from NADH to CoQ. The  $\Delta E_0'$  for the latter transfer is 0.42 V, corresponding to a  $\Delta G'$  of -19 kcal/mol of electrons transferred. With its highly exergonic free energy change, the flow of electrons through Complex I is more than adequate to drive ATP synthesis. Complex II is also known as succinate dehydrogenase or succinate:CoQ oxidoreductase. The  $\Delta E_0'$  for electron flow through Complex II is about 0.05 V, corresponding to a  $\Delta G'$  of -2.3 kcal/mol of electrons transferred, which is insufficient to drive ATP synthesis. The difference in free energy of electron flow through Complexes I and II accounts for the fact that a pair of electrons originating from NADH and passing to oxygen supports production of 3 equivalents of ATP, while 2 electrons from succinate support the production of only 2 equivalents of ATP.

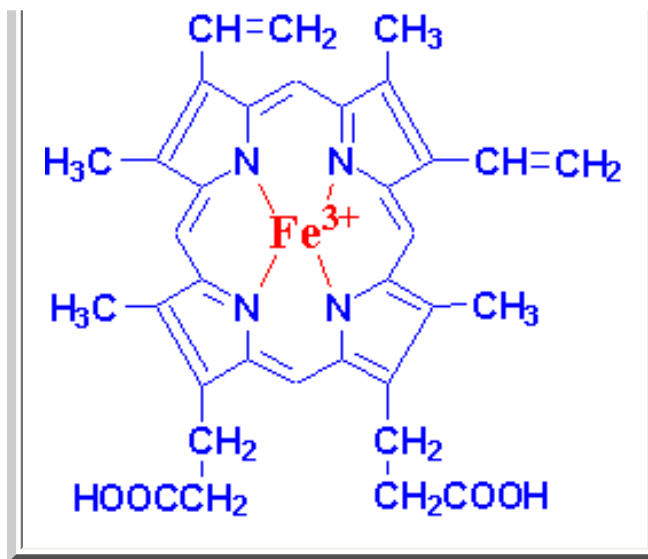
Reduced CoQ (CoQH<sub>2</sub>) diffuses in the lipid phase of the membrane and donates its electrons to Complex III, whose principal components are the heme proteins known as cytochromes *b* and *c1* and a non-heme-iron protein, known as the **Rieske iron sulfur protein**. In contrast to the heme of hemoglobin and myoglobin, the heme iron of all cytochromes participates in the cyclic redox reactions of electron transport, alternating between the oxidized (Fe<sup>+3</sup>) and reduced (Fe<sup>+2</sup>) forms. The electron carrier from Complex III to Complex IV is the smallest of the cytochromes, cytochrome *c* (molecular weight 12,000). Complex IV, also known as cytochrome oxidase, contains the heme proteins known as cytochrome *a* and cytochrome *a3*, as well as copper-containing proteins in which the copper undergoes a transition from Cu<sup>+</sup> to Cu<sup>2+</sup> during the transfer of electrons through the complex to molecular oxygen. Oxygen is the final electron acceptor, with water being the final product of oxygen reduction.

Normal oxidation of NADH or succinate is always a 2-electron reaction, with the transfer of 2 hydride ions to a flavin. A hydride ion is composed of 1 proton and 1 electron. Unlike NADH and succinate, flavins can participate in either 1-electron or 2-electron reactions; thus, flavin that is fully reduced by the dehydrogenase reactions can subsequently be oxidized by 2 sequential 1-hydride reactions. The fully reduced form of a flavin is known as the **quinol** form and the fully oxidized form is known as the **quinone** form; the intermediate containing a single electron is known as the **semiquinone** or **semiquinol** form.

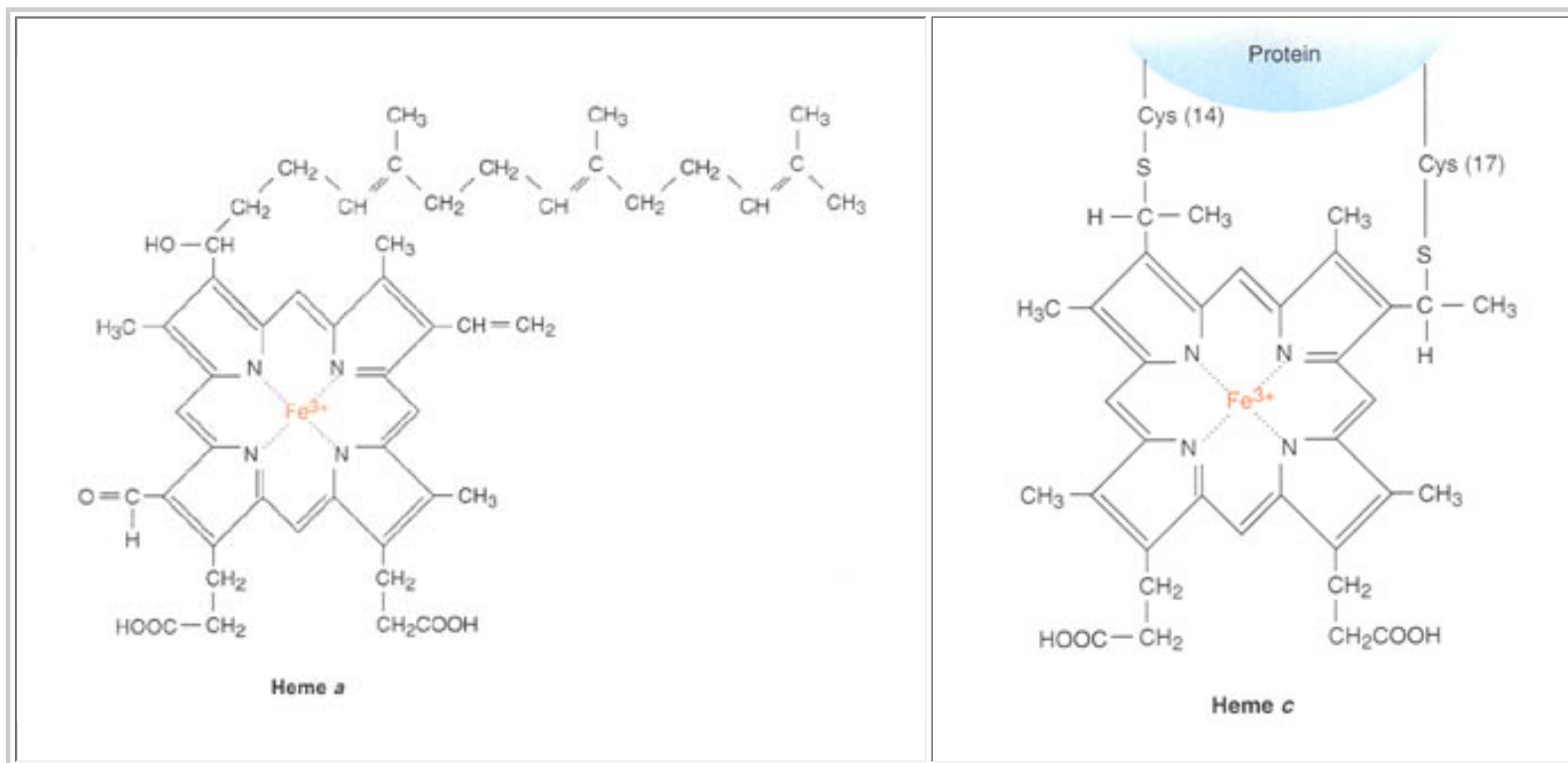
Like flavins, CoQ (also known as **ubiquinone**) can undergo either 1- or 2-electron reactions leading to formation of the reduced quinol, the oxidized quinone, and the semiquinone intermediate. The ability of flavins and CoQ to form semiquinone intermediates is a key feature of the mitochondrial electron transport systems, since these cofactors link the obligatory 2-electron reactions of NADH and succinate with the obligatory 1-electron reactions of the cytochromes.

The cytochromes are heme proteins. Like hemoglobin and myoglobin, the cytochromes generally contain 1 heme group per polypeptide---except for cytochrome *b*, which has 2 heme residues in 1 polypeptide chain. There are 3 forms of heme found in heme proteins, each of which are derived from **iron-protoporphyrin IX** also called **heme b**.





Cytochromes vary in the structure of the heme and in its binding to apoprotein. Cytochromes of the *c* type contain a modified iron protoporphyrin IX known as **heme c**. In heme *c* the 2 vinyl ( $C=C$ ) side chains are covalently bonded to cysteine sulfhydryl residues of the apoprotein. Only cytochromes of the *c* type contain covalently bound heme. **Heme a** is also a modified iron protoporphyrin IX. Heme *a* is found in cytochromes of the *a* type and in the chlorophyll of green plants.



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## Oxidative Phosphorylation

The free energy available as a consequence of transferring 2 electrons from NADH or succinate to molecular oxygen is -57 and -36 kcal/mol, respectively. Oxidative phosphorylation traps this energy as the high-energy phosphate of ATP. In order for oxidative phosphorylation to proceed, two principal conditions must be

met. First, the inner mitochondrial membrane must be physically intact so that protons can only reenter the mitochondrion by a process coupled to ATP synthesis. Second, a high concentration of protons must be developed on the outside of the inner membrane.

The energy of the proton gradient is known as the **chemiosmotic potential**, or **proton motive force (PMF)**. This potential is the sum of the concentration difference of protons across the membrane and the difference in electrical charge across the membrane. The 2 electrons from NADH generate a 6-proton gradient. Thus, oxidation of 1 mole of NADH leads to the availability of a PMF with a free energy of about -31.2 kcal ( $6 \times -5.2$  kcal). The energy of the gradient is used to drive ATP synthesis as the protons are transported back down their thermodynamic gradient into the mitochondrion.

Electrons return to the mitochondrion through the integral membrane protein known as ATP synthase (or Complex V). ATP synthase is a multiple subunit complex that binds ADP and inorganic phosphate at its catalytic site inside the mitochondrion, and requires a proton gradient for activity in the forward direction. ATP synthase is composed of 3 fragments:  $F_0$ , which is localized in the membrane;  $F_1$ , which protrudes from the inside of the inner membrane into the matrix; and oligomycin sensitivity--conferring protein (OSCP), which connects  $F_0$  to  $F_1$ . In damaged mitochondria, permeable to protons, the ATP synthase reaction is active in the reverse direction acting as a very efficient ATP hydrolase or ATPase.

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## Stoichiometry of Oxidative Phosphorylation

For each pair of electrons originating from NADH, 3 equivalents of ATP are synthesized, requiring 22.4 kcal of energy. Thus, with 31.2 kcal of available energy, it is clear that the proton gradient generated by electron transport contains sufficient energy to drive normal ATP synthesis. Electrons from succinate have about 2/3 the energy of NADH electrons: they generate PMFs that are about 2/3 as great as NADH electrons and lead to the synthesis of only 2 moles of ATP per mole of succinate oxidized.

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## Regulation of Oxidative Phosphorylation

Since electron transport is directly coupled to proton translocation, the flow of electrons through the electron transport system is regulated by the magnitude of the PMF. The higher the PMF the lower the rate of electron transport, and vice versa. Under resting conditions, with a high cell energy charge, the demand for new synthesis of ATP is limited and, although the PMF is high, flow of protons back into the mitochondria through ATP synthase is minimal. When energy demands are increased, such as during vigorous muscle activity, cytosolic ADP rises and is exchanged with intramitochondrial ATP via the transmembrane adenine nucleotide carrier ADP/ATP translocase. Increased intramitochondrial concentrations of ADP cause the PMF to become discharged as protons pour through ATP synthase, regenerating the ATP pool. Thus, while the rate of electron transport is dependent on the PMF, the magnitude of the PMF at any moment simply reflects the energy charge of the cell. In turn the energy charge, or more precisely ADP concentration, normally determines the rate of electron transport by mass action principles. The rate of electron transport is usually measured by assaying the rate of oxygen consumption and is referred to as the **cellular respiratory rate**. The respiratory rate is known as the **state 4** rate when the energy charge is high, the concentration of ADP is low, and electron transport is limited by ADP. When ADP levels rise and inorganic phosphate is available, the flow of protons through ATP synthase is elevated and higher rates of electron transport are observed; the resultant respiratory rate is known as the **state 3** rate. Thus, under physiological conditions mitochondrial respiratory activity cycles between state 3 and state 4 rates.

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## Inhibitors of Oxidative Phosphorylation



The pathway of electron flow through the electron transport assembly, and the unique properties of the PMF, have been determined through the uses of a number of important antimetabolites. Some of these agents are inhibitors of electron transport at specific sites in the electron transport assembly, while others stimulate electron transport by discharging the proton gradient. For example, antimycin A is a specific inhibitor of cytochrome *b*. In the presence of antimycin A, cytochrome *b* can be reduced but not oxidized. As expected, in the presence of cytochrome *c* remains oxidized in the presence of antimycin A, as do the downstream cytochromes *a* and *a<sub>3</sub>*.

An important class of antimetabolites are the uncoupling agents exemplified by 2,4-dinitrophenol (DNP). Uncoupling agents act as lipophilic weak acids, associating with protons on the exterior of mitochondria, passing through the membrane with the bound proton, and dissociating the proton on the interior of the mitochondrion. These agents cause maximum respiratory rates but the electron transport generates no ATP, since the translocated protons do not return to the interior through ATP synthase.

## Inhibitors of Oxidative Phosphorylation

| Name               | Function                           | Site of Action                       |
|--------------------|------------------------------------|--------------------------------------|
| Rotenone           | e <sup>-</sup> transport inhibitor | Complex I                            |
| Amytal             | e <sup>-</sup> transport inhibitor | Complex I                            |
| Antimycin A        | e <sup>-</sup> transport inhibitor | Complex III                          |
| Cyanide            | e <sup>-</sup> transport inhibitor | Complex IV                           |
| Carbon Monoxide    | e <sup>-</sup> transport inhibitor | Complex IV                           |
| Azide              | e <sup>-</sup> transport inhibitor | Complex IV                           |
| 2,4,-dinitrophenol | Uncoupling agent                   | transmembrane H <sup>+</sup> carrier |
| Pentachlorophenol  | Uncoupling agent                   | transmembrane H <sup>+</sup> carrier |
| Oligomycin         | Inhibits ATP synthase              | OSCP fraction of ATP synthase        |

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## Energy from Cytosolic NADH

In contrast to oxidation of mitochondrial NADH, cytosolic NADH when oxidized via the electron transport system gives rise to 2 equivalents of ATP if it is oxidized by the [glycerol phosphate shuttle](#) and 3 ATPs if it



proceeds via the [malate aspartate shuttle](#). The glycerol phosphate shuttle is coupled to an inner mitochondrial membrane, FAD-linked dehydrogenase, of low energy potential like that found in Complex II. Thus, cytosolic NADH oxidized by this pathway can generate only 2 equivalents of ATP. The shuttle involves two different glycerol-3-phosphate dehydrogenases: one is cytosolic, acting to produce glycerol-3-phosphate, and one is an integral protein of the inner mitochondrial membrane that acts to oxidize the glycerol-3-phosphate produced by the cytosolic enzyme. The net result of the process is that reducing equivalents from cytosolic NADH are transferred to the mitochondrial electron transport system. The catalytic site of the mitochondrial glycerol phosphate dehydrogenase is on the outer surface of the inner membrane, allowing ready access to the product of the second, or cytosolic, glycerol-3-phosphate dehydrogenase.

In some tissues, such as that of heart and muscle, mitochondrial glycerol-3-phosphate dehydrogenase is present in very low amounts, and the malate aspartate shuttle is the dominant pathway for aerobic oxidation of cytosolic NADH. In contrast to the glycerol phosphate shuttle, the malate aspartate shuttle generates 3 equivalents of ATP for every cytosolic NADH oxidized.

In action, NADH efficiently reduces oxaloacetate (OAA) to malate via cytosolic malate dehydrogenase (MDH). Malate is transported to the interior of the mitochondrion via the  $\alpha$ -ketoglutarate/malate antiporter. Inside the mitochondrion, malate is oxidized by the MDH of the TCA cycle, producing OAA and NADH. In this step the cytosolic, NADH-derived reducing equivalents become available to the NADH dehydrogenase of the inner mitochondrial membrane and are oxidized, giving rise to 3 ATPs as described earlier. The mitochondrial transaminase uses glutamate to convert membrane-impermeable OAA to aspartate and  $\alpha$ -ketoglutarate. This provides a pool of  $\alpha$ -ketoglutarate for the aforementioned antiporter. The aspartate which is also produced is translocated out of the mitochondrion.

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## Other Biological Oxidations

Oxidase complexes, like cytochrome oxidase, transfer electrons directly from NADH and other substrates to oxygen, producing water. Oxygenases, widely localized in membranes of the endoplasmic reticulum, catalyze the addition of molecular oxygen to organic molecules. There are 2 kinds of oxygenase complexes, monooxygenases and dioxygenases. Dioxygenases add the 2 atoms of molecular oxygen ( $O_2$ ) to carbon and nitrogen of organic compounds. Monooxygenase complexes play a key role in detoxifying drugs and other compounds (e.g., PCBs and dioxin) and in the normal metabolism of steroids, fatty acids and fat soluble vitamins. Monooxygenases act by sequentially transferring 2 electrons from NADH or NADPH to 1 of the 2 atoms of oxygen in  $O_2$ , generating  $H_2O$  from 1 oxygen atom and incorporating the other oxygen atom into an organic compound as a hydroxyl group (R-OH). The hydroxylated products are markedly more water-soluble than their precursors and are much more readily excreted from the body. Widely used synonyms for the monooxygenases are: mixed function oxidases, hydroxylases, and mixed function hydroxylases.

The chief components of monooxygenase complexes include cytochrome *b5*, cytochrome P450, and cytochrome P450 reductase, which contains FAD plus FMN. There are many P450 isozymes; for example, up to 50 different P450 gene products can be found in liver, where the bulk of drug metabolism occurs. Some of these same gene products are also found in other tissues, where they are responsible for tissue-specific oxygenase activities. P450 reducing equivalents arise either from NADH via cytochrome *b5* or from NADPH via cytochrome P450 reductase, both of which are associated with cytochrome P450 in the membrane-localized complexes.

Enzymatic reactions involving molecular oxygen usually produce water or organic oxygen in well regulated reactions having specific products. However, under some metabolic conditions (e.g., reperfusion of anaerobic tissues) unpaired electrons gain access to molecular oxygen in unregulated, non-enzymatic reactions. The products, called **free radicals**, are quite toxic. These free radicals, especially hydroxy radical, randomly attack all cell components, including proteins, lipids and nucleic acids, potentially causing extensive cellular damage. Tissues are replete with enzymes to protect against the random chemical reactions that these

free radicals initiate. Several free radical scavenging enzymes have been identified.

Superoxide dismutases (SODs) in animals contain either zinc ( $\text{Zn}^{2+}$ ) and copper ( $\text{Cu}^{2+}$ ), known as CuZnSOD, or manganese ( $\text{Mn}^{2+}$ ) as in the case of the mitochondrial form. These SODs convert superoxide to peroxide and thereby minimizes production of hydroxy radical, the most potent of the oxygen free radicals. Peroxides produced by SOD are also toxic. They are detoxified by conversion to water via the enzyme peroxidase. The best known mammalian peroxidase is glutathione peroxidase, which contains the modified amino acid [selenocysteine](#) in its reactive center.

Glutathione (see the [Pentose Phosphate Page](#)) is important in maintaining the normal reduction potential of cells and provides the reducing equivalents for glutathione peroxidase to convert hydrogen peroxide to water. In red blood cells the lack of glutathione leads to extensive peroxide attack on the plasma membrane, producing fragile red blood cells that readily undergo hemolysis.

Catalase (located in peroxisomes) provides a reductant route for the degradation of hydrogen peroxide. Mammalian catalase has the highest turnover number of any documented enzyme.

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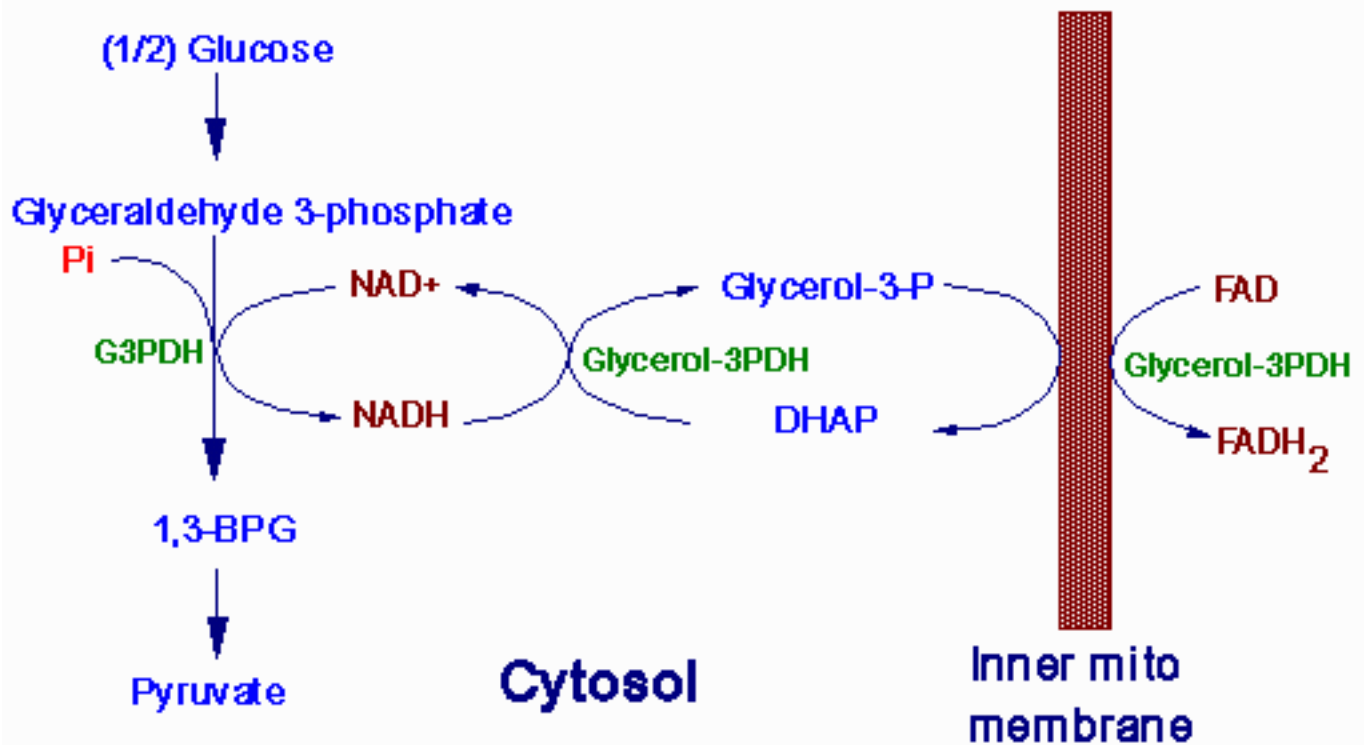
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## Glycerol Phosphate Shuttle



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The glycerol phosphate shuttle is a secondary mechanism for the transport of electrons from cytosolic NADH to mitochondrial carriers of the oxidative phosphorylation pathway. The primary cytoplasmic NADH electron shuttle is the [malate-aspartate shuttle](#). Two enzymes are involved in this shuttle. One is the cytosolic version of the enzyme **glycerol-3-phosphate dehydrogenase (glycerol-3-PDH)** which has as one substrate, NADH. The second is the mitochondrial form of the enzyme which has as one of its' substrates,  $\text{FAD}^+$ . The net result is that there is a continual conversion of the glycolytic intermediate, DHAP and glycerol-3-phosphate with the concomitant transfer of the electrons from reduced cytosolic NADH to mitochondrial oxidized  $\text{FAD}^+$ . Since the electrons from mitochondrial  $\text{FADH}_2$  feed into the [oxidative phosphorylation pathway](#) at coenzyme Q (as opposed to **NADH-ubiquinone oxidoreductase** [complex I]) only 2 moles of ATP will be generated from glycolysis. G3PDH is **glyceraldehyde-3-phosphate dehydrogenase**.

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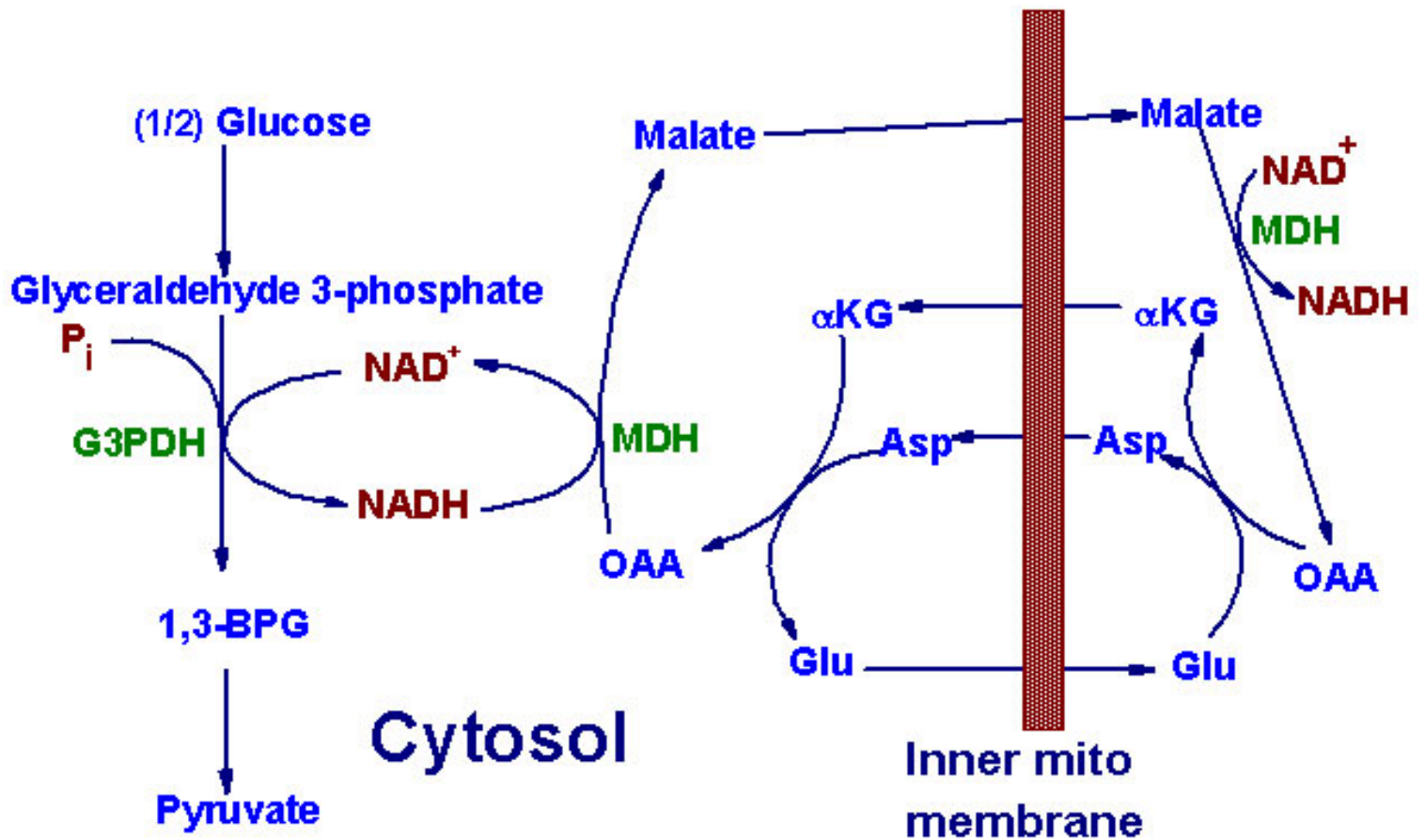
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Last modified: **Tuesday, 12-Aug-2003 20:02:03 EST**



The malate/aspartate shuttle is the principal mechanism for the movement of reducing equivalents (in the form of NADH) from the cytoplasm to the mitochondria. The glycolytic pathway is a primary source of NADH. Within the mitochondria the electrons of NADH can be coupled to ATP production during the process of **oxidative phosphorylation**. The electrons are "carried" into the mitochondria in the form of malate. Cytoplasmic **malate dehydrogenase** (MDH) reduces oxaloacetate (OAA) to malate while oxidizing NADH to NAD<sup>+</sup>. Malate then enters the mitochondria where the reverse reaction is carried out by mitochondrial MDH. Movement of mitochondrial OAA to the cytoplasm to maintain this cycle requires it be transaminated to aspartate (Asp) with the amino group being donated by glutamate (Glu). The Asp then leaves the mitochondria and enters the cytoplasm. The deamination of glutamate generates  $\alpha$ -ketoglutarate ( $\alpha$ -KG) which leaves the mitochondria for the cytoplasm. All the participants in the cycle are present in the proper cellular compartment for the shuttle to function due to concentration dependent movement. When the energy level of the cell rises the rate of mitochondrial oxidation of NADH to NAD<sup>+</sup> declines and therefore, the shuttle slows. G3PDH is **glyceraldehyde-3-phosphate dehydrogenase**.

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- [Fructose-1,6-bisphosphate to Fructose-6-phosphate, Bypass 2](#)
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- [Substrates for Gluconeogenesis](#)
- [Regulation of Gluconeogenesis](#)

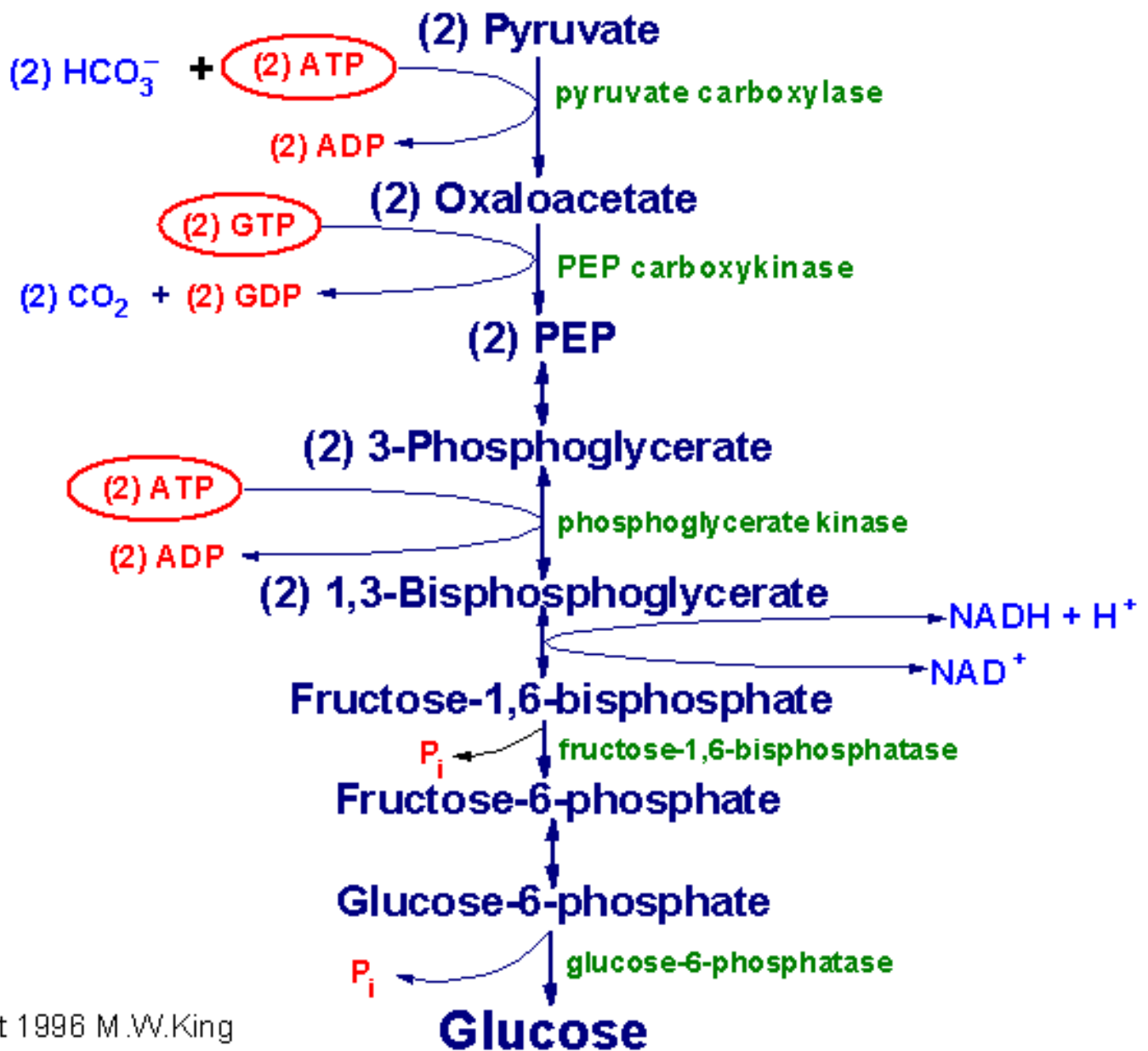
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## Introduction

Gluconeogenesis is the biosynthesis of new glucose, (i.e. not glucose from glycogen). The production of glucose from other metabolites is necessary for use as a fuel source by the **brain, testes, erythrocytes** and **kidney medulla** since glucose is the sole energy source for these organs. During starvation, however, the brain can derive energy from **ketone bodies** which are converted to acetyl-CoA.

Synthesis of glucose from three and four carbon precursors is essentially a reversal of **glycolysis**. The relevant features of the pathway of gluconeogenesis are diagrammed below.



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The relevant reactions of gluconeogenesis are depicted. The enzymes of the 3 bypass steps are indicated in green along with phosphoglycerate kinase. This latter enzyme is included since when functioning in the gluconeogenic direction the reaction consumes energy. Gluconeogenesis from 2 moles of pyruvate to 2 moles of 1,3-bisphosphoglycerate consumes 6 moles of ATP. This makes the process of gluconeogenesis very costly from an energy standpoint considering that glycolysis to 2 moles of pyruvate only yields 2 moles of ATP. Note that several steps are required in going from 2 moles of 1,3-bisphosphoglycerate to 1 mole of fructose-1,6-bisphosphate. First there is a reversal of the glyceraldehyde-3-phosphate dehydrogenase reaction which requires a supply of NADH. When lactate is the gluconeogenic substrate the NADH is supplied by the lactate dehydrogenase reaction, and it is supplied by the malate dehydrogenase reaction when pyruvate is the substrate. Secondly, 1 mole of glyceraldehyde-3-phosphate must be isomerized to DHAP and then a mole of DHAP can be condensed to a mole of glyceraldehyde-3-phosphate to form 1 mole of fructose-1,6-bisphosphate in a reversal of the aldolase reaction. Most non-hepatic tissues lack glucose-6-phosphatase and so the glucose-6-phosphate generated in these tissues would be a substrate for glycogen synthesis. In hepatocytes the glucose-6-phosphatase reactions allows

the liver to supply the blood with free glucose. Remember that due to the high  $K_m$  of liver glucokinase most of the glucose will not be phosphorylated and will flow down its' concentration gradient out of hepatocytes and into the blood.

The three reactions of glycolysis that proceed with a large negative free energy change are bypassed during gluconeogenesis by using different enzymes. These three are the pyruvate kinase, phosphofructokinase-1(PFK-1) and hexokinase/glucokinase catalyzed reactions. In the liver or kidney cortex and in some cases skeletal muscle, the glucose-6-phosphate (G6P) produced by gluconeogenesis can be incorporated into glycogen. In this case the third bypass occurs at the glycogen phosphorylase catalyzed reaction. Since skeletal muscle lacks glucose-6-phosphatase it cannot deliver free glucose to the blood and undergoes gluconeogenesis exclusively as a mechanism to generate glucose for storage as glycogen.

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## Pyruvate to Phosphoenolpyruvate (PEP), Bypass 1

Conversion of pyruvate to PEP requires the action of two mitochondrial enzymes. The first is an ATP-requiring reaction catalyzed by pyruvate carboxylase, (PC). As the name of the enzyme implies, **pyruvate** is carboxylated to form **oxaloacetate (OAA)**. The  $\text{CO}_2$  in this reaction is in the form of **bicarbonate** ( $\text{HCO}_3^-$ ). This reaction is an **anaplerotic** reaction since it can be used to fill-up the **TCA cycle**. The second enzyme in the conversion of pyruvate to PEP is PEP carboxykinase (PEPCK). PEPCK requires GTP in the decarboxylation of OAA to yield PEP. Since PC incorporated  $\text{CO}_2$  into pyruvate and it is subsequently released in the PEPCK reaction, no net fixation of carbon occurs. Human cells contain almost equal amounts of mitochondrial and cytosolic PEPCK so this second reaction can occur in either cellular compartment.

For gluconeogenesis to proceed, the OAA produced by PC needs to be transported to the cytosol. However, no transport mechanism exist for its' direct transfer and OAA will not freely diffuse. Mitochondrial OAA can become cytosolic via three pathways, conversion to PEP (as indicated above through the action of the mitochondrial PEPCK), transamination to aspartate or reduction to malate, all of which are transported to the cytosol.

If OAA is converted to PEP by mitochondrial PEPCK, it is transported to the cytosol where it is a direct substrate for gluconeogenesis and nothing further is required. Transamination of OAA to aspartate allows the aspartate to be transported to the cytosol where the reverse transamination occurs yielding cytosolic OAA. This transamination reaction requires continuous transport of glutamate into, and  $\alpha$ -ketoglutarate out of, the mitochondrion. Therefore, this process is limited by the availability of these other substrates. Either of these latter two reactions will predominate when the substrate for gluconeogenesis is **lactate**. Whether mitochondrial decarboxylation or transamination occurs is a function of the availability of PEPCK or transamination intermediates.

Mitochondrial OAA can also be reduced to malate in a reversal of the TCA cycle reaction catalyzed by malate dehydrogenase (MDH). The reduction of OAA to malate requires NADH, which will be accumulating in the mitochondrion as the energy charge increases. The increased energy charge will allow cells to carry out the ATP costly process of gluconeogenesis. The resultant malate is transported to the cytosol where it is oxidized to OAA by cytosolic MDH which requires  $\text{NAD}^+$  and yields NADH. The NADH produced during the cytosolic oxidation of malate to OAA is utilized during the

glyceraldehyde-3-phosphate dehydrogenase reaction of glycolysis. The coupling of these two oxidation-reduction reactions is required to keep gluconeogenesis functional when pyruvate is the principal source of carbon atoms. The conversion of OAA to malate predominates when pyruvate (derived from glycolysis or amino acid catabolism) is the source of carbon atoms for gluconeogenesis. When in the cytoplasm, OAA is converted to PEP by the cytosolic version of PEPCK. Hormonal signals control the level of PEPCK protein as a means to regulate the flux through gluconeogenesis (see below).

The net result of the PC and PEPCK reactions is:



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## Fructose-1,6-bisphosphate to Fructose-6-phosphate, Bypass 2

Fructose-1,6-bisphosphate (F1,6BP) conversion to fructose-6-phosphate (F6P) is the reverse of the rate limiting step of glycolysis. The reaction, a simple hydrolysis, is catalyzed by fructose-1,6-bisphosphatase (F1,6BPase). Like the [regulation of glycolysis](#) occurring at the PFK-1 reaction, the F1,6BPase reaction is a major point of control of gluconeogenesis (see below).

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## Glucose-6-phosphate (G6P) to Glucose (or Glycogen), Bypass 3

G6P is converted to glucose through the action of glucose-6-phosphatase (G6Pase). This reaction is also a simple hydrolysis reaction like that of F1,6BPase. Since the brain and skeletal muscle, as well as most non-hepatic tissues, lack G6Pase activity, any gluconeogenesis that occurs in these tissues is not utilized for blood glucose supply. In the kidney, muscle and especially the liver, G6P can be shunted toward glycogen if blood glucose levels are adequate. The reactions necessary for glycogen synthesis are an alternate **bypass 3** series of reactions.

Phosphorolysis of glycogen is carried out by glycogen phosphorylase, whereas, [glycogen synthesis](#) is catalyzed by glycogen synthase. The G6P produced from gluconeogenesis can be converted to glucose-1-phosphate (G1P) by phosphoglucose mutase (PGM). G1P is then converted to UDP-glucose (the substrate for glycogen synthase) by UDP-glucose pyrophosphorylase, a reaction requiring hydrolysis of UTP.

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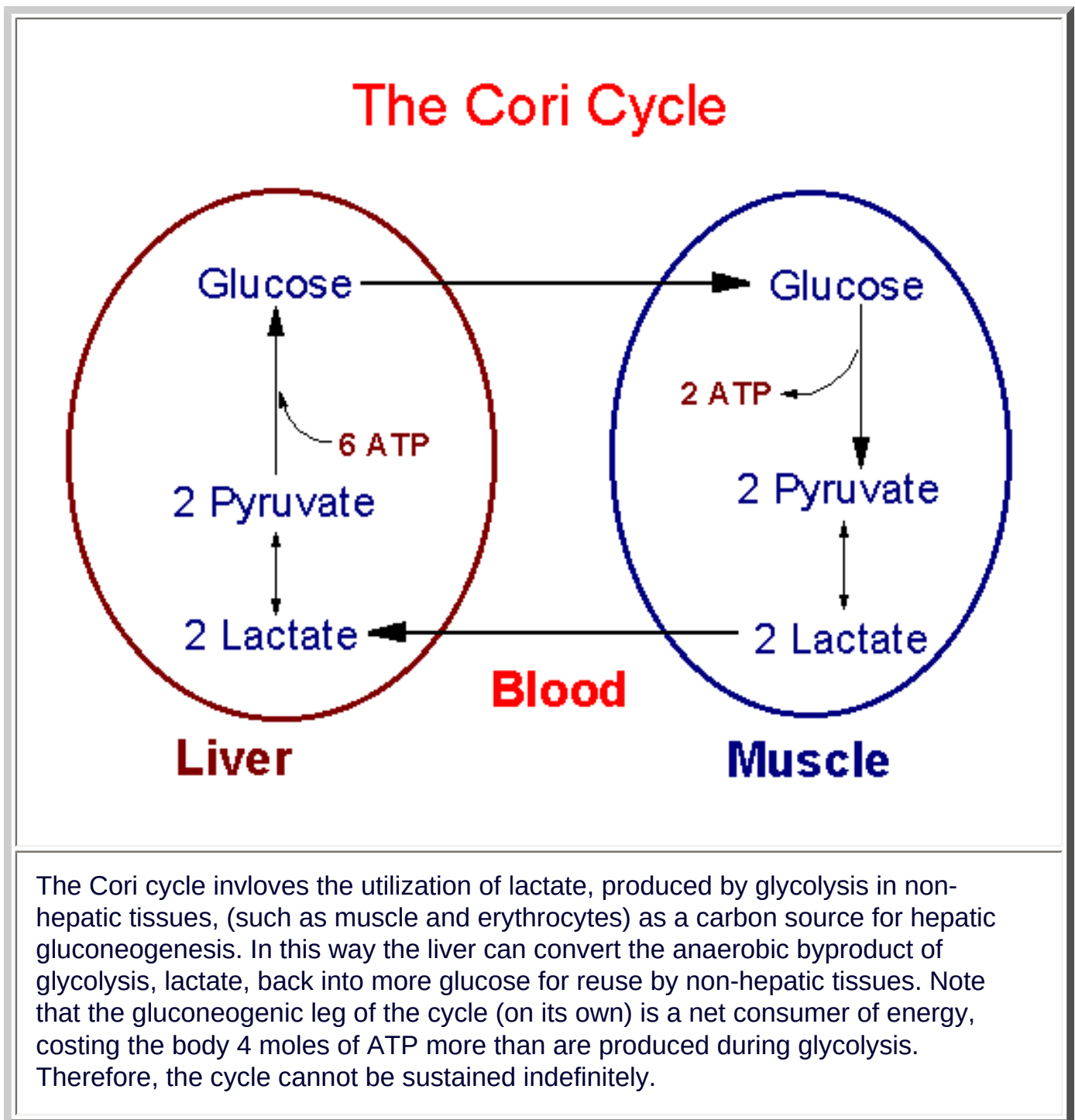
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## Substrates for Gluconeogenesis

### Lactate:

Lactate is a predominate source of carbon atoms for glucose synthesis by gluconeogenesis. During anaerobic glycolysis in skeletal muscle, pyruvate is reduced to lactate by lactate dehydrogenase (LDH). This reaction serves two critical functions during anaerobic glycolysis. First, in the direction of lactate formation the LDH reaction requires NADH and yields NAD<sup>+</sup> which is then available for use by

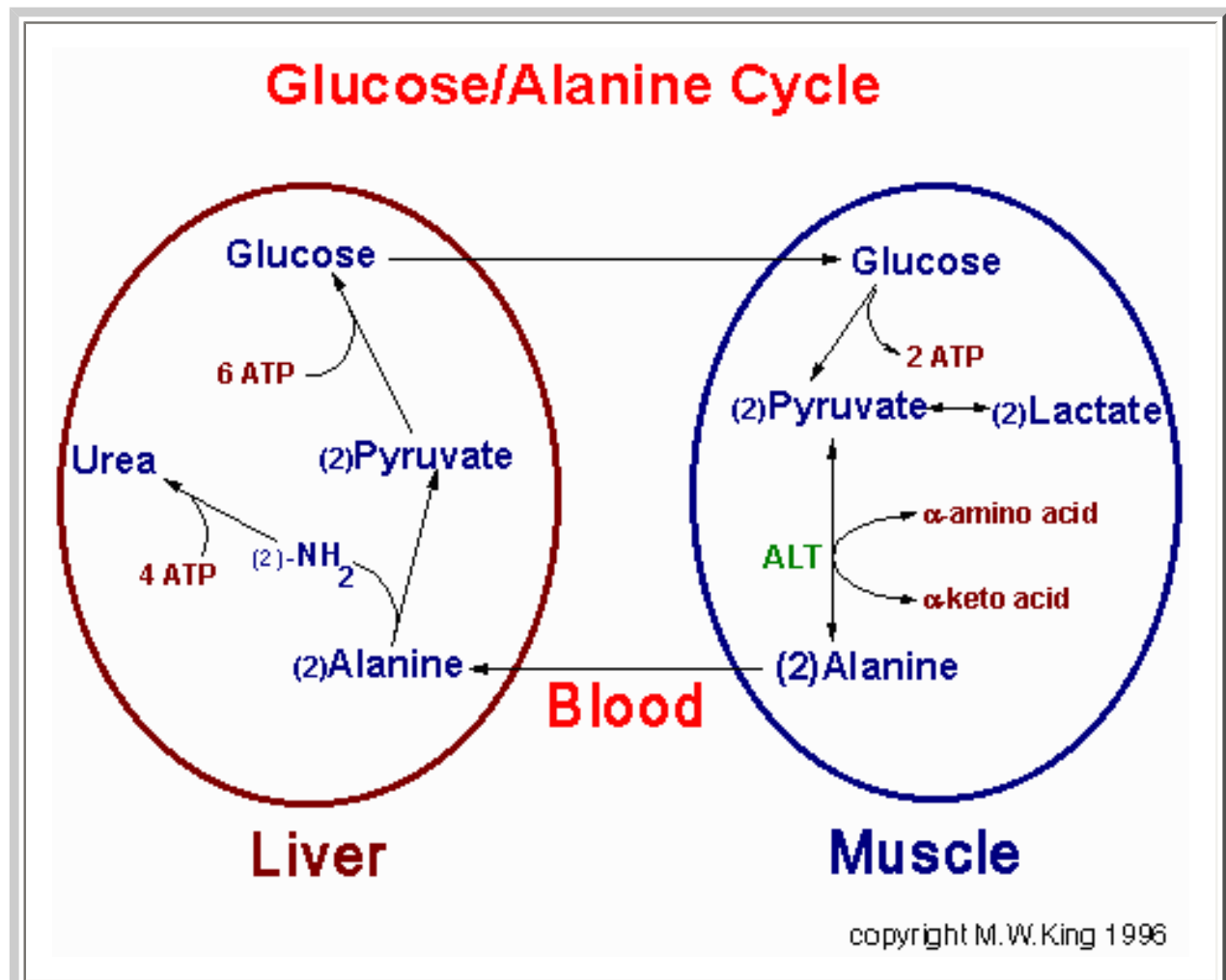
the glyceraldehyde-3-phosphate dehydrogenase reaction of glycolysis. These two reaction are, therefore, intimately coupled during anaerobic glycolysis. Secondly, the lactate produced by the LDH reaction is released to the blood stream and transported to the liver where it is converted to glucose. The glucose is then returned to the blood for use by muscle as an energy source and to replenish glycogen stores. This cycle is termed the **Cori cycle**.



## Pyruvate:

Pyruvate, generated in muscle and other peripheral tissues, can be transaminated to alanine which is returned to the liver for gluconeogenesis. The transamination reaction requires an  $\alpha$ -amino

acid as donor of the amino group, generating an  $\alpha$ -keto acid in the process. This pathway is termed the **glucose-alanine cycle**. Although the majority of amino acids are degraded in the liver some are deaminated in muscle. The glucose-alanine cycle is, therefore, an indirect mechanism for muscle to eliminate nitrogen while replenishing its energy supply. However, the major function of the glucose-alanine cycle is to allow non-hepatic tissues to deliver the amino portion of catabolized amino acids to the liver for excretion as urea. Within the liver the alanine is converted back to pyruvate and used as a gluconeogenic substrate (if that is the hepatic requirement) or oxidized in the TCA cycle. The amino nitrogen is converted to urea in the **urea cycle** and excreted by the kidneys.



The glucose-alanine cycle is used primarily as a mechanism for skeletal muscle to eliminate nitrogen while replenishing its energy supply. Glucose oxidation produces pyruvate which can undergo transamination to alanine. This reaction is catalyzed by alanine transaminase, ALT (ALT used to be referred to a serum glutamate-pyruvate transaminase, SGPT). Additionally, during periods of fasting, skeletal muscle protein is degraded for the energy value of the amino acid carbons and alanine is a major amino acid in protein. The alanine then enters the blood stream and is transported to the liver. Within the liver alanine is converted back to pyruvate which is then a source of carbon atoms for gluconeogenesis. The newly formed glucose can then enter the blood for delivery back to the muscle. The amino group transported from the muscle to the liver in the form of alanine is converted to urea in the



urea cycle and excreted.

## Amino Acids:

All 20 of the amino acids, excepting leucine and lysine, can be degraded to TCA cycle intermediates as discussed in the [metabolism of amino acids](#). This allows the carbon skeletons of the amino acids to be converted to those in oxaloacetate and subsequently into pyruvate. The pyruvate thus formed can be utilized by the gluconeogenic pathway. When glycogen stores are depleted, in muscle during exertion and liver during fasting, catabolism of muscle proteins to amino acids contributes the major source of carbon for maintenance of blood glucose levels.

## Glycerol:

Oxidation of fatty acids yields enormous amounts of energy on a molar basis, however, the carbons of the fatty acids cannot be utilized for net synthesis of glucose. The two carbon unit of acetyl-CoA derived from [β-oxidation of fatty acids](#) can be incorporated into the TCA cycle, however, during the TCA cycle two carbons are lost as CO<sub>2</sub>. Thus, explaining why fatty acids do not undergo net conversion to carbohydrate.

The glycerol backbone of lipids can be used for gluconeogenesis. This requires phosphorylation to **glycerol-3-phosphate** by glycerol kinase and dehydrogenation to dihydroxyacetone phosphate (DHAP) by glyceraldehyde-3-phosphate dehydrogenase (G3PDH). The G3PDH reaction is the same as that used in the transport of cytosolic reducing equivalents into the mitochondrion for use in oxidative phosphorylation. This transport pathway is called the [glycerol-phosphate shuttle](#). The glycerol backbone of adipose tissue stored triacylglycerols is ensured of being used as a gluconeogenic substrate since adipose cells lack glycerol kinase. In fact adipocytes require a basal level of glycolysis in order to provide them with DHAP as an intermediate in the [synthesis of triacylglycerols](#).

## Propionate:

Oxidation of [fatty acids with an odd number of carbon atoms](#) and the [oxidation of some amino acids](#) generates as the terminal oxidation product, **propionyl-CoA**. Propionyl-CoA is converted to the TCA intermediate, **succinyl-CoA**. This conversion is carried out by the ATP-requiring enzyme, propionyl-CoA carboxylase then methylmalonyl-CoA epimerase and finally the [vitamin B<sub>12</sub>](#) requiring enzyme, methylmalonyl-CoA mutase. The utilization of propionate in gluconeogenesis only has quantitative significance in ruminants.

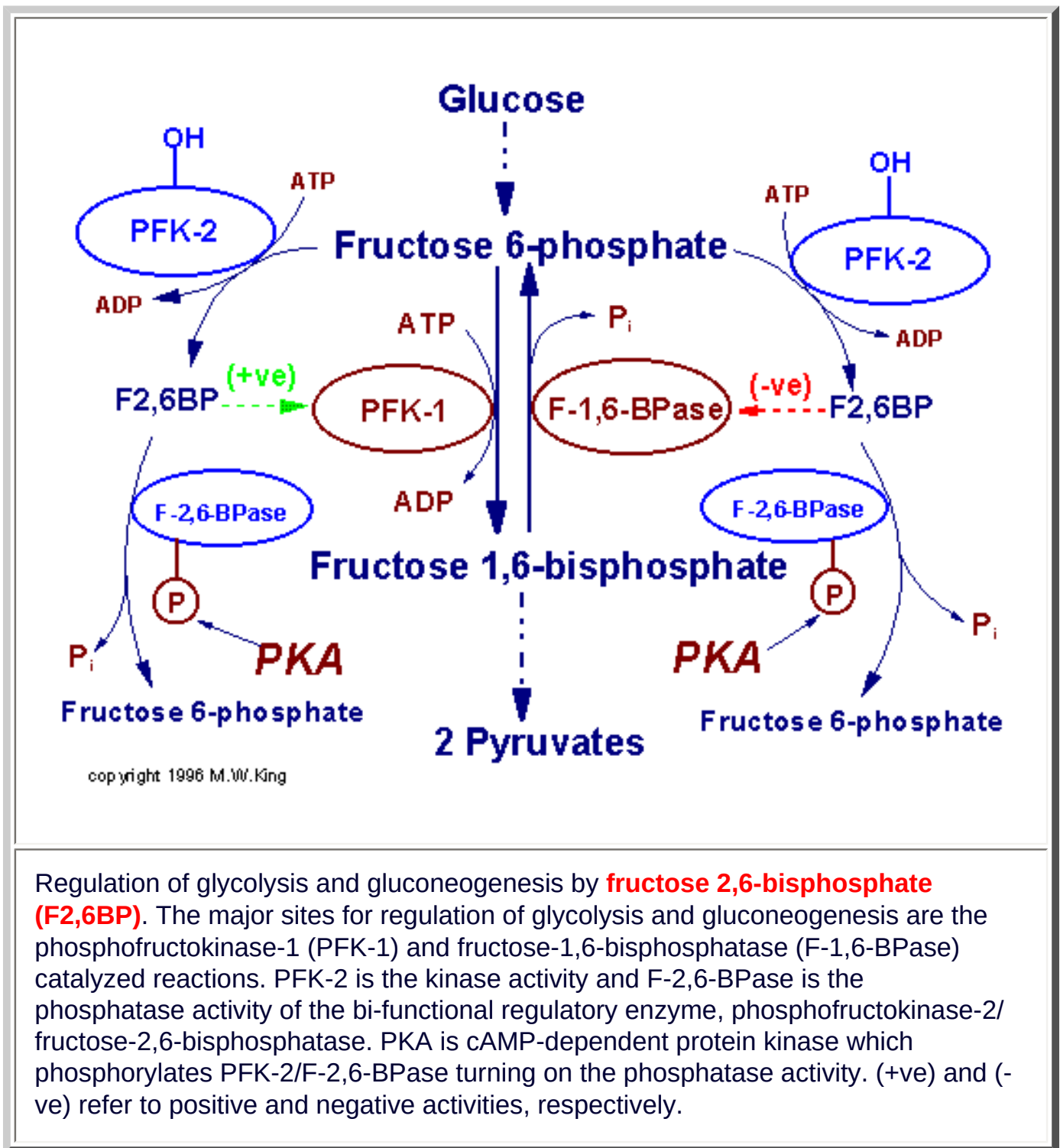
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## Regulation of Gluconeogenesis

Obviously the regulation of gluconeogenesis will be in direct contrast to the regulation of glycolysis. In general, negative effectors of glycolysis are positive effectors of gluconeogenesis. Regulation of the activity of PFK-1 and F1,6BPase is the most significant site for controlling the flux

toward glucose oxidation or glucose synthesis. As described in [control of glycolysis](#), this is predominantly controlled by **fructose-2,6-bisphosphate, F2,6BP** which is a powerful negative allosteric effector of F1,6BPase activity.



The level of F2,6BP will decline in hepatocytes in response to glucagon stimulation as well as stimulation by catecholamines. Each of these signals is elicited through activation of cAMP-dependent protein kinase (PKA). One substrate for PKA is PFK-2, the bifunctional enzyme responsible for the synthesis and hydrolysis of F2,6BP. When PFK-2 is phosphorylated by PKA it acts as a phosphatase leading to the dephosphorylation of F2,6BP with a concomitant increase in F1,6BPase activity and a

decrease in PFK-1 activity. Secondly, F1,6Bpase activity is regulated by the ATP/ADP ratio. When this is high, gluconeogenesis can proceed maximally.

Gluconeogenesis is also controlled at the level of the pyruvate to PEP bypass. The hepatic signals elicited by glucagon or epinephrine lead to phosphorylation and inactivation of pyruvate kinase (PK) which will allow for an increase in the flux through gluconeogenesis. PK is also allosterically inhibited by ATP and alanine. The former signals adequate energy and the latter that sufficient substrates for gluconeogenesis are available. Conversely, a reduction in energy levels as evidenced by increasing concentrations of ADP lead to inhibition of both PK and PEPCK. Allosteric activation of PK occurs through acetyl-CoA. Each of these regulations occurs on a short time scale, whereas long-term regulation can be effected at the level of PEPCK. The amount of this enzyme increases in response to prolonged glucagon stimulation. This situation would occur in a starving individual or someone with an inadequate diet.

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## Introduction

Utilization of dietary lipids requires that they first be absorbed through the intestine. As these molecules are oils they would be essentially insoluble in the aqueous intestinal environment. Solubilization (emulsification) of dietary lipid is accomplished via bile salts that are synthesized in the liver and secreted from the gallbladder.

The emulsified fats can then be degraded by pancreatic lipases (lipase and phospholipase A<sub>2</sub>). These enzymes, secreted into the intestine from the pancreas, generate free fatty acids and a mixture of mono- and diacylglycerols from dietary triacylglycerols. Pancreatic lipase degrades triacylglycerols at the 1 and 3 positions sequentially to generate 1,2-diacylglycerols and 2-acylglycerols. Phospholipids are degraded at the 2 position by pancreatic phospholipase A<sub>2</sub> releasing a free fatty acid and the lysophospholipid.

Following absorption of the products of pancreatic lipase by the intestinal mucosal cells, the resynthesis of triacylglycerols occurs. The triacylglycerols are then solubilized in lipoprotein complexes (complexes of lipid and protein) called **chylomicrons**. A chylomicron contains lipid droplets surrounded by the more polar lipids and finally a layer of proteins. Triacylglycerols synthesized in the liver are packaged into VLDLs and released into the blood directly. Chylomicrons from the intestine are then released into the blood via the lymph system for delivery to the various tissues for storage or production of energy through oxidation.

The triacylglycerol components of VLDLs and chylomicrons are hydrolyzed to free fatty acids and glycerol in the capillaries of adipose tissue and skeletal muscle by the action of lipoprotein lipase. The free fatty acids are then absorbed by the cells and the glycerol is returned via the blood to the liver (and kidneys). The glycerol is then converted to the glycolytic

intermediate DHAP.

The classification of blood lipids is distinguished based upon the density of the different lipoproteins. As lipid is less dense than protein, the lower the density of lipoprotein the less protein there is.

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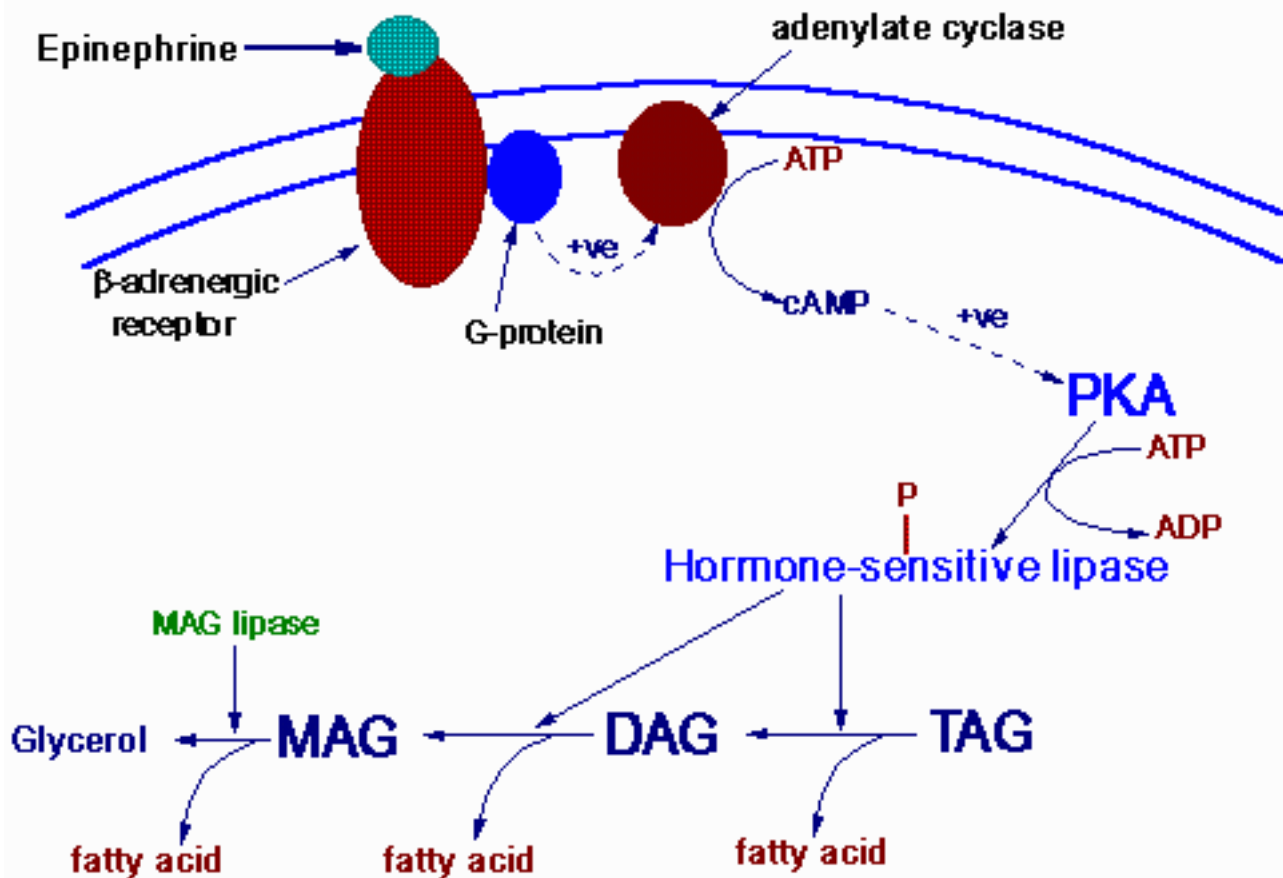
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## Mobilization of Fat Stores

The primary sources of fatty acids for oxidation are dietary and mobilization from cellular stores. Fatty acids from the diet can be delivered from the gut to cells via transport in the blood. Fatty acids are stored in the form of **triacylglycerols** primarily within adipocytes of adipose tissue. In response to energy demands, the fatty acids of stored triacylglycerols can be mobilized for use by peripheral tissues. The release of metabolic energy, in the form of fatty acids, is controlled by a complex series of interrelated cascades that result in the activation of hormone-sensitive lipase.

The stimulus to activate this cascade, in adipocytes, can be glucagon, epinephrine or  $\beta$ -corticotropin. These hormones bind cell-surface receptors that are coupled to the activation of adenylate cyclase upon ligand binding. The resultant increase in cAMP leads to activation of PKA, which in turn phosphorylates and activates hormone-sensitive lipase. This enzyme hydrolyzes fatty acids from carbon atoms 1 or 3 of triacylglycerols. The resulting diacylglycerols are substrates for either hormone-sensitive lipase or for the non-inducible enzyme diacylglycerol lipase. Finally the monoacylglycerols are substrates for monoacylglycerol lipase. The net result of the action of these enzymes is three moles of free fatty acid and one mole of glycerol. The free fatty acids diffuse from adipose cells, combine with albumin in the blood, and are thereby transported to other tissues, where they passively diffuse into cells.

## Hormone-Induced Fatty Acid Mobilization in Adipocytes



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Model for the activation of hormone-sensitive lipase by epinephrine. Epinephrine binds its' receptor and leads to the activation of adenylate cyclase. The resultant increase in cAMP activates PKA which then phosphorylates and activates hormone-sensitive lipase. Hormone-sensitive lipase hydrolyzes fatty acids from triacylglycerols and diacylglycerols. The final fatty acid is released from monoacylglycerols through the action of monoacylglycerol lipase, an enzyme active in the absence of hormonal stimulation.

In contrast to the hormonal activation of adenylate cyclase and (subsequently) hormone-sensitive lipase in adipocytes, the mobilization of fat from adipose tissue is inhibited by numerous stimuli. The most significant inhibition is that exerted upon adenylate cyclase by insulin. When an individual is well fed state, insulin released from the pancreas prevents the inappropriate mobilization of stored fat. Instead, any excess fat and carbohydrate are incorporated into the triacylglycerol pool within adipose tissue.

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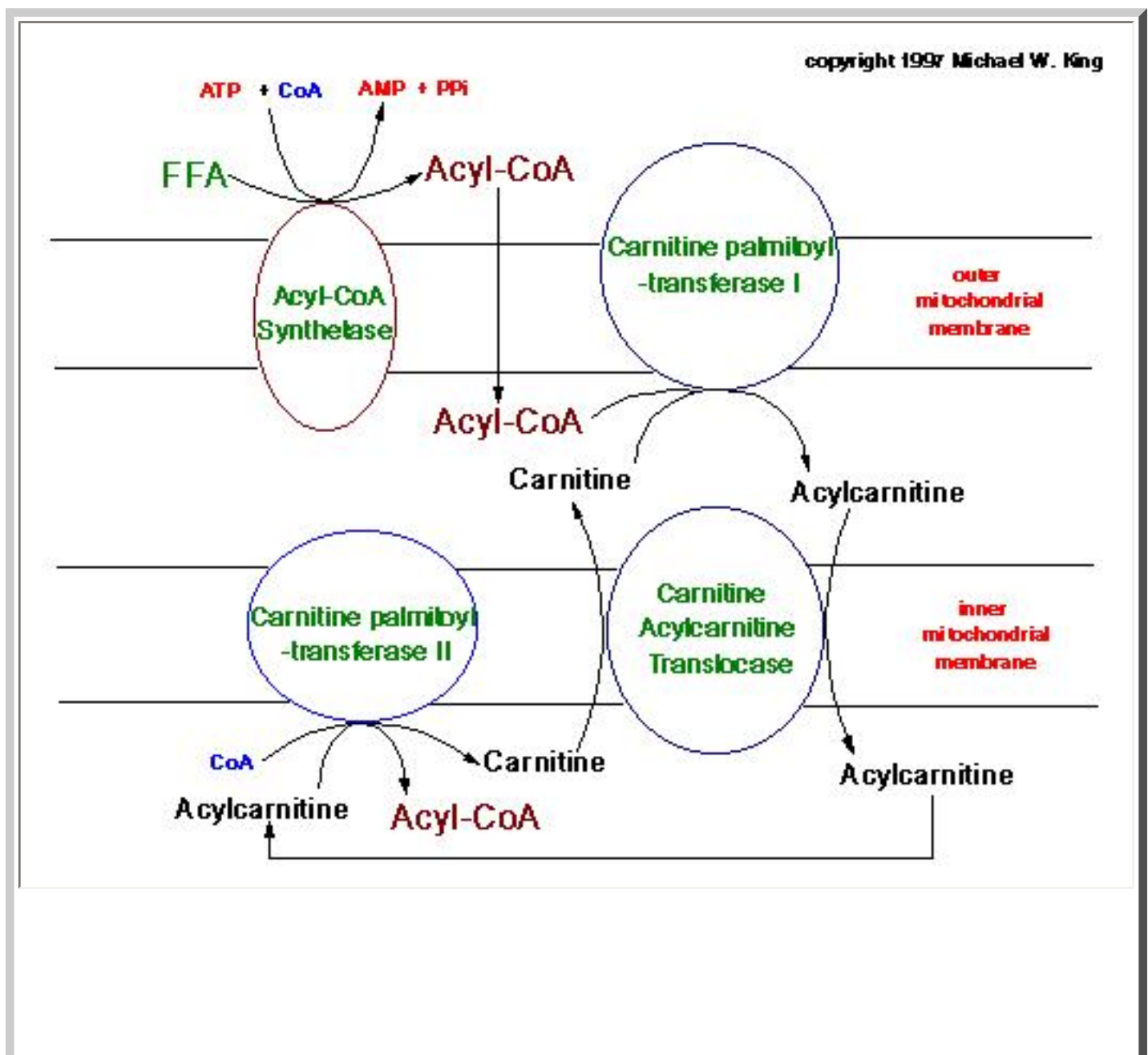
## Reactions of Oxidation



Fatty acids must be activated in the cytoplasm before being oxidized in the mitochondria. Activation is catalyzed by fatty acyl-CoA ligase (also called acyl-CoA synthetase or thiokinase). The net result of this activation process is the consumption of 2 molar equivalents of ATP.



Oxidation of fatty acids occurs in the mitochondria. The transport of fatty acyl-CoA into the mitochondria is accomplished via an **acyl-carnitine** intermediate, which itself is generated by the action of carnitine acyltransferase I, an enzyme that resides in the outer mitochondrial membrane. The acyl-carnitine molecule then is transported into the mitochondria where carnitine acyltransferase II catalyzes the regeneration of the fatty acyl-CoA molecule.



Transport of fatty acids from the cytoplasm to the inner mitochondrial space for oxidation. Following activation to a fatty-CoA, the CoA is exchanged for carnitine by carnitine-palmitoyltransferase I. The fatty-carnitine is then transported to the inside of the mitochondrion where a reversal exchange takes place through the action of carnitine-palmitoyltransferase II. Once inside the mitochondrion the fatty-CoA is a substrate for the  $\beta$ -oxidation machinery.

The process of fatty acid oxidation is termed  $\beta$ -oxidation since it occurs through the sequential removal of 2-carbon units by oxidation at the  $\beta$ -carbon position of the fatty acyl-CoA molecule.

Each round of  $\beta$ -oxidation produces one mole of NADH, one mole of  $\text{FADH}_2$  and one mole of acetyl-CoA. The acetyl-CoA--- the end product of each round of  $\beta$ -oxidation--- then enters the TCA cycle, where it is further oxidized to  $\text{CO}_2$  with the concomitant generation of three moles of NADH, one mole of  $\text{FADH}_2$  and one mole of ATP. The NADH and  $\text{FADH}_2$  generated during the fat oxidation and acetyl-CoA oxidation in the TCA cycle then can enter the respiratory pathway for the production of ATP.

The oxidation of fatty acids yields significantly more energy per carbon atom than does the oxidation of carbohydrates. The net result of the oxidation of one mole of oleic acid (an 18-carbon fatty acid) will be 146 moles of ATP (2 mole equivalents are used during the activation of the fatty acid), as compared with 114 moles from an equivalent number of glucose carbon atoms.

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## Alternative Oxidation Pathways

The majority of natural lipids contain an even number of carbon atoms. A small proportion that contain odd numbers; upon complete  $\beta$ -oxidation, these yield acetyl-CoA units plus a single mole of propionyl-CoA. The propionyl-CoA is converted, in an ATP-dependent pathway, to succinyl-CoA. The succinyl-CoA can then enter the TCA cycle for further oxidation.

The oxidation of unsaturated fatty acids is essentially the same process as for saturated fats, except when a double bond is encountered. In such a case, the bond is isomerized by a specific enoyl-CoA isomerase and oxidation continues. In the case of linoleate, the presence of the  $\Delta$ -12 unsaturation results in the formation of a dienoyl-CoA during oxidation. This molecule is the substrate for an additional oxidizing enzyme, the NADPH requiring 2,4-dienoyl-CoA reductase.

**Phytanic acid** is a fatty acid present in the tissues of ruminants and in dairy products and is, therefore, an important dietary component of fatty acid intake. Because phytanic acid is methylated, it cannot act as a substrate for the first enzyme of the  $\beta$ -oxidation pathway (acyl-

CoA dehydrogenase). An additional mitochondrial enzyme,  $\alpha$ -hydroxylase, adds a hydroxyl group to the  $\alpha$ -carbon of phytanic acid, which then serves as a substrate for the remainder of the normal oxidative enzymes. This process is termed  $\alpha$ -oxidation.

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## Regulation of Fatty Acid Metabolism

In order to understand how the synthesis and degradation of fats needs to be exquisitely regulated, one must consider the energy requirements of the organism as a whole. The blood is the carrier of triacylglycerols in the form of VLDLs and chylomicrons, fatty acids bound to albumin, amino acids, lactate, ketone bodies and glucose. The pancreas is the primary organ involved in sensing the organism's dietary and energetic states by monitoring glucose concentrations in the blood. Low blood glucose stimulates the secretion of glucagon, whereas, elevated blood glucose calls for the secretion of insulin.

The metabolism of fat is regulated by two distinct mechanisms. One is **short-term regulation**, which can come about through events such as substrate availability, allosteric effectors and/or enzyme modification. The other mechanism, **long-term regulation**, is achieved by alteration of the rate of enzyme synthesis and turn-over.

ACC is the rate-limiting (committed) step in fatty acid synthesis. This enzyme is activated by citrate and inhibited by palmitoyl-CoA and other long-chain fatty acyl-CoAs. ACC activity can also be affected by phosphorylation. For instance, glucagon-stimulated increases in PKA activity result in the phosphorylation of certain serine residues in ACC leading to decreased activity of the enzyme. By contrast, insulin leads to PKA-independent phosphorylation of ACC at sites distinct from glucagon, which bring about increased ACC activity. Both of these reaction chains are examples of short-term regulation.

Insulin, a product of the well-fed state, stimulates ACC and FAS synthesis, whereas starvation leads to a decrease in the synthesis of these enzymes. Adipose tissue levels of lipoprotein lipase also are increased by insulin and decreased by starvation. However, the effects of insulin and starvation on lipoprotein lipase in the heart are just the inverse of those in adipose tissue. This sensitivity allows the heart to absorb any available fatty acids in the blood in order to oxidize them for energy production. Starvation also leads to increases in the levels of cardiac enzymes of fatty acid oxidation, and to decreases in FAS and related enzymes of synthesis.

Adipose tissue contains hormone-sensitive lipase, which is activated by PKA-dependent phosphorylation; this activation increases the release of fatty acids into the blood. This in turn leads to the increased oxidation of fatty acids in other tissues such as muscle and liver. In the liver, the net result (due to increased acetyl-CoA levels) is the production of ketone bodies (see below). This would occur under conditions in which the carbohydrate stores and gluconeogenic precursors available in the liver are not sufficient to allow increased glucose production. The increased levels of fatty acid that become available in response to glucagon or epinephrine are assured of being completely oxidized, because PKA also phosphorylates ACC; the synthesis of fatty acid is thereby inhibited.

Insulin has the opposite effect to glucagon and epinephrine: it increases the synthesis of triacylglycerols (and glycogen). One of the many effects of insulin is to lower cAMP levels, which leads to increased dephosphorylation through the enhanced activity of protein phosphatases such as PP-1. With respect to fatty acid metabolism, this yields dephosphorylated and inactive hormone-sensitive lipase. Insulin also stimulates certain phosphorylation events. This occurs through activation of several cAMP-independent kinases, one of which phosphorylates and thereby stimulates the activity of ACC.

Fat metabolism can also be regulated by malonyl-CoA-mediated inhibition of carnitine acyltransferase I. Such regulation serves to prevent *de novo* synthesized fatty acids from entering the mitochondria and being oxidized.

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## Clinical Significance of Fatty Acids

The majority of clinical problems related to fatty acid metabolism are associated with processes of oxidation.

These disorders fall into four main groups:

- **1. Deficiencies in Carnitine:** Deficiencies in carnitine lead to an inability to transport fatty acids into the mitochondria for oxidation. This can occur in newborns and particularly in pre-term infants. Carnitine deficiencies also are found in patients undergoing hemodialysis or exhibiting organic aciduria. Carnitine deficiencies may manifest systemic symptomology or may be limited to only muscles. Symptoms can range from mild occasional muscle cramping to severe weakness or even death. Treatment is by oral carnitine administration.
- **2. Carnitine Palmitoyltransferase I (CPT I) Deficiency:** Deficiencies in this enzyme affect primarily the liver and lead to reduced fatty acid oxidation and ketogenesis. [Carnitine Palmitoyltransferase II \(CPT II\) deficiency](#) results in recurrent muscle pain and fatigue and myoglobinuria following strenuous exercise. Carnitine acyltransferases may also be inhibited by sulfonylurea drugs such as **tolbutamide** and **glyburide**.
- **3. Deficiencies in Acyl-CoA Dehydrogenases:** A group of inherited diseases that impair  $\beta$ -oxidation result from deficiencies in acyl-CoA dehydrogenases. The enzymes affected may belong to one of four categories:
  - very long-chain acyl-CoA dehydrogenase (VLCAD) [Disease description](#)
  - long-chain acyl-CoA dehydrogenase (LCAD) [Disease description](#)
  - medium-chain acyl-CoA dehydrogenase (MCAD) [Disease description](#)
  - short-chain acyl-CoA dehydrogenase (SCAD) [Disease description](#)

[MCAD deficiency](#) is the most common form of this disease. In the first years of life this deficiency will become apparent following a prolonged fasting period. Symptoms include

vomiting, lethargy and frequently coma. Excessive urinary excretion of medium-chain dicarboxylic acids as well as their glycine and carnitine esters is diagnostic of this condition. In the case of this enzyme deficiency, taking care to avoid prolonged fasting is sufficient to prevent clinical problems.

- **4. Refsum's Disease:** Refsum's disease is a rare inherited disorder in which patients lack the mitochondrial  $\alpha$ -oxidizing enzyme. As a consequence, they accumulate large quantities of phytanic acid in their tissues and serum. This leads to severe symptoms, including cerebellar ataxia, retinitis pigmentosa, nerve deafness and peripheral neuropathy. As expected, the restriction of dairy products and ruminant meat from the diet can ameliorate the symptoms of this disease.

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## Ketogenesis

During high rates of fatty acid oxidation, primarily in the liver, large amounts of acetyl-CoA are generated. These exceed the capacity of the TCA cycle, and one result is the synthesis of ketone bodies, or **ketogenesis**. The ketone bodies are **acetoacetate**,  **$\beta$ -hydroxybutyrate**, and **acetone**.

The formation of acetoacetyl-CoA occurs by condensation of two moles of acetyl-CoA through a reversal of the thiolase catalyzed reaction of fat oxidation. Acetoacetyl-CoA and an additional acetyl-CoA are converted to  **$\beta$ -hydroxy- $\beta$ -methylglutaryl-CoA (HMG-CoA)** by HMG-CoA synthase, an enzyme found in large amounts only in the liver. Some of the HMG-CoA leaves the mitochondria, where it is converted to mevalonate (the precursor for cholesterol synthesis) by HMG-CoA reductase. HMG-CoA in the mitochondria is converted to acetoacetate by the action of HMG-CoA lyase. Acetoacetate can undergo spontaneous decarboxylation to acetone, or be enzymatically converted to  $\beta$ -hydroxybutyrate through the action of  $\beta$ -hydroxybutyrate dehydrogenase. When the level of glycogen in the liver is high the production of  $\beta$ -hydroxybutyrate increases.

When carbohydrate utilization is low or deficient, the level of oxaloacetate will also be low, resulting in a reduced flux through the TCA cycle. This in turn leads to increased release of ketone bodies from the liver for use as fuel by other tissues. In early stages of starvation, when the last remnants of fat are oxidized, heart and skeletal muscle will consume primarily ketone bodies to preserve glucose for use by the brain. Acetoacetate and  $\beta$ -hydroxybutyrate, in particular, also serve as major substrates for the biosynthesis of neonatal cerebral lipids.

Ketone bodies are utilized by extrahepatic tissues through the conversion of  $\beta$ -hydroxybutyrate to acetoacetate and of acetoacetate to acetoacetyl-CoA. The first step involves the reversal of the  $\beta$ -hydroxybutyrate dehydrogenase reaction, and the second involves the action (shown below) of acetoacetate:succinyl-CoA transferase, also called ketoacyl-CoA-transferase.





The latter enzyme is present in all tissues except the liver. Importantly, its absence allows the liver to produce ketone bodies but not to utilize them. This ensures that extrahepatic tissues have access to ketone bodies as a fuel source during prolonged starvation.

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## Regulation of Ketogenesis

The fate of the products of fatty acid metabolism is determined by an individual's physiological status. Ketogenesis takes place primarily in the liver and may be affected by several factors:

- **1.** Control in the release of free fatty acids from adipose tissue directly affects the level of ketogenesis in the liver. This is, of course, substrate-level regulation.
- **2.** Once fats enter the liver, they have two distinct fates. They may be activated to acyl-CoAs and oxidized, or esterified to glycerol in the production of triacylglycerols. If the liver has sufficient supplies of glycerol-3-phosphate, most of the fats will be turned to the production of triacylglycerols.
- **3.** The generation of acetyl-CoA by oxidation of fats can be completely oxidized in the TCA cycle. Therefore, if the demand for ATP is high the fate of acetyl-CoA is likely to be further oxidation to CO<sub>2</sub>.
- **4.** The level of fat oxidation is regulated hormonally through phosphorylation of ACC, which may activate it (in response to glucagon) or inhibit it (in the case of insulin).

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## Clinical Significance of Ketogenesis

The production of ketone bodies occurs at a relatively low rate during normal feeding and under conditions of normal physiological status. Normal physiological responses to carbohydrate shortages cause the liver to increase the production of ketone bodies from the acetyl-CoA generated from fatty acid oxidation. This allows the heart and skeletal muscles primarily to use ketone bodies for energy, thereby preserving the limited glucose for use by the brain.

The most significant disruption in the level of ketosis, leading to profound clinical manifestations, occurs in untreated insulin-dependent diabetes mellitus. This physiological state, **diabetic ketoacidosis (DKA)**, results from a reduced supply of glucose (due to a significant decline in circulating insulin) and a concomitant increase in fatty acid oxidation (due



to a concomitant increase in circulating glucagon). The increased production of acetyl-CoA leads to ketone body production that exceeds the ability of peripheral tissues to oxidize them. Ketone bodies are relatively strong acids ( $pK_a$  around 3.5), and their increase lowers the pH of the blood. This acidification of the blood is dangerous chiefly because it impairs the ability of hemoglobin to bind oxygen.

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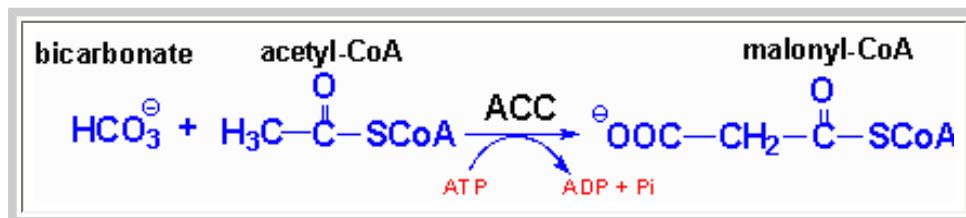
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## Fatty Acid Synthesis

One might predict that the pathway for the synthesis of fatty acids would be the reversal of the oxidation pathway. However, this would not allow distinct regulation of the two pathways to occur even given the fact that the pathways are separated within different cellular compartments.

The pathway for fatty acid synthesis occurs in the cytoplasm, whereas, oxidation occurs in the mitochondria. The other major difference is the use of nucleotide co-factors. Oxidation of fats involves the reduction of  $\text{FADH}^+$  and  $\text{NAD}^+$ . Synthesis of fats involves the oxidation of  $\text{NADPH}$ . However, the essential chemistry of the two processes are reversals of each other. Both oxidation and synthesis of fats utilize an activated two carbon intermediate, acetyl-CoA. However, the acetyl-CoA in fat synthesis exists temporarily bound to the enzyme complex as malonyl-CoA.

The synthesis of malonyl-CoA is the first committed step of fatty acid synthesis and the enzyme that catalyzes this reaction, acetyl-CoA carboxylase (ACC), is the major site of regulation of fatty acid synthesis. Like other enzymes that transfer  $\text{CO}_2$  to substrates, ACC requires a [biotin](#) co-factor.



The rate of fatty acid synthesis is controlled by the equilibrium between monomeric ACC and polymeric ACC. The activity of ACC requires polymerization. This conformational change is enhanced by citrate and inhibited by long-chain fatty acids. ACC is also controlled through hormone mediated phosphorylation (see below).

The acetyl groups that are the products of fatty acid oxidation are linked to CoASH. As you should recall, CoA contains a **phosphopantetheine** group coupled to AMP. The carrier of acetyl groups (and elongating acyl groups) during fatty acid synthesis is also a phosphopantetheine prosthetic group, however, it is attached a serine hydroxyl in the synthetic enzyme complex. The carrier portion of the synthetic complex is called **acyl carrier protein, ACP**. This

is somewhat of a misnomer in eukaryotic fatty acid synthesis since the ACP portion of the synthetic complex is simply one of many domains of a single polypeptide. The acetyl-CoA and malonyl-CoA are transferred to ACP by the action of acetyl-CoA transacylase and malonyl-CoA transacylase, respectively. The attachment of these carbon atoms to ACP allows them to enter the fatty acid synthesis cycle.

The synthesis of fatty acids from acetyl-CoA and malonyl-CoA is carried out by fatty acid synthase, FAS. The active enzyme is a dimer of identical subunits.

All of the reactions of fatty acid synthesis are carried out by the multiple enzymatic activities of FAS. Like fat oxidation, fat synthesis involves 4 enzymatic activities. These are,  $\beta$ -keto-ACP synthase,  $\beta$ -keto-ACP reductase, 3-OH acyl-ACP dehydratase and enoyl-CoA reductase. The two reduction reactions require NADPH oxidation to NADP<sup>+</sup>.

The primary fatty acid synthesized by FAS is palmitate. Palmitate is then released from the enzyme and can then undergo separate elongation and/or unsaturation to yield other fatty acid molecules.

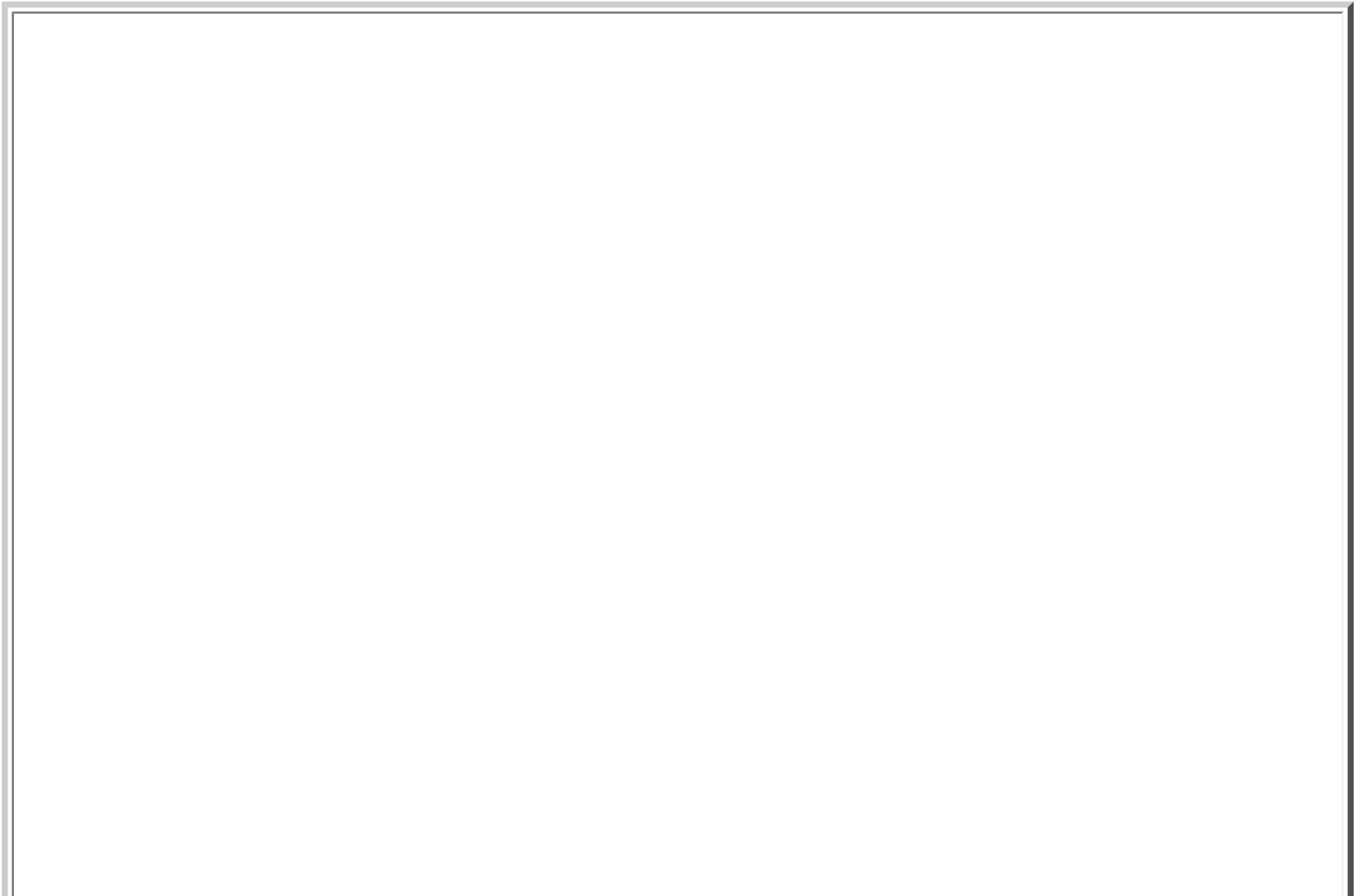
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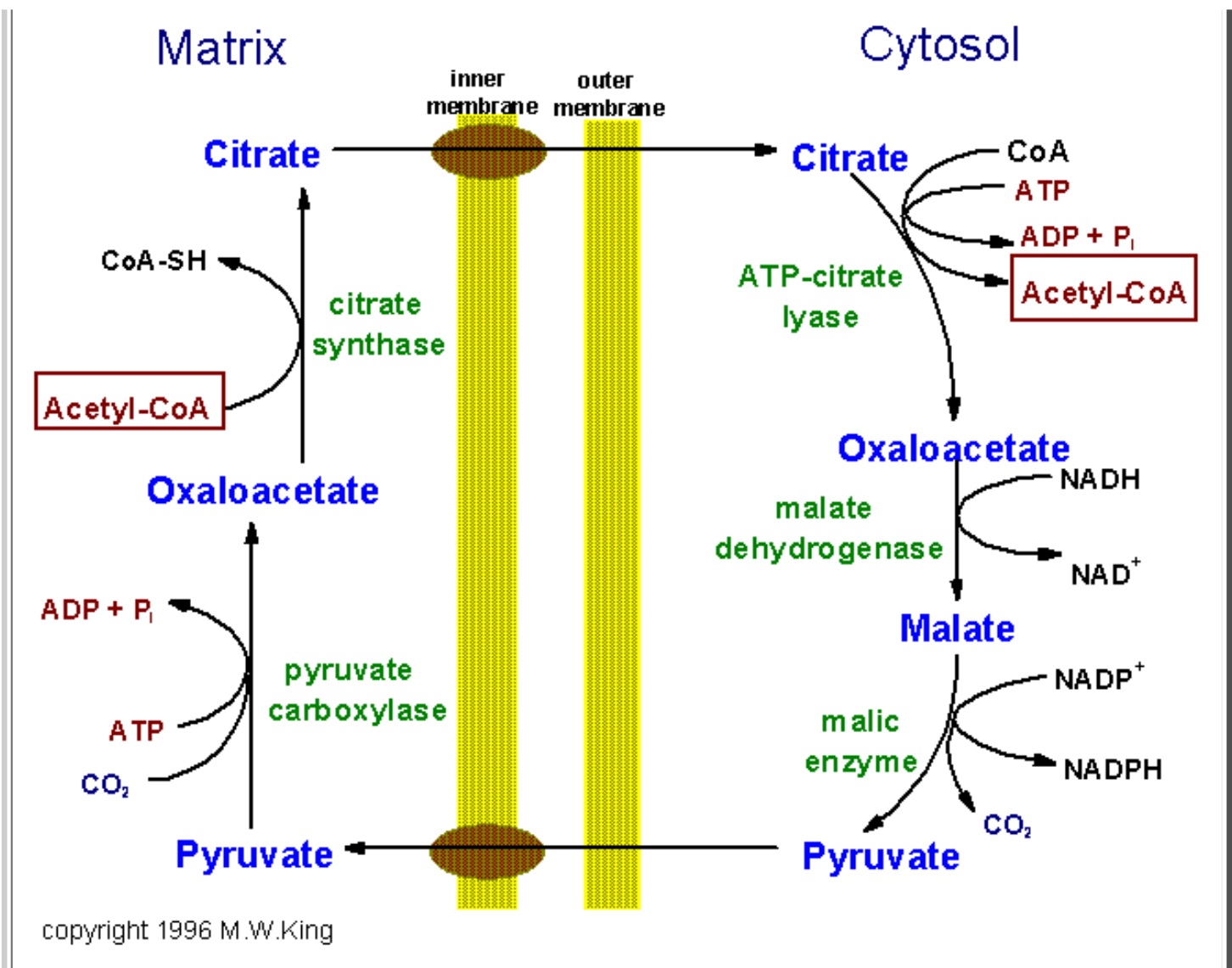
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## Origin of Cytoplasmic Acetyl-CoA

Acetyl-CoA is generated in the mitochondria primarily from two sources, the [pyruvate dehydrogenase](#) (PDH) reaction and [fatty acid oxidation](#). In order for these acetyl units to be utilized for fatty acid synthesis they must be present in the cytoplasm. The shift from fatty acid oxidation and glycolytic oxidation occurs when the need for energy diminishes. This results in reduced oxidation of acetyl-CoA in the [TCA cycle](#) and the [oxidative phosphorylation](#) pathway. Under these conditions the mitochondrial acetyl units can be stored as fat for future energy demands.

Acetyl-CoA enters the cytoplasm in the form of citrate via the tricarboxylate transport system (see Figure). In the cytoplasm, citrate is converted to oxaloacetate and acetyl-CoA by the ATP driven ATP-citrate lyase reaction. This reaction is essentially the reverse of that catalyzed by the TCA enzyme citrate synthase except it requires the energy of ATP hydrolysis to drive it forward. The resultant oxaloacetate is converted to malate by malate dehydrogenase (MDH).





Pathway for the movement of acetyl-CoA units from within the mitochondrion to the cytoplasm for use in lipid and cholesterol biosynthesis. Note that the cytoplasmic malic enzyme catalyzed reaction generates NADPH which can be used for reductive biosynthetic reactions such as those of fatty acid and cholesterol synthesis.

The malate produced by this pathway can undergo oxidative decarboxylation by malic enzyme. The co-enzyme for this reaction is NADP<sup>+</sup> generating NADPH. The advantage of this series of reactions for converting mitochondrial acetyl-CoA into cytoplasmic acetyl-CoA is that the NADPH produced by the malic enzyme reaction can be a major source of reducing co-factor for the fatty acid synthase activities.

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## Regulation of Fatty Acid Metabolism

One must consider the global organismal energy requirements in order to effectively understand how the synthesis and degradation of fats (and also carbohydrates) needs to be exquisitely regulated. The blood is the carrier of triacylglycerols in the form of VLDLs and chylomicrons, fatty acids bound to albumin, amino acids, lactate, ketone bodies and glucose. The pancreas is the primary organ involved in sensing the organisms dietary and energetic states via glucose concentrations in the blood. In response to low blood glucose, glucagon is secreted, whereas, in response to elevated blood glucose insulin is secreted.

The regulation of fat metabolism occurs via two distinct mechanisms. One is **short term regulation** which is

regulation effected by events such as substrate availability, allosteric effectors and/or enzyme modification. ACC is the rate limiting (committed) step in fatty acid synthesis. This enzyme is activated by citrate and inhibited by palmitoyl-CoA and other long chain fatty acyl-CoAs. ACC activity also is affected by phosphorylation. The primary phosphorylation of ACC occurs through the action of AMP-activated protein kinase, AMPK (this is not the same as cAMP-dependent protein kinase, PKA). Glucagon stimulated increases in PKA activity result in phosphorylation and inhibition of ACC. Additionally, glucagon activation of PKA leads to phosphorylation and activation of phosphoprotein phosphatase inhibitor-1, PPI-1 which results in a reduced ability to dephosphorylate ACC maintaining the enzyme in a less active state. On the other hand insulin leads to activation of phosphatases, thereby leading to dephosphorylation of ACC that results in increased ACC activity. These forms of regulation are all defined as short term regulation.

Control of a given pathways' regulatory enzymes can also occur by alteration of enzyme synthesis and turn-over rates. These changes are **long term regulatory effects**. Insulin stimulates ACC and FAS synthesis, whereas, starvation leads to decreased synthesis of these enzymes. Adipose tissue lipoprotein lipase levels also are increased by insulin and decreased by starvation. However, in contrast to the effects of insulin and starvation on adipose tissue, their effects on heart lipoprotein lipase are just the inverse. This allows the heart to absorb any available fatty acids in the blood in order to oxidize them for energy production. Starvation also leads to increases in the levels of fatty acid oxidation enzymes in the heart as well as a decrease in FAS and related enzymes of synthesis.

Adipose tissue contains hormone-sensitive lipase, that is activated by PKA-dependent phosphorylation leading to increased fatty acid release to the blood. The activity of hormone-sensitive lipase is also affected positively through the action of AMPK. Both of these effects lead to increased fatty acid oxidation in other tissues such as muscle and liver. In the liver the net result (due to increased acetyl-CoA levels) is the production of ketone bodies. This would occur under conditions where insufficient carbohydrate stores and gluconeogenic precursors were available in liver for increased glucose production. The increased fatty acid availability in response to glucagon or epinephrine is assured of being completely oxidized since both PKA and AMPK also phosphorylate (and as a result inhibits) ACC, thus inhibiting fatty acid synthesis.

Insulin, on the other hand, has the opposite effect to glucagon and epi leading to increased glycogen and triacylglyceride synthesis. One of the many effects of insulin is to lower cAMP levels which leads to increased dephosphorylation through the enhanced activity of protein phosphatases such as PP-1. With respect to fatty acid metabolism this yields dephosphorylated and inactive hormone sensitive lipase. Insulin also stimulates certain phosphorylation events. This occurs through activation of several cAMP-independent kinases. Insulin stimulated phosphorylation of ACC activates this enzyme.

Regulation of fat metabolism also occurs through **malonyl-CoA** induced inhibition of carnitine acyltransferase I. This functions to prevent the newly synthesized fatty acids from entering the mitochondria and being oxidized.

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## Elongation and Desaturation

The fatty acid product released from FAS is palmitate (via the action of palmitoyl thioesterase) which is a 16:0 fatty acid, i.e. 16 carbons and no sites of unsaturation. Elongation and unsaturation of fatty acids occurs in both the mitochondria and endoplasmic reticulum (microsomal membranes). The predominant site of these processes is in the ER membranes. Elongation involves condensation of acyl-CoA groups with malonyl-CoA. The resultant product is two carbons longer ( $\text{CO}_2$  is released from malonyl-CoA as in the FAS reaction) which undergoes reduction, dehydration and reduction yielding a saturated fatty acid. The reduction reactions of elongation require NADPH as co-factor just as for the similar reactions catalyzed by FAS. Mitochondrial elongation involves acetyl-CoA units and is a reversal of oxidation except that the final reduction utilizes NADPH instead of  $\text{FADH}_2$  as co-factor.

Desaturation occurs in the ER membranes as well and in mammalian cells involves 4 broad specificity fatty acyl-CoA desaturases (non-heme iron containing enzymes). These enzymes introduce unsaturation at C4, C5, C6 or C9. The electrons transferred from the oxidized fatty acids during desaturation are transferred from the desaturases to cytochrome b5 and then NADH-cytochrome b5 reductase. These electrons are un-coupled from mitochondrial oxidative-phosphorylation and, therefore, do not yield ATP.

Since these enzymes cannot introduce sites of unsaturation beyond C9 they cannot synthesize either **linoleate** ( $18:2^{\Delta 9, 12}$ ) or **linolenate** ( $18:3^{\Delta 9, 12, 15}$ ). These fatty acids must be acquired from the diet and are, therefore, referred to as **essential fatty acids**. Linoleic is especially important in that it required for the synthesis of **arachidonic acid**.

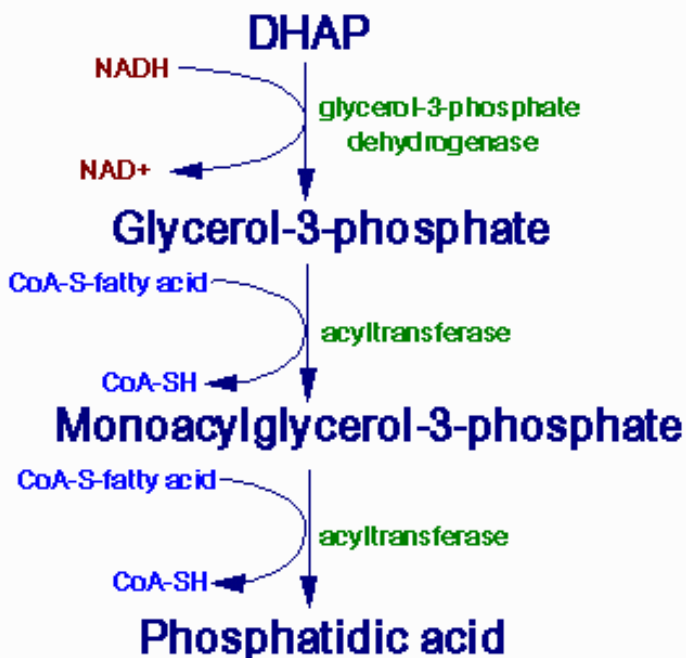
As we shall encounter later, arachidonate is a precursor for the **eicosanoids** (the **prostaglandins** and **thromboxanes**). It is this role of fatty acids in eicosanoid synthesis that leads to poor growth, wound healing and dermatitis in persons on fat free diets. Also, linoleic acid is a constituent of epidermal cell sphingolipids that function as the skins water permeability barrier.

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## Synthesis of Triglycerides

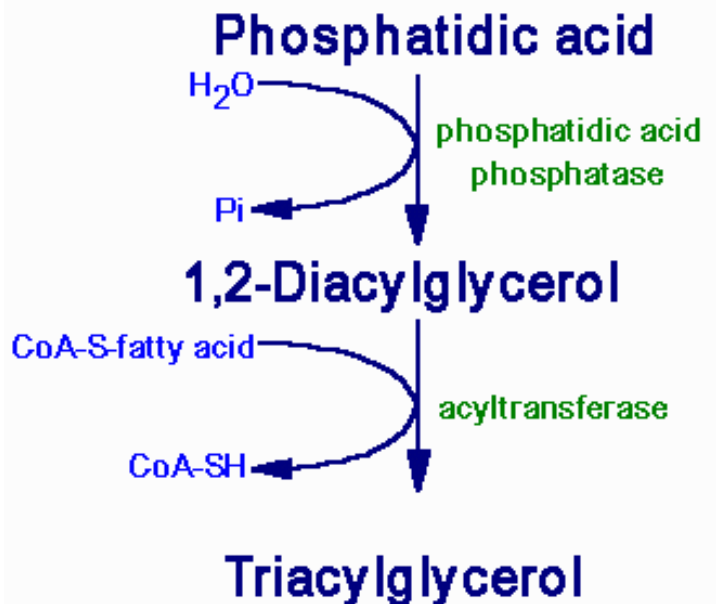
Fatty acids are stored for future use as triacylglycerols in all cells, but primarily in adipocytes of adipose tissue. Triacylglycerols constitute molecules of glycerol to which three fatty acids have been esterified. The fatty acids present in triacylglycerols are predominantly saturated. The major building block for the synthesis of triacylglycerols, in tissues other than adipose tissue, is glycerol. Adipocytes lack glycerol kinase, therefore, dihydroxyacetone phosphate (DHAP), produced during glycolysis, is the precursor for triacylglycerol synthesis in adipose tissue. This means that adipocytes must have glucose to oxidize in order to store fatty acids in the form of triacylglycerols. DHAP can also serve as a backbone precursor for triacylglycerol synthesis in tissues other than adipose, but does so to a much lesser extent than glycerol.

### Phosphatidic Acid Synthesis



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### Triacylglycerol Synthesis



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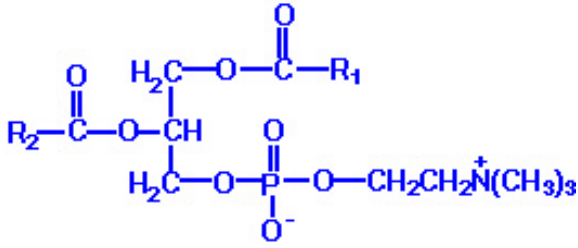
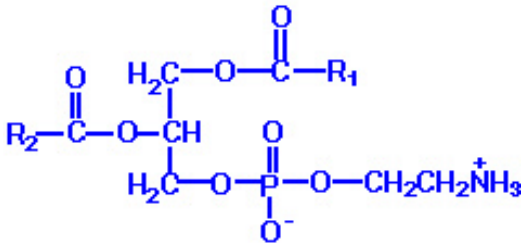
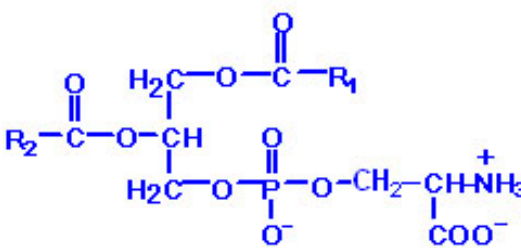
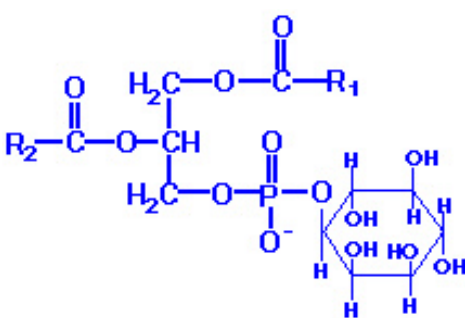
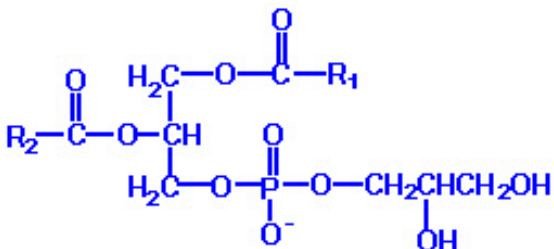
The glycerol backbone of triacylglycerols is activated by phosphorylation at the C-3 position by glycerol kinase. The utilization of DHAP for the backbone is carried out through the action of glycerol-3-phosphate dehydrogenase, a reaction that requires NADH (the same reaction as that used in the [glycerol-phosphate shuttle](#)). The fatty acids incorporated into triacylglycerols are activated to acyl-CoAs through the action of acyl-CoA synthetases. Two molecules of acyl-CoA are esterified to glycerol-3-phosphate to yield 1,2-diacylglycerol phosphate (commonly identified as **phosphatidic acid**). The phosphate is then removed, by phosphatidic acid phosphatase, to yield 1,2-diacylglycerol, the substrate for addition of the third fatty acid. Intestinal monoacylglycerols, derived from the hydrolysis of dietary fats, can also serve as substrates for the synthesis of 1,2-diacylglycerols.

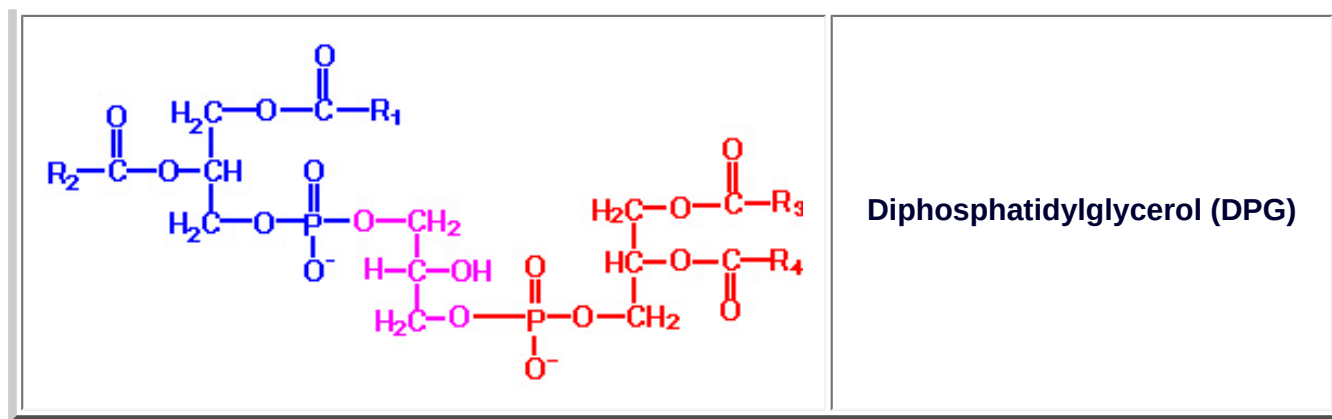
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## Phospholipid Structures

Phospholipids are synthesized by esterification of an alcohol to the phosphate of phosphatidic acid (1,2-diacylglycerol 3-phosphate). Most phospholipids have a saturated fatty acid on C-1 and an unsaturated fatty acid on C-2 of the glycerol backbone. The most commonly added alcohols (serine, ethanolamine and choline) also contain nitrogen that may be positively charged, whereas, glycerol and inositol do not. The major classifications of phospholipids are:

|                                                                                     |                               |
|-------------------------------------------------------------------------------------|-------------------------------|
|    | Phosphatidylcholine (PC)      |
|    | Phosphatidylethanolamine (PE) |
|  | Phosphatidylserine (PS)       |
|  | Phosphatidylinositol (PI)     |
|  | Phosphatidylglycerol (PG)     |



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## Phospholipid Synthesis

Phospholipids can be synthesized by two mechanisms. One utilizes a CDP-activated polar head group for attachment to the phosphate of phosphatidic acid. The other utilizes CDP-activated 1,2-diacylglycerol and an inactivated polar head group.

**PC:** This class of phospholipids is also called the **lecithins**. At physiological pH, phosphatidylcholines are neutral zwitterions. They contain primarily palmitic or stearic acid at carbon 1 and primarily oleic, linoleic or linolenic acid at carbon 2. The lecithin **dipalmitoyllecithin** is a component of **lung** or **pulmonary surfactant**. It contains palmitate at both carbon 1 and 2 of glycerol and is the major (80%) phospholipid found in the extracellular lipid layer lining the pulmonary alveoli.

Choline is activated first by phosphorylation and then by coupling to CDP prior to attachment to phosphatidic acid. PC is also synthesized by the addition of choline to CDP-activated 1,2-diacylglycerol. A third pathway to PC synthesis, involves the conversion of either PS or PE to PC. The conversion of PS to PC first requires decarboxylation of PS to yield PE; this then undergoes a series of three methylation reactions utilizing S-adenosylmethionine (SAM) as methyl group donor.

**PE:** These molecules are neutral zwitterions at physiological pH. They contain primarily palmitic or stearic acid on carbon 1 and a long chain unsaturated fatty acid (e.g. 18:2, 20:4 and 22:6) on carbon 2.

Synthesis of PE can occur by two pathways. The first requires that ethanolamine be activated by phosphorylation and then by coupling to CDP. The ethanolamine is then transferred from CDP-ethanolamine to phosphatidic acid to yield PE. The second involves the decarboxylation of PS.

**PS:** Phosphatidylserines will carry a net charge of -1 at physiological pH and are composed of fatty acids similar to the phosphatidylethanolamines.

The pathway for PS synthesis involves an exchange reaction of serine for ethanolamine in PE. This exchange occurs when PE is in the lipid bilayer of the a membrane. As indicated above, PS can serve as a source of PE through a decarboxylation reaction.

**PI:** These molecules contain almost exclusively stearic acid at carbon 1 and arachidonic acid at carbon 2. Phosphatidylinositols composed exclusively of non-phosphorylated inositol exhibit a net charge of -1 at physiological pH. These molecules exist in membranes with various levels of phosphate esterified to the hydroxyls of the inositol. Molecules with phosphorylated inositol are termed **polyphosphoinositides**. The polyphosphoinositides are important intracellular transducers of signals emanating from the plasma membrane.

The synthesis of PI involves CDP-activated 1,2-diacylglycerol condensation with *myo*-inositol. PI subsequently undergoes a series of phosphorylations of the hydroxyls of inositol leading to the production of polyphosphoinositides. One polyphosphoinositide (**phosphatidylinositol 4,5-bisphosphate, PIP<sub>2</sub>**) is a critically important membrane phospholipid involved in the transmission of signals for cell growth and differentiation from outside the cell to inside.

**PG:**Phosphatidylglycerols exhibit a net charge of -1 at physiological pH. These molecules are found in high concentration in mitochondrial membranes and as components of **pulmonary surfactant**. Phosphatidylglycerol also is a precursor for the synthesis of cardiolipin.

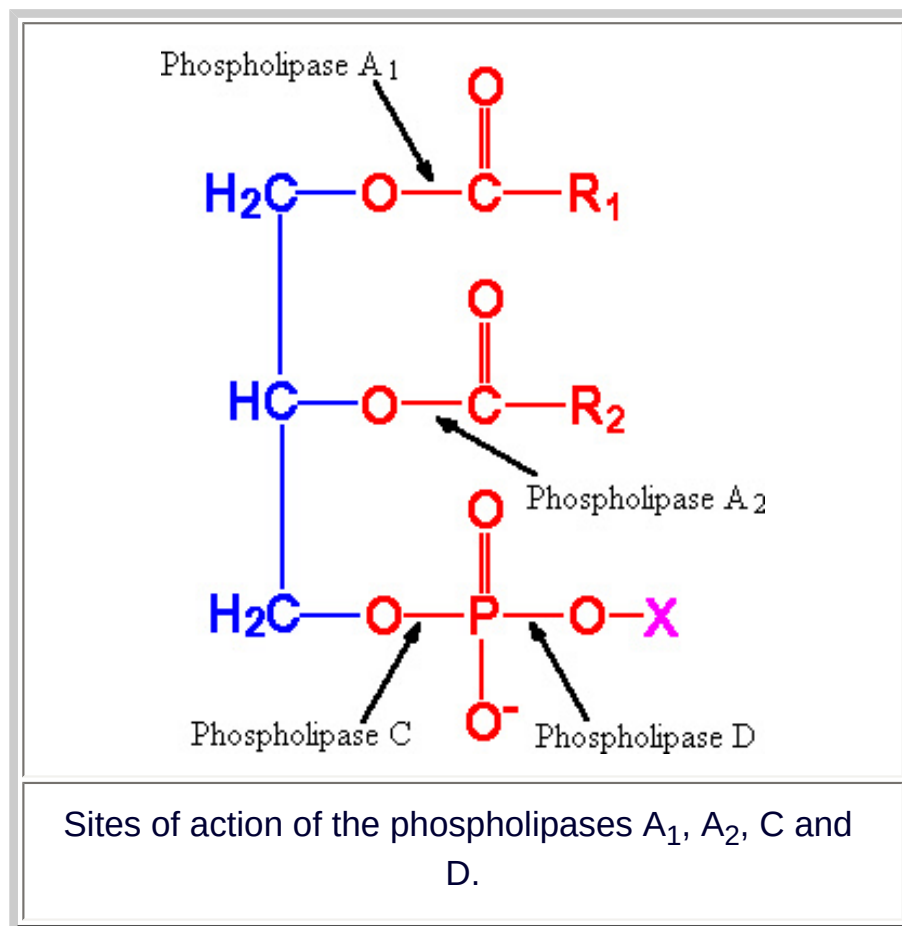
PG is synthesized from CDP-diacylglycerol and glycerol-3-phosphate. The vital role of PG is to serve as the precursor for the synthesis of diphosphatidylglycerols (DPGs).

**DPG:**These molecules are very acidic, exhibiting a net charge of -2 at physiological pH. They are found primarily in the inner mitochondrial membrane and also as components of **pulmonary surfactant**.

One important class of diphosphatidylglycerols is the **cardiolipins**. These molecules are synthesized by the condensation of CDP-diacylglycerol with PG.

The fatty acid distribution at the C-1 and C-2 positions of glycerol within phospholipids is continually in flux, owing to phospholipid degradation and the continuous phospholipid remodeling that occurs while these molecules are in membranes. Phospholipid degradation results from the action of phospholipases. There are various phospholipases that exhibit substrate specificities for different positions in phospholipids.

In many cases the acyl group which was initially transferred to glycerol, by the action of the acyl transferases, is not the same acyl group present in the phospholipid when it resides within a membrane. The remodeling of acyl groups in phospholipids is the result of the action of phospholipase A<sub>1</sub> and phospholipase A<sub>2</sub>.



The products of these phospholipases are called **lysophospholipids** and can be substrates for acyl transferases utilizing different acyl-CoA groups. Lysophospholipids can also accept acyl groups from other phospholipids in an exchange reaction catalyzed by lysolecithin:lecithin acyltransferase (LLAT).

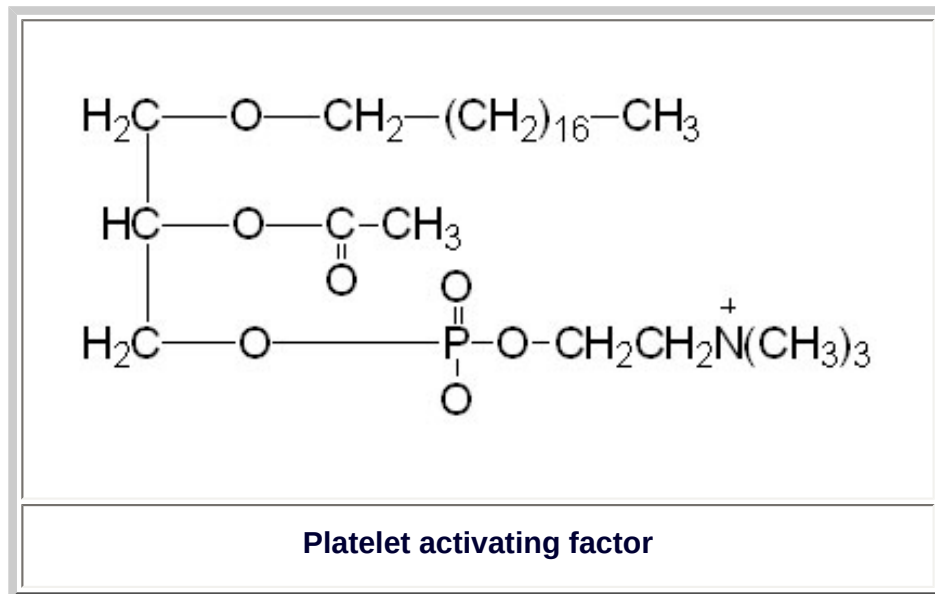
Phospholipase A<sub>2</sub> is also an important enzyme, whose activity is responsible for the release of arachidonic acid from the C-2 position of membrane phospholipids. The released arachidonate is then a substrate for the synthesis of the prostaglandins and leukotrienes.

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## Plasmalogens

Plasmalogens are glycerol ether phospholipids. They are of two types, alkyl ether ( $\text{-O-CH}_2\text{-}$ ) and alkenyl ether ( $\text{-O-CH=CH-}$ ). Dihydroxyacetone phosphate serves as the glycerol precursor for the synthesis of glycerol ether phospholipids. Three major classes of plasmalogens have been identified: **choline**, **ethanolamine** and **serine plasmalogens**. Ethanolamine plasmalogen is prevalent in myelin. Choline plasmalogen is abundant in cardiac tissue.

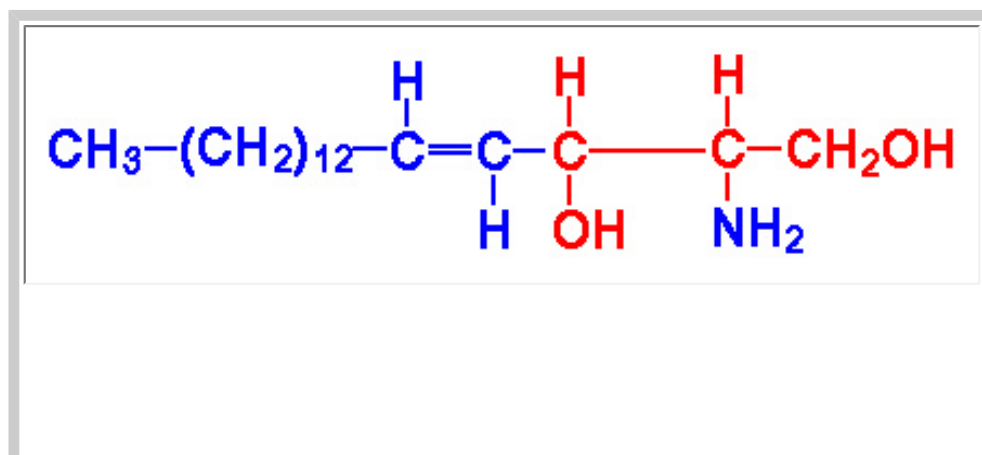
One particular choline alkyl ether plasmalogen (1-O-1'-enyl-2-acetyl-*sn*-glycero-3-phosphocholine) has been identified as an extremely powerful biological mediator, capable of inducing cellular responses at concentrations as low as  $10^{-11}$  M. This molecule is called **platelet activating factor, PAF**. PAF functions as a mediator of hypersensitivity, acute inflammatory reactions and anaphylactic shock. PAF is synthesized in response to the formation of antigen-IgE complexes on the surfaces of basophils, neutrophils, eosinophils, macrophages and monocytes. The synthesis and release of PAF from cells leads to platelet aggregation and the release of serotonin from platelets. PAF also produces responses in liver, heart, smooth muscle, and uterine and lung tissues.

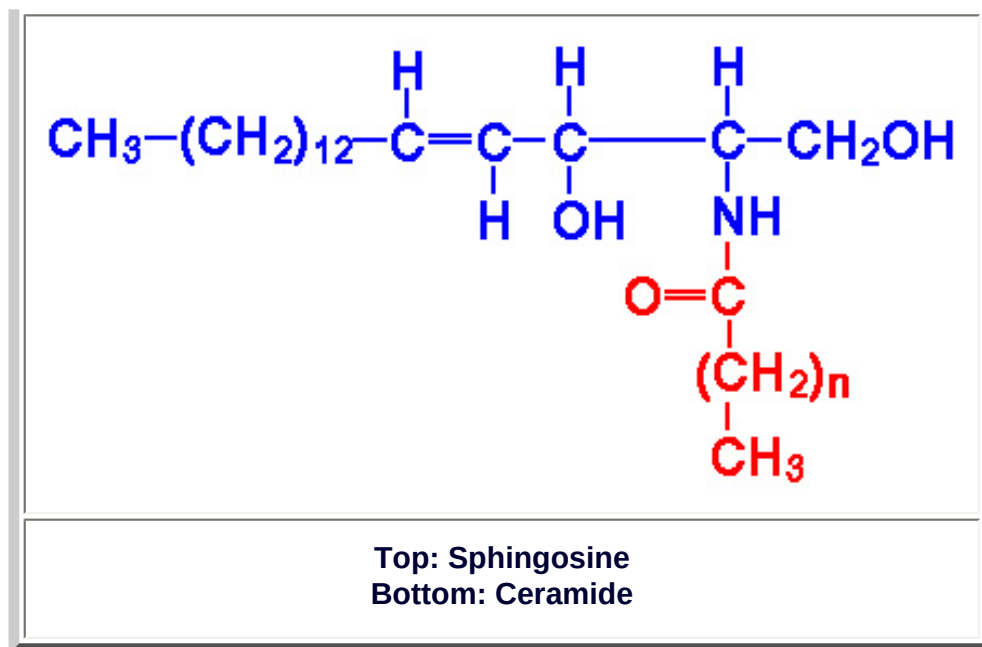


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## Metabolism of the Sphingolipids

The sphingolipids, like the phospholipids, are composed of a polar head group and two nonpolar tails. The core of sphingolipids is the long-chain amino alcohol, **sphingosine**. Amino acylation, with a long chain fatty acid, at carbon 2 of sphingosine yields a **ceramide**.

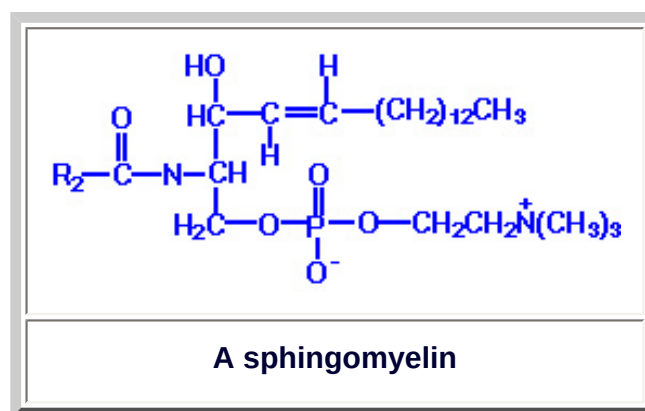




The sphingolipids include the **sphingomyelins** and **glycosphingolipids** (the cerebroside, sulfatides, globosides and gangliosides). Sphingomyelins are the only sphingolipid that are phospholipids. Sphingolipids are a component of all membranes but are particularly abundant in the myelin sheath.

**Sphingomyelins** are sphingolipids that are also phospholipids. Sphingomyelins are important structural lipid components of nerve cell membranes. The predominant sphingomyelins contain palmitic or stearic acid N-acylated at carbon 2 of sphingosine.

The sphingomyelins are synthesized by the transfer of phosphorylcholine from phosphatidylcholine to a ceramide in a reaction catalyzed by sphingomyelin synthase.



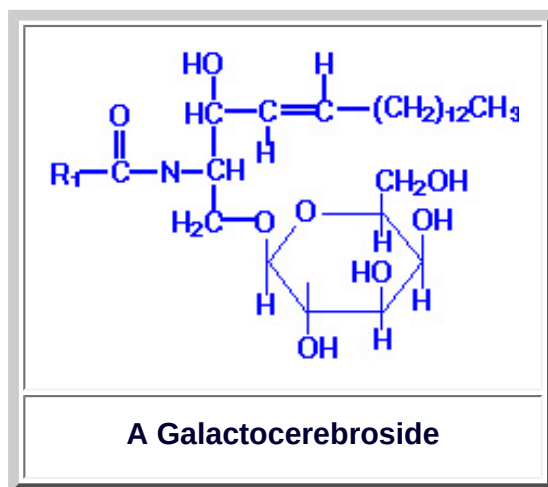
Defects in the enzyme acid sphingomyelinase result in the lysosomal storage disease known as **Niemann-Pick disease**. There are at least 4 related disorders identified as Niemann-Pick disease **Type A and B** (both of which result from defects in acid sphingomyelinase), **Type C1** and a related **C2** and **Type D**. Types C1, C2 and D do not result from defects in acid sphingomyelinase. More information on Niemann-Pick sub-type C1 is presented below in the section on **Clinical Significances of Sphingolipids**.

**Glycosphingolipids**, or glycolipids, are composed of a ceramide backbone with a wide variety of carbohydrate groups (mono- or oligosaccharides) attached to carbon 1 of sphingosine. The four principal classes of glycosphingolipids are the **cerebrosides**, **sulfatides**, **globosides** and **gangliosides**.

**Cerebrosides** have a single sugar group linked to ceramide. The most common of these is galactose

(**galactocerebrosides**), with a minor level of glucose (**glucocerebrosides**). Galactocerebrosides are found predominantly in neuronal cell membranes. By contrast glucocerebrosides are not normally found in membranes, especially neuronal membranes; instead, they represent intermediates in the synthesis or degradation of more complex glycosphingolipids.

Galactocerebrosides are synthesized from ceramide and UDP-galactose. Excess accumulation of glucocerebrosides is observed in **Gaucher's disease**.



**Sulfatides:** The sulfuric acid esters of galactocerebrosides are the sulfatides. Sulfatides are synthesized from galactocerebrosides and activated sulfate, **3'-phosphoadenosine 5'-phosphosulfate (PAPS)**. Excess accumulation of sulfatides is observed in **sulfatide lipidosis** (metachromatic leukodystrophy).

**Globosides:** Globosides represent cerebrosides that contain additional carbohydrates, predominantly galactose, glucose or GalNAc. Lactosyl ceramide is a globoside found in erythrocyte plasma membranes. Globotriaosylceramide (also called ceramide trihexoside) contains glucose and two moles of galactose and accumulates, primarily in the kidneys, of patients suffering from **Fabry's disease**.

**Gangliosides:** Gangliosides are very similar to globosides except that they also contain NANA in varying amounts. The specific names for gangliosides are a key to their structure. The letter G refers to ganglioside, and the subscripts M, D, T and Q indicate that the molecule contains mono-, di-, tri and quatra(tetra)-sialic acid. The numerical subscripts 1, 2 and 3 refer to the carbohydrate sequence that is attached to ceramide; 1 stands for GalGalNAcGalGlc-ceramide, 2 for GalNAcGalGlc-ceramide and 3 for GalGlc-ceramide.

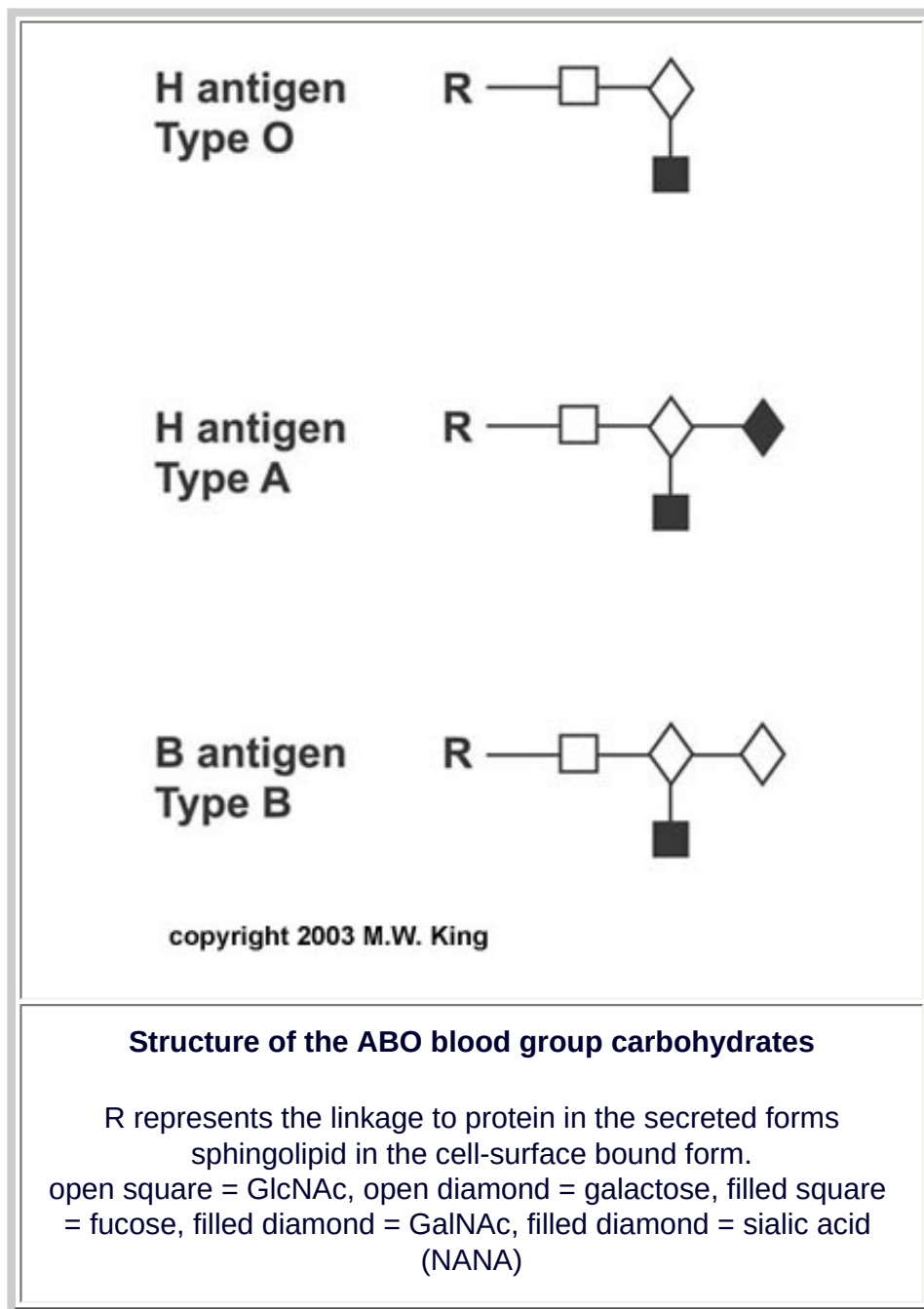
Deficiencies in lysosomal enzymes, which normally are responsible for the degradation of the carbohydrate portions of various gangliosides, underlie the symptoms observed in rare autosomally inherited diseases termed **lipid storage diseases**, many of which are listed below.

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## Clinical Significances of Sphingolipids

One of the most clinically important classes of sphingolipids are those that confer antigenic determinants on the surfaces of cells, particularly the erythrocytes. The **ABO blood group antigens** are the carbohydrate moieties of glycolipids on the surface of cells as well as the carbohydrate portion of serum glycoproteins. When present on the surface of cells the ABO carbohydrates are linked to sphingolipid and are therefore of the glycosphingolipid class. When the ABO carbohydrates are associated with protein in the form of glycoproteins they are found in the serum and are referred to as the secreted forms. Some individuals produce the glycoprotein forms of the ABO antigens while others do not. This property distinguishes **secretors** from **non-secretors**, a property that has forensic importance such as in cases of rape.





A significant cause of death in premature infants and, on occasion, in full term infants is [respiratory distress syndrome \(RDS\)](#) or hyaline membrane disease. This condition is caused by an insufficient amount of pulmonary surfactant. Under normal conditions the surfactant is synthesized by type II endothelial cells and is secreted into the alveolar spaces to prevent atelectasis following expiration during breathing. Surfactant is comprised primarily of dipalmitoyllecithin; additional lipid components include phosphatidylglycerol and phosphatidylinositol along with proteins of 18 and 36 kDa (termed surfactant proteins). During the third trimester the fetal lung synthesizes primarily sphingomyelin, and type II endothelial cells convert the majority of their stored glycogen to fatty acids and then to dipalmitoyllecithin. Fetal lung maturity can be determined by measuring the ratio of lecithin to sphingomyelin (L/S ratio) in the amniotic fluid. An L/S ratio less than 2.0 indicates a potential risk of RDS. The risk is nearly 75-80% when the L/S ratio is 1.5.

The carbohydrate portion of the ganglioside,  $G_{M1}$ , present on the surface of intestinal epithelial cells, is the site of attachment of cholera toxin, the protein secreted by *Vibrio cholerae*.

These are just a few examples of how sphingolipids and glycosphingolipids are involved in various recognition functions at the surface of cells. As with the complex glycoproteins, an understanding of all of the functions of the glycolipids is far from complete.

## Disorders Associated with Abnormal Sphingolipid Metabolism

| Disorder                                                                                                                                               | Enzyme Deficiency                                    | Accumulating Substance                                              | Symptoms                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <a href="#">Tay-Sachs disease</a><br>see below table                                                                                                   | HEXA                                                 | G <sub>M2</sub> ganglioside                                         | rapidly progressing mental retardation, blindness, early mortality                 |
| <a href="#">Sandhoff-Jatzkewitz disease</a><br>see below table                                                                                         | HEXB                                                 | Globoside, G <sub>M2</sub> ganglioside                              | same symptoms as Tay-Sachs, progresses more rapidly                                |
| <a href="#">Tay-Sachs AB variant</a><br>see below table                                                                                                | G <sub>M2</sub> activator (GM2A)                     | G <sub>M2</sub> ganglioside                                         | same symptoms as Tay-Sachs                                                         |
| <a href="#">Gaucher's disease</a>                                                                                                                      | Glucocerebrosidase                                   | Glucocerebroside                                                    | hepatosplenomegaly, mental retardation in infantile form, long bone degeneration   |
| <a href="#">Fabry's disease</a>                                                                                                                        | α-Galactosidase A                                    | Globotriaosylceramide; also called ceramide trihexoside (CTH)       | kidney failure, skin rashes                                                        |
| Niemann-Pick disease, more info below<br><a href="#">Types A and B</a><br><a href="#">Type C1</a><br><a href="#">Type C2</a><br><a href="#">Type D</a> | Sphingomyelinase<br>see info below<br>see info below | Sphingomyelin<br>LDL-derived cholesterol<br>LDL-derived cholesterol | all types lead to mental retardation, hepatosplenomegaly, early fatality potential |
| <a href="#">Krabbe's disease; globoid leukodystrophy</a>                                                                                               | Galactocerebrosidase                                 | Galactocerebroside                                                  | mental retardation, myelin deficiency                                              |
| <a href="#">G<sub>M1</sub> gangliosidosis</a>                                                                                                          | G <sub>M1</sub> ganglioside:β - galactosidase        | G <sub>M1</sub> ganglioside                                         | mental retardation, skeletal abnormalities, hepatomegaly                           |
| <a href="#">Sulfatide lipodosis; metachromatic leukodystrophy</a>                                                                                      | Arylsulfatase A                                      | Sulfatide                                                           | mental retardation, metachromasia of nerves                                        |
| <a href="#">Fucosidosis</a>                                                                                                                            | α-L-Fucosidase                                       | Pentahexosylfucoglycolipid                                          | cerebral degeneration, thickened skin, muscle spasticity                           |
| <a href="#">Farber's lipogranulomatosis</a>                                                                                                            | Acid ceramidase                                      | Ceramide                                                            | hepatosplenomegaly, painful swollen joints                                         |

The  $G_{M2}$  gangliosidoses include Tay-Sachs disease, the Sandhoff diseases and the  $G_{M2}$  activator deficiencies.  $G_{M2}$  ganglioside degradation requires the enzyme  $\beta$ -hexosaminidase and the  $G_{M2}$  activator protein (GM2A). Hexosaminidase is a dimer composed of 2 subunits, either  $\alpha$  and/or  $\beta$ . The HexS protein is  $\alpha\alpha$ , HexA is  $\alpha\beta$  and HexB is  $\beta\beta$ . It is the  $\alpha$ -subunit that carries out the catalysis of  $G_{M2}$  gangliosides. The activator first binds to  $G_{M2}$  gangliosides followed by hexosaminidase and then digestion occurs.

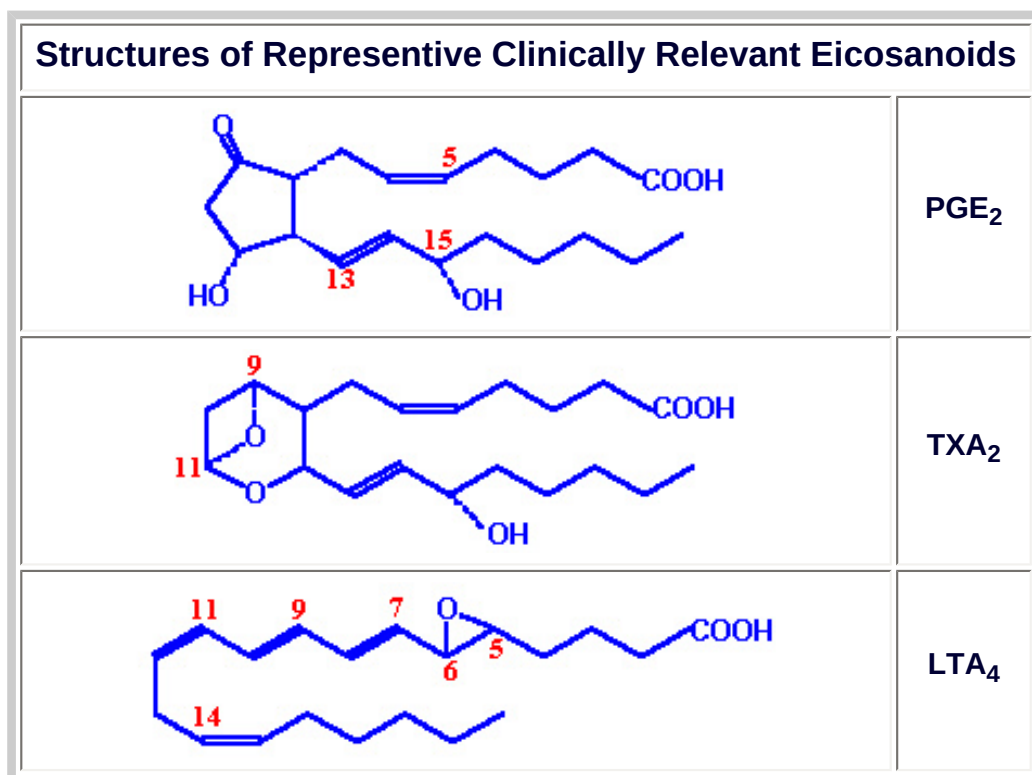
Based upon genetic linkage analyses as well as enzyme studies and the characterization of accumulating lysosomal substances, Niemann Pick disease should be divided into type I and type II; type I has 2 subtypes, A and B (NPA and NPB), which show deficiency of acid sphingomyelinase. Niemann Pick disease type II likewise has 2 subtypes, type C1 and C2 (NPC) and type D (NPD). It is obviously confusing to use the abbreviation NPD for Niemann Pick disease in some cases and for subtype D of Niemann Pick disease in other cases.

Recent studies (**Science** vol. 277 pp. 228-231 and 232-235: July 11, 1997) identified the gene for NPC1. This gene contains regions of homology to mediators of cholesterol homeostasis suggesting why LDL-cholesterol accumulates in lysosomes of afflicted individuals. The encoded protein product of *NPC1* gene is 1278 amino acids long. Within the protein are regions of homology to the transmembrane domain of the morphogen receptor patched (of *Drosophila melanogaster*), the sterol-sensing domain (SSD) of **SREBP** (sterol regulated element binding protein) cleavage-activating protein, **SCAP** and HMG-CoA reductase.

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## Metabolism of the Eicosanoids

The eicosanoids consist of the **prostaglandins (PGs)**, **thromboxanes (TXs)** and **leukotrienes (LTs)**. The PGs and TXs are collectively identified as prostanoids. Prostaglandins were originally shown to be synthesized in the prostate gland, thromboxanes from platelets (thrombocytes) and leukotrienes from leukocytes, hence the derivation of their names.

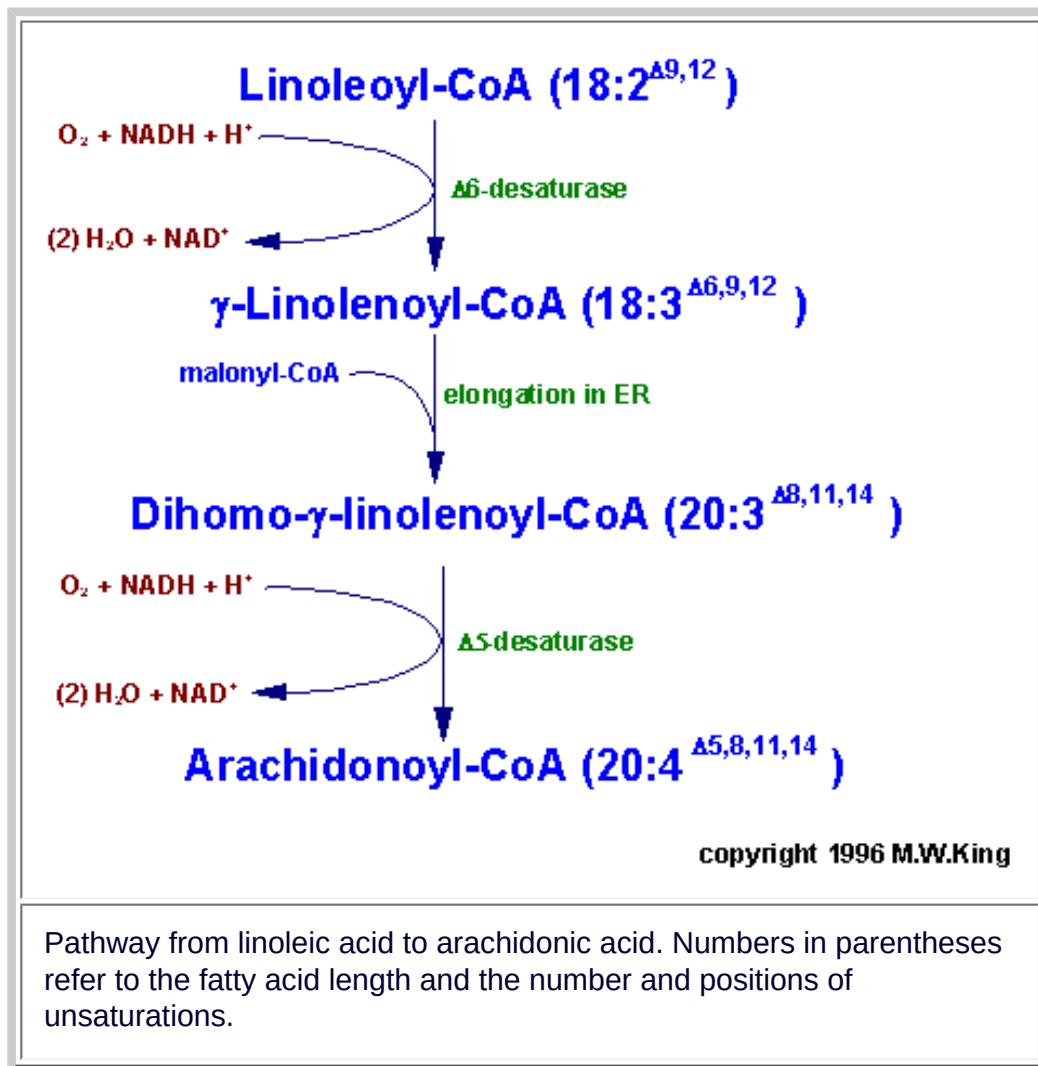


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The eicosanoids produce a wide range of biological effects on inflammatory responses (predominantly those of

the joints, skin and eyes), on the intensity and duration of pain and fever, and on reproductive function (including the induction of labor). They also play important roles in inhibiting gastric acid secretion, regulating blood pressure through vasodilation or constriction, and inhibiting or activating platelet aggregation and thrombosis.

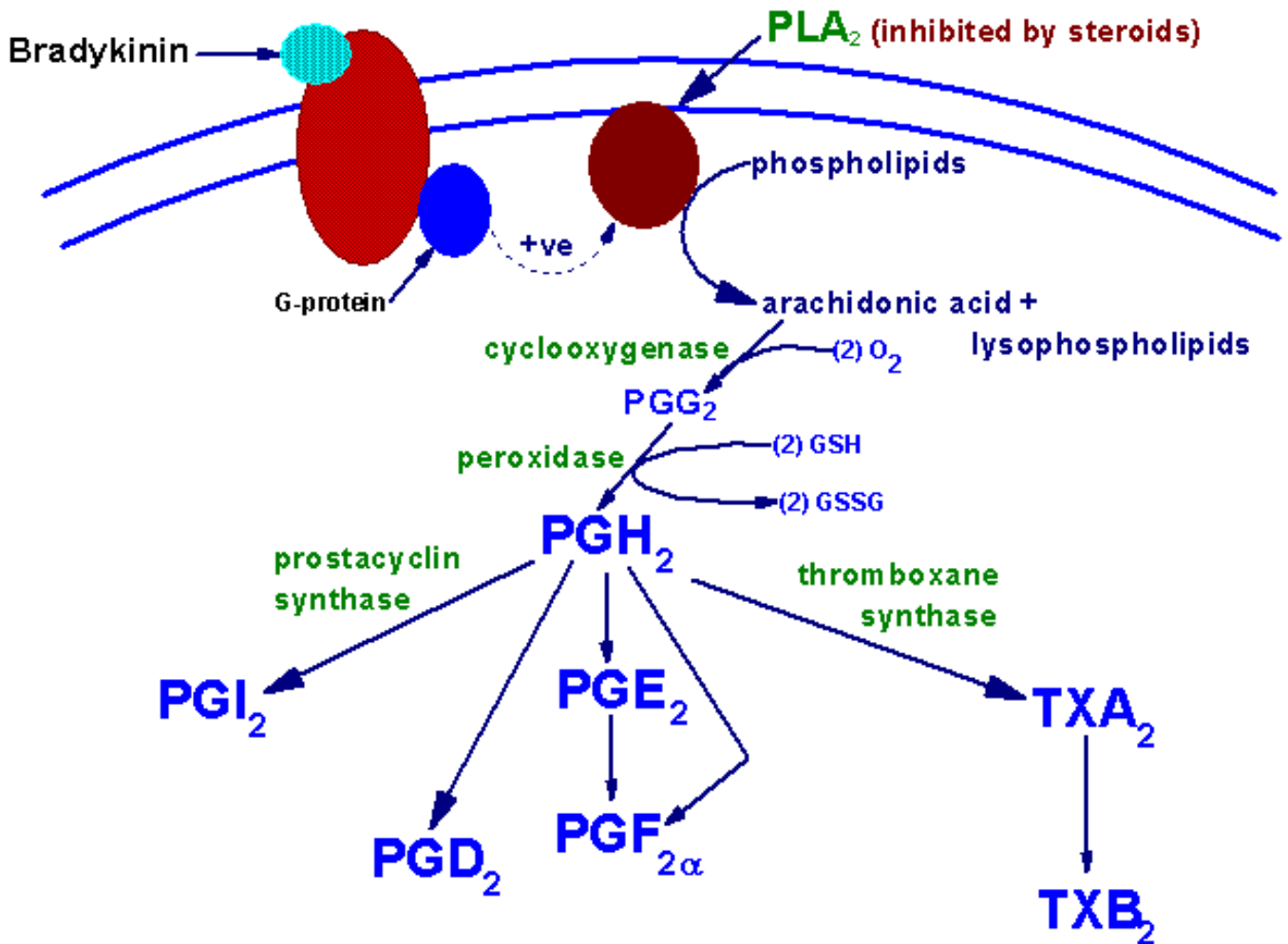
The principal eicosanoids of biological significance to humans are a group of molecules derived from the C<sub>20</sub> fatty acid, **arachidonic acid**. Minor eicosanoids are derived from eicosapentaenoic acid which is itself derived from α-linolenic acid obtained in the diet. The major source of arachidonic acid is through its release from cellular stores. Within the cell, it resides predominantly at the C-2 position of membrane phospholipids and is released from there upon the activation of phospholipase A<sub>2</sub> (see diagram above). The immediate dietary precursor of arachidonate is linoleic acid. Linoleic acid is converted to arachidonic acid through the steps outlined in the figure below. Linoleic acid (arachidonate precursor) and α-linolenic acid (eicosapentaenoate precursor) are essential fatty acids, therefore, their absence from the diet would seriously threaten the body's ability to synthesize eicosanoids.



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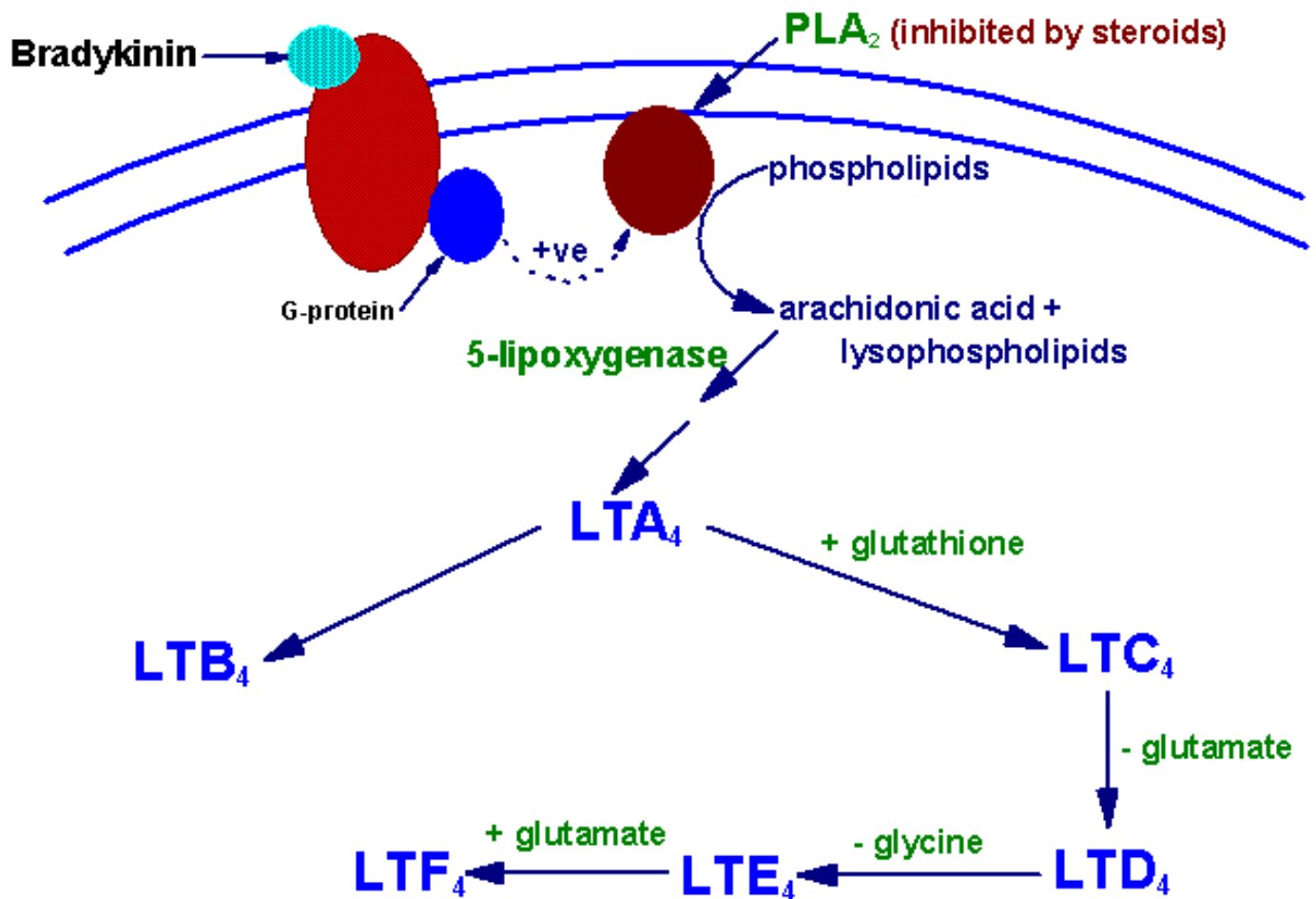
All mammalian cells except erythrocytes synthesize eicosanoids. These molecules are extremely potent, able to cause profound physiological effects at very dilute concentrations. All eicosanoids function locally at the site of synthesis, through receptor-mediated G-protein linked signaling pathways leading to an increase in cAMP levels.

Two main pathways are involved in the biosynthesis of eicosanoids. The prostaglandins and thromboxanes are synthesized by the **cyclic pathway**, the leukotrienes by the **linear pathway**.



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Synthesis of the clinically relevant prostaglandins and thromboxanes from arachidonic acid. Numerous stimuli (e.g. epinephrine, thrombin and bradykinin) activate phospholipase A<sub>2</sub> which hydrolyzes arachidonic acid from membrane phospholipids. The prostaglandins are identified as PG and the thromboxanes as TX. Prostaglandin PGI<sub>2</sub> is also known as **prostacyclin**. The subscript 2 in each molecule refers to the number of -C=C- present.



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Synthesis of the clinically relevant leukotrienes from arachidonic acid. Numerous stimuli (e.g. epinephrine, thrombin and bradykinin) activate phospholipase A<sub>2</sub> which hydrolyzes arachidonic acid from membrane phospholipids. The leukotrienes are identified as LT. The leukotrienes, LTC<sub>4</sub>, LTD<sub>4</sub>, LTE<sub>4</sub> and LTF<sub>4</sub> are known as the **peptidoleukotrienes** because of the presence of amino acids. The peptidoleukotrienes, LTC<sub>4</sub>, LTD<sub>4</sub> and LTE<sub>4</sub> are components of **slow-reacting substance of anaphylaxis**. The subscript 4 in each molecule refers to the number of -C=C- present.

The cyclic pathway is initiated through the action of prostaglandin G/H synthase, PGS (also called prostaglandin endoperoxide synthetase). This enzyme possesses two activities, cyclooxygenase (COX) and peroxidase. There are 2 forms of the COX activity. COX-1 (PGS-1) is expressed constitutively in gastric mucosa, kidney, platelets, and vascular endothelial cells. COX-2 (PGS-2) is inducible and is expressed in macrophages and monocytes in response to inflammation. The primary trigger for COX-2 induction in monocytes and macrophages is **platelet-activating factor, PAF** and **interleukin-1, IL-1**. Both COX-1 and COX-2 catalyze the 2-step conversion of arachidonic acid to PGG<sub>2</sub> and then to PGH<sub>2</sub>.

The linear pathway is initiated through the action of lipoxygenases. It is the enzyme, 5-lipoxygenase that gives rise to the leukotrienes.

A widely used class of drugs, the non-steroidal anti-inflammatory drugs (**NSAIDs**) such as ibuprofen, indomethacin, naproxen, phenylbutazone and aspirin, all act upon the cyclooxygenase activity, inhibiting both COX-1 and COX-2. Because inhibition of COX-1 activity in the gut is associated with NSAID-induced ulcerations, pharmaceutical companies have developed drugs targeted exclusively against the inducible COX-2 activity [e.g.



Celebrex (celecoxib), Bextra (valdecoxib), Prexige (lumiracoxib) and the recently removed Vioxx (rofecoxib)]. Another class, the corticosteroidal drugs, act to inhibit phospholipase A<sub>2</sub>, thereby inhibiting the release of arachidonate from membrane phospholipids and the subsequent synthesis of eicosinoids.

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## Properties of Significant Eicosanoids

| Eicosanoid                          | Major site(s) of synthesis                                                               | Major biological activities                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PGD <sub>2</sub>                    | mast cells                                                                               | inhibits platelet and leukocyte aggregation, decreases T-cell proliferation and lymphocyte migration and secretion of IL-1 $\alpha$ and IL-2; induces vasodilation and production of cAMP                                                                                                                                          |
| PGE <sub>2</sub>                    | kidney, spleen, heart                                                                    | increases vasodilation and cAMP production, enhancement of the effects of bradykinin and histamine, induction of uterine contractions and of platelet aggregation, maintaining the open passageway of the fetal ductus arteriosus; decreases T-cell proliferation and lymphocyte migration and secretion of IL-1 $\alpha$ and IL-2 |
| PGF <sub>2<math>\alpha</math></sub> | kidney, spleen, heart                                                                    | increases vasoconstriction, bronchoconstriction and smooth muscle contraction                                                                                                                                                                                                                                                      |
| PGH <sub>2</sub>                    |                                                                                          | precursor to thromboxanes A <sub>2</sub> and B <sub>2</sub> , induction of platelet aggregation and vasoconstriction                                                                                                                                                                                                               |
| PGI <sub>2</sub>                    | heart, vascular endothelial cells                                                        | inhibits platelet and leukocyte aggregation, decreases T-cell proliferation and lymphocyte migration and secretion of IL-1 $\alpha$ and IL-2; induces vasodilation and production of cAMP                                                                                                                                          |
| TXA <sub>2</sub>                    | platelets                                                                                | induces platelet aggregation, vasoconstriction, lymphocyte proliferation and bronchoconstriction                                                                                                                                                                                                                                   |
| TXB <sub>2</sub>                    | platelets                                                                                | induces vasoconstriction                                                                                                                                                                                                                                                                                                           |
| LTB <sub>4</sub>                    | monocytes, basophils, neutrophils, eosinophils, mast cells, epithelial cells             | induces leukocyte chemotaxis and aggregation, vascular permeability, T-cell proliferation and secretion of INF- $\gamma$ , IL-1 and IL-2                                                                                                                                                                                           |
| LTC <sub>4</sub>                    | monocytes and alveolar macrophages, basophils, eosinophils, mast cells, epithelial cells | component of SRS-A, microvascular vasoconstrictor, vascular permeability and bronchoconstriction and secretion of INF- $\gamma$                                                                                                                                                                                                    |

|                  |                                                                               |                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| LTD <sub>4</sub> | monocytes and alveolar macrophages, eosinophils, mast cells, epithelial cells | predominant component of SRS-A, microvascular vasoconstrictor, vascular permeability and bronchoconstriction and secretion of INF- $\gamma$ |
| LTE <sub>4</sub> | mast cells and basophils                                                      | component of SRS-A, microvascular vasoconstrictor and bronchoconstriction                                                                   |

**\*\*SRS-A = slow-reactive substance of anaphylaxis**

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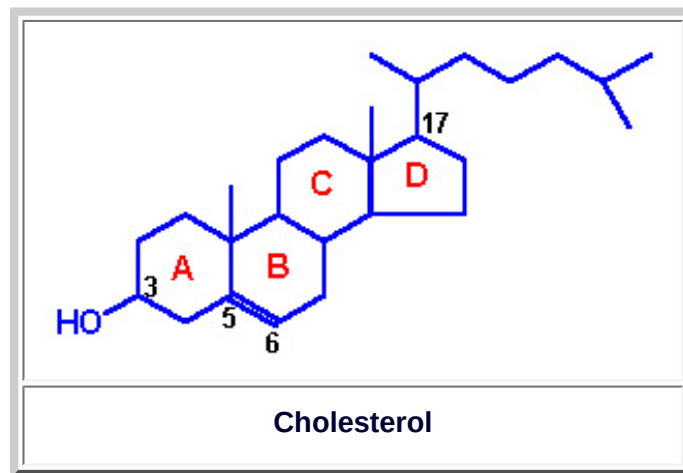
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## Introduction to Cholesterol Metabolism

Cholesterol is an extremely important biological molecule that has roles in membrane structure as well as being a precursor for the synthesis of the [steroid hormones](#) and bile acids. Both dietary cholesterol and that synthesized *de novo* are transported through the circulation in [lipoprotein particles](#). The same is true of cholesteryl esters, the form in which cholesterol is stored in cells.

The synthesis and utilization of cholesterol must be tightly regulated in order to prevent over-accumulation and abnormal deposition within the body. Of particular importance clinically is the abnormal deposition of cholesterol and cholesterol-rich lipoproteins in the coronary arteries. Such deposition, eventually leading to atherosclerosis, is the leading contributory factor in diseases of the coronary arteries.



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## Biosynthesis of Cholesterol

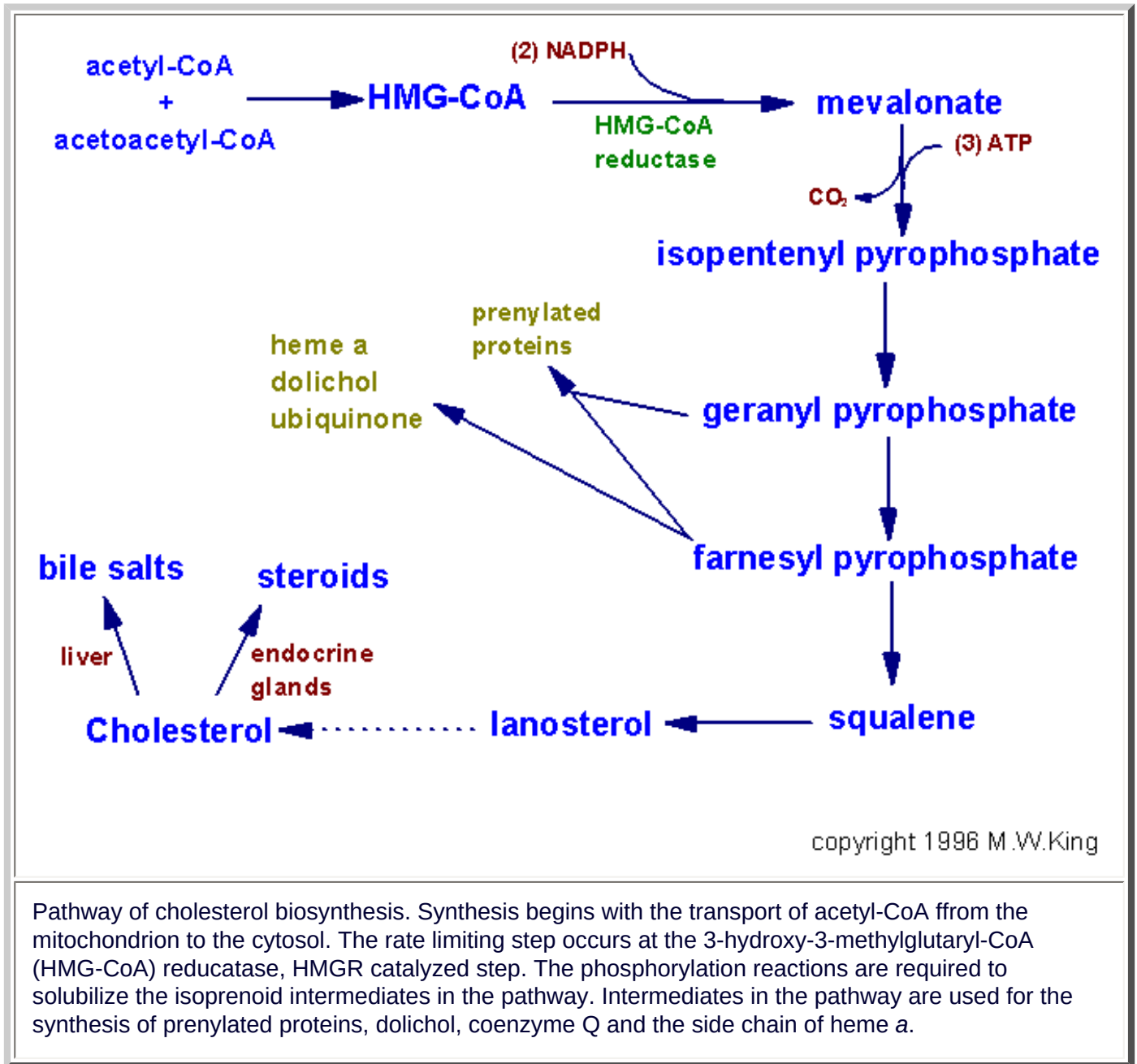
Slightly less than half of the cholesterol in the body derives from biosynthesis *de novo*. Biosynthesis in the liver accounts for approximately 10%, and in the intestines approximately 15%, of the amount produced each day. Cholesterol synthesis occurs in the cytoplasm and microsomes from the two-carbon acetate group of acetyl-CoA.

The process has five major steps:

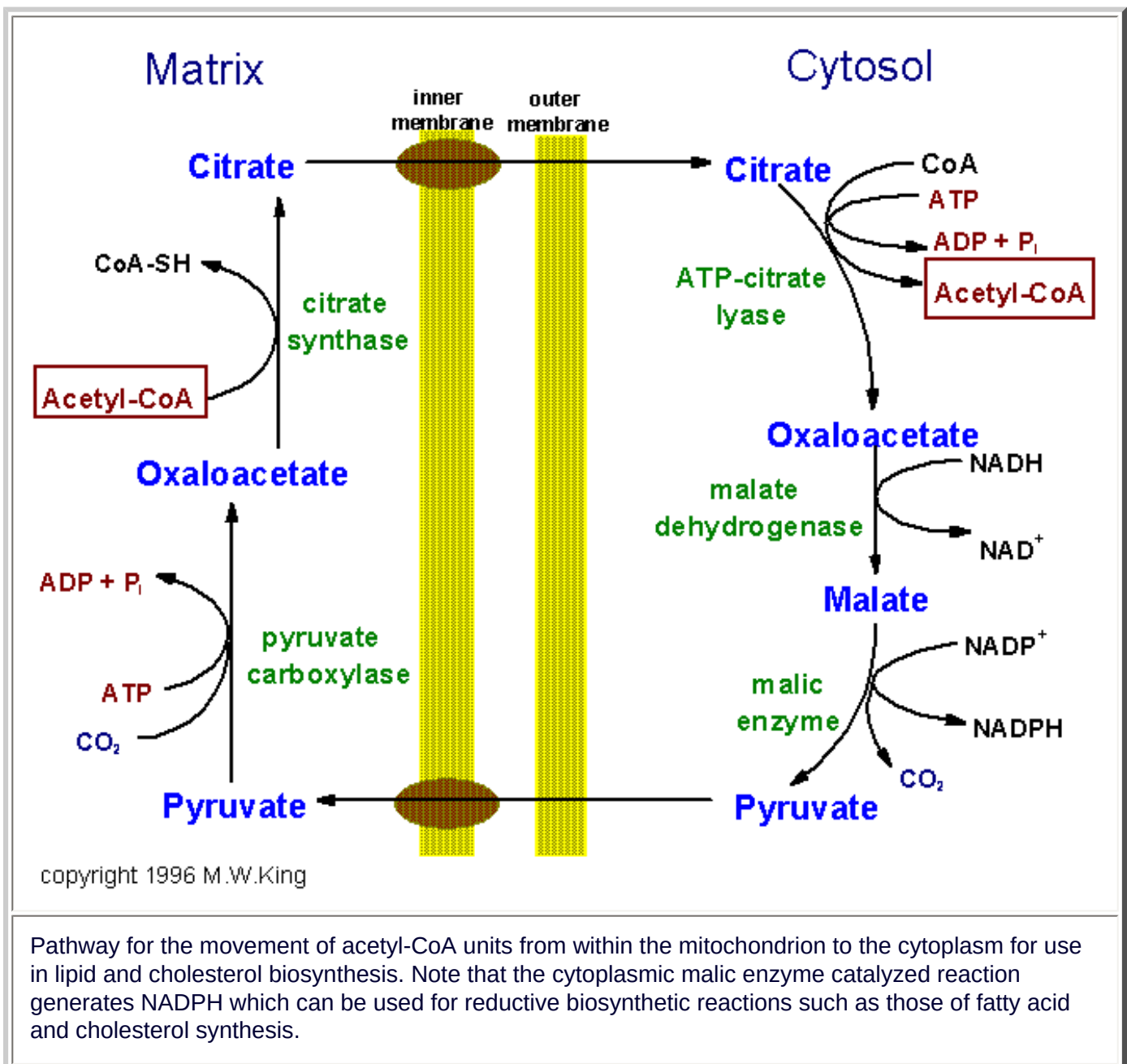
- **1.** Acetyl-CoAs are converted to 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA)
- **2.** HMG-CoA is converted to mevalonate
- **3.** Mevalonate is converted to the isoprene based molecule, isopentenyl pyrophosphate (IPP), with the concomitant

loss of CO<sub>2</sub>

- 4. IPP is converted to squalene
- 5. Squalene is converted to cholesterol.



The acetyl-CoA utilized for cholesterol biosynthesis is derived from an oxidation reaction (eg, fatty acids or pyruvate) in the mitochondria and is transported to the cytoplasm by the same process as that described for [fatty acid synthesis](#) (see the Figure below). Acetyl-CoA can also be derived from cytoplasmic oxidation of ethanol by acetyl-CoA synthetase. All the reduction reactions of cholesterol biosynthesis use NADPH as a cofactor. The isoprenoid intermediates of cholesterol biosynthesis can be diverted to other synthesis reactions, such as those for dolichol (used in the synthesis of [N-linked glycoproteins](#), coenzyme Q (of the [oxidative phosphorylation](#)) pathway or the side chain of heme a. Additionally, these intermediates are used in the [lipid modification of some proteins](#).



Acetyl-CoA units are converted to mevalonate by a series of reactions that begins with the formation of **HMG-CoA**. Unlike the HMG-CoA formed during ketone body synthesis in the mitochondria, this form is synthesized in the cytoplasm. However, the pathway and the necessary enzymes are the same as those in the mitochondria. Two moles of acetyl-CoA are condensed in a reversal of the thiolase reaction, forming acetoacetyl-CoA. Acetoacetyl-CoA and a third mole of acetyl-CoA are converted to HMG-CoA by the action of HMG-CoA synthase. HMG-CoA is converted to **mevalonate** by HMG-CoA reductase, HMGR (this enzyme is bound in the endoplasmic reticulum, ER). HMGR absolutely requires NADPH as a cofactor and two moles of NADPH are consumed during the conversion of HMG-CoA to mevalonate.

The reaction catalyzed by HMGR is the rate limiting step of cholesterol biosynthesis, and this enzyme is subject to complex regulatory controls.

Mevalonate is then activated by three successive phosphorylations, yielding **5-pyrophosphomevalonate**. In addition to activating mevalonate, the phosphorylations maintain its solubility, since otherwise it is insoluble in water. After phosphorylation, an ATP-dependent decarboxylation yields **isopentenyl pyrophosphate**, IPP, an activated isoprenoid molecule. Isopentenyl pyrophosphate is in equilibrium with its isomer, **dimethylallyl pyrophosphate**, **DMPP**. One molecule of IPP condenses with one molecule of DMPP to generate **geranyl pyrophosphate**, **GPP**. GPP further condenses with another IPP molecule to yield **farnesyl pyrophosphate**, **FPP**. Finally, the NADPH-requiring enzyme, squalene synthase

catalyzes the head-to-tail condensation of two molecules of FPP, yielding squalene (squalene synthase also is tightly associated with the endoplasmic reticulum). Squalene undergoes a two step cyclization to yield **lanosterol**. The first reaction is catalyzed by squalene monooxygenase. This enzyme uses NADPH as a cofactor to introduce molecular oxygen as an epoxide at the 2,3 position of squalene. Through a series of 19 additional reactions, lanosterol is converted to cholesterol.

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## Regulating Cholesterol Synthesis

Normal healthy adults synthesize cholesterol at a rate of approximately 1g/day and consume approximately 0.3g/day. A relatively constant level of cholesterol in the body (150 - 200 mg/dL) is maintained primarily by controlling the level of *de novo* synthesis. The level of cholesterol synthesis is regulated in part by the dietary intake of cholesterol. Cholesterol from both diet and synthesis is utilized in the formation of membranes and in the synthesis of the steroid hormones and bile acids (see below). The greatest proportion of cholesterol is used in bile acid synthesis.

The cellular supply of cholesterol is maintained at a steady level by three distinct mechanisms:

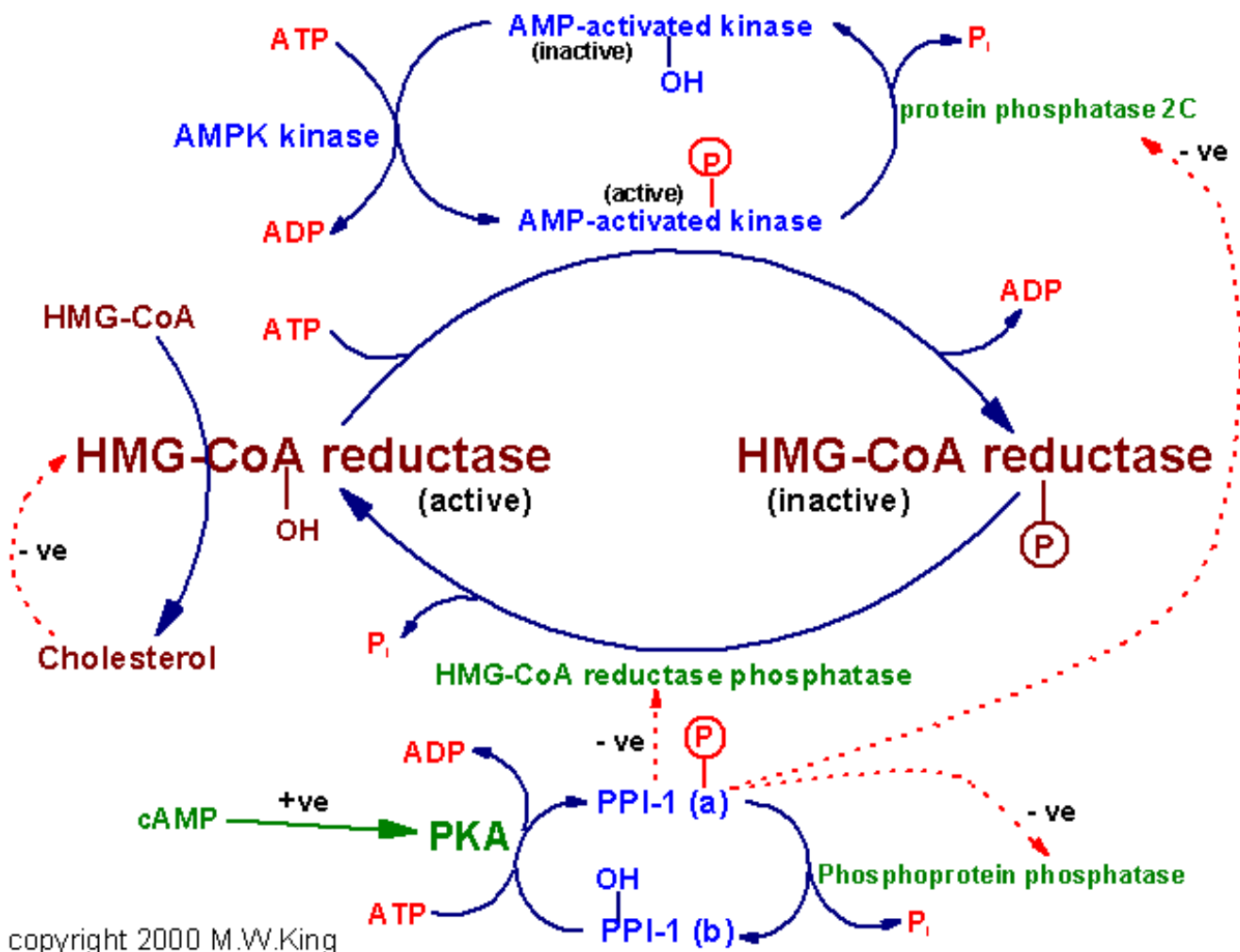
- **1.** Regulation of HMGR activity and levels
- **2.** Regulation of excess intracellular free cholesterol through the activity of acyl-CoA:cholesterol acyltransferase, ACAT
- **3.** Regulation of plasma cholesterol levels via LDL receptor-mediated uptake and HDL-mediated reverse transport.

Regulation of HMGR activity is the primary means for controlling the level of cholesterol biosynthesis. The enzyme is controlled by four distinct mechanisms: feed-back inhibition, control of gene expression, rate of enzyme degradation and phosphorylation-dephosphorylation.

The first three control mechanisms are exerted by cholesterol itself. Cholesterol acts as a feed-back inhibitor of pre-existing HMGR as well as inducing rapid degradation of the enzyme. The latter is the result of cholesterol-induced polyubiquitination of HMGR and its degradation in the proteosome (see [proteolytic degradation](#) below). This ability of cholesterol is a consequence of the **sterol sensing domain, SSD** of HMGR. In addition, when cholesterol is in excess the amount of mRNA for HMGR is reduced as a result of decreased expression of the gene. The mechanism by which cholesterol (and other sterols) affect the transcription of the HMGR gene is described below under [regulation of sterol content](#).

Regulation of HMGR through covalent modification occurs as a result of phosphorylation and dephosphorylation. The enzyme is most active in its unmodified form. Phosphorylation of the enzyme decreases its activity. HMGR is phosphorylated by AMP-activated protein kinase, **AMPK** (this is not the same as cAMP-dependent protein kinase, PKA). AMPK itself is activated via phosphorylation. The phosphorylation of AMPK is catalyzed by one or more AMPK kinases (AMPKKs). Visit [AMPK: The Master Metabolic Regulator](#) for more detailed information on the role of AMPK in regulating metabolism.





Regulation of HMGR by covalent modification. HMGR is most active in the dephosphorylated state. Phosphorylation is catalyzed by AMP-activated protein kinase, AMPK, (used to be termed HMGR kinase), an enzyme whose activity is also regulated by phosphorylation. Phosphorylation of AMPK is catalyzed by AMPK kinase (AMPKK). Hormones such as glucagon and epinephrine negatively affect cholesterol biosynthesis by increasing the activity of the inhibitor of phosphoprotein phosphatase inhibitor-1, PPI-1. Conversely, insulin stimulates the removal of phosphates and, thereby, activates HMGR activity. Additional regulation of HMGR occurs through an inhibition of its' activity as well as of its' synthesis by elevation in intracellular cholesterol levels. This latter phenomenon involves the transcription factor SREBP described below.

The activity of HMGR is additionally controlled by the cAMP signaling pathway. Increases in cAMP lead to activation of cAMP-dependent protein kinase, PKA. In the context of HMGR regulation, PKA phosphorylates phosphoprotein phosphatase inhibitor-1 (PPI-1) leading to an increase in its' activity. PPI-1 can inhibit the activity of numerous phosphatases including protein phosphatase 2C and HMG-CoA reductase phosphatase which remove phosphates from AMPK and HMGR, respectively. This maintains AMPK in the phosphorylated and active state, and HMGR in the phosphorylated and inactive state. As the stimulus leading to increased cAMP production is removed, the level of phosphorylations decreases and that of dephosphorylations increases. The net result is a return to a higher level of HMGR activity.

Since the intracellular level of cAMP is regulated by hormonal stimuli, regulation of cholesterol biosynthesis is hormonally controlled. Insulin leads to a decrease in cAMP, which in turn activates cholesterol synthesis. Alternatively, glucagon and epinephrine, which increase the level of cAMP, inhibit cholesterol synthesis.

The ability of insulin to stimulate, and glucagon to inhibit, HMGR activity is consistent with the effects of these hormones on other metabolic pathways. The basic function of these two hormones is to control the availability and delivery of energy to all cells of the body.

Long-term control of HMGR activity is exerted primarily through control over the synthesis and degradation of the enzyme. When levels of cholesterol are high, the level of expression of the HMGR gene is reduced. Conversely, reduced levels of cholesterol activate expression of the gene. Insulin also brings about long-term regulation of cholesterol metabolism by increasing the level of HMGR synthesis.

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## Proteolytic Regulation of HMG-CoA Reductase

The stability of HMGR is regulated as the rate of flux through the mevalonate synthesis pathway changes. When the flux is high the rate of HMGR degradation is also high. When the flux is low, degradation of HMGR decreases. This phenomenon can easily be observed in the presence of the [statin drugs](#).

HMGR is localized to the ER and like SREBP (see below) contains a sterol-sensing domain, SSD. When sterol levels increase in cells there is an concomitant increase in the rate of HMGR degradation. The degradation of HMGR occurs within the proteosome, a multiprotein complex dedicated to protein degradation. The primary signal directing proteins to the proteosome is ubiquitination. Ubiquitin is a 7.6kDa protein that is covalently attached to proteins targeted for degradation by ubiquitin ligases. These enzymes attach multiple copies of ubiquitin allowing for recognition by the proteosome. HMGR has been shown to be ubiquitinated prior to its degradation. The primary sterol regulating HMGR degradation is cholesterol itself. As the levels of free cholesterol increase in cells, the rate of HMGR degradation increases.

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## The Utilization of Cholesterol

Cholesterol is transported in the plasma predominantly as cholesteryl esters associated with [lipoproteins](#). Dietary cholesterol is transported from the small intestine to the liver within chylomicrons. Cholesterol synthesized by the liver, as well as any dietary cholesterol in the liver that exceeds hepatic needs, is transported in the serum within LDLs. The liver synthesizes VLDLs and these are converted to LDLs through the action of endothelial cell-associated lipoprotein lipase. Cholesterol found in plasma membranes can be extracted by HDLs and esterified by the HDL-associated enzyme LCAT. The cholesterol acquired from peripheral tissues by HDLs can then be transferred to VLDLs and LDLs via the action of cholesteryl ester transfer protein (apo-D) which is associated with HDLs. **Reverse cholesterol transport** allows peripheral cholesterol to be returned to the liver in LDLs. Ultimately, cholesterol is excreted in the bile as free cholesterol or as bile salts following conversion to bile acids in the liver.

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## Regulation of Cellular Sterol Content

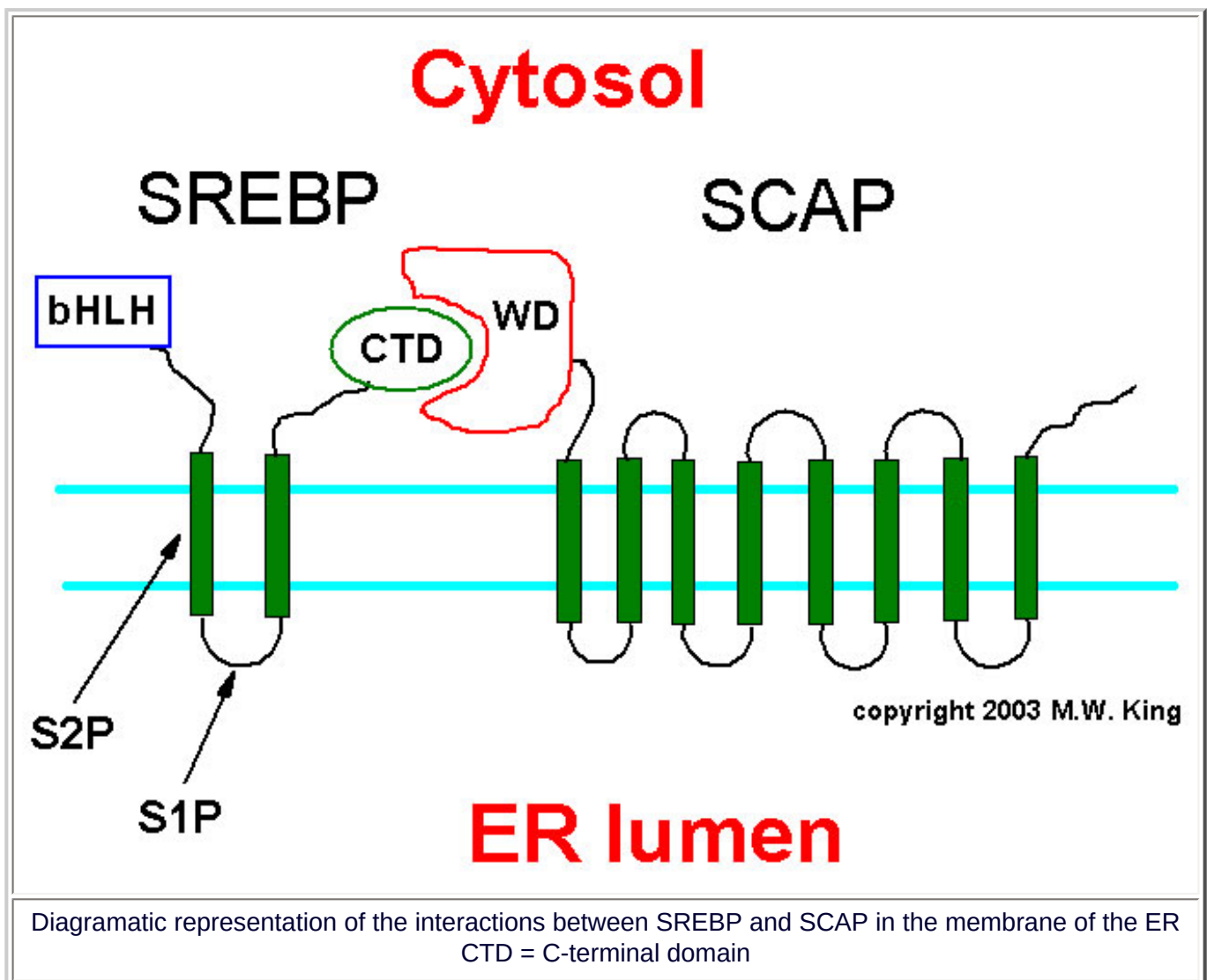
The continual alteration of the intracellular sterol content occurs through the regulation of key sterol synthetic enzymes as well as by altering the levels of cell-surface LDL receptors. As cells need more sterol they will induce their synthesis and uptake, conversely when the need declines synthesis and uptake are decreased. Regulation of these events is brought about primarily by sterol-regulated transcription of key rate limiting enzymes and by the regulated degradation of HMGR. Activation of transcriptional control occurs through the regulated cleavage of the membrane-bound transcription factor **sterol regulated element binding protein, SREBP**. As discussed above, degradation of HMGR is controlled by the ubiquitin-mediated pathway for proteolysis.

Sterol control of transcription affects more than 30 genes involved in the biosynthesis of cholesterol, triacylglycerols, phospholipids and fatty acids. Transcriptional control requires the presence of an octamer sequence in the gene termed the **sterol regulatory element, SRE-1**. It has been shown that SREBP is the transcription factor that binds to SRE-1 elements. It turns out that there are 2 distinct SREBP gene, SREBP-1 and SREBP-2. In addition, the SREBP-1 gene encodes 2 proteins, SREBP-1a and SREBP-1c as a consequence of alternative exon usage. All 3 proteins are proteolytically regulated by sterols. Full-length SREBPs have several domains and are embedded in the membrane of the endoplasmic reticulum (ER). The N-terminal domain contains a [transcription factor motif](#) of the basic helix-loop-helix (bHLH) type that is exposed to the cytoplasmic side of the ER. There are 2 transmembrane spanning domains followed by a large C-terminal domain also exposed to the cytosolic side. When sterols are scarce, cleavage of the full-length SREBP takes place with the result being that the N-terminal bHLH motif is released into the cytosol. The bHLH domain then migrates to the nucleus to direct transcription. Conversely, when sterols are abundant, cleavage of SREBP is inhibited. To control the level of SREBP-mediated transcription, the soluble bHLH domain is itself subject to rapid proteolysis.

The cleavage of SREBP is carried out by 2 distinct enzymes, one of which is regulated by sterols. The regulated cleavage occurs in the luminal loop between the 2 transmembrane domains. This cleavage is catalyzed by **site-1 protease, S1P**. High sterol content blocks the activity of S1P. The second cleavage, catalyzed by **site-2 protease, S2P**, occurs in the first transmembrane span, leading to release of active SREBP. In order for S2P to act on SREBP, site-1 must already have been cleaved.

Additional studies on sterol-regulated gene expression demonstrated that cleavage of SREBP by S1P is controlled by the level and action of an additional protein termed, **SREBP cleavage-activating protein, SCAP**. SCAP is a large protein also found in the ER membrane and contains at least 8 transmembrane spans. The C-terminal portion, which extends into the cytosol, has been shown to interact with the C-terminal domain of SREBP. This C-terminal region of SCAP contains 4 motifs called **WD40 repeats**. The WD40 repeats are required for interaction of SCAP with SREBP. Interestingly, the N-terminus of SCAP, including membrane spans 2-6, resembles HMGR which itself is subject to sterol-stimulated degradation (see above). This shared motif is called the **sterol sensing domain, SSD**. Several proteins whose functions involve sterols also contain the SSD. These include **patched**, an important development regulating receptor whose ligand, **hedgehog**, is modified by attachment of cholesterol and the **Neimann Pick C1 (NPC1)** protein which is involved in cholesterol transport in the secretory pathway. NPC1 is one of several genes whose activities, when disrupted, lead to severe neurological dysfunction.

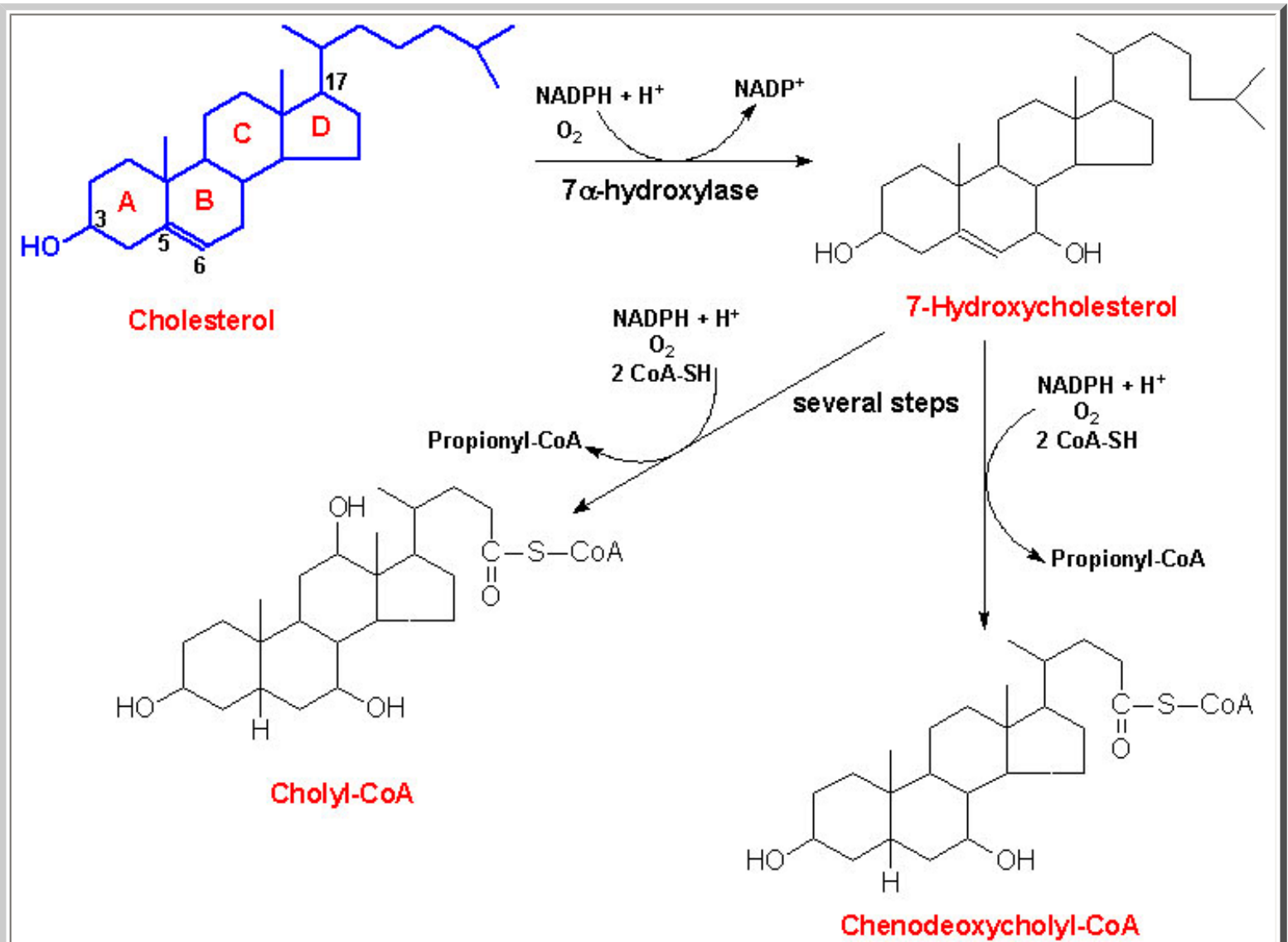
The function of SCAP is to positively stimulate S1P-mediated cleavage of SREBP. The function of sterols is to inhibit this positive action of SCAP. The activity of SCAP involves movement from the ER to the Golgi and back. Because the C-terminus of SCAP interacts with SREBP, movement of SCAP takes SREBP along for the ride. When sterols are low, SCAP and SREBP move to the Golgi. This transit is required for SREBP cleavage as S1P is Golgi-localized. When sterols are high, movement of SCAP is halted. Thus, the overall effect of sterols is to regulate the ability of SCAP to present SREBP to S1P.



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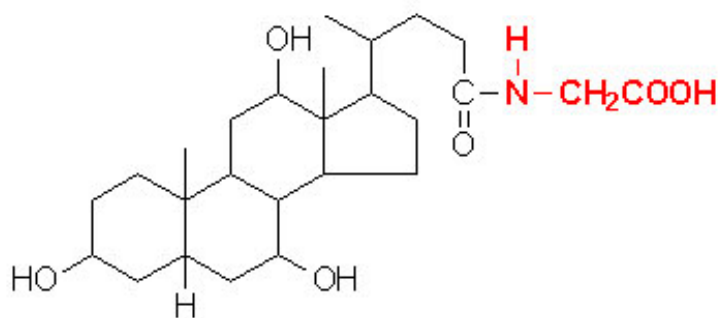
## Bile Acids Synthesis and Utilization

The end products of cholesterol utilization are the bile acids, synthesized in the liver. Synthesis of bile acids is one of the predominant mechanisms for the excretion of excess cholesterol. However, the excretion of cholesterol in the form of bile acids is insufficient to compensate for an excess dietary intake of cholesterol.

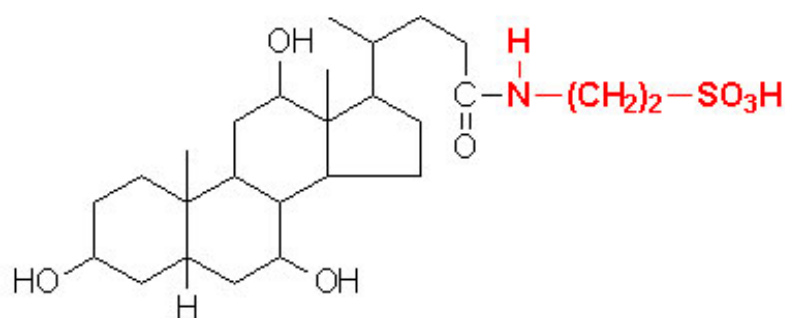


Synthesis of the 2 primary bile acids, cholic acid and chenodeoxycholic acid. The reaction catalyzed by the 7 $\alpha$ -hydroxylase is the rate limiting step in bile acid synthesis. Conversion of 7 $\alpha$ -hydroxycholesterol to the bile acids requires several steps not shown in detail in this image. Only the relevant co-factors needed for the synthesis steps are shown.

The most abundant bile acids in human bile are **chenodeoxycholic acid** (45%) and **cholic acid** (31%). These are referred to as the **primary bile acids**. Within the intestines the primary bile acids are acted upon by bacteria and converted to the **secondary bile acids**, identified as deoxycholate (from cholate) and lithocholate (from chenodeoxycholate). Both primary and secondary bile acids are reabsorbed by the intestines and delivered back to the liver via the portal circulation.



Glycocholic acid



Taurocholic acid

Structure of the conjugated cholic acids

Within the liver the carboxyl group of primary and secondary bile acids is conjugated via an amide bond to either glycine or taurine before their being resecreted into the bile canaliculi. These conjugation reactions yield **glycoconjugates** and **tauroconjugates**, respectively. The bile canaliculi join with the bile ductules, which then form the bile ducts. Bile acids are carried from the liver through these ducts to the gallbladder, where they are stored for future use. The ultimate fate of bile acids is secretion into the intestine, where they aid in the emulsification of dietary lipids. In the gut the glycine and taurine residues are removed and the bile acids are either excreted (only a small percentage) or reabsorbed by the gut and returned to the liver. This process of secretion from the liver to the gallbladder, to the intestines and finally reabsorption is termed the **enterohepatic circulation**.

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## Clinical Significance of Bile Acid Synthesis

Bile acids perform four physiologically significant functions:

- 1. their synthesis and subsequent excretion in the feces represent the only significant mechanism for the elimination of excess cholesterol.
- 2. bile acids and phospholipids solubilize cholesterol in the bile, thereby preventing the precipitation of cholesterol in the gallbladder.
- 3. they facilitate the digestion of dietary triacylglycerols by acting as emulsifying agents that render fats accessible to pancreatic lipases.
- 4. they facilitate the intestinal absorption of fat-soluble vitamins.

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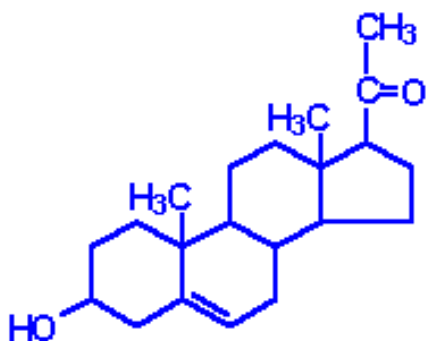
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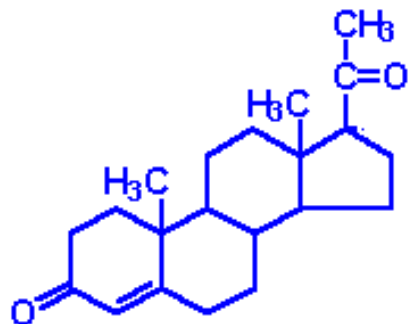
## Introduction

With the exception of [retinoic acid](#), the steroid hormones are all derived from cholesterol. Moreover, with the exception of [vitamin D](#), they all contain the same cyclopentanophenanthrene ring and atomic numbering system as cholesterol. The conversion of C<sub>27</sub> cholesterol to the 18-, 19-, and 21-carbon steroid hormones (designated by the nomenclature C with a subscript number indicating the number of carbon atoms, e.g. C<sub>19</sub> for androstanes) involves the rate-limiting, irreversible cleavage of a 6-carbon residue from cholesterol, producing **pregnenolone (C<sub>21</sub>)** plus isocaproaldehyde. Common names of the steroid hormones are widely recognized, but systematic nomenclature is gaining acceptance and familiarity with both nomenclatures is increasingly important. Steroids with 21 carbon atoms are known systematically as **pregnanes**, whereas those containing 19 and 18 carbon atoms are known as **androstanes** and **estrans**, respectively. The important mammalian steroid hormones are shown below along with the structure of the precursor, pregnenolone. Retinoic acid and vitamin D are not derived from pregnenolone, but from vitamin A and cholesterol respectively.

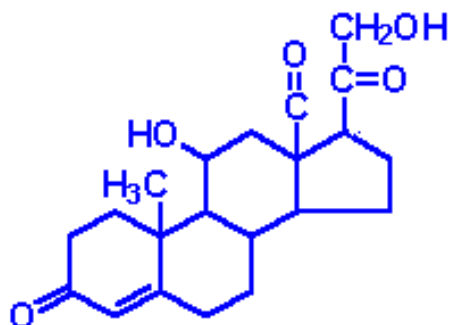


**Pregnenolone:** produced directly from cholesterol, the precursor molecule for all C<sub>18</sub>, C<sub>19</sub> and C<sub>21</sub> steroids

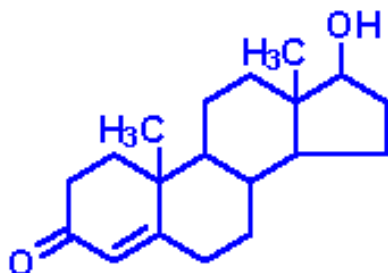




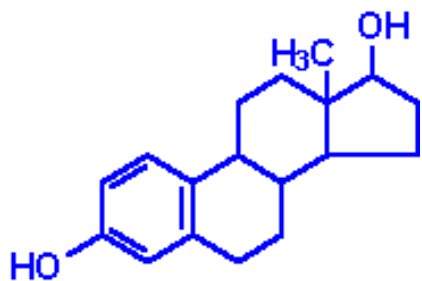
**Progesterone:** a progestin, produced directly from pregnenolone and secreted from the *corpus luteum*, responsible for changes associated with luteal phase of the menstrual cycle, differentiation factor for mammary glands



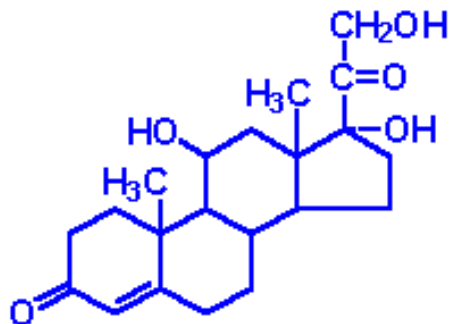
**Aldosterone:** the principal mineralocorticoid, produced from progesterone in the *zona glomerulosa* of adrenal cortex, raises blood pressure and fluid volume, increases Na<sup>+</sup> uptake



**Testosterone:** an androgen, male sex hormone synthesized in the testes, responsible for secondary male sex characteristics, produced from progesterone



**Estradiol:** an estrogen, principal female sex hormone, produced in the ovary, responsible for secondary female sex characteristics



**Cortisol:** dominant glucocorticoid in humans, synthesized from progesterone in the *zona fasciculata* of the adrenal cortex, involved in stress adaptation, elevates blood pressure and  $\text{Na}^+$  uptake, numerous effects on the immune system

All the steroid hormones exert their action by passing through the plasma membrane and binding to intracellular receptors. The mechanism of action of the thyroid hormones is similar; they interact with intracellular receptors. Both the steroid and thyroid hormone-receptor complexes exert their action by binding to specific nucleotide sequences in the DNA of responsive genes. These DNA sequences are identified as **hormone response elements, HREs**. The interaction of steroid-receptor complexes with DNA leads to altered rates of transcription of the associated genes.

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## Steroid Hormone Biosynthesis Reactions

The particular steroid hormone class synthesized by a given cell type depends upon its complement of peptide hormone receptors, its response to peptide hormone stimulation and its genetically expressed complement of enzymes. The following indicates which peptide hormone is responsible for stimulating the synthesis of which steroid hormone:

- Luteinizing Hormone (LH):
  - **progesterone and testosterone**
- Adrenocorticotrophic hormone (ACTH):
  - **cortisol**
- Follicle Stimulating Hormone (FSH):
  - **estradiol**
- Angiotensin II/III:
  - **aldosterone**

The first reaction in converting cholesterol to  $\text{C}_{18}$ ,  $\text{C}_{19}$  and  $\text{C}_{21}$  steroids involves the cleavage of a 6-carbon group from cholesterol and is the principal committing, regulated, and rate-limiting step in steroid biosynthesis. The enzyme system that catalyzes the cleavage reaction is known as P450-linked side chain cleaving enzyme (P450ssc), or desmolase, and is found in the mitochondria of steroid-producing cells, but not in significant quantities in other cells.

Mitochondrial desmolase is a complex enzyme system consisting of cytochrome P450, and adrenodoxin (a P450 reductant). The

activity of each of these components is increased by 2 principal cAMP- and PKA-dependent processes. First, cAMP stimulates PKA, leading to the phosphorylation of a cholesteryl-ester esterase and generating increased concentrations of cholesterol, the substrate for desmolase. Second, long-term regulation is effected at the level the gene for desmolase. This gene contains a cAMP regulatory element (CRE) that binds cAMP and increases the level of desmolase RNA transcription, thereby leading to increased levels of the enzyme. Finally, cholesterol is a negative feedback regulator of HMG CoA reductase activity (see [regulation of cholesterol synthesis](#)). Thus, when cytosolic cholesterol is depleted, *de novo* cholesterol synthesis is stimulated by freeing HMG CoA reductase of its feedback constraints. Subsequent to desmolase activity, pregnenolone moves to the cytosol, where further processing depends on the cell (tissue) under consideration.

The various hydroxylases involved in the synthesis of the steroid hormones have a nomenclature that indicates the site of hydroxylation (e.g. 17 $\alpha$ -hydroxylase introduces a hydroxyl group to carbon 17). These hydroxylase enzymes are members of the cytochrome P450 class of enzymes and as such also have a nomenclature indicative of the site of hydroxylation in addition to being identified as P450 class enzymes (e.g. the 17 $\alpha$ -hydroxylase is also identified as P450c17).

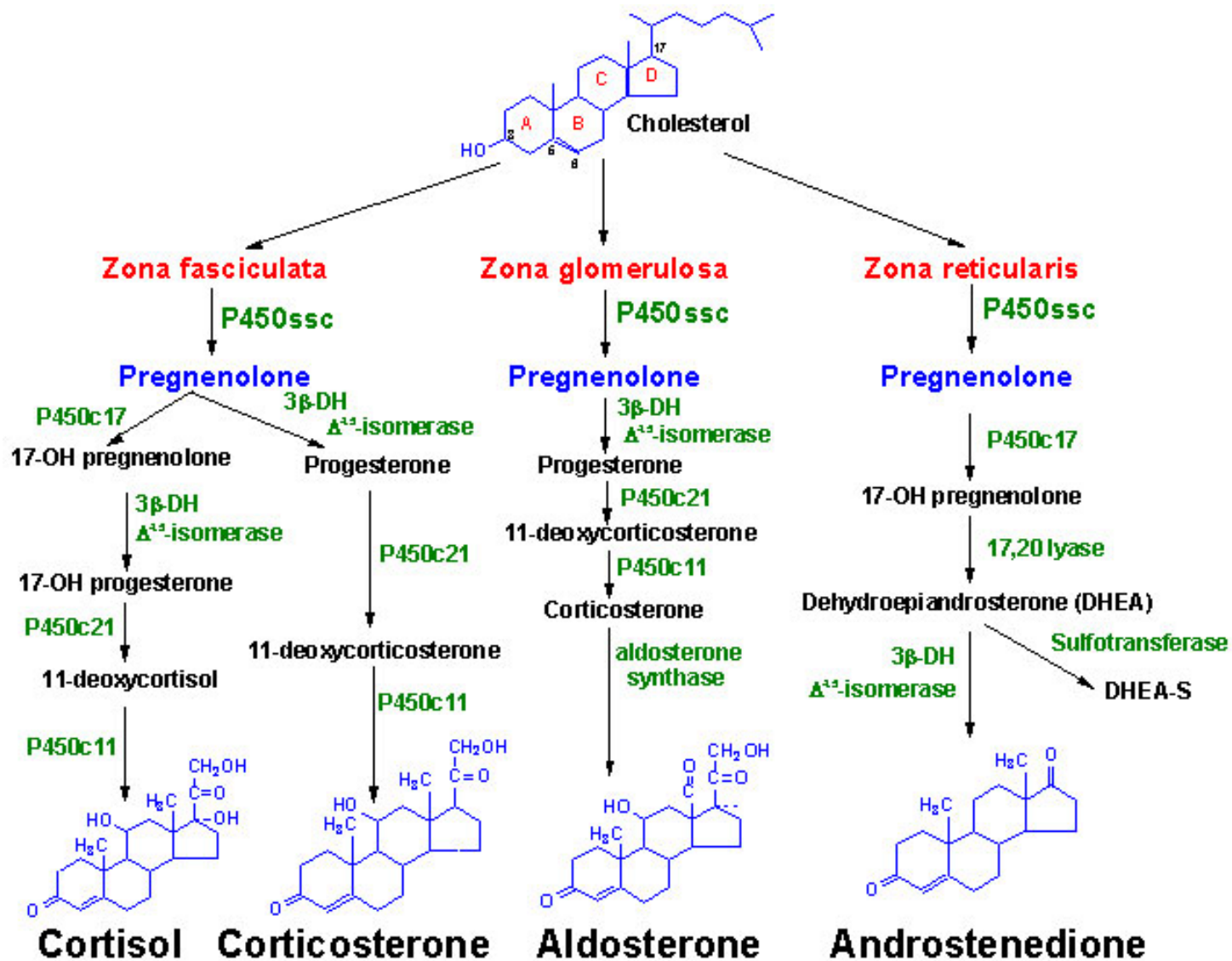
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## Steroids of the Adrenal Cortex

The adrenal cortex is responsible for production of 3 major classes of steroid hormones: **glucocorticoids**, which regulate carbohydrate metabolism; **mineralocorticoids**, which regulate the body levels of sodium and potassium; and **androgens**, whose actions are similar to that of steroids produced by the male gonads. Adrenal insufficiency is known as [Addison disease](#), and in the absence of steroid hormone replacement therapy can rapidly cause death (in 1--2 weeks).

The adrenal cortex is composed of 3 main tissue regions: **zona glomerulosa**, **zona fasciculata**, and **zona reticularis**. Although the pathway to pregnenolone synthesis is the same in all zones of the cortex, the zones are histologically and enzymatically distinct, with the exact steroid hormone product dependent on the enzymes present in the cells of each zone.



Synthesis of the various adrenal steroid hormones from cholesterol. Only the terminal hormone structures are included. 3 $\beta$ -DH is 3 $\beta$ -dehydrogenase, P450c11 is 11 $\beta$ -hydroxylase, P450c17 is 17 $\alpha$ -hydroxylase, P450c21 is 21 $\beta$ -hydroxylase.

Zona glomerulosa cells lack the P450c17 that converts pregnenolone and progesterone to their C<sub>17</sub> hydroxylated analogs. Thus, the pathways to the glucocorticoids (deoxycortisol and cortisol) and the androgens [dehydroepiandrosterone (DHEA) and androstenedione] are blocked in these cells. Zona glomerulosa cells are unique in the adrenal cortex in containing the enzyme responsible for converting corticosterone to aldosterone, the principal and most potent mineralocorticoid. This enzyme is P450c18 (or 18 $\alpha$ -hydroxylase), also called aldosterone synthase. The result is that the zona glomerulosa is mainly responsible for the conversion of cholesterol to the weak mineralocorticoid, corticosterone and the principal mineralocorticoid, aldosterone.

Cells of the zona fasciculata and zona reticularis lack aldosterone synthase (P450c18) that converts corticosterone to aldosterone, and thus these tissues produce only the weak mineralocorticoid corticosterone. However, both these zones do contain the P450c17 missing in zona glomerulosa and thus produce the major glucocorticoid, cortisol. Zona fasciculata and zona reticularis cells also contain the C17,20 lyase, whose activity is responsible for producing the androgens, dehydroepiandrosterone (DHEA) and androstenedione. Thus, fasciculata and reticularis cells can make corticosteroids and the adrenal androgens, but not aldosterone.

As noted earlier, P450ssc is a mitochondrial activity. Its product, pregnenolone, moves to the cytosol, where it is converted either to androgens or to 11-deoxycortisol and 11-deoxycorticosterone by enzymes of the endoplasmic reticulum. The latter 2 compounds then re-enter the mitochondrion, where the enzymes are located for tissue-specific conversion to glucocorticoids or mineralocorticoids, respectively.

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## Regulation of Adrenal Steroid Synthesis

Adrenocorticotrophic hormone (ACTH) of the hypothalamus regulates the hormone production of the zona fasciculata and zona reticularis. ACTH receptors in the plasma membrane activate adenylate cyclase with production of the second messenger, cAMP. The effect of ACTH on the production of cortisol is particularly important, with the result that a classic feedback loop is prominent in regulating the circulating levels of corticotropin releasing hormone, (CRH), ACTH, and cortisol.

Mineralocorticoid secretion from the zona glomerulosa is stimulated by an entirely different mechanism. Angiotensins II and III, derived from the action of the kidney protease renin on liver-derived **angiotensinogen**, stimulate zona glomerulosa cells by binding a plasma membrane receptor coupled to phospholipase C. Thus, angiotensin II and III binding to their receptor leads to the activation of PKC and elevated intracellular  $\text{Ca}^{2+}$  levels. These events lead to increased P450ssc activity and increased production of aldosterone. In the kidney, aldosterone regulates sodium retention by stimulating gene expression of mRNA for the  $\text{Na}^+/\text{K}^+$ -ATPase responsible for the reaccumulation of sodium from the urine.

The interplay between renin from the kidney and plasma angiotensinogen is important in regulating plasma aldosterone levels, sodium and potassium levels, and ultimately blood pressure. Among the drugs most widely employed used to lower blood pressure are the **angiotensin converting enzyme (ACE) inhibitors**. These compounds are potent competitive inhibitors of the enzyme that converts angiotensin I to the physiologically active angiotensins II and III. This feedback loop is closed by potassium, which is a potent stimulator of aldosterone secretion. Changes in plasma potassium of as little as 0.1 millimolar can cause wide fluctuations (+/- 50%) in plasma levels of aldosterone. Potassium increases aldosterone secretion by depolarizing the plasma membrane of zona glomerulosa cells and opening a voltage-gated calcium channel, with a resultant increase in cytoplasmic calcium and the stimulation of calcium-dependent processes.

Although fasciculata and reticularis cells each have the capability of synthesizing androgens and glucocorticoids, the main pathway normally followed is that leading to glucocorticoid production. However, when genetic defects occur in the 3 enzyme complexes leading to glucocorticoid production, large amounts of the most important androgen, dehydroepiandrosterone (DHEA), are produced. These lead to **hirsutism** and other masculinizing changes in secondary sex characteristics.

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## Gonadal Steroid Hormones

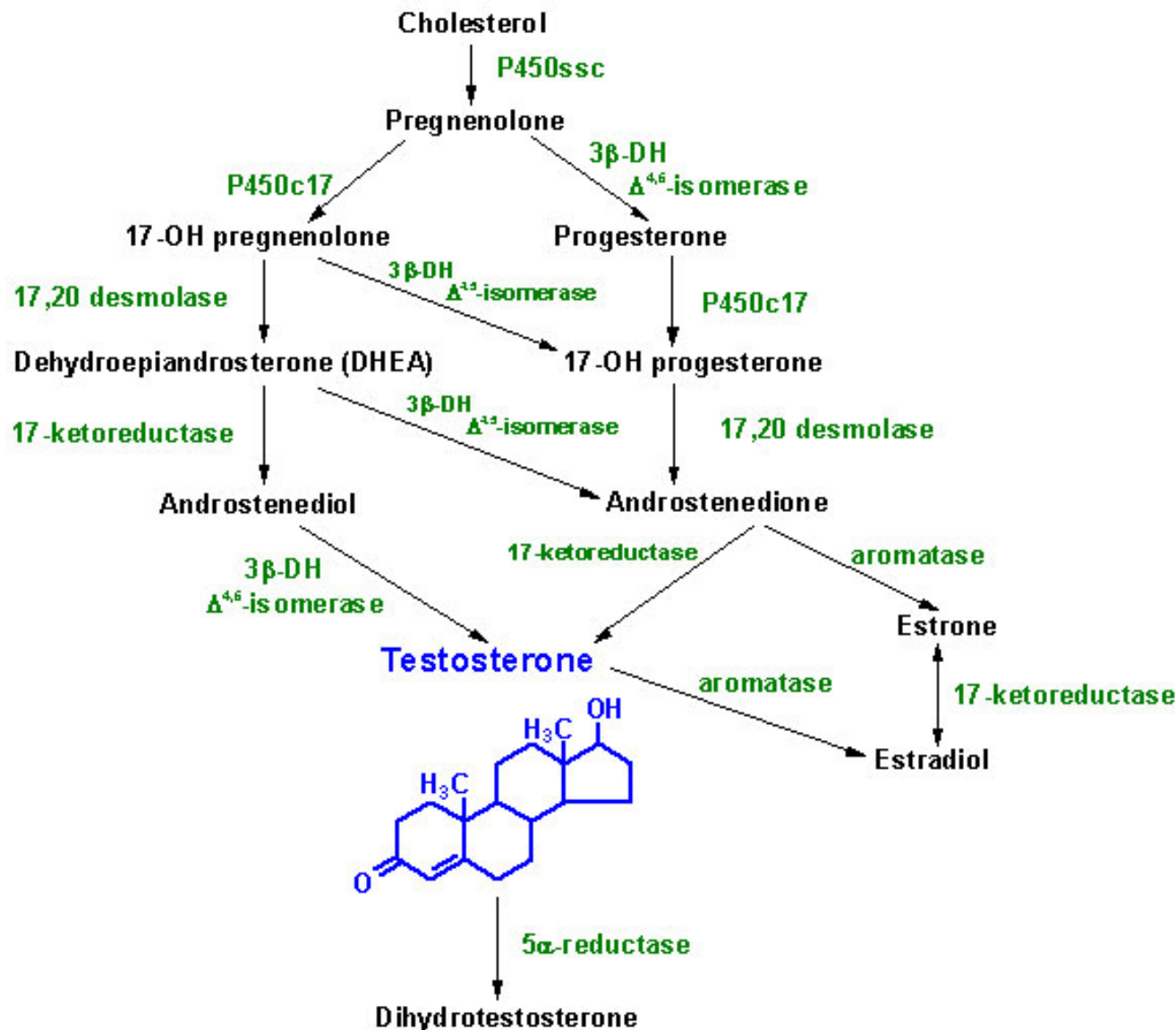
Although many steroids are produced by the testes and the ovaries, the two most important are testosterone and estradiol. These

compounds are under tight biosynthetic control, with short and long negative feedback loops that regulate the secretion of **follicle stimulating hormone (FSH)** and **luteinizing hormone (LH)** by the pituitary and **gonadotropin releasing hormone (GnRH)** by the hypothalamus. Low levels of circulating sex hormone reduce feedback inhibition on GnRH synthesis (the long loop), leading to elevated FSH and LH. The latter peptide hormones bind to gonadal tissue and stimulate P450ssc activity, resulting in sex hormone production via cAMP and PKA mediated pathways. The roles of cAMP and PKA in gonadal tissue are the same as that described for glucocorticoid production in the adrenals, but in this case adenylate cyclase activation is coupled to the binding of LH to plasma membrane receptors.

The biosynthetic pathway to sex hormones in male and female gonadal tissue includes the production of the androgens--- androstenedione and dehydroepiandrosterone. Testes and ovaries contain an additional enzyme, a 17 $\beta$ -hydroxysteroid dehydrogenase, that enables androgens to be converted to testosterone

In males, LH binds to Leydig cells, stimulating production of the principal Leydig cell hormone, testosterone. Testosterone is secreted to the plasma and also carried to Sertoli cells by **androgen binding protein (ABP)**. In Sertoli cells the  $\Delta$ -4 double bond of testosterone is reduced, producing dihydrotestosterone. Testosterone and dihydrotestosterone are carried in the plasma, and delivered to target tissue, by a specific **gonadal-steroid binding globulin (GBG)**. In a number of target tissues, testosterone can be converted to dihydrotestosterone (DHT). DHT is the most potent of the male steroid hormones, with an activity that is 10 times that of testosterone. Because of its relatively lower potency, testosterone is sometimes considered to be a prohormone.

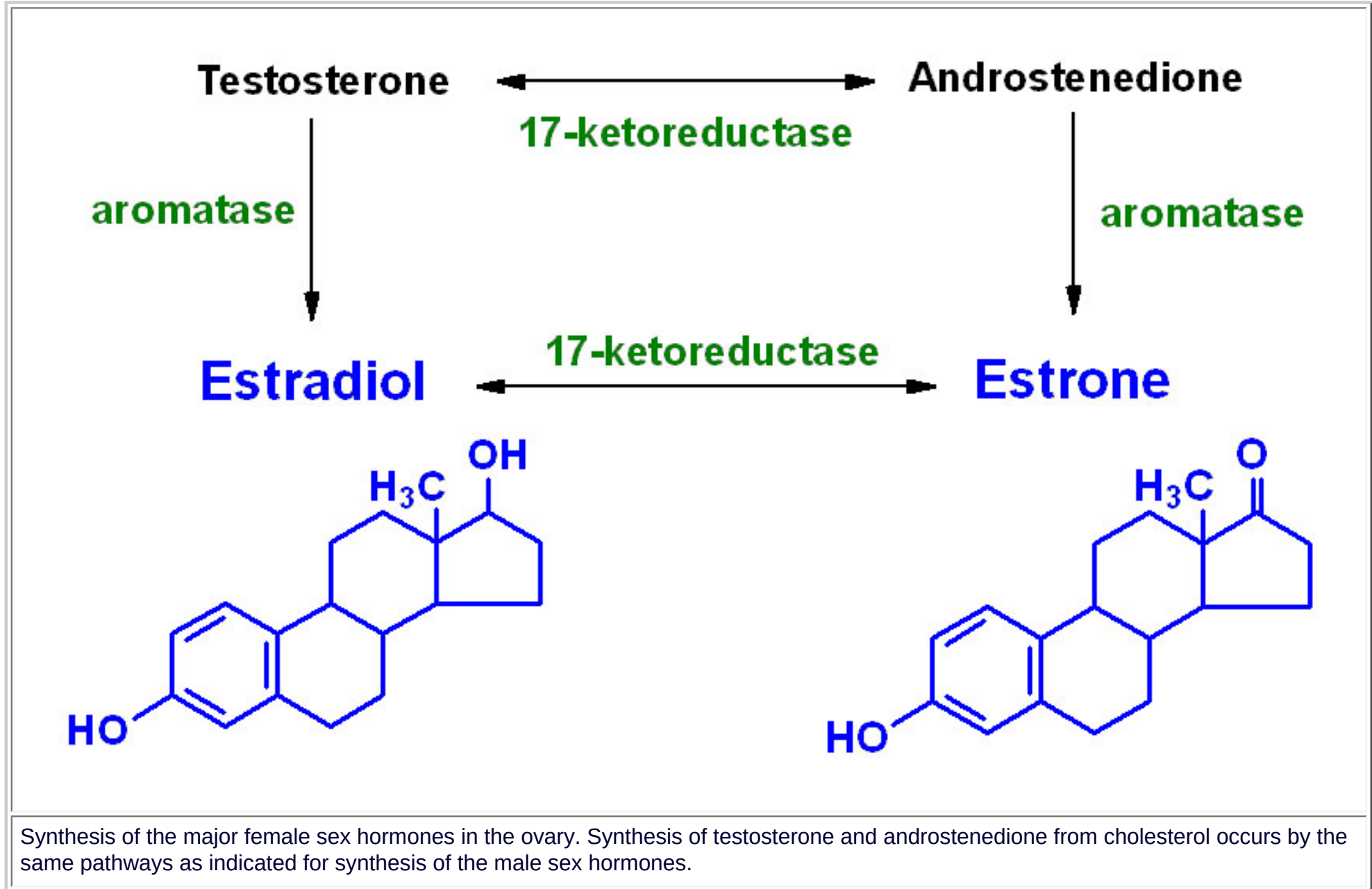




Synthesis of the male sex hormones in Leydig cells of the testis. P450SSC, 3β-DH, and P450c17 are the same enzymes as those needed for adrenal steroid hormone synthesis. 17,20-desmolase is the same as 17,20-lyase of adrenal hormone synthesis..

Testosterone is also produced by Sertoli cells but in these cells it is regulated by FSH, again acting through a cAMP- and PKA-regulatory pathway. In addition, FSH stimulates Sertoli cells to secrete **androgen-binding protein (ABP)**, which transports testosterone and DHT from Leydig cells to sites of spermatogenesis. There, testosterone acts to stimulate protein synthesis and sperm development. In females, LH binds to thecal cells of the ovary, where it stimulates the synthesis of androstenedione and testosterone by the usual

cAMP- and PKA-regulated pathway. An additional enzyme complex known as aromatase is responsible for the final conversion of the latter 2 molecules into the estrogens. Aromatase is a complex endoplasmic reticulum enzyme found in the ovary and in numerous other tissues in both males and females. Its action involves hydroxylations and dehydrations that culminate in aromatization of the A ring of the androgens.



Aromatase activity is also found in granulosa cells, but in these cells the activity is stimulated by FSH. Normally, thecal cell

androgens produced in response to LH diffuse to granulosa cells, where granulosa cell aromatase converts these androgens to estrogens. As granulosa cells mature they develop competent large numbers of LH receptors in the plasma membrane and become increasingly responsive to LH, increasing the quantity of estrogen produced from these cells. Granulosa cell estrogens are largely, if not all, secreted into follicular fluid. Thecal cell estrogens are secreted largely into the circulation, where they are delivered to target tissue by the same globulin (GBG) used to transport testosterone.

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## Steroid Hormone Receptors

The receptors to which steroid hormones bind are ligand-activated proteins that regulate transcription of selected genes. Unlike [peptide hormone receptors](#), that span the plasma membrane and bind ligand outside the cell, steroid hormone receptors are found in the cytosol and the nucleus. The steroid hormone receptors belong to the steroid and thyroid hormone receptor super-family of proteins, that includes receptors for steroid hormones, thyroid hormones, vitamin D and vitamin A (retinoic acid).

When these receptors bind ligand they undergo a conformational change that renders them activated to recognize and bind to specific nucleotide sequences. These specific nucleotide sequences in the DNA are referred to as **hormone-response elements** (HREs). When ligand-receptor complexes interact with DNA they alter the transcriptional level (responses can be either activating or repressing) of the associated gene. Thus, the steroid-thyroid family of receptors all have three distinct domains: a ligand-binding domain, a DNA-binding domain and a transcriptional regulatory domain. Although there is the commonly observed effect of altered transcriptional activity in response to hormone-receptor interaction, there are family member-specific effects with ligand-receptor interaction. Binding of thyroid hormone to its receptor results in release of the receptor from DNA. Several receptors are induced to interact with other transcriptional mediators in response to ligand binding. Binding of glucocorticoid leads to translocation of the ligand-receptor complex from the cytosol to the nucleus.

The receptors for the retinoids ([vitamin A and its derivatives](#)) are identified as **RARs** (for retinoic acid, RA receptors) and exist in at least three subtypes, RAR $\alpha$ , RAR $\beta$  and RAR $\gamma$ . In addition, there is another family of nuclear receptors termed the **retinoid X receptors** (RXRs) that represents a second class of retinoid-responsive transcription factors. The RXRs have been shown to enhance the DNA-binding activity of RARs and the thyroid hormone receptors (TRs). There are also three distinct RXRs ( $\alpha$ ,  $\beta$  and  $\gamma$ ). The major difference between the RARs and RXRs is that the former exhibit highest affinity for all-*trans*-retinoic acid (all-*trans*-RA) and the latter for 9-*cis*-RA.

Additional super-family members are the **peroxisome proliferator-activated receptors** (PPARs). These receptors were originally discovered as proteins activated by agents that stimulate proliferation of peroxisomes in rat liver. An intracellular lipid-binding protein identified as aP2 is expressed exclusively in differentiated adipocytes. An adipocyte-specific enhancer of the aP2 gene is a target for peroxisome proliferators, fatty acids and 9-*cis*-RA. Subsequent to these observations it was found that there is an adipocyte-specific PPAR family identified as PPAR $\gamma$ . The PPAR $\gamma$  proteins form heterodimers with RXRs to activate adipocyte-specific enhancers such as the one in the aP2 gene.

Recent evidence has demonstrated a role for PPAR $\gamma$  proteins in the etiology of [type 2 diabetes](#). A relatively new class of drugs used to increase the sensitivity of the body to insulin are the **thiazolidinedione** drugs. These compounds bind to and alter the function of PPAR $\gamma$ . Mutations in the gene for PPAR $\gamma$  have been correlated with insulin resistance. It is still not completely clear how impaired PPAR $\gamma$  signaling can affect the sensitivity of the body to insulin or indeed if the observed mutations are a direct or indirect cause of the symptoms of insulin resistance.

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- Structure and Function of Hormones
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- Basics of Peptide Hormones
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## Structure and Function of Hormones

The integration of body functions in humans and other higher organisms is carried out by the nervous system, the immune system, and the endocrine system. The endocrine system is composed of a number of tissues that secrete their products, called **endocrine hormones**, into the circulatory system; from there they are disseminated throughout the body, regulating the function of distant tissues and maintaining homeostasis. In a separate but related system, **exocrine** tissues secrete their products into ducts and then to the outside of the body or to the intestinal tract. Classically, endocrine hormones are considered to be derived from amino acids, peptides, or sterols and to act at sites distant from their tissue of origin. However, the latter definition has begun to blur as it is found that some secreted substances act at

a distance (classical endocrines), close to the cells that secrete them (**paracrines**), or directly on the cell that secreted them (**autocrines**). **Insulin-like growth factor-I (IGF-I)**, which behaves as an endocrine, paracrine, and autocrine, provides a prime example of this difficulty.

Hormones are normally present in the plasma and interstitial tissue at concentrations in the range of  $10^{-7}\text{M}$  to  $10^{-10}\text{M}$ . Because of these very low physiological concentrations, sensitive protein receptors have evolved in target tissues to sense the presence of very weak signals. In addition, systemic feedback mechanisms have evolved to regulate the production of endocrine hormones.

Once a hormone is secreted by an endocrine tissue, it generally binds to a specific plasma protein carrier, with the complex being disseminated to distant tissues. Plasma carrier proteins exist for all classes of endocrine hormones. Carrier proteins for peptide hormones prevent hormone destruction by plasma proteases. Carriers for steroid and thyroid hormones allow these very hydrophobic substances to be present in the plasma at concentrations several hundred-fold greater than their solubility in water would permit. Carriers for small, hydrophilic amino acid-derived hormones prevent their filtration through the renal glomerulus, greatly prolonging their circulating half-life.

Tissues capable of responding to endocrines have 2 properties in common: they possess a receptor having very high affinity for hormone, and the receptor is coupled to a process that regulates metabolism of the target cells. Receptors for most amino acid--derived hormones and all peptide hormones are located on the plasma membrane. Activation of these receptors by hormones (the first messenger) leads to the intracellular production of a **second messenger**, such as cAMP, which is responsible for initiating the intracellular biological response. Steroid and thyroid hormones are hydrophobic and diffuse from their binding proteins in the plasma, across the plasma membrane to intracellularly localized receptors. The resultant complex of steroid and receptor bind to response elements of nuclear DNA, regulating the production of mRNA for specific proteins.

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## Receptors for Peptide Hormones

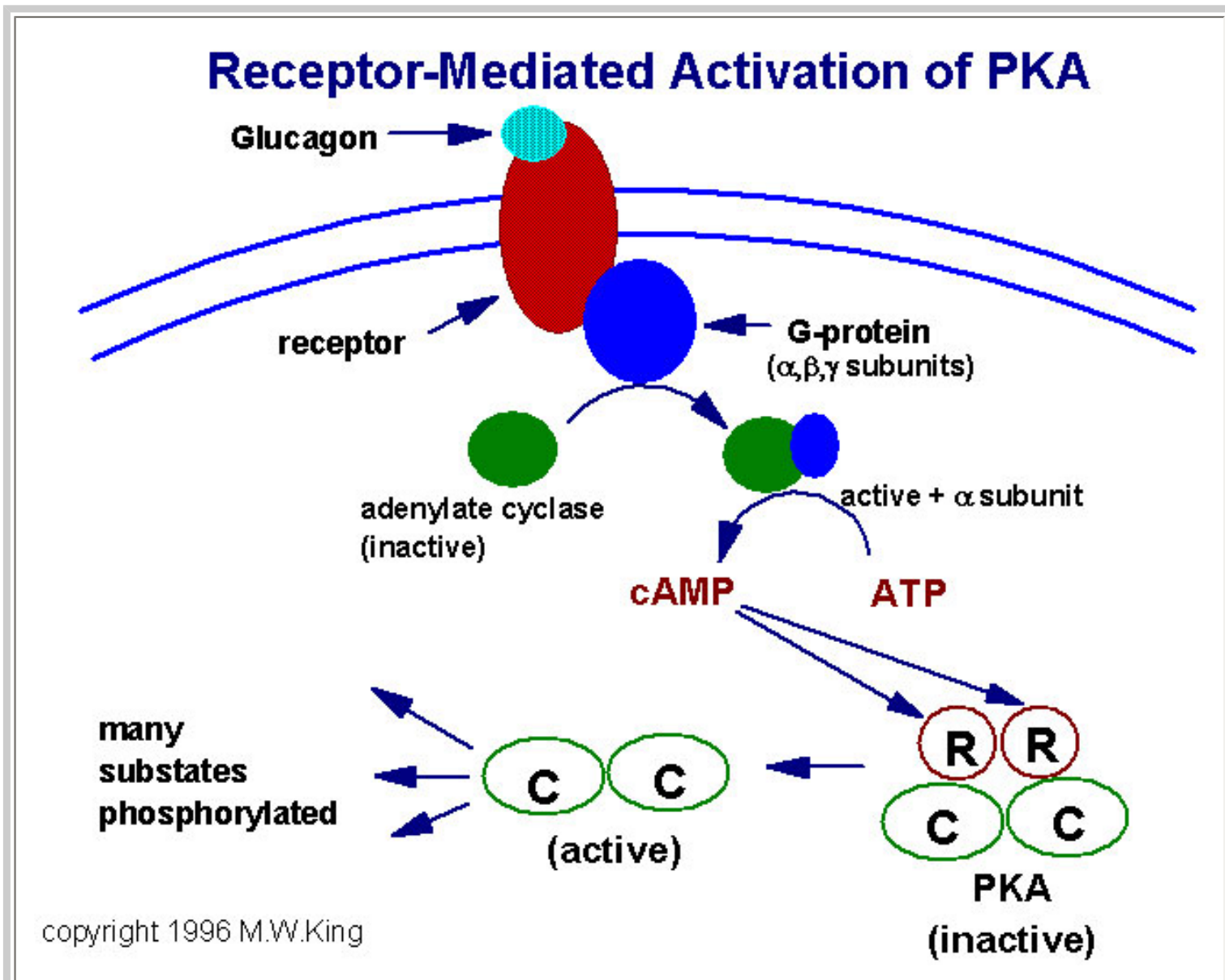
With the exception of the thyroid hormone receptor, the receptors for amino acid-derived and peptide hormones are located in the plasma membrane. Receptor structure is varied: some receptors consist of a single polypeptide chain with a domain on either side of the membrane, connected by a membrane-spanning domain. Some receptors are comprised of a single polypeptide chain that is passed back and forth in serpentine fashion across the membrane, giving multiple intracellular, transmembrane, and extracellular domains. Other receptors are composed of multiple polypeptides. For example, the insulin receptor is a disulfide-linked tetramer with the  $\beta$  subunits spanning the membrane and the  $\alpha$  subunits located on the exterior surface.

Subsequent to hormone binding, a signal is transduced to the interior of the cell, where second messengers and phosphorylated proteins generate appropriate metabolic responses. The main second messengers are **cAMP**,  **$\text{Ca}^{2+}$** , **inositol triphosphate ( $\text{IP}_3$ )**, and **diacylglycerol (DAG)**. Proteins are phosphorylated on serine and threonine by cAMP-dependent protein kinase (PKA) and DAG-activated protein kinase C (PKC). Additionally a series of membrane-associated and intracellular tyrosine kinases phosphorylate specific tyrosine residues on target enzymes and other regulatory proteins.

The hormone-binding signal of most, but not all, plasma membrane receptors is transduced to the interior of cells by the binding of receptor-ligand complexes to a series of membrane-localized GDP/GTP binding proteins known as **G-proteins**. The classic interactions between receptors, G-protein transducer, and membrane-localized adenylate cyclase are illustrated below using the pancreatic hormone **glucagon** as an example. When G-proteins bind to receptors, GTP exchanges with GDP bound to the  $\alpha$  subunit of the G-protein. The  $\text{G}_{\alpha}$ -GTP complex binds adenylate cyclase, activating the enzyme. The activation of adenylate cyclase leads to cAMP production in the cytosol and to the activation of PKA, followed by regulatory phosphorylation of numerous enzymes. Stimulatory G-proteins are designated  $\text{G}_s$ ,



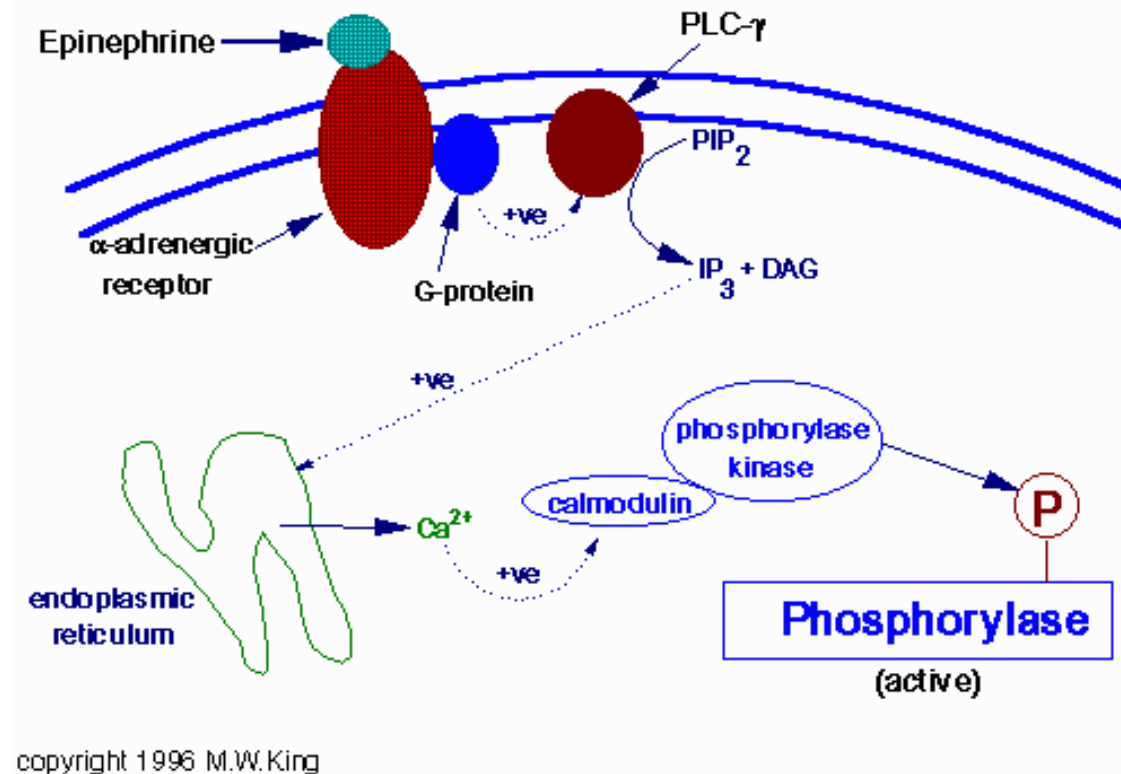
inhibitory G-proteins are designated  $G_i$ .



Representative pathway for the activation of **cAMP-dependent protein kinase, PKA**. In this example glucagon binds to its' cell-surface receptor, thereby activating the receptor. Activation of the receptor is coupled to the activation of a receptor-coupled G-protein (GTP-binding and hydrolyzing protein) composed of 3 subunits. Upon activation the  $\alpha$ -subunit dissociates and binds to and activates adenylate cyclase. Adenylate cyclase then converts ATP to cyclic-AMP (cAMP). The cAMP thus produced then binds to the regulatory subunits of PKA leading to dissociation of the associated catalytic subunits. The catalytic subunits are inactive until dissociated from the regulatory subunits. Once released the catalytic subunits of PKA phosphorylate numerous substrate using ATP as the phosphate donor.

A second class of peptide hormones induces the transduction of 2 second messengers, DAG and  $IP_3$  (diagrammed below for  $\alpha$ -adrenergic stimulation by epinephrine). Hormone binding is followed by interaction with a stimulatory G-protein which is followed in turn by G-protein activation of membrane-localized phospholipase C- $\gamma$ , (PLC- $\gamma$ ). PLC- $\gamma$  hydrolyzes phosphatidylinositol 4,5-bisphosphate ( $PIP_2$ ) to produce 2 messengers:  $IP_3$ , which is soluble in the cytosol, and DAG, which remains in the membrane phase. Cytosolic  $IP_3$  binds to sites on the endoplasmic reticulum, opening  $Ca^{2+}$  channels and allowing stored  $Ca^{2+}$  to flood the cytosol. There it activates numerous enzymes, many by activating their calmodulin or calmodulin-like subunits. DAG has 2 roles: it binds and activates protein kinase C (PKC), and it opens  $Ca^{2+}$  channels in the plasma membrane, reinforcing the effect of  $IP_3$ . Like PKA, PKC phosphorylates serine and threonine residues of many proteins, thus modulating their catalytic activity.

## $\alpha$ -Receptor-Mediated Responses on Phosphorylase



Pathways involved in the regulation of glycogen phosphorylase by epinephrine activation of  $\alpha$ -adrenergic receptors (see [Glycogen](#) page for details of the regulatory mechanisms). PLC- $\gamma$  is phospholipase C- $\gamma$ .

Only 1 receptor class, that for the **natriuretic factors** (e.g. atrial natriuretic factor, ANF), has been shown to be coupled to the production of intracellular cGMP. ANF, a peptide secreted by cardiac atrial tissue, is much like other peptide hormones in that it is secreted into the circulatory system and has effects on distant tissue. The principal site of ANF action is the kidney glomerulus, where it modulates the rate of filtration, increasing Na<sup>+</sup> excretion in the urine. The receptors for the natriuretic factors are integral plasma membrane proteins, whose intracellular domains catalyze the formation of cGMP following natriuretic factor-binding. Intracellular cGMP activates a protein kinase G (PKG), which phosphorylates and modulates enzyme activity, leading to the biological effects of the natriuretic factors.

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## Basics of Peptide Hormones

Many amino acid and peptide hormones are elaborated by neural tissue, with ultimate impact on the entire system. When their

composition was still unknown, hypothalamic secretory products were known as releasing factors, since their effect was to release endocrine hormones from the pituitary. More recently the releasing factors have been renamed **releasing hormones**. Currently, both names are in common use.

Releasing hormones are synthesized in neural cell bodies of the hypothalamus and secreted at the axon terminals into the portal hypophyseal circulation, which directly bathes the anterior pituitary. These peptides initiate a cascade of biochemical reactions that culminate in hormone-regulated, whole-body biological end points. Cells of the anterior pituitary, with specific receptors for individual releasing hormones, generally respond through a  $\text{Ca}^{2+}$ ,  $\text{IP}_3$ , PKC-linked pathway that stimulates exocytosis of preexisting vesicles containing the various anterior pituitary hormones. The pituitary hormones are carried via the systemic circulation to target tissues throughout the body. At the target tissues they generate unique biological activities.

The secretion of hypothalamic, pituitary, and target tissue hormones is under tight regulatory control by a series of feedback and feed-forward loops. This complexity can be demonstrated using the growth hormone (GH) regulatory system as an example. The stimulatory substance **growth hormone releasing hormone (GRH)**, and the inhibitory substance **somatostatin (SS)** (somatostatin is also called **growth hormone-inhibiting hormone, GIH**), both products of the hypothalamus, control pituitary GH secretion. Under the influence of GRH, growth hormone is released into the systemic circulation, causing the target tissue to secrete **insulin-like growth factor-1, IGF-1**. Growth hormone also has other more direct metabolic effects; it is both hyperglycemic and lipolytic. The principal source of systemic IGF-1 is the liver, although most other tissues secrete and contribute to systemic IGF-1. Liver IGF-1 is considered to be the principal regulator of tissue growth. In particular, the IGF-1 secreted by the liver is believed to synchronize growth throughout the body, resulting in a homeostatic balance of tissue size and mass. IGF-1 secreted by peripheral tissues is generally considered to be autocrine or paracrine in its biological action.

Systemic IGF-1 also has hypothalamic and pituitary regulatory targets. The negative feedback loops cause down-regulation of GH secretion directly at the pituitary. The longer positive feedback loop, involving IGF-1 regulation at the hypothalamus, stimulates the secretion of GIH, which in turn inhibits the secretion of growth hormone by the pituitary. The latter is a relatively unusual negative feed-forward regulatory process. In addition, a shorter negative feedback loop is shown that involves direct IGF-1 action on the pituitary, leading to down-regulation of GH secretion. Similar feedback loops exist for all the major endocrine hormones, and many subtle nuances modulate each regulatory loop.

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## Growth Hormone

Human placental lactogen (hPL), GH, and prolactin (PRL) comprise the growth hormone family. All have about 200 amino acids, 2 disulfide bonds, and no glycosylation. Although each has special receptors and unique characteristics to their activity, they all possess growth-promoting and lactogenic activity. Mature GH (22,000 daltons) is synthesized in acidophilic pituitary somatotropes as a single polypeptide chain. Because of alternate RNA splicing, a small amount of a somewhat smaller molecular form is also secreted.

While details of the method of signal transduction by the members of the GH family of tropic hormones remain unclear, PKC activity has been demonstrated to correlate directly with the biological effects of PRL and GH. This appears to indicate that the PKC signal transduction pathway is operative in transducing signals for the GH family of hormones.

The role of growth hormone in regulating IGF-1 production was noted above. Humans respond to natural or recombinant human or primate growth hormone with appropriate secretion of IGF-1, but growth hormone of other species has no normal biological effect in man. The latter is puzzling because interspecies GH homologies are quite high in many cases, and most other species respond well to human

growth hormone. In humans, growth hormone promotes gluconeogenesis and is consequently hyperglycemic. It promotes amino acid uptake by cells, with the result that GH therapy puts an organism into positive nitrogen balance, similar to that seen in growing children. Finally, growth hormone is lipolytic, inducing the breakdown of tissue lipids and thus providing energy supplies that are used to support the stimulated protein synthesis induced by increased amino acid uptake.

There are a number of genetic deficiencies associated with GH. **GH-deficient dwarfs** lack the ability to synthesize or secrete GH, and these short-statured individuals respond well to GH therapy. **Pygmies** lack the IGF-1 response to GH but not its metabolic effects; thus in pygmies the deficiency is post-receptor in nature. Finally, **Laron dwarfs** have normal or excess plasma GH, but lack liver GH receptors and have low levels of circulating IGF-1. The defect in these individuals is clearly related to an inability to respond to GH by the production of IGF-1. The production of excessive amounts of GH before epiphyseal closure of the long bones leads to **gigantism**, and when GH becomes excessive after epiphyseal closure, acral bone growth leads to the characteristic features of **acromegaly**.

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## Prolactin (PRL)

Prolactin is produced by acidophilic pituitary lactotropes. Prolactin is the lone tropic hormone of the pituitary that is routinely under negative control by **prolactin inhibiting hormone (PIH)**, which is now known to be **dopamine**. Decreased hypophyseal dopamine production, or damage to the hypophyseal stalk, leads to rapid up-regulation of PRL secretion. A number of other hypothalamic releasing hormones induce increased prolactin secretion; as a result, it is unclear whether a specific PRH exists for up-regulating PRL secretion. PRL initiates and maintains lactation in mammals, but normally only in mammary tissue that has been primed with estrogenic sex hormones.

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## Human Placental Lactogen (hPL)

Human placental lactogen is produced by the placenta late in gestation. At its height it is secreted at a rate of about 1 g/day, the highest secretory rate of any known human hormone. However, little hPL reaches the fetal circulation, and hPL has only about 1% the activity of PRL or GH in producing biological effects, leading some to question its functional importance in humans.

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## The Gonadotropins

The glycoprotein hormones are the most chemically complex family of the peptide hormones. All members of the family are highly glycosylated. Each of the glycoprotein hormones is an ( $\alpha$ : $\beta$ ) heterodimer, with the  $\alpha$  subunit being identical in all members of the family. The biological activity of the hormone is determined by the  $\beta$ -subunit, which is not active in the absence of the  $\alpha$  subunit. The molecular weight of the gonadotropins (follicle stimulating hormone, FSH; luteinizing hormone, LH, and chorionic gonadotropin, CG) is about 25,000 Daltons, whereas that of the thyroid tropic hormone thyroid stimulating hormone, TSH is about 30,000. All members of the glycoprotein family transduce their intracellular effects via the receptor, G-protein, adenylate cyclase, second-messenger system.

The gonadotropins (LH, FSH and CG) bind to cells in the ovaries and testes, stimulating the production of the steroid sex hormones estrogen, testosterone (T), and dihydrotestosterone (DHT). In males, luteinizing hormone (LH) binds to Leydig cells of the testes to induce

the secretion of T, while follicle stimulating hormone (FSH) binds to Sertoli cells and induces the secretion of T and DHT. In females, LH induces thecal cells to secrete estradiol, and FSH stimulates estrogen synthesis by granulosa cells. The synthesis of the sex hormones is reviewed is covered in the [steroid hormone page](#).

Human chorionic gonadotropin (hCG) is a placental hormone. The production of hCG increases markedly after implantation; its appearance in the plasma and urine is one of the earliest signals of pregnancy and the basis of many pregnancy tests.

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## Thyroid Stimulating Hormone (TSH)

Secretion of TSH (also called **thyrotropin**), the final member of the glycoprotein hormone family, is stimulated by **thyrotropin-releasing hormone, TRH** from the hypothalamus. While cAMP causes increased secretion of TSH by thyrotropes, it is not yet certain that cAMP is the physiological signal that regulates TSH production.

Circulating TSH binds to receptors on the basal membrane of thyroid follicles. The receptors are coupled through a G-protein to adenylate cyclase. The result is that ligand binding increases thyrocyte cAMP and PKA, leading in the short term to increased secretion of **thyroxine (T4)** and **triiodothyronine (T3)**. Chronic stimulation of the receptor causes an increase in the synthesis of a major thyroid hormone precursor, **thyroglobulin**.

Thyroglobulin produced on rough endoplasmic reticulum has a molecular weight of 660,000. It is glycosylated and contains more than 100 tyrosine residues, which become iodinated and are used to synthesize T3 and T4. Thyroglobulin is exocytosed through the apical membrane into the closed lumen of thyroid follicles, where it accumulates as the major protein of the thyroid and where maturation takes place. Briefly, a Na<sup>+</sup>/K<sup>+</sup>-ATPase-driven pump concentrates iodide (I<sup>-</sup>) in thyroid cells, and the iodide is transported to the follicle lumen. There it is oxidized to I<sup>+</sup> by a thyroperoxidase found only in thyroid tissue. The addition of I<sup>+</sup> to tyrosine residues of thyroglobulin is catalyzed by the same enzyme, leading to the production of thyroglobulin containing monoiodotyrosyl (MIT) and diiodotyrosyl (DIT) residues. The thyronines, T3 and T4, are formed by combining MIT and DIT residues on thyroglobulin.

Mature, iodinated thyroglobulin is taken up in vesicles by thyrocytes and fuses with lysosomes. Lysosomal proteases degrade thyroglobulin releasing amino acids and T3 and T4, which are secreted into the circulation. These compounds are very hydrophobic and require a carrier protein for delivery to target tissues. In the plasma, T3 and T4 are bound to a carrier glycoprotein known as thyroxin-binding globulin and are disseminated throughout the body in this form.

Thyroid hormones act by binding to cytosolic receptors very similar to steroid hormone receptors, and for this reason T3 and T4 are often classified along with the hydrophobic steroid hormones. The principal role of thyroid hormones is also like that of steroid hormones. In adults, the ligand receptor combination binds to thyroid hormone response elements in nuclear DNA and is responsible for up-regulating general protein synthesis and inducing a state of positive nitrogen balance. In the embryo, thyroid hormone is necessary for normal development. Hypothyroidism in the embryo is responsible for **cretinism**, which is characterized by multiple congenital defects and mental retardation.

Thyroid stimulating autoantibodies (TSAb) also activate the human thyroid TSH receptor, leading to the hyperthyroidism of **Graves' disease**. TSAb bind to the TSH receptor and mimic the TSH stimulation of the gland by increasing intracellular cAMP.

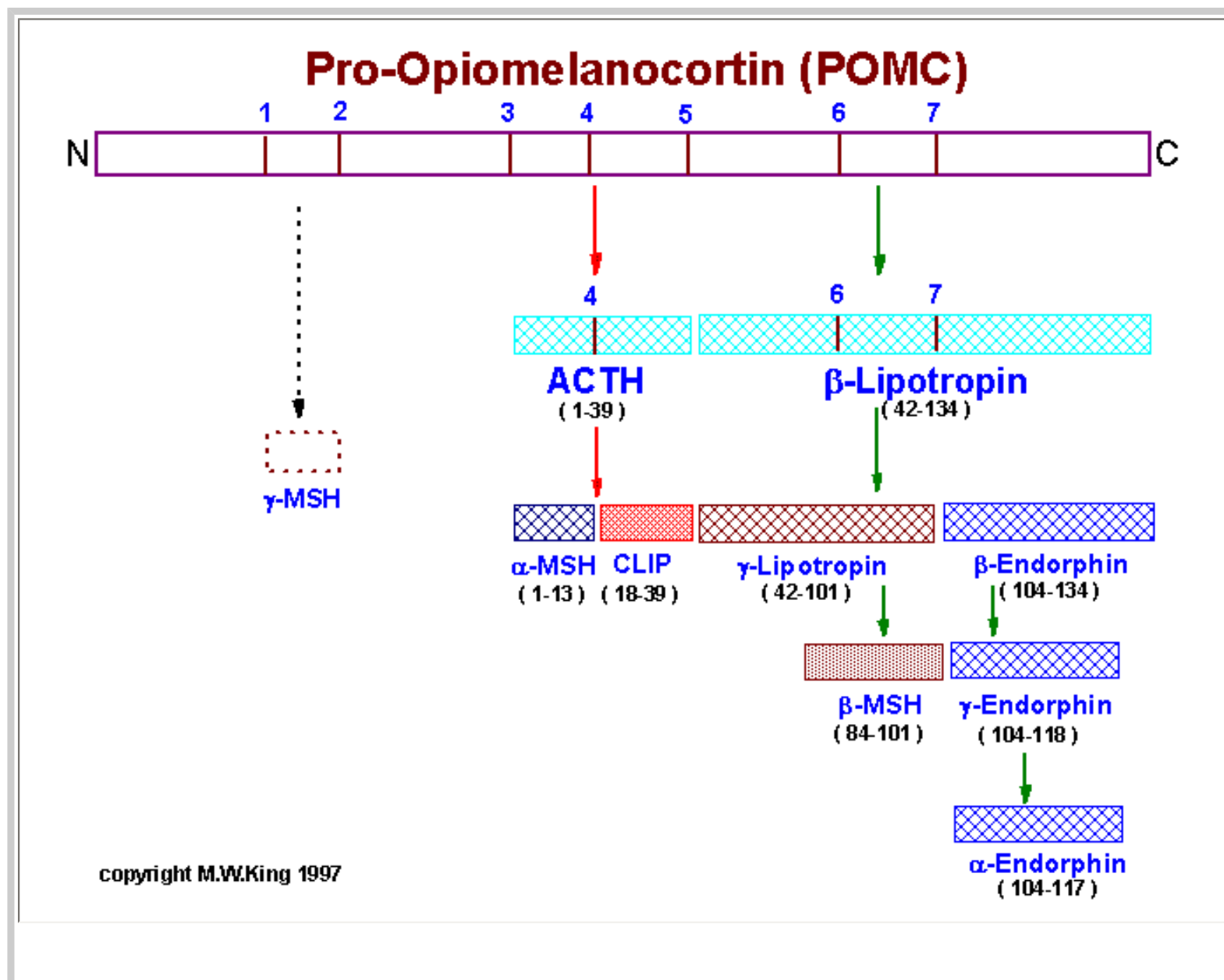
The feedback loop that regulates T3 and T4 production is a single short negative loop, with the T3 and T4 being responsible for down-regulating pituitary TSH secretion. Meanwhile, continuously secreted hypothalamic TRH is responsible for up-regulating TSH production. The TSH actually secreted by thyrotropes is the net result of the negative effects of T3 and T4 and the positive effect of TRH.

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## The Pro-Opiomelanocortin Family

The POMC gene is expressed in both the anterior and intermediate lobes of the pituitary gland. The primary protein product of the POMC gene is a 285 amino acid precursor that can undergo differential processing to yield at least 8 peptides, dependent upon the location of synthesis and the stimulus leading to their production.



Processing of the POMC precursor protein. Cleavage sites are indicated by the numbers 1 to 7 and consist of the sequences, Arg-Lys, Lys-Arg or Lys-Lys. Adrenocorticotrophic hormone (ACTH) and  $\beta$ -lipotropin are products generated in the corticotrophic cells of the anterior pituitary under the control of corticotropin releasing hormone (CRH). Alpha-melanocyte stimulating hormone ( $\alpha$ -MSH), corticotropin-like intermediate lobe peptide (CLIP),  $\gamma$ -lipotropin and  $\beta$ -endorphin are products generated in the intermediate lobe of the pituitary under the control of dopamine.  $\alpha$ -,  $\beta$ - and  $\gamma$ -MSH are collectively referred to as melanotropin or intermedin. The numbers in parentheses below each hormone indicate the amino acids of POMC present in each. The N-terminus of ACTH is given as amino acid number 1. The presence and function of  $\gamma$ -MSH is unclear hence the dotted lines. Actions, if any, of CLIP and  $\beta$ -lipotropin are also unclear.

Corticotropin releasing hormone (CRH) induces rapid secretion of adrenocorticotrophic hormone (ACTH, also called corticotropin) and a variety of other peptides from corticotropes of the anterior pituitary. ACTH, a 39 amino acid peptide, is the main physiologically active product of CRH activity. ACTH is derived by post-translational modification from a 241 amino acid precursor known as **pro-opiomelanocortin (POMC)**. The rapid CRH-stimulated secretion of ACTH is associated with induction of adenylate cyclase activity and an increase in cAMP and PKA in corticotropes. Longer-term responses of corticotropes to CRH include a marked increase in POMC mRNA.

The processing of POMC involves glycosylations, acetylations, and extensive proteolytic cleavage at sites shown to contain regions of basic protein sequences. The proteases that recognize these cleavage sites are tissue-specific; thus, the physiologically active product of the anterior pituitary is ACTH. The peptides  $\beta$ -lipotropin ( $\beta$ -LPH) and CLIP and have unknown activity in humans.

Many of the other POMC products are synthesized in other neural tissue that contains proteases with appropriate specificity. In human embryos and in pregnant women, the intermediate lobe is active and leads to the production of endorphins and enkephalins. These same endorphin-producing pathways are active in other neural tissue, and since they bind to the opiate receptors in other parts of the brain they are assumed to represent natural opiate-like analgesic compounds.

The biological role of ACTH is to stimulate the production of adrenal cortex steroids, principally cortisol and corticosterone. The mechanism of action of ACTH involves activation of adenylate cyclase, elevation of cAMP, and increased PKA activity of adrenal cortex tissue. The main effect of these events is to increase the activity of the P450-linked side chain-cleaving enzyme (P450SSC), which converts cholesterol to pregnenolone during **steroid hormone synthesis**. Because of the distribution of enzymes in the various adrenal cortex subdivisions, the principal physiological effect of ACTH is production of the glucocorticosteroids.

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## Vasopressin and Oxytocin

The principal hormones of the posterior pituitary are the nonapeptides **oxytocin** and **vasopressin**. These substances are synthesized as prohormones in neural cell bodies of the hypothalamus and mature as they pass down axons in association with carrier proteins termed **neurophysins**. The axons terminate in the posterior pituitary, and the hormones are secreted directly into the systemic circulation. Vasopressin is also known as **antidiuretic hormone (ADH)**, because it is the main regulator of body fluid osmolarity. The secretion of vasopressin is regulated in the hypothalamus by osmoreceptors, which sense water concentration and stimulate increased vasopressin

secretion when plasma osmolarity increases. The secreted vasopressin increases the reabsorption rate of water in kidney tubule cells, causing the excretion of urine that is concentrated in  $\text{Na}^+$  and thus yielding a net drop in osmolarity of body fluids. Vasopressin deficiency leads to watery urine and polydipsia, a condition known as [diabetes insipidus](#). Vasopressin binds plasma membrane receptors and acts through G-proteins to activate the cAMP/PKA regulatory system.

The mechanism of action of oxytocin is unknown. Oxytocin secretion in nursing women is stimulated by direct neural feedback obtained by stimulation of the nipple during suckling. Its physiological effects include the contraction of mammary gland myoepithelial cells, which induces the ejection of milk from mammary glands, and the stimulation of uterine smooth muscle contraction leading to childbirth. The remaining hormones reviewed in this chapter are not coupled to tropic neuropeptides, but each has its own regulatory feedback loop that involves the hormone, a plasma component, and a target tissue.

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## Natriuretic Hormones

Natriuresis refers to enhanced urinary excretion of sodium. This can occur in certain disease states and through the action of diuretic drugs. At least 3 natriuretic hormones have been identified. **Atrial natriuretic peptide (ANP)** was the first cardiac natriuretic hormone identified. This hormone is secreted by cardiac muscle when sodium chloride intake is increased and when the volume of the extracellular fluid expands. Active ANP is a 28-amino acid peptide containing a 17-amino acid ring formed by intrachain disulfide bonding. Two smaller forms of ANP have also been isolated from the brain. A **brain natriuretic peptide (BNP)** (first isolated from porcine brain) has been identified and found in human heart and blood (but not human brain). BNP has different amino acids in its 17-amino acid ring and is encoded for by a different gene. In humans, a third natriuretic peptide (CNP) is present in the brain but not in the heart.

The action of ANP is to cause natriuresis presumably by increasing glomerular filtration rate (its exact mechanism of action remains unclear). ANP induces relaxation of the mesangial cells of the glomeruli and thus may increase the surface area of these cells so that filtration is increased. Alternatively, ANP might act on tubule cells to increase sodium excretion. Other effects of ANP include reducing blood pressure, decreasing the responsiveness of adrenal glomerulosa cells to stimuli that result in aldosterone production and secretion, inhibit secretion of vasopressin and decreasing vascular smooth muscle cell responses to vasoconstrictive agents. These latter actions of ANP are counter to the effects of angiotensin II. In fact, ANP also lowers renin secretion by the kidneys thus lowering circulating angiotensin II levels.

Three different ANP receptors have been identified: ANPR-A, ANPR-B and ANPR-C. When ANP, BNP or CNP bind to receptor, an increase in guanylate cyclase activity results leading to production of cyclic GMP (cGMP). Both ANPR-A and ANPR-B proteins span the plasma membrane and their intracellular domains possess intrinsic guanylate cyclase activity. The exact function of the ANPR-C protein is unclear as this receptor does not contain an intracellular domain with intrinsic guanylate cyclase activity. It is hypothesized that it may act through a G-protein that activates PLC- $\gamma$  and inhibits adenylate cyclase or that it acts simply as a clearance receptor removing natriuretic peptides from the blood.

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## Renin-Angiotensin System

The renin-angiotensin system is responsible for regulation of blood pressure. The intrarenal baroreceptor system is a key mechanism for regulating renin secretion. A drop in pressure results in the release of renin from the juxtaglomerular cells of the kidneys. Renin secretion is also regulated by the rate of  $\text{Na}^+$  and  $\text{Cl}^-$  transport across the macula densa. The higher the transport of these ions the lower the rate of

renin secretion. The only function for renin is to cleave a 10-amino acid peptide from the N-terminal end of **angiotensinogen**. This decapeptide is called **angiotensin I**. Angiotensin I is then cleaved by the action of **angiotensin-converting enzyme, ACE** to the active hormone, **angiotensin II**. ACE removes 2 amino acids from the C-terminal end of angiotensin I.

Angiotensin II was also referred to as hypertensin and angiotonin. It is one of the most potent naturally occurring vasoconstrictors. The vasoconstrictive action of angiotensin II is primarily exerted on the arterioles and leads to a rise in both systolic and diastolic blood pressure. In individuals that are depleted of sodium or who have liver disease (e.g. cirrhosis), the pressive actions of angiotensin II are greatly reduced. These conditions lead to increased circulating levels of angiotensin II which in turn leads to a down-regulation in the numbers of angiotensin II receptors on smooth muscle cells. As a consequence, administration of exogenous angiotensin II to these individuals has little effect.

Other physiological responses to angiotensin II include induction of adrenal cortex synthesis and secretion of aldosterone. Angiotensin II also acts on the brain leading to increased blood pressure, vasopressin and ACTH secretion and increased water intake. Angiotensin II affects the contractility of the mesangial cells of the kidney leading to decreased glomerular filtration rate. One additional effect of angiotensin II is to potentiate the release of norepinephrine.

Two distinct types of angiotensin II receptors have been identified, AT<sub>1</sub> and AT<sub>2</sub>. The AT<sub>1</sub> receptors are classical serpentine (7 transmembrane spanning) receptors. The AT<sub>1</sub> receptors are coupled to a G-protein that leads to activation of PLC- $\gamma$ . Although the AT<sub>2</sub> receptors are also serpentine, they do not appear to be coupled to activation of G-proteins.

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## Parathyroid Hormone (PTH)

Parathyroid hormone (PTH, molecular weight 9,500) is synthesized and secreted by **chief cells** of the parathyroid in response to systemic Ca<sup>2+</sup> levels. The Ca<sup>2+</sup> receptor of the parathyroid gland responds to Ca<sup>2+</sup> by increasing intracellular levels of PKC, Ca<sup>2+</sup> and IP<sub>3</sub>; this stage is followed, after a period of protein synthesis, by PTH secretion. The synthesis and secretion of PTH in chief cells is constitutive, but Ca<sup>2+</sup> regulates the level of PTH in chief cells (and thus its secretion) by increasing the rate of PTH proteolysis when plasma Ca<sup>2+</sup> levels rise and by decreasing the proteolysis of PTH when Ca<sup>2+</sup> levels fall. The role of PTH is to regulate Ca<sup>2+</sup> concentration in extracellular fluids. The feedback loop that regulates PTH secretion therefore involves the parathyroids, Ca<sup>2+</sup>, and the target tissues described below.

PTH acts by binding to cAMP-coupled plasma membrane receptors, initiating a cascade of reactions that culminates in the biological response. The body response to PTH is complex but is aimed in all tissues at increasing Ca<sup>2+</sup> levels in extracellular fluids. PTH induces the dissolution of bone by stimulating osteoclast activity, which leads to elevated plasma Ca<sup>2+</sup> and phosphate. In the kidney, PTH reduces renal Ca<sup>2+</sup> clearance by stimulating its reabsorption; at the same time, PTH reduces the reabsorption of phosphate and thereby increases its clearance. Finally, PTH acts on the liver, kidney, and intestine to stimulate the production of the steroid hormone **1,25-**

**dihydroxycholecalciferol (calcitriol)**, which is responsible for Ca<sup>2+</sup> absorption in the intestine.

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## Calcitonin (CT)

Calcitonin (CT) is a 32-amino acid peptide secreted by C cells of the thyroid gland. Calcitonin is employed therapeutically to relieve

the symptoms of osteoporosis, although details of its mechanism of action remain unclear. However, it has been observed that CT induces the synthesis of PTH in isolated cells, which leads *in vivo* to increased plasma  $\text{Ca}^{2+}$  levels. In addition, CT has been shown to reduce the synthesis of **osteoporin (Opn)**, a protein made by osteoclasts and responsible for attaching osteoclasts to bone. Thus, it appears that CT elevates plasma  $\text{Ca}^{2+}$  via PTH induction and reduces bone reabsorption by decreasing osteoclast binding to bone.

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## Insulin, Glucagon and Somatostatin

The primary function of the pancreatic hormones is the regulation of whole-body energy metabolism, principally by regulating the concentration and activity of numerous enzymes involved in catabolism and anabolism of the major cell energy supplies.

The earliest of these hormones recognized was **insulin**, whose major function is to counter the concerted action of a number of hyperglycemia-generating hormones and to maintain low blood glucose levels. Because there are numerous hyperglycemic hormones, untreated disorders associated with insulin generally lead to severe hyperglycemia and shortened life span. Insulin is a member of a family of structurally and functionally similar molecules that include the **insulin-like growth factors** (IGF-1 and IGF-2), and **relaxin**. The tertiary structure of all 4 molecules is similar, and all have growth-promoting activities, but the dominant role of insulin is metabolic while the dominant roles of the IGFs and relaxin are in the regulation of cell growth and differentiation. For an extended discussion of the actions of insulin see the [Diabetes page](#).

Insulin is synthesized as a preprohormone in the  $\beta$  cells of the islets of Langerhans. Its signal peptide is removed in the cisternae of the endoplasmic reticulum and it is packaged into secretory vesicles in the Golgi, folded to its native structure, and locked in this conformation by the formation of 2 disulfide bonds. Specific protease activity cleaves the center third of the molecule, which dissociates as **C peptide**, leaving the amino terminal **B peptide** disulfide bonded to the carboxy terminal **A peptide**.

Insulin secretion from  $\beta$  cells is principally regulated by plasma glucose levels, but the precise mechanism by which the glucose signal is transduced remains unclear. One possibility is that the increased uptake of glucose by pancreatic  $\beta$ -cells leads to a concomitant increase in metabolism. The increase in metabolism leads to an elevation in the ATP/ADP ratio. This in turn leads to an inhibition of an ATP-sensitive  $\text{K}^+$  channel. The net result is a depolarization of the cell leading to  $\text{Ca}^{2+}$  influx and insulin secretion.

Chronic increases in numerous other hormones (including GH, hPL, estrogens, and progestins), up-regulate insulin secretion, probably by increasing the preproinsulin mRNA and enzymes involved in processing the increased preprohormone. In contrast, epinephrine diminishes insulin secretion by a cAMP-coupled regulatory path. In addition, epinephrine counters the effect of insulin in liver and peripheral tissue, where it binds to  $\beta$ -adrenergic receptors, induces adenylate cycles activity, increases cAMP, and activates PKA. The latter events induce glycogenolysis and gluconeogenesis, both of which are hyperglycemic and which thus counter insulin's effect on blood glucose levels.

Insulin secreted by the pancreas is directly infused via the portal vein to the liver, where it exerts profound metabolic effects. In most other tissues insulin increases the number of plasma membrane glucose transporters, but in liver glucose uptake is dramatically increased because of increased activity of the enzymes glucokinase, phosphofructokinase-1 (PFK-1), and pyruvate kinase (PK), the key regulatory enzymes of [glycolysis](#). The latter effects are induced by insulin-dependent activation of phosphodiesterase, with decreased PKA activity and diminished phosphorylation of the regulatory glycolytic enzymes. In addition, phosphatases specific for the phosphorylated forms of the glycolytic enzymes increase in activity under the influence of insulin. All these events lead to conversion of the glycolytic enzymes to their active forms and consequently a significant increase in glycolysis. In addition, glucose-6-phosphatase activity is down-regulated. The net effect is an increase in the content of hepatocyte glucose and its phosphorylated derivatives, with diminished blood glucose.



In addition to the latter events, diminished cAMP and elevated phosphatase activity combine to convert glycogen phosphorylase to its inactive form and glycogen synthase to its active form, with the result that not only is glucose funneled to glycolytic products, but glycogen content is increased as well.

Insulin generates its intracellular effects by binding to a plasma membrane receptor, which is the same in all cells. The receptor is a disulfide-bonded glycoprotein. One function of insulin (aside from its role in [signal transduction](#)) is to increase glucose transport in extrahepatic tissue is by increasing the number of glucose transport molecules in the plasma membrane. Glucose transporters are in a continuous state of turnover. Increases in the plasma membrane content of transporters stem from an increase in the rate of recruitment of new transporters into the plasma membrane, deriving from a special pool of preformed transporters localized in the cytoplasm.

In addition to its role in regulating glucose metabolism, insulin stimulates lipogenesis, diminishes lipolysis, and increases amino acid transport into cells. Insulin also modulates transcription, altering the cell content of numerous mRNAs. It stimulates growth, DNA synthesis, and cell replication, effects that it holds in common with the IGFs and relaxin.

Glucagon is a 29-amino acid hormone synthesized by the  $\alpha$  cells of the islets of Langerhans as a very much larger proglucagon molecule. Like insulin, glucagon lacks a plasma carrier protein, and like insulin its circulating half life is also about 5 minutes. As a consequence of the latter trait, the principal effect of glucagon is on the liver, which is the first tissue perfused by blood containing pancreatic secretions. The role of glucagon is well established. It binds to plasma membrane receptors and is coupled through G-proteins to adenylate cyclase. The resultant increases in cAMP and PKA reverse all of the effects described above that insulin has on liver. The increases also lead to a marked elevation of circulating glucose, with the glucose being derived from liver gluconeogenesis and liver glycogenolysis.

Somatostatin, secreted by  $\delta$  cells of the pancreas, is a 14--amino acid peptide, identical to somatostatin secreted by the hypothalamus. In neural tissue somatostatin inhibits GH secretion and thus has systemic effects. In the pancreas, somatostatin acts a paracrine inhibitor of other pancreatic hormones and thus also has systemic effects. It has been speculated that somatostatin secretion responds principally to blood glucose levels, increasing as blood glucose levels rise and thus leading to down-regulation of glucagon secretion.

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## Gastrointestinal Hormones and Peptides

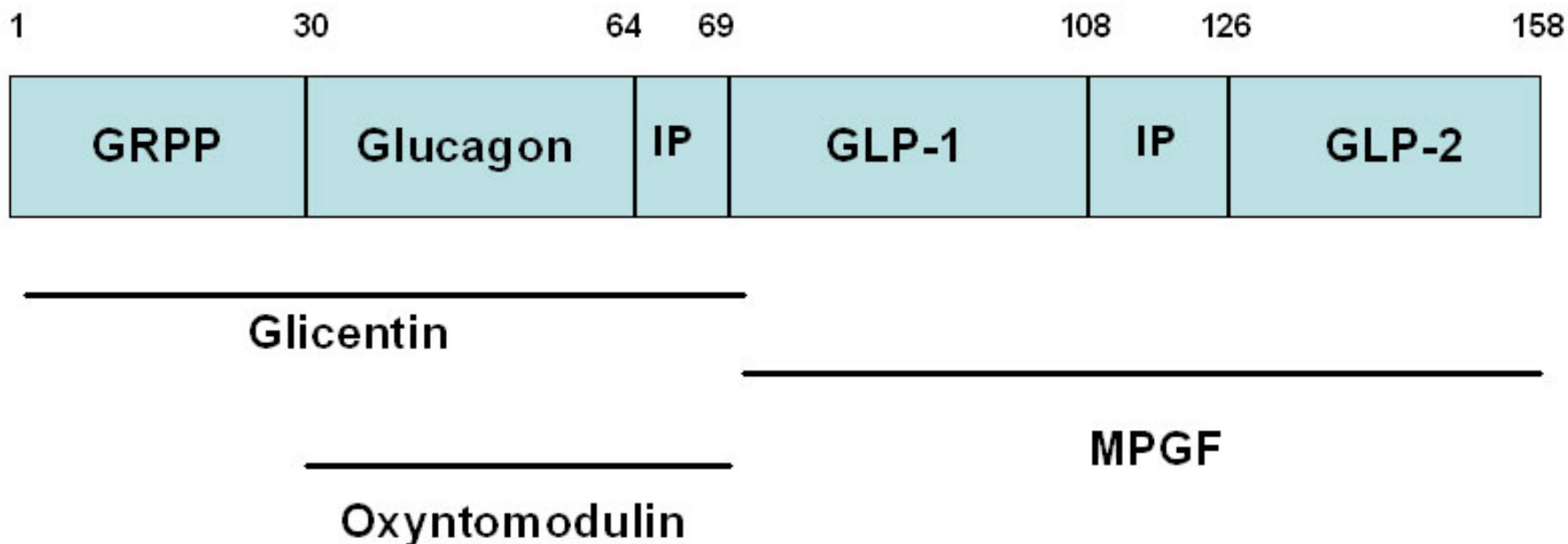
There are more than 30 peptides currently identified as being expressed within the digestive tract, making the gut the largest endocrine organ in the body. The regulatory peptides synthesized by the gut include hormones, peptide neurotransmitters and growth factors. Indeed, several hormones and neurotransmitters first identified in the central nervous system and other endocrine organs have subsequently been found in endocrine cells and/or neurons of the gut.

Although the many gastrointestinal hormones and peptides each have significant physiological roles (for more info visit the [Table of Vertebrate Hormones](#) page), the focus here will be a discussion of the glucagon-like peptides (principally glucagon-like peptide-1, GLP-1) and glucose-dependent insulintropic peptide (GIP). Both of these gut hormones constitute the class of molecules referred to as the **incretins**. Incretins are molecules associated with food intake-stimulation of insulin secretion from the pancreas.

GLP-1 is derived from the product of the proglucagon gene. This gene encodes a preproprotein that is differentially cleaved dependent upon the tissue in which it is synthesized. For example, in pancreatic  $\alpha$ -cells prohormone convertase 2 action leads to the release of glucagon. In the gut prohormone convertase 1/3 action leads to release of several peptides including GLP-1. Upon nutrient ingestion GLP-1 is secreted from intestinal enteroendocrine L-cells that are found predominantly in the ileum and colon with some production from these cell types in the duodenum and jejunum. Bioactive GLP-1 consists of 2 forms; GLP-1(7-37) and GLP-1(7-36)amide,



where the latter form constitutes the majority (80%) of the circulating hormone.



Structure of the mammalian preproglucagon product. GRPP=glucitin-related pancreatic peptide. IP=intervening peptide. GLP-2=glucagon-related peptide-2. Additional peptides are derived from the preproprotein including glicentin which is composed of amino acids 1-69, oxyntomodulin is composed of amino acids 30-69 and the major proglucagon fragment (MPGF) comprises amino acids 72-158.

The primary physiological responses to GLP-1 are glucose-dependent insulin secretion, inhibition of glucagon secretion and inhibition of gastric acid secretion and gastric emptying. The latter effect will lead to increased satiety with reduced food intake along with a reduced desire to ingest food. The action of GLP-1 at the level of insulin and glucagon secretion results in significant reduction in circulating levels of glucose following nutrient intake. This activity has significance in the context of [diabetes](#). The glucose lowering activity of GLP-1 is highly transient as the half-life of this hormone in the circulation is less than 2 minutes. Removal of bioactive GLP-1 is a consequence of N-terminal proteolysis catalyzed by dipeptidyl peptidase IV (DPP IV). DPP IV is also known as the lymphocyte surface antigen CD26 and has numerous activities unrelated to incretin inactivation (see [Diabetes](#) section for more information on DPP IV).)

All of the effects of GLP-1 are mediated following activation of the GLP-1 receptor (GLP-1R). The GLP-1R is a typical seven-

transmembrane spanning receptor coupled to G-protein activation, increased cAMP production and activation of PKA. However, there are also PKA-independent responses initiated through the GLP-1R. Other major responses to the actions of GLP-1 include pancreatic  $\beta$ -cell proliferation and expansion concomitant with a reduction of  $\beta$ -cell apoptosis (death). In addition, GLP-1 activity results in increased expression of the glucose transporter-2 (GLUT-2) and glucokinase genes in pancreatic cells.

Glucose-dependent insulintropic peptide (GIP) is derived from a 153-amino acid proprotein encoded by the GIP gene and circulates as a biologically active 42-amino acid peptide. GIP is synthesized by enteroendocrine K-cells whose locations are primarily in the duodenum and proximal jejunum. The original activity associated with GIP was the inhibition of gastric acid secretion and was thus, originally called gastric inhibitory peptide. However, subsequent research demonstrated that this gut hormone possessed potent stimulation of glucose-dependent insulin secretion. In addition, GIP has significant effects on fat metabolism exerted at the level of adipocytes. These actions include stimulation of lipoprotein lipase activity leading to increased uptake and incorporation of fatty acids by adipocytes. Whereas GIP exerts positive effects on pancreatic  $\beta$ -cell proliferation and survival similar to that shown for GLP-1, the hormone does not affect glucagon secretion nor gastric emptying. Like GLP-1, GIP is inactivated through the action of DPP IV.

The GIP receptor (GIPR) is a seven-transmembrane G-protein coupled protein found on pancreatic  $\beta$ -cells. Responses to GIP have been shown to be defective in type 2 diabetic patients. Interestingly, in gene knock-out mice it has been shown that loss of the GIPR is correlated to resistance to obesity even if the animals are fed a high fat diet.

| Hormone                                                                                               | Location                                                     | Major Action                                                                                             |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Glucagon-like peptide 1 (GLP-1)                                                                       | enteroendocrine L cells predominantly in the ileum and colon | potentiates glucose-dependent insulin secretion, inhibits glucagon secretion, inhibits gastric emptying  |
| Glucose-dependent insulintropic polypeptide (GIP)<br>originally called gastric inhibitory polypeptide | enteroendocrine K cells of the duodenum and proximal jejunum | inhibits secretion of gastric acid, enhances insulin secretion                                           |
| Gastrin                                                                                               | gastric antrum, duodenum                                     | gastric acid and pepsin secretion                                                                        |
| Cholecystokinin (CCK)                                                                                 | duodenum, jejunum                                            | stimulates gallbladder contraction and bile flow, increases secretion of digestive enzymes from pancreas |
| Secretin                                                                                              | duodenum, jejunum                                            | pancreatic bicarbonate secretion                                                                         |
| Vasoactive intestinal peptide (VIP)                                                                   | pancreas                                                     | smooth muscle relaxation; stimulates pancreatic bicarbonate secretion                                    |
| Motilin                                                                                               | small bowel                                                  | initiates interdigestive intestinal motility                                                             |

|                                      |                                |                                                                                                                                                                               |
|--------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pancreatic polypeptide (PP)          | pancreas                       | inhibits pancreatic bicarbonate and protein secretion                                                                                                                         |
| Enkephalins                          | stomach, duodenum, gallbladder | opiate-like actions                                                                                                                                                           |
| Substance P                          | entire gastrointestinal tract  | CNS function in pain (nociception),<br>involved in vomit reflex, stimulates<br>salivary secretions, induces<br>vasodilation<br>antagonists have anti-depressant<br>properties |
| Bombesin-like immunoreactivity (BLI) | stomach, duodenum              | stimulates release of gastrin and CCK                                                                                                                                         |

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## Common Vertebrate Hormones

| Hormone                   | Structure                                                                | Functions                                                                                                                                     |
|---------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pituitary Hormones</b> |                                                                          |                                                                                                                                               |
| Oxytocin                  | polypeptide of 9 amino acids <b>CYIQNCPLG</b> (C's are disulfide bonded) | uterine contraction, causes milk ejection in lactating females, responds to suckling reflex and estradiol, lowers steroid synthesis in testes |

|                                                       |                                                                                                                        |                                                                                                                                                       |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vasopressin</b><br>(antidiuretic hormone, ADH)     | polypeptide of 9 amino acids <b>CYFQNCPRG</b> (C's are disulfide bonded)                                               | responds to osmoreceptor which senses extracellular $[Na^+]$ , blood pressure regulation, increases $H_2O$ reabsorption from distal tubules in kidney |
| <b>Melanocyte-stimulating hormones (MSH)</b>          | $\alpha$ polypeptide = 13 amino acids<br>$\beta$ polypeptide = 18 amino acids<br>$\gamma$ polypeptide = 12 amino acids | pigmentation                                                                                                                                          |
| <b>Corticotropin</b><br>(adrenocorticotropin, ACTH)   | polypeptide = 39 amino acids                                                                                           | stimulates cells of adrenal gland to increase steroid synthesis and secretion                                                                         |
| Lipotropin (LPH)                                      | $\beta$ polypeptide = 93 amino acids<br>$\gamma$ polypeptide = 60 amino acids                                          | increases fatty acid release from adipocytes                                                                                                          |
| <b>Thyrotropin</b> (thyroid-stimulating hormone, TSH) | 2 proteins: $\alpha$ is 96 amino acids; $\beta$ is 112                                                                 | acts on thyroid follicle cells to stimulate thyroid hormone synthesis                                                                                 |
| <b>Growth hormone</b> (GH, or somatotropin)           | protein of 191 amino acids                                                                                             | general anabolic stimulant, increases release of insulin-like growth factor-I (IGF-I), cell growth and bone sulfation                                 |
| <b>Prolactin</b> (PRL)                                | protein of 197 amino acids                                                                                             | stimulates differentiation of secretory cells of mammary gland and stimulates milk synthesis                                                          |

|                                                                                                            |                                                        |                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Luteinizing hormone (LH);</b><br>human chorionic gonadotropin (hCG) is similar and produced in placenta | 2 proteins: $\alpha$ is 96 amino acids; $\beta$ is 121 | increases ovarian progesterone synthesis, luteinization; acts on Leydig cells of testes to increase testosterone synthesis and release and increases interstitial cell development |
| <b>Follicle-stimulating hormone (FSH)</b>                                                                  | 2 proteins: $\alpha$ is 96 amino acids; $\beta$ is 120 | ovarian follicle development and ovulation, increases estrogen production; acts on Sertoli cells of semiferous tubule to increase spermatogenesis                                  |

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## Hypothalamic Hormones

|                                                                                      |                                                    |                                                                          |
|--------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------|
| <b>Corticotropin-releasing factor (CRF or CRH)</b>                                   | protein of 41 amino acids                          | acts on corticotrope to release ACTH and $\beta$ -endorphin (lipotropin) |
| <b>Gonadotropin-releasing factor (GnRF or GnRH)</b>                                  | polypeptide of 10 amino acids                      | acts on gonadotrope to release LH and FSH                                |
| <b>Prolactin-releasing factor (PRF)</b>                                              | this may be TRH                                    | acts on lactotrope to release prolactin                                  |
| <b>Prolactin-release inhibiting factor (PIF)</b>                                     | may be derived from GnRH precursor, 56 amino acids | acts on lactotrope to inhibit prolactin release                          |
| <b>Growth hormone-releasing factor (GRF or GRH)</b>                                  | protein of 40 and 44 amino acids                   | stimulates GH secretion                                                  |
| <b>Somatostatin (SIF, also called growth hormone-release inhibiting factor, GIF)</b> | polypeptide of 14 and 28 amino acids               | inhibits GH and TSH secretion                                            |
| <b>Thyrotropin-releasing factor (TRH or TRF)</b>                                     | polypeptide of 3 amino acids: <b>EHP</b>           | stimulates TSH and prolactin secretion                                   |



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## Thyroid Hormones

|                                                |                                                                                                                              |                                                                                                               |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <a href="#">Thyroxine and triiodothyronine</a> | iodinated dityrosin derivatives                                                                                              | responds to TSH and stimulates oxidations in many cells                                                       |
| Calcitonin                                     | protein of 32 amino acids                                                                                                    | produced in parafollicular C cells of the thyroid, regulation of $\text{Ca}^{2+}$ and $\text{P}_i$ metabolism |
| Calcitonin gene-related peptide (CGRP)         | protein of 37 amino acids, product of the calcitonin gene derived by alternative splicing of the precursor mRNA in the brain | acts as a vasodilator                                                                                         |

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## Parathyroid Hormone

|                                           |                           |                                                                                                                                                                                |
|-------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Parathyroid hormone (PTH)</a> | protein of 84 amino acids | regulation of $\text{Ca}^{2+}$ and $\text{P}_i$ metabolism, stimulates bone resorption thus increasing serum $[\text{Ca}^{2+}]$ , stimulates $\text{P}_i$ secretion by kidneys |
|-------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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## Hormones and Peptides of the Gut

[additional dicussion of gastrointestinal hormones](#)

|                                                                                                        |                                                                              |                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Glucagon-like peptide 1<br>(GLP-1)<br>formerly called enteroglucagon                                   | Two forms: 31 amino acids, GLP-1(7-37) and 30 amino acids, GLP-1(7-36) amide | potentiates glucose-dependent insulin secretion, inhibits glucagon secretion, inhibits gastric emptying                                                                                                                                      |
| Glucose-dependent insulinotropic polypeptide (GIP)<br>originally called gastric inhibitory polypeptide | polypeptide of 42 amino acids                                                | inhibits secretion of gastric acid, enhances insulin secretion                                                                                                                                                                               |
| Gastrin                                                                                                | 17 amino acids                                                               | produced by stomach antrum, stimulates acid and pepsin secretion, also stimulates pancreatic secretions                                                                                                                                      |
| Secretin                                                                                               | 27 amino acids                                                               | secreted from duodenum at pH values below 4.5, stimulates pancreatic acinar cells to release bicarbonate and H <sub>2</sub> O                                                                                                                |
| Cholecystokinin, CCK                                                                                   | 33 amino acids                                                               | stimulates gallbladder contraction and bile flow, increases secretion of digestive enzymes from pancreas                                                                                                                                     |
| Motilin                                                                                                | 22 amino acids                                                               | controls gastrointestinal muscles                                                                                                                                                                                                            |
| Vasoactive intestinal peptide (VIP)                                                                    | 28 amino acids                                                               | produced by hypothalamus and GI tract, relaxes the GI, inhibits acid and pepsin secretion, acts as a neurotransmitter in peripheral autonomic nervous system, increases secretion of H <sub>2</sub> O and electrolytes from pancreas and gut |

|                                                                                                                                                                       |                                                                                                          |                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Somatostatin                                                                                                                                                          | 14 amino acid version                                                                                    | inhibits release and action of numerous gut peptides, e.g. CCK, gastrin, secretin, motilin, GIP; also inhibits insulin and glucagon secretion from pancreas       |
| Substance P<br>a member of the tachykinin family that includes neurokinin A (NKA) and neurokinin B (NKB)                                                              | 11 amino acids                                                                                           | CNS function in pain (nociception), involved in vomit reflex, stimulates salivary secretions, induces vasodilation<br>antagonists have anti-depressant properties |
| PP, PYY and NPY constitute the Pancreatic Polypeptide family of 36 amino acid peptides.<br>PP and PYY exhibit endocrine functions.<br>NPY functions as a neuropeptide |                                                                                                          |                                                                                                                                                                   |
| Pancreatic Polypeptide, PP                                                                                                                                            | 36 amino acids                                                                                           | suppresses glucose-induced insulin secretion, inhibits bicarbonate and protein secretion from pancreas                                                            |
| Peptide Tyrosine Tyrosine, PYY                                                                                                                                        | 36 amino acids                                                                                           | inhibits gastric motility by inhibiting cholinergic neurotransmission, inhibits gastric acid secretion                                                            |
| Neuropeptide Tyrosine, NPY                                                                                                                                            | 36 amino acids<br>6 receptors                                                                            | effects on hypothalamic function in appetite, controls feeding behavior and energy homeostasis, levels increase during starvation to induce food intake           |
| Amphiregulin                                                                                                                                                          | 2 peptides: 78 amino acid truncated form and 84 amino acid form with 6 additional N-terminal amino acids | homology to EGF and binds to the EGF receptor (EGFR)                                                                                                              |

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## Pancreatic Hormones

|                        |                                                     |                                                                                                                                        |
|------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Insulin                | disulfide bonded dipeptide of 21 and 30 amino acids | produced by $\beta$ -cells of the pancreas, increases glucose uptake and utilization, increases lipogenesis, general anabolic effects  |
| Glucagon               | polypeptide of 29 amino acids                       | produced by $\alpha$ -cells of the pancreas, increases lipid mobilization and glycogenolysis in order to increase blood glucose levels |
| Pancreatic polypeptide | polypeptide of 36 amino acids                       | increases glycogenolysis, regulation of gastrointestinal activity                                                                      |
| Somatostatin           | 14 amino acid version                               | inhibition of glucagon and somatotropin release                                                                                        |

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## Placental Hormones

|                        |                                                        |                                                                                                           |
|------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Estrogens              | steroids                                               | maintenance of pregnancy                                                                                  |
| Progestins             | steroids                                               | mimic action of progesterone                                                                              |
| Chorionic gonadotropin | 2 proteins: $\alpha$ is 96 amino acids; $\beta$ is 147 | activity similar to LH                                                                                    |
| Placental lactogen     | protein of 191 amino acids                             | acts like prolactin and GH                                                                                |
| Relaxin                | 2 proteins of 22 and 32 amino acids                    | produced in ovarian corpus luteum, inhibits myometrial contractions, secretion increases during gestation |

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## Gonadal Hormones

|                        |                                                                              |                                                        |
|------------------------|------------------------------------------------------------------------------|--------------------------------------------------------|
| Estrogens (ovarian)    | steroids; estradiol and estrone                                              | maturation and function of female secondary sex organs |
| Progestins (ovarian)   | steroid; progesterone                                                        | implantation of ovum and maintenance of pregnancy      |
| Androgens (testicular) | steroid; testosterone                                                        | maturation and function of male secondary sex organs   |
| Inhibins A and B       | 1 protein ( $\alpha$ is 134 amino acids; $\beta$ is 115 and 116 amino acids) | inhibition of FSH secretion                            |

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## Adrenal Cortical Hormones

|                                    |                                       |                                                       |
|------------------------------------|---------------------------------------|-------------------------------------------------------|
| <a href="#">Glucocorticoids</a>    | steroids; cortisol and corticosterone | diverse effects on inflammation and protein synthesis |
| <a href="#">Mineralocorticoids</a> | steroids; aldosterone                 | maintenance of salt balance                           |

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## Adrenal Medullary Hormones

|                               |                       |                                                                                 |
|-------------------------------|-----------------------|---------------------------------------------------------------------------------|
| Epinephrine (adrenalin)       | derived from tyrosine | glycogenolysis, lipid mobilization, smooth muscle contraction, cardiac function |
| Norepinephrine (noradrenalin) | tyrosine derivative   | lipid mobilization, arteriole contraction                                       |

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## Liver Hormones

### Angiotensin II

polypeptide of 8 amino acids derived from angiotensinogen (present in the  $\alpha_2$ -globin fraction of plasma) which is cleaved by the kidney enzyme **renin** to give the decapeptide, angiotensin I, the C-terminal 2 amino acids are then released (by action of angiotensin-converting enzyme, ACE) to yield angiotensin II

responsible for essential hypertension through stimulated synthesis and release of aldosterone from adrenal cells

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## Kidney Hormones

### Calcitriol

[1,25-(OH)<sub>2</sub>-vitamin D<sub>3</sub>]

derived from 7-dehydrocholesterol

responsible for maintenance of calcium and phosphorous hoemostasis, increases intestinal Ca<sup>2+</sup> uptake, regulates bone mineralization

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## Cardiac Hormones

### Atrial natriuretic peptide (ANP)

several active peptides cleaved from a 126 amino acid precursor

released from heart atria in response to hypovolemia, acts on outer adrenal cells to decrease aldosterone production; smooth muscle relaxation

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## Pineal Hormones

Melatonin

*N*-acetyl-5-methoxytryptamine

regulation of circadian rhythms

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**Last modified: Tuesday, 05-Oct-2004 15:40:16 EST**

## Post-Translational Modification

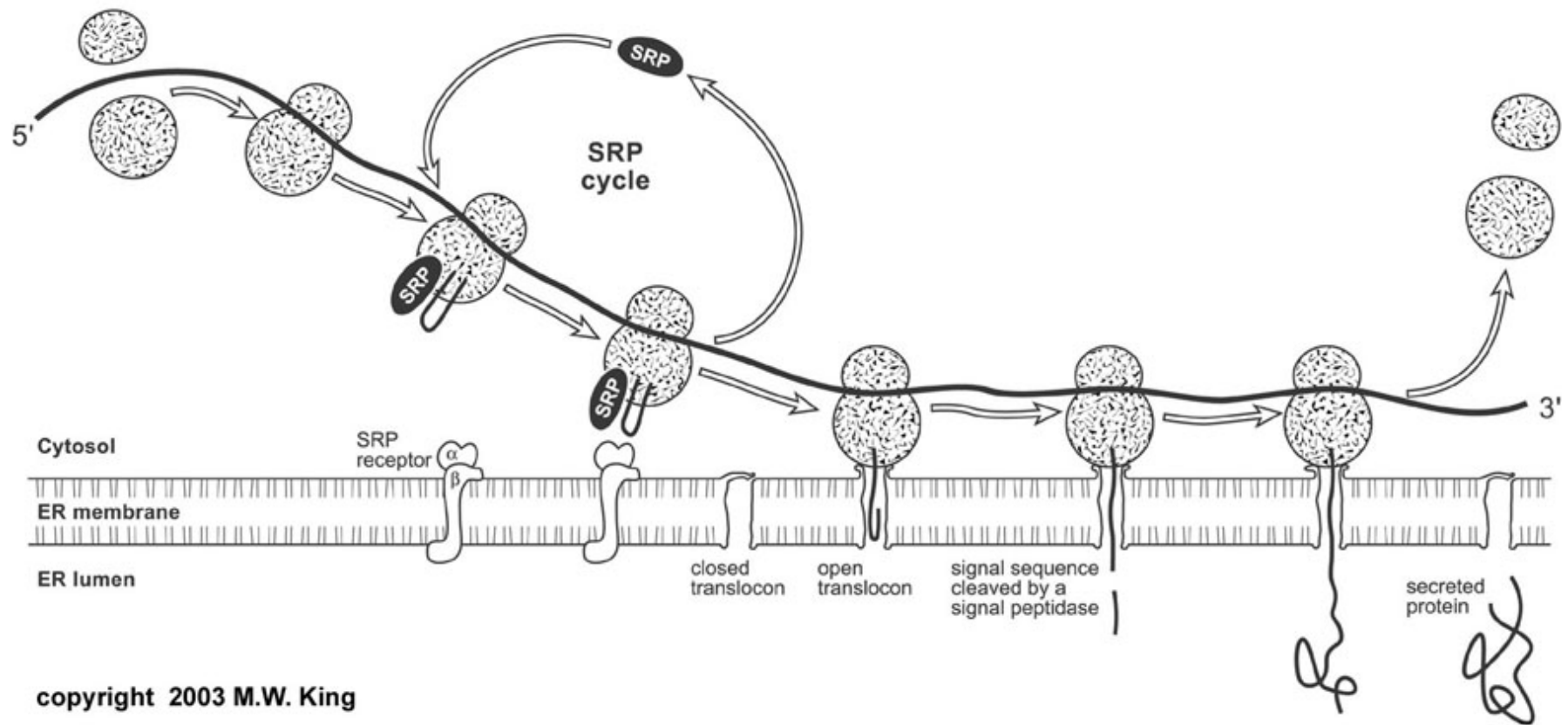
- Secreted and Membrane-Associated Proteins
- Proteolytic Cleavage of Proteins
- Glycoproteins
  - Mechanism of Sugar Linkage
- Lysosomal Targeting of Enzymes
- Clinical Significances of Glycoproteins
  - Defects in Glycoprotein Degradation
- Acylation
- Methylation
- Phosphorylation
- Sulfation
- Prenylation
- Vitamin C-Dependent Modifications
- Vitamin K-Dependent Modifications
- Selenoproteins

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### Secreted and Membrane-Associated Proteins

Proteins that are membrane bound or are destined for excretion are synthesized by ribosomes associated with the membranes of the endoplasmic reticulum (ER). The ER associated with ribosomes is termed rough ER (RER). This class of proteins all contain an N-terminus termed a **signal sequence** or **signal peptide**. The signal peptide is usually 13-36 predominantly hydrophobic residues. The signal peptide is recognized by a multi-protein complex termed the **signal recognition particle (SRP)**. This signal peptide is removed following passage through the endoplasmic reticulum membrane. The removal of the signal peptide is catalyzed by signal peptidase. Proteins that contain a signal peptide are called **preproteins** to distinguish them from proproteins. However, some proteins that are destined for secretion are also further proteolyzed following secretion and, therefore contain pro sequences. This class of proteins is termed **preproproteins**.



Mechanism of synthesis of membrane bound or secreted proteins. Ribosomes engage the ER membrane through interaction of the **signal recognition particle, SRP** in the ribosome with the SRP receptor in the ER membrane. As the protein is synthesized the **signal sequence** is passed through the ER membrane into the lumen of the ER. After sufficient synthesis the signal peptide is removed by the action of signal peptidase. Synthesis will continue and if the protein is secreted it will end up completely in the lumen of the ER. If the protein is membrane associated a **stop transfer** motif in the protein will stop the transfer of the protein through the ER membrane. This will become the membrane spanning domain of the protein.

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## Proteolytic Cleavage

Most proteins undergo proteolytic cleavage following translation. The simplest form of this is the removal of the initiation methionine. Many proteins are synthesized as inactive precursors that are activated under proper physiological conditions by limited proteolysis. Pancreatic enzymes and enzymes involved in clotting are examples of the latter. Inactive precursor proteins that are activated by removal of polypeptides are termed, **proproteins**.

A complex example of post-translational processing of a preproprotein is the cleavage of **prepro-opiomelanocortin (POMC)** synthesized in the pituitary (see the [Peptide Hormones](#) page for discussion of POMC). This preproprotein undergoes complex cleavages, the pathway of which differs depending upon the cellular location of POMC synthesis.

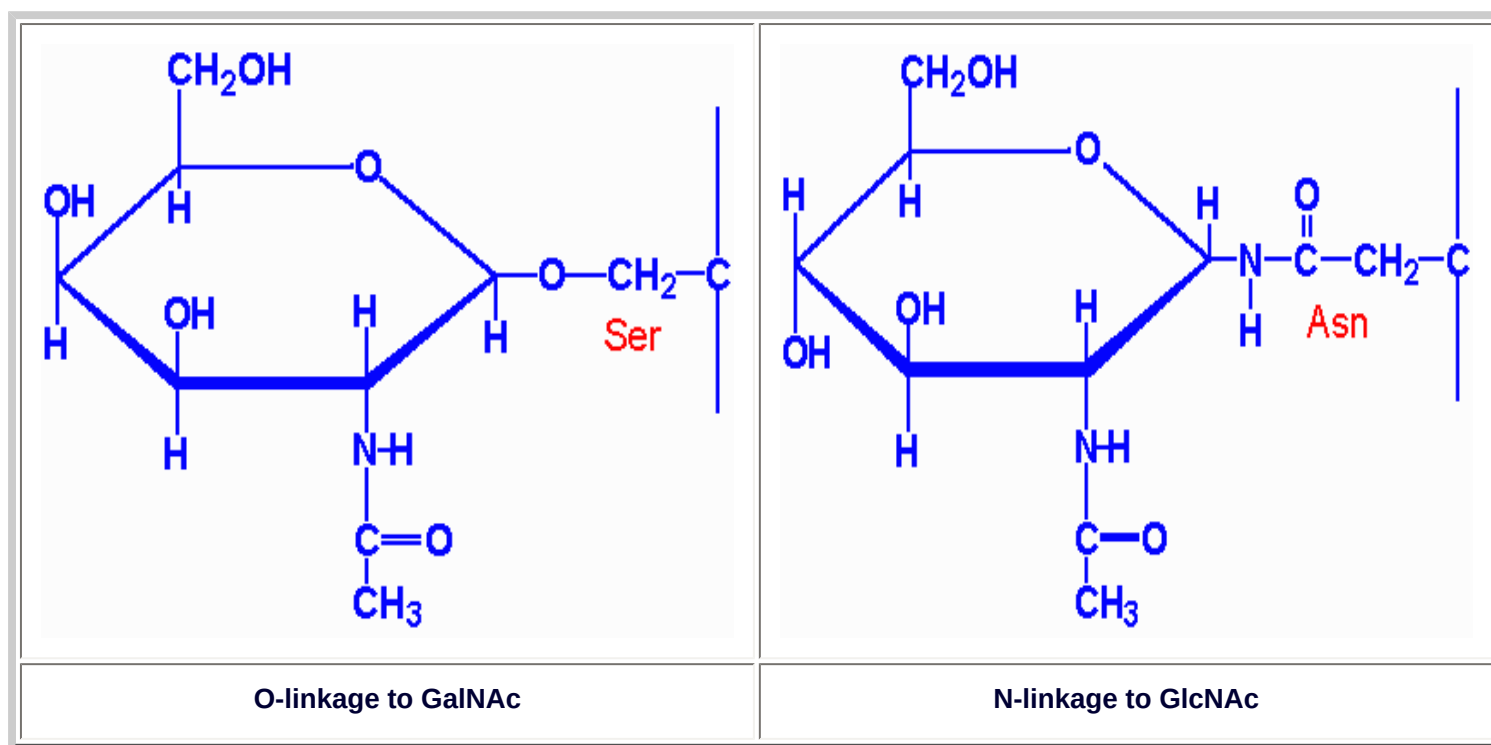
Another example of a preproprotein is **insulin**. Since insulin is secreted from the pancreas it has a prepeptide. Following cleavage of the 24 amino acid signal peptide the protein folds into proinsulin. Proinsulin is further cleaved yielding active insulin which is composed of two peptide chains linked together through disulfide bonds..

Still other proteins (of the enzyme class) are synthesized as inactive precursors called **zymogens**. Zymogens are activated by proteolytic cleavage such as is the situation for several proteins of the **blood clotting** cascade.

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## Glycoproteins

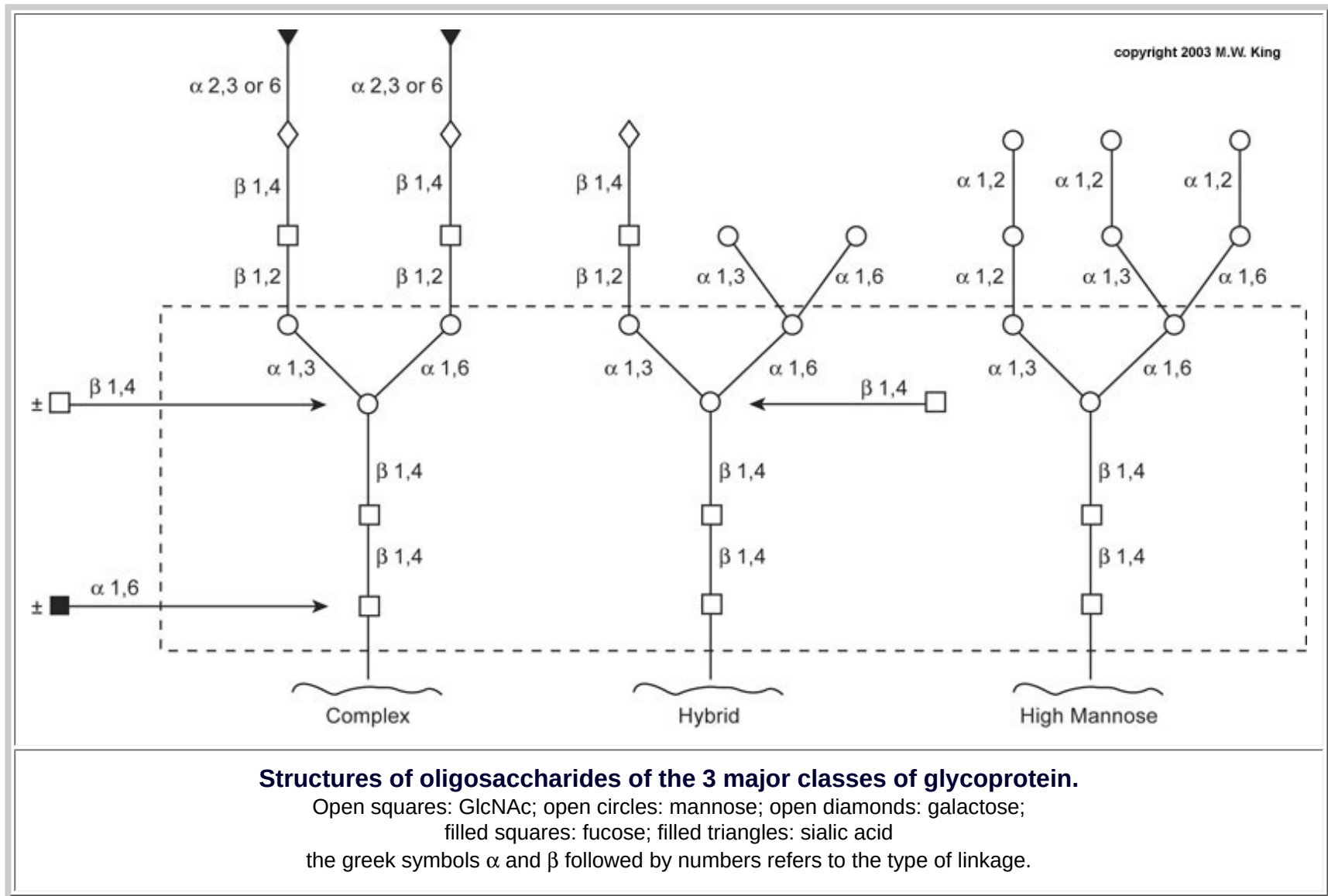
Membrane associated carbohydrate is exclusively in the form of oligosaccharides covalently attached to proteins forming **glycoproteins**, and to a lesser extent covalently attached to lipid forming the **glycolipids**. Glycoproteins consist of proteins covalently linked to carbohydrate. The predominant sugars found in glycoproteins are glucose, galactose, mannose, fucose, GalNAc, GlcNAc and NANA. The distinction between **proteoglycans** and glycoproteins resides in the level and types of carbohydrate modification. The carbohydrate modifications found in glycoproteins are rarely complex: carbohydrates are linked to the protein component through either **O-glycosidic** or **N-glycosidic** bonds. The N-glycosidic linkage is through the amide group of asparagine. The O-glycosidic linkage is to the hydroxyl of serine, threonine or hydroxylysine. The linkage of carbohydrate to hydroxylysine is generally found only in the collagens. The linkage of carbohydrate to 5-hydroxylysine is either the single sugar galactose or the disaccharide glucosylgalactose. In ser- and thr-type O-linked glycoproteins, the carbohydrate directly attached to the protein is GalNAc. In N-linked glycoproteins, it is GlcNAc.



The predominant carbohydrate attachment in glycoproteins of mammalian cells is via N-glycosidic linkage. The site of carbohydrate attachment to N-linked glycoproteins is found within a consensus sequence of amino acids, **N-X-S(T)**, where X is any amino acid except proline. When an analysis of proteins in the public databases is carried out, it can be shown that approximately 65% of all the proteins contain at least one occurrence of the Asn-X-Ser/Thr consensus. N-

linked glycoproteins all contain a common core of carbohydrate attached to the polypeptide. This core consists of three mannose residues and two GlcNAc. A variety of other sugars are attached to this core and comprise three major N-linked families:

- 1. **High-mannose type** contains all mannose outside the core in varying amounts.
- 2. **Hybrid type** contains various sugars and amino sugars.
- 3. **Complex type** is similar to the hybrid type, but in addition, contains sialic acids to varying degrees.



Most proteins that are secreted, or bound to the plasma membrane, are modified by carbohydrate attachment. The part that is modified, in plasma membrane-bound proteins, is the extracellular portion of the protein that is modified. Intracellular proteins are less frequently modified by carbohydrate attachment. However, the attachment of carbohydrate to intracellular proteins confers unique functional activities on these proteins. Linkage of carbohydrate to

cytosolic and/or nuclear proteins occurs via O-linkage and involves attachment of GlcNAc to serine or threonine residues. The linkage is catalyzed by the enzyme O-GlcNAc transferase, OGT. Several transcription factors and RNA polymerase II have been shown to be modified by O-GlcNAc linkage.

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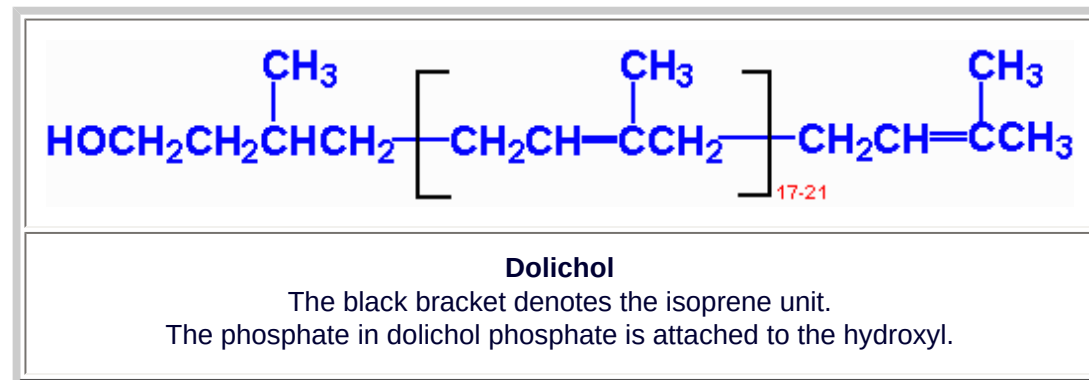
## Mechanism of Carbohydrate Linkage to Protein

The protein component of all glycoproteins is synthesized from polyribosomes that are bound to the endoplasmic reticulum (ER). The processing of the sugar groups occurs **cotranslationally** in the lumen of the ER and continues in the Golgi apparatus for N-linked glycoproteins. Attachment of sugars in O-linked glycoproteins occurs **post-translationally** in the Golgi apparatus. Sugars used for glycoprotein synthesis (both N-linked and O-linked) are activated by coupling to nucleotides. Glucose and GlcNAc are coupled to UDP and mannose is coupled to GDP.

**O-linked sugars:** The synthesis of O-linked glycoproteins occurs via the stepwise addition of nucleotide-activated sugars directly onto the polypeptide. The nucleotide-activated sugars are coupled to either UDP, GDP (as with mannose) or CMP (for instance, NANA). The attachment of sugars is catalyzed by specific glycoprotein glycosyltransferases. Evidence indicates that each specific type of carbohydrate linkage in O-linked glycoproteins is the result of a different glycosyltransferase.

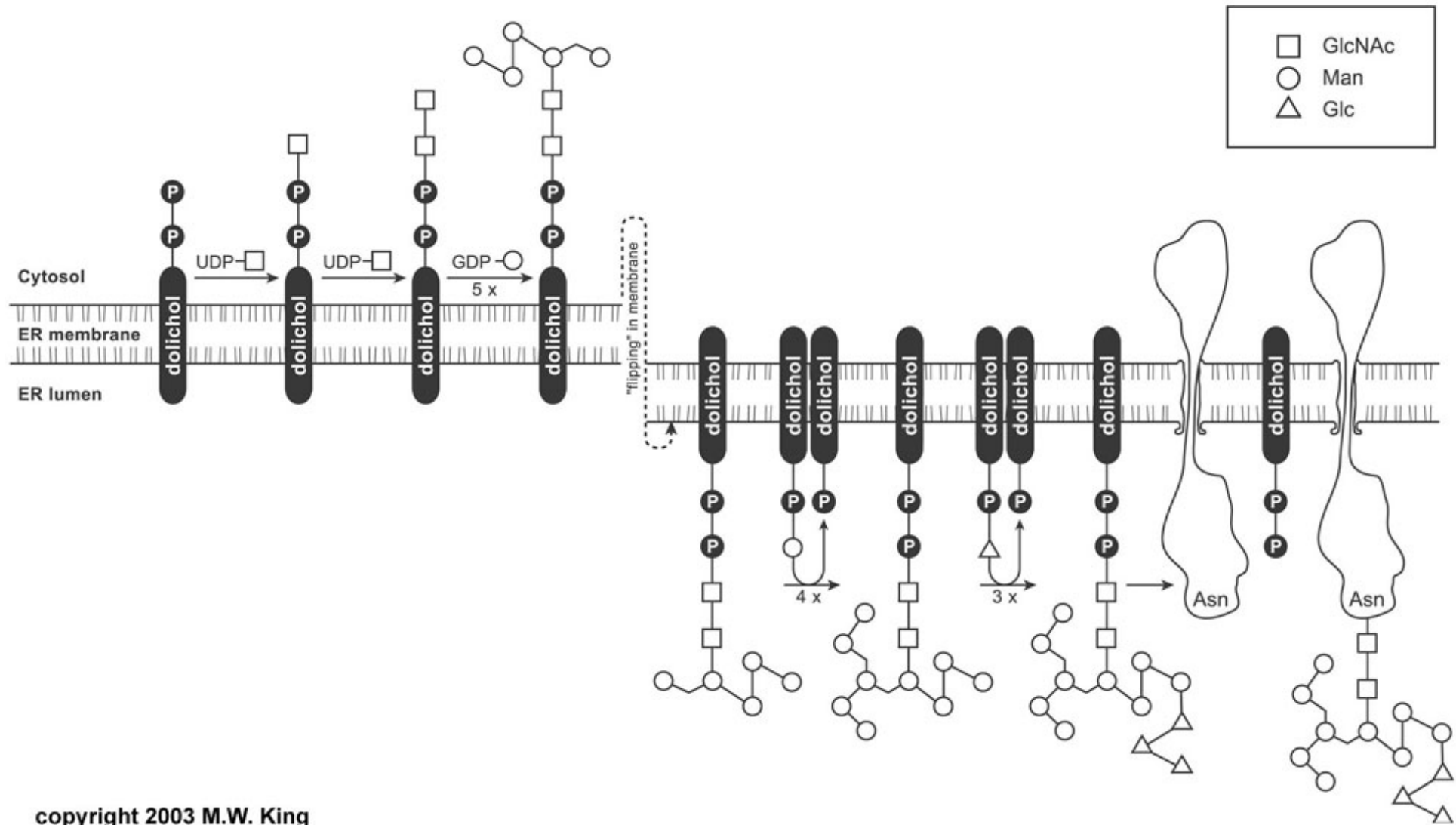
**N-linked sugars:** As indicated earlier, the three major classes of N-linked carbohydrate modifications are high-mannose, hybrid and complex. The major distinguishing feature of the complex class is the presence of sialic acid, whereas the hybrid class contains no sialic acid.

In contrast to the step-wise addition of sugar groups to the O-linked class of glycoproteins, N-linked glycoprotein synthesis requires a lipid intermediate: **dolichol phosphate**. Dolichols are polyprenols (C80-C100) containing 17 to 21 isoprene units, in which the terminal unit is saturated.



As indicated, the formation of the GlcNAc-β-Asn linkage in proteins occurs in the endoplasmic reticulum (ER) through cotranslational addition of a preassembled carbohydrate core structure that is delivered via the carbohydrate-dolichol lipid intermediate. The preassembled carbohydrate core structure comprises three terminal residues of glucose attached to a branched cluster of nine mannose residues that are in turn attached to two GlcNAc residues attached to dolicholpyrophosphate. The structure is abbreviated Glc<sub>3</sub>Man<sub>9</sub>GlcNAc<sub>2</sub>-P-P-dolichol. This structure is commonly referred to as the **lipid-linked oligosaccharide (LLO)**, whereas the oligosaccharide structure itself is termed the **en bloc oligosaccharide**. In mammalian cells the importance of the terminal glucose residues is evident from the fact that transfer of Man<sub>9</sub>GlcNAc<sub>2</sub>-P-P-dolichol is some 25-times less efficient than the complete structure. In addition, structures that contain three terminal glucose residues, but not the complete Man<sub>9</sub>GlcNAc<sub>2</sub> structure, are efficiently transferred to protein by oligosaccharyltransferase. Synthesis of the *en bloc* dolichol-P-P-oligosaccharide unit begins on the cytoplasmic face of the ER membrane and prior to transfer to the protein, the structure “flips” to the luminal side.





Pathway by which the synthesis and transfer of the lipid-linked oligosaccharide unit takes place at the membrane of the ER.

Immediately following transfer of the *en bloc* oligosaccharide unit to the protein, processing and alteration of the composition of the oligosaccharide ensues and continues as the protein passes through the ER then into and through the Golgi apparatus. Initially, the terminal glucose is removed through the action of glucosidase I (GI), a membrane bound enzyme recognizing  $\alpha 1,2$ -linked glucose. The remaining two glucose residues are then removed by glucosidase II (GII), a soluble enzyme recognizing  $\alpha 1,3$ -linked glucose. After removal of the glucose residues, the action of  $\alpha$ -mannosidases removes several mannose residues as the protein progresses to the Golgi. The action of the various glucosidases and mannosidases leaves N-linked glycoproteins containing a common core of

carbohydrate consisting of three mannose residues and two GlcNAc. Through the action of a wide range of glycosyltransferases and glycosidases a variety of other sugars are attached to this core as the protein progresses through the Golgi. These latter reactions generate the three major families of N-linked glycoproteins described above.

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## Lysosomal Targeting of Enzymes

Enzymes that are destined for the lysosomes (lysosomal enzymes) are directed there by a specific carbohydrate modification. During transit through the Golgi apparatus a residue of GlcNAc-1-phosphate (GlcNAc-1-P) is added to the carbon-6 hydroxyl group of one or more specific mannose residues that have been added to these enzymes. The GlcNAc is activated by coupling to UDP and is transferred by UDP-GlcNAc:lysosomal enzyme GlcNAc-1-phosphotransferase (GlcNAc-phosphotransferase), yielding a phosphodiester intermediate: GlcNAc-1-P-6-Man-protein. A second reaction (catalyzed by GlcNAc 1-phosphodiester-*N*-acetylglucosaminidase) removes the GlcNAc leaving mannose residues phosphorylated in the 6 position: Man-6-P-protein. A specific Man-6-P receptor (MPR) is present in the membranes of the Golgi apparatus. Binding of Man-6-P to this receptor targets proteins to the lysosomes.

Two distinct MPRs have been identified and both are members of the P-type lectin family. Both are type I integral membrane glycoproteins that contain an N-terminal extracellular domain, a single transmembrane domain and a C-terminal cytoplasmic domain. One receptor is large with a molecular weight of approximately 300kDa, the other receptor is smaller with a molecular weight of approximately 46kDa. Structural similarities between these two receptors indicates they are derived from a single ancestral gene with the larger receptor arising through multiple gene duplications. The extracellular portion of the larger receptor contains 15 repeating elements, each of which is highly similar to the extracellular domain of the smaller receptor. Both receptors exist as dimers embedded in the membrane.

The large receptor binds two moles of Man-6-P and the smaller binds one mole of Man-6-P per subunit, thus 4 and 2 moles of Man-6-P per dimer, respectively. The bovine and murine versions of the smaller receptors require divalent cations for ligand binding and thus the receptor has been termed the **cation-dependent Man-6-P receptor (CD-MPR)**. However, the human counterpart may not require cations for its activity. The larger receptor does not require divalent cations for ligand binding and is therefore, commonly referred to as the **cation-independent Man-6-P receptor (CI-MPR)**. However, the CI-MPR has been shown to bind the nonglycosylated polypeptide hormone, insulin-like growth factor II (IGF-II) and as such the larger MPR is more frequently identified as IGF-II/MPR. The IGF-II/MPR is available at the cell surface and its role in binding IGF-II is to target this hormone for degradation in the lysosomes. In addition to IGF-II, the IGF-II/MPR has been shown to bind a diverse array of Man-6-P-containing proteins as well as several nonglycosylated proteins. Although IGF-II/MPR and CD-MPR exhibit distinct activities, both receptors function to target newly synthesized lysosomal enzymes to the lysosomes.

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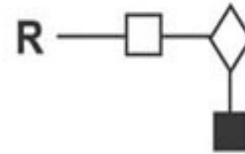
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## Clinical Significances of Glycoproteins

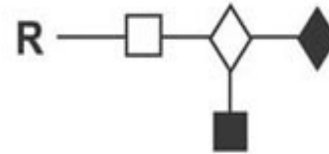
Glycoproteins on cell surfaces are important for communication between cells, for maintaining cell structure and for self-recognition by the immune system. The alteration of cell-surface glycoproteins can, therefore, produce profound physiological effects, of which several are listed below.

- 1. The **ABO blood group antigens** are the carbohydrate moieties of glycolipids on the surface of cells as well as the carbohydrate portion of serum glycoproteins. When present on the surface of cells the ABO carbohydrates are linked to sphingolipid and are therefore of the glycosphingolipid class. When the ABO carbohydrates are associated with protein in the form of glycoproteins they are found in the serum and are referred to as the secreted forms. Some individuals produce the glycoprotein forms of the ABO antigens while others do not. This property distinguishes **secretors** from **non-secretors**, a property that has forensic importance such as in cases of rape. For more information of blood group antigens, including ABO visit the blood groups page at the [SCARF](#) site.

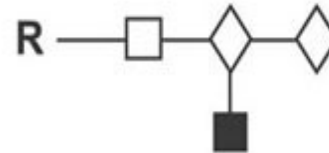
H antigen  
Type O



H antigen  
Type A



B antigen  
Type B



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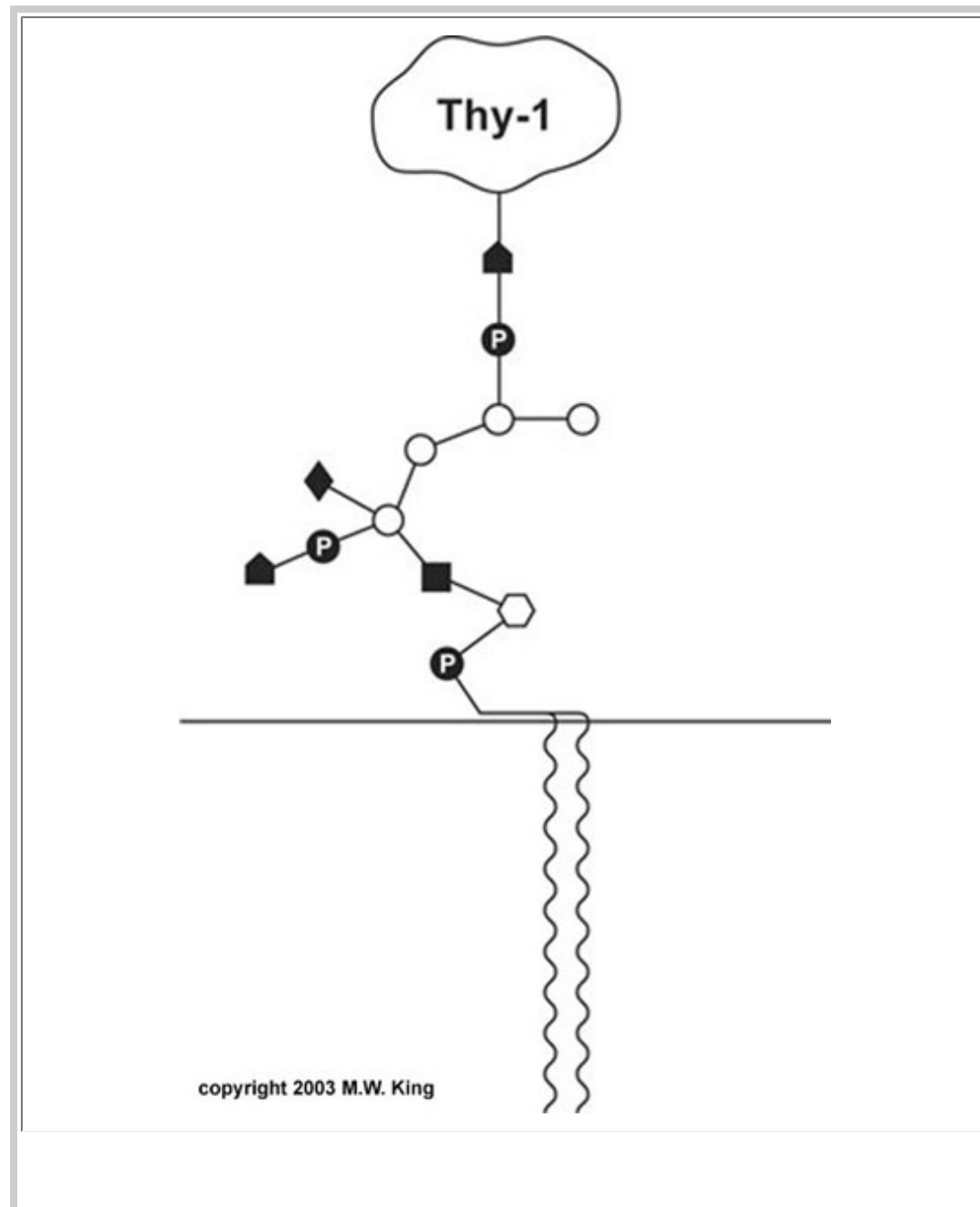
### Structure of the ABO blood group carbohydrates,

R represents the linkage to protein in the secreted forms, sphingolipid in the cell-surface bound form.  
open square = GlcNAc, open diamond = galactose, filled square = fucose, filled diamond = GalNAc, filled diamond = sialic acid (NANA)

- **2.** The truncation of erythrocyte surface glycoproteins leads to cell clumping, as in [congenital dyserythropoietic anemia type II](#). Also referred to as HEMPAS (hereditary erythroblastic multinuclearity with positive acidified-serum test).
- **3.** Several viruses, bacteria and parasites have exploited the presence of cell-surface carbohydrates, principally associated with protein (glycoproteins), using them as portals of entry into the cell.

- **A. Human immunodeficiency virus (HIV)**, the causative agent of **AIDS**, gains entry into cells of the immune system by attaching to a class of cellular receptors known as the **chemokine receptors**, most notably CXCR4 and CCR5. For more information on chemokines and their receptors visit the [C.O.P.E](#) site.
- **B.** Members of the **poxvirus** family of viruses gain entry into cells, most frequently migratory leukocytes, by attaching to chemokine receptors including CCR1, CCR5 and CXCR4 (Science [1999] vol. 286 pp. 1968-1971).
- **C. Dystroglycan (DG)** is a component of the dystrophin-glycoprotein complex. It is a laminin receptor encoded by a single gene and cleaved by postranslational processing into two proteins, peripheral membrane  $\alpha$ -DG and transmembrane  $\beta$ -DG.  $\alpha$ -DG interacts with laminin-2 in the basal lamina and  $\beta$ -DG binds to dystrophin containing cytoskeletal proteins in muscle and peripheral nerves. DG is involved in agrin- and laminin-induced acetylcholine receptor clustering at neuromuscular junctions, morphogenesis, early development, and the pathogenesis of muscular dystrophies. Recent evidence (Science (1998) vol. 282 pp. 2076-2079 and 2079-2081) demonstrates that  $\alpha$ -DG present on Schwann cell membranes is the receptor for *Mycobacterium leprae* and also serves as the receptor for the arenavirus class of pathogens. Arenaviruses cause hemorrhagic fever in humans. Lymphocytic choriomeningitis virus (LCMV), Lassa fever virus (LFV), Oliveros and Mobala (all members of the arenavirus family) all bind to  $\alpha$ -DG. The specificity of this interaction was demonstrated by the resistance to LCMV infection of cells harboring a null mutation in DG.
- **D.** Rhinoviruses utilize attachment to **ICAM-1 (intercellular adhesion molecule-1)** to gain entry into cells.
- **E.** The pathogenic human parvovirus, B19, attaches to the erythrocyte-specific cell-surface globoside identified as **erythrocyte P antigen** to infect erythrocytes.
- **F.** The malarial parasite Plasmodium vivax, binds to the erythrocyte **chemokine receptor** known as the **Duffy blood group antigen** (also known as the erythrocyte receptor for interleukin-8) to infect erythrocytes.
- **G.** The **MN blood group system** is a well-characterized set of erythrocyte surface antigens that represent the variable carbohydrate modifications of the trans-membrane glycoprotein, **glycophorin**. Glycophorin is the cellular receptor for influenza virus as well as the receptor for erythrocyte invasion by the malarial parasite Plasmodium falciparum.
- **H.** Helicobacter pylori is the bacterium responsible for chronic active gastritis and gastric and duodenal ulcers; it is also the causative agent for one of the most common forms of cancer in humans, adenocarcinoma. This bacterium attaches to the **Lewis blood group antigen** on the surfaces of gastric mucous cells.
- **I.** Rabies virus binds to cells through interactions with **neural cell adhesion molecule (N-CAM)**.
- **J.** The receptor for **fibroblast growth factor (FGF)** has been reported to be the portal of entry for human herpes virus Type I. Recent new evidence indicates that the portal of entry for human herpes simplex Type I viruses is **3-O-sulfated heparan sulfate** (Cell 99:13-22, 1999).
- **K.** Human herpesvirus 6 (HHV-6) infection occurs in virtually all persons within the first 2 years of life and persists the entire lifetime. In immunocompromised patients HHV-6 causes opportunistic infections and is the causative agent of *exanthema subitum*. HHV-6 has been linked to multiple sclerosis and to the progression of AIDS. The cellular receptor for HHV-6 is the cell-surface type-I glycoprotein, **CD46** (Cell 99:817-827, 1999).
- **4.** Some glycoproteins are tethered to the membrane by a lipid linkage: the protein is attached to the carbohydrate through phosphatidylethanolamine (PE)

linkage, and the carbohydrate is in turn attached to the membrane via linkage to phosphatidylinositol (PI), which anchors the structure within the membrane. The linkage is called a **glycosylphosphatidylinositol (GPI)** anchor, and proteins that are anchored in this way are termed glypiated proteins. The disease, **paroxysmal nocturnal hemoglobinuria**, results from the loss of the erythrocyte surface glycoprotein, **decay-accelerating factor, (DAF)**. DAF prevents erythrocyte lysis by complement. When this factor is lost from the erythrocyte surface, abnormal hemolysis occurs, with the end result of hemoglobin accumulation in the urine.



### The GPI linkage of the T-cell marker Thy-1

Line represents the outer surface of the membrane

Squiggles represent the lipid portion of the GPI linkage embedded in the membrane

Open circles = mannose, filled diamonds = GalNAc, filled squares = fucose

Filled pentagons = ethanolamine, solid circles with P = phosphates

Other important GPI linked proteins are the enzymes acetylcholinesterase, intestinal and placental alkaline phosphatase and 5'-nucleotidase, the cell adhesion molecule **N-CAM (neural cell adhesion molecule)** and the T-cell markers **Thy-1** and **LFA-3 (lymphocyte function associated antigen-3)**.

- 5. The proper degradation of glycoproteins has medical relevance. Degradation occurs within lysosomes and requires specific lysosomal hydrolases, termed glycosidases. Exoglycosidases remove sugars sequentially from the non-reducing end and exhibit restricted substrate specificities. In contrast, endoglycosidases cleave carbohydrate linkages from within and exhibit broader substrate specificities. Several inherited disorders involving the abnormal storage of glycoprotein degradation products have been identified in humans. These disorders result from defects in the genes encoding specific glycosidases, leading to incomplete degradation and subsequent over-accumulation of partially degraded glycoproteins. As a general class, such disorders are known as **lysosomal storage diseases** and include the diseases known as **mucopolidoses** that result from incomplete degradation of the carbohydrate portions of glycolipids.
- 6. Defects in the proper targeting of glycoproteins to the lysosomes can also lead to clinical complications. Deficiencies in the enzyme responsible for the transfer of GlcNAc-1-P to Man residues (GlcNAc phosphotransferase) in lysosomal enzymes leads to the formation of dense inclusion bodies formation in the fibroblasts. Two disorders related to deficiencies in the targeting of lysosomal enzymes are termed **I-cell disease (mucopolidosis II)** and **pseudo-Hurler polydystrophy (mucopolidosis III, also called mucopolidosis-HI)**. I-cell disease is characterized by severe psychomotor retardation, skeletal abnormalities, coarse facial features, painful restricted joint movement, and early mortality. Pseudo-Hurler polydystrophy is less severe; it progresses more slowly, and afflicted individuals live to adulthood.

### Enzyme Defects in Degradation of Asn-GlcNAc Type Glycoproteins

| Disease                        | Enzyme Deficiency       | Symptoms/Comments                                                                            |
|--------------------------------|-------------------------|----------------------------------------------------------------------------------------------|
| aspartylglycosaminuria         | aspartylglycosaminidase | progressive mental retardation, delayed speech and motor development, coarse facial features |
| $\beta$ -Mannosidosis          | $\beta$ -Mannosidase    | primarily neurological defects, speech impairment                                            |
| $\alpha$ -Mannosidosis         | $\alpha$ -Mannosidase   | mental retardation, dystosis multiplex, hepatosplenomegaly, hearing loss, delayed speech     |
| GM <sub>1</sub> Gangliosidosis | $\beta$ -Galactosidase  | also identified as a glycosphingolipid storage disease                                       |



|                                                                 |                                   |                                                                                                                                  |
|-----------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| GM <sub>2</sub> Gangliosidosis<br>(Sandhoff-Jatzkewitz disease) | β-N-acetylhexosaminidases A and B | also identified as a glycosphingolipid storage disease                                                                           |
| Sialidosis<br>(also identified as Mucopolipidosis I)            | Neuraminidase<br>(sialidase)      | myoclonus, congenital ascites, hepatosplenomegaly, coarse facial features, delayed mental and motor development                  |
| Fucosidosis                                                     | α-Fucosidase                      | progressive motor and mental deterioration, growth retardation, coarse facial features, recurrent sinus and pulmonary infections |

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OMIM links for additional [Defects in Glycoprotein Degradation](#)

## Acylation

Many proteins are modified at their N-termini following synthesis. In most cases the initiator methionine is hydrolyzed and an acetyl group is added to the new N-terminal amino acid. Acetyl-CoA is the acetyl donor for these reactions. Some proteins have the 14 carbon myristoyl group added to their N-termini. The donor for this modification is myristoyl-CoA. This latter modification allows association of the modified protein with membranes. The catalytic subunit of **cyclicAMP-dependent protein kinase (PKA)** is myristoylated.

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## Methylation

Post-translational methylation occurs at lysine residues in some proteins such as calmodulin and cytochrome c. The activated methyl donor is **S-adenosylmethionine**.

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## Phosphorylation

Post-translational phosphorylation is one of the most common protein modifications that occurs in animal cells. The vast majority of phosphorylations occur as a mechanism to regulate the biological activity of a protein and as such are transient. In other words a phosphate (or more than one in many cases) is added and later removed.

Physiologically relevant examples are the phosphorylations that occur in glycogen synthase and glycogen phosphorylase in hepatocytes in response to glucagon release from the pancreas. Phosphorylation of synthase inhibits its activity, whereas, the activity of phosphorylase is increased. These two events lead to increased hepatic glucose delivery to the blood.

The enzymes that phosphorylate proteins are termed **kinases** and those that remove phosphates are termed **phosphatases**. Protein kinases catalyze reactions of the following type:



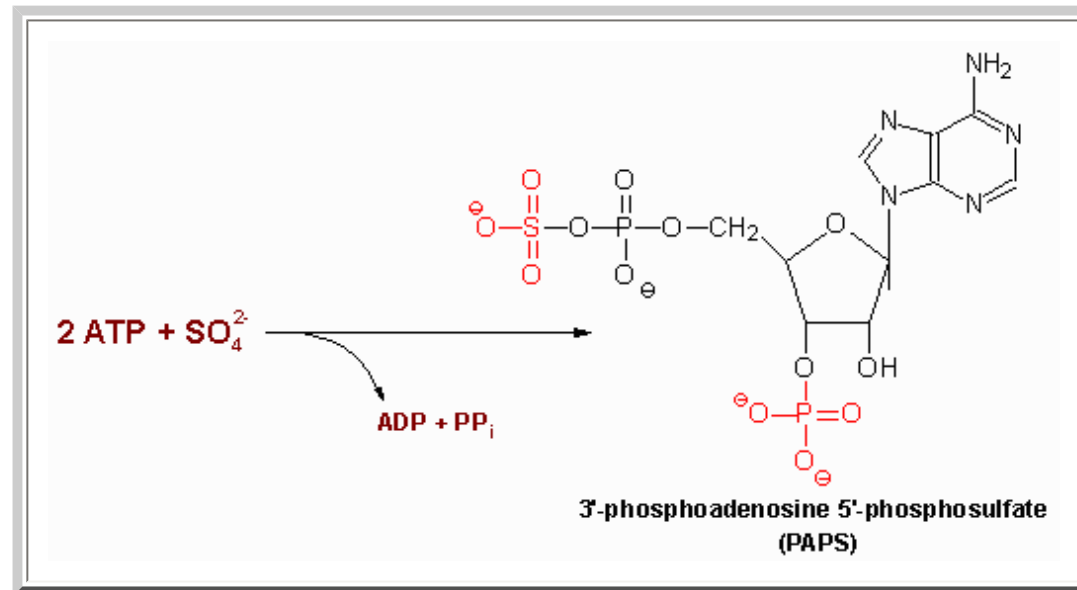
In animal cells serine, threonine and tyrosine are the amino acids subject to phosphorylation. The largest group of kinases are those that phosphorylate either serines or threonines and as such are termed serine/threonine kinases. The ratio of phosphorylation of the three different amino acids is approximately 1000/100/1 for serine/threonine/tyrosine.

Although the level of tyrosine phosphorylation is minor, the importance of phosphorylation of this amino acid is profound. As an example, the activity of numerous growth factor receptors is controlled by tyrosine phosphorylation.

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## Sulfation

Sulfate modification of proteins occurs at tyrosine residues such as in fibrinogen and in some secreted proteins (eg **gastrin**). The universal sulfate donor is **3'-phosphoadenosyl-5'-phosphosulphate (PAPS)**.



Since sulfate is added permanently it is necessary for the biological activity and not used as a regulatory modification like that of tyrosine phosphorylation.

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## Prenylation

Prenylation refers to the addition of the 15 carbon farnesyl group or the 20 carbon geranylgeranyl group to acceptor proteins, both of which are isoprenoid compounds derived from the cholesterol biosynthetic pathway. The isoprenoid groups are attached to cysteine residues at the carboxy terminus of proteins in a thioether linkage (C-S-C). A common consensus sequence at the C-terminus of prenylated proteins has been identified and is composed of **CAAX**, where C is cysteine, A is any aliphatic amino acid (except alanine) and X is the C-terminal amino acid. In order for the prenylation reaction to occur the three C-terminal amino acids (AAX) are first removed and the cysteine activated by methylation in a reaction utilizing S-adenosylmethionine as the methyl donor.

Important examples of prenylated proteins include the oncogenic GTP-binding and hydrolyzing protein **Ras** and the  $\gamma$ -subunit of the visual protein **transducin**, both of which are farnesylated. Numerous GTP-binding and hydrolyzing proteins (termed G-proteins) of **signal transduction** cascades have  $\gamma$ -subunits modified by geranylgeranylation.

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## Vitamin C-Dependent Modifications

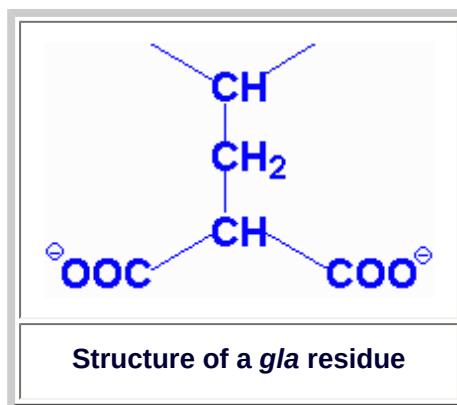
Modifications of proteins that depend upon [vitamin C](#) as a cofactor include proline and lysine hydroxylations and carboxy terminal amidation. The hydroxylating enzymes are identified as prolyl hydroxylase and lysyl hydroxylase. The donor of the amide for C-terminal amidation is glycine.

The most important hydroxylated proteins are the **collagens**. Several peptide hormones such as **oxytocin** and **vasopressin** have C-terminal amidation.

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## Vitamin K-Dependent Modifications

[Vitamin K](#) is a cofactor in the carboxylation of glutamic acid residues. The result of this type of reaction is the formation of a  $\gamma$ -carboxyglutamate (gamma-carboxyglutamate), referred to as a **gla residue**.



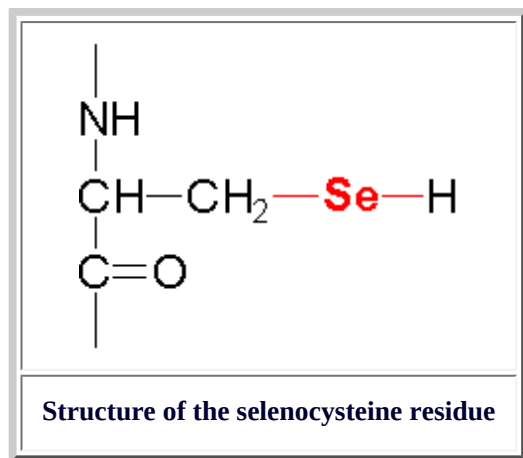
The formation of gla residues within several proteins of the [blood clotting](#) cascade is critical for their normal function. The presence of gla residues allows the protein to chelate calcium ions and thereby render an altered conformation and biological activity to the protein. The coumarin-based anticoagulants, **warfarin** and **dicumarol** function by inhibiting the carboxylation reaction.

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## Selenoproteins

Selenium is a trace element and is found as a component of several prokaryotic and eukaryotic enzymes that are involved in redox reactions. The selenium in these selenoproteins is incorporated as a unique amino acid, **selenocysteine**, during translation. A particularly important eukaryotic selenoenzyme is glutathione peroxidase. This enzyme is required during the oxidation of [glutathione](#) by hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) and organic hydroperoxides.



Incorporation of selenocysteine by the translational machinery occurs via an interesting and unique mechanism. The tRNA for selenocysteine is charged with serine and then enzymatically selenylated to produce the selenocysteinyl-tRNA. The anticodon of selenocysteinyl-tRNA interacts with a stop codon in the mRNA (UGA) instead of a serine codon. The selenocysteinyl-tRNA has a unique structure that is not recognized by the termination machinery and is brought into the ribosome by a dedicated specific elongation factor. An element in the 3' non-translated region (UTR) of selenoprotein mRNAs determines whether UGA is read as a stop codon or as a selenocysteine codon.

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- Platelet Activation and von Willebrand Factor (vWF)
- Description of Clotting Factors
- Image of the Clotting Cascade
- Intrinsic Clotting Cascade
- Extrinsic Clotting Cascade
- Activation of Thrombin
- Regulation of Thrombin Levels
- Activation of Fibrin
- Dissolution of Fibrin Clots
- Clinical Significances of Hemostasis: The Bleeding Disorders
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## Introduction

The ability of the body to control the flow of blood following vascular injury is paramount to continued survival. The process of blood clotting and then the subsequent dissolution of the clot, following repair of the injured tissue, is termed **hemostasis**. Hemostasis, composed of 4 major events that occur in a set order following the loss of vascular integrity:

1. The initial phase of the process is **vascular constriction**. This limits the flow of blood to the area of injury.
2. Next, platelets become activated by **thrombin** and aggregate at the site of injury, forming a temporary, loose **platelet plug**. The protein **fibrinogen** is primarily responsible for stimulating platelet clumping. Platelets clump by binding to collagen that becomes exposed following rupture of the endothelial lining of vessels. Upon activation, platelets release **adenosine-5'-diphosphate, ADP** and **TXA<sub>2</sub>** (which activate additional platelets), **serotonin, phospholipids, lipoproteins**, and other proteins important for the coagulation cascade. In addition to induced secretion, activated platelets change their shape to accommodate the formation of the plug.
3. To insure stability of the initially loose platelet plug, a **fibrin mesh** (also called the **clot**) forms and entraps the plug. If the plug contains only platelets it is termed a **white thrombus**; if red blood cells are present it is called a **red thrombus**.
4. Finally, the clot must be dissolved in order for normal blood flow to resume following tissue repair. The dissolution of the clot occurs through the action of **plasmin**.

Two pathways lead to the formation of a fibrin clot: the intrinsic and extrinsic pathway. Although they are initiated by distinct mechanisms, the two converge on a common pathway that leads to clot formation. The formation of a red thrombus or a clot in response to an abnormal vessel wall in the absence of tissue

injury is the result of the [intrinsic pathway](#). Fibrin clot formation in response to tissue injury is the result of the [extrinsic pathway](#). Both pathways are complex and involve numerous different proteins termed [clotting factors](#).

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## Platelet Activation and von Willebrand Factor (vWF)

In order for hemostasis to occur, platelets must adhere to exposed collagen, release the contents of their granules, and aggregate. The adhesion of platelets to the collagen exposed on endothelial cell surfaces is mediated by **von Willebrand factor (vWF)**. Inherited deficiencies of vWF are the causes of [von Willebrand disease, \(vWD\)](#) (see [below](#) for more details). The function of vWF is to act as a bridge between a specific glycoprotein on the surface of platelets ([GPIb/IX](#)) and collagen fibrils. In addition to its role as a bridge between platelets and exposed collagen on endothelial surfaces, vWF binds to and stabilizes coagulation factor VIII. Binding of factor VIII by vWF is required for normal survival of factor VIII in the circulation.

von Willebrand factor is a complex multimeric glycoprotein that is produced by and stored in the  $\alpha$ -granules of platelets. It is also synthesized by megakaryocytes and found associated with subendothelial connective tissue.

The initial activation of platelets is induced by thrombin binding to specific receptors on the surface of platelets, thereby initiating a signal transduction cascade. The thrombin receptor is coupled to a G-protein that, in turn, activates **phospholipase C- $\gamma$**  (PLC- $\gamma$ ). PLC- $\gamma$  hydrolyzes **phosphatidylinositol-4,5-bisphosphate** (PIP<sub>2</sub>) leading to the formation of **inositol trisphosphate** (IP<sub>3</sub>) and **diacylglycerol** (DAG). IP<sub>3</sub> induces the release of intracellular Ca<sup>2+</sup> stores, and DAG activates protein kinase C (PKC).

The collagen to which platelets adhere as well as the release of intracellular Ca<sup>2+</sup> leads to the activation of phospholipase A<sub>2</sub> (PLA<sub>2</sub>), which then hydrolyzes membrane phospholipids, leading to liberation of **arachidonic acid**. The arachidonic acid release leads to an increase in the production and subsequent release of **thromboxane A<sub>2</sub>** (TXA<sub>2</sub>). This is another platelet activator that functions through the PLC- $\gamma$  pathway. Another enzyme activated by the released intracellular Ca<sup>2+</sup> stores is myosin light chain kinase (MLCK). Activated MLCK phosphorylates the light chain of myosin which then interacts with actin, resulting in altered platelet morphology and motility.

One of the many effects of PKC is the phosphorylation and activation of a specific 47,000-Dalton platelet protein. This activated protein induces the release of platelet granule contents; one of which is **ADP**. ADP further stimulates platelets increasing the overall activation cascade; it also modifies the platelet membrane in such a way as to allow **fibrinogen** to adhere to two platelet surface glycoproteins, [GPIIb](#) and [GPIIIa](#), resulting in fibrinogen-induced **platelet aggregation**.

Activation of platelets is required for their consequent aggregation to a platelet plug. However, equally significant is the role of activated platelet surface phospholipids in the activation of the coagulation cascade.

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## Primary Factors

| Factor        | Trivial Name(s) | Pathway   | Characteristic |
|---------------|-----------------|-----------|----------------|
| Prekallikrein | Fletcher factor | Intrinsic |                |



|                                        |                                                                                  |           |                                        |
|----------------------------------------|----------------------------------------------------------------------------------|-----------|----------------------------------------|
| High molecular weight kininogen (HMWK) | contact activation cofactor; Fitzgerald, Flaujeac Williams factor                | Intrinsic |                                        |
| I                                      | Fibrinogen                                                                       | Both      |                                        |
| II                                     | Prothrombin                                                                      | Both      | Contains N-term. <i>gla</i> segment    |
| III                                    | Tissue Factor                                                                    | Extrinsic |                                        |
| IV                                     | Calcium                                                                          | Both      |                                        |
| V                                      | Proaccelerin, labile factor, accelerator (Ac-) globulin                          | Both      | Protein cofactor                       |
| VI (Va)                                | Accelerin                                                                        |           | This is Va, redundant to Factor V      |
| VII                                    | Proconvertin, serum prothrombin conversion accelerator (SPCA), cothromboplastin  | Extrinsic | Endopeptidase with <i>gla</i> residues |
| VIII                                   | Antihemophiliac factor A, antihemophilic globulin (AHG)                          | Intrinsic | Protein cofactor                       |
| IX                                     | Christmas Factor, antihemophilic factor B, plasma thromboplastin component (PTC) | Intrinsic | Endopeptidase with <i>gla</i> residues |
| X                                      | Stuart-Prower Factor                                                             | Both      | Endopeptidase with <i>gla</i> residues |
| XI                                     | Plasma thromboplastin antecedent (PTA)                                           | Intrinsic | Endopeptidase                          |
| XII                                    | Hageman Factor                                                                   | Intrinsic | Endopeptidase                          |
| XIII                                   | Protransglutaminase, fibrin stabilizing factor (FSF), fibrinoligase              | Both      | Transpeptidase                         |

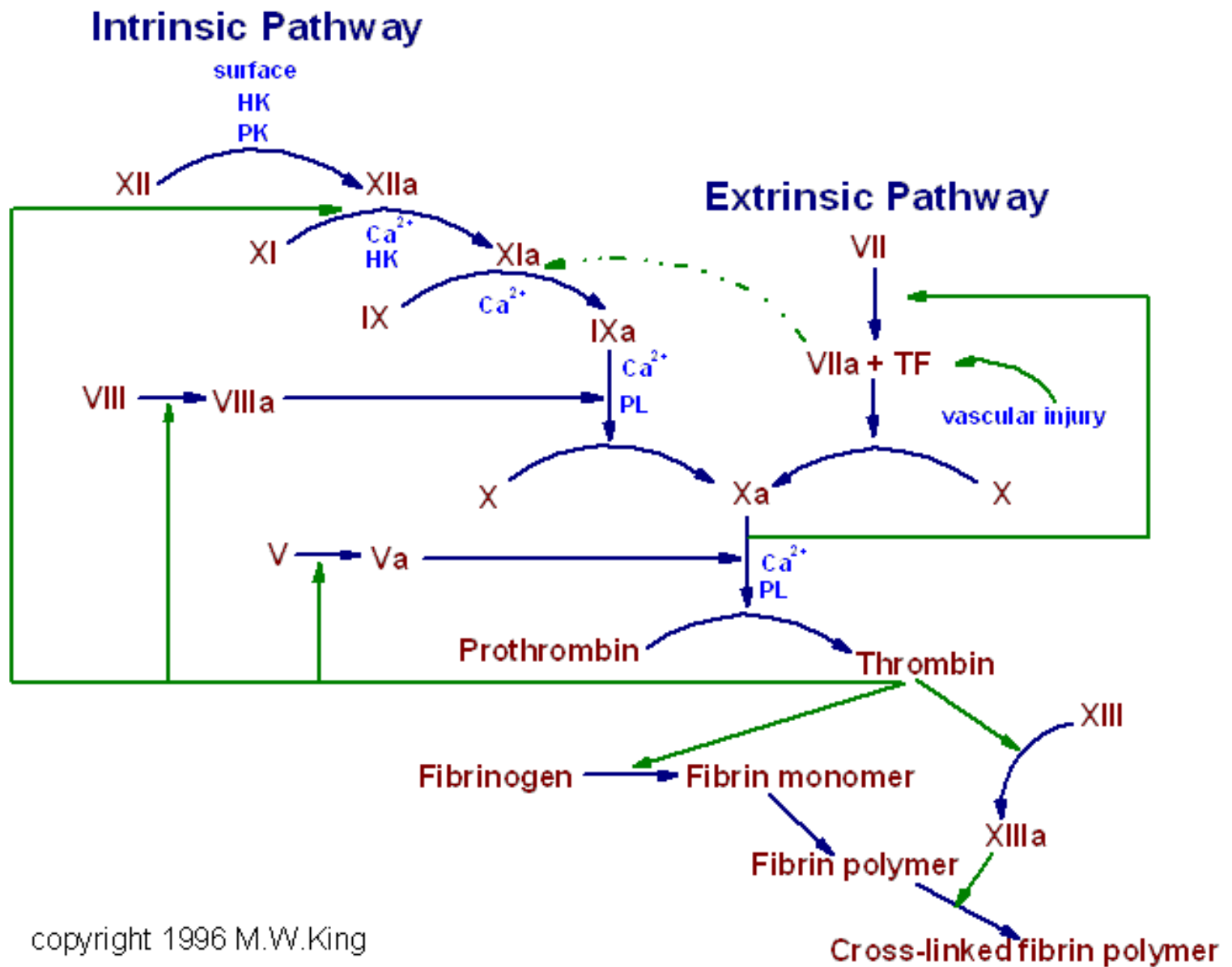
## Functional Classification of Clotting Factors

| Zymogens of Serine Proteases | Activities                                                                                             |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| Factor XII                   | binds to exposed collagen at site of vessel wall injury, activated by high-MW kininogen and kallikrein |

|                                      |                                                                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Factor XI                            | activated by factor XIIa                                                                                                           |
| Factor IX                            | activated by factor XIa in presence of $\text{Ca}^{2+}$                                                                            |
| Factor VII                           | activated by thrombin in presence of $\text{Ca}^{2+}$                                                                              |
| Factor X                             | activated on surface of activated platelets by tenase complex and by factor VIIa in presence of tissue factor and $\text{Ca}^{2+}$ |
| Factor II                            | activated on surface of activated platelets by prothrombinase complex                                                              |
| <b>Cofactors</b>                     | <b>Activities</b>                                                                                                                  |
| Factor VIII                          | activated by thrombin; factor VIIIa is a cofactor in the activation of factor X by factor IXa                                      |
| Factor V                             | activated by thrombin; factor Va is a cofactor in the activation of prothrombin by factor Xa                                       |
| Factor III (tissue factor)           | a subendothelial cell-surface glycoprotein that acts as a cofactor for factor VII                                                  |
| <b>Fibrinogen</b>                    | <b>Activity</b>                                                                                                                    |
| Factor I                             | cleaved by thrombin to form fibrin clot                                                                                            |
| <b>Transglutaminase</b>              | <b>Activity</b>                                                                                                                    |
| Factor XIII                          | activated by thrombin in presence of $\text{Ca}^{2+}$ ; stabilizes fibrin clot by covalent cross-linking                           |
| <b>Regulatory and other proteins</b> | <b>Activities</b>                                                                                                                  |
| von Willebrand factor                | associated with subendothelial connective tissue; serves as a bridge between platelet glycoprotein GPIb/IX and collagen            |
| Protein C                            | activated to protein Ca by thrombin bound to thrombomodulin; then degrades factors VIIIa and Va                                    |
| Protein S                            | acts as a cofactor of protein C; both proteins contain <i>gla</i> residues                                                         |
| Thrombomodulin                       | protein on the surface of endothelial cells; binds thrombin, which then activates protein C                                        |
| Antithrombin III                     | most important coagulation inhibitor, controls activities of thrombin, and factors IXa, Xa, XIa and XIIa                           |

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## The Clotting Cascades



The clotting cascades: The intrinsic cascade is initiated when contact is made between blood and exposed endothelial cell surfaces. The extrinsic pathway is initiated upon vascular injury which leads to exposure of **tissue factor (TF)** (also identified as **factor III**), a subendothelial cell-surface glycoprotein that binds phospholipid. The green dotted arrow represents a point of cross-over between the extrinsic and intrinsic pathways. The two pathways converge at the activation of factor X to Xa. Factor Xa has a role in the further activation of factor VII to VIIa as depicted by the green arrow. Active factor Xa hydrolyzes and activates prothrombin to thrombin. Thrombin can then activate factors XI, VIII and V furthering the cascade. Ultimately the role of thrombin is to convert fibrinogen to fibrin and to activate factor XIII to XIIIa. Factor XIIIa (also termed **transglutaminase**) cross-links fibrin polymers solidifying the clot. HK = high molecular weight kininogen. PK = prekallikrein. PL = phospholipid.

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## The Intrinsic Clotting Cascade

The intrinsic pathway requires the clotting factors VIII, IX, X, XI, and XII. Also required are the proteins prekallikrein and high-molecular-weight kininogen, as well as calcium ions and phospholipids secreted from platelets. Each of these pathway constituents leads to the conversion of factor X (inactive) to factor Xa (a signifies active). Initiation of the intrinsic pathway occurs when prekallikrein, high-molecular-weight kininogen, factor XI and factor XII are exposed to a negatively charged surface. This is termed the **contact phase**. Exposure of collagen to a vessel surface is the primary stimulus for the contact phase.

The assemblage of contact phase components results in conversion of prekallikrein to **kallikrein**, which in turn activates factor XII to **factor XIIa**. Factor XIIa can then hydrolyze more prekallikrein to kallikrein, establishing a reciprocal activation cascade. Factor XIIa also activates factor XI to **factor XIa** and leads to the release of **bradykinin**, a potent vasodilator, from high-molecular-weight kininogen.

In the presence of  $\text{Ca}^{2+}$ , factor XIa activates factor IX to **factor IXa**. Factor IX is a proenzyme that contains **vitamin K-dependent  $\gamma$ -carboxyglutamate (gla)** residues, whose serine protease activity is activated following  $\text{Ca}^{2+}$  binding to these gla residues. Several of the serine proteases of the cascade (II, VII, IX, and X) are gla-containing proenzymes. Active factor IXa cleaves factor X at an internal arg-ile bond leading to its activation to **factor Xa**.

The activation of factor Xa requires assemblage of the **tenase complex** ( $\text{Ca}^{2+}$  and factors VIIIa, IXa and X) on the surface of activated platelets. One of the responses of platelets to activation is the presentation of phosphatidylserine and phosphatidylinositol on their surfaces. The exposure of these phospholipids allows the tenase complex to form. The role of factor VIII in this process is to act as a receptor, in the form of factor VIIIa, for factors IXa and X. Factor VIIIa is termed a **cofactor** in the clotting cascade. The activation of factor VIII to **factor VIIIa** (the actual receptor) occurs in the presence of minute quantities of thrombin. As the concentration of thrombin increases, factor VIIIa is ultimately cleaved by thrombin and inactivated. This dual action of thrombin, upon factor VIII, acts to limit the extent of tenase complex formation and thus the extent of the coagulation cascade.

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## Extrinsic Clotting Cascade

Activated factor Xa is the site at which the intrinsic and extrinsic coagulation cascades converge. The extrinsic pathway is initiated at the site of injury in response to the release of **tissue factor** (factor III). Tissue factor is a **cofactor** in the factor VIIa-catalyzed activation of factor X. Factor VIIa, a gla residue containing serine protease, cleaves factor X to factor Xa in a manner identical to that of factor IXa of the intrinsic pathway. The activation of factor VII occurs through the action of thrombin or factor Xa. The ability of factor Xa to activate factor VII creates a link between the intrinsic and extrinsic pathways. An additional link between the two pathways exists through the ability of tissue factor and factor VIIa to activate factor IX. The formation of complex between factor VIIa and tissue factor is believed to be a principal step in the overall clotting cascade. Evidence for this stems from the fact that persons with hereditary deficiencies in the components of the **contact phase** of the intrinsic pathway do not exhibit clotting problems. A major mechanism for the inhibition of the extrinsic pathway occurs at the tissue factor--factor VIIa-- $\text{Ca}^{2+}$ --Xa complex. The protein, **lipoprotein-associated coagulation inhibitor, LACI** specifically binds to this complex. LACI is also referred to as extrinsic pathway inhibitor, EPI or tissue factor pathway inhibitor, TFPI and was formerly named anticonvertin. LACI is composed of 3 tandem protease inhibitor domains. Domain 1 binds to factor Xa and domain 2 binds to factor VIIa only in the presence of factor Xa.

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## Activation of Prothrombin to Thrombin

The common point in both pathways is the activation of factor X to factor Xa. Factor Xa activates **prothrombin** (factor II) to **thrombin** (factor IIa). Thrombin, in turn, converts fibrinogen to fibrin. The activation of thrombin occurs on the surface of activated platelets and requires formation of a **prothrombinase complex**. This complex is composed of the platelet phospholipids, phosphatidylinositol and phosphatidylserine,  $\text{Ca}^{2+}$ , factors Va and Xa, and prothrombin. Factor V is a **cofactor** in the formation of the prothrombinase complex, similar to the role of factor VIII in tenase complex formation. Like factor VIII activation, factor V is activated to factor Va by means of minute amounts and is inactivated by increased levels of thrombin. Factor Va binds to specific receptors on the surfaces of activated platelets and forms a complex with prothrombin and factor Xa.

Prothrombin is a 72,000-Dalton, single-chain protein containing ten gla residues in its N-terminal region. Within the prothrombinase complex, prothrombin is cleaved at 2 sites by factor Xa. This cleavage generates a 2-chain active thrombin molecule containing an A and a B chain which are held together by a single disulfide bond.

In addition to its role in activation of fibrin clot formation, thrombin plays an important regulatory role in coagulation. Thrombin combines with **thrombomodulin** present on endothelial cell surfaces forming a complex that converts protein C to **protein Ca**. The cofactor **protein S** and protein Ca degrade factors Va and VIIIa, thereby limiting the activity of these 2 factors in the coagulation cascade.

Thrombin also binds to and leads to the release of G-protein-coupled protease activated receptors (PARs), specifically PAR-1, -3 and -4. The release of these proteins leads to the activation of numerous signaling cascades that in turn increase release of the **interleukins**, **ILs**, IL-1 and IL-6, increases secretion of intercellular adhesion molecule-1 (ICAM-1) and vascular cell adhesion molecule-1 (VCAM-1). The thrombin-induced signaling also leads to increased platelet activation and leukocyte adhesion. Thrombin also activates thrombin-activatable fibrinolysis inhibitor (TAFI) thus modulating fibrinolysis (degradation of fibrin clots). TAFI is also known as carboxypeptidase U (CPU) whose activity leads to removal of C-terminal lysines from partially degraded fibrin. This leads to an impairment of plasminogen activation, thereby reducing the rate of fibrin clot dissolution (i.e. fibrinolysis).

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## Control of Thrombin Levels

The inability of the body to control the circulating level of active thrombin would lead to dire consequences. There are 2 principal mechanisms by which thrombin activity is regulated. The predominant form of thrombin in the circulation is the inactive prothrombin, whose activation requires the pathways of proenzyme activation described above for the coagulation cascade. At each step in the cascade, feedback mechanisms regulate the balance between active and inactive enzymes.

The activation of thrombin is also regulated by 4 specific thrombin inhibitors. **Antithrombin III** is the most important since it can also inhibit the activities of factors IXa, Xa, XIa and XIIa. The activity of antithrombin III is potentiated in the presence of heparin by the following means: heparin binds to a specific site on antithrombin III, producing an altered conformation of the protein, and the new conformation has a higher affinity for thrombin as well as its other substrates. This effect of heparin is the basis for its clinical use as an anticoagulant. The naturally occurring heparin activator of antithrombin III is present as heparan and heparan sulfate on the surface of vessel endothelial cells. It is this feature that controls the activation of the intrinsic coagulation cascade.

However, thrombin activity is also inhibited by  **$\alpha_2$ -macroglobulin**, **heparin cofactor II** and  **$\alpha_1$ -antitrypsin**. Although a minor player in thrombin regulation  $\alpha_1$ -antitrypsin is the primary serine protease inhibitor of human plasma. Its physiological significance is demonstrated by the fact that lack of this protein plays a causative role in the development of **emphysema**.

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## Activation of Fibrinogen to Fibrin

Fibrinogen (factor I) consists of 3 pairs of polypeptides  $([A-\alpha][B-\beta][\gamma])_2$ . The 6 chains are covalently linked near their N-terminals through disulfide bonds. The A and B portions of the A- $\alpha$  and B- $\beta$  chains comprise the **fibrinopeptides**, A and B, respectively. The fibrinopeptide regions of fibrinogen contain several glutamate and aspartate residues imparting a high negative charge to this region and aid in the solubility of fibrinogen in plasma. Active thrombin is a serine protease that hydrolyses fibrinogen at four **arg-gly** bonds between the fibrinopeptide and the **a** and **b** portions of the protein.

Thrombin-mediated release of the fibrinopeptides generates fibrin monomers with a subunit structure  $(\alpha-\beta-\gamma)_2$ . These monomers spontaneously aggregate in a regular array, forming a somewhat weak fibrin clot. In addition to fibrin activation, thrombin converts factor XIII to **factor XIIIa**, a highly specific transglutaminase that introduces cross-links composed of covalent bonds between the amide nitrogen of glutamines and  $\epsilon$ -amino group of lysines in the fibrin monomers.

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## Dissolution of Fibrin Clots

Degradation of fibrin clots is the function of **plasmin**, a serine protease that circulates as the inactive proenzyme, **plasminogen**. Any free circulating plasmin is rapidly inhibited by  **$\alpha_2$ -antiplasmin**. Plasminogen binds to both fibrinogen and fibrin, thereby being incorporated into a clot as it is formed. **Tissue plasminogen activator (tPA)** and, to a lesser degree, urokinase are serine proteases which convert plasminogen to plasmin. Inactive tPA is released from vascular endothelial cells following injury; it binds to fibrin and is consequently activated. Urokinase is produced as the precursor, prourokinase by epithelial cells lining excretory ducts. The role of urokinase is to activate the dissolution of fibrin clots that may be deposited in these ducts.

Active tPA cleaves plasminogen to plasmin which then digests the fibrin; the result is soluble degradation product to which neither plasmin nor plasminogen can bind. Following the release of plasminogen and plasmin they are rapidly inactivated by their respective inhibitors. The inhibition of tPA activity results from binding to specific inhibitory proteins. At least 4 distinct inhibitors have been identified, of which 2-- **plasminogen activator-inhibitors type 1 (PAI-1)** and **type 2 (PAI-2)** are of greatest physiological significance.

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## Clinical Significances of Hemostasis: The Bleeding Disorders

Defects in the process of hemostasis, leading to bleeding disorders, have been identified at the level of the proteins of the clotting cascades, platelet activation and function, contact activation and antithrombin function.

### Hemophilia A

Hemophilia A is classic hemophilia (a disease referring to the inability to clot blood). It is an X-linked



disorder resulting from a deficiency in factor VIII, a key component of the coagulation cascade. There are severe, moderate and mild forms of hemophilia A that reflect the level of active factor VIII in the plasma.

Hemophilia A arises from a variety of mutations. Some 150 different point mutations have been characterized in the factor VIII gene in hemophilia A. Inheritance of the disorder occurs with a frequency of 1:5,000 to 1:10,000 males in all populations. Factor VIII is a cofactor in the activation of factor X to factor Xa in a reaction catalyzed by factor IXa. Activation of factor VIII occurs via proteolytic cleavage by thrombin and factor Xa. Inactivation of factor VIIIa occurs by limited proteolysis by factor Xa or activated protein C.

Individuals with deficiencies in factor VIII suffer joint and muscle hemorrhage, easy bruising and prolonged bleeding from wounds. Treatment of hemophilia A is accomplished by infusion of factor VIII concentrates prepared from either human plasma or by recombinant DNA technology.

## Hemophilia B

Hemophilia B results from deficiencies in factor IX. The prevalence of hemophilia B is approximately one-tenth that of hemophilia A. All patients with hemophilia B have prolonged coagulation time and decreased factor IX clotting activity. Like hemophilia A, there are severe, moderate and mild forms of hemophilia B and reflect the factor IX activity in plasma.

At least 300 unique factor IX mutations have been identified, 85% are point mutations, 3% are short nucleotide deletions or insertions and 12% are gross gene alterations.

## Disorders of Fibrinogen and Factor XIII

Several cardiovascular risk factors are associated with abnormalities in fibrinogen. As a result of the acute-phase response or through other poorly understood mechanisms, elevated plasma fibrinogen levels have been observed in patients with coronary artery disease, diabetes, hypertension, peripheral artery disease, hyperlipoproteinemia and hypertriglyceridemia. In addition, pregnancy, menopause, hypercholesterolemia, use of oral contraceptives and smoking lead to increased plasma fibrinogen levels.

Although rare, there are inherited disorders in fibrinogen. These disorders include afibrinogenemia (a complete lack of fibrinogen), hypofibrinogenemia (reduced levels of fibrinogen) and dysfibrinogenemia (presence of dysfunctional fibrinogen). Afibrinogenemia is characterized by neonatal umbilical cord hemorrhage, ecchymoses, mucosal hemorrhage, internal hemorrhage, and recurrent abortion. The disorder is inherited in an autosomal recessive manner. Hypofibrinogenemia is characterized by fibrinogen levels below 100mg/dL (normal is 250-350mg/dL) and can be either acquired or inherited. Symptoms of hypofibrinogenemia are similar to, but less severe than, afibrinogenemia. Dysfibrinogenemias are extremely heterogeneous affecting any of the functional properties of fibrinogen. Clinical consequences of dysfibrinogenemias include hemorrhage, spontaneous abortion and thromboembolism.

Factor XIII is the proenzyme form of plasma transglutaminase and is activated by thrombin in the presence of calcium ions. Active factor XIII catalyzes the cross-linking of fibrin monomers. Factor XIII is a tetramer of two two different peptides, a and b (forming  $a_2b_2$ ). Hereditary deficiencies (autosomal recessive) occur resulting in the absence of either subunit. Clinical manifestation of factor XIII deficiency is delayed bleeding although primary hemostasis is normal. Deficiency leads to neonatal umbilical cord bleeding, intracranial hemorrhage and soft tissue hematomas.

## von Willebrand Disease

von Willebrand disease (vWD) is due to inherited deficiency in von Willebrand factor (vWF). vWD is the most common inherited bleeding disorder of humans. Using sensitive laboratory testing, abnormalities

in vWF can be detected in approximately 8000 people per million. Clinically significant vWD occurs in approximately 125 people per million. This is a frequency at least twice that of hemophilia A.

Deficiency of vWF results in defective platelet adhesion and causes a secondary deficiency in factor VIII. The result is that vWF deficiency can cause bleeding that appears similar to that caused by platelet dysfunction or hemophilia. vWD is an extremely heterogeneous disorder that has been classified into several major subtypes. Type I vWD is the most common and is inherited as an autosomal dominant trait. This variant is due to simple quantitative deficiency of all vWF multimers. Type 2 vWD is also subdivided further dependent upon whether the dysfunctional protein has decreased or paradoxically increased function in certain laboratory tests of binding to platelets. Type 3 vWD is clinically severe and is characterized by recessive inheritance and virtual absence of vWF.

## Factor XI and Contact Activation

When blood makes contact with negatively charged surfaces it triggers a series of interactions that involve factor XI, prekallikrein and high molecular weight kininogen leading to blood coagulation. This process is referred to as **contact activation**. Deficiency in factor XI confers an injury-related bleeding tendency. This deficiency was identified in 1953 and originally termed hemophilia C. Factor XI deficiency is very common in Ashkenazic Jews and is inherited as an autosomal disorder with either homozygosity or compound heterozygosity. Three independent point mutations in factor XI have been identified.

## Antithrombin Deficiency

Antithrombin functions to inhibit several activated coagulation factors including thrombin, factor IXa and factor Xa, by forming a stable complex with the various factors.. Heparin and heparan sulfates increase the activity of antithrombin at least 1000 fold.

Deficiency in antithrombin is seen in approximately 2% of patients with venous thromboembolic disease. Inheritance occurs as an autosomal dominant trait. The prevalence of symptomatic antithrombin deficiency ranges from 1 per 2000 to 1 per 5000 in the general population. Deficiencies results from mutations that affect synthesis or stability of antithrombin or from mutations that affect the protease and/or heparin binding sites of antithrombin.

Clinical manifestations of antithrombin deficiency include deep vein thrombosis and pulmonary embolism. Arterial thrombosis is rare in antithrombin deficiency. Thrombosis may occur spontaneously or in association with surgery, trauma or pregnancy. Treatment of acute episodes of thrombosis is by infusion of heparin (for 5-7 days) followed by oral anticoagulant therapy.

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## Pharmacological Intervention in Bleeding

Coumarin drugs, such as **warfarin** as well as the **glycosaminoglycans**, **heparin** and **heparan sulfate**, are useful as anticoagulants. Heparin is useful as an anticoagulant because it binds to, and activates, antithrombin III which then inhibits the serine proteases of the coagulation cascade. Heparin is abundant in granules of the mast cells that line the vasculature. In response to injury, the heparin is released and inhibits coagulation. The coumarin drugs inhibit coagulation by inhibiting the vitamin K-dependent  $\gamma$ -carboxylation reactions necessary to the function of thrombin, and factors VII, IX, and X as well as proteins C and S. These drugs act by inhibiting the reduction of the quinone derivatives of vitamin K to their active hydroquinone forms. Because of the mode of action of coumarin drugs, it takes several days for their maximum effect to be realized. For this reason, heparin is normally administered first followed by

warfarin or warfarin-related drugs.

The plasminogen activators also are useful for controlling coagulation. Because tPA is highly selective for the degradation of fibrin in clots, it is extremely useful in restoring the patency of the coronary arteries following thrombosis, in particular during the short period following myocardial infarct. Streptokinase (an enzyme from the *Streptococci* bacterium) is another plasminogen activator useful from a therapeutic standpoint. However, it is less selective than tPA, being able to activate circulating plasminogen as well as that bound to a fibrin clot.

Aspirin is an important inhibitor of platelet activation. By virtue of inhibiting the activity of cyclooxygenase, aspirin reduces the production of TXA<sub>2</sub>. Aspirin also reduces endothelial cell production of prostacyclin (PGI<sub>2</sub>), an inhibitor of platelet aggregation and a vasodilator. Localized to the site of coagulation is a balance between the levels of platelet derived TXA<sub>2</sub> and endothelial cell derived PGI<sub>2</sub>. This allows for platelet aggregation and clot formation but preventing excessive accumulation of the clot, thus maintaining blood flow around the site of the clot. Endothelial cells regenerate active cyclooxygenase faster than platelets because mature platelets cannot synthesize the enzyme, requiring new platelets to enter the circulation (platelet half-life is approximately 4 days). Therefore, PGI<sub>2</sub> synthesis is greater than that of TXA<sub>2</sub>. The net effect of aspirin is more in favor of endothelial cell-mediated inhibition of the coagulation cascade. This reflects the cardiovascular benefits to low dose administration of aspirin.

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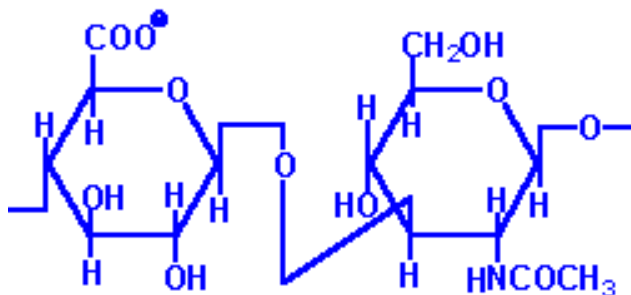
- Glycosaminoglycans
- Characteristics of GAGs
- Proteoglycans
- Clinical Significances
- Mucopolysaccharidoses

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## Glycosaminoglycans

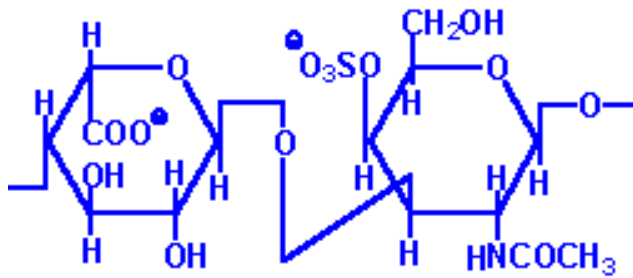
The most abundant heteropolysaccharides in the body are the **glycosaminoglycans (GAGs)**. These molecules are long unbranched polysaccharides containing a repeating disaccharide unit. The disaccharide units contain either of two modified sugars--- **N-acetylgalactosamine (GalNAc)** or **N-acetylglucosamine (GlcNAc)** and a uronic acid such as **glucuronate** or **iduronate**. GAGs are highly negatively charged molecules, with extended conformation that imparts high viscosity to the solution. GAGs are located primarily on the surface of cells or in the extracellular matrix (ECM). Along with the high viscosity of GAGs comes low compressibility, which makes these molecules ideal for a lubricating fluid in the joints. At the same time, their rigidity provides structural integrity to cells and provides passageways between cells, allowing for cell migration. The specific GAGs of physiological significance are **hyaluronic acid**, **dermatan sulfate**, **chondroitin sulfate**, **heparin**, **heparan sulfate**, and **keratan sulfate**. Although each of these GAGs has a predominant disaccharide component (see Table below), heterogeneity does exist in the sugars present in the make-up of any given class of GAG.

Hyaluronic is unique among the GAGs in that it does not contain any sulfate and is not found covalently attached to proteins as a proteoglycan. It is, however, a component of non-covalently formed complexes with proteoglycans in the ECM. Hyaluronic acid polymers are very large (with molecular weights of 100,000 - 10,000,000) and can displace a large volume of water. This property makes them excellent lubricators and shock absorbers.



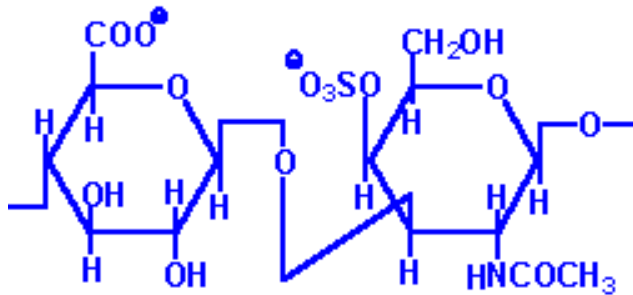
**Hyaluronates:**  
composed of **D-glucuronate** +  
**GlcNAc**

linkage is  $\beta(1, 3)$



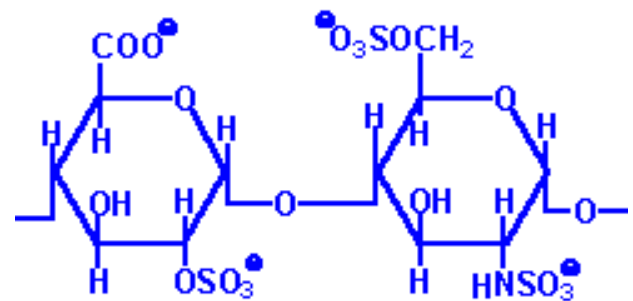
**Dermatan sulfates:**  
composed of L-iduronate  
(many are sulfated) +  
GalNAc-4-sulfate

linkages is  $\beta(1, 3)$



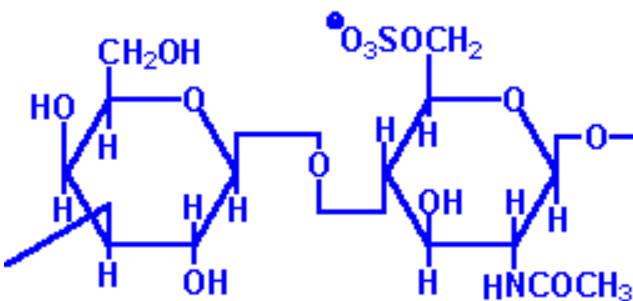
**Chondroitin 4- and 6-sulfates :**  
composed of D-glucuronate +  
GalNAc-4- or 6-sulfate

linkage is  $\beta(1, 3)$   
(the figure contains GalNAc 4-sulfate)



**Heparin and Heparan sulfates:**  
composed of D-glucuronate-  
2-sulfate (or iduronate-2-sulfate) + N-sulfo-D-glucosamine-6-sulfate

linkage is  $\alpha(1, 4)$   
(heparans have less sulfate than heparins)



**Keratan sulfates:**  
composed of galactose +  
GlcNAc-6-sulfate

linkage is  $\beta(1, 4)$

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## Characteristics of GAGs

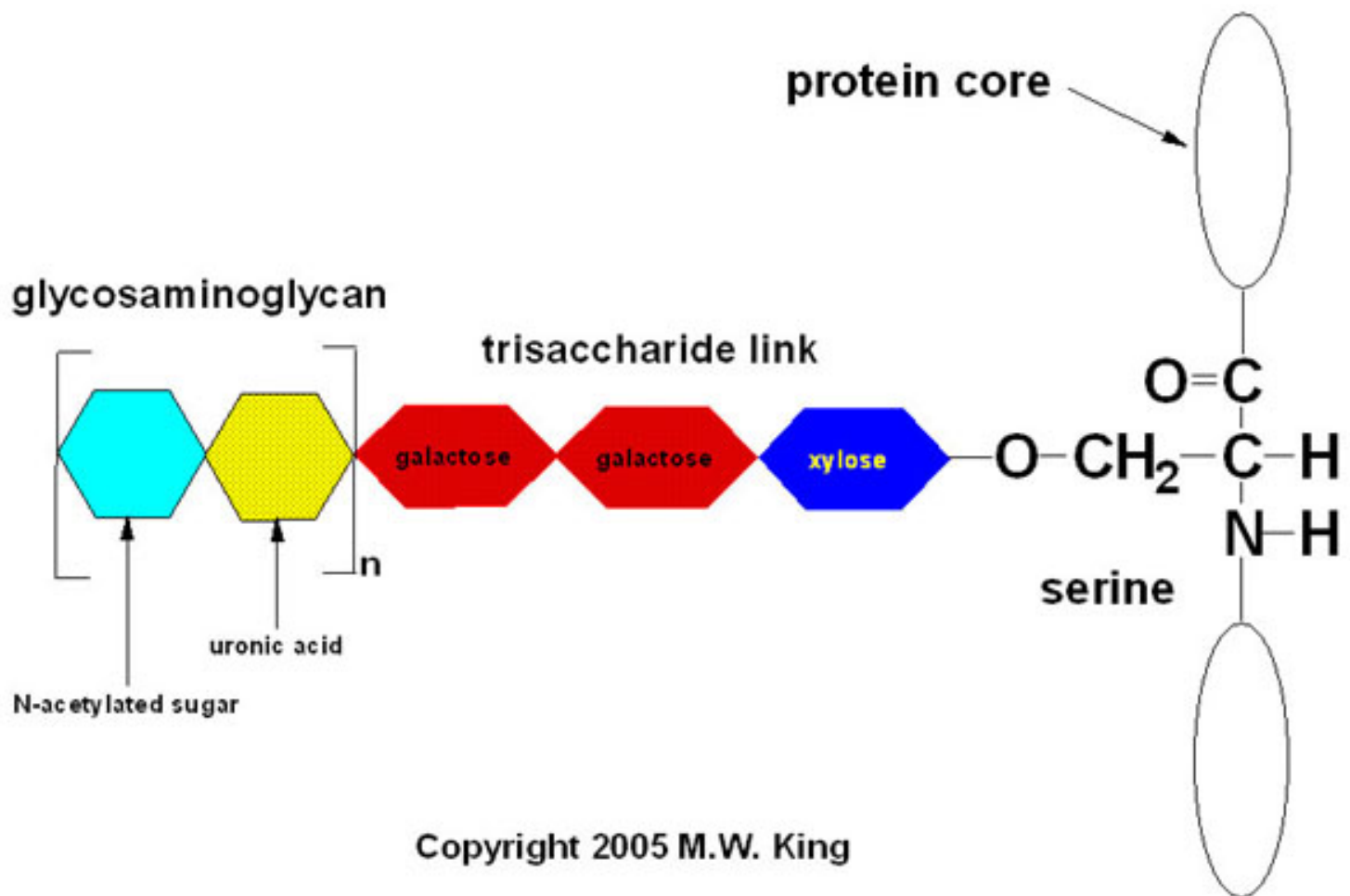
| GAG                 | Localization                                                                                       | Comments                                            |
|---------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Hyaluronate         | synovial fluid, vitreous humor,<br>ECM of loose connective tissue                                  | large polymers, shock absorbing                     |
| Chondroitin sulfate | cartilage, bone, heart valves                                                                      | most abundant GAG                                   |
| Heparan sulfate     | basement membranes, components of cell surfaces                                                    | contains higher acetylated glucosamine than heparin |
| Heparin             | component of intracellular granules of mast cells lining the arteries of the lungs, liver and skin | more sulfated than heparan sulfates                 |
| Dermatan sulfate    | skin, blood vessels, heart valves                                                                  |                                                     |
| Keratan sulfate     | cornea, bone, cartilage aggregated with chondroitin sulfates                                       |                                                     |

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## Proteoglycans

The majority of GAGs in the body are linked to core proteins, forming **proteoglycans** (also called **mucopolysaccharides**). The GAGs extend perpendicularly from the core in a brush-like structure. The linkage of GAGs to the protein core involves a specific trisaccharide composed of two galactose residues and a xylulose residue (**GAG-GalGalXyl-O-CH<sub>2</sub>-protein**). The trisaccharide linker is coupled to the protein core through an O-glycosidic bond to a S residue in the protein. Some forms of keratan sulfates are linked to the protein core through an N-asparaginyl bond. The protein cores of proteoglycans are rich in S and T residues, which allows multiple GAG attachments.





**Structure of the GAG linkage to protein in proteoglycans**

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## Clinical Significance

Proteoglycans and GAGs perform numerous vital functions within the body, some of which still remain to be studied. One well-defined function of the GAG heparin is its role in preventing coagulation of the blood. Heparin is abundant in granules of mast cells that line blood vessels. The release of heparin from these granules, in response to injury, and its subsequent entry into the serum leads to an inhibition of [blood clotting](#), in the following manner. Free heparin complexes with and activates antithrombin III, which in turn inhibits all the serine proteases of the coagulation cascade. This phenomenon has been clinically exploited in the use of heparin injection for anti-coagulation therapies.

Several genetically inherited diseases, for example the **lysosomal storage diseases**, result from defects in the lysosomal enzymes responsible for the metabolism of complex membrane-associated GAGs. These specific diseases, termed **mucopolysaccharidoses (MPS)** (in reference to

the earlier term, mucopolysaccharide, used for glycosaminoglycans) lead to an accumulation of GAGs within cells. There are at least 14 known types of lysosomal storage diseases that affect GAG catabolism; some of the more commonly encountered examples are [Hurler's syndrome](#), [Hunter's syndrome](#), [Sanfilippo syndrome](#), [Maroteaux-Lamy syndrome](#) and [Morquio's syndrome](#). All of these disorders, excepting Hunter's syndrome, are inherited in an autosomal recessive manner.

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## Diseases of Glycosaminoglycan (Mucopolysaccharide) Metabolism

| Type:<br>Syndrome          | Enzyme Defect               | Affected<br>GAG                            | Symptoms                                                                                                                                                                                                                                            |
|----------------------------|-----------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MPS I H:<br>Hurler's       | $\alpha$ -L-iduronidase     | dermatan<br>sulfate,<br>heparan<br>sulfate | corneal clouding,<br>dystosis multiplex,<br>organomegaly, heart<br>disease, dwarfism,<br>mental retardation;<br>early mortality                                                                                                                     |
| MPS I S:<br>Scheie's       | $\alpha$ -L-iduronidase     | dermatan<br>sulfate,<br>heparan<br>sulfate | corneal clouding;<br>aortic valve disease;<br>joint stiffening; normal<br>intelligence and life<br>span                                                                                                                                             |
| MPS I H/S:<br>Huler/Scheie | $\alpha$ -L-iduronidase     | dermatan<br>sulfate,<br>heparan<br>sulfate | intermediate between<br>I H and I S                                                                                                                                                                                                                 |
| MPS II:<br>Hunter's        | L-iduronate-2-<br>sulfatase | dermatan<br>sulfate,<br>heparan<br>sulfate | mild and severe<br>forms, only X-linked<br>MPS, dystosis<br>multiplex,<br>organomegaly, facial<br>and physical<br>deformities, no<br>corneal clouding,<br>mental retardation,<br>death before 15<br>except in mild form<br>then survival to 20 - 60 |

|                                        |                                                                              |                                                               |                                                                                                                                               |
|----------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MPS III A:</b><br>Sanfilippo A      | Heparan <i>N</i> -sulfatase                                                  | heparan sulfate                                               | profound mental deterioration, hyperactivity, skin, brain, lungs, heart and skeletal muscle are affected in all 4 types of MPS-III            |
| <b>MPS III B:</b><br>Sanfilippo B      | $\alpha$ - <i>N</i> -acetyl- <b>D</b> -glucosaminidase                       | heparan sulfate                                               | phenotype similar to III A                                                                                                                    |
| <b>MPS III C:</b><br>Sanfilippo C      | AcetylCoA:<br>$\alpha$ -glucosaminide-acetyltransferase                      | heparan sulfate                                               | phenotype similar to III A                                                                                                                    |
| <b>MPS III D:</b><br>Sanfilippo D      | <i>N</i> -acetylglucosamine-6-sulfatase                                      | heparan sulfate                                               | phenotype similar to III A                                                                                                                    |
| <b>MPS IV A:</b><br>Morquio A          | Galactose-6-sulfatase                                                        | keratan sulfate, chondroitin 6-sulfate                        | corneal clouding, odontoid hypoplasia, aortic valve disease, distinctive skeletal abnormalities                                               |
| <b>MPS IV B:</b><br>Morquio B          | $\beta$ -Galactosidase                                                       | keratan sulfate                                               | severity of disease similar to IV A                                                                                                           |
| MPS V, a designation no longer used    |                                                                              |                                                               |                                                                                                                                               |
| <b>MPS VI:</b><br>Maroteaux-Lamy       | <i>N</i> -acetylgalactosamine-4-sulfatase<br><br>also called arylsulfatase B | dermatan sulfate                                              | 3 distinct forms from mild to severe, aortic valve disease, dystosis multiplex, normal intelligence, corneal clouding, coarse facial features |
| <b>MPS VII:</b><br>Sly                 | $\beta$ -Glucuronidase                                                       | heparan sulfate, dermatan sulfate, chondroitin 4-, 6-sulfates | hepatosplenomegaly, dystosis multiplex, wide spectrum of severity, hydrops fetalis                                                            |
| MPS VIII, a designation no longer used |                                                                              |                                                               |                                                                                                                                               |

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# Mucopolysaccharidoses

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- [Type I \(MPS I\)](#) 3 forms: Hurler=Type I H; Scheie=Type I S; Hurler/Scheie=Type I H/S
- [Type II \(MPS II\)](#) - Hunter's Syndrome
- Mucopolysaccharidoses Type III (MPS III)
  - [Sanfilippo Type A](#)
  - [Sanfilippo Type B](#)
  - [Sanfilippo Type C](#)
  - [Sanfilippo Type D](#)
- Mucopolysaccharidoses Type IV (MPS IV)
  - [Morquio's Type A](#)
  - [Morquio's Type B](#)
- Type V (MPS V) a designation that is no longer used
- [Type VI \(MPS VI\)](#) - Maroteaux-Lamy Syndrome
- [Type VII \(MPS VII\)](#) - Sly Syndrome
- Type VIII (MPS VIII) a designation that is no longer used

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# Disorders of Lysosomal Enzyme Transport

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- [Mucopolidosis I](#) - Sialidosis
  - [Mucopolidosis II](#) - I-Cell Disease
  - [Mucopolidosis III](#) - Pseudo-Hurler polydystrophy
- 

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# Oligosaccharidoses

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- [Fucosidosis](#)
  - [Mucopolidosis VI](#) - Sialolipidosis
  - [α-Mannosidosis](#)
  - [β-Mannosidosis](#)
  - [Sialidoses Types I and II](#)
  - [Galactosialidosis](#) - Goldberg syndrome
  - [Aspartylglucosaminuria](#)
- 

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- [Tyrosine-Derived Neurotransmitters](#)
- [Tryptophan-Derived Neurotransmitters](#)
- [Creatine Biosynthesis](#)
- [Glutathione Functions](#)
- [Polyamine Biosynthesis](#)
- [Nitric Oxide \(NO\) Synthesis and Function](#)

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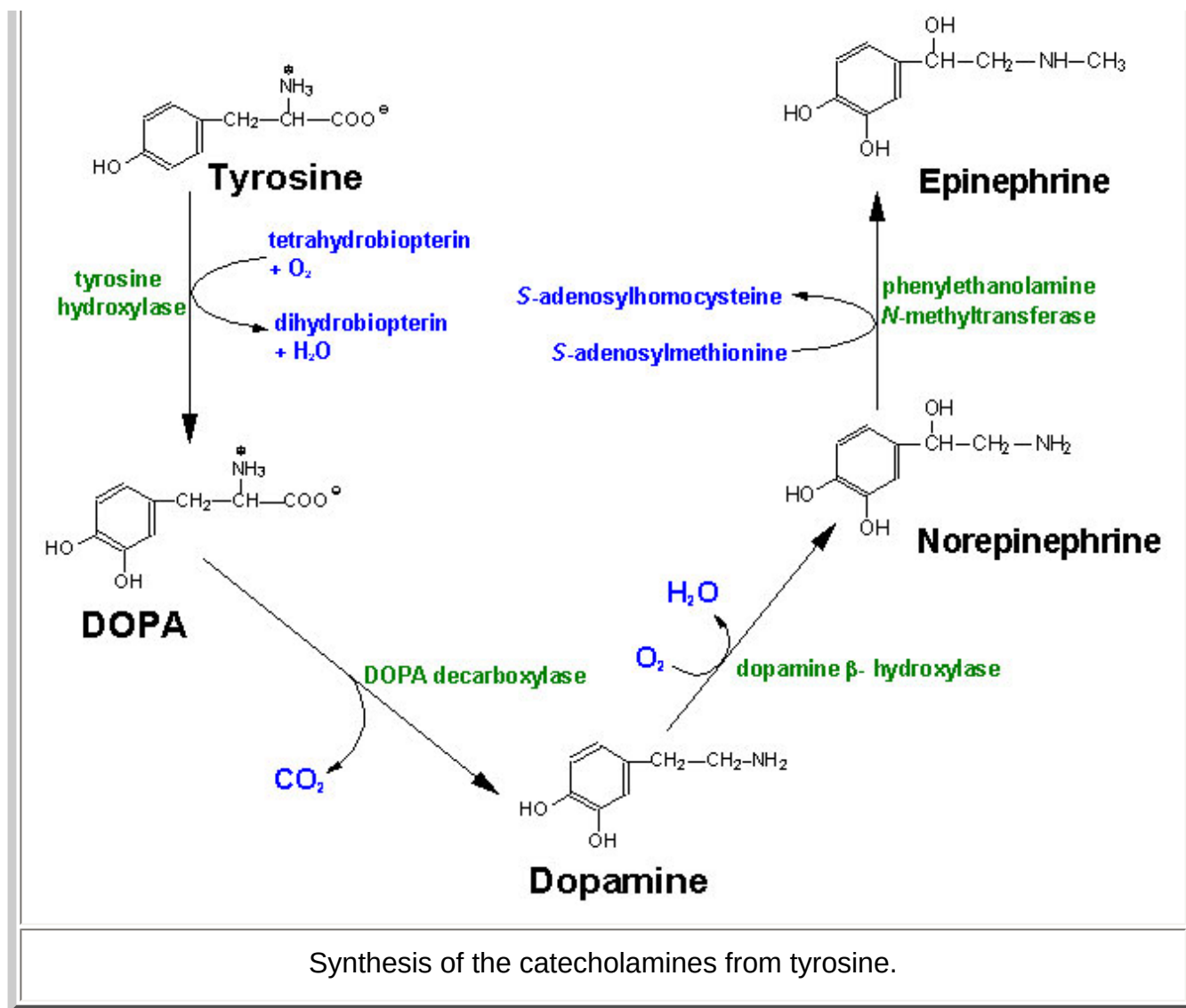
## Tyrosine-Derived Neurotransmitters

The majority of tyrosine that does not get incorporated into proteins is catabolized for energy production. One other significant fate of tyrosine is conversion to the **catecholamines**. The catecholamine neurotransmitters are **dopamine**, **norepinephrine**, and **epinephrine** (see also [Biochemistry of Nerve Transmission](#)).

Norepinephrine is the principal neurotransmitter of sympathetic postganglionic endings. Both norepinephrine and the methylated derivative, epinephrine are stored in synaptic knobs of neurons that secrete it, however, epinephrine is not a mediator at postganglionic sympathetic endings.

Tyrosine is transported into catecholamine-secreting neurons and adrenal medullary cells where catecholamine synthesis takes place. The first step in the process requires tyrosine hydroxylase, which like phenylalanine hydroxylase requires tetrahydrobiopterin (H<sub>4</sub>B) as cofactor. The dependence of tyrosine hydroxylase on H<sub>4</sub>B necessitates the coupling to the action of dihydropteridine reductase (DHPR) as is the situation for phenylalanine hydroxylase and tryptophan hydroxylase (see below).

The hydroxylation reaction generates DOPA (3,4-dihydrophenylalanine). DOPA decarboxylase converts DOPA to dopamine, dopamine  $\beta$ -hydroxylase converts dopamine to norepinephrine and phenylethanolamine N-methyltransferase converts norepinephrine to epinephrine. This latter reaction is one of several in the body that uses SAM as a methyl donor generating S-adenosylhomocysteine. Within the substantia nigra and some other regions of the brain, synthesis proceeds only to dopamine. Within the adrenal medulla dopamine is converted to norepinephrine and epinephrine.



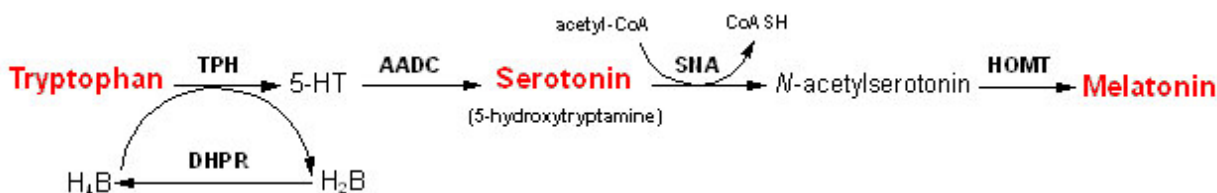
Once synthesized, dopamine, norepinephrine and epinephrine are packaged in granulated vesicles. Within these vesicles, norepinephrine and epinephrine are bound to ATP and a protein called **chromogranin A**.

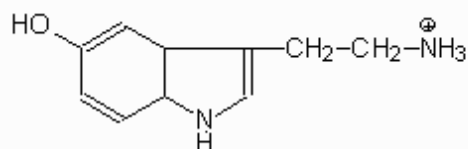
Metabolism of the catecholamines occurs through the actions of catecholamine-O-methyltransferase, (COMT) and monoamine oxidase, (MAO). Both of these enzymes are widely distributed throughout the body. However, COMT is not found in nerve endings as is MAO.

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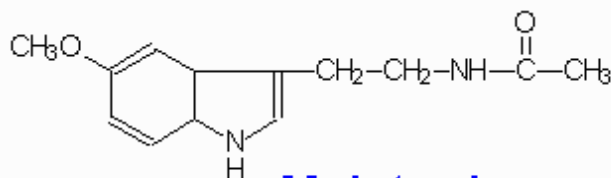
## Tryptophan-Derived Neurotransmitters

Tryptophan serves as the precursor for the synthesis of **serotonin** (5-hydroxytryptamine, 5-HT, see also [Biochemistry of Nerve Transmission](#)) and **melatonin** (*N*-acetyl-5-methoxytryptamine).





**Serotonin**  
(5-hydroxytryptamine)



**Melatonin**

Pathway for serotonin and melatonin synthesis from tryptophan. Abbreviations: THP = tryptophan hydroxylase, DHPR = dihydropteridine reductase, H<sub>2</sub>B = dihydrobiopterin, H<sub>4</sub>B = tetrahydrobiopterin, 5-HT = 5-hydroxytryptophan, AADC = aromatic L-amino acid decarboxylase, SNA = serotonin *N*-acetylase, HOMT = hydroxyindole-*O*-methyltransferase.

Serotonin is synthesized through 2-step process involving a tetrahydrobiopterin-dependent hydroxylation reaction (catalyzed by tryptophan-5-monoxygenase, also called tryptophan hydroxylase) and then a decarboxylation catalyzed by aromatic L-amino acid decarboxylase. The hydroxylase is normally not saturated and as a result, an increased uptake of tryptophan in the diet will lead to increased brain serotonin content.

Serotonin is present at highest concentrations in platelets and in the gastrointestinal tract. Lesser amounts are found in the brain and the retina. Serotonin containing neurons have their cell bodies in the midline raphe nuclei of the brain stem and project to portions of the hypothalamus, the limbic system, the neocortex and the spinal cord. After release from serotonergic neurons, most of the released serotonin is recaptured by an active reuptake mechanism. The function of the antidepressant, **Prozac** is to inhibit this reuptake process, thereby, resulting in prolonged serotonin presence in the synaptic cleft.

The function of serotonin is exerted upon its interaction with specific receptors. Several serotonin receptors have been cloned and are identified as 5HT<sub>1</sub>, 5HT<sub>2</sub>, 5HT<sub>3</sub>, 5HT<sub>4</sub>, 5HT<sub>5</sub>, 5HT<sub>6</sub>, and 5HT<sub>7</sub>. Within the 5HT<sub>1</sub> group there are subtypes 5HT<sub>1A</sub>, 5HT<sub>1B</sub>, 5HT<sub>1D</sub>, 5HT<sub>1E</sub>, and 5HT<sub>1F</sub>. There are three 5HT<sub>2</sub> subtypes, 5HT<sub>2A</sub>, 5HT<sub>2B</sub>, and 5HT<sub>2C</sub> as well as two 5HT<sub>5</sub> subtypes, 5HT<sub>5a</sub> and 5HT<sub>5B</sub>. Most of these receptors are coupled to G-proteins that affect the activities of either adenylate cyclase or phospholipase Cγ (PLCγ). The 5HT<sub>3</sub> class of receptors are ion channels.

Some serotonin receptors are presynaptic and others postsynaptic. The 5HT<sub>2A</sub> receptors mediate platelet aggregation and smooth muscle contraction. The 5HT<sub>2C</sub> receptors are suspected in control of food intake as mice lacking this gene become obese from increased food intake and are also subject to fatal seizures. The 5HT<sub>3</sub> receptors are present in the gastrointestinal tract and are related to vomiting. Also present in the gastrointestinal tract are 5HT<sub>4</sub> receptors where they function in secretion and peristalsis. The 5HT<sub>6</sub> and 5HT<sub>7</sub> receptors are distributed throughout the limbic system of the brain and the 5HT<sub>6</sub> receptors have high affinity for antidepressant drugs.

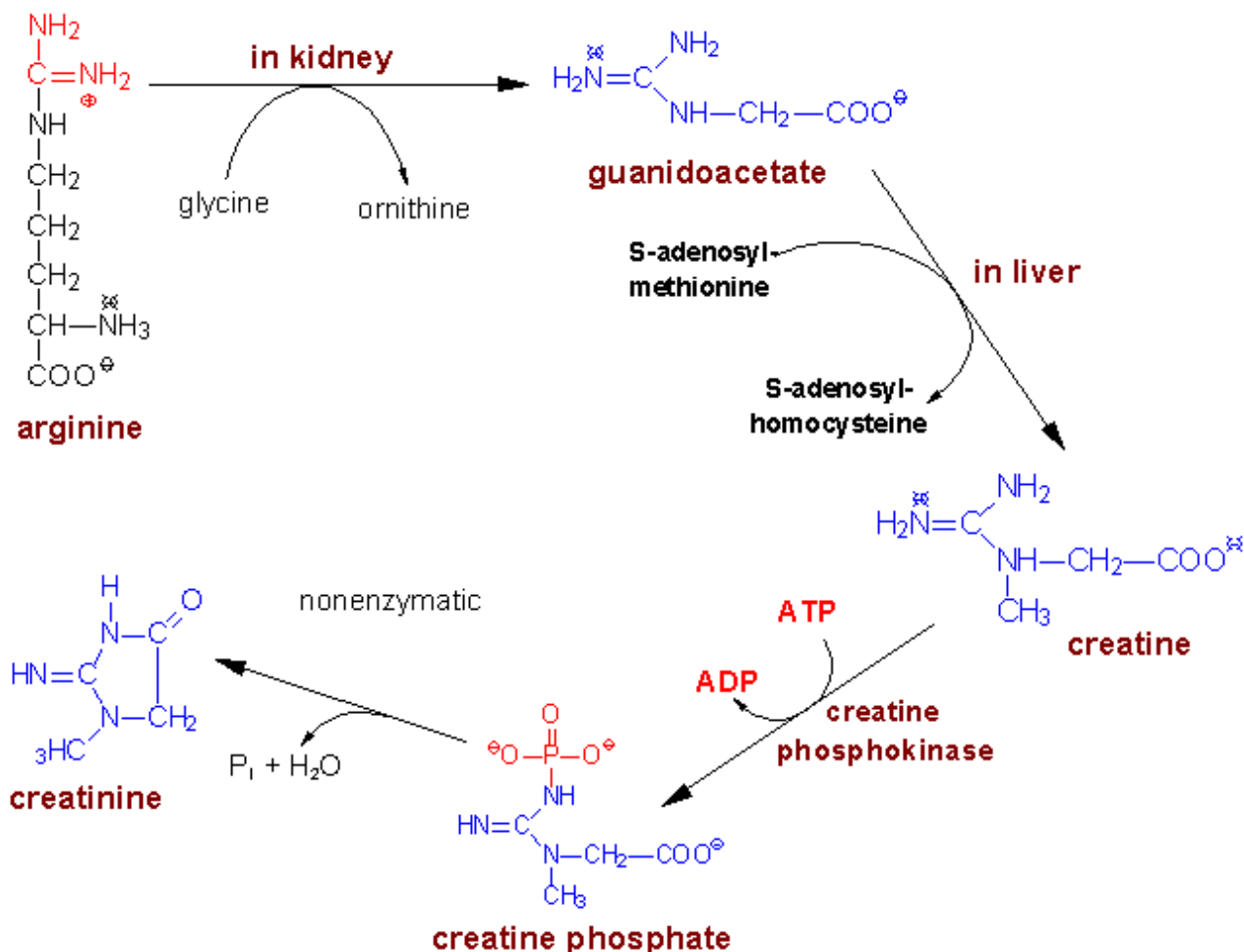
Melatonin is derived from serotonin within the pineal gland and the retina, where the necessary N-acetyltransferase enzyme is found. The pineal parenchymal cells secrete melatonin into the blood and cerebrospinal fluid. Synthesis and secretion of melatonin increases during the dark period of the day and is maintained at a low level during daylight hours. This diurnal variation in melatonin synthesis is brought about by norepinephrine secreted by the postganglionic sympathetic nerves that innervate the pineal gland. The effects of norepinephrine are exerted through interaction with β-adrenergic receptors. This leads to increased levels of cAMP, which in turn activate the N-acetyltransferase required for melatonin synthesis. Melatonin functions by inhibiting the synthesis and secretion of other neurotransmitters such as dopamine and GABA.

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## Creatine Biosynthesis

Creatine is synthesized in the liver by methylation of guanidoacetate using SAM as the methyl donor. Guanidoacetate itself

is formed in the kidney from the amino acids arginine and glycine.



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### Synthesis of creatine and creatinine

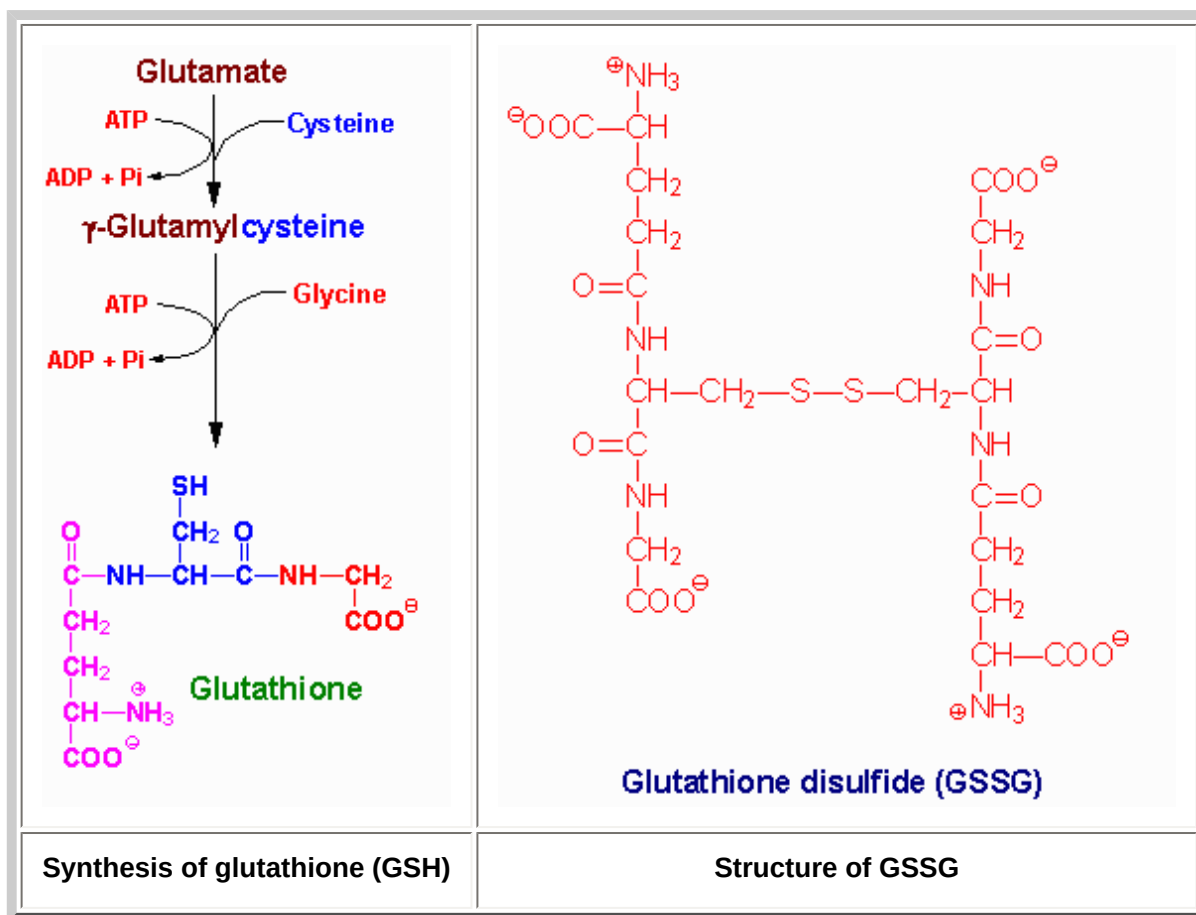
Creatine is used as a storage form of high energy phosphate. The phosphate of ATP is transferred to creatine, generating creatine phosphate, through the action of creatine phosphokinase. The reaction is reversible such that when energy demand is high (e.g. during muscle exertion) creatine phosphate donates its phosphate to ADP to yield ATP.

Both creatine and creatine phosphate are found in muscle, brain and blood. Creatinine is formed in muscle from creatine phosphate by a nonenzymatic dehydration and loss of phosphate. The amount of creatinine produced is related to muscle mass and remains remarkably constant from day to day. Creatinine is excreted by the kidneys and the level of excretion (creatinine clearance rate) is a measure of renal function.

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## Glutathione Functions

Glutathione (abbreviated GSH) is a tripeptide composed of glutamate, cysteine and glycine that has numerous important functions within cells. It serves as a reductant, is conjugated to drugs to make them more water soluble, is involved in amino acid transport across cell membranes (the **γ-glutamyl cycle**), is a part of the peptidoleukotrienes, serves as a cofactor for some enzymatic reactions and as an aid in the rearrangement of protein disulfide bonds.



The role of GSH as a reductant is extremely important particularly in the highly oxidizing environment of the erythrocyte. The sulfhydryl of GSH can be used to reduce peroxides formed during oxygen transport. The resulting oxidized form of GSH consists of two molecules disulfide bonded together (abbreviated GSSG). The enzyme glutathione reductase utilizes NADPH as a cofactor to reduce GSSG back to two moles of GSH. Hence, the [pentose phosphate pathway](#) is an extremely important pathway of erythrocytes for the continuing production of the NADPH needed by glutathione reductase. In fact as much as 10% of glucose consumption, by erythrocytes, may be mediated by the pentose phosphate pathway.

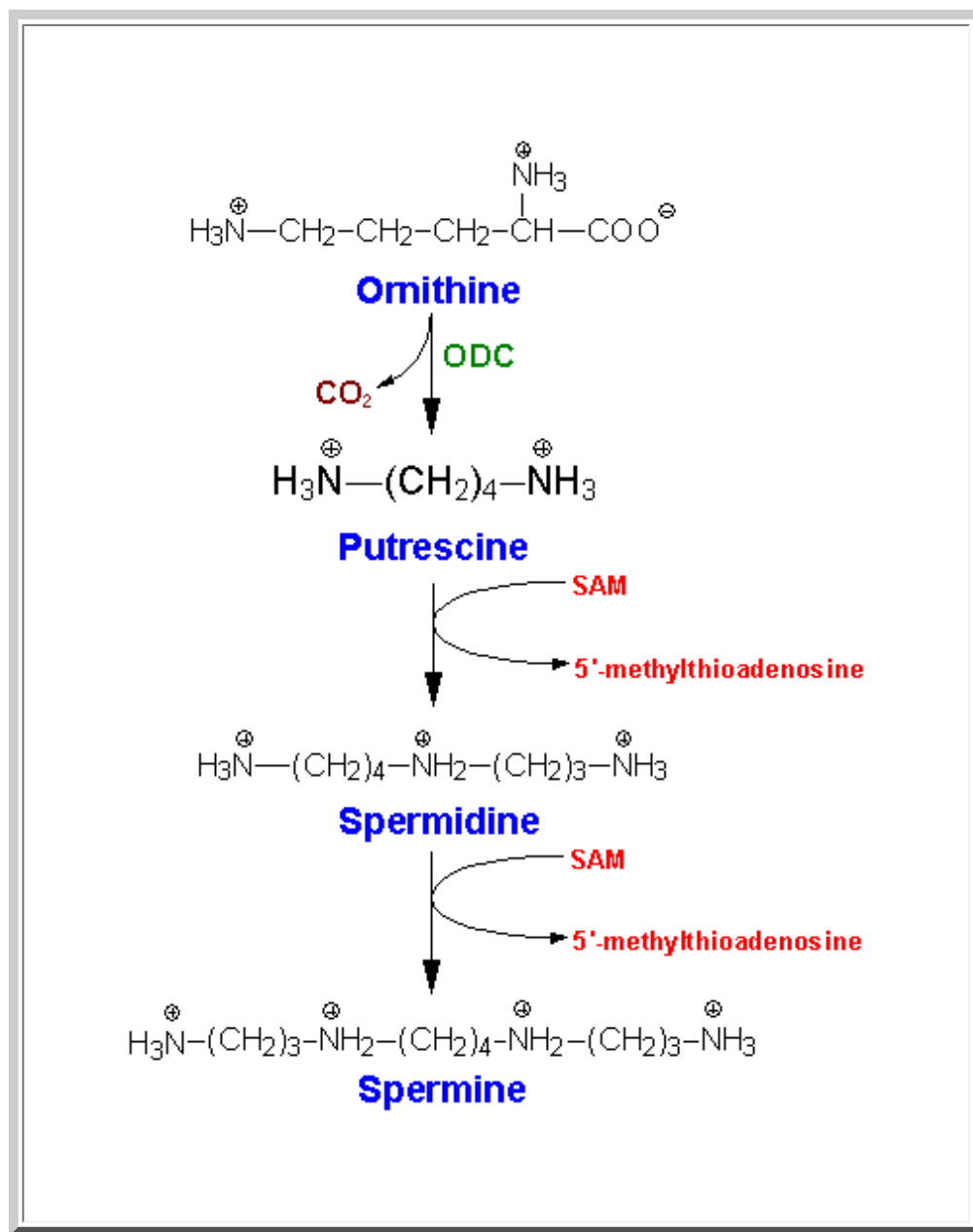
Several mechanisms exist for the transport of amino acids across cell membranes. Many are symport or antiport mechanisms that couple amino acid transport to sodium transport. The **γ-glutamyl cycle** is an example of a group transfer mechanism of amino acid transport. Although this mechanism requires more energy input, it is rapid and has a high capacity. The cycle functions primarily in the kidney, particularly renal epithelial cells. The enzyme γ-glutamyl transpeptidase is located in the cell membrane and shuttles GSH to the cell surface to interact with an amino acid. Reaction with an amino acid liberates cysteinylglycine and generates a γ-glutamyl-amino acid which is transported into the cell and hydrolyzed to release the amino acid. Glutamate is released as 5-oxoproline and the cysteinylglycine is cleaved to its component amino acids. Regeneration of GSH requires an ATP-dependent conversion of 5-oxoproline to glutamate and then the 2 additional moles of ATP that are required during the normal generation of GSH.

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## Polyamine Biosynthesis

One of the earliest signals that cells have entered their replication cycle is the appearance of elevated levels of mRNA for ornithine decarboxylase (ODC), and then increased levels of the enzyme, which is the first enzyme in the pathway to synthesis of the polyamines. Because of the latter, and because the polyamines are highly cationic and tend to bind nucleic acids with high affinity, it is believed that the polyamines are important participants in DNA synthesis, or in the regulation of that process.





The key features of the pathway are that it involves **putrescine**, an ornithine catabolite, and S-adenosylmethionine (SAM) as a donor of 2 propylamine residues. The first propylamine conjugation yields **spermidine** and addition of another to spermidine yields **spermine**.

The function of ODC is to produce the 4-carbon saturated diamine, putrescine. At the same time, SAM decarboxylase cleaves the SAM carboxyl residue, producing decarboxylated SAM (S-adenosylmethionine), which retains the methyl group usually involved in SAM methyltransferase activity. SAM decarboxylase activity is regulated by product inhibition and allosterically stimulated by putrescine. Spermidine synthase catalyzes the condensation reaction, producing spermidine and 5'-methylthioadenosine. A second propylamine residue is added to spermidine producing spermine.

The signal for regulating ODC activity is unknown, but since the product of its activity, putrescine, regulates SAM decarboxylase activity, it appears that polyamine production is principally regulated by ODC concentration.

The butylamino group of spermidine is used in a posttranslational modification reaction important to the process of **translation**. A specific lysine residue in the translational initiation factor **eIF-4D** is modified. Following the modification the residue is hydroxylated yielding a residue in the protein termed **hypusine**.

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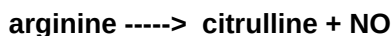
## Nitric Oxide Synthesis and Function

Vasodilators, such as acetylcholine and bradykinin, do not exert their effects upon the vascular smooth muscle cell in the absence of the overlying endothelium. When acetylcholine (or bradykinin) binds its receptor on the surface of endothelial cells, a

signal cascade, coupled to the activation phospholipase C- $\gamma$  (PLC $\gamma$ ), is initiated. The PLC $\gamma$ -mediated release of inositol trisphosphate, IP<sub>3</sub> (from membrane associated phosphatidylinositol-4,5-bisphosphate, PIP<sub>2</sub>), leads to the release of intracellular stores of Ca<sup>2+</sup>. In turn, the elevation in Ca<sup>2+</sup> leads to the liberation of **endothelium-derived relaxing factor (EDRF)** which then diffuses into the adjacent smooth muscle. Within smooth muscle cells, EDRF reacts with the heme moiety of a soluble guanylyl cyclase, resulting in activation of the latter and a consequent elevation of intracellular levels of cGMP. The net effect is the activation of cGMP-dependent protein kinase (PKG) and the phosphorylation of substrates leading to smooth muscle cell relaxation. The coronary artery vasodilator, nitroglycerin, acts to increase intracellular release of EDRF and thus of activation of the cGMP signal cascade.

Quite unexpectedly, EDRF was found to be the free radical diatomic gas, **nitric oxide, NO**. So stunning was the elucidation of the pathway to and actions of NO that Drs. Murad, Ignarro and Furchgott were awarded the Nobel Prize in 1998 for their work on this system.

NO is formed by the action of NO synthase, (NOS) on the amino acid arginine. In fact there are 3 isozymes of NOS in mammalian cells. These are neuronal NOS (nNOS, also called NOS-1), inducible or macrophage NOS (iNOS, also called NOS-2), and endothelial NOS (eNOS, also called NOS-3).



Nitric oxide synthases are very complex enzymes, employing five redox cofactors: NADPH, FAD, FMN, heme and tetrahydrobiopterin (H<sub>4</sub>B). NO can also be formed from nitrite, derived from vasodilators such as glycerin trinitrate (nitroglycerin) during their metabolism. The half-life of NO is extremely short, lasting only 2-4 seconds. This is because it is a highly reactive free radical and interacts with oxygen and superoxide. NO is inhibited by hemoglobin and other heme proteins which bind it tightly.

Both eNOS and nNOS are constitutively expressed and regulated by Ca<sup>2+</sup>. The calcium regulation is imparted by the associated calmodulin subunits, thus explaining how vasodilators such as acetylcholine effect smooth muscle relaxation as a consequence of increasing intracellular endothelial cell calcium levels. Although iNOS contains calmodulin subunits, its activity is unaffected by changes in Ca<sup>2+</sup> concentration. iNOS is transcriptionally activated in macrophages, neutrophils, and smooth muscle cells.

The major functions of NO production through activation of iNOS are associated with the bactericidal and tumoricidal actions of macrophages. Overproduction of NO via iNOS is associated with cytokine-induced septic shock such as occurs post-operatively in patients with bacterial infections. Bacteria produce endotoxins such as lipopolysaccharide (LPS) that activate iNOS in macrophages.

Nitric oxide is involved in a number of other important cellular processes in addition to its impact on vascular smooth muscle cells. Events initiated by NO that are important for **blood coagulation** include inhibition of platelet aggregation and adhesion and inhibition of neutrophil adhesion to platelets and to the vascular endothelium. NO is also generated by cells of the immune system and as such is involved in non-specific host defense mechanisms and macrophage-mediated killing. NO also inhibits the proliferation of tumor cells and microorganisms. Additional cellular responses to NO include induction of apoptosis (programmed cell death), DNA breakage and mutation.

Chemical inhibitors of NOS are available and can markedly decrease production of NO. The effect is a dramatic increase in blood pressure due to vasoconstriction. Another important cardiovascular effect of NO is exerted through the production of cGMP, which acts to inhibit platelet aggregation.

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## Table of Neurotransmitters

| Transmitter Molecule                                               | Derived From | Site of Synthesis                               |
|--------------------------------------------------------------------|--------------|-------------------------------------------------|
| <a href="#">Acetylcholine</a>                                      | Choline      | CNS, parasympathetic nerves                     |
| <a href="#">Serotonin</a><br>5-Hydroxytryptamine (5-HT)            | Tryptophan   | CNS, chromaffin cells of the gut, enteric cells |
| <a href="#">GABA</a>                                               | Glutamate    | CNS                                             |
| Glutamate                                                          |              | CNS                                             |
| Aspartate                                                          |              | CNS                                             |
| Glycine                                                            |              | spinal cord                                     |
| Histamine                                                          | Histidine    | hypothalamus                                    |
| <a href="#">Epinephrine</a><br><a href="#">synthesis pathway</a>   | Tyrosine     | adrenal medulla, some CNS cells                 |
| <a href="#">Norpinephrine</a><br><a href="#">synthesis pathway</a> | Tyrosine     | CNS, sympathetic nerves                         |
| <a href="#">Dopamine</a><br><a href="#">synthesis pathway</a>      | Tyrosine     | CNS                                             |

|                  |          |                                         |
|------------------|----------|-----------------------------------------|
| Adenosine        | ATP      | CNS, peripheral nerves                  |
| ATP              |          | sympathetic, sensory and enteric nerves |
| Nitric oxide, NO | Arginine | CNS, gastrointestinal tract             |

Many other neurotransmitters are derived from precursor proteins, the so-called **peptide neurotransmitters**. As many as 50 different peptides have been shown to exert their effects on neural cell function. Several of these peptide transmitters are derived from the larger protein **pre-opiomelanocortin (POMC)**. Neuropeptides are responsible for mediating sensory and emotional responses including hunger, thirst, sex drive, pleasure and pain.  
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## Synaptic Transmission

Synaptic transmission refers to the propagation of nerve impulses from one nerve cell to another. This occurs at a specialized cellular structure known as the **synapse**--- a junction at which the axon of the presynaptic neuron terminates at some location upon the postsynaptic neuron. The end of a presynaptic axon, where it is juxtaposed to the postsynaptic neuron, is enlarged and forms a structure known as the **terminal button**. An axon can make contact anywhere along the second neuron: on the dendrites (an **axodendritic synapse**), the cell body (an **axosomatic synapse**) or the axons (an **axo-axonal synapse**).

Nerve impulses are transmitted at synapses by the release of chemicals called **neurotransmitters**. As a nerve impulse, or **action potential**, reaches the end of a presynaptic axon, molecules of neurotransmitter are released into the synaptic space. The neurotransmitters are a diverse group of chemical compounds ranging from simple amines such as **dopamine** and amino acids such as  **$\gamma$ -aminobutyrate (GABA)**, to polypeptides such as the **enkephalins**. The mechanisms by which they elicit responses in both presynaptic and postsynaptic neurons are as diverse as the mechanisms employed by growth factor and cytokine receptors.

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## Neuromuscular Transmission

A different type of nerve transmission occurs when an axon terminates on a skeletal muscle fiber, at a specialized structure called the **neuromuscular junction**. An action potential occurring at this site is known as **neuromuscular transmission**. At a neuromuscular junction, the axon subdivides into numerous terminal buttons that reside within depressions formed in the **motor end-plate**. The particular transmitter in use at the neuromuscular junction is acetylcholine.

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## Neurotransmitter Receptors

Once the molecules of neurotransmitter are released from a cell as the result of the firing of an action potential, they bind to specific receptors on the surface of the postsynaptic cell. In all cases in which these receptors have been cloned and characterized in detail, it has been shown that there are numerous subtypes of receptor for any given neurotransmitter. As well as being present on the surfaces of postsynaptic neurons, neurotransmitter receptors are found on presynaptic neurons. In general, presynaptic neuron receptors act to inhibit further release of neurotransmitter.

The vast majority of neurotransmitter receptors belong to a class of proteins known as the **serpentine receptors**. This class exhibits a characteristic transmembrane structure: that is, it spans the cell membrane, not once but seven times. The link between neurotransmitters and intracellular signaling is carried out by association either with G-proteins (small GTP-binding and hydrolyzing proteins) or with protein kinases, or by the receptor itself

in the form of a ligand-gated ion channel (for example, the acetylcholine receptor). One additional characteristic of neurotransmitter receptors is that they are subject to **ligand-induced desensitization**: That is, they can become unresponsive upon prolonged exposure to their neurotransmitter.

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## Acetylcholine

Acetylcholine (**ACh**) is a simple molecule synthesized from choline and acetyl-CoA through the action of choline acetyltransferase. Neurons that synthesize and release ACh are termed **cholinergic neurons**. When an action potential reaches the terminal button of a presynaptic neuron a voltage-gated calcium channel is opened. The influx of calcium ions,  $Ca^{2+}$ , stimulates the exocytosis of presynaptic vesicles containing ACh, which is thereby released into the synaptic cleft. Once released, ACh must be removed rapidly in order to allow repolarization to take place; this step, hydrolysis, is carried out by the enzyme, acetylcholinesterase. The acetylcholinesterase found at nerve endings is anchored to the plasma membrane through a glycolipid.

ACh receptors are ligand-gated cation channels composed of four different polypeptide subunits arranged in the form  $[(\alpha 2)(\beta)(\gamma)(\delta)]$ . Two main classes of ACh receptors have been identified on the basis of their responsiveness to the toadstool alkaloid, muscarine, and to nicotine, respectively: the **muscarinic receptors** and the **nicotinic receptors**. Both receptor classes are abundant in the human brain. Nicotinic receptors are further divided into those found at neuromuscular junctions and those found at neuronal synapses. The activation of ACh receptors by the binding of ACh leads to an influx of  $Na^{+}$  into the cell and an efflux of  $K^{+}$ , resulting in a depolarization of the postsynaptic neuron and the initiation of a new action potential.

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## Cholinergic Agonists and Antagonists

Numerous compounds have been identified that act as either agonists or antagonists of cholinergic neurons. The principal action of cholinergic agonists is the excitation or inhibition of autonomic effector cells that are innervated by postganglionic parasympathetic neurons and as such are referred to as **parasympathomimetic agents**. The cholinergic agonists include choline esters (such as ACh itself) as well as protein- or alkaloid-based compounds. Several naturally occurring compounds have been shown to affect cholinergic neurons, either positively or negatively.

The responses of cholinergic neurons can also be enhanced by administration of cholinesterase (ChE) inhibitors. ChE inhibitors have been used as components of nerve gases but also have significant medical application in the treatment of disorders such as glaucoma and myasthenia gravis as well as in terminating the effects of neuromuscular blocking agents such as atropine.

## Natural Cholinergic Agonist and Antagonists

|          | Source of Compound                      | Mode of Action                                                     |
|----------|-----------------------------------------|--------------------------------------------------------------------|
| Agonists |                                         |                                                                    |
| Nicotine | Alkaloid prevalent in the tobacco plant | Activates nicotinic class of ACh receptors, locks the channel open |

|                                             |                                                                      |                                                                                 |
|---------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Muscarine                                   | Alkaloid produced by <i>Amanita muscaria</i> mushrooms               | Activates muscarinic class of ACh receptors                                     |
| $\alpha$ -Latrotoxin                        | Protein produced by the black widow spider                           | Induces massive ACh release, possibly by acting as a $\text{Ca}^{2+}$ ionophore |
| <b>Antagonists</b>                          |                                                                      |                                                                                 |
| Atropine (and related compound Scopolamine) | Alkaloid produced by the deadly nightshade, <i>Atropa belladonna</i> | Blocks ACh actions only at muscarinic receptors                                 |
| Botulinus Toxin                             | Eight proteins produced by <i>Clostridium botulinum</i>              | Inhibits the release of ACh                                                     |
| $\alpha$ -Bungarotoxin                      | Protein produced by <i>Bungarus</i> genus of snakes                  | Prevents ACh receptor channel opening                                           |
| <i>d</i> -Tubocurarine                      | Active ingredient of curare                                          | Prevents ACh receptor channel opening at motor end-plate                        |

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## Catecholamines

The principal catecholamines are **norepinephrine**, **epinephrine** and **dopamine**. These compounds are formed from phenylalanine and tyrosine. Tyrosine is produced in the liver from phenylalanine through the action of phenylalanine hydroxylase. The tyrosine is then transported to catecholamine-secreting neurons where a series of reactions convert it to dopamine, to norepinephrine and finally to epinephrine (see [Specialized Products of Amino Acids](#)).

Catecholamines exhibit peripheral nervous system excitatory and inhibitory effects as well as actions in the CNS such as respiratory stimulation and an increase in psychomotor activity. The excitatory effects are exerted upon smooth muscle cells of the vessels that supply blood to the skin and mucous membranes. Cardiac function is also subject to excitatory effects, which lead to an increase in heart rate and in the force of contraction. Inhibitory effects, by contrast, are exerted upon smooth muscle cells in the wall of the gut, the bronchial tree of the lungs, and the vessels that supply blood to skeletal muscle.

In addition to their effects as neurotransmitters, norepinephrine and epinephrine can influence the rate of metabolism. This influence works both by modulating endocrine function such as insulin secretion and by increasing the rate of glycogenolysis and fatty acid mobilization.

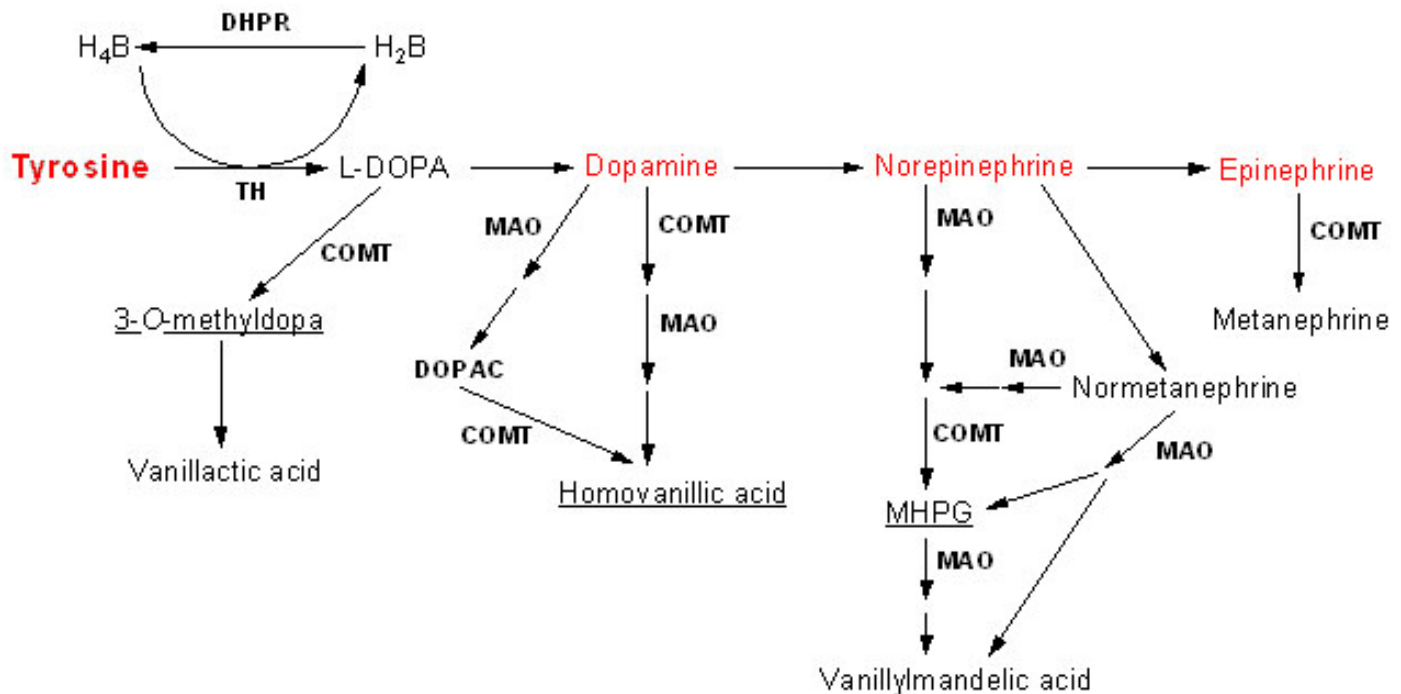
The catecholamines bind to two different classes of receptors termed the  $\alpha$ - and  $\beta$ -adrenergic receptors. The catecholamines therefore are also known as **adrenergic neurotransmitters**; neurons that secrete them are **adrenergic neurons**. Norepinephrine-secreting neurons are **noradrenergic**. The adrenergic receptors are classical



serpentine receptors that couple to intracellular G-proteins. Some of the norepinephrine released from presynaptic noradrenergic neurons recycled in the presynaptic neuron by a reuptake mechanism.

## Catecholamine Catabolism

Epinephrine and norepinephrine are catabolized to inactive compounds through the sequential actions of catecholamine-O-methyltransferase (COMT) and monoamine oxidase (MAO). Compounds that inhibit the action of MAO have been shown to have beneficial effects in the treatment of clinical depression, even when tricyclic antidepressants are ineffective. The utility of MAO inhibitors was discovered serendipitously when patients treated for tuberculosis with isoniazid showed signs of an improvement in mood; isoniazid was subsequently found to work by inhibiting MAO.



Metabolism of the catecholamine neurotransmitters. Only clinically important enzymes are included in this diagram. The catabolic byproducts of the catecholamines, whose levels in the cerebrospinal fluid are indicative of defects in catabolism, are underlined. Abbreviations: TH = tyrosine hydroxylase, DHPR = dihydropteridine reductase, H<sub>2</sub>B = dihydrobiopterin, H<sub>4</sub>B = tetrahydrobiopterin, MAO = monoamine oxidase, COMT = catecholamine-O-methyltransferase, MHPG = 3-methoxy-4-hydroxyphenylglycol, DOPAC = dihydroxyphenylacetic acid.

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## Serotonin

Serotonin (5-hydroxytryptamine, **5HT**) is formed by the hydroxylation and decarboxylation of tryptophan (see [Specialized Products of Amino Acids](#)). The greatest concentration of 5HT (90%) is found in the enterochromaffin cells of the gastrointestinal tract. Most of the remainder of the body's 5HT is found in platelets and the CNS. The

effects of 5HT are felt most prominently in the cardiovascular system, with additional effects in the respiratory system and the intestines. Vasoconstriction is a classic response to the administration of 5HT.

Neurons that secrete 5HT are termed **serotonergic**. Following the release of 5HT, a portion is taken back up by the presynaptic serotonergic neuron in a manner similar to that of the reuptake of norepinephrine.

The function of serotonin is exerted upon its interaction with specific receptors. Several serotonin receptors have been cloned and are identified as 5HT<sub>1</sub>, 5HT<sub>2</sub>, 5HT<sub>3</sub>, 5HT<sub>4</sub>, 5HT<sub>5</sub>, 5HT<sub>6</sub>, and 5HT<sub>7</sub>. Within the 5HT<sub>1</sub> group there are subtypes 5HT<sub>1A</sub>, 5HT<sub>1B</sub>, 5HT<sub>1D</sub>, 5HT<sub>1E</sub>, and 5HT<sub>1F</sub>. There are three 5HT<sub>2</sub> subtypes, 5HT<sub>2A</sub>, 5HT<sub>2B</sub>, and 5HT<sub>2C</sub> as well as two 5HT<sub>5</sub> subtypes, 5HT<sub>5a</sub> and 5HT<sub>5B</sub>. Most of these receptors are coupled to G-proteins that affect the activities of either adenylate cyclase or phospholipase C $\gamma$ . The 5HT<sub>3</sub> class of receptors are ion channels.

Some serotonin receptors are presynaptic and others postsynaptic. The 5HT<sub>2A</sub> receptors mediate platelet aggregation and smooth muscle contraction. The 5HT<sub>2C</sub> receptors are suspected in control of food intake as mice lacking this gene become obese from increased food intake and are also subject to fatal seizures. The 5HT<sub>3</sub> receptors are present in the gastrointestinal tract and are related to vomiting. Also present in the gastrointestinal tract are 5HT<sub>4</sub> receptors where they function in secretion and peristalsis. The 5HT<sub>6</sub> and 5HT<sub>7</sub> receptors are distributed throughout the limbic system of the brain and the 5HT<sub>6</sub> receptors have high affinity for antidepressant drugs.

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## GABA

Several amino acids have distinct excitatory or inhibitory effects upon the nervous system. The amino acid derivative,  $\gamma$ -aminobutyrate, also called 4-aminobutyrate, (**GABA**) is a well-known inhibitor of presynaptic transmission in the CNS, and also in the retina. The formation of GABA occurs by the decarboxylation of glutamate catalyzed by glutamate decarboxylase (**GAD**). GAD is present in many nerve endings of the brain as well as in the  $\beta$ -cells of the pancreas. Neurons that secrete GABA are termed **GABAergic**.

GABA exerts its effects by binding to two distinct receptors, GABA-A and GABA-B. The GABA-A receptors form a Cl<sup>-</sup> channel. The binding of GABA to GABA-A receptors increases the Cl<sup>-</sup> conductance of presynaptic neurons. The anxiolytic drugs of the benzodiazepine family exert their soothing effects by potentiating the responses of GABA-A receptors to GABA binding. The GABA-B receptors are coupled to an intracellular G-protein and act by increasing conductance of an associated K<sup>+</sup> channel.

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## Introduction

The pentose phosphate pathway is primarily an anabolic pathway that utilizes the 6 carbons of glucose to generate 5 carbon sugars and reducing equivalents. However, this pathway does oxidize glucose and under certain conditions can completely oxidize glucose to CO<sub>2</sub> and water. The primary functions of this pathway are:

- **1.** To generate reducing equivalents, in the form of NADPH, for reductive biosynthesis reactions within cells.
- **2.** To provide the cell with ribose-5-phosphate (R5P) for the synthesis of the nucleotides and nucleic acids.
- **3.** Although not a significant function of the PPP, it can operate to metabolize dietary pentose sugars derived from the digestion of nucleic acids as well as to rearrange the carbon skeletons of dietary carbohydrates into glycolytic/gluconeogenic intermediates

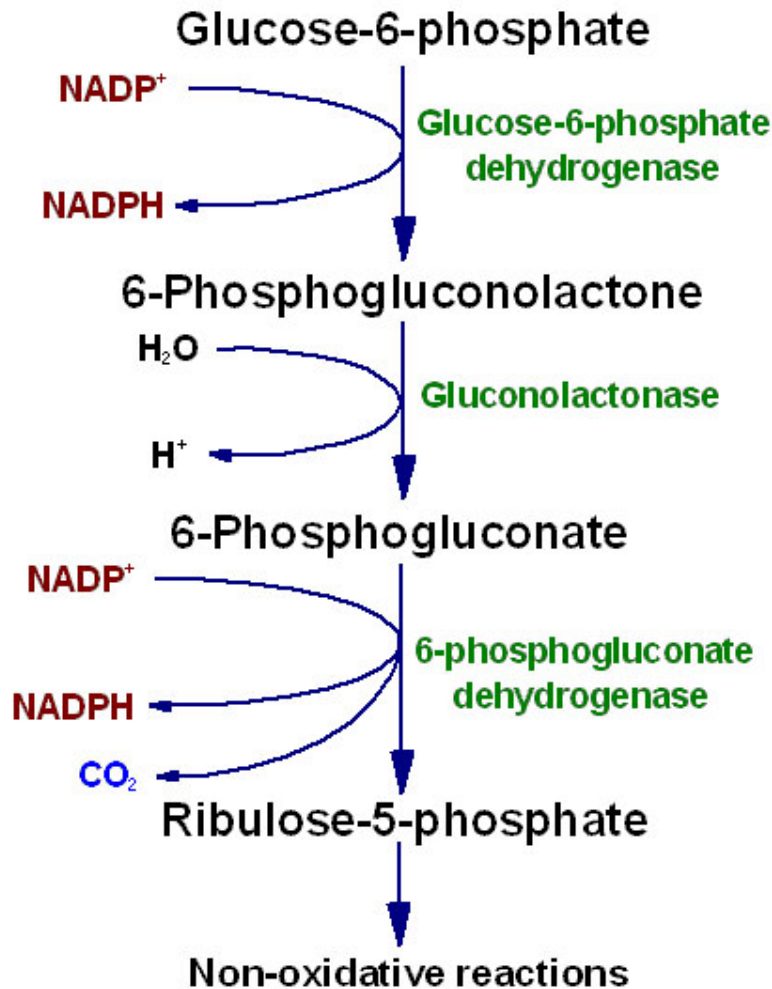
Enzymes that function primarily in the reductive direction utilize the NADP<sup>+</sup>/NADPH cofactor pair as co-factors as opposed to oxidative enzymes that utilize the NAD<sup>+</sup>/NADH cofactor pair. The reactions of fatty acid biosynthesis and steroid biosynthesis utilize large amounts of NADPH. As a consequence, cells of the **liver, adipose tissue, adrenal cortex, testis** and **lactating mammary gland** have high levels of the PPP enzymes. In fact 30% of the oxidation of glucose in the liver occurs via the PPP. Additionally, **erythrocytes** utilize the reactions of the PPP to generate large amounts of NADPH used in the reduction of glutathione (see below). The conversion of ribonucleotides to deoxyribonucleotides (through the action of ribonucleotide reductase) requires NADPH as the electron source, therefore, any rapidly proliferating cell needs large quantities of NADPH.

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## Reactions of the Pentose Phosphate Pathway

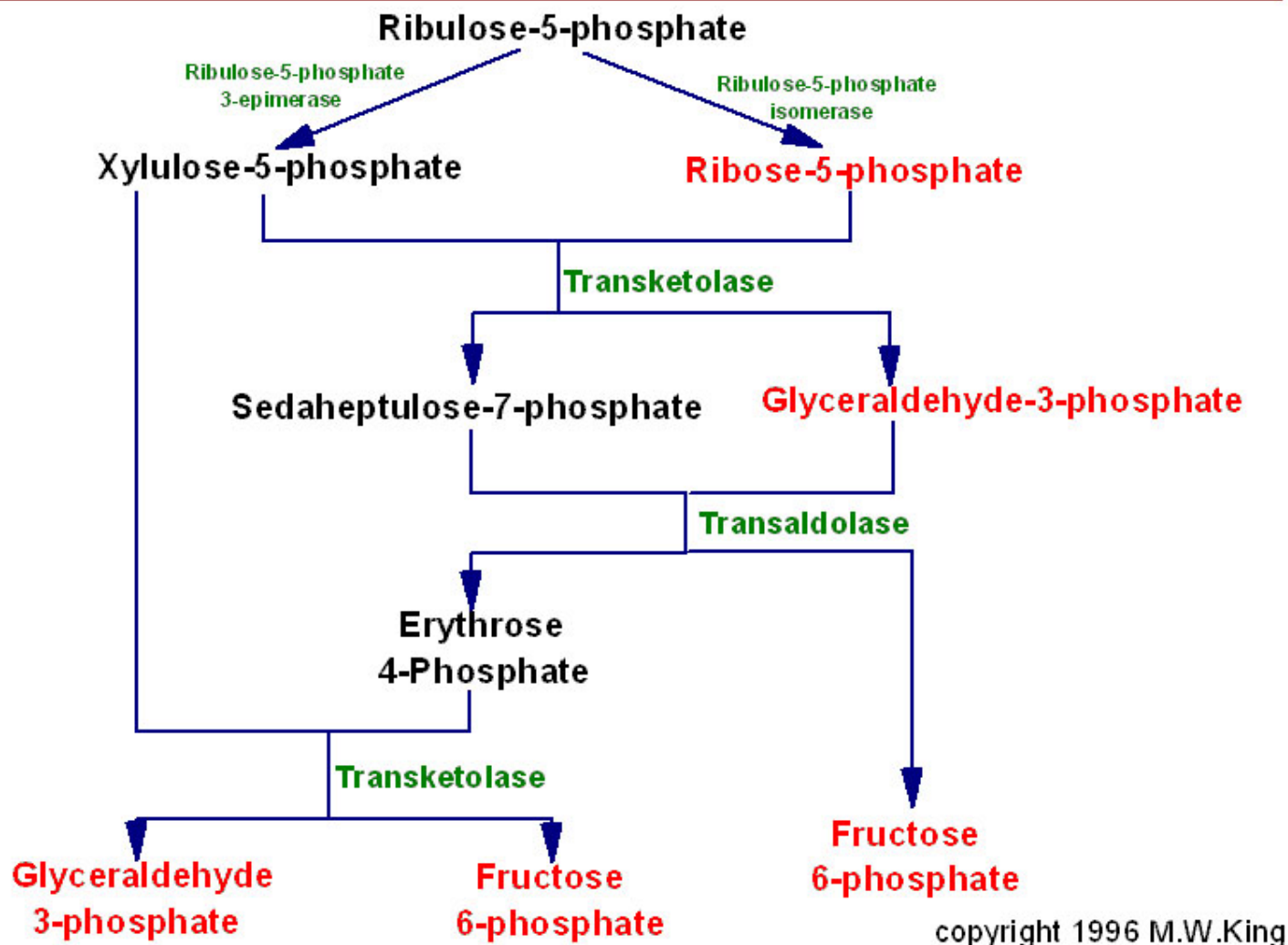
## Oxidative Stage of Pentose Phosphate Pathway



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The reactions of oxidative portion of the pentose phosphate pathway are shown. The enzymes are in green.

## Non-Oxidative Stage of Pentose Phosphate Pathway



The reactions of the non-oxidative portion of the pentose phosphate pathway are shown. Enzymes are in green. Relevant carbohydrate intermediates of this portion of the pathway are in red.

The reactions of the PPP operate exclusively in the cytoplasm. From this perspective it is understandable that fatty acid synthesis (as opposed to oxidation) takes place in the cytoplasm. The pentose phosphate pathway has both an **oxidative** and a **non-oxidative** arm. The oxidation steps, utilizing glucose-6-phosphate (G6P) as the substrate, occur at the beginning of the pathway and are the reactions that generate NADPH. The reactions catalyzed by glucose-6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase generate one mole of NADPH each for every mole of glucose-6-phosphate (G6P) that enters the PPP.

The non-oxidative reactions of the PPP are primarily designed to generate R5P. Equally important reactions of the PPP are to convert dietary 5 carbon sugars into both 6 (**fructose-6-phosphate**) and 3 (**glyceraldehyde-3-phosphate**) carbon sugars which can then be utilized by the pathways of glycolysis.

The primary enzymes involved in the non-oxidative steps of the PPP are transaldolase and transketolase.

- Transketolase functions to transfer 2 carbon groups from substrates of the PPP, thus rearranging the carbon atoms that enter this pathway. Like other enzymes that transfer 2 carbon groups, transketolase requires **thiamine pyrophosphate (TPP)** as a co-factor in the transfer reaction.
- Transaldolase transfers 3 carbon groups and thus is also involved in a rearrangement of the carbon skeletons of the substrates of the PPP. The transaldolase reaction involves **Schiff base** formation between the substrate and a lysine residue in the enzyme.

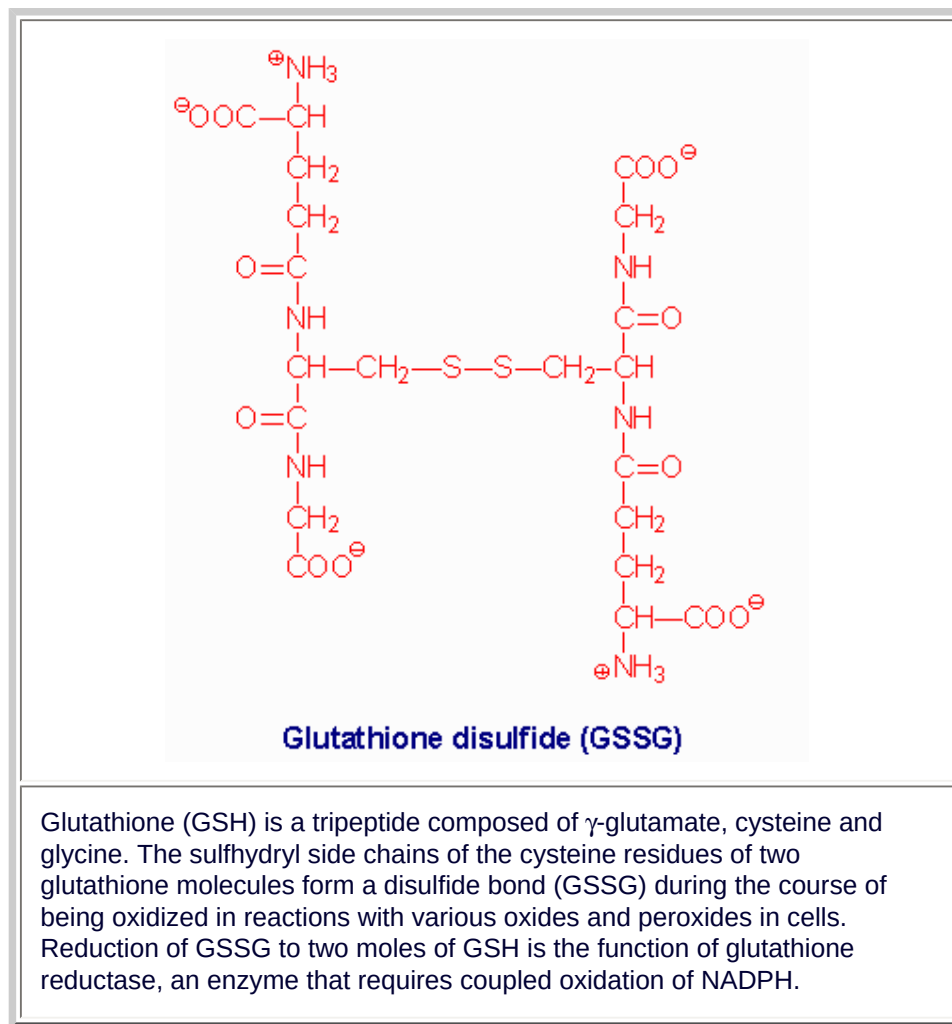
The net result of the PPP, if not used solely for R5P production, is the oxidation of G6P, a 6 carbon sugar, into a 5 carbon sugar.

In turn, 3 moles of 5 carbon sugar are converted, via the enzymes of the PPP, back into two moles of 6 carbon sugars and one mole of 3 carbon sugar. The 6 carbon sugars can be recycled into the pathway in the form of G6P, generating more NADPH. The 3 carbon sugar generated is glyceraldehyde-3-phosphate which can be shunted to glycolysis and oxidized to pyruvate. Alternatively, it can be utilized by the gluconeogenic enzymes to generate more 6 carbon sugars (fructose-6-phosphate or glucose-6-phosphate).

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## Metabolic Disorders Associated with the Pentose Phosphate Pathway

Oxidative stress within cells is controlled primarily by the action of the peptide, **glutathione, GSH**. See [Specialized Products of Amino Acids](#) for the synthesis of GSH.



Glutathione is the tripeptide  **$\gamma$ -glutamylcysteinylglycine**. The cysteine thiol plays the role in reducing oxidized thiols in other proteins. Oxidation of 2 cysteine thiols forms a disulfide bond. Although this bond plays a very important role in protein structure and function, inappropriately introduced disulfides can be detrimental. Glutathione can reduce disulfides nonenzymatically. Oxidative stress also generates peroxides that in turn can be reduced by glutathione to generate water and an alcohol, or 2 waters if the peroxide were hydrogen peroxide.

Regeneration of reduced glutathione is carried out by the enzyme, glutathione reductase. This enzyme requires the co-factor NADPH when operating in the direction of glutathione reduction which is the thermodynamically favored direction of the reaction.

It should be clear that any disruption in the level of NADPH may have a profound effect upon a cells ability to deal with oxidative stress. No other cell than the erythrocyte is exposed to greater oxidizing conditions. After all it is the oxygen carrier of the body.

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## Erythrocytes and the Pentose Phosphate Pathway

The predominant pathways of carbohydrate metabolism in the red blood cell (RBC) are glycolysis, the PPP and 2,3-



bisphosphoglycerate (2,3-BPG) metabolism (refer to discussion of [hemoglobin](#) for review of the synthesis and role of 2,3-BPG). Glycolysis provides ATP for membrane ion pumps and NADH for re-oxidation of methemoglobin. The PPP supplies the RBC with NADPH to maintain the reduced state of glutathione. The inability to maintain reduced glutathione in RBCs leads to increased accumulation of peroxides, predominantly  $\text{H}_2\text{O}_2$ , that in turn results in a weakening of the cell wall and concomitant hemolysis. Accumulation of  $\text{H}_2\text{O}_2$  also leads to increased rates of oxidation of hemoglobin to methemoglobin that also weakens the cell wall. Glutathione removes peroxides via the action of glutathione peroxidase. The PPP in erythrocytes is essentially the only pathway for these cells to produce NADPH. Any defect in the production of NADPH could, therefore, have profound effects on erythrocyte survival. Several deficiencies in the level of activity (not function) of glucose-6-phosphate dehydrogenase have been observed to be associated with resistance to the malarial parasite, *Plasmodium falciparum*, among individuals of Mediterranean and African descent. The basis for this resistance is the weakening of the red cell membrane (the erythrocyte is the host cell for the parasite) such that it cannot sustain the parasitic life cycle long enough for productive growth.

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## Myoglobin

Myoglobin and hemoglobin are hemeproteins whose physiological importance is principally related to their ability to bind molecular oxygen. Myoglobin is a monomeric heme protein found mainly in muscle tissue where it serves as an intracellular storage site for oxygen. During periods of oxygen deprivation **oxymyoglobin** releases its bound oxygen which is then used for metabolic purposes.

The tertiary structure of myoglobin is that of a typical water soluble globular protein. Its secondary structure is unusual in that it contains a very high proportion (75%) of  $\alpha$ -helical secondary structure. A myoglobin polypeptide is comprised of 8 separate right handed  $\alpha$ -helices, designated A through H, that are connected by short non helical regions. Amino acid R-groups packed into the interior of the molecule are predominantly hydrophobic in character while those exposed on the surface of the molecule are generally hydrophilic, thus making the molecule relatively water soluble.

Each myoglobin molecule contains one heme prosthetic group inserted into a hydrophobic cleft in the protein. Each heme residue contains one central coordinately bound iron atom that is normally in the  $\text{Fe}^{2+}$ , or ferrous, oxidation state. The oxygen carried by hemeproteins is bound directly to the ferrous iron atom of the heme prosthetic group. Oxidation of the iron to the  $\text{Fe}^{3+}$ , ferric, oxidation state renders the molecule incapable of normal oxygen binding. Hydrophobic interactions between the tetrapyrrole ring and hydrophobic amino acid R groups on the interior of the cleft in the protein strongly stabilize the heme protein conjugate. In addition a nitrogen atom from a histidine R group located above the plane of the heme ring is coordinated with the iron atom further stabilizing the interaction between the heme and the protein. In oxymyoglobin the remaining bonding site on the iron atom (the 6th coordinate position) is occupied by the oxygen, whose binding is stabilized by a second histidine residue.

Carbon monoxide also binds coordinately to heme iron atoms in a manner similar to that of oxygen, but the binding of carbon monoxide to heme is much stronger than that of oxygen. The preferential binding of carbon monoxide to heme iron is largely responsible for the asphyxiation that results from carbon monoxide poisoning.

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## Hemoglobin

Adult hemoglobin is a  $[\alpha(2):\beta(2)]$  tetrameric hemeprotein found in erythrocytes where it is responsible for binding oxygen in the lung and transporting the bound oxygen throughout the body where it is used in aerobic metabolic pathways. For a description of the different types of hemoglobin tetramers see the section below on [Hemoglobin Genes](#). Each subunit of a hemoglobin tetramer has a heme prosthetic group identical to that described for myoglobin. The common peptide subunits are designated  $\alpha$ ,  $\beta$ ,  $\gamma$  and  $\delta$  which are

arranged into the most commonly occurring functional hemoglobins.

Although the secondary and tertiary structure of various hemoglobin subunits are similar, reflecting extensive homology in amino acid composition, the variations in amino acid composition that do exist impart marked differences in hemoglobin's oxygen carrying properties. In addition, the quaternary structure of hemoglobin leads to physiologically important allosteric interactions between the subunits, a property lacking in monomeric myoglobin which is otherwise very similar to the  $\alpha$ -subunit of hemoglobin.

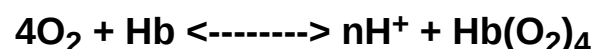
Comparison of the oxygen binding properties of myoglobin and hemoglobin illustrate the allosteric properties of hemoglobin that results from its quaternary structure and differentiate hemoglobin's oxygen binding properties from that of myoglobin. The curve of oxygen binding to hemoglobin is sigmoidal typical of allosteric proteins in which the substrate, in this case oxygen, is a positive homotropic effector. When oxygen binds to the first subunit of deoxyhemoglobin it increases the affinity of the remaining subunits for oxygen. As additional oxygen is bound to the second and third subunits oxygen binding is further, incrementally, strengthened, so that at the oxygen tension in lung alveoli, hemoglobin is fully saturated with oxygen. As oxyhemoglobin circulates to deoxygenated tissue, oxygen is incrementally unloaded and the affinity of hemoglobin for oxygen is reduced. Thus at the lowest oxygen tensions found in very active tissues the binding affinity of hemoglobin for oxygen is very low allowing maximal delivery of oxygen to the tissue. In contrast the oxygen binding curve for myoglobin is hyperbolic in character indicating the absence of allosteric interactions in this process.

The allosteric oxygen binding properties of hemoglobin arise directly from the interaction of oxygen with the iron atom of the heme prosthetic groups and the resultant effects of these interactions on the quaternary structure of the protein. When oxygen binds to an iron atom of deoxyhemoglobin it pulls the iron atom into the plane of the heme. Since the iron is also bound to histidine F8, this residue is also pulled toward the plane of the heme ring. The conformational change at histidine F8 is transmitted throughout the peptide backbone resulting in a significant change in tertiary structure of the entire subunit. Conformational changes at the subunit surface lead to a new set of binding interactions between adjacent subunits. The latter changes include disruption of salt bridges and formation of new hydrogen bonds and new hydrophobic interactions, all of which contribute to the new quaternary structure.

The latter changes in subunit interaction are transmitted, from the surface, to the heme binding pocket of a second deoxy subunit and result in easier access of oxygen to the iron atom of the second heme and thus a greater affinity of the hemoglobin molecule for a second oxygen molecule. The tertiary configuration of low affinity, deoxygenated hemoglobin (Hb) is known as the taut (T) state. Conversely, the quaternary structure of the fully oxygenated high affinity form of hemoglobin (HbO<sub>2</sub>) is known as the relaxed (R) state.

In the context of the affinity of hemoglobin for oxygen there are four primary regulators, each of which has a negative impact. These are CO<sub>2</sub>, hydrogen ion (H<sup>+</sup>), chloride ion (Cl<sup>-</sup>), and 2,3-bisphosphoglycerate (2,3BPG, or also just BPG). Some older texts abbreviate 2,3BPG as DPB. Although they can influence O<sub>2</sub> binding independent of each other, CO<sub>2</sub>, H<sup>+</sup> and Cl<sup>-</sup> primarily function as a consequence of each other on the affinity of hemoglobin for O<sub>2</sub>. We shall consider the transport of O<sub>2</sub> from the lungs to the tissues first.

In the high O<sub>2</sub> environment (high pO<sub>2</sub>) of the lungs there is sufficient O<sub>2</sub> to overcome the inhibitory nature of the T state. During the O<sub>2</sub> binding-induced alteration from the T form to the R form several amino acid side groups on the surface of hemoglobin subunits will dissociate protons as depicted in the equation below. This proton dissociation plays an important role in the expiration of the CO<sub>2</sub> that arrives from the tissues (see below). However, because of the high pO<sub>2</sub>, the pH of the blood in the lungs (~7.4 - 7.5) is not sufficiently low enough to exert a negative influence on hemoglobin binding O<sub>2</sub>. When the oxyhemoglobin reaches the tissues the pO<sub>2</sub> is sufficiently low, as well as the pH (~7.2), that the T state is favored and the O<sub>2</sub> released.

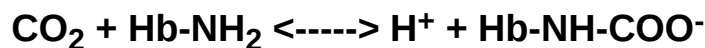


If we now consider what happens in the tissues, it is possible to see how  $\text{CO}_2$ ,  $\text{H}^+$ , and  $\text{Cl}^-$  exert their negative effects on hemoglobin binding  $\text{O}_2$ . Metabolizing cells produce  $\text{CO}_2$  which diffuses into the blood and enters the circulating red blood cells (RBCs). Within RBCs the  $\text{CO}_2$  is rapidly converted to carbonic acid through the action of carbonic anhydrase as shown in the equation below:

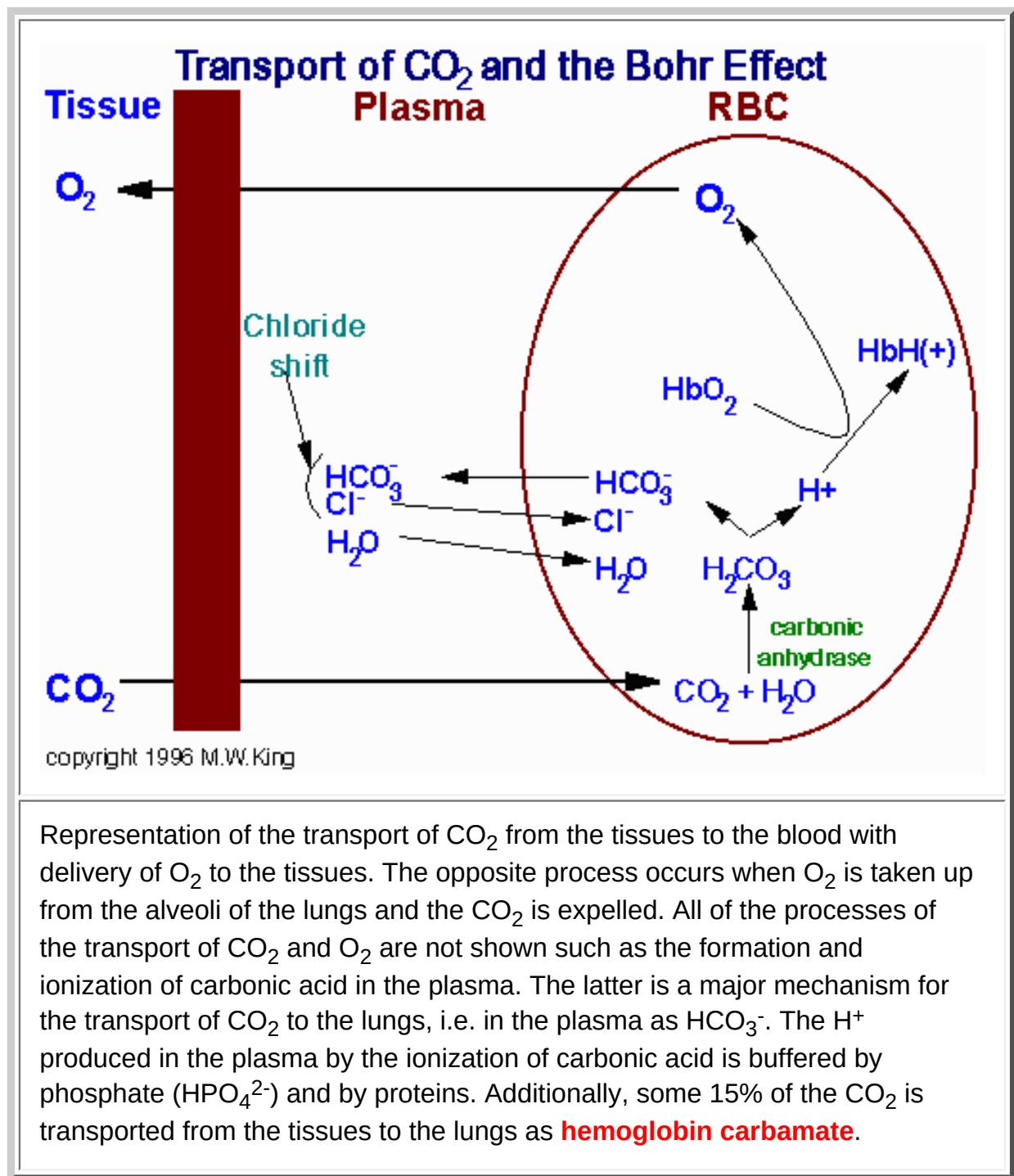


The bicarbonate ion produced in this dissociation reaction diffuses out of the RBC and is carried in the blood to the lungs. This effective  $\text{CO}_2$  transport process is referred to as **isohydric transport**. Approximately 80% of the  $\text{CO}_2$  produced in metabolizing cells is transported to the lungs in this way. A small percentage of  $\text{CO}_2$  is transported in the blood as a dissolved gas. In the tissues, the  $\text{H}^+$  dissociated from carbonic acid is buffered by hemoglobin which exerts a negative influence on  $\text{O}_2$  binding forcing release to the tissues. As indicated above, within the lungs the high  $\text{pO}_2$  allows for effective  $\text{O}_2$  binding by hemoglobin leading to the T to R state transition and the release of protons. The protons combine with the bicarbonate that arrived from the tissues forming carbonic acid which then enters the RBCs. Through a reversal of the carbonic anhydrase reaction,  $\text{CO}_2$  and  $\text{H}_2\text{O}$  are produced. The  $\text{CO}_2$  diffuses out of the blood, into the lung alveoli and is released on expiration.

In addition to isohydric transport, as much as 15% of  $\text{CO}_2$  is transported to the lungs bound to N-terminal amino groups of the T form of hemoglobin. This reaction, depicted below, forms what is called **carbamino-hemoglobin**. As indicated this reaction also produces  $\text{H}^+$ , thereby lowering the pH in tissues where the  $\text{CO}_2$  concentration is high. The formation of  $\text{H}^+$  leads to release of the bound  $\text{O}_2$  to the surrounding tissues. Within the lungs, the high  $\text{O}_2$  content results in  $\text{O}_2$  binding to hemoglobin with the concomitant release of  $\text{H}^+$ . The released protons then promote the dissociation of the carbamino to form  $\text{CO}_2$  which is then released with expiration.



As the above discussion demonstrates, the conformation of hemoglobin and its oxygen binding are sensitive to hydrogen ion concentration. These effects of hydrogen ion concentration are responsible for the well known **Bohr effect** in which increases in hydrogen ion concentration decrease the amount of oxygen bound by hemoglobin at any oxygen concentration (partial pressure). Coupled to the diffusion of bicarbonate out of RBCs in the tissues there must be ion movement into the RBCs to maintain electrical neutrality. This is the role of  $\text{Cl}^-$  and is referred to as the **chloride shift**. In this way,  $\text{Cl}^-$  plays an important role in bicarbonate production and diffusion and thus also negatively influences  $\text{O}_2$  binding to hemoglobin.

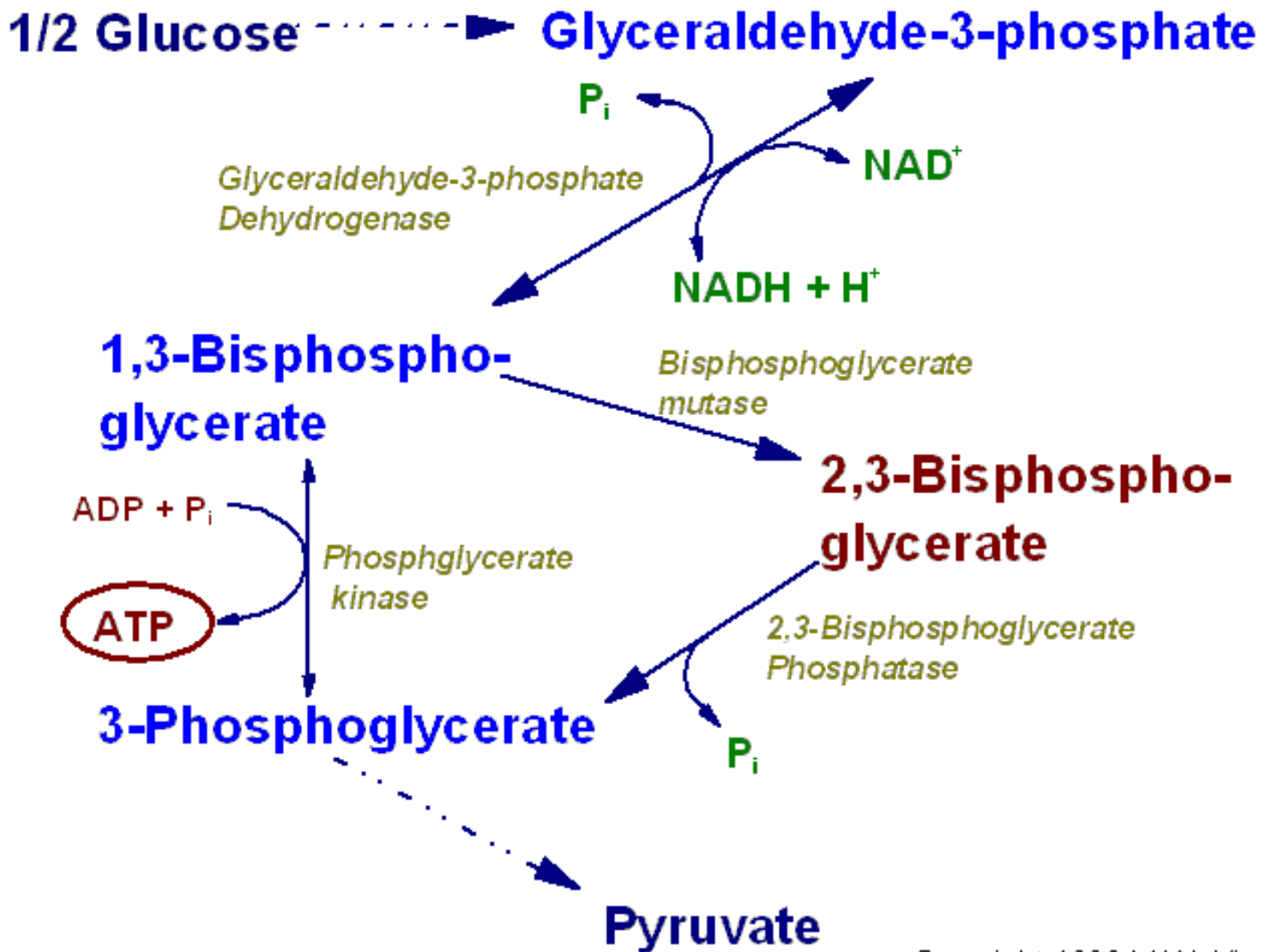


Representation of the transport of CO<sub>2</sub> from the tissues to the blood with delivery of O<sub>2</sub> to the tissues. The opposite process occurs when O<sub>2</sub> is taken up from the alveoli of the lungs and the CO<sub>2</sub> is expelled. All of the processes of the transport of CO<sub>2</sub> and O<sub>2</sub> are not shown such as the formation and ionization of carbonic acid in the plasma. The latter is a major mechanism for the transport of CO<sub>2</sub> to the lungs, i.e. in the plasma as HCO<sub>3</sub><sup>-</sup>. The H<sup>+</sup> produced in the plasma by the ionization of carbonic acid is buffered by phosphate (HPO<sub>4</sub><sup>2-</sup>) and by proteins. Additionally, some 15% of the CO<sub>2</sub> is transported from the tissues to the lungs as **hemoglobin carbamate**.

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## Role of 2,3-bisphosphoglycerate (2,3-BPG)

The compound **2,3-bisphosphoglycerate (2,3-BPG)**, derived from the glycolytic intermediate 1,3-bisphosphoglycerate, is a potent allosteric effector on the oxygen binding properties of hemoglobin. The pathway to 2,3BPG synthesis is diagrammed in the figure below.



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The pathway for 2,3-bisphosphoglycerate (**2,3-BPG**) synthesis within erythrocytes. Synthesis of 2,3-BPG represents a major reaction pathway for the consumption of glucose in erythrocytes. The synthesis of 2,3-BPG in erythrocytes is critical for controlling hemoglobin affinity for oxygen. Note that when glucose is oxidized by this pathway the erythrocyte loses the ability to gain 2 moles of ATP from glycolytic oxidation of 1,3-BPG to 3-phosphoglycerate via the phosphoglycerate kinase reaction.

In the deoxygenated T conformer, a cavity capable of binding 2,3-BPG forms in the center of the molecule. 2,3-BPG can occupy this cavity stabilizing the T state. Conversely, when 2,3-BPG is not available, or not bound in the central cavity, Hb can be converted to  $\text{HbO}_2$  more readily. Thus, like increased hydrogen ion concentration, increased 2,3-BPG concentration favors conversion of R form Hb to T form Hb and decreases the amount of oxygen bound by Hb at any oxygen concentration. Hemoglobin molecules differing in subunit composition are known to have different 2,3-BPG binding properties with correspondingly different allosteric responses to 2,3-BPG. For example, HbF (the fetal form of hemoglobin) binds 2,3-BPG much less avidly than HbA (the adult form of hemoglobin) with the result that HbF in fetuses of pregnant women binds oxygen with greater affinity than the mothers HbA, thus giving the fetus preferential access to oxygen carried by the mothers circulatory system.



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## The Hemoglobin Genes

The  $\alpha$ - and  $\beta$ -globin proteins contained in functional hemoglobin tetramers are derived from gene clusters. The  $\alpha$ -globin genes are on chromosome 16 and the  $\beta$ -globin genes are on chromosome 11. Both gene clusters contain not only the major adult genes,  $\alpha$  and  $\beta$ , but other expressed sequences that are utilized at different stages of development. The orientation of the genes in both clusters is in the same 5' to 3' direction with the earliest expressed genes at the 5' end of both clusters. In addition to functional genes, both clusters contain non-functional pseudogenes.

Hemoglobin synthesis begins in the first few weeks of embryonic development within the yolk sac. The major hemoglobin at this stage of development is a tetramer composed of 2 zeta ( $\zeta$ ) chains encoded within the  $\alpha$  cluster and 2 eta ( $\epsilon$ ) chains from the  $\beta$  cluster. By 6-8 weeks of gestation the expression of this version of hemoglobin declines dramatically coinciding with the change in hemoglobin synthesis from the yolk sac to the liver. Expression from the  $\alpha$  cluster consists of identical proteins from the  $\alpha 1$  and  $\alpha 2$  genes. Expression from these genes in the  $\alpha$  cluster remains on throughout life.

Within the  $\beta$ -globin cluster there is an additional set of genes, the fetal  $\beta$ -globin genes identified as the gamma ( $\gamma$ ) genes. The 2 fetal genes called  $G\gamma$  and  $A\gamma$ , the derivation of which stems from the single amino acid difference between the 2 fetal genes: glycine in  $G\gamma$  and alanine in  $A\gamma$  at position 136. These fetal  $\gamma$  genes are expressed as the embryonic genes are turned off.

Shortly before birth there is a smooth switch from fetal  $\gamma$ -globin gene expression to adult  $\beta$ -globin gene expression. The switch from fetal  $\gamma$ - to adult  $\beta$ -globin does not directly coincide with the switch from hepatic synthesis to bone marrow synthesis since at birth it can be shown that both  $\gamma$  and  $\beta$  synthesis is occurring in the marrow.

Given the pattern of globin gene activity throughout fetal development and in the adult the composition of the hemoglobin tetramers is of course distinct. Fetal hemoglobin is identified as HbF and includes both  $\alpha_2G\gamma_2$  and  $\alpha_2A\gamma_2$ . Fetal hemoglobin has a slightly higher affinity for oxygen than does adult hemoglobin. This allows the fetus to extract oxygen more efficiently from the maternal circulation. In adults the major hemoglobin is identified as HbA (more commonly HbA<sub>1</sub>) and is a tetramer of 2  $\alpha$  and 2  $\beta$  chains as indicated earlier. A minor adult hemoglobin, identified as HbA<sub>2</sub>, is a tetramer of 2  $\alpha$  chains and 2  $\delta$  chains. The  $\delta$  gene is expressed with a timing similar to the  $\beta$  gene but because the promoter has acquired a number of mutations its' efficiency of transcription is reduced.

The overall hemoglobin composition in a normal adult is approximately 97.5% HbA<sub>1</sub>, 2% HbA<sub>2</sub> and 0.5% HbF.

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## Hemoglobinopathies

A large number of mutations have been described in the globin genes. These mutations can be divided into two distinct types: those that cause **qualitative abnormalities** (e.g. sickle cell anemia) and those that cause **quantitative abnormalities** (the thalassemias). Taken together these disorders are referred to as the **hemoglobinopathies**.

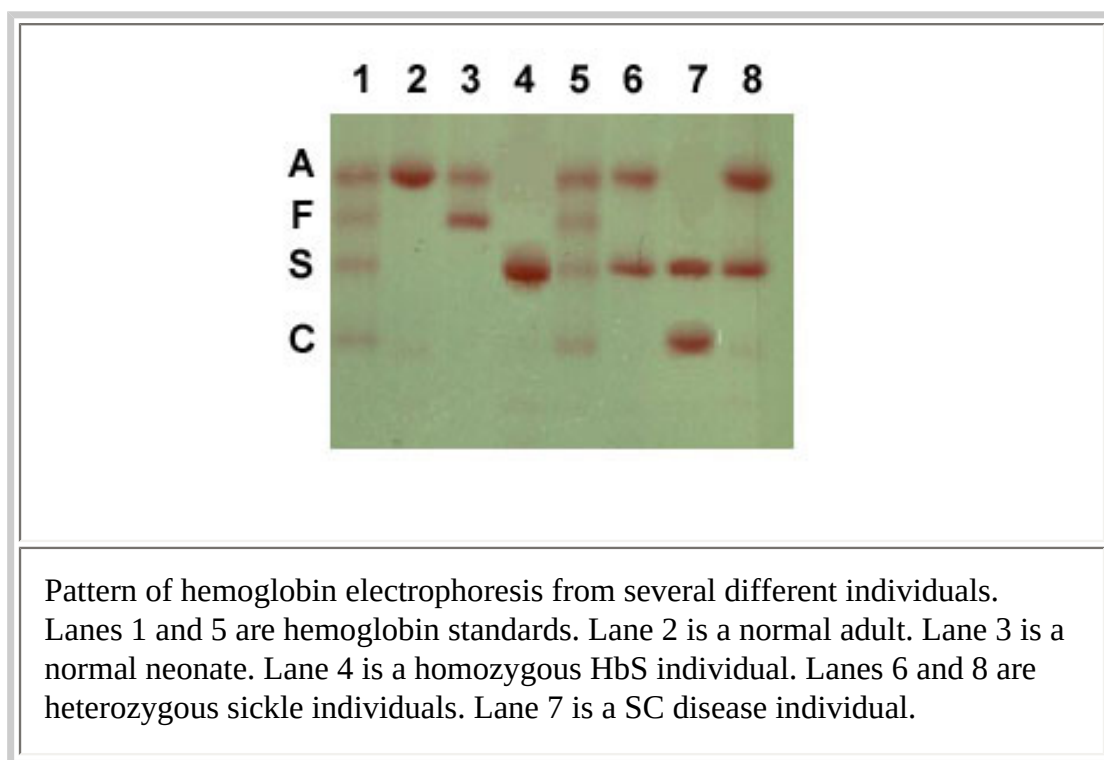
Of the mutations leading to qualitative alterations in hemoglobin, the missense mutation in the  $\beta$ -globin gene that causes **sickle cell anemia** is the most common. The mutation causing sickle cell anemia is a single nucleotide substitution (A to T) in the codon for amino acid 6. The change converts a glutamic acid codon (GAG) to a valine codon (GTG). The form of hemoglobin in persons with sickle cell anemia is referred

to as HbS.

The underlying problem in sickle cell anemia is that the valine for glutamic acid substitution results in hemoglobin tetramers that aggregate into arrays upon deoxygenation in the tissues. This aggregation leads to deformation of the red blood cell making it relatively inflexible and unable to traverse the capillary beds. Repeated cycles of oxygenation and deoxygenation leads to irreversible sickling. The end result is clogging of the fine capillaries. Because bones are particularly affected by the reduced blood flow, frequent and severe bone pain results. This is the typical symptom during a sickle cell "crisis". Long term the recurrent clogging of the capillary beds leads to damage to the internal organs, in particular the kidneys, heart and lungs. The continual destruction of the sickled red blood cells leads to chronic anemia and episodes of hyperbilirubinemia.

An additional relatively common mutation at codon 6 is the conversion to a lysine codon (AAG) which results in the generation of HbC.

Electrophoresis of hemoglobin proteins from individuals suspected of having sickle cell anemia (or several other types of hemoglobin disorders) is an effective diagnostic tool because the variant hemoglobins have different charges. An example of this technique is shown in the Figure below.



Another effective tool to identify the genotype of individuals suspected of having sickle cell disease as well as for prenatal diagnosis is to either carry out RFLP mapping or to use PCR. An example of the use of these tools can be seen in the [Molecular Tools of Medicine](#) page.

In addition to the missense mutations that lead to HbS and HbC, a number of frameshift mutations leading to qualitative abnormalities in hemoglobin have been identified. A 2-nucleotide insertion between codons 144 and 145 in the  $\beta$ -globin gene results in the generation of **hemoglobin Cranston**. The insertion, which is near the C-terminus of the  $\beta$ -globin protein, results in the normal stop codon being out of frame and synthesis proceeding into the 3'-untranslated region to a fortuitous stop codon. The result is a  $\beta$ -globin protein of 157 amino acids.

In the **hemoglobin Constant Spring** variant, a mutation in the  $\alpha$ -globin gene converts the stop codon (UAA) to a glutamine codon (CAA) so that the protein ends up being 31 amino acids longer than normal. The resultant  $\alpha$ -globin protein in hemoglobin Constant Spring is not only qualitatively altered but because it is

unstable it is a quantitative abnormality as well.

Because the globin gene loci contain clusters of similar genes there is the potential for **unequal cross-over** between the sister chromatids during meiosis. The generation of **hemoglobin Gun Hill** and **Lepore hemoglobins** are both the result of unequal cross over events. Hemoglobin Gun Hill is the result of a deletion of 15 nucleotides caused by unequal cross over between codons 91-94 of one  $\beta$ -globin gene and codons 96-98 of the other. Generation of Lepore hemoglobins results from unequal cross over between the  $\delta$ -globin and  $\beta$ -globin genes. The resultant hybrid  $\delta\beta$  gene is called Lepore and the  $\beta\delta$  hybrid gene is called anti-Lepore. As indicated earlier, the promoter of the  $\delta$ -globin gene is inefficient so the consequences of this unequal cross over event are both qualitative and quantitative.

The **thalassemias** are the result of abnormalities in hemoglobin synthesis and affect both clusters. Deficiencies in  $\beta$ -globin synthesis results in the  **$\beta$ -thalassemias** and deficiencies in  $\alpha$ -globin synthesis results in the  **$\alpha$ -thalassemias**. The term thalassemia is derived from the Greek thalassa meaning "sea" and was applied to these disorders because of the high frequency of their occurrence in individuals living around the Mediterranean Sea.

In normal individuals an equal amount of both  $\alpha$ - and  $\beta$ -globin proteins are made allowing them to combine stoichiometrically to form the correct hemoglobin tetramers. In the  $\alpha$ -thalassemias normal amounts of  $\beta$ -globin are made. The  $\beta$ -globin proteins are capable of forming homotetramers ( $\beta_4$ ) and these tetramers are called **hemoglobin H, HbH**. An excess of HbH in red blood cells leads to the formation of inclusion bodies commonly seen in patients with  $\alpha$ -thalassemia. In addition, the HbH tetramers have a markedly reduced oxygen carrying capacity. In  $\beta$ -thalassemia, where the  $\beta$ -globins are deficient, the  $\alpha$ -globins are in excess and will form  $\alpha$ -globin homotetramers. The  $\alpha$ -globin homotetramers are extremely insoluble which leads to premature red cell destruction in the bone marrow and spleen.

With the  $\alpha$ -thalassemias the level of  $\alpha$ -globin production can range from none to very nearly normal levels. This is due in part to the fact that there are 2 identical  $\alpha$ -globin genes on chromosome 16. Thus, the  $\alpha$ -thalassemias involve inactivation of 1 to all 4  $\alpha$ -globin genes. If 3 of the 4  $\alpha$ -globin genes are functional, individuals are completely asymptomatic. This situation is identified as the "silent carrier" state or sometimes as  $\alpha$ -thalassemia 2. Genotypically this situation is designated  $\alpha\alpha/\alpha-$  or  $\alpha-/ \alpha\alpha$ . If 2 of the 4 genes is inactivated individuals are designated as " $\alpha$ -thalassemia trait" or as  $\alpha$ -thalassemia 1. Genotypically this situation is designated  $\alpha\alpha/-$ . In individuals of African descent with  $\alpha$ -thalassemia 1, the disorder usually results from the inactivation of 1  $\alpha$ -globin gene on each chromosome and is designated  $\alpha-/ \alpha-$ . This means that these individuals are homozygous for the  $\alpha$ -thalassemia 2 chromosome. The phenotype of  $\alpha$ -thalassemia 1 is relatively benign. The mean red cell volume (designated MCV in clinical tests) is reduced in  $\alpha$ -thalassemia 1 but individuals are generally asymptomatic. The clinical situation becomes more severe if only 1 of the 4  $\alpha$ -globin genes is functional. Because of the dramatic reduction in  $\alpha$ -globin chain production in this latter situation, a high level of  $\beta_4$  tetramer is present. clinically this is referred to as **hemoglobin H disease**. Afflicted individuals have moderate to marked anemia and their MCV is quite low, but the disease is not fatal. The most severe situation results when no  $\alpha$ -globin chains are made (genotypically designated  $-/-$ ). This leads to prenatal lethality or early neonatal death. The predominant fetal hemoglobin in afflicted individuals is a tetramer of  $\gamma$ -chains and is referred to as **hemoglobin Bart's**. This hemoglobin has essentially no oxygen carrying capacity resulting in oxygen starvation in the fetal tissues. Heart failure results as the heart tries to pump the unoxygenated blood to oxygen starved tissues leading to marked edema. This latter situation is called **hydrops fetalis**.

A large number of mutations have been identified leading to decreased or absent production of  $\beta$ -globin chains resulting in the  $\beta$ -thalassemias. In the most severe situation mutations in both the maternal and paternal  $\beta$ -globin genes leads to loss of normal amounts of  $\beta$ -globin protein. A complete lack of HbA is denoted as  $\beta^0$ -thalassemia. If one or the other mutations allows production of a small amount of functional  $\beta$ -

globin then the disorder is denoted as  $\beta^+$ -thalassemia.

Both  $\beta^0$ - and  $\beta^+$ -thalassemias are referred to as **thalassemia major**, also called **Cooley's anemia** after Dr. Thomas Cooley who first described the disorder. Afflicted individuals suffer from severe anemia beginning in the first year of life leading to the need for blood transfusions. As a consequence of the anemia the bone marrow dramatically increases its' effort at blood production. The cortex of the bone becomes thinned leading to pathologic fracturing and distortion of the bones in the face and skull. In addition, there is marked hepatosplenomegaly as the liver and spleen act as additional sites of blood production. Without intervention these individuals will die within the decade of life. As indicated,  $\beta$ -thalassemia major patient require blood transfusions, however, in the long term these transfusions lead to the accumulation of iron in the organs, particularly the heart, liver and pancreas. Organ failure ensues with death in the teens to early twenties. Iron chelation therapies appear to improve the outlook for  $\beta$ -thalassemia major patients but this requires continuous infusion of the chelating agent.

Individuals heterozygous for  $\beta$ -thalassemia have what is termed **thalassemia minor**. Afflicted individuals harbor one normal  $\beta$ -globin gene and one that harbors a mutation leading to production of reduced or no  $\beta$ -globin. Individuals that do not make any functional  $\beta$ -globin protein from 1 gene are termed  $\beta^0$  heterozygotes. If  $\beta$ -globin production is reduced at one locus the individuals are termed  $\beta^+$  heterozygotes. Thalassemia minor individuals are generally asymptomatic.

The term **thalassemia intermedia** is used to designate individuals with significant anemia and who are symptomatic but unlike thalassemia major do not require transfusions. This syndrome results in individuals where both  $\beta$ -globin genes express reduced amounts of protein or where one gene makes none and the other makes a mildly reduced amount. A person who is a compound heterozygote with  $\alpha$ -thalassemia and  $\beta^+$ -thalassemia will also manifest as thalassemia intermedia.

The primary cause of  $\alpha$ -thalassemias is deletion, whereas, for  $\beta$ -thalassemias the mutations are more subtle. In  $\beta$ -thalassemias, point mutations in the promoter, mutations in the translational initiation codon, a point mutation in the polyadenylation signal and an array of mutations leading to splicing abnormalities have been characterized.

An interesting and common (up to 30% of persons from Southeast Asia) hemoglobinopathy that has both quantitative and qualitative characteristics is caused by the synthesis of **hemoglobin E**. Hemoglobin E arises due to a point mutation in codon 26 that changes glutamic acid (GAG) to lysine (AAG). Individuals with this mutation make only around 60% of the normal amount of  $\beta$ -globin protein. The reason for this is that the mutation creates a cryptic splice site such that 40% of the hemoglobin E mRNA is shorter by 16 nucleotides and does not give rise to detectable  $\beta$ -globin protein.

There are some individuals in whom the developmental timing of globin production is altered as a consequence of mutation. Persons with **hereditary persistence of fetal hemoglobin, HPFH** continue to make HbF as adults. Because the syndrome is benign most individuals do not even know they carry a hemoglobin abnormality. Many HPFH individuals harbor large deletions of the  $\delta$ - and  $\beta$ -coding region of the cluster. There is no deletion of the fetal globin genes and by an as yet uncharacterized mechanism expression of these genes persists in adulthood.

As discussed functional hemoglobin is a heterotetramer. In addition, mutations in either the  $\alpha$ -globin or the  $\beta$ -globin genes lead to quantitative and qualitative abnormalities in hemoglobin. Therefore, it should not be surprising that complex compound heterozygosities can result in offspring of individuals harboring different mutations.

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## Introduction

Modern molecular medicine encompasses the utilization of many molecular biological techniques in the analysis of disease, disease genes and disease gene function. The study of disease genes and their function in an unaffected individual has been possible by the development of **recombinant DNA** and **cloning** techniques. The basis of the term recombinant DNA refers to the recombining of different segments of DNA. Cloning refers to the process of preparing multiple copies of an individual type of recombinant DNA molecule. The classical mechanisms for producing recombinant molecules involves the insertion of exogenous fragments of DNA into either bacterially derived **plasmid** (circular double stranded



autonomously replicating DNAs found in bacteria) vectors or **bacteriophage** (viruses that infect bacteria) based vectors. The term **vector** refers to the DNA molecule used to **carry** or **transport** DNA of interest into cells.

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## Common Enzymes used in Molecular Biology

| Enzyme(s)                  | Activity                                                                                                | Comments                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Restriction endonucleases  | recognize specific nucleotide sequences and cleaves the DNA within or near to the recognition sequences | <a href="#">see below</a>                                                                                                              |
| Reverse transcriptase (RT) | retrovirally encoded RNA-dependent DNA polymerase                                                       | used to convert mRNA into a complimentary DNA ( <b>cDNA</b> ) copy for the purpose of cloning cDNAs                                    |
| RNase H                    | recognizes RNA-DNA duplexes and randomly cleaves the phosphodiester backbone of the RNA                 | used primarily to cleave the mRNA strand that is annealed to the first strand of cDNA generated by reverse transcription               |
| DNA polymerase             | synthesis of DNA                                                                                        | used during most procedures where DNA synthesis is required, also used in in vitro mutagenesis                                         |
| Klenow DNA polymerase      | proteolytic fragment of DNA polymerase that lacks the 5' ----> 3' exonuclease activity                  | used to incorporate radioactive nucleotides into restriction enzyme generated ends of DNA, also can be used in place of DNA polymerase |

|                         |                                                                     |                                                                                                                                                                                                                                                  |
|-------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNA ligase              | covalently attaches a free 5' phosphate to a 3' hydroxyl            | used in all procedures where to molecules of DNA need to be covalently attached                                                                                                                                                                  |
| Alkaline phosphatase    | removes phosphates from 5' ends of DNA molecules                    | used to allow 5' ends to be subsequently radiolabeled with the $\gamma$ -phosphate of ATP in the presence of polynucleotide kinase, also used to prevent self-ligation of restriction enzyme digested plasmids and lambda vectors                |
| Polynucleotide kinase   | introduces $\gamma$ -phosphate of ATP to 5' ends of DNA             | see above for alkaline phosphatase                                                                                                                                                                                                               |
| DNase I                 | randomly hydrolyzes the phosphodiester bonds of double-stranded DNA | is used in the identification of regions of DNA that are bound by protein and thereby protected from DNase I digestion, also used to identify transcriptionally active regions of chromatin since they are more susceptible to DNase I digestion |
| S <sub>1</sub> Nuclease | exonuclease that recognizes single-stranded regions of DNA          | used to remove regions of single strandedness in DNA or RNA-DNA duplexes                                                                                                                                                                         |
| Exonuclease III         | exonuclease that removes nucleotides from the 3' end of DNAs        | used to generate deletions in DNA for sequencing, or to map functional domains of DNA duplexes                                                                                                                                                   |

|                                 |                                                                                                                 |                                                                                                                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal transferase            | DNA polymerase that requires only a 3'-OH, lengthens 3' ends with any dNTP                                      | used to introduce homopolymeric (same dNTP) <b>tails</b> onto the 3' ends of DNA duplexes, also used to introduce radiolabeled nucleotides on the 3' ends of DNA |
| T3, T7, and SP6 RNA polymerases | bacterial virus encoded RNA polymerase, recognize specific nucleotide sequences for initiation of transcription | used to synthesize RNA <i>in vitro</i>                                                                                                                           |
| Taq and Vent DNA polymerase     | thermostable DNA polymerases                                                                                    | used in <b>PCR</b>                                                                                                                                               |
| Taq and Vent DNA ligases        | thermostable DNA ligases                                                                                        | used in <b>LCR</b>                                                                                                                                               |

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## Restriction Endonucleases

Restriction endonuclease are enzymes that will recognize, bind to and hydrolyze specific nucleic acid sequences in double-stranded DNA. The term **restriction endonuclease** was given to this class of bacterially derived enzymes since they were identified as being involved in **restricting** the growth of certain bacteriophages. Bacteria are capable of modifying specific sequences within their genomes by methylation which prevents their own DNA from being recognized by the restriction enzymes encoded by their genomes. This process is termed **modification and restriction**. Infecting bacteriophage DNA is not modified and, hence, will be digested by the restriction endonucleases present in the bacterium.

The key to the *in vitro* utilization of restriction endonucleases is their strict nucleotide sequence specificity. The different enzymes are identified by being given a name indicating the bacteria from which they were isolated, e.g. the enzymes **EcoRI** which recognizes the sequences, **5'-GAATTC-3'**, was isolated from *Escherichia coli*. One unique feature of restriction enzymes is that the nucleotide sequences they recognize are palindromic, i.e. they are the same sequences in the 5' ---> 3' direction of both strands. Some restriction endonucleases make staggered symmetrical cuts away from the center of their recognition site within the DNA duplex, some make symmetrical cuts in the middle of their recognition site

while still others cleave the DNA at a distance from the recognition sequence. Enzymes that make staggered cuts leave the resultant DNA with cohesive or sticky ends. Enzymes that cleave the DNA at the center of the recognition sequence leave blunt-ended fragments of DNA.

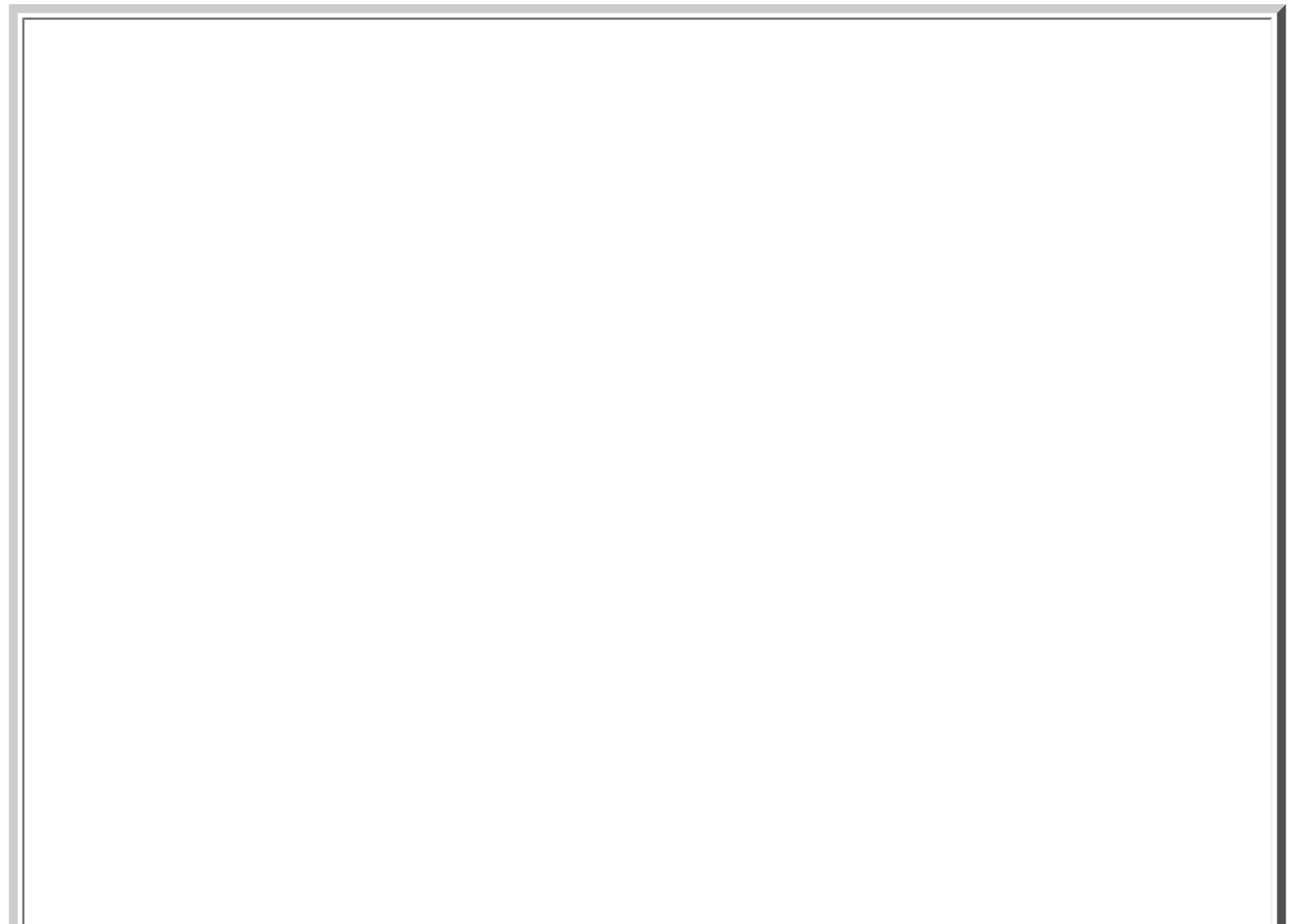
Any two pieces of DNA containing the same sequences within their sticky ends can anneal together and be covalently ligated together in the presence of **DNA ligase**. Any two blunt-ended fragments of DNA can be ligated together irrespective of the sequences at the ends of the duplexes.

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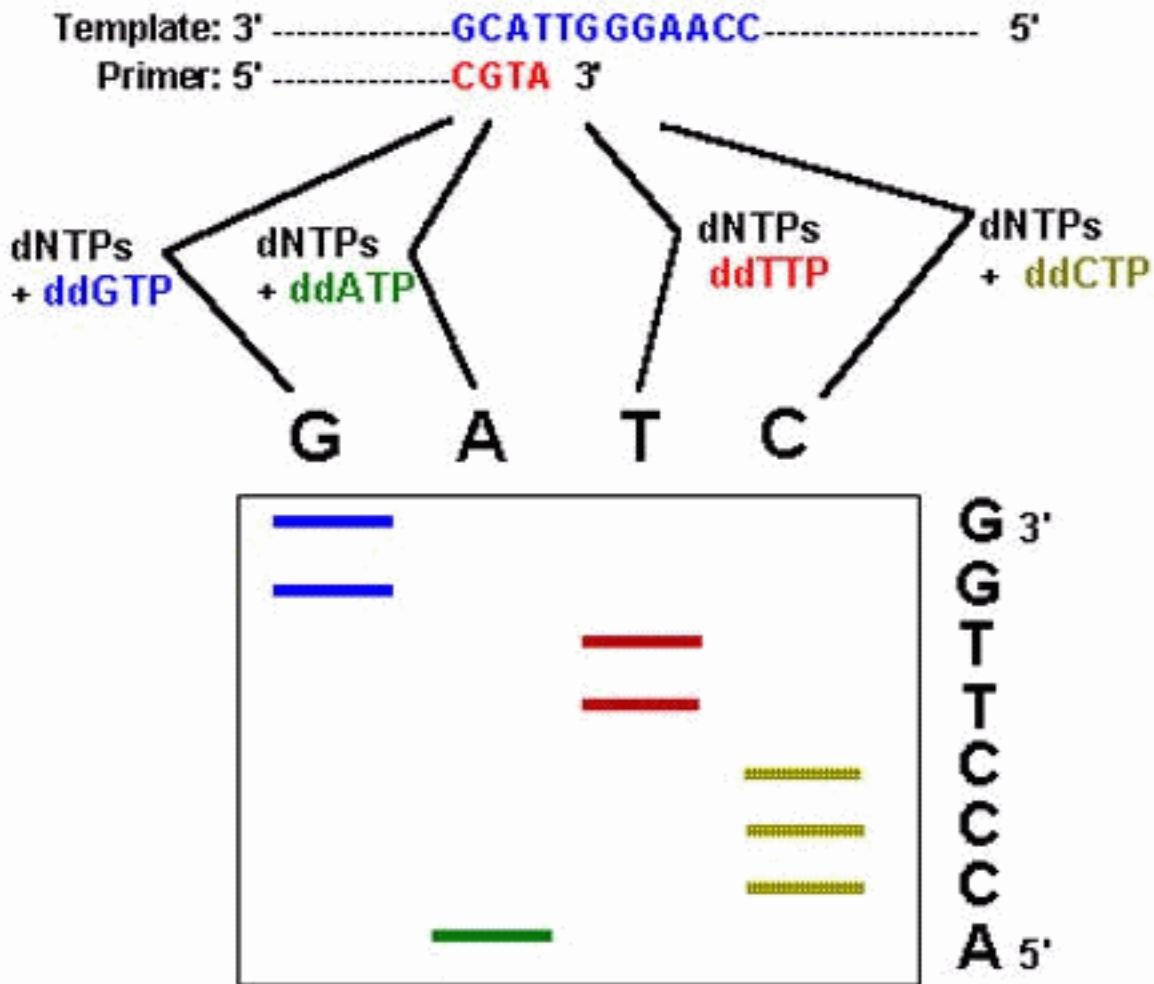
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## DNA Sequencing

Sequencing of DNA can be accomplished by either chemical or enzymatic means. The original technique for sequencing, **Maxam and Gilbert sequencing**, relies on the nucleotide-specific chemical cleavage of DNA and is not routinely used any more. The enzymatic technique, **Sanger sequencing**, involves the use of dideoxynucleotides (2',3'-dideoxy) that terminate DNA synthesis and is, therefore, also called dideoxy chain termination sequencing.



# Sanger ddNTP Chain Termination Sequencing



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The Sanger DNA sequencing protocol utilizes dideoxynucleotides (ddNTPs) to terminate chain elongation during the *in vitro* synthesis of DNA from a cloned template. Synthesis is initiated using a specific oligonucleotide primer. During the synthesis reaction a radioactive nucleotide (usually dATP) is incorporated into the elongating strands. Four separate reactions are carried out simultaneously, each of which contains all 4 dNTPs and a single ddNTP. The higher the concentration of ddNTP the more frequently chain elongation will terminate. Therefore, one can regulate the extent of sequence information obtainable by varying the dNTP/ddNTP ratio. Following the extension reactions the products are resolved by electrophoresis in a denaturing (urea) polyacrylamide gel. The results are obtained when the gel is dried and exposed to x-ray film. Bands near the bottom of the gel represent short reaction products (ie closest to the 3'-end of the primer) and those near the top the longest products.

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## Background to Cloning

Any fragment of DNA can be cloned once it is introduced into a suitable vector for transforming a bacterial host. Cloning refers to the production of large quantities of identical DNA molecules and usually involves the use of a bacterial cell as a host for the DNA, although cloning can be done in eukaryotic cells as well. **cDNA cloning** refers to the production of a library of cloned DNAs that represent all mRNAs present in a particular cell or tissue. **Genomic cloning** refers to the production of a library of cloned DNAs representing the entire genome of a particular organism. From either of these types of libraries one can isolate (by a variety of screening protocols) a single cDNA or gene clone.

In order to clone either cDNAs or copies of genes a **vector** is required to **carry** the cloned DNA. Vectors used in molecular biology are of two basic classes. One class of vectors is derived from bacterial **plasmids**. Plasmids are circular DNAs found in bacteria that replicate autonomously from the host genome. These DNAs were first identified because they harbored genes that conferred antibiotic resistance to the bacteria. The antibiotic resistance genes found on the original plasmids are used in modern in vitro engineered plasmids to allow selection of bacteria that have taken up the plasmids containing the DNAs of interest. Plasmids are limited in that in general fragments of DNA less than 10,000 base pairs (bp) can be cloned. In practice fragments of around 5,000 bp are the limit.

The second class of vectors are derived from the bacteriophage (bacterial virus) lambda. This virus is capable of both **lysogeny** (integration into the host genome) and **lysis** (infection followed by lysis of the infected host). The genes required for lysogeny have been removed from the lambda based vectors in order to allow only the lytic life cycle to take place. The advantage to lambda-based vectors is that they can carry fragments of DNA up to 25,000 bp. In the analysis of the human genome even lambda-based vectors are limiting and a **yeast artificial chromosome (YAC)** vector system has been developed for the cloning of DNA fragments up to 500,000 bp (see below).

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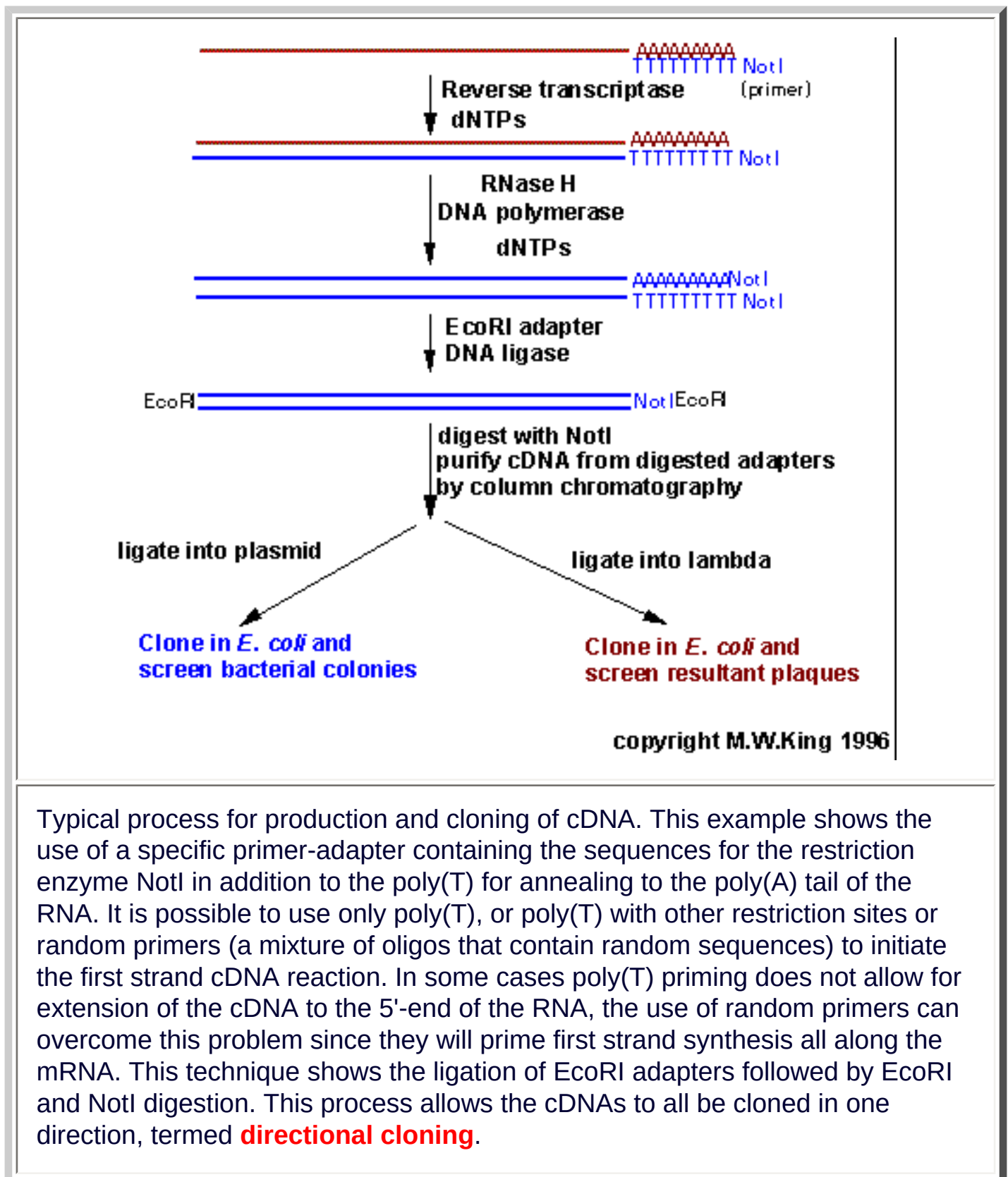
## cDNA Cloning

cDNAs are made from the mRNAs of a cell by any number of related techniques. Each technique consists of first reverse transcription of the mRNA followed by synthesis of the second strand of DNA and insertion of the double-stranded cDNA into either a plasmid or lambda vector for cloning. This process creates a **library** of cloned cDNA representing each mRNA species. Screening of the library for a particular cDNA clone is accomplished using nucleic acid or protein-based (proteins or antibodies) probes. cDNA libraries can also be screened by biological assay of the products produced by the cloned cDNAs. Screening with proteins, antibodies or by biological assay are mechanisms for analysis of the expression of proteins from cloned cDNAs and is given the term **expression cloning**. Nucleic acids probes can be generated from DNA (including synthetic oligonucleotides, oligos) or RNA. Nucleic acid



probes can be radioactively labeled or labeled with modified nucleotides that are recognizable by specific antibodies and detected by colorimetric or chemiluminescent assays.

A typical cDNA cloning protocol is diagrammed below.



Typical process for production and cloning of cDNA. This example shows the use of a specific primer-adapter containing the sequences for the restriction enzyme NotI in addition to the poly(T) for annealing to the poly(A) tail of the RNA. It is possible to use only poly(T), or poly(T) with other restriction sites or random primers (a mixture of oligos that contain random sequences) to initiate the first strand cDNA reaction. In some cases poly(T) priming does not allow for extension of the cDNA to the 5'-end of the RNA, the use of random primers can overcome this problem since they will prime first strand synthesis all along the mRNA. This technique shows the ligation of EcoRI adapters followed by EcoRI and NotI digestion. This process allows the cDNAs to all be cloned in one direction, termed **directional cloning**.

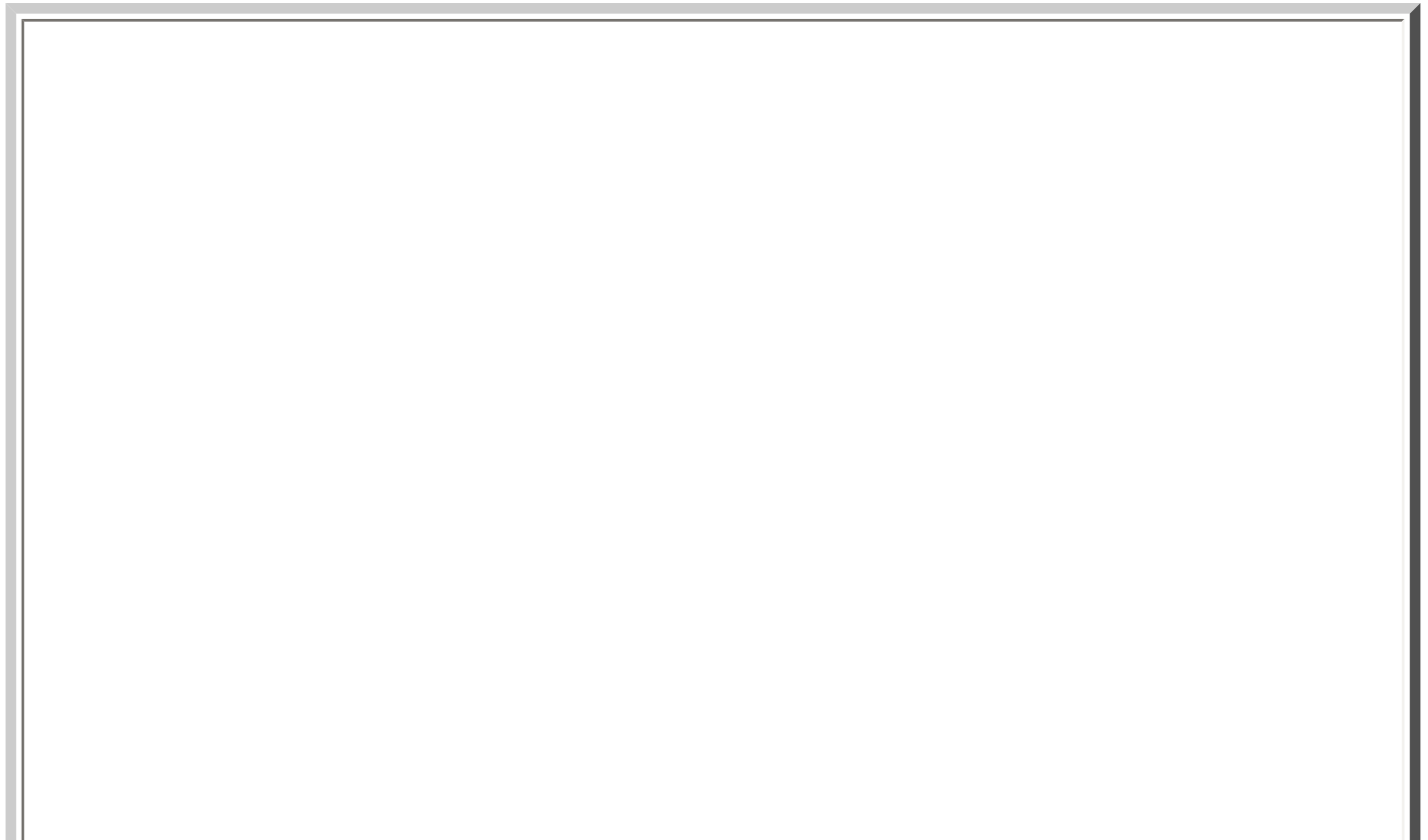
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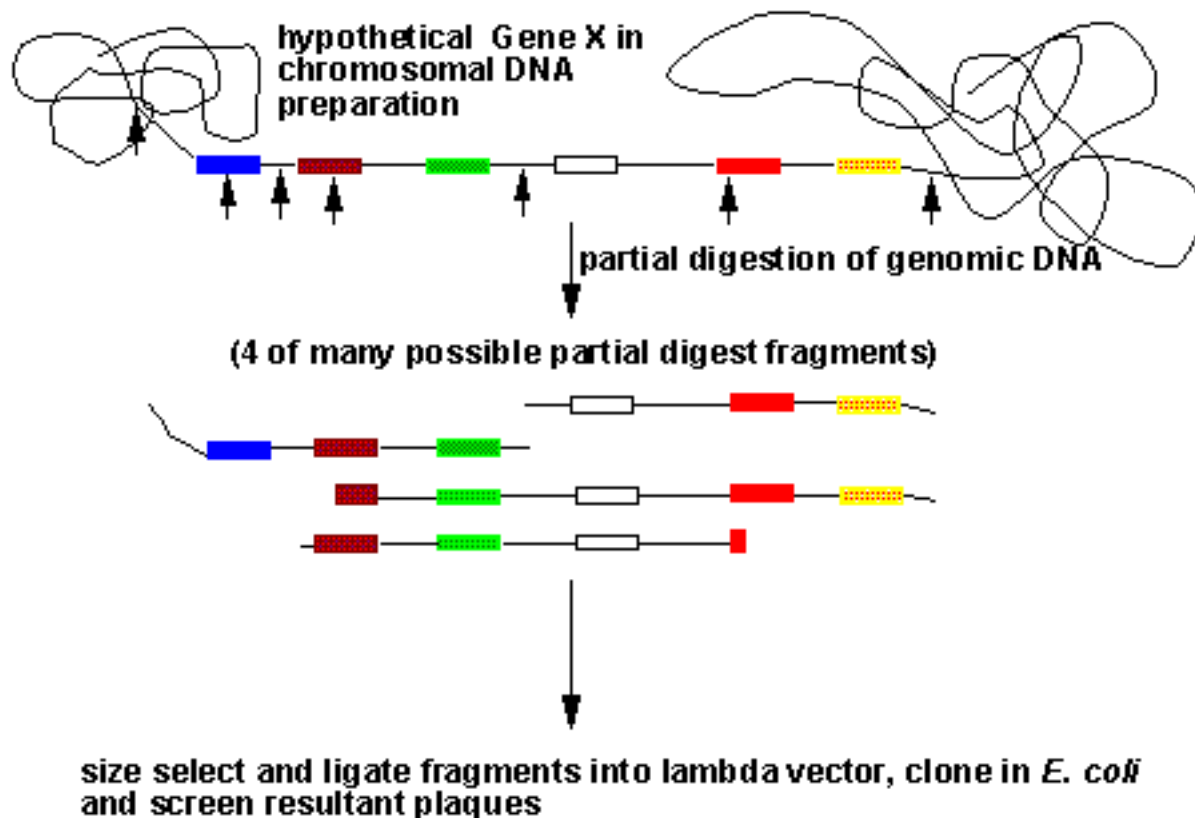
## Genomic Cloning

The majority of genomic cloning utilizes lambda-based vector systems. These vector systems are capable of carrying 15-25,000 bp of DNA. Cloning slightly larger fragments of genomic DNA can be accomplished using a chimeric plasmid-lambda vector system termed a **cosmid**. Cosmid vectors contain only the **cos** (cohesive) ends of the lambda genome (required for packaging the DNA into infectious virus particles) along with a plasmid antibiotic resistance gene and origin of DNA replication. Since approximately 30,000 bp of lambda DNA have been removed from cosmid vectors, larger genomic DNA fragments can be cloned. Still larger genomic DNA fragments can be cloned into **YAC** vectors (see below).

Genomic DNA can be isolated from any cell or tissue for cloning. The genomic DNA is first digested with restriction enzymes to generate fragments in the size range that are optimal for the vector being utilized for cloning. Given that some genes encompass many more base pairs than can be inserted into conventional lambda or cosmid vectors, the clones that are present in a genomic library must be overlapping. In order to generate overlapping clones, the DNA is only partially digested with restriction enzymes. This means that not every restriction site, present in all the copies of a given gene in the preparation of DNA, is cleaved. The partially digested DNA is then size-selected by a variety of techniques (e.g. gel electrophoresis or gradient centrifugation) prior to cloning. Screening of genomic libraries is accomplished primarily with nucleic acid-based probes. However, they can be screened with proteins that are known to bind specific sequences of DNA (e.g. transcription factors).

A typical genomic DNA cloning protocol is diagrammed below.





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Diagrammatic representation of a hypothetical gene present in a preparation of genomic DNA. The boxes indicate exons and the lines separating the boxes represent introns. The bold arrows indicate the positions of restriction enzyme sites, e.g. *Sau3A*I. Following partial enzyme digestion a wide range of different fragments of the gene will be generated, 4 possible fragments are indicated. Fragments in the size range of 15 - 25 kilobase pairs (kbp) are purified by gel electrophoresis or gradient centrifugation and ligated into a lambda vector. The DNA is packaged into phage particles in vitro and used to infect *E. coli*.

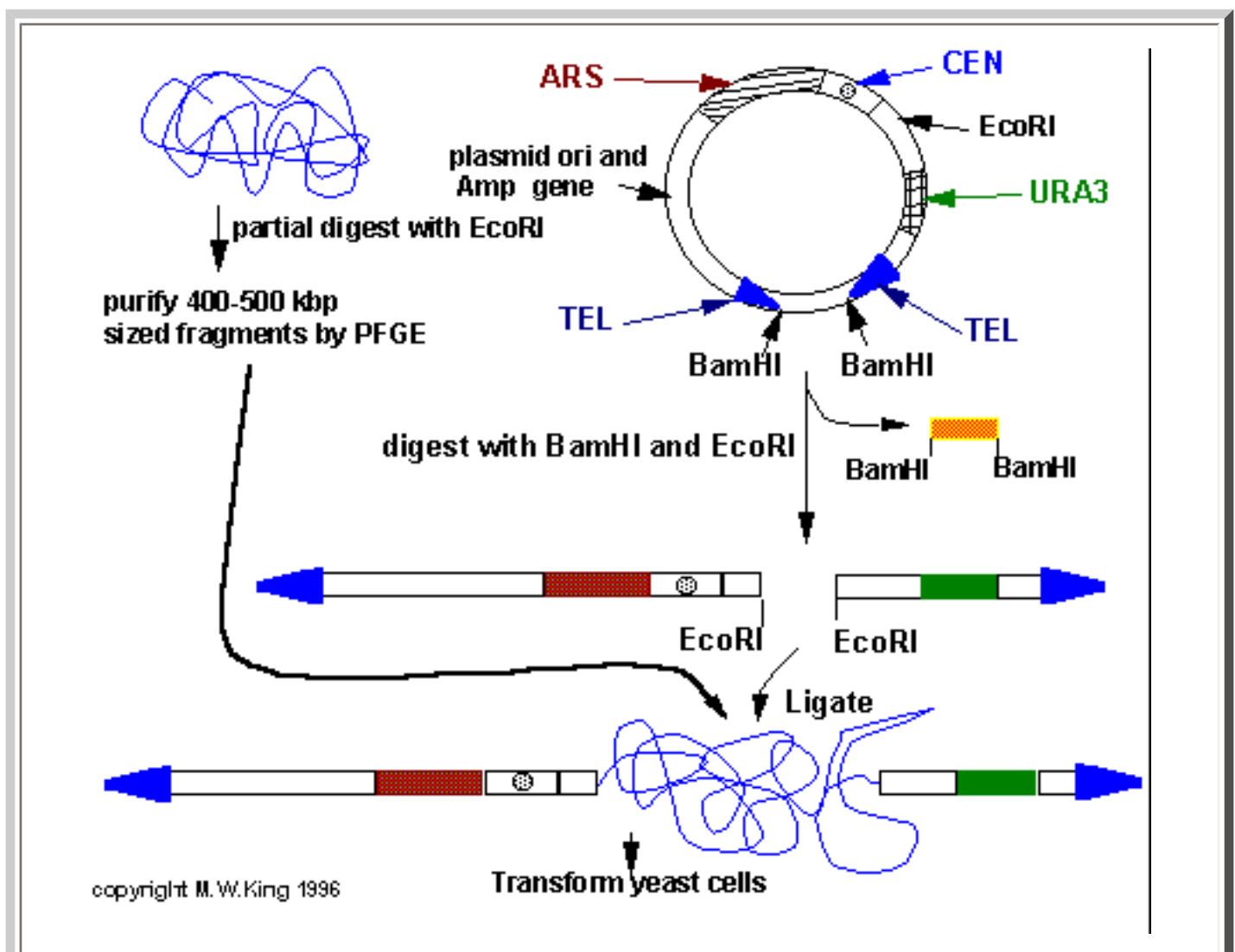
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## Cloning Genomic DNA in YAC Vectors

YAC vectors allow the cloning, within yeast cells, fragments of genomic DNA that approach 500,000 bp. These vectors contain several elements of typical yeast chromosomes, hence the term YAC. The YAC vectors contain a yeast **centromere (CEN)**, yeast **telomeres (TEL)**, telomeres are the specific sequences that are present at the ends of chromosomes and

are necessary for replication) and a yeast **autonomously replicating sequence (ARS)**. Yeast ARSes are essentially origins of replication that function in yeast cells autonomously from the replication of yeast chromosomal replication origins. YAC vectors also contain genes, (e.g. URA3, a gene involved in uracil synthesis) that allow selection of yeast cells that have taken up the vector. In order to propagate the vector in bacterial cells, prior to insertion of genomic DNA, YAC vectors contain a bacterial replication origin and a bacterial selectable marker such as the gene for ampicillin resistance.

In the cloning of genomic DNA in a typical YAC vector, the genomic DNA is partially digested with EcoRI and fragments in the range of 400 - 500 kilobase pairs (kbp) are purified by **pulsed field gel electrophoresis, PFGE**. The YAC vector is digested with EcoRI and BamHI which places the telomere sequences at the ends of the linearized vector. The small BamHI fragment is separated from the rest of the YAC vector by standard gel electrophoresis. The genomic DNA is then ligated into the vector and then used to transform yeast cells.



Diagrammatic representation of a typical YAC vector used to clone genomic DNA. The vector contains yeast telomeres (TEL), a centromere (CEN), a selectable marker (URA3), and autonomously replicating sequences (ARS) as well as bacterial plasmid sequences for antibiotic selection and replication in *E. coli*.

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## Analysis of Cloned Products

The analysis of cloned cDNAs and genes involves a number of techniques. The initial characterization usually involves mapping of the number and location of different restriction enzyme sites. This information is useful for DNA sequencing since it provides a means to digest the clone into specific fragments for sub-cloning, a process involving the cloning of fragments of a particular cloned DNA. Once the DNA is fully characterized cDNA clones can be used to produce RNA in vitro and the RNA translated in vitro to characterize the protein. Clones of cDNAs also can be used as probes to analyze the structure of a gene by **Southern blotting** or to analyze the size of the RNA and pattern of its expression by **Northern blotting**. Northern blotting is also a useful tool in the analysis of the exon-intron organization of gene clones since only fragments of a gene that contain exons will hybridize to the RNA on the blot.

### Southern Blotting

Southern blotting is the analysis of DNA structure following its attachment to a solid support. The DNA to be analyzed is first digested with a given restriction enzyme then the resultant DNA fragments are separated in an agarose gel. The gel is treated with NaOH to denature the DNA, then the NaOH is neutralized. The DNA is transferred from the gel to nitrocellulose or nylon filter paper by either capillary diffusion or under electric current. The DNA is fixed to the filter by baking or ultraviolet light treatment. The filter can then be probed for the presence of a given fragment of DNA by various radioactive or non-radioactive means.

### Northern Blotting

Northern blotting involves the analysis of RNA following its attachment to a solid support. The RNA is sized by gel electrophoresis then transferred to nitrocellulose or nylon filter paper as for Southern blotting. Probing the filter for a particular RNA is done similarly to probing of Southern blots.

### Western Blotting

Western blotting involves the analysis of proteins following attachment to a solid support. The proteins are separated by size SDS-PAGE and electrophoretically transferred to nitrocellulose or nylon filters. The filter is then probed with antibodies raised against a particular protein.

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## Restriction Fragment Length Polymorphism (RFLP) Analysis

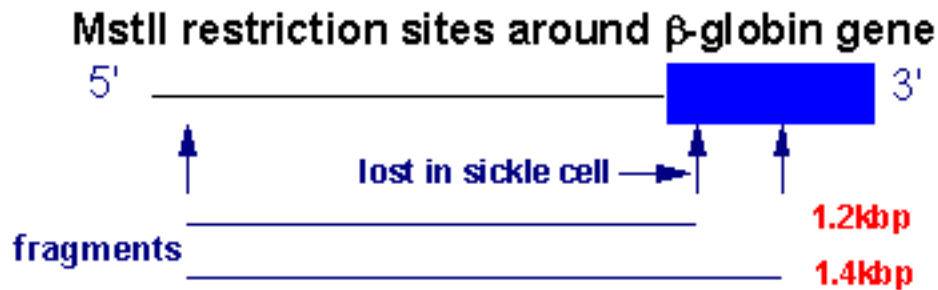
The genetic variability at a particular locus (gene) due to even minor base changes can alter the pattern of restriction enzyme digestion fragments that can be generated. Pathogenic alterations to the genotypic can be due to deletions or insertions within the gene being analyzed or even single nucleotide substitutions that can create or delete a restriction enzyme recognition site. RFLP analysis takes advantage of this and utilizes **Southern blotting** of restriction enzyme digested genomic DNA to detect familial patterns of the fragments of a given gene, detectable by screening the Southern blot with a probe corresponding to the gene of interest. A classic example of a disease detectable by RFLP is **sickle cell anemia**.

Sickle cell anemia results (at the level of the gene) from a single nucleotide change (A --> T) at codon 6 within the  $\beta$ -globin gene. This alteration leads to a **glu ----> val** amino acid substitution, while at the same time abolishing a *MstII* restriction site. As a result a  $\beta$ -globin gene probe can be used to detect the different *MstII* restriction fragments. It should be recalled that there are two copies of each gene in all human cells, therefore, RFLP analysis detects both copies: the affected allele and the unaffected allele.

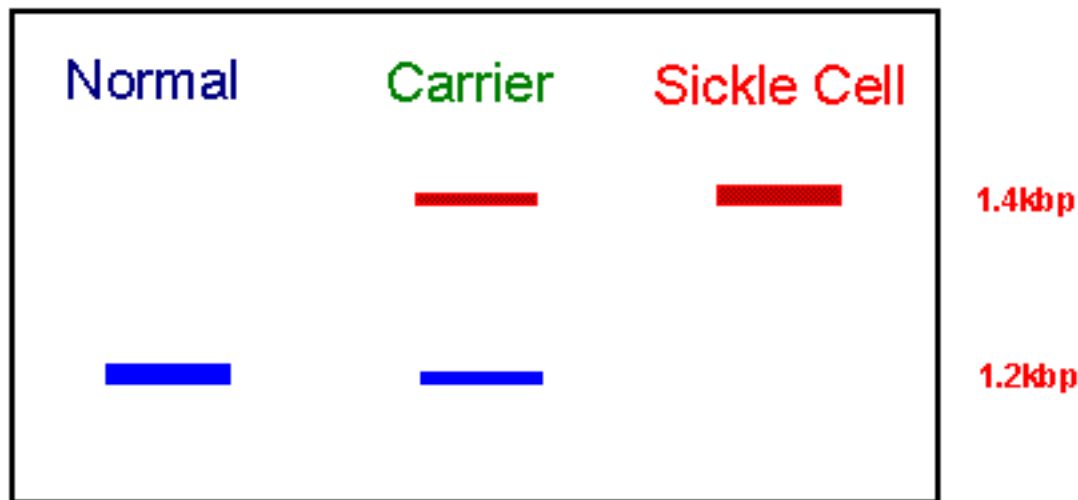
Size variability in detectable fragments within a family pedigree indicate differences in the pattern of restriction sites within and around the gene being analyzed. RFLP patterns are inherited and segregate in Mendelian fashion thus, allowing their use in genotyping such as in cases of paternity dispute or in criminal investigations.

Another form of DNA polymorphism detectable by classical RFLP mapping results from the inherited variations in the number of tandemly repeated DNA sequence elements that are from 2 to 60 bp in length. The number of repeats is also variable from 2 to 40 copies. These elements are termed **variable number tandem repeats (VNTRs)**. When restriction enzyme digestion cuts DNA flanking the VNTRs, the lengths of the resultant fragments will be variable depending upon the number of repeats at a given locus. Many different VNTR loci have been identified and are extremely useful for **DNA fingerprint analysis** such as in forensic and paternity identity cases.





### Possible RFLP Data



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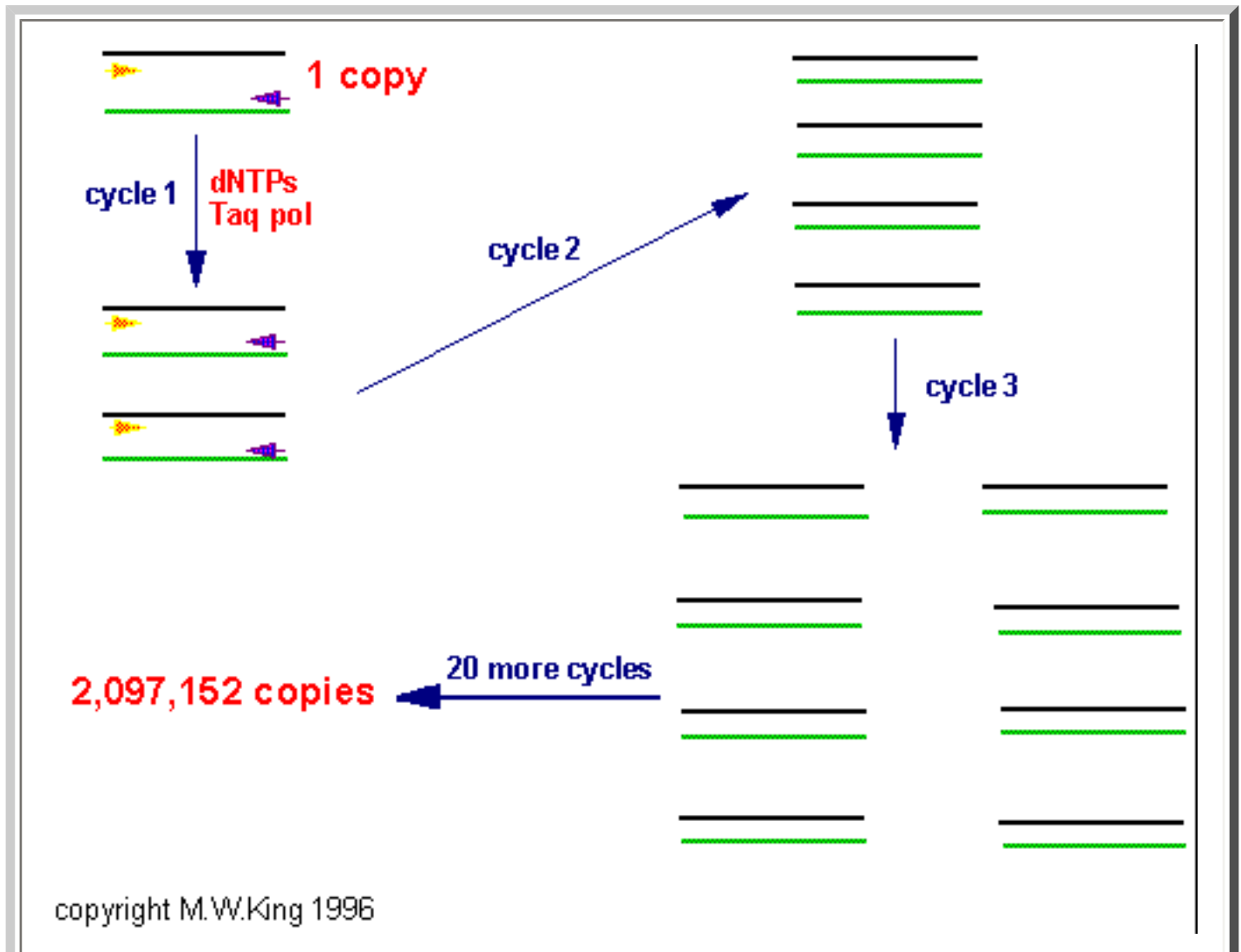
Diagrammatic representation of an RFLP analysis for the presence of the sickle-cell locus. Genomic DNA is isolated and digested with the restriction enzyme MstII. One MstII site is lost at the sickle-cell locus. The DNA is then Southern blotted and analyzed with a  $\beta$ -globin-specific probe corresponding to sequences at the 5'-end of the gene. Individuals homozygous for the normal globin genes will exhibit a single hybridization band since both maternal and paternal genes are unaffected. Heterozygotes will exhibit the normal band and the larger sickle-cell gene band. Homozygous sickle-cell individuals will exhibit a single larger hybridization band.

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## The Polymerase Chain Reaction (PCR)

The PCR is a powerful technique used to amplify DNA millions of fold, by repeated replication of a template, in a short period of time. The process utilizes sets of specific in vitro

synthesized oligonucleotides to prime DNA synthesis. The design of the primers is dependent upon the sequences of the DNA that is desired to be analyzed. The technique is carried out through many cycles (usually 20 - 50) of melting the template at high temperature, allowing the primers to anneal to complementary sequences within the template and then replicating the template with DNA polymerase. The process has been automated with the use of **thermostable DNA polymerases** isolated from bacteria that grow in thermal vents in the ocean or hot springs. During the first round of replication a single copy of DNA is converted to two copies and so on resulting in an exponential increase in the number of copies of the sequences targeted by the primers. After just 20 cycles a single copy of DNA is amplified over 2,000,000 fold.



The polymerase chain reaction can be used to amplify both double and single stranded (eg the products of a reverse transcription reaction, RT-PCR) DNA. The template is mixed with specific or degenerate primers, dNTPs, polymerase buffer including  $\text{MgCl}_2$  and thermostable DNA polymerase. The template is denatured at high temperature (eg  $95^\circ\text{C}$ ) and then cooled to a temperature that will allow optimal primer binding. The reaction temperature is then raised to that optimal for the DNA polymerase (eg  $72^\circ\text{C}$ ) whereby the primers are extended along the template. This series of steps is carried out 20 - 30 times leading to exponential amplification of the target template. The amplification is so great that the reaction products can be visualized following gel electrophoresis.

The products of PCR reactions are analyzed by separation in agarose gels followed by ethidium bromide staining and visualization with *uv* transillumination. Alternatively, radioactive dNTPs can be added to the PCR in order to incorporate label into the products. In this case the products of the PCR are visualized by exposure of the gel to x-ray film. The added advantage of radiolabeling PCR products is that the levels of individual amplification products can be quantitated.

PCR can be used in the analysis of disease genes by being able to amplify detectable amounts of specific fragments of DNA. The amplified fragments from disease genes may be larger, due to insertions, or smaller, due to deletions. The dramatic amplification of DNA by PCR allows the analysis of disease genes in extremely small samples of DNA. For example, only a small number of fetal cells need be extracted from amniotic fluid in order to analyze for the presence of specific disease genes. Additionally, single point mutations can be detected by modified PCR techniques such as the **ligase chain reaction (LCR)** and **PCR-single-strand conformational polymorphisms (PCR-SSCP) analysis**. The PCR technique also can be used to identify the level of expression of genes in extremely small samples of material, e.g. tissues or cells from the body. This technique is termed **reverse transcription-PCR (RT-PCR)**.

## Examples of Inherited Disorders Detectable by PCR

| Disease                        | Affected Gene                                          |
|--------------------------------|--------------------------------------------------------|
| Adenosine deaminase deficiency | Adenosine deaminase (ADA)                              |
| Lesch-Nyhan syndrome           | Hypoxanthine-guanine phosphoribosyltransferase (HGPRT) |

|                                              |                                                                               |
|----------------------------------------------|-------------------------------------------------------------------------------|
| $\alpha$ -1-Antitrypsin deficiency           | $\alpha$ -1-Antitrypsin                                                       |
| Cystic fibrosis                              | cystic fibrosis transmembrane conductance (CFTR) protein                      |
| Fabry disease                                | $\alpha$ -Galactosidase                                                       |
| Gaucher's disease                            | Glucocerebrosidase                                                            |
| Sandhoff-Jatzkewitz disease                  | Hexosaminidase A and B                                                        |
| Tay-Sachs disease                            | Hexosaminidase A                                                              |
| Familial hypercholesterolemia (FH)           | LDL receptor                                                                  |
| Glucose-6-phosphate dehydrogenase deficiency | Glucose-6-phosphate dehydrogenase                                             |
| Maple syrup urine disease                    | branched-chain $\alpha$ -keto acid dehydrogenase                              |
| Phenylketonuria (PKU)                        | Phenylalanine hydroxylase                                                     |
| Ornithine transcarbamylase deficiency        | Ornithine transcarbamylase                                                    |
| Retinoblastoma (Rb)                          | RB gene product, pRB                                                          |
| Sickle-cell anemia                           | point mutation in $\beta$ -globin                                             |
| $\beta$ -Thalassemia                         | mutations in $\beta$ -globin gene that result in loss of synthesis of protein |
| Hemophilia A                                 | Factor VIII                                                                   |

Hemophilia B

Factor IX

von Willebrand disease

von Willebrand factor (vWF)

## Reverse Transcription-PCR (RT-PCR)

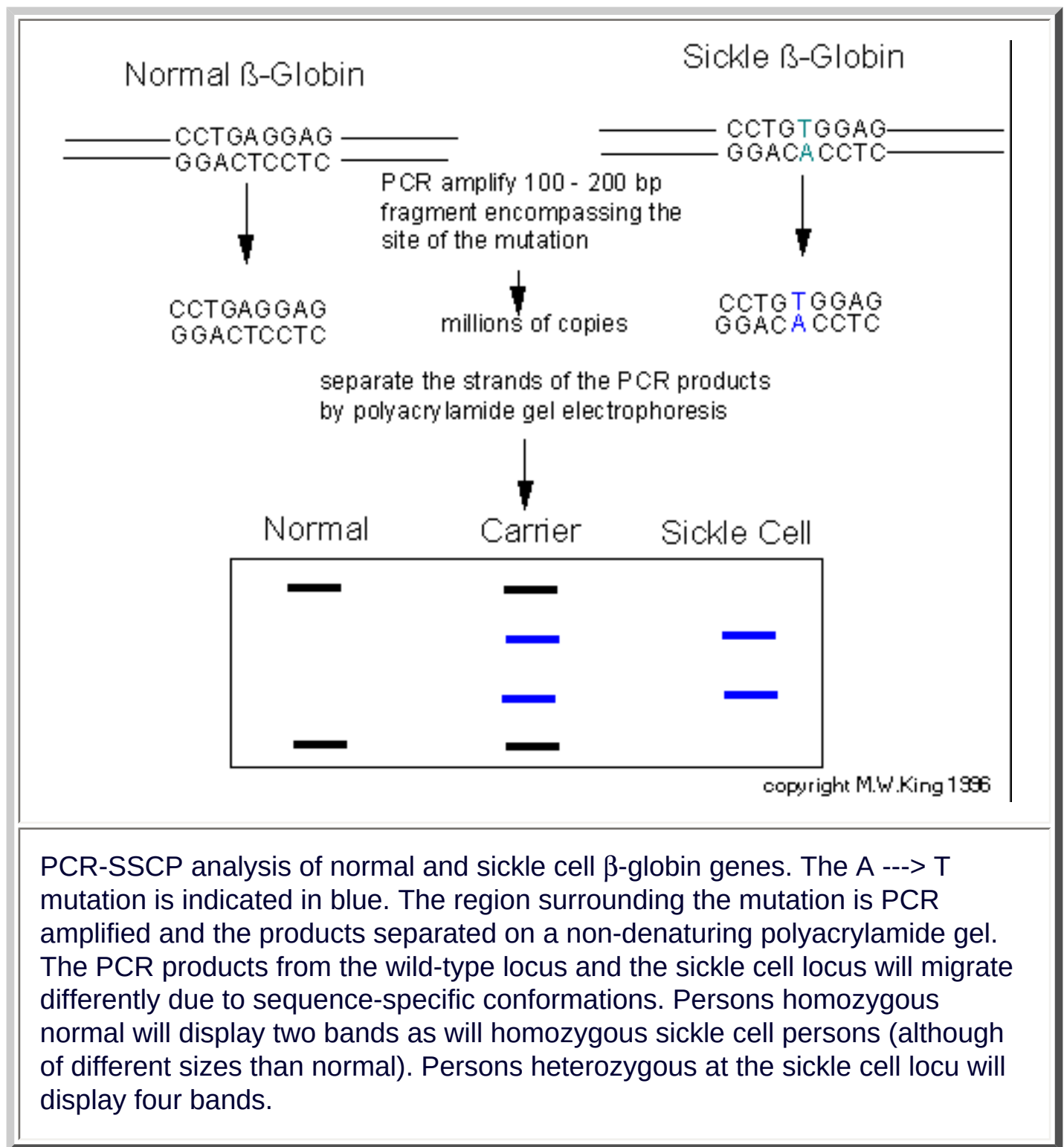
RT-PCR is a rapid and quantitative procedure for the analysis of the level of expression of genes. This technique utilizes the ability of **reverse transcriptase** (RT) to convert RNA into single-stranded cDNA and couples it with the PCR-mediated amplification of specific types of cDNAs present in the RT reaction. The cDNAs that are produced during the RT reaction represent a **window** into the pattern of genes that are being expressed at the time the RNA was extracted.

Total cellular RNA can be extracted from tissues or cells by any of several techniques and used as a template for RT. In most cases the RNA is primed using random primers. A small aliquot of the RT reaction is then added to a PCR reaction containing primers specific to the sequences one wishes to amplify. The products of the RT-PCR can be then be visualized as described above for standard PCR.

## PCR-Single-Strand Conformation Polymorphism (PCR-SSCP)

Many inherited disorders are due to single nucleotide changes within critical regions of the affected gene (eg sickle cell anemia). The PCR-SSCP technique can detect single mutations in genes due to the altered conformation mobility of the single strands of DNA (within an **electrophoresis gel**) harboring the mutation relative to the wild-type strands that do not. Specific PCR primers are made that span the sequences of a given disease gene where a mutation is known to exist and the region amplified by PCR. The same region of the wild-type gene is PCR amplified. The two strands of wild-type PCR product will migrate differently than the two strands of mutant PCR product. Even single point mutations lead to the strands of amplified DNA existing in different conformations which alter their mobility when subjected to electrophoresis in non-denaturing gels.

In order to accurately visualize the PCR products following gel electrophoresis either the primers are radioactively labeled or radioactive nucleotides are incorporated into the PCR products. The PCR products are separated in a polyacrylamide gel and visualized by exposure of the gel to x-ray film. Individuals that are homozygous wild-type at the locus being analyzed will exhibit two distinct bands in the gel as will those individuals that are homozygous mutant. However, due to the nucleotide change the mutant PCR products will migrate with different mobilities in the gel. Individuals that are heterozygous will exhibit a pattern consisting of all four bands.



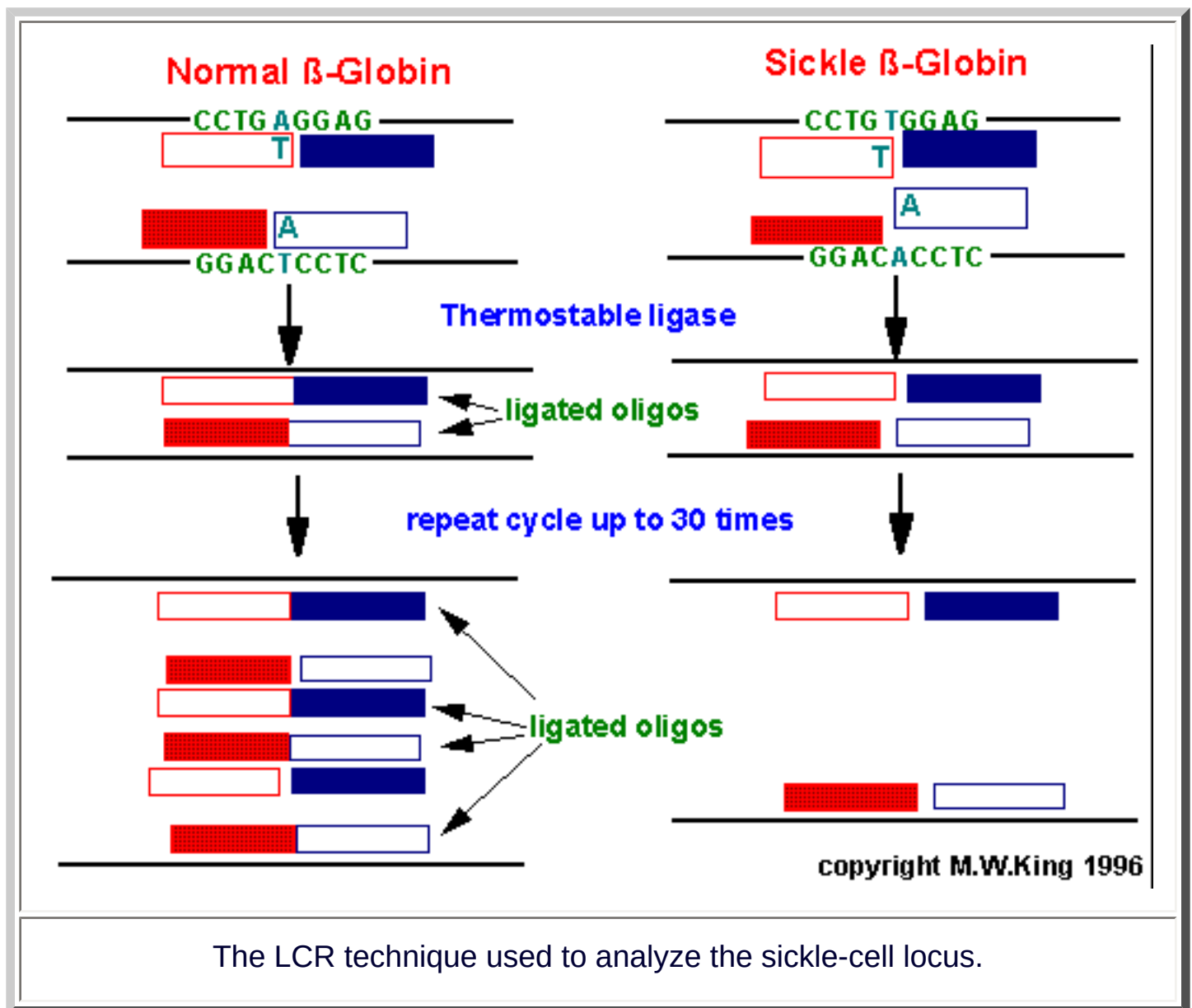
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## The Ligase Chain Reaction (LCR)

The LCR is another technique that allows detection of single point mutations in disease genes. The technique utilizes a thermostable DNA ligase to ligate together perfectly adjacent oligos. Two sets of oligos are designed to anneal to one strand of the gene at the site of the



mutation, a second set of two oligos anneals to the other strand. The oligos are designed such that they will only completely anneal to the wild-type sequences. In the example shown below for the sickle-cell mutation, the 3' nucleotide of one oligo in each pair is mismatched. This mismatch prevents the annealing of the oligos directly adjacent to each other. Therefore, DNA ligase will not ligate the two oligos of each pair together. With the wild-type sequence the oligo pairs that are ligated together become targets for annealing the oligos and, therefore, result in an exponential amplification of the wild-type target. Given that prior sequence knowledge is required in order to detect point mutations in disease genes, the LCR technique is utilized for the diagnosis of the presence of a mutant allele in high risk patients.



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## Microarray Analysis

This section is to be completed soon, sorry for the inconvenience.

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## Transgenesis

Transgenesis refers to the process of introducing exogenous genes into the germ line of an organism. The first successful transgenesis experiments were carried out in mice. One relatively well known experiment involved the introduction of the rat growth hormone gene into the germ line of mice. These transgenic mice grew to twice their normal size.

To create a transgenic animal the gene of interest must be passed from generation to generation, i.e. it must be inherited in the germ line. To accomplish this with mice or livestock animals, vectors containing the gene of interest with appropriate regulatory elements (e.g. the  $\beta$ -lactoglobulin promoter if expression of the transgene in the milk is desired) are injected into the nucleus of fertilized eggs. The eggs are then transplanted into the uterus of receptive females for development of the potential transgenic offspring. In order to test the resultant animal for germ line transmission of the transgene the chromosomal DNA of their offspring is tested for the presence of the transgene. If the transgene exhibits Mendelian inheritance then it is being transmitted in the germ line.

Currently the process of transgenesis is being utilized in both the plant and livestock industries. The aim of the majority of these experiments is to generate plants and animals that are more resistant to diseases and infections. However, some transgenic farm animal such as sheep and cows are being developed in order to obtain high levels of expression of therapeutically important proteins during milk synthesis. This allows large amounts of the protein of interest to be purified from the milk of the transgenic animals.

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## Gene Therapy

Transgenesis with humans would allow for the elimination of disease genes in a population of offspring, however, technical as well as ethical issues likely will prevent any transgenic experiments to be carried out with human eggs. Therefore, the ability to replace known disease genes with normal copies in afflicted humans is the ultimate goal of gene therapy. Human gene therapy protocols aim to introduce correcting copies of disease genes into somatic cells of the affected individual. Expression of a correct copy of an affected gene in somatic cells prevents transmission through the germ line, thereby, avoiding many of the ethical issues of transgenesis. This is analogous to treatment of individuals by organ or tissue transplantation.

The most common techniques utilized in gene therapy studies is the introduction of the corrected gene into bone marrow cells, skin fibroblasts or hepatocytes. The vectors most commonly utilized are derived from retroviruses and utilize only the transcriptional promoter regions of these viruses (the LTRs) to drive expression of the gene of interest. The advantage of retroviral-based vector systems is that expression occurs in most cell types.

A number of human inherited disorders have been corrected in cultured cells and several diseases (e.g. malignant melanoma and severe combined immunodeficiency disease, SCID) are currently being treated by gene therapy techniques indicating that gene therapy is likely to be a powerful therapeutic technique against a host of diseases in coming years.

## Human Disorders Treated in Cultured Cells by Gene Therapy

| Disorder                           | Affected Gene                                          |
|------------------------------------|--------------------------------------------------------|
| SCID                               | Adenosine deaminase (ADA)                              |
| SCID                               | Purine nucleoside phosphorylase (PNP)                  |
| Lesch-Nyhan syndrome               | Hypoxanthine-guanine phosphoribosyltransferase (HGPRT) |
| Gaucher disease                    | Glucocerebrosidase                                     |
| Familial hypercholesterolemia (FH) | LDL receptor                                           |
| Phenylketonuria (PKU)              | Phenylalanine hydroxylase                              |
| $\beta$ -Thalassemia               | $\beta$ -Globin                                        |
| Hemophilia B                       | Factor IX                                              |

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- [Nucleic Acid Structure and Nomenclature](#)
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## Introduction

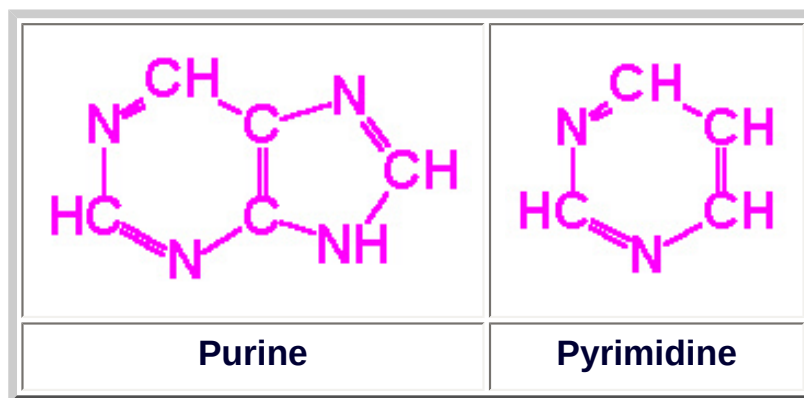
As a class, the nucleotides may be considered one of the most important metabolites of the cell. Nucleotides are found primarily as the monomeric units comprising the major nucleic acids of the cell, RNA and DNA. However, they also are required for numerous other important functions within the cell. These functions include:

- **1.** serving as energy stores for future use in phosphate transfer reactions. These reactions are predominantly carried out by ATP.
- **2.** forming a portion of several important coenzymes such as NAD<sup>+</sup>, NADP<sup>+</sup>, FAD and coenzyme A.
- **3.** serving as mediators of numerous important cellular processes such as **second messengers** in signal transduction events. The predominant second messenger is cyclic-AMP (cAMP), a cyclic derivative of AMP formed from ATP.
- **4.** controlling numerous enzymatic reactions through allosteric effects on enzyme activity.
- **5.** serving as activated intermediates in numerous biosynthetic reactions. These activated intermediates include S-adenosylmethionine (S-AdoMet) involved in methyl transfer reactions as well as the many sugar coupled nucleotides involved in [glycogen](#) and [glycoprotein](#) synthesis.

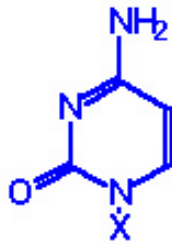
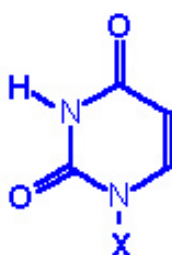
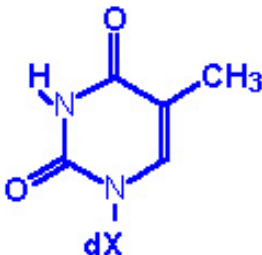
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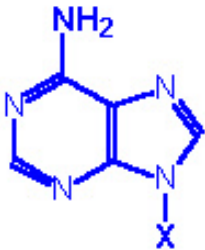
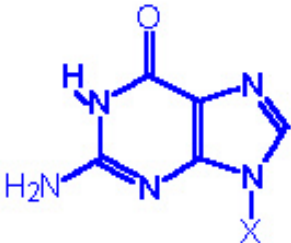
## Nucleoside and Nucleotide Structure and Nomenclature

The nucleotides found in cells are derivatives of the heterocyclic highly basic, compounds, **purine** and **pyrimidine**.



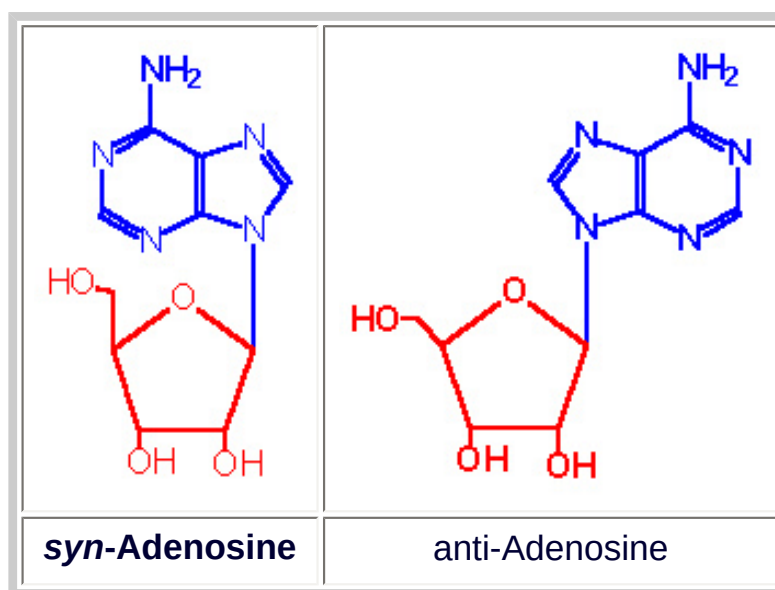
It is the chemical basicity of the nucleotides that has given them the common term "bases" as they are associated with nucleotides present in DNA and RNA. There are five major bases found in cells. The derivatives of purine are called **adenine** and **guanine**, and the derivatives of pyrimidine are called **thymine**, **cytosine** and **uracil**. The common abbreviations used for these five bases are, **A**, **G**, **T**, **C** and **U**.

| Base Formula                                                                        | Base (X=H)  | Nucleoside<br>X=ribose or deoxyribose | Nucleotide<br>X=ribose phosphate |
|-------------------------------------------------------------------------------------|-------------|---------------------------------------|----------------------------------|
|   | Cytosine, C | Cytidine, A                           | Cytidine monophosphate<br>CMP    |
|  | Uracil, U   | Uridine, U                            | Uridine monophosphate<br>UMP     |
|  | Thymine, T  | Thymidine, T                          | Thymidine monophosphate<br>TMP   |

|                                                                                   |            |              |                                |
|-----------------------------------------------------------------------------------|------------|--------------|--------------------------------|
|   | Adenine, A | Adenosine, A | Adenosine monophosphate<br>AMP |
|  | Guanine, G | Guanosine, A | Guanosine monophosphate<br>GMP |

The purine and pyrimidine bases in cells are linked to carbohydrate and in this form are termed, **nucleosides**. The nucleosides are coupled to D-ribose or 2'-deoxy-D-ribose through a  $\beta$ -N-glycosidic bond between the anomeric carbon of the ribose and the N<sup>9</sup> of a purine or N<sup>1</sup> of a pyrimidine.

The base can exist in 2 distinct orientations about the N-glycosidic bond. These conformations are identified as, **syn** and **anti**. It is the anti conformation that predominates in naturally occurring nucleotides.



Nucleosides are found in the cell primarily in their phosphorylated form. These are termed **nucleotides**. The most common site of phosphorylation of nucleotides found in cells is the hydroxyl group attached to the 5'-carbon of the ribose. The carbon atoms of the ribose present in nucleotides are designated with a prime (') mark to distinguish them from the backbone numbering in the bases. Nucleotides can exist in the mono-, di-, or tri-phosphorylated forms.

Nucleotides are given distinct abbreviations to allow easy identification of their structure and state of phosphorylation. The monophosphorylated form of adenosine (adenosine-5'-monophosphate) is written as, **AMP**. The di- and tri-phosphorylated forms are written as, **ADP** and **ATP**, respectively. The use of these abbreviations assumes that the nucleotide is in the 5'-phosphorylated form. The di- and tri-phosphates of nucleotides are linked by acid anhydride bonds. Acid anhydride bonds have a high  $\Delta G^0$  for hydrolysis imparting upon them a high potential to



transfer the phosphates to other molecules. It is this property of the nucleotides that results in their involvement in group transfer reactions in the cell.

The nucleotides found in DNA are unique from those of RNA in that the ribose exists in the 2'-deoxy form and the abbreviations of the nucleotides contain a **d** designation. The monophosphorylated form of adenosine found in DNA (deoxyadenosine-5'-monophosphate) is written as **dAMP**.

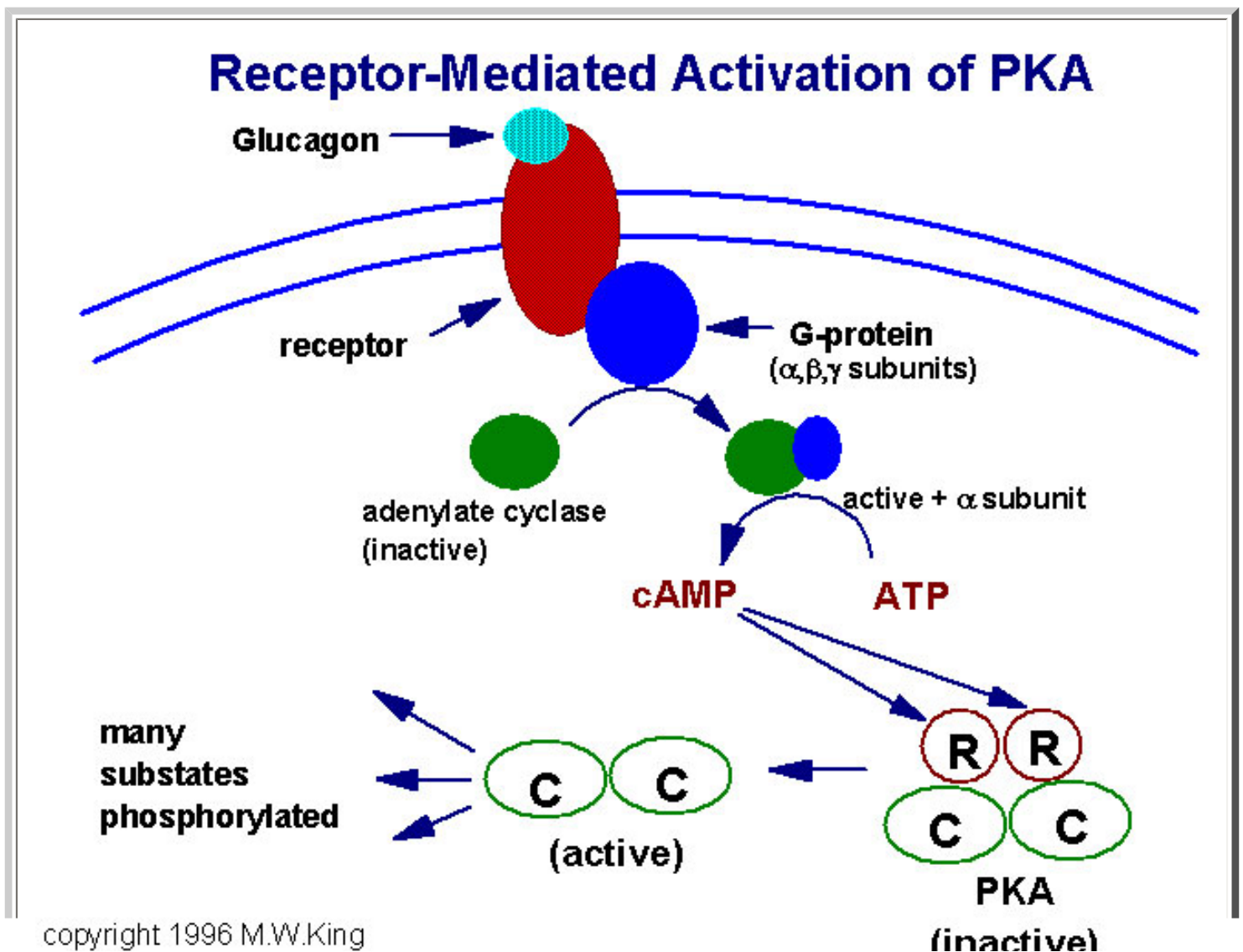
The nucleotide uridine is never found in DNA and thymine is almost exclusively found in DNA. Thymine is found in tRNAs but not rRNAs nor mRNAs. There are several less common bases found in DNA and RNA. The primary modified base in DNA is 5-methylcytosine. A variety of modified bases appear in the tRNAs. Many modified nucleotides are encountered outside of the context of DNA and RNA that serve important biological functions.

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## Adenosine Derivatives

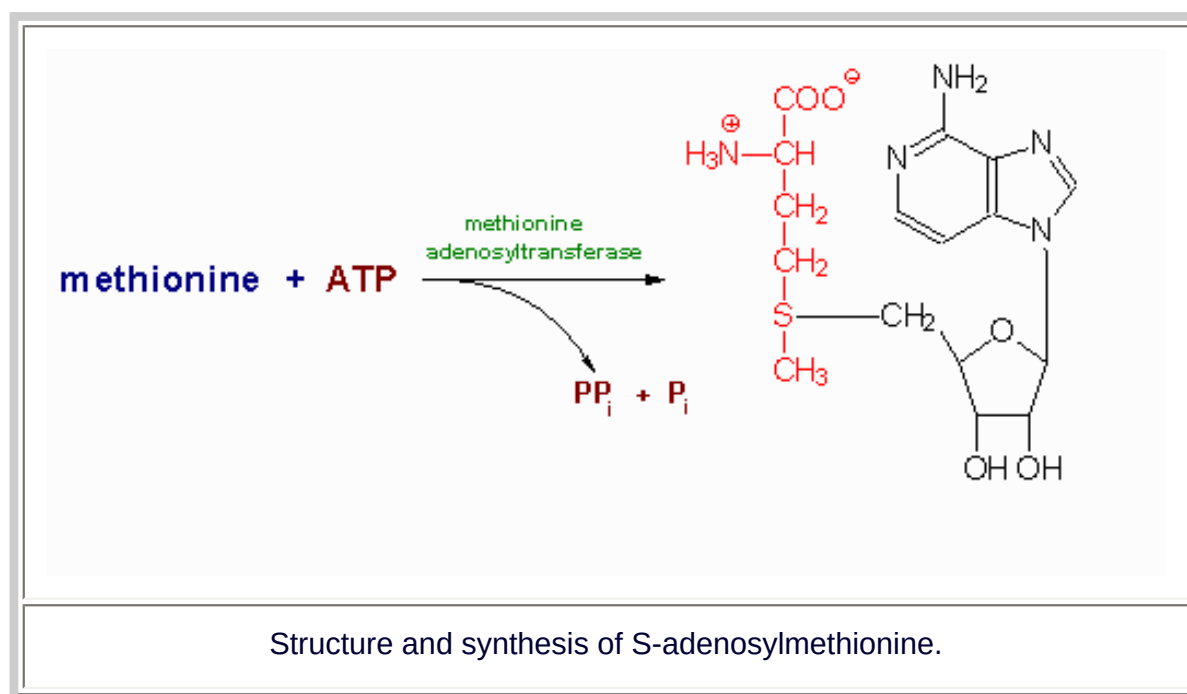
The most common adenosine derivative is the cyclic form, **3'-5'-cyclic adenosine monophosphate, cAMP**. This compound is a very powerful **second messenger** involved in passing **signal transduction** events from the cell surface to internal proteins, e.g. **cAMP-dependent protein kinase, PKA** (see Figure below). PKA phosphorylates a number of proteins, thereby, affecting their activity either positively or negatively. Cyclic-AMP is also involved in the regulation of ion channels by direct interaction with the channel proteins, e.g. in the activation of odorant receptors by odorant molecules.

Formation of cAMP occurs in response to activation of receptor coupled adenylate cyclase. These receptors can be of any type, e.g. hormone receptors or odorant receptors.



Representative pathway for the activation of cAMP-dependent protein kinase, PKA). In this example glucagon binds to its' cell-surface receptor, thereby activating the receptor. Activation of the receptor is coupled to the activation of a receptor-coupled G-protein (GTP-binding and hydrolyzing protein) composed of 3 subunits. Upon activation the  $\alpha$  subunit dissociates and binds to and activates adenylate cyclase. Adenylate cyclase then converts ATP to cyclic-AMP (cAMP). The cAMP thus produced then binds to the regulatory subunits of PKA leading to dissociation of the associated catalytic subunits. The catalytic subunits are inactive until dissociated from the regulatory subunits. Once released the catalytic subunits of PKA phosphorylate numerous substrate using ATP as the phosphate donor.

**S-adenosylmethionine** is a form of activated methionine which serves as a methyl donor in methylation reactions and as a source of propylamine in the synthesis of polyamines.



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## Guanosine Derivatives

A cyclic form of GMP (**cGMP**) also is found in cells involved as a second messenger molecule. In many cases its' role is to antagonize the effects of cAMP. Formation of cGMP occurs in response to receptor mediated signals similar to those for activation of adenylate cyclase. However, in this case it is guanylate cyclase that is coupled to the receptor.

The most important cGMP coupled signal transduction cascade is that [photoreception](#). However, in this case activation of **rhodopsin** (in the rods) or other **opsins** (in the cones) by the absorption of a photon of light (through 11-*cis*-retinal covalently associated with rhodopsin and opsins) activates transducin which in turn activates a cGMP specific phosphodiesterase that hydrolyzes cGMP to GMP. This lowers the effective concentration of cGMP bound

to gated ion channels resulting in their closure and a concomitant hyperpolarization of the cell.

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## Synthetic Nucleotide Analogs

Many nucleotide analogues are chemically synthesized and used for their therapeutic potential. The nucleotide analogues can be utilized to inhibit specific enzymatic activities. A large family of analogues are used as anti-tumor agents, for instance, because they interfere with the synthesis of DNA and thereby preferentially kill rapidly dividing cells such as tumor cells. Some of the nucleotide analogues commonly used in chemotherapy are 6-mercaptopurine, 5-fluorouracil, 5-iodo-2'-deoxyuridine and 6-thioguanine. Each of these compounds disrupts the normal replication process by interfering with the formation of correct Watson-Crick base-pairing.

Nucleotide analogs also have been targeted for use as antiviral agents. Several analogs are used to interfere with the replication of HIV, such as **AZT (azidothymidine)** and **ddI (dideoxyinosine)**.

Several purine analogs are used to treat gout. The most common is **allopurinol**, which resembles hypoxanthine. Allopurinol inhibits the activity of xanthine oxidase, an enzyme involved in *de novo* [purine biosynthesis](#). Additionally, several nucleotide analogues are used after organ transplantation in order to suppress the immune system and reduce the likelihood of transplant rejection by the host.

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## Polynucleotides

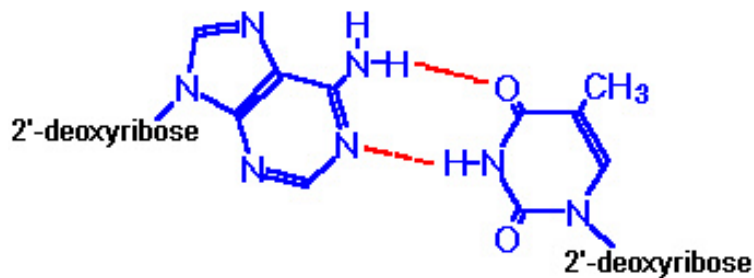
Polynucleotides are formed by the condensation of two or more nucleotides. The condensation most commonly occurs between the alcohol of a 5'-phosphate of one nucleotide and the 3'-hydroxyl of a second, with the elimination of H<sub>2</sub>O, forming a **phosphodiester bond**. The formation of phosphodiester bonds in DNA and RNA exhibits directionality. The primary structure of DNA and RNA (the linear arrangement of the nucleotides) proceeds in the 5' ----> 3' direction. The common representation of the primary structure of DNA or RNA molecules is to write the nucleotide sequences from left to right synonymous with the 5' -----> 3' direction as shown:

**5'-pGpApTpC-3'**

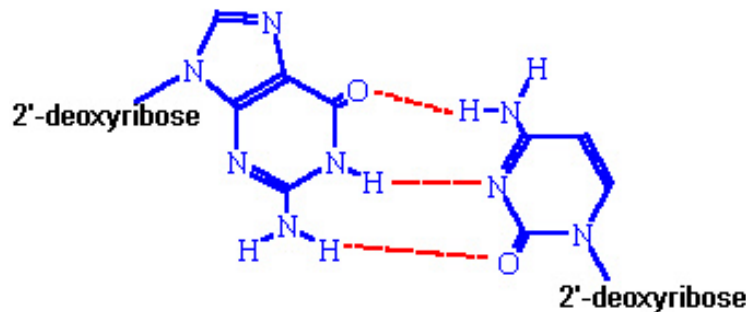
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## Structure of DNA

Utilizing X-ray diffraction data, obtained from crystals of DNA, James Watson and Francis Crick proposed a model for the structure of DNA. This model (subsequently verified by additional data) predicted that DNA would exist as a helix of two complementary antiparallel strands, wound around each other in a rightward direction and stabilized by H-bonding between bases in adjacent strands. In the **Watson-Crick model**, the bases are in the interior of the helix aligned at a nearly 90 degree angle relative to the axis of the helix. Purine bases form hydrogen bonds with pyrimidines, in the crucial phenomenon of **base pairing**. Experimental determination has shown that, in any given molecule of DNA, the concentration of adenine (A) is equal to thymine (T) and the concentration of cytidine (C) is equal to guanine (G). This means that A will only base-pair with T, and C with G. According to this pattern, known as **Watson-Crick base-pairing**, the base-pairs composed of G and C contain three H-bonds, whereas those of A and T contain two H-bonds. This makes G-C base-pairs more stable than A-T base-pairs.



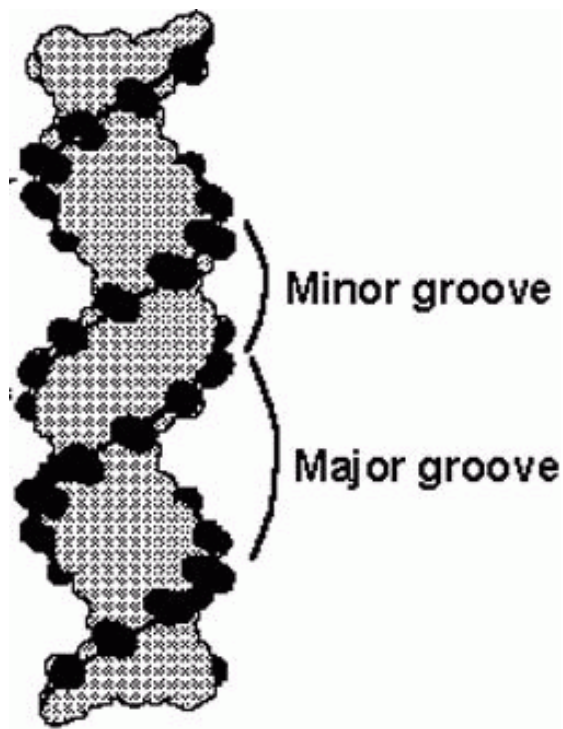
**A-T Base Pair**



**G-C Base Pair**

The antiparallel nature of the helix stems from the orientation of the individual strands. From any fixed position in the helix, one strand is oriented in the 5' → 3' direction and the other in the 3' → 5' direction. On its exterior surface, the double helix of DNA contains two deep grooves between the ribose-phosphate chains. These two grooves are of unequal size and termed the **major** and **minor grooves**. The difference in their size is due to the asymmetry of the deoxyribose rings and the structurally distinct nature of the upper surface of a base-pair relative to the bottom surface.

The double helix of DNA has been shown to exist in several different forms, depending upon sequence content and ionic conditions of crystal preparation. The **B-form** of DNA prevails under physiological conditions of low ionic strength and a high degree of hydration. Regions of the helix that are rich in pCpG dinucleotides can exist in a novel left-handed helical conformation termed **Z-DNA**. This conformation results from a 180 degree change in the orientation of the bases relative to that of the more common A- and B-DNA.



## Structure of B-DNA

## Structure of Z-DNA

### Parameters of Major DNA Helices

| Parameters                                    | A Form           | B Form                      | Z-Form                                                          |
|-----------------------------------------------|------------------|-----------------------------|-----------------------------------------------------------------|
| Direction of helical rotation                 | Right            | Right                       | Left                                                            |
| Residues per turn of helix                    | 11               | 10                          | 12 base pairs                                                   |
| Rotation of helix per residue (in degrees)    | 33               | 36                          | -30                                                             |
| Base tilt relative to helix axis (in degrees) | 20               | 6                           | 7                                                               |
| Major groove                                  | narrow and deep  | wide and deep               | Flat                                                            |
| Minor groove                                  | wide and shallow | narrow and deep             | narrow and deep                                                 |
| Orientation of N-glycosidic Bond              | Anti             | Anti                        | Anti for Py, Syn for Pu                                         |
| Comments                                      |                  | most prevalent within cells | occurs in stretches of alternating purine-pyrimidine base pairs |

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## Thermal Properties of DNA

As cells divide it is a necessity that the DNA be copied (replicated), in such a way that each daughter cell acquires the same amount of genetic material. In order for this process to proceed the two strands of the helix must first be separated, in a process termed denaturation. This process can also be carried out in vitro. If a solution of DNA is subjected to high temperature, the H-bonds between bases become unstable and the strands of the helix separate in a process of **thermal denaturation**.

The base composition of DNA varies widely from molecule to molecule and even within different regions of the same molecule. Regions of the duplex that have predominantly A-T base-pairs will be less thermally stable than those rich in G-C base-pairs. In the process of thermal denaturation, a point is reached at which 50% of the DNA molecule exists as single strands. This point is the **melting temperature ( $T_M$ )**, and is characteristic of the base composition of that DNA molecule. The  $T_M$  depends upon several factors in addition to the base composition. These include the chemical nature of the solvent and the identities and concentrations of ions in the solution.

When thermally melted DNA is cooled, the complementary strands will again re-form the correct base pairs, in a process termed **annealing** or **hybridization**. The rate of annealing is dependent upon the nucleotide sequence of the two strands of DNA.

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## Analysis of DNA Structure

**Chromatography:** Several of the chromatographic techniques available for the [characterization of proteins](#) can also be applied to the characterization of DNA. The most commonly used technique is HPLC (high performance liquid chromatography). Affinity chromatographic techniques also can be employed. One common affinity matrix is hydroxyapatite (a form of calcium phosphate), which binds double-stranded DNA with a higher affinity than single-stranded DNA.

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**Electrophoresis:** This procedure can serve the same function with regard to DNA molecules as it does for the [analysis of proteins](#). However, since DNA molecules have much higher molecular weights than proteins, the molecular sieve used in electrophoresis of DNA must be different as well. The material of choice is agarose, a carbohydrate polymer purified from a salt water algae. It is a copolymer of mannose and galactose that when melted and re-cooled forms a gel with pores sizes dependent upon the concentration of agarose. The phosphate backbone of DNA is highly negatively charged, therefore DNA will migrate in an electric field. The size of DNA fragments can then be determined by comparing their migration in the gel to known size standards. Extremely large molecules of DNA (in excess of 106 base pairs) are effectively separated in agarose gels using **pulsed-field gel electrophoresis (PFGE)**. This technique employs two or more electrodes, placed orthogonally with respect to the gel, that receive short alternating pulses of current. PFGE allows whole chromosomes and large portions of chromosomes to be analyzed.

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## **Protein Primary Structure**

The primary structure of peptides and proteins refers to the linear number and order of the amino acids present. The convention for the designation of the order of amino acids is that the N-terminal end (i.e. the end bearing the residue with the free  $\alpha$ -amino group) is to the left (and the number 1 amino acid) and the C-terminal end (i.e. the end with the residue containing a free  $\alpha$ -carboxyl group) is to the right.

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## **Protein Secondary Structure**

The ordered array of amino acids in a protein confer regular conformational forms upon that protein. These conformations constitute the secondary structures of a protein. In general proteins fold into two broad classes of structure termed, **globular proteins** or **fibrous proteins**. Globular proteins are compactly folded and coiled, whereas, fibrous proteins are more filamentous or elongated. It is the partial double-bond character of the peptide bond that defines the conformations a polypeptide chain may assume. Within a single protein different regions of the polypeptide chain may assume different conformations determined by the primary sequence of the amino acids.

## The Alpha-Helix

The  $\alpha$ -helix is a common secondary structure encountered in proteins of the globular class. The formation of the  $\alpha$ -helix is spontaneous and is stabilized by H-bonding between amide nitrogens and carbonyl carbons of peptide bonds spaced four residues apart.

This orientation of H-bonding produces a helical coiling of the peptide backbone such that the R-groups lie on the exterior of the helix and perpendicular to its axis.

Not all amino acids favor the formation of the  $\alpha$ -helix due to steric constraints of the R-groups. Amino acids such as A, D, E, I, L and M favor the formation of  $\alpha$ -helices, whereas, G and P favor disruption of the helix. This is particularly true for P since it is a pyrrolidine based imino acid ( $\text{HN}=\text{}$ ) whose structure significantly restricts movement about the peptide bond in which it is present, thereby, interfering with extension of the helix. The disruption of the helix is important as it introduces additional folding of the polypeptide backbone to allow the formation of globular proteins.

## $\beta$ -Sheets

Whereas an  $\alpha$ -helix is composed of a single linear array of helically disposed amino acids,  $\beta$ -sheets are composed of 2 or more different regions of stretches of at least 5-10 amino acids. The folding and alignment of stretches of the polypeptide backbone aside one another to form  $\beta$ -sheets is stabilized by H-bonding between amide nitrogens and carbonyl carbons. However, the H-bonding residues are present in adjacently opposed stretches of the polypeptide backbone as opposed to a linearly contiguous region of the backbone in the  $\alpha$ -helix.

$\beta$ -Sheets are said to be pleated. This is due to positioning of the  $\alpha$ -carbons of the peptide bond which alternates above and below the plane of the sheet.

$\beta$ -Sheets are either parallel or antiparallel. In parallel sheets adjacent peptide chains proceed in the same direction (i.e. the direction of N-terminal to C-terminal ends is the same), whereas, in antiparallel sheets adjacent chains are aligned in opposite directions.

## Super-secondary Structure

Some proteins contain an ordered organization of secondary structures that form distinct

functional domains or structural motifs. Examples include the helix-turn-helix domain of bacterial proteins that regulate transcription and the leucine zipper, helix-loop-helix and zinc finger domains of eukaryotic [transcriptional regulators](#). These domains are termed **super-secondary structures**.

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## Tertiary Structure

Tertiary structure refers to the complete three-dimensional structure of the polypeptide units of a given protein. Included in this description is the spatial relationship of different secondary structures to one another within a polypeptide chain and how these secondary structures themselves fold into the three-dimensional form of the protein. Secondary structures of proteins often constitute distinct **domains**. Therefore, tertiary structure also describes the relationship of different domains to one another within a protein. The interactions of different domains is governed by several forces: These include **hydrogen bonding, hydrophobic interactions, electrostatic interactions** and **van der Waals forces**.

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## Forces Controlling Protein Structure

### Hydrogen Bonding:

Polypeptides contain numerous proton donors and acceptors both in their backbone and in the R-groups of the amino acids. The environment in which proteins are found also contains the ample H-bond donors and acceptors of the water molecule. H-bonding, therefore, occurs not only within and between polypeptide chains but with the surrounding aqueous medium.

### Hydrophobic Forces:

Proteins are composed of amino acids that contain either hydrophilic or hydrophobic R-groups. It is the nature of the interaction of the different R-groups with the aqueous environment that plays the major role in shaping protein structure. The spontaneous folded state of globular proteins is a reflection of a balance between the opposing energetics of H-bonding between hydrophilic R-groups and the aqueous environment and the repulsion from the aqueous environment by the hydrophobic R-groups. The hydrophobicity of certain amino acid R-groups tends to drive them away from the exterior of proteins and into the interior. This driving force restricts the available conformations into which a protein may fold.

### Electrostatic Forces:

Electrostatic forces are mainly of three types; **charge-charge, charge-dipole** and

**dipole-dipole.** Typical charge-charge interactions that favor protein folding are those between oppositely charged R-groups such as K or R and D or E. A substantial component of the energy involved in protein folding is charge-dipole interactions. This refers to the interaction of ionized R-groups of amino acids with the dipole of the water molecule. The slight dipole moment that exist in the polar R-groups of amino acid also influences their interaction with water. It is, therefore, understandable that the majority of the amino acids found on the exterior surfaces of globular proteins contain charged or polar R-groups.

### van der Waals Forces:

There are both attractive and repulsive van der Waals forces that control protein folding. Attractive van der Waals forces involve the interactions among induced dipoles that arise from fluctuations in the charge densities that occur between adjacent uncharged non-bonded atoms. Repulsive van der Waals forces involve the interactions that occur when uncharged non-bonded atoms come very close together but do not induce dipoles. The repulsion is the result of the electron-electron repulsion that occurs as two clouds of electrons begin to overlap.

Although van der Waals forces are extremely weak, relative to other forces governing conformation, it is the huge number of such interactions that occur in large protein molecules that make them significant to the folding of proteins.

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## Quaternary Structure

Many proteins contain 2 or more different polypeptide chains that are held in association by the same non-covalent forces that stabilize the tertiary structures of proteins. Proteins with multiple polypeptide chains are termed **oligomeric** proteins. The structure formed by monomer-monomer interaction in an oligomeric protein is known as **quaternary structure**.

Oligomeric proteins can be composed of multiple identical polypeptide chains or multiple distinct polypeptide chains. Proteins with identical subunits are termed **homooligomers**. Proteins containing several distinct polypeptide chains are termed **heterooligomers**.

**Hemoglobin**, the oxygen carrying protein of the blood, contains two  $\alpha$  and two  $\beta$  subunits arranged with a quaternary structure in the form,  $\alpha_2\beta_2$ . Hemoglobin is, therefore, a hetero-oligomeric protein.

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## Complex Protein Structures

Proteins also are found to be covalently conjugated with carbohydrates. These modifications occur following the synthesis ([translation](#)) of proteins and are, therefore, termed post-translational modifications. These forms of modification impart specialized functions upon the resultant proteins. Proteins covalently associated with carbohydrates are termed

**glycoproteins.** Glycoproteins are of two classes, N-linked and O-linked, referring to the site of covalent attachment of the sugar moieties. N-linked sugars are attached to the amide nitrogen of the R-group of asparagine; O-linked sugars are attached to the hydroxyl groups of either serine or threonine and occasionally to the hydroxyl group of the modified amino acid, hydroxylysine.

There are extremely important glycoproteins found on the surface of erythrocytes. It is the variability in the composition of the carbohydrate portions of many glycoproteins and glycolipids of erythrocytes that determines blood group specificities. There are at least 100 blood group determinants, most of which are due to carbohydrate differences. The most common blood groups, A, B, and O, are specified by the activity of specific gene products whose activities are to incorporate distinct sugar groups onto RBC membrane glycosphingolipids as well as secreted glycoproteins.

Structural complexes involving protein associated with lipid via noncovalent interactions are termed lipoproteins. The distinct roles of **lipoproteins** are described on the linked page. Their major function in the body is to aid in the storage transport of lipid and cholesterol.

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## Clinical Significances

Visit the **Inborn Errors** page for a more complete listing of diseases related to abnormal proteins. Several brief examples are presented below.

The substitution of a hydrophobic amino acid (V) for an acidic amino acid (E) in the  $\beta$ -chain of hemoglobin results in **sickle cell anemia** (HbS). This change of a single amino acid alters the structure of hemoglobin molecules in such a way that the deoxygenated proteins polymerize and precipitate within the erythrocyte, leading to their characteristic sickle shape.

Collagens are the most abundant proteins in the body. Alterations in collagen structure arising from abnormal genes or abnormal processing of collagen proteins results in numerous diseases, including Larsen syndrome, scurvy, osteogenesis imperfecta and Ehlers-Danlos syndrome.

**Ehlers-Danlos syndrome** (see **OMIM links**) is actually the name associated with at least ten distinct disorders that are biochemically and clinically distinct yet all manifest structural weakness in connective tissue as a result of defective collagen structure.

**Osteogenesis imperfecta** (see **OMIM links**) also encompasses more than one disorder. At least four biochemically and clinically distinguishable maladies have been identified as osteogenesis imperfecta, all of which are characterized by multiple fractures and resultant bone deformities. **Marfan's syndrome** manifests itself as a disorder of the connective tissue and was originally believed to be the result of abnormal collagens. However, recent evidence has shown that Marfan's syndrome results from mutations in the extracellular protein, **fibrillin**, which is an integral constituent of the non-collagenous microfibrils of the extracellular matrix.

Several forms of **familial hypercholesterolemia** (see also **OMIM links**) are the result of genetic defects in the gene encoding the receptor for low-density lipoprotein (LDL). These

defects result in the synthesis of abnormal LDL receptors that are incapable of binding to LDLs, or that bind LDLs but the receptor/LDL complexes are not properly internalized and degraded. The outcome is an elevation in serum cholesterol levels and increased propensity toward the development of atherosclerosis.

A number of proteins can contribute to cellular transformation and carcinogenesis when their basic structure is disrupted by mutations in their genes. These genes are termed **proto-oncogenes**. For some of these proteins, all that is required to convert them to the **oncogenic** form is a single amino acid substitution. The cellular gene, **c-Ras**, is observed to sustain single amino acid substitutions at positions 12 or 61 with high frequency in colon carcinomas. Mutations in c-Ras are most frequently observed genetic alterations in colon cancer.  
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## Amino-Terminal Sequence Determination

Prior to sequencing peptides it is necessary to eliminate disulfide bonds within peptides and between peptides. Several different chemical reactions can be used in order to permit separation of peptide strands and prevent protein conformations that are dependent upon disulfide bonds. The most common treatments are to use either **2-mercaptoethanol** or **dithiothreitol**. Both of these chemicals reduce disulfide bonds. To prevent reformation of the disulfide bonds the peptides are treated with **iodoacetic acid** in order to alkylate the free sulfhydryls.

There are three major chemical techniques for sequencing peptides and proteins from the N-terminus. These are the **Sanger, Dansyl chloride** and **Edman techniques**.

**Sanger's Reagent:** This sequencing technique utilizes the compound, 2,4-dinitrofluorobenzene (DNF) which reacts with the N-terminal residue under alkaline conditions. The derivatized amino acid can be hydrolyzed and will be labeled with a dinitrobenzene group that imparts a yellow color to the amino acid. Separation of the modified amino acids (DNP-derivative) by electrophoresis and comparison with the migration of DNP-derivative standards allows for the identification of the N-terminal amino acid.

**Dansyl chloride:** Like DNF, dansyl chloride reacts with the N-terminal residue under alkaline conditions. Analysis of the modified amino acids is carried out similarly to the Sanger method except that the dansylated amino acids are detected by fluorescence. This imparts a higher sensitivity into this technique over that of the Sanger method.

**Edman degradation:** The utility of the Edman degradation technique is that it allows for additional amino acid sequence to be obtained from the N-terminus inward. Using this method it is possible to obtain the entire sequence of peptides. This method utilizes phenylisothiocyanate to react with the N-terminal residue under alkaline conditions. The resultant phenylthiocarbamyl derivatized amino acid is hydrolyzed in anhydrous acid. The hydrolysis reaction results in a rearrangement of the released N-terminal residue to a



phenylthiohydantoin derivative. As in the Sanger and Dansyl chloride methods, the N-terminal residue is tagged with an identifiable marker, however, the added advantage of the Edman process is that the remainder of the peptide is intact. The entire sequence of reactions can be repeated over and over to obtain the sequences of the peptide. This process has subsequently been automated to allow rapid and efficient sequencing of even extremely small quantities of peptide.

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## Protease Digestion

Due to the limitations of the Edman degradation technique, peptides longer than around 50 residues can not be sequenced completely. The ability to obtain peptides of this length, from proteins of greater length, is facilitated by the use of enzymes, endopeptidases, that cleave at specific sites within the primary sequence of proteins. The resultant smaller peptides can be chromatographically separated and subjected to Edman degradation sequencing reactions.

### Specificities of Several Endoproteases

| Enzyme       | Source          | Specificity                                             | Additional Points                                                |
|--------------|-----------------|---------------------------------------------------------|------------------------------------------------------------------|
| Trypsin      | Bovine pancreas | peptide bond C-terminal to R, K, but not if next to P   | highly specific for positively charged residues                  |
| Chymotrypsin | Bovine pancreas | peptide bond C-terminal to F, Y, W but not if next to P | prefers bulky hydrophobic residues, cleaves slowly at N, H, M, L |

|                  |                                     |                                                                   |                                                          |
|------------------|-------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------|
| Elastase         | Bovine pancreas                     | peptide bond C-terminal to A, G, S, V, but not if next to P       |                                                          |
| Thermolysin      | <i>Bacillus thermoproteolyticus</i> | peptide bond N-terminal to I, M, F, W, Y, V, but not if next to P | prefers small neutral residues, can cleave at A, D, H, T |
| Pepsin           | Bovine gastric mucosa               | peptide bond N-terminal to L, F, W, Y, but when next to P         | exhibits little specificity, requires low pH             |
| Endopeptidase V8 | <i>Staphylococcus aureus</i>        | peptide bond C-terminal to E                                      |                                                          |

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## Carboxy-Terminal Sequence Determination

No reliable chemical techniques exist for sequencing the C-terminal amino acid of peptides. However, there are enzymes, exopeptidases, that have been identified that cleave peptides at the C-terminal residue which can then be analyzed chromatographically and compared to standard amino acids. This class of exopeptidases are called, carboxypeptidases.

## Specificities of Several Exopeptidases

| Enzyme | Source | Specificity |
|--------|--------|-------------|
|--------|--------|-------------|

|                    |                 |                                                                                              |
|--------------------|-----------------|----------------------------------------------------------------------------------------------|
| Carboxypeptidase A | Bovine pancreas | Will not cleave when C-terminal residue = R, K or P or if P resides next to terminal residue |
| Carboxypeptidase B | Bovine pancreas | Cleaves when C-terminal residue = R or K; not when P resides next to terminal residue        |
| Carboxypeptidase C | Citrus leaves   | All free C-terminal residues, pH optimum = 3.5                                               |
| Carboxypeptidase Y | Yeast           | All free C-terminal residues, slowly at G residues                                           |

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## Chemical Digestion of Proteins

The most commonly utilized chemical reagent that cleaves peptide bonds by recognition of specific amino acid residues is **cyanogen bromide (CNBr)**. This reagent causes specific cleavage at the C-terminal side of M residues. The number of peptide fragments that result from CNBr cleavage is equivalent to one more than the number of M residues in a protein.

The most reliable chemical technique for C-terminal residue identification is **hydrazinolysis**. A peptide is treated with **hydrazine,  $\text{NH}_2\text{-NH}_2$** , at high temperature ( $90^\circ\text{C}$ ) for an extended length of time (20-100hr). This treatment cleaves all of the peptide bonds yielding amino-acyl hydrazides of all the amino acids excluding the C-terminal residue which can be identified chromatographically compared to amino acid standards. Due to the high percentage of hydrazine induced side reactions this technique is only used on carboxypeptidase resistant peptides.

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## Size Exclusion Chromatography

This chromatographic technique is based upon the use of a porous gel in the form of insoluble beads placed into a column. As a solution of proteins is passed through the column, small proteins can penetrate into the pores of the beads and, therefore, are retarded in their rate of travel through the column. The larger proteins a protein is the less likely it will enter the pores. Different beads with different pore sizes can be used depending upon the desired protein size separation profile.

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## Ion Exchange Chromatography

Each individual protein exhibits a distinct overall net charge at a given pH. Some proteins will be negatively charged and some will be positively charged at the same pH. This property of proteins is the basis for ion exchange chromatography. Fine cellulose resins are used that are either negatively (**cation exchanger**) or positively (**anion exchanger**) charged. Proteins of opposite charge to the resin are retained as a solution of proteins is passed through the column. The bound proteins are then eluted by passing a solution of ions bearing a charge opposite to that of the column. By utilizing a gradient of increasing ionic strength, proteins with increasing affinity for the resin are progressively eluted.

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## Affinity Chromatography

Proteins have high affinities for their substrates or co-factors or prosthetic groups or receptors or antibodies raised against them. This affinity can be exploited in the purification of proteins. A column of beads bearing the high affinity compound can be prepared and a solution of protein passed through the column. The bound proteins are then eluted by passing a solution of unbound soluble high affinity compound through the column.

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## High Performance Liquid Chromatography (HPLC)

In column chromatography the smaller and more tightly packed a resin is the greater the separation capability of the column. In gravity flow columns the limitation column packing is the time it takes to pass the solution of proteins through the column. HPLC utilizes tightly packed fine diameter resins to impart increased resolution and overcomes the flow limitations by pumping the solution of proteins through the column under high pressure. Like standard column chromatography, HPLC columns can be used for size exclusion or charge separation. An additional separation technique commonly used with HPLC is to utilize hydrophobic resins to retard the movement of nonpolar proteins. The proteins are then eluted from the column with a gradient of increasing concentration of an organic solvent. This latter form of HPLC is termed **reversed-phase HPLC**.

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## Electrophoresis of Proteins

Proteins also can be characterized according to size and charge by separation in an electric current (electrophoresis) within solid sieving gels made from polymerized and cross-

linked acrylamide. The most commonly used technique is termed **SDS polyacrylamide gel electrophoresis (SDS-PAGE)**. The gel is a thin slab of acrylamide polymerized between two glass plates. This technique utilizes a negatively charged detergent (sodium dodecyl sulfate) to denature and solubilize proteins. SDS denatured proteins have a uniform negative charge such that all proteins will migrate through the gel in the electric field based solely upon size. The larger the protein the more slowly it will move through the matrix of the polyacrylamide. Following electrophoresis the migration distance of unknown proteins relative to known standard proteins is assessed by various staining or radiographic detection techniques.

The use of polyacrylamide gel electrophoresis also can be used to determine the isoelectric charge of proteins (**pI**). This technique is termed **isoelectric focusing**. Isoelectric focusing utilizes a thin tube of polyacrylamide made in the presence of a mixture of small positively and negatively charged molecules termed ampholytes. The ampholytes have a range of pIs that establish a pH gradient along the gel when current is applied. Proteins will, therefore, cease migration in the gel when they reach the point where the ampholytes have established a pH equal to the proteins pI.

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## Centrifugation of Proteins

Proteins will sediment through a solution in a centrifugal field dependent upon their mass. Analytical centrifugation measure the rate that proteins sediment. The most common solution utilized is a linear gradient of sucrose (generally from 5-20%). Proteins are layered atop the gradient in an ultracentrifuge tube then subjected to centrifugal fields in excess of 100,000 x g. The sizes of unknown proteins can then be determined by comparing their migration distance in the gradient with those of known standard proteins.

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- Control of Gene Expression in Prokaryotes
- The *lac* Operon of *E. coli*
- The *trp* Operon of *E. coli*
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## Introduction

The controls that act on gene expression (i.e., the ability of a gene to produce a biologically active protein) are much more complex in eukaryotes than in prokaryotes. A major difference is the presence in eukaryotes of a nuclear membrane, which prevents the simultaneous transcription and translation that occurs in prokaryotes. Whereas, in prokaryotes, control of transcriptional initiation is the major point of regulation, in eukaryotes the regulation of gene expression is controlled nearly equivalently from many different points.

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## Gene Control in Prokaryotes

In bacteria, genes are clustered into **operons**: gene clusters that encode the proteins necessary to perform coordinated function, such as biosynthesis of a given amino acid. RNA that is transcribed from a prokaryotic operon is **polycistronic** a term implying that multiple proteins are encoded in a single transcript.

In bacteria, control of the rate of transcriptional initiation is the predominant site for control of gene expression. As with the majority of prokaryotic genes, initiation is controlled by two DNA sequence elements that are approximately 35 bases and 10 bases, respectively, upstream of the site of transcriptional initiation and as such are identified as the -35 and -10 positions. These 2 sequence elements are termed **promoter sequences**, because they *promote* recognition of transcriptional start sites by RNA polymerase. The consensus sequence for the -35 position is TTGACA, and for the -10 position, TATAAT. (The -10 position is also known as the **Pribnow-box**.) These promoter sequences are recognized and contacted by RNA polymerase.

The activity of RNA polymerase at a given promoter is in turn regulated by interaction with accessory proteins, which affect its ability to recognize start sites. These regulatory proteins can act both positively (**activators**) and negatively (**repressors**). The accessibility of promoter regions of prokaryotic DNA is in many cases regulated by the interaction of proteins with sequences termed **operators**. The operator region is adjacent to the promoter elements in most operons and in most cases the sequences of the operator bind a repressor protein. However, there are several operons in *E. coli* that contain overlapping sequence



elements, one that binds a repressor and one that binds an activator.

As indicated above, prokaryotic genes that encode the proteins necessary to perform coordinated function are clustered into operons. Two major modes of transcriptional regulation function in bacteria (*E. coli*) to control the expression of operons. Both mechanisms involve repressor proteins. One mode of regulation is exerted upon operons that produce gene products necessary for the utilization of energy; these are **catabolite-regulated operons**. The other mode regulates operons that produce gene products necessary for the synthesis of small biomolecules such as amino acids. Expression from the latter class of operons is **attenuated** by sequences within the transcribed RNA.

A classic example of a catabolite-regulated operon is the **lac operon**, responsible for obtaining energy from  $\beta$ -galactosides such as lactose. A classic example of an attenuated operon is the **trp operon**, responsible for the biosynthesis of tryptophan.

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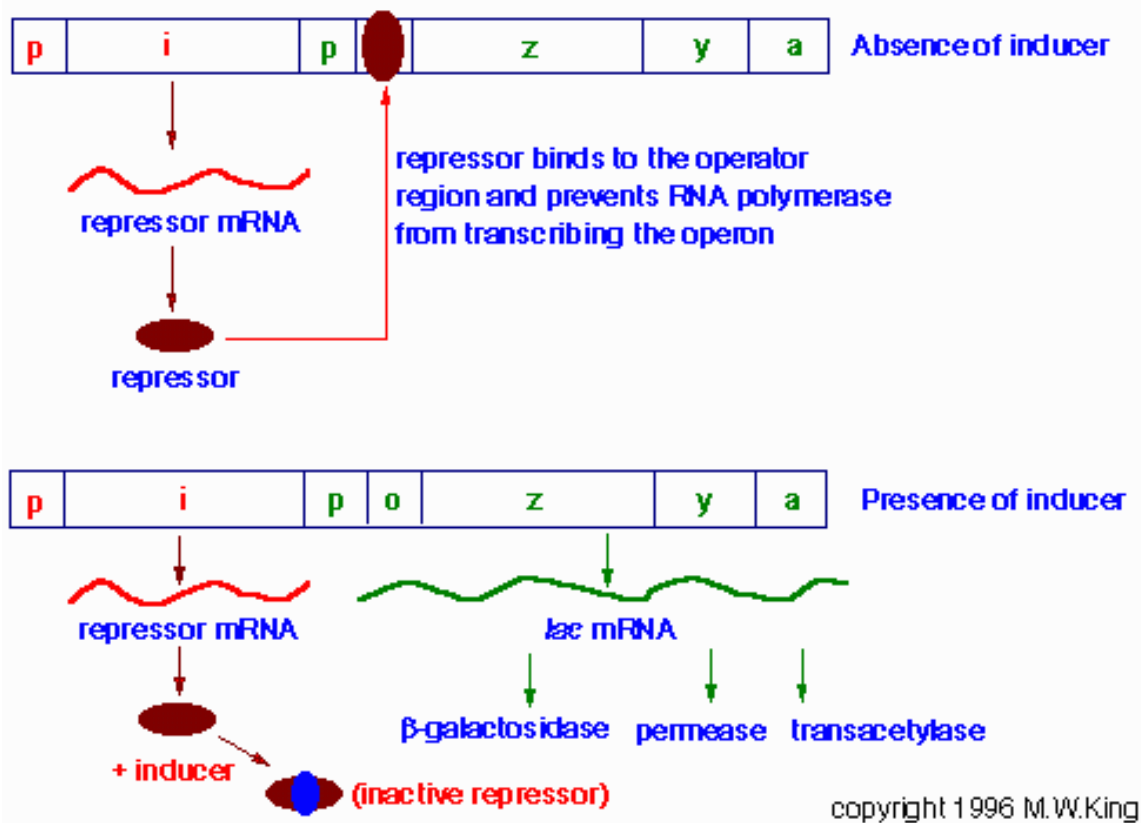
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## The *lac* Operon

The *lac* operon (see diagram below) consists of one regulatory gene (the *i* gene) and three structural genes (*z*, *y*, and *a*). The *i* gene codes for the repressor of the *lac* operon. The *z* gene codes for  $\beta$ -galactosidase ( $\beta$ -gal), which is primarily responsible for the hydrolysis of the disaccharide, lactose into its monomeric units, galactose and glucose. The *y* gene codes for permease, which increases permeability of the cell to  $\beta$ -galactosides. The *a* gene encodes a transacetylase. During normal growth on a glucose-based medium, the *lac* repressor is bound to the operator region of the *lac* operon, preventing transcription. However, in the presence of an inducer of the *lac* operon, the repressor protein binds the inducer and is rendered incapable of interacting with the operator region of the operon. RNA polymerase is thus able to bind at the promoter region, and transcription of the operon ensues. The *lac* operon is repressed, even in the presence of lactose, if glucose is also present. This repression is maintained until the glucose supply is exhausted. The repression of the *lac* operon under these conditions is termed **catabolite repression** and is a result of the low levels of cAMP that result from an adequate glucose supply. The repression of the *lac* operon is relieved in the presence of glucose if excess cAMP is added. As the level of glucose in the medium falls, the level of cAMP increases. Simultaneously there is an increase in inducer binding to the *lac* repressor. The net result is an increase in transcription from the operon. The ability of cAMP to activate expression from the *lac* operon results from an interaction of cAMP with a protein termed **CRP** (for **cAMP receptor protein**). The protein is also called **CAP** (for **catabolite activator protein**). The cAMP-CRP complex binds to a region of the *lac* operon just upstream of the region bound by RNA polymerase and that somewhat overlaps that of the repressor binding site of the operator region. The binding of the cAMP-CRP complex to the *lac* operon stimulates RNA polymerase activity 20-to-50-fold.



## The *lac* Operon



Regulation of the *lac* operon in *E. coli*. The repressor of the operon is synthesized from the *i* gene. The repressor protein binds to the operator region of the operon and prevents RNA polymerase from transcribing the operon. In the presence of an inducer (such as the natural inducer, allolactose) the repressor is inactivated by interaction with the inducer. This allows RNA polymerase access to the operon and transcription proceeds. The resultant mRNA encodes the  $\beta$ -galactosidase, permease and transacetylase activities necessary for utilization of  $\beta$ -galactosides (such as lactose) as an energy source. The *lac* operon is additionally regulated through binding of the cAMP-receptor protein, CRP (also termed the catabolite activator protein, CAP) to sequences near the promoter domain of the operon. The result is a 50 fold enhancement of polymerase activity.

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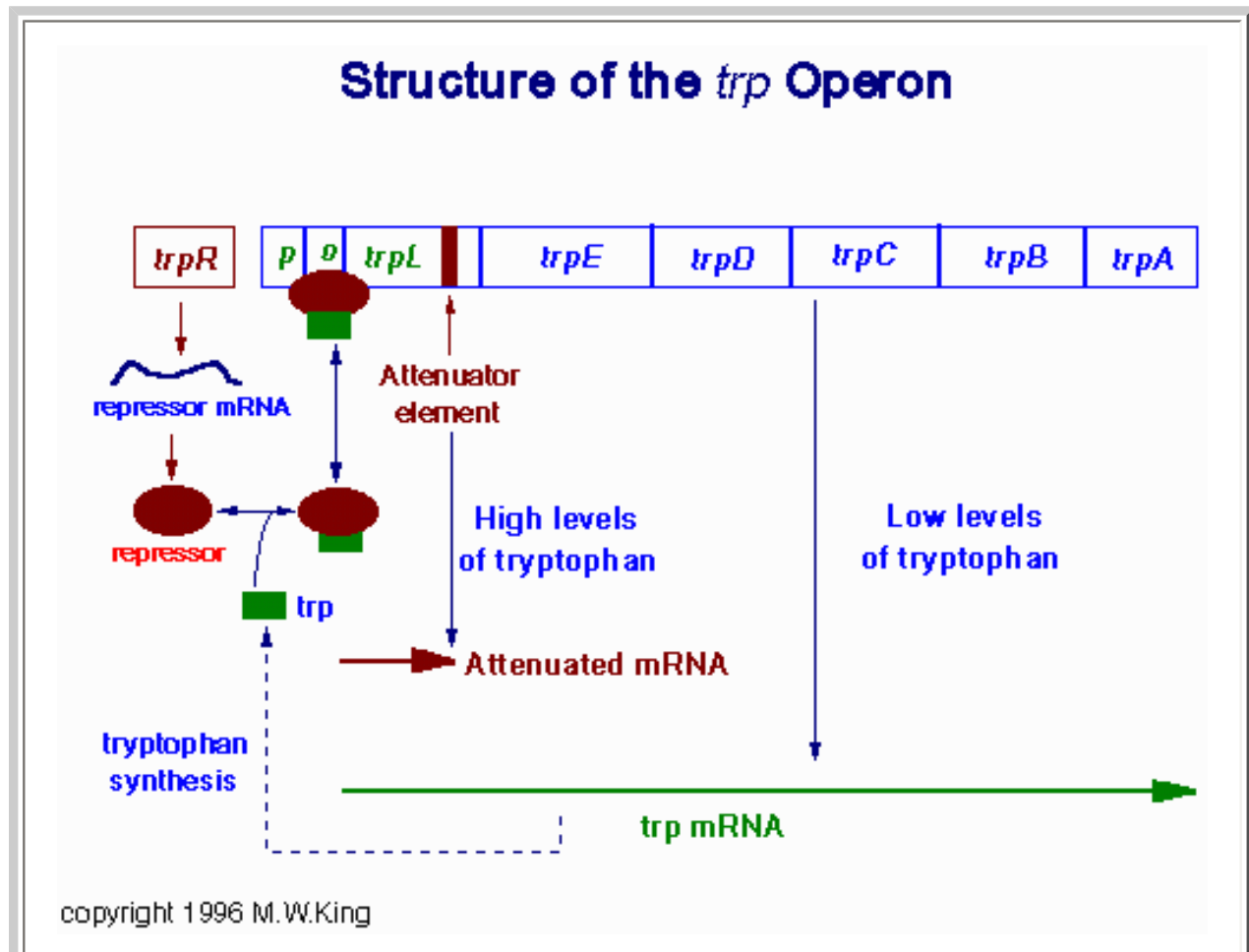
## The *trp* Operon

The *trp* operon (see diagram below) encodes the genes for the synthesis of tryptophan. This cluster of genes, like the *lac* operon, is regulated by a repressor that binds to the operator sequences. The activity of the *trp* repressor for binding the operator region is enhanced when it binds tryptophan; in this capacity, tryptophan is known as a **corepressor**. Since the activity of the *trp* repressor is enhanced in the presence of tryptophan, the rate of expression of the *trp* operon is graded in response to the level of tryptophan in the cell.

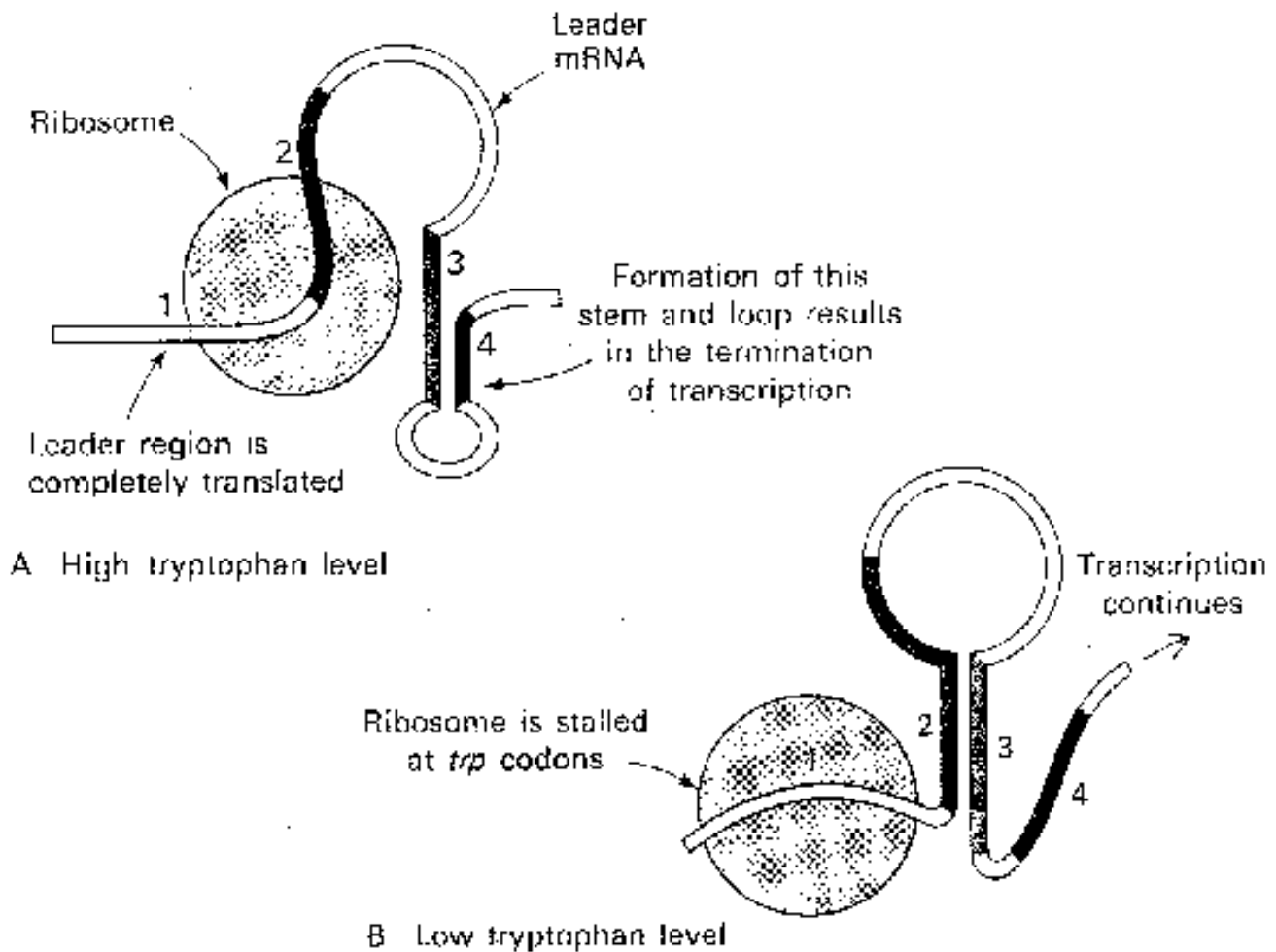
Expression of the *trp* operon is also regulated by **attenuation**. The attenuator region, which is composed of sequences found within the transcribed RNA, is involved in controlling transcription from the operon after RNA polymerase has initiated synthesis. The attenuator of sequences of the RNA are found near the 5' end of the RNA termed the leader region of the RNA. The leader sequences are located prior to the start of the coding region for the first gene of the operon (the *trpE* gene). The attenuator region contains codons for a small leader polypeptide, that contains tandem tryptophan codons. This region of the RNA is also capable of forming several different stable stem-loop structures.

Depending on the level of tryptophan in the cell---and hence the level of charged *trp*-tRNAs---the position of ribosomes on the leader polypeptide and the rate at which they are translating allows different stem-loops to form. If tryptophan is abundant, the ribosome prevents stem-loop 1-2 from forming and thereby favors stem-loop 3-4. The latter is found near a region rich in uracil and acts as the transcriptional terminator loop as described in the [RNA synthesis](#) page. Consequently, RNA polymerase is dislodged from the template.

The operons coding for genes necessary for the synthesis of a number of other amino acids are also regulated by this attenuation mechanism. It should be clear, however, that this type of transcriptional regulation is not feasible for eukaryotic cells.



Regulation of the *trp* operon in *E. coli*. The *trp* operon is controlled by both a repressor protein binding to the operator region as well as by translation-induced transcriptional attenuation. The *trp* repressor binds the operator region of the *trp* operon only when bound to tryptophan. This makes tryptophan a **co-repressor** of the operon. The *trpL* gene encodes a non-functional **leader** peptide which contains several adjacent *trp* codons. The structural genes of the operon responsible for tryptophan biosynthesis are *trpE*, *D*, *C*, *B* and *A*. When tryptophan level are high some binds to the repressor which then binds to the operator region and inhibits transcription. The mechanism of **attenuation** of the *trp* operon is diagrammed below.



Attenuation of the *trp* operon. The attenuation region of the *trp* operon contains sequences that allow the resulting mRNA to form several different stem-loop structures. These regions are identified as 1 through 4. The stem-loops that are significant as to whether transcription is attenuated or not are formed between regions 2 and 3 or between regions 3 and 4. When tryptophan levels are high there is plenty of charged trp-tRNAs available and ribosomes translating the leader peptide encoded by the *trpL* gene do not stall at the repeated trp codons in the leader peptide. Under these conditions the ribosomes rapidly cover regions 1 and 2 of the mRNA which allows the stem-loop composed of regions 3 and 4 to form. The stem-loop formed by regions 3-4 results in a transcriptional termination structure and transcription of the *trp* operon ceases, i.e. is attenuated. Conversely, when tryptophan levels are low the level of charged trp-tRNAs will also be low. This leads to a stalling of the ribosomes within the leader peptide when they encounter the trp codon repeats. The ribosome stalls over region 1 of the mRNA which allows stem-loop 2-3 to form and prevents the transcriptional termination stem-loop 3-4 from forming. The inability of this structure to form allows the entire operon to be transcribed and the tryptophan biosynthetic enzymes to be produced.

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## Gene Control in Eukaryotes

In eukaryotic cells, the ability to express biologically active proteins comes under regulation at several points:

- **1. Chromatin Structure:** The physical structure of the DNA, as it exists compacted into [chromatin](#), can affect the ability of transcriptional regulatory proteins (termed **transcription factors**) and RNA polymerases to find access to specific genes and to activate transcription from them. The presence of the histones and CpG methylation most affect accessibility of the chromatin to RNA polymerases and transcription factors.
- **2. Transcriptional Initiation:** This is the most important mode for control of eukaryotic gene expression (see below for more details). Specific factors that exert control include the strength of **promoter elements** within the DNA sequences of a given gene, the presence or absence of **enhancer sequences** (which enhance the activity of RNA polymerase at a given promoter by binding specific transcription factors), and the interaction between multiple activator proteins and inhibitor proteins.
- **3. Transcript Processing and Modification:** Eukaryotic mRNAs must be capped and polyadenylated, and the introns must be accurately removed (see [RNA Synthesis Page](#)). Several genes have been identified that undergo tissue-specific patterns of alternative splicing, which generate biologically different proteins from the same gene.
- **4. RNA Transport:** A fully processed mRNA must leave the nucleus in order to be translated into protein.
- **5. Transcript Stability:** Unlike prokaryotic mRNAs, whose half-lives are all in the range of 1--5 minutes, eukaryotic mRNAs can vary greatly in their stability. Certain unstable transcripts have

sequences (predominately, but not exclusively, in the 3'-non-translated regions) that are signals for rapid degradation.

- **6. Translational Initiation:** Since many mRNAs have multiple methionine codons, the ability of ribosomes to recognize and initiate synthesis from the correct AUG codon can affect the expression of a gene product. Several examples have emerged demonstrating that some eukaryotic proteins initiate at non-AUG codons. This phenomenon has been known to occur in *E. coli* for quite some time, but only recently has it been observed in eukaryotic mRNAs.
- **7. Post-Translational Modification:** Common modifications include glycosylation, acetylation, fatty acylation, disulfide bond formations, etc.
- **8. Protein Transport:** In order for proteins to be biologically active following translation and processing, they must be transported to their site of action.
- **9. Control of Protein Stability:** Many proteins are rapidly degraded, whereas others are highly stable. Specific amino acid sequences in some proteins have been shown to bring about rapid degradation.

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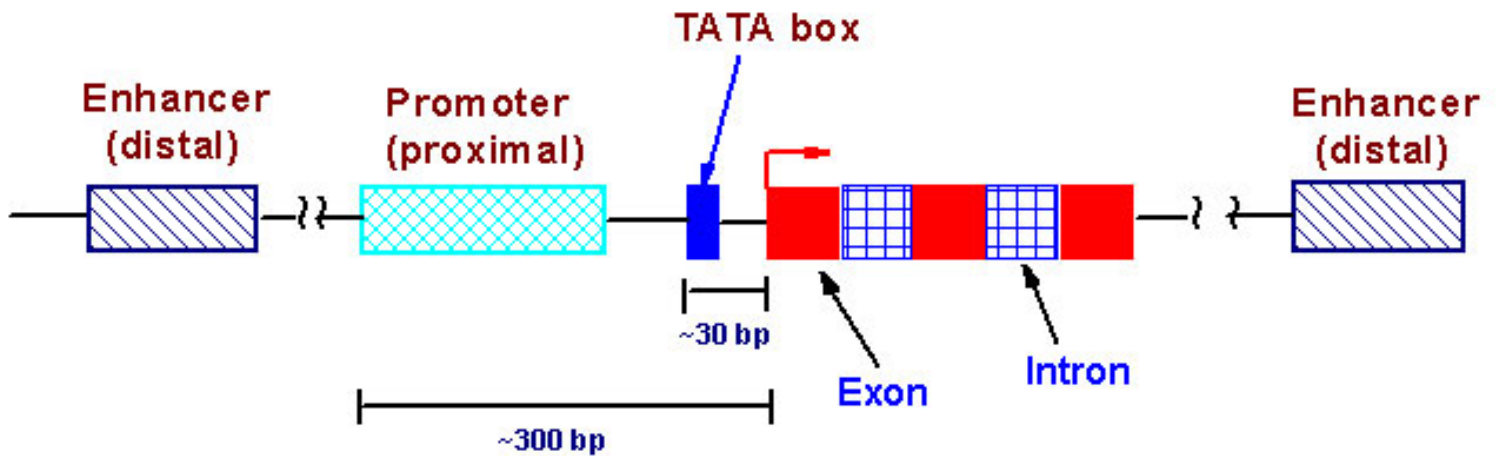
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## Control of Eukaryotic Transcription Initiation

Transcription of the different classes of RNAs in eukaryotes is carried out by three different polymerases (see [RNA Synthesis Page](#)). RNA pol I synthesizes the rRNAs, except for the 5S species. RNA pol II synthesizes the mRNAs and some small nuclear RNAs (snRNAs) involved in RNA splicing. RNA pol III synthesizes the 5S rRNA and the tRNAs. The vast majority of eukaryotic RNAs are subjected to post-transcriptional processing.

The most complex controls observed in eukaryotic genes are those that regulate the expression of RNA pol II-transcribed genes, the mRNA genes. Almost all eukaryotic mRNA genes contain a basic structure consisting of coding exons and non-coding introns and basal promoters of two types and any number of different transcriptional regulatory domains (see diagrams below). The basal promoter elements are termed **CCAAT-boxes** (pronounced cat) and **TATA-boxes** because of their sequence motifs. The TATA-box resides 20 to 30 bases upstream of the transcriptional start site and is similar in sequence to the prokaryotic Pribnow-box (consensus  $\text{TATA}^{\text{T/A}}\text{A}^{\text{T/A}}$ , where  $^{\text{T/A}}$  indicates that either base may be found at that position).

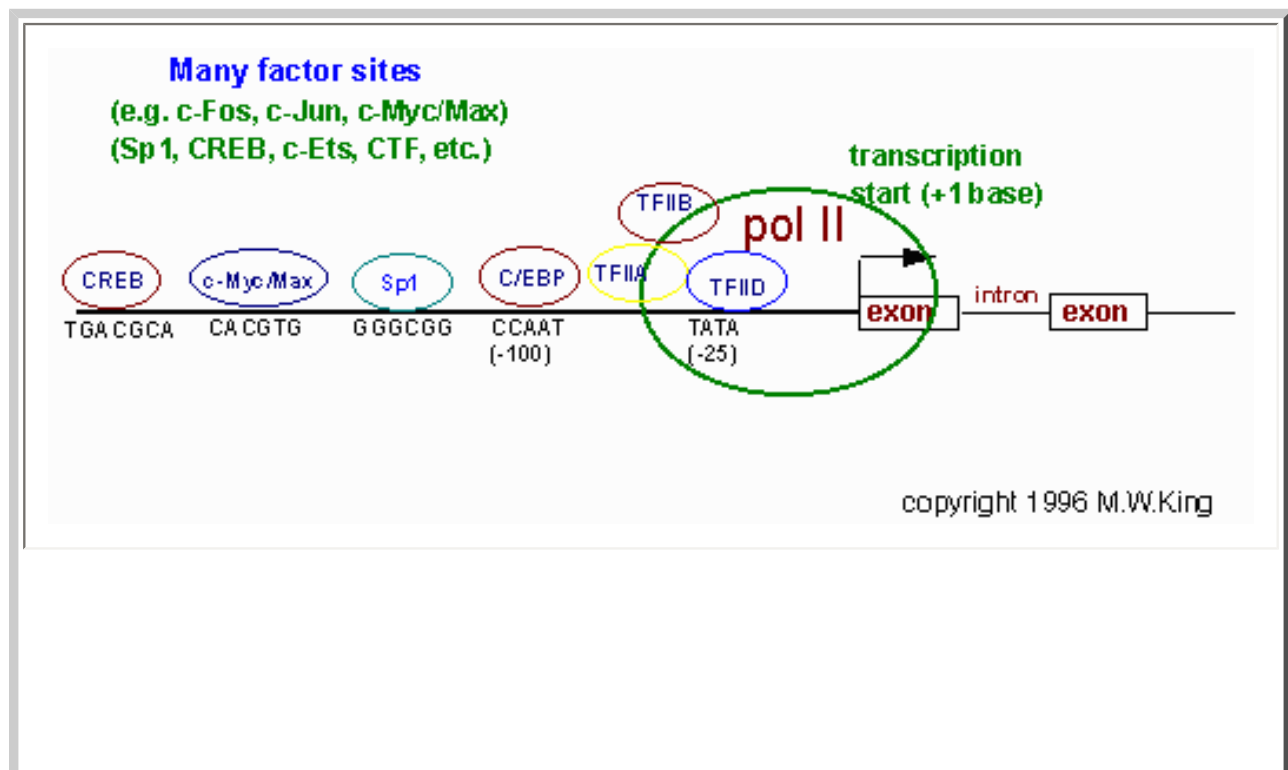




Typical structure of a eukaryotic mRNA gene.

Numerous proteins identified as **TFIIA, B, C**, etc. (for **t**ranscription **f**actors regulating RNA pol **II**), have been observed to interact with the TATA-box. The CCAAT-box (consensus  $GG^T/C$ CAATCT) resides 50 to 130 bases upstream of the transcriptional start site. The protein identified as **C/EBP** (for **C**CAAT-box/**E**nhancer **B**inding **P**rotein) binds to the CCAAT-box element.

There are many other regulatory sequences in mRNA genes, as well, that bind various transcription factors (see diagram below). These regulatory sequences are predominantly located upstream (5') of the transcription initiation site, although some elements occur downstream (3') or even within the genes themselves. The number and type of regulatory elements to be found varies with each mRNA gene. Different combinations of transcription factors also can exert differential regulatory effects upon transcriptional initiation. The various cell types each express characteristic combinations of transcription factors; this is the major mechanism for cell-type specificity in the regulation of mRNA gene expression.



Structure of the upstream region of a typical eukaryotic mRNA gene that hypothetically contains 2 exons and a single intron. The diagram indicates the TATA-box and CCAAT-box basal elements at positions -25 and -100, respectively. The transcription factor TFIID has been shown to be the **TATA-box binding protein, TBP**. Several additional transcription factor binding sites have been included and shown to reside upstream of the 2 basal elements and of the transcriptional start site. The location and order of the variously indicated transcription factor-binding sites is only diagrammatic and not indicative as being typical of all eukaryotic mRNA genes. There exists a vast array of different transcription factors that regulate the transcription of all 3 classes of eukaryotic gene encoding the mRNAs, tRNAs and rRNAs. [CREB=cAMP response element binding protein] [C/EBP=CCAAT-box/enhancer binding protein]. The large green circle represents RNA polymerase II.

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## Structural Motifs in Eukaryotic Transcription Factors

- Homeodomain:** The homeodomain is a highly conserved domain of 60 amino acids found in a large family of transcription factors. This family was first identified in *Drosophila* as a group of genes that, when altered, would cause transformations of one body part for another (eg legs for antenna), so called homeotic transformations. This class of genes has been identified in both invertebrate and vertebrate organisms. The homeodomain itself forms a structure highly similar to the bacterial helix-turn-helix proteins. The principal function of all homeodomain containing proteins is in the establishment of pattern in an organism such as that of the spinal column in vertebrates.
- POU Domain:** The POU domain is a domain that is a hybrid between a domain related to the homeodomain and an POU-specific domain. The term POU was derived from the names of the first 3 factors shown to have a region of similarity, **P**it-1 (a pituitary-specific transcription factor), **O**ct-1 (an octamer binding protein first shown to regulate immunoglobulin gene transcription) and **u**nc-86 (a nematode gene).
- Helix-Loop-Helix (HLH):** The HLH domain is involved in protein dimerization. The HLH motif is composed of two regions of  $\alpha$ -helix separated by a region of variable length which forms a loop between the 2  $\alpha$ -helices. This motif is quite similar to the Helix-turn-helix motif found in several prokaryotic transcription factors such as the CRP protein involved in the regulation of the *lac* operon. The  $\alpha$ -helical domains are structurally similar and are necessary for protein interaction with sequence elements that exhibit a twofold axis of symmetry. This class of transcription factor most often contains a region of basic amino acids located on the N-terminal side of the HLH domain (termed bHLH proteins) that is necessary in order for the protein to bind DNA at specific sequences. The HLH domain is necessary for homo- and heterodimerization. Examples of bHLH proteins include MyoD (a myogenesis inducing transcription factor) and c-Myc (originally identified as a retroviral oncogene). Several HLH proteins that do not contain the basic region act as repressors because of this lack. These HLH proteins repress the activity of other bHLH proteins by forming heterodimers with them and preventing DNA binding.

- Zinc Fingers:** The zinc finger domain is a DNA-binding motif consisting of specific spacings of cysteine and histidine residues that allow the protein to bind zinc atoms. The metal atom coordinates the sequences around the cysteine and histidine residues into a finger-like domain. The finger domains can interdigitate into the major groove of the DNA helix. The spacing of the zinc finger domain in this class of transcription factor coincides with a half-turn of the double helix. The classic example is the RNA pol III transcription factor, TFIIIA. Proteins of the steroid/thyroid hormone family of transcription factors also contain zinc fingers.
- Leucine Zipper:** The leucine zipper domain is necessary for protein dimerization. It is a motif generated by a repeating distribution of leucine residues spaced 7 amino acids apart within  $\alpha$ -helical regions of the protein. These leucine residues end up with their R-groups protruding from the  $\alpha$ -helical domain in which the leucine residues reside. The protruding R-groups are thought to interdigitate with leucine R groups of another leucine zipper domain, thus stabilizing homo- or heterodimerization. The leucine zipper domain is present in many DNA-binding proteins, such as c-Myc, and C/EBP.
- Winged Helix:** The winged helix is a DNA-binding motif composed of an  $\alpha/\beta$  structure. This structure contains 3 N-terminal  $\alpha$ -helices and a 3-stranded antiparallel  $\beta$ -sheet. The folding of the  $\beta$ -sheet region about the  $\alpha$ -helices give the appearance of wings on the helices, hence the term **winged-helix**. This motif was first identified in the transcription factor HNF-3 $\gamma$ . HNF-3 $\gamma$  is a member of a large family of transcription factors that are related to the *Drosophila* gene *fork head*, hence the gene family is termed the fork head (FKH) family. The nomenclature of the fork head family of transcription factors has been changed so that all members have names that initiate with **Fox**.

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### Table of Representative Transcription Factors

| Factor          | Sequence Motif                                                                                | Comments                                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| c-Myc and Max   | CACGTG                                                                                        | c-Myc first identified as retroviral oncogene; Max specifically associates with c-Myc in cells                                                       |
| c-Fos and c-Jun | TGAC <sup>C</sup> / <sub>G</sub> T <sup>C</sup> / <sub>A</sub> A                              | both first identified as retroviral oncogenes; associate in cells, also known as the factor AP-1                                                     |
| CREB            | TGACG <sup>C</sup> / <sub>T</sub> C <sup>C</sup> / <sub>A</sub> G <sup>G</sup> / <sub>A</sub> | binds to the cAMP response element; family of at least 10 factors resulting from different genes or alternative splicing; can form dimers with c-Jun |

|                                               |                         |                                                                                                                            |
|-----------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| c-ErbA; also TR<br>(thyroid hormone receptor) | GTGTCAAAGGTCA           | first identified as retroviral oncogene; member of the steroid/thyroid hormone receptor superfamily; binds thyroid hormone |
| c-Ets                                         | $G/C^A/C GGAA/T GT/C$   | first identified as retroviral oncogene; predominates in B- and T-cells                                                    |
| GATA                                          | $T/A GATA$              | family of erythroid cell-specific factors,<br>GATA-1 to -6                                                                 |
| c-Myb                                         | $T/C AAC^G/T G$         | first identified as retroviral oncogene; hematopoietic cell-specific factor                                                |
| MyoD                                          | CAACTGAC                | controls muscle differentiation                                                                                            |
| NF-(kappa)B and c-Rel                         | $GGGA^A/C TN^T/C CC(1)$ | both factors identified independently; c-Rel first identified as retroviral oncogene; predominate in B- and T-cells        |
| RAR (retinoic acid receptor)                  | ACGTCATGACCT            | binds to elements termed RAREs (retinoic acid response elements) also binds to c-Jun/c-Fos site                            |
| SRF (serum response factor)                   | GGATGTCCATATTAGGACATCT  | exists in many genes that are inducible by the growth factors present in serum                                             |

The list is only representative of the hundreds of identified factors, some emphasis is placed on several factors that exhibit oncogenic potential.

<sup>1</sup>N signifies that any base can occupy that position.

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## Introduction

Transcription is the mechanism by which a template strand of DNA is utilized by specific RNA polymerases to generate one of the three different classifications of RNA. These 3 RNA classes are:

- **1. Messenger RNAs (mRNAs):** This class of RNAs are the genetic **coding** templates used by the translational machinery to determine the order of amino acids incorporated into an elongating polypeptide in the process of [translation](#).
- **2. Transfer RNAs (tRNAs):** This class of small RNAs form covalent attachments to individual amino acids and recognize the encoded sequences of the mRNAs to allow correct insertion of amino acids into the elongating polypeptide chain.
- **3. Ribosomal RNAs (rRNAs):** This class of RNAs are assembled, together with numerous ribosomal proteins, to form the **ribosomes**. Ribosomes engage the mRNAs and form a catalytic domain into which the tRNAs enter with their attached amino acids. The proteins of the ribosomes catalyze all of the functions of polypeptide synthesis.

All RNA polymerases are dependent upon a DNA template in order to synthesize RNA. The resultant RNA is, therefore, complimentary to the template strand of the DNA duplex and identical to the non-template strand. The non-template strand is called the coding strand because its' sequences are identical to those of the mRNA. However, in RNA, U is substituted for T.

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## Classes of RNA Polymerases

In prokaryotic cells, all 3 RNA classes are synthesized by a single polymerase. In eukaryotic cells there are 3 distinct classes of RNA polymerase, **RNA polymerase (pol) I, II and III**. Each polymerase is responsible for the synthesis of a different class of RNA. The capacity of the various polymerases to synthesize different RNAs was shown with the toxin  $\alpha$ -amanitin. At low concentrations of  $\alpha$ -amanitin synthesis of mRNAs are affected but not rRNAs nor tRNAs. At high concentrations, both mRNAs and tRNAs are affected. These observations have allowed the identification of which polymerase synthesizes which class of RNAs. RNA pol I is responsible for rRNA synthesis (excluding the 5S rRNA). There are 4 major rRNAs in eukaryotic cells designated by their sedimentation size. The 28S, 5S and 5.8S RNAs are associated with the large ribosomal subunit and the 18S rRNA is associated with the small ribosomal subunit. RNA pol II synthesizes the mRNAs and some of the small nuclear RNAs (snRNAs) involved in RNA splicing. RNA pol III synthesizes the tRNAs, the 5S rRNA and some snRNAs.

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## Mechanism of RNA Polymerases

Synthesis of RNA exhibits several features that are synonymous with DNA replication. RNA synthesis requires accurate and efficient initiation, elongation proceeds in the 5' → 3' direction (i.e. the polymerase moves along the template strand of DNA in the 3' → 5' direction), and RNA synthesis requires distinct and accurate termination. Transcription exhibits several features that are distinct from replication.

1. Transcription initiates, both in prokaryotes and eukaryotes, from many more sites than replication.
- 2. There are many more molecules of RNA polymerase per cell than DNA polymerase.
- 3. RNA polymerase proceeds at a rate much slower than DNA polymerase (approximately 50-100 bases/sec for RNA versus near 1000 bases/sec for DNA).



- **4.** Finally the fidelity of RNA polymerization is much lower than DNA. This is allowable since the aberrant RNA molecules can simply be turned over and new correct molecules made.

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## Processes of Transcription

Signals are present within the DNA template that act in *cis* to stimulate the initiation of transcription. These sequence elements are termed **promoters**. Promoter sequences promote the ability of RNA polymerases to recognize the nucleotide at which initiation begins. Additional sequence elements are present within genes that act in *cis* to enhance polymerase activity even further. These sequence elements are termed **enhancers**. Transcriptional promoter and enhancer elements are important sequences used in the [control of gene expression](#).

*E. coli* RNA polymerase is composed of 5 distinct polypeptide chains. Association of several of these generates the **RNA polymerase holoenzyme**. The **sigma** subunit is only transiently associated with the holoenzyme. This subunit is required for accurate initiation of transcription by providing polymerase with the proper cues that a start site has been encountered.

In both prokaryotic and eukaryotic transcription the first incorporated ribonucleotide is a purine and it is incorporated as a triphosphate. In *E. coli* several additional nucleotides are added before the sigma subunit dissociates.

Elongation involves the addition of the 5'-phosphate of ribonucleotides to the 3'-OH of the elongating RNA with the concomitant release of pyrophosphate. Nucleotide addition continues until specific termination signals are encountered. Following termination the core polymerase dissociates from the template. The core and sigma subunit can then reassociate forming the holoenzyme again ready to initiate another round of transcription.

In *E. coli* transcriptional termination occurs by both factor-dependent and factor-independent means. Two structural features of all *E. coli* factor-independently terminating genes have been identified. One feature is the presence of 2 symmetrical GC-rich segments that are capable of forming a stem-loop structure in the RNA and the second is a downstream A rich sequence in the template. The formation of the stem-loop in the RNA destabilizes the association between polymerase and the DNA template. This is further destabilized by the weaker nature of the AU base pairs that are formed, between the template and the RNA, following the stem-loop. This leads to dissociation of polymerase and termination of transcription. Most genes in *E. coli* terminate by this method.

Factor-dependent termination requires the recognition of termination sequences by the termination protein, **rho**. The rho factor recognizes and binds to sequences in the 3' portion of the RNA. This binding destabilizes the polymerase-template interaction leading to dissociation of the polymerase and termination of transcription.

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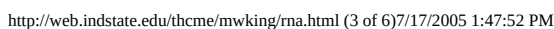
## Posttranscriptional Processing of RNAs

When transcription of bacterial rRNAs and tRNAs is completed they are immediately ready for use in translation. No additional processing takes place. Translation of bacterial mRNAs can begin even before transcription is completed due to the lack of the nuclear-cytoplasmic separation that exists in eukaryotes. The ability to initiate translation of prokaryotic RNAs while transcription is still in progress affords a unique opportunity for regulating the transcription of certain genes. An additional feature of bacterial mRNAs is that most are **polycistronic**. This means that multiple polypeptides can be synthesized from a single primary transcript.

Polycistronic mRNAs are very rare in eukaryotic cells but have been identified. The mitochondrial genomes in mammals and the slime mold, *Dictyostelium discoideum*, encode polycistronic mRNAs that are processed into primarily mono-, di-, and tricistronic transcripts. In addition, several viruses encode polycistronic RNAs.

In contrast to bacterial transcripts, eukaryotic RNAs (all 3 classes) undergo significant post-transcriptional processing. All 3 classes of RNA are transcribed from genes that contain **introns**. The sequences encoded by the intronic DNA must be removed from the primary transcript prior to the RNAs being biologically active. The process of intron removal is called **RNA splicing**. Additional processing occurs to mRNAs. The 5' end of all eukaryotic mRNAs are **capped** with a unique 5' -----> 5' linkage to a 7-methylguanosine residue. The capped end of the mRNA is thus, protected from exonucleases and more importantly is recognized by specific proteins of the translational machinery.

Messenger RNAs also are polyadenylated at the 3' end. A specific sequence, **AUAAA**, is recognized by the endonuclease which cleaves the primary transcript approximately 11 - 30 bases 3' of the sequence. A stretch of 20 - 250 A residues is then added to the 3' end by the polyadenylate polymerase activity.



In addition to intron removal in tRNAs, extra nucleotides at both the 5' and 3' ends are cleaved, the sequence **5'-CCA-3'** is added to the 3' end of all tRNAs and several nucleotides undergo modification. There have been more than 60 different modified bases identified in tRNAs.

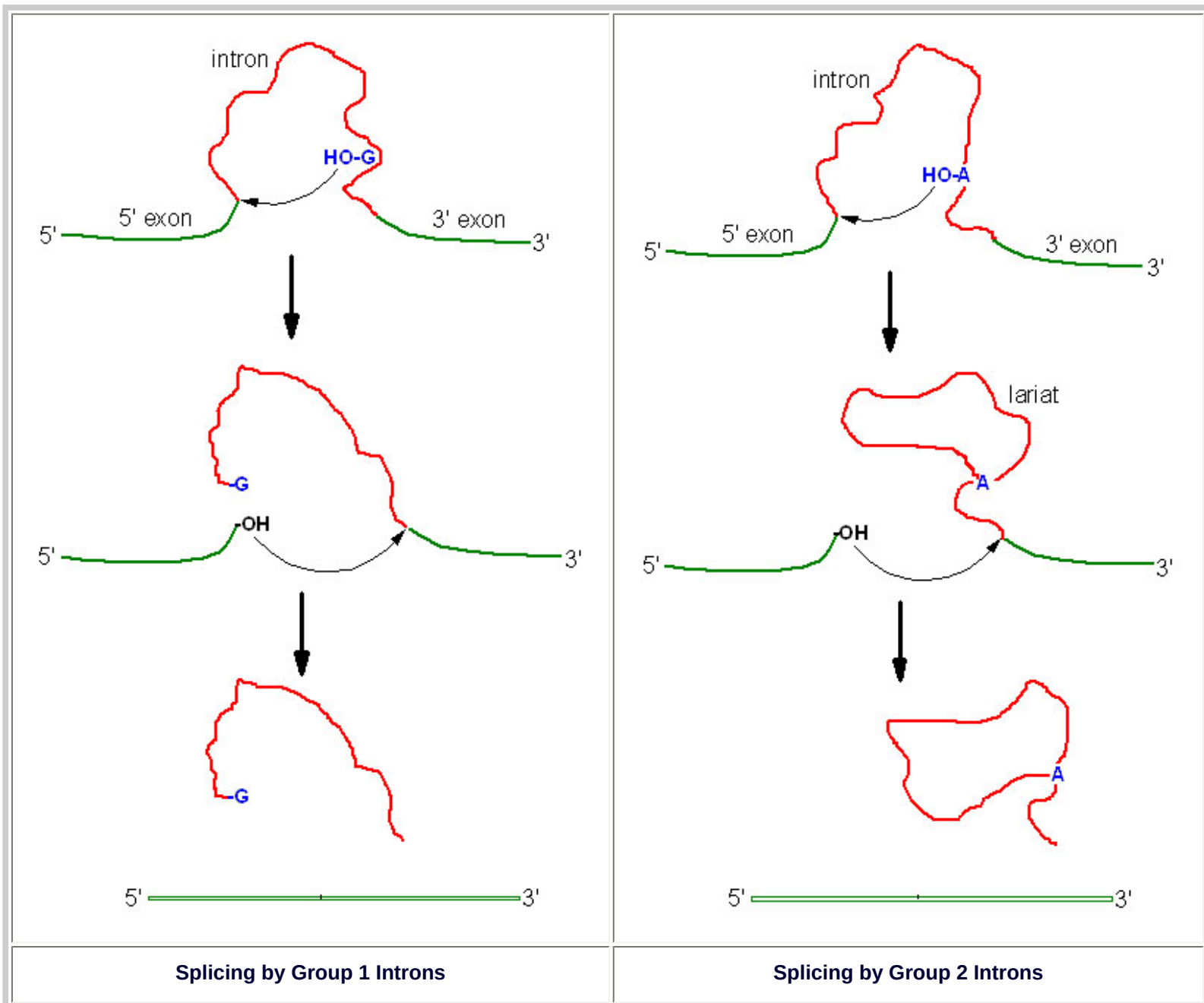
Both prokaryotic and eukaryotic rRNAs are synthesized as long precursors termed **preribosomal RNAs**. In eukaryotes a 45S preribosomal RNA serves as the precursor for the 18S, 28S and 5.8S rRNAs.

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## Splicing of RNAs

There are several different classes of reactions involved in intron removal. The 2 most common are the **group I** and **group II** introns. Group I introns are found in nuclear, mitochondrial and chloroplast rRNA genes, group II in mitochondrial and chloroplast mRNA genes. Many of the group I and group II introns are **self-splicing**, i.e. no additional protein factors are necessary for the intron to be accurately and efficiently spliced out.

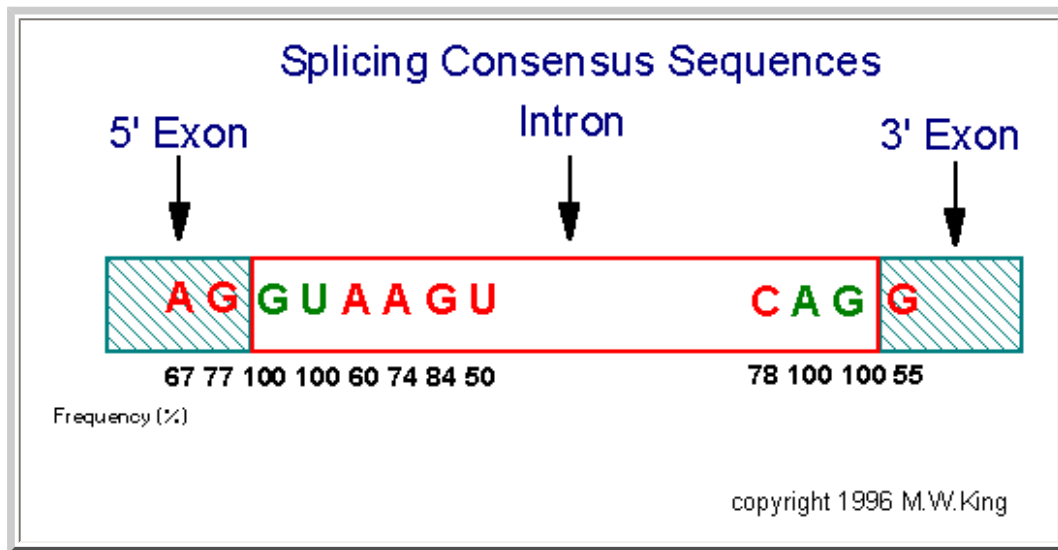
Group I introns require an external guanosine nucleotide as a cofactor. The 3'-OH of the guanosine nucleotide acts as a nucleophile to attack the 5'-phosphate of the 5' nucleotide of the intron. The resultant 3'-OH at the 3' end of the 5' exon then attacks the 5' nucleotide of the 3' exon releasing the intron and covalently attaching the two exons together. The 3' end of the 5' exon is termed the **splice donor site** and the 5' end of the 3' exon is termed the **splice acceptor site**.



Group II introns are spliced similarly except that instead of an external nucleophile the 2'-OH of an adenine residue within the intron is the nucleophile. This residue attacks the 3' nucleotide of the 5' exon forming an internal loop called a **lariat structure**. The 3' end of the 5' exon then attacks the 5' end of the 3' exon as in group I splicing releasing the intron and covalently attaching the two exons together.

The third class of introns is also the largest class found in nuclear mRNAs. This class of introns undergoes a splicing reaction similar to group II introns in that an internal lariat structure is formed. However, the splicing is catalyzed by specialized RNA-protein complexes called **small nuclear ribonucleoprotein particles** (snRNPs, pronounced **snurps**). The RNAs found in snRNPs are identified as U1, U2, U4, U5 and U6. The genes encoding these snRNAs are highly conserved in vertebrate and insects and are also found in yeasts and slime molds indicating their importance.

Analysis of a large number of mRNA genes has led to the identification of highly conserved consensus sequences at the 5' and 3' ends of essentially all mRNA introns.



The U1 RNA has sequences that are complementary to sequences near the 5' end of the intron. The binding of U1 RNA distinguishes the GU at the 5' end of the intron from other randomly placed GU sequences in mRNAs. The U2 RNA also recognizes sequences in the intron, in this case near the 3' end. The addition of U4, U5 and U6 RNAs forms a complex identified as the **spliceosome** that then removes the intron and joins the two exons together.

An additional mechanism of intron removal is the process of tRNA splicing. These introns are spliced by a specific splicing endonuclease that involves a cut-and-paste mechanism. In order for tRNA intron removal to occur the tRNA must first be properly folded into its characteristic cloverleaf shape. Misfolded precursor tRNAs are not processed which allows the splicing reaction to serve as a control step in the generation of mature tRNAs.

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## Clinical Significances of Alternative and Aberrant Splicing

The presence of introns in eukaryotic genes would appear to be an extreme waste of cellular energy when considering the number of nucleotides incorporated into the primary transcript only to be removed later as well as the energy utilized in the synthesis of the splicing machinery. However, the presence of introns can protect the genetic makeup of an organism from genetic damage by outside influences such as chemical or radiation. An additionally important function of introns is to allow **alternative splicing** to occur, thereby, increasing the genetic diversity of the genome without increasing the overall number of genes. By altering the pattern of exons, from a single primary transcript, that are spliced together different proteins can arise from the processed mRNA from a single gene. Alternative splicing can occur either at specific developmental stages or in different cell types.

This process of alternative splicing has been identified to occur in the primary transcripts from at least 40 different genes. Depending upon the site of transcription, the calcitonin gene yields an RNA that synthesizes **calcitonin (thyroid)** or **calcitonin-gene related peptide (CGRP, brain)**. Even more complex is the alternative splicing that occurs in the  $\alpha$ -tropomyosin transcript. At least 8 different alternatively spliced  $\alpha$ -tropomyosin mRNAs have been identified.

Abnormalities in the splicing process can lead to various disease states. Many defects in the  $\beta$ -globin genes are known to exist leading to  **$\beta$ -thalassemias**. Some of these defects are caused by mutations in the sequences of the gene required for intron recognition and, therefore, result in abnormal processing of the  $\beta$ -globin primary transcript.

Patients suffering from a number of different connective tissue diseases exhibit humoral auto-antibodies that recognize

cellular RNA-protein complexes. Patients suffering from [systemic lupus erythematosus](#) have auto-antibodies that recognize the U1 RNA of the spliceosome.

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## Introduction

Translation is the RNA directed synthesis of polypeptides. This process requires all three classes of RNA. Although the chemistry of peptide bond formation is relatively simple, the processes leading to the ability to form a peptide bond are exceedingly complex. The **template** for correct addition of individual amino acids is the mRNA, yet both tRNAs and rRNAs are involved in the process. The tRNAs carry activated amino acids into the ribosome which is composed of rRNA and ribosomal proteins. The ribosome is associated with the mRNA ensuring correct access of activated tRNAs and containing the necessary enzymatic activities to catalyze peptide bond formation.

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# Historical Perspectives

Early genetic experiments demonstrated:

- **1.** the colinearity between the DNA and protein encoded by the DNA. Yanofsky showed that order of observed mutations in the *E. coli* tryptophan synthetase gene was the same as the corresponding amino acid changes in the protein.
- **2.** Crick and Brenner demonstrated, from a large series of double mutants of the bacteriophage T4, that the genetic code is read in a sequential manner starting from a fixed point in the gene, the code was most likely a triplet and that all 64 possible combinations of the 4 nucleotides code for amino acids, i.e. the **code is degenerate** since there are only 20 amino acids.

The above mentioned experiments only indicated deductive correlation's regarding the genetic code. The precise dictionary of the genetic code was originally determined by the use of *in vitro* translation systems derived from *E. coli* cells. Synthetic polyribonucleotides were added to these translation system along with all twenty amino acids. One amino acid at a time was radiolabeled. The first demonstration of the dictionary of the genetic code was with the use of poly(U). This synthetic polyribonucleotide encoded the amino acid phenylalanine, i.e. the resulting polypeptide was poly(F).

The utilization of a variety of repeating di- tri- and tetra polyribonucleotides established the entire genetic code. These results of these experiments confirmed that some amino acids are encoded for by more than one triplet codon, hence the degeneracy of the genetic code. These experiments also established the identity of translational termination codons.

An additional important point to come from these early experiments was that the 5' end of the RNA corresponded to the amino terminus of the polypeptide. This was important since previous labeling experiments had demonstrated that the N-terminus is the beginning of the elongating polypeptide. Therefore, *in vitro* translation experiments established that the RNA is read in the 5' to 3' direction.

Crick first postulated that translation of the genetic code would be carried out through mediation of adapter molecules. Each adapter was postulated to carry a specific amino acid and to recognize the corresponding codon. He suggested that the adapters contain RNA because codon recognition could then occur by complementarity to the sequences of the codons in the mRNA.

During the course of *in vitro* protein synthesis and labeling experiments it was shown that the amino acids became transiently bound to a low molecular weight mass fraction of RNA. This fraction of RNAs have been termed transfer RNAs (tRNAs) since they transfer amino acids to the elongating polypeptide. These results indicate that accurate translation requires two equally important recognition steps:

- **1.** the correct choice of amino acid needs to be made for attachment to the correspondingly correct tRNA.
- **2.** selection of the correct amino acid-charged tRNA by the mRNA. This process is

facilitated by the ribosomes which we will discuss below.

### Summary of Experiments to Determine the Genetic Code

- 1. The genetic code is read in a sequential manner starting near the 5' end of the mRNA. This means that translation proceeds along the mRNA in the 5' → 3' direction which corresponds to the N-terminal to C-terminal direction of the amino acid sequences within proteins.
- 2. The code is composed of a triplet of nucleotides.
- 3. That all 64 possible combinations of the 4 nucleotides code for amino acids, i.e. the code is degenerate since there are only 20 amino acids.

The precise dictionary of the genetic code was determined with the use of *in vitro* translation systems and polyribonucleotides. The results of these experiments confirmed that some amino acids are encoded by more than one triplet codon, hence the degeneracy of the genetic code. These experiments also established the identity of translational termination codons.

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### The Genetic Code

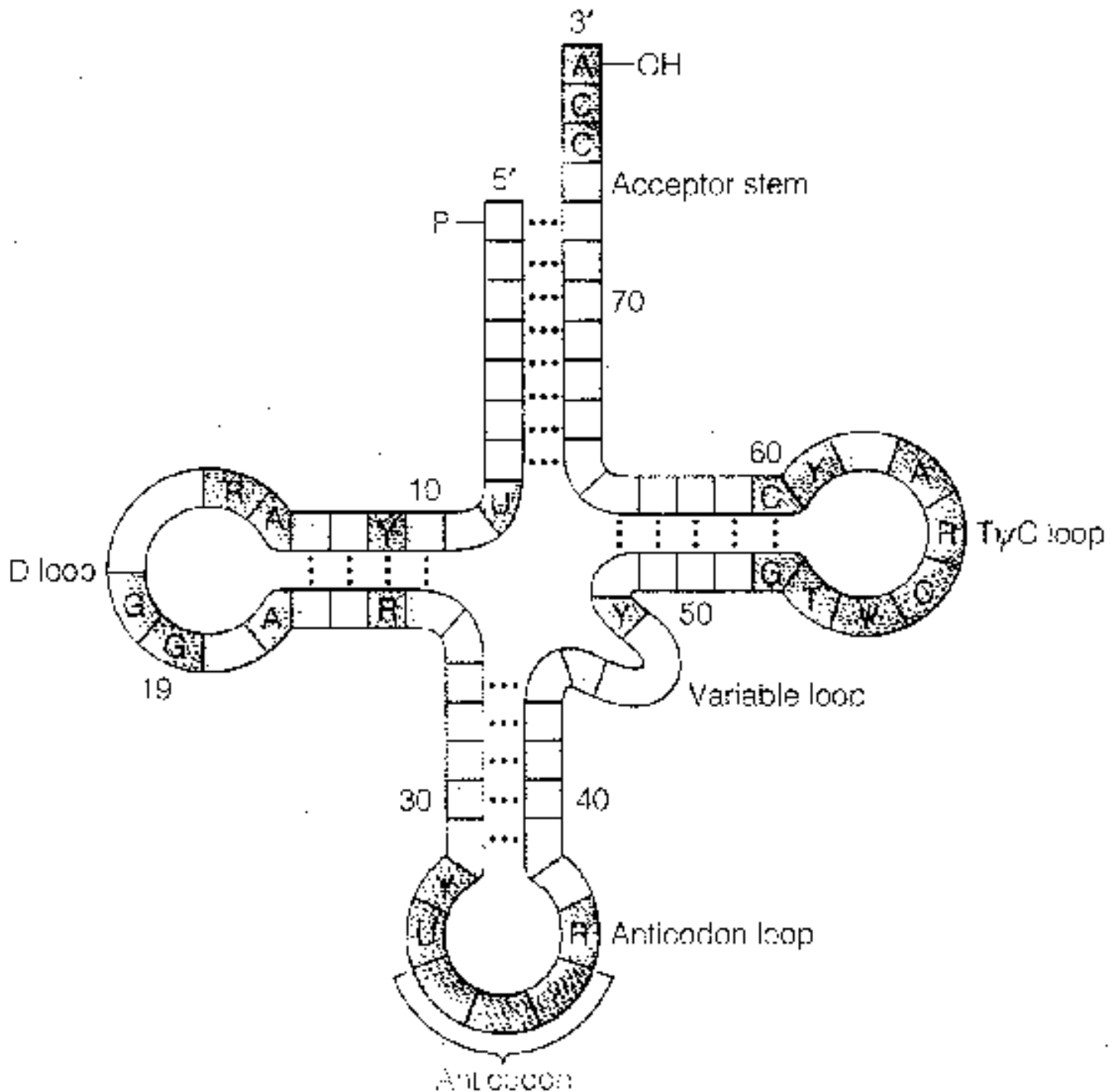
|   | U                                        | C                                        | A                                        | G                                        |
|---|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| U | UUU Phe<br>UUC Phe<br>UUA Leu<br>UUG Leu | UCU Ser<br>UCC Ser<br>UCA Ser<br>UCG Ser | UAU Tyr<br>UAC Tyr<br>UAA End<br>UAG End | UGU Cys<br>UGC Cys<br>UGA End<br>UGG Trp |
| C | CUU Leu<br>CUC Leu<br>CUA Leu<br>CUG Leu | CCU Pro<br>CCC Pro<br>CCA Pro<br>CCG Pro | CAU His<br>CAC His<br>CAA Gln<br>CAG Gln | CGU Arg<br>CGC Arg<br>CGA Arg<br>CGG Arg |
| A | AUU Ile<br>AUC Ile<br>AUA Ile<br>AUG Met | ACU Thr<br>ACC Thr<br>ACA Thr<br>ACG Thr | AAU Asn<br>AAC Asn<br>AAA Lys<br>AAG Lys | AGU Ser<br>AGC Ser<br>AGA Arg<br>AGG Arg |
| G | GUU Val<br>GUC Val<br>GUA Val<br>GUG Val | GCU Ala<br>GCC Ala<br>GCA Ala<br>GCG Ala | GAU Asp<br>GAC Asp<br>GAA Glu<br>GAG Glu | GGU Gly<br>GGC Gly<br>GGA Gly<br>GGG Gly |

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## Characteristics of tRNAs

More than 300 different tRNAs have been sequenced, either directly or from their corresponding DNA sequences. tRNAs vary in length from 60 - 95 nucleotides (18 - 28 kD). The majority contain 76 nucleotides. Evidence has shown that the role of tRNAs in translation is to **carry** activated amino acids to the elongating polypeptide chain. All tRNAs:

- **1.** exhibit a cloverleaf-like secondary structure.
- **2.** have a 5'-terminal phosphate.
- **3.** have a 7 bp stem that includes the 5'-terminal nucleotide and may contain non-Watson-Crick base pairs, e.g. GU. This portion of the tRNA is called the **acceptor** since the amino acid is carried by the tRNA while attached to the 3'-terminal OH group.
- **4.** have a D loop and a T $\phi$ C loop.
- **5.** have an anti-codon loop.
- **6.** terminate at the 3'-end with the sequence 5'-CCA-3'.
- **7.** contain 13 invariant positions and 8 semi-variant positions.
- **8.** contain numerous modified nucleotide bases.



Typical cloverleaf diagram of a tRNA

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## Activation of Amino Acids

Activation of amino acids is carried out by a two step process catalyzed by **aminoacyl-tRNA synthetases**. Each tRNA, and the amino acid it carries, are recognized by individual

aminoacyl-tRNA synthetases. This means there exists at least 20 different aminoacyl-tRNA synthetases, there are actually at least 21 since the initiator met-tRNA of both prokaryotes and eukaryotes is distinct from non-initiator met-tRNAs.

Activation of amino acids requires energy in the form of ATP and occurs in a two step reaction catalyzed by the aminoacyl-tRNA synthetases. First the enzyme attaches the amino acid to the  $\alpha$ -phosphate of ATP with the concomitant release of pyrophosphate. This is termed an **aminoacyl-adenylate intermediate**. In the second step the enzyme catalyzes transfer of the amino acid to either the 2'- or 3'-OH of the ribose portion of the 3'-terminal adenosine residue of the tRNA generating the activated aminoacyl-tRNA. Although these reaction are freely reversible, the forward reaction is favored by the coupled hydrolysis of  $PP_i$ .

Accurate recognition of the correct amino acid as well as the correct tRNA is different for each aminoacyl-tRNA synthetase. Since the different amino acids have different R groups, the enzyme for each amino acid has a different binding pocket for its specific amino acid. It is not the anticodon that determines the tRNA utilized by the synthetases. Although the exact mechanism is not known for all synthetases, it is likely to be a combination of the presence of specific modified bases and the secondary structure of the tRNA that is correctly recognized by the synthetases.

It is absolutely necessary that the discrimination of correct amino acid and correct tRNA be made by a given synthetase prior to release of the aminoacyl-tRNA from the enzyme. Once the product is released there is no further way to **proof-read** whether a given tRNA is coupled to its corresponding tRNA. Erroneous coupling would lead to the wrong amino acid being incorporated into the polypeptide since the discrimination of amino acid during protein synthesis comes from the recognition of the anticodon of a tRNA by the codon of the mRNA and not by recognition of the amino acid. This was demonstrated by reductive desulfuration of cys-tRNA<sup>cys</sup> with Raney nickel generating ala-tRNA<sup>cys</sup>. Alanine was then incorporated into an elongating polypeptide where cysteine should have been.

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## The Wobble Hypothesis

As discussed above, 3 of the possible 64 triplet codons are recognized as translational termination codons. The remaining 61 codons might be considered as being recognized by individual tRNAs. Most cells contain **isoaccepting tRNAs**, different tRNAs that are specific for the same amino acid, however, many tRNAs bind to two or three codons specifying their cognate amino acids. As an example yeast tRNA<sup>phe</sup> has the anticodon 5'-GmAA-3' and can recognize the codons 5'-UUC-3' and 5'-UUU-3'. It is, therefore, possible for non-Watson-Crick base pairing to occur at the third codon position, i.e. the 3' nucleotide of the mRNA codon and the 5' nucleotide of the tRNA anticodon. This has phenomenon been termed the **wobble hypothesis**.

### Wobble Positions in Anticodon and Codon Interactions



### Wobble Positions in Codon and Anticodon Interactions



Diagram showing the various modified nucleotides of tRNAs that are found in the wobble position in the anticodon. The top half shows the wobble nucleotides of the anticodon in blue and the various nucleotides (in red) of the wobble position of the codon that can be found in non-Watson-Crick base-pairs. The lower panel illustrates the opposite showing the wobble nucleotides of the codon in blue and the associated wobble nucleotides of the anticodon in red.

Now that we have charged aminoacyl-tRNAs and the mRNAs to convert nucleotide sequences to amino acid sequences we need to bring the two together accurately and efficiently. This is the job of the ribosomes. Ribosomes are composed of proteins and rRNAs.

All living organisms need to synthesis proteins and all cells of an organism need to synthesize proteins, therefore, it is not hard to imagine that ribosomes are a major constituent of all cells of all organisms. The make up of the ribosomes, both rRNA and associated proteins are slightly different between prokaryotes and eukaryotes.

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## Order of Events in Translation

The ability to begin to identify the roles of the various ribosomal proteins in the processes of ribosome assembly and translation was aided by the discovery that the ribosomal subunits will self assemble in vitro from their constituent parts.

Following assembly of both the small and large subunits onto the mRNA, and given the presence of charged tRNAs, protein synthesis can take place. To reiterate the process of protein synthesis:

- **1.** synthesis proceeds from the N-terminus to the C-terminus of the protein.
- **2.** the ribosomes "read" the mRNA in the 5' to 3' direction.



- **3.** active translation occurs on polyribosomes (also termed polysomes). This means that more than one ribosome can be bound to and translate a given mRNA at any one time.
- **4.** chain elongation occurs by sequential addition of amino acids to the C-terminal end of the ribosome bound polypeptide.

Translation proceeds in an ordered process. First accurate and efficient **initiation** occurs, then chain **elongation** and finally accurate and efficient **termination** must occur. All three of these processes require specific proteins, some of which are ribosome associated and some of which are separate from the ribosome, but may be temporarily associated with it.

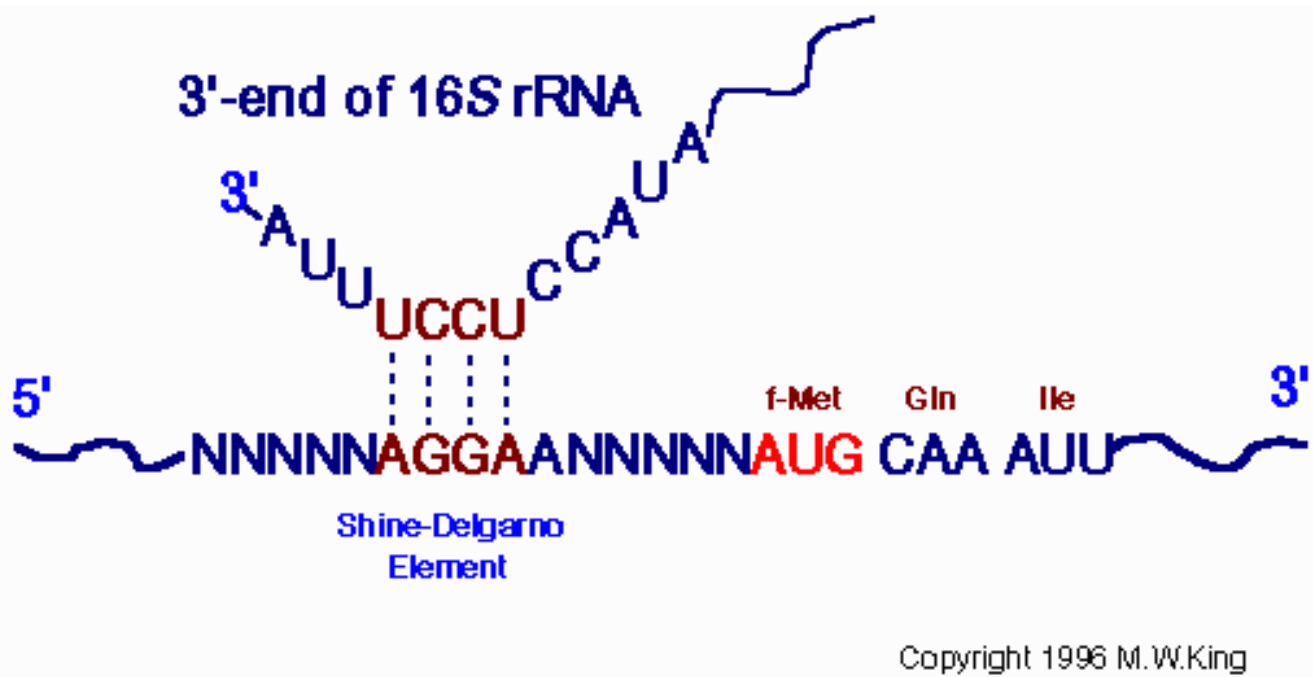
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## Initiation

Initiation of translation in both prokaryotes and eukaryotes requires a specific initiator tRNA, **tRNA<sup>met</sup><sub>i</sub>**, that is used to incorporate the initial methionine residue into all proteins. In *E. coli* a specific version of tRNA<sup>met</sup><sub>i</sub> is required to initiate translation, [**tRNA<sup>fmet</sup><sub>i</sub>**]. The methionine attached to this initiator tRNA is formylated. Formylation requires N<sup>10</sup>-formyl-THF and is carried out after the methionine is attached to the tRNA. The fmet-tRNA<sup>fmet</sup><sub>i</sub> still recognizes the same codon, AUG, as regular tRNA<sup>met</sup>. Although tRNA<sup>met</sup><sub>i</sub> is specific for initiation in eukaryotes it is not a formylated tRNA<sup>met</sup>.

The initiation of translation requires recognition of an **AUG codon**. In the polycistronic prokaryotic RNAs this AUG codon is located adjacent to a **Shine-Delgarno element** in the mRNA. The **Shine-Delgarno element** is recognized by complementary sequences in the small subunit rRNA (16S in *E. coli*). In eukaryotes initiator AUGs are generally, but not always, the first encountered by the ribosome. A specific sequence context surrounding the initiator AUG aids ribosomal discrimination. This context is **A/GCCA/GCCAUGA/G** in most mRNAs.



The Shine-Delgarno element is found at the 5' side of each initiator AUG codon in prokaryotic polycistronic mRNAs. This element is complementary to sequences present near the 3'-end of the 16S rRNA of the prokaryotic ribosome.

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## Eukaryotic Initiation Factors and Their Functions

The specific non-ribosomally associated proteins required for accurate translational initiation are termed **initiation factors**. In *E. coli* they are IFs in eukaryotes they are **eIFs**. Numerous eIFs have been identified:

| Initiation Factor | Activity                                                                   |
|-------------------|----------------------------------------------------------------------------|
| eIF-1             | repositioning of met-tRNA to facilitate mRNA binding                       |
| eIF-2             | ternary complex formation                                                  |
| eIF-2A            | AUG-dependent met-tRNA <sub>met</sub> <sup>i</sup> binding to 40S ribosome |

|                                                                                                                                                                               |                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eIF-2B (also called GEF)<br>guanine nucleotide exchange factor                                                                                                                | GTP/GDP exchange during eIF-2 recycling                                                                                                                                                                    |
| eIF-3<br>composed of ~10 subunits                                                                                                                                             | ribosome subunit antiassociation, binding to 40S subunit                                                                                                                                                   |
| Initiation factor complex<br>often referred to as eIF-4F<br>composed of 3 primary subunits:<br>eIF-4E, eIF-4A, eIF-4G and at least 2 additional factors: PABP, Mnk1 (or Mnk2) | mRNA binding to 40S subunit, ATPase-dependent RNA helicase activity, interaction between polyA tail and cap structure                                                                                      |
| PABP:<br>polyA-binding protein                                                                                                                                                | binds to the polyA tail of mRNAs and provides a link to eIF-4G                                                                                                                                             |
| Mnk1 and Mnk2<br>eIF-4E kinases                                                                                                                                               | phosphorylate eIF-4E increasing association with cap structure                                                                                                                                             |
| eIF-4A                                                                                                                                                                        | ATPase-dependent RNA helicase                                                                                                                                                                              |
| eIF-4E                                                                                                                                                                        | 5' cap recognition                                                                                                                                                                                         |
| 4E-BP (also called PHAS)<br>3 known forms                                                                                                                                     | when de-phosphorylated 4E-BP binds eIF-4E and represses its' activity, phosphorylation of 4E-BP occurs in response many growth stimuli leading to release of eIF-4E and increased translational initiation |
| eIF-4G                                                                                                                                                                        | acts as a scaffold for the assembly of eIF-4E and -4A in the eIF-4F complex, interaction with PABP allows 5'-end and 3'-ends of mRNAs to interact                                                          |
| eIF-4B                                                                                                                                                                        | stimulates helicase, binds simultaneously with eIF-4F                                                                                                                                                      |

|       |                                                       |
|-------|-------------------------------------------------------|
| eIF-5 | release of eIF-2 and eIF-3, ribosome-dependent GTPase |
| eIF-6 | ribosome subunit antiassociation                      |

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## Specific Steps in Translational Initiation

Initiation of translation requires 4 specific steps:

- **1.** A ribosome must dissociate into its' 40S and 60S subunits.
- **2.** A ternary complex termed the **preinitiation complex** is formed consisting of the initiator, GTP, eIF-2 and the 40S subunit.
- **3.** The mRNA is bound to the preinitiation complex.
- **4.** The 60S subunit associates with the preinitiation complex to form the 80S initiation complex.

The initiation factors eIF-1 and eIF-3 bind to the 40S ribosomal subunit favoring antiassociation to the 60S subunit. The prevention of subunit reassociation allows the preinitiation complex to form.

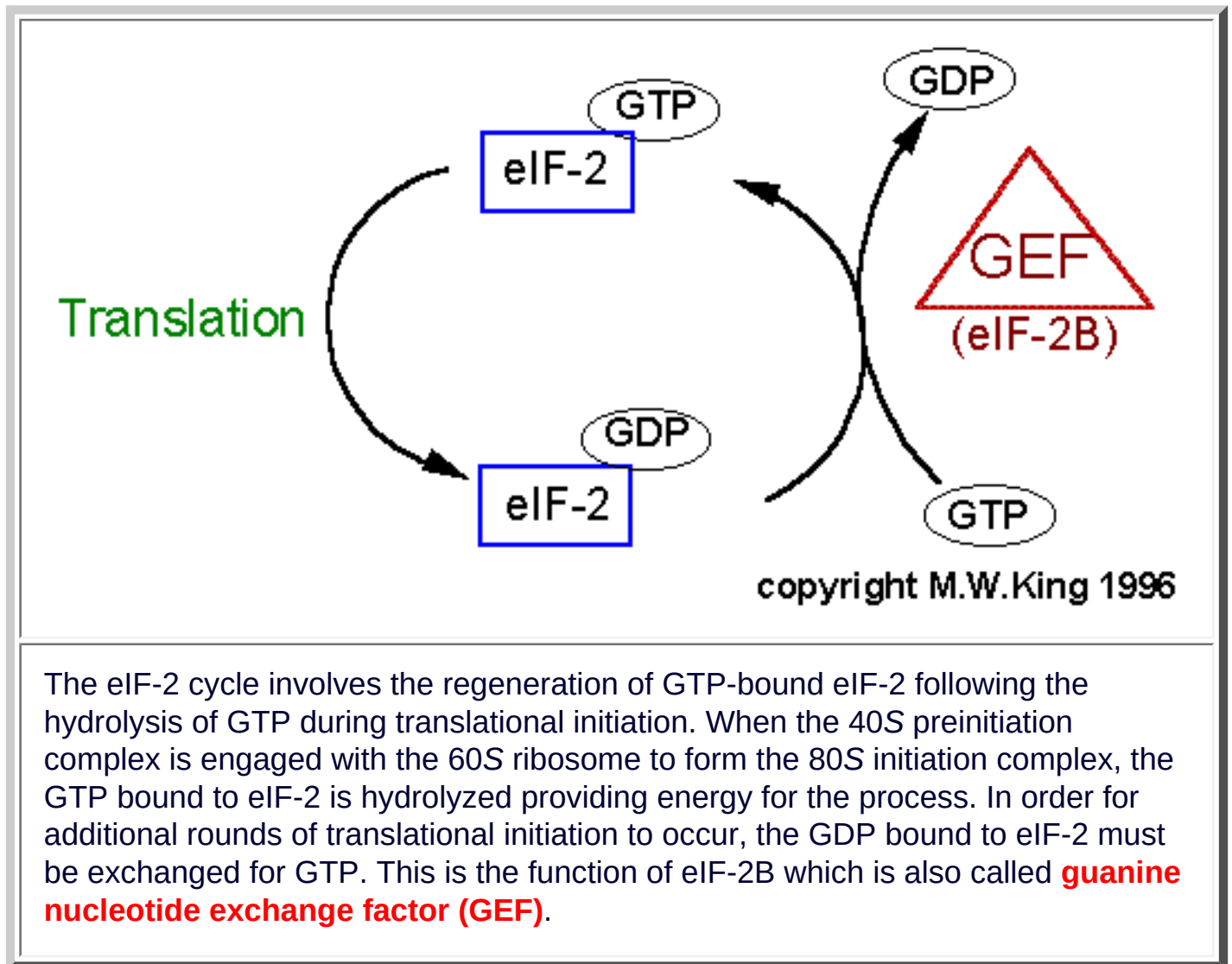
The first step in the formation of the preinitiation complex is the binding of GTP to eIF-2 to form a binary complex. eIF-2 is composed of three subunits,  $\alpha$ ,  $\beta$  and  $\gamma$ . The binary complex then binds to the activated initiator tRNA, met-tRNA<sup>met</sup> forming a ternary complex that then binds to the 40S subunit forming the 43S preinitiation complex. The preinitiation complex is stabilized by the earlier association of eIF-3 and eIF-1 to the 40S subunit.

The cap structure of eukaryotic mRNAs is bound by specific eIFs prior to association with the preinitiation complex. Cap binding is accomplished by the initiation factor **eIF-4F**. This factor is actually a complex of 3 proteins; eIF-4E, A and G. The protein eIF-4E is a 24 kDa protein which physically recognizes and binds to the cap structure. eIF-4A is a 46 kDa protein which binds and hydrolyzes ATP and exhibits RNA helicase activity. Unwinding of mRNA secondary structure is necessary to allow access of the ribosomal subunits. eIF-4G aids in binding of the mRNA to the 43S preinitiation complex.

Once the mRNA is properly aligned onto the preinitiation complex and the initiator met-tRNA<sup>met</sup> is bound to the initiator AUG codon (a process facilitated by eIF-1) the 60S subunit associates with the complex. The association of the 60S subunit requires the activity of eIF-5 which has first bound to the preinitiation complex. The energy needed to stimulate the formation of the 80S initiation complex comes from the hydrolysis of the GTP bound to eIF-2. The GDP bound form of eIF-2 then binds to eIF-2B which stimulates the exchange of GTP for GDP on eIF-2. When GTP is exchanged eIF-2B dissociates from eIF-2. This is termed the **eIF-2 cycle** (see diagram below). This cycle is absolutely required in order for eukaryotic translational initiation to

occur. The GTP exchange reaction can be affected by phosphorylation of the  $\alpha$ -subunit of eIF-2.

At this stage the initiator met-tRNA<sup>met</sup> is bound to the mRNA within a site of the ribosome termed the **P-site**, for peptide site. The other site within the ribosome to which incoming charged tRNAs bind is termed the **A-site**, for amino acid site.



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## Elongation

The process of elongation, like that of initiation requires specific non-ribosomal proteins. In *E. coli* these are EFs and in eukaryotes **eEFs**. Elongation of polypeptides occurs in a cyclic manner such that at the end of one complete round of amino acid addition the A site will be empty and ready to accept the incoming aminoacyl-tRNA dictated by the next codon of the mRNA. This means that not only does the incoming amino acid need to be attached to the peptide chain but the ribosome must move down the mRNA to the next codon. Each incoming aminoacyl-tRNA is brought to the ribosome by an **eEF-1 $\alpha$ -GTP complex**. When the correct tRNA is deposited into the A site the GTP is hydrolyzed and the eEF-1 $\alpha$ -GDP complex dissociates. In order for additional

translocation events the GDP must be exchanged for GTP. This is carried out by **eEF-1 $\beta$**  similarly to the GTP exchange that occurs with eIF-2 catalyzed by eIF-2B.

The peptide attached to the tRNA in the P site is transferred to the amino group at the aminoacyl-tRNA in the A site. This reaction is catalyzed by **peptidyltransferase**. This process is termed **transpeptidation**. The elongated peptide now resides on a tRNA in the A site. The A site needs to be freed in order to accept the next aminoacyl-tRNA. The process of moving the peptidyl-tRNA from the A site to the P site is termed, **translocation**. Translocation is catalyzed by **eEF-2** coupled to GTP hydrolysis. In the process of translocation the ribosome is moved along the mRNA such that the next codon of the mRNA resides under the A site. Following translocation eEF-2 is released from the ribosome. The cycle can now begin again.

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## Termination

Like initiation and elongation, translational termination requires specific protein factors identified as releasing factors, RFs in *E. coli* and **eRFs** in eukaryotes. There are 2 RFs in *E. coli* and one in eukaryotes. The signals for termination are the same in both prokaryotes and eukaryotes. These signals are termination codons present in the mRNA. There are 3 termination codons, UAG, UAA and UGA.

In *E. coli* the termination codons UAA and UAG are recognized by RF-1, whereas RF-2 recognizes the termination codons UAA and UGA. The eRF binds to the A site of the ribosome in conjunction with GTP. The binding of eRF to the ribosome stimulates the peptidyltransferase activity to transfer the peptidyl group to water instead of an aminoacyl-tRNA. The resulting uncharged tRNA left in the P site is expelled with concomitant hydrolysis of GTP. The inactive ribosome then releases its mRNA and the 80S complex dissociates into the 40S and 60S subunits ready for another round of translation.

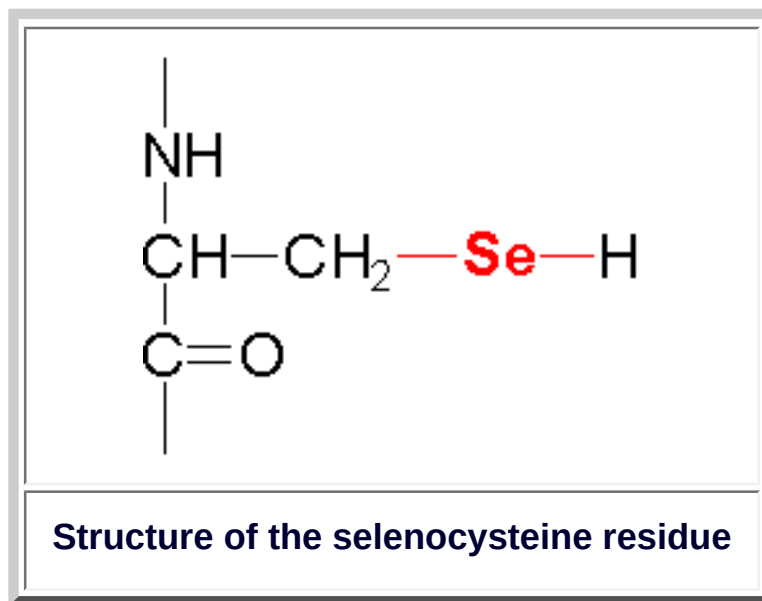
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## Selenoproteins

Selenium is a trace element and is found as a component of several prokaryotic and eukaryotic enzymes that are involved in redox reactions. The selenium in these selenoproteins is incorporated as a unique amino acid, **selenocysteine**, during translation. A particularly important eukaryotic selenoenzyme is **glutathione peroxidase**. This enzyme is required during the oxidation of **glutathione** by hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) and organic hydroperoxides.





Incorporation of selenocysteine by the translational machinery occurs via an interesting and unique mechanism. The tRNA for selenocysteine is charged with serine and then enzymatically selenylated to produce the selenocysteinyl-tRNA. The anticodon of selenocysteinyl-tRNA interacts with a stop codon in the mRNA (UGA) instead of a serine codon. The selenocysteinyl-tRNA has a unique structure that is not recognized by the termination machinery and is brought into the ribosome by a dedicated specific elongation factor. An element in the 3' non-translated region (UTR) of selenoprotein mRNAs determines whether UGA is read as a stop codon or as a selenocysteine codon.

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## Regulation of eIF-4E Activity

The cellular levels of eIF-4E are the lowest of all eukaryotic initiation factors which makes this factor a prime target for regulation. Indeed, at least 3 distinct mechanisms are known to exist that regulate the level and activity of eIF-4E. These include regulation of the level of transcription of the eIF-4E gene, post-translational modification via phosphorylation and inhibition by interaction with binding proteins.

Although the exact mechanisms used to upregulate the transcription of the eIF-4E gene are not yet well understood, it is known that exposure of cells to growth factors as well as activation of T cells leads to increased expression of eIF-4E. The [proto-oncogene](#) MYC is believed to play a role in the transcriptional activation of eIF-4E as 2 functional MYC-binding sites have been found in the promoter region of the eIF-4E gene. Of significant note is the finding that cells that are stably over-expressing the MYC gene also have enhanced levels of eIF-4E. Quite strikingly it has been shown that promiscuous elevation in the levels of eIF-4E lead to tumorigenesis placing this translation factor in the category of [proto-oncogene](#).

Numerous extracellular stimuli (e.g. insulin, EGF, angiotensin II and gastrin) that exert a portion of their effects at the level of enhanced translation do so by affecting the state of eIF-4E phosphorylation. However, it should be noted that not all signals that lead to increased eIF-4E phosphorylation lead to increased rates of translation. Changes in eIF-4E phosphorylation

correlate well with progression through the cell cycle. In resting ( $G_0$ ) cells eIF-4E phosphorylation is low, it increases during  $G_1$  and S phase and then declines again in M phase. Phosphorylation of eIF-4E occurs at one major site which is Ser209 (in the human and mouse proteins).

The primary signal transduction pathway leading to eIF-4E phosphorylation is that involving the RAS gene. Many growth factors stimulate activation of RAS in response to binding their cognate receptors. Subsequently, RAS activation leads to the phosphorylation and activation of MAP-interacting kinase-1 (Mnk1) which in turn phosphorylates eIF-4E. Although the exact effect of eIF-4E phosphorylation is not clearly defined, it may be necessary to increase affinity of eIF-4E for the mRNA cap structure and for eIF-4G.

The principal mechanism utilized in the regulation of eIF-4E activity is through its interaction with a family of binding/repressor proteins termed 4E-BPs (4E binding proteins) which are widely distributed in numerous vertebrate and invertebrate organisms. In mammalian cells 3 related 4E-BPs have been found where 4E-BP1 and 4E-BP2 are also identified as PHAS-I and PHAS-II (PHAS refers to **p**roperties of **h**eat and **a**cid **s**tability).

Binding of 4E-BPs to eIF-4E does not alter the affinity of eIF-4E for the cap structure but prevents the interaction of eIF-4E with eIF-4G which in turn suppresses the formation of the eIF-4F complex (see Table of Initiation Factors above). The ability of 4E-BPs to interact with eIF-4E is controlled via the phosphorylation of specific Ser and Thr residues in 4E-BP. When hypophosphorylated, 4E-BPs bind with high efficiency to eIF-4E but lose their binding capacity when phosphorylated. Numerous growth and signal transduction stimulating effectors lead to phosphorylation of 4E-BPs just as these same responses can lead to phosphorylation of eIF-4E.

There are several signal transduction pathways whose activations lead to phosphorylation of 4E-BPs. These include pathways that lead to activation of phosphatidylinositol 3-kinase (PI3K), the Akt Ser/Thr kinase which is also called protein kinase B, PKB (Akt was originally identified as a virally encoded oncogene) and the FKBP12-rapamycin-associated protein/mammalian target of rapamycin (FRAP/mTOR) family of proteins. The mammalian TOR proteins are homologs of the yeast TOR proteins that were identified in a screen for yeast mutants resistant to rapamycin. Rapamycin is an immunosuppressant used primarily in the prevention of tissue rejection following organ transplantation. Rapamycin functions within cells by binding to the immunophilin (an intracellular protein that binds to immunosuppressive drugs, e.g. FK506 and rapamycin) FK506-binding protein 12 (FKBP12). The net effect of phosphorylation of 4E-BPs is that they are released from eIF-4E allowing eIF-4E to actively bind eIF-4G.

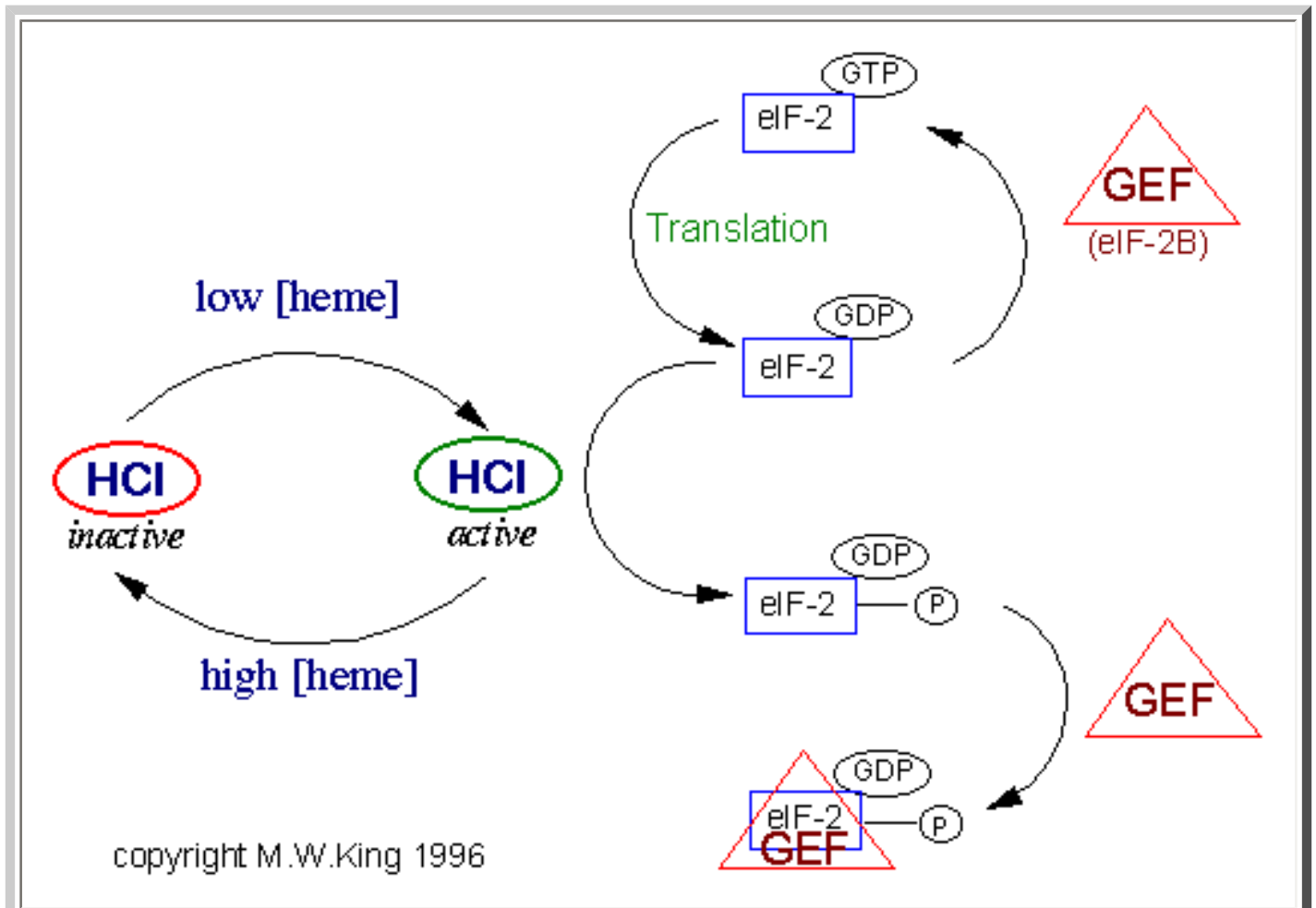
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## Heme Control of Translation

Regulation of initiation in eukaryotes is effected by phosphorylation of a ser(S) residue in the  $\alpha$  subunit of **eIF-2**. Phosphorylated eIF-2 in the absence of eIF-2B is just as active an initiator as non-phosphorylated eIF-2. However, when eIF-2 is phosphorylated the GDP-bound complex is stabilized and exchange for GTP is inhibited. The exchange of GDP for GTP is mediated by **eIF-2B** (also called guanine nucleotide exchange factor - **GEF**). When eIF-2 is phosphorylated it binds eIF-2B more tightly thus slowing the rate of exchange. It is this inhibited exchange that affects the rate of initiation.

The phosphorylation of eIF-2 is the result of an activity called **heme-controlled inhibitor (HCI)** which functions as diagrammed below. HCI is generated in the absence of heme, a mitochondrial product. Removal of phosphate is catalyzed by a specific eIF-2 phosphatase which is unaffected by heme. The presence of HCI was first seen in *in vitro* translation system derived from lysates of reticulocytes. Reticulocytes synthesize almost exclusively hemoglobin at an extremely high rate. In an intact reticulocyte eIF-2 is protected from phosphorylation by a specific 67 kDa protein.



The regulation of translation by **heme controlled inhibitor (HCI)**. Control of translation by heme is clinically important only in erythrocytes. Erythrocytes are enucleate and contain primarily globin mRNA. When the level of heme (required for the synthesis of biologically active hemoglobin) is low it would be inefficient for erythrocytes to synthesize globin protein. As the level of heme falls the activity of HCI increases. HCI is a kinase which phosphorylates eIF-2. When phosphorylated, eIF-2 still hydrolyzes bound GTP to GDP and still interacts with eIF-2B (GEF). However, the rate of eIF-2B-mediated GTP exchange is greatly reduced. This renders eIF-2 incapable of being used to form an new ternary initiation complex and translational initiation is reduced. When the level of heme again rises the activity of HCI is reduced and translational initiation is once again active.

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## Interferon Control of Translation

Regulation of translation can also be induced in virally infected cells. It would benefit a virally infected cell to turn off protein synthesis to prevent propagation of the viruses. This is accomplished by the induced synthesis of **interferons (IFs)**. There are 3 classes of IFs. The **leukocyte** or  **$\alpha$ -IFs**, the **fibroblast** or  **$\beta$ -IFs** and the **lymphocyte** or  **$\gamma$ -IFs**. IFs are induced by dsRNAs and themselves induce a specific kinase termed **RNA-dependent protein kinase (PKR)** that phosphorylates eIF-2 thereby shutting off translation in a similar manner to that of heme control of translation. Additionally, IFs induce the synthesis of **2'-5'-oligoadenylate, pppA(2'p5'A)n**, that activates a pre-existing ribonuclease, RNase L. RNase L degrades all classes of mRNAs thereby shutting off translation.

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## Iron Control of Translation

Regulation of the translation of certain mRNAs occurs through the action of specific RNA-binding proteins. Protein of this class have been identified that bind to sequences in either the 5' non-translated region (5'-UTR) or 3'-UTR. Two particularly interesting and important regulatory schemes related to iron metabolism encompass RNA binding proteins that bind to either the 5'-UTR of one mRNA or the 3'-UTR of another.

The transferrin receptor is a protein located in the plasma membrane that binds the protein transferrin. Transferrin is the major iron transport protein in the plasma. When iron levels are low the rate of synthesis of the transferrin receptor mRNA increases so that cells can take up more iron. This regulation occurs through the action of an **iron response element binding protein (IRBP)** that binds to specific **iron response elements (IREs)** in the 3'-UTR of the transferrin receptor mRNA. These IREs form hair-pin loop structures that are recognized by IRBP. This IRBP is an iron-deficient form of aconitase, the iron-requiring enzyme of the **TCA cycle**. When iron levels are low, IRBP is free of iron and can therefore, interact with the IREs in the 3'-UTR of the transferrin receptor mRNA. Transferrin receptor mRNA with IRBP bound is stabilized from degradation. Conversely, when iron levels are high, IRBP binds iron then cannot interact with the IREs in the transferrin receptor mRNA. The effect is an increase in degradation of the transferrin receptor mRNA.

A related, but opposite, phenomenon controls the translation of the ferritin mRNA. Ferritin is an iron-binding protein that prevents toxic levels of ionized iron ( $\text{Fe}^{2+}$ ) from building up in cells. The ferritin mRNA has an IRE in its 5'-UTR. As with the transferrin receptor story, when iron levels are high, IRBP cannot bind to the IRE in the 5'-UTR of the ferritin mRNA. This allows the ferritin mRNA to be translated. Conversely, when iron levels are low, the IRBP binds to the IRE in the ferritin mRNA preventing its translation.

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## Protein Synthesis Inhibitors

Many of the antibiotics utilized for the treatment of bacterial infections as well as certain toxins function through the inhibition of translation. Inhibition can be effected at all stages of translation from initiation to elongation to termination.

### Several Antibiotic and Toxin inhibitors of Translation

| Inhibitor        | Comments                                                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Chloramphenicol  | inhibits prokaryotic <i>peptidyl transferase</i>                                                                                    |
| Streptomycin     | inhibits prokaryotic peptide chain initiation, also induces mRNA misreading                                                         |
| Tetracycline     | inhibits prokaryotic aminoacyl-tRNA binding to the ribosome small subunit                                                           |
| Neomycin         | similar in activity to streptomycin                                                                                                 |
| Erythromycin     | inhibits prokaryotic translocation through the ribosome large subunit                                                               |
| Fusidic acid     | similar to erythromycin only by preventing EF-G from dissociating from the large subunit                                            |
| Puromycin        | resembles an aminoacyl-tRNA, interferes with peptide transfer resulting in premature termination in both prokaryotes and eukaryotes |
| Diphtheria toxin | catalyzes ADP-ribosylation of and inactivation of eEF-2                                                                             |
| Ricin            | found in castor beans, catalyzes cleavage of the eukaryotic large subunit rRNA                                                      |

Cycloheximide

inhibits eukaryotic ***peptidyltransferase***

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## Introduction

All cells undergo a division cycle during their life span. Some cells are continually dividing (e.g. stem cells), others divide a specific number of times until cell death (**apoptosis**) occurs, and still others divide a few times before entering a terminally differentiated or quiescent state. Most cells of the body fall into the latter category of cells. During the process of cell division everything within the cell must be duplicated in order to ensure the survival of the two resulting daughter cells. Of particular importance for cell survival is the accurate, efficient and rapid duplication of the cellular genome. This process is termed **DNA replication**.

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## Eukaryotic Genomes

The size of eukaryotic genomes is vastly larger than those of prokaryotes. This is partly due to the complexity of eukaryotic organisms compared to prokaryotes. However, the size of a particular eukaryotic genome is not directly correlated to the organism's complexity. This is the result of the presence of a large amount of non-coding DNA. The functions of these non-coding nucleic acid sequences are only partly understood. Some sequences are involved in the control of gene expression while others may simply be present in the genome to act as an evolutionary buffer able to withstand nucleotide mutation without disrupting the integrity of the organism.

One abundant class of DNA is termed **repetitive DNA**. There are 2 different sub-classes of repetitive DNA, **highly repetitive** and **moderately repetitive**. Highly repetitive DNA consists of short sequences 6-10 bp long reiterated from 100,000- 1,000,000 times. The DNA of the genome consisting of the **genes** (coding sequences) is identified as **non-repetitive DNA** since most genes occur but once in an organism's haploid genome. However, it should be pointed out that several genes exist as tandem clusters of multiple copies of the same gene ranging from 50 to 10,000 copies such as is the case for the rRNA genes and the histone genes.

Another characteristic feature that distinguishes eukaryotic from prokaryotic genes is the presence of **introns**. Introns are stretches of nucleic acid sequences that separate the coding exons of a gene. The existence of introns in prokaryotes is extremely rare. Essentially all human genes contain introns. A notable exception are the histone genes which are intronless. In many genes the presence of introns separates exons into coding regions exhibiting distinct functional domains.

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## Chromatin Structure

Chromatin is a term designating the structure in which DNA exists within cells. The structure of chromatin is determined and stabilized through the interaction of the DNA with DNA-binding proteins. There are 2 classes of DNA-binding proteins. The **histones** are the major class of DNA-binding proteins involved in maintaining the compacted structure of chromatin. There are 5 different histone proteins identified as **H1, H2A, H2B, H3** and **H4**.

The other class of DNA-binding proteins is a diverse group of proteins called simply, **non-histone proteins**. This class of proteins includes the various transcription factors, polymerases, hormone receptors and other nuclear enzymes. In any given cell there are greater than 1000 different types of non-histone proteins bound to the DNA.

The binding of DNA by the histones generates a structure called the nucleosome. The nucleosome core contains an octamer protein structure consisting of 2 subunits each of H2A, H2B, H3 and H4. Histone H1 occupies the internucleosomal DNA and is identified as the linker histone. The nucleosome core contains approximately 150 bp of DNA. The linker DNA between each nucleosome can vary from 20 to more than 200 bp. These nucleosomal core structures would appear as beads on a string if the DNA were pulled into a linear structure.

The nucleosome cores themselves coil into a solenoid shape which itself coils to further compact the DNA. These final coils are compacted further into the characteristic chromatin seen in a karyotyping spread. The protein-DNA structure of chromatin is stabilized by attachment to a non-histone protein scaffold called the nuclear matrix.

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## Eukaryotic Cell Cycles

The cell cycle is defined as the sequence of events that occurs during the lifetime of a



cell. The eukaryotic cell cycle is divided into 4 major periods. During each period a specific sequence of events occurs. The ultimate conclusion of one cell cycle is **cytokinesis** resulting in two identical daughter cells.

The 4 phases of a typical cell cycle and the events occurring during each phase are outlined:

- **1. M phase** is the period when cells prepare for and then undergo cytokinesis. M phase stands for **mitotic phase** or **mitosis**. During mitosis the chromosomes are paired and then divided prior to cell division. The events in this stage of the cell cycle leading to cell division are prophase, metaphase, anaphase and telophase.
- **2. G<sub>1</sub> phase** corresponds to the gap in the cell cycle that occurs following cytokinesis. During this phase cells make a decision to either exit the cell cycle and become quiescent or terminally differentiated or to continue dividing. Terminal differentiation is identified as a non-dividing state for a cell. Quiescent and terminally differentiated cells are identified as being in G<sub>0</sub> phase. Cells in G<sub>0</sub> can remain in this state for extended periods of time. Specific stimuli may induce the G<sub>0</sub> cell to re-enter the cell cycle at the G<sub>1</sub> phase or alternatively may induce permanent terminal differentiation.

During G<sub>1</sub> cells begin synthesizing all the cellular components needed in order to generate two identically complimented daughter cells. As a result the size of cells begins to increase during G<sub>1</sub>.

- **3. S phase** is the phase of the cell cycle during which the DNA is replicated. This is the DNA synthesis phase. Additionally, some specialized proteins are synthesized during S phase, particularly the histones.
- **4. G<sub>2</sub> phase** is reached following completion of DNA replication. During G<sub>2</sub> the chromosomes begin condensing, the nucleoli disappear and two microtubule organizing centers begin polymerizing tubulins for eventual production of the spindle poles.

Typical eukaryotic cell cycles occupy approximately 16 - 24 hrs when grown in culture. However, in the context of the multicellular organization of organisms the cell cycles can be as short as 6 - 8 hrs to greater than 100 days. The high variability of cell cycle times is due to the variability of the G<sub>1</sub> phase of the cycle.

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## DNA Replication

Replication of DNA occurs during the process of normal cell division cycles. Because the genetic complement of the resultant daughter cells must be the same as the parental cell, DNA replication must possess a very high degree of fidelity. The entire process of DNA replication is complex and involves multiple enzymatic activities.

The mechanics of DNA replication was originally characterized in the bacterium, *E. coli* which contains 3 distinct enzymes capable of catalyzing the replication of DNA. These have

been identified as DNA polymerase (pol) I, II, and III. Pol I is the most abundant replicating activity in *E. coli* but has as its primary role to ensure the fidelity of replication through the repair of damaged and mismatched DNA. Replication of the *E. coli* genome is the job of pol III. This enzyme is much less abundant than pol I, however, its activity is nearly 100 times that of pol I.

There have been 5 distinct eukaryotic DNA polymerases identified,  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$  and  $\epsilon$ . The identity of the individual enzymes relates to its subcellular localization as well as its primary replicative activity. The polymerase of eukaryotic cells that is the equivalent of *E. coli* pol III is pol- $\alpha$ . The pol I equivalent in eukaryotes is pol- $\beta$ . Polymerase- $\gamma$  is responsible for replication of mitochondrial DNA.

The ability of DNA polymerases to replicate DNA requires a number of additional accessory proteins. The combination of polymerases with several of the accessory proteins yields an activity identified as **DNA polymerase holoenzyme**. These accessory proteins include (not ordered with respect to importance):

- 1. Primase
- 2. Processivity accessory proteins
- 3. Single strand binding proteins
- 4. Helicase
- 5. DNA ligase
- 6. Topoisomerases
- 7. Uracil-DNA N-glycosylase

The process of DNA replication begins at specific sites in the chromosomes termed **origins of replication**, requires a primer bearing a free 3'-OH, proceeds specifically in the **5' -----> 3' direction** on both strands of DNA concurrently and results in the copying of the template strands in a **semiconservative** manner. The semiconservative nature of DNA replication means that the newly synthesized daughter strands remain associated with their respective parental template strands.

The large size of eukaryotic chromosomes and the limits of nucleotide incorporation during DNA synthesis, make it necessary for multiple origins of replication to exist in order to complete replication in a reasonable period of time. The precise nature of origins of replication in higher eukaryotic organisms is unclear. However, it is clear that at a replication origin the strands of DNA must dissociate and unwind in order to allow access to DNA polymerase. Unwinding of the duplex at the origin as well as along the strands as the replication process proceeds is carried out by helicases. The resultant regions of single-stranded DNA are stabilized by the binding of **single-strand binding proteins**. The stabilized single-stranded regions are then accessible to the enzymatic activities required for replication to proceed. The site of the unwound template strands is termed the **replication fork**.

In order for DNA polymerases to synthesize DNA they must encounter a free 3'-OH which is the substrate for attachment of the 5'-phosphate of the incoming nucleotide. During repair of damaged DNA the 3'-OH can arise from the hydrolysis of the backbone of one of the



two strands. During replication the 3'-OH is supplied through the use of an **RNA primer**, synthesized by the **primase activity**. The primase utilizes the DNA strands as templates and synthesizes a short stretch of RNA generating a primer for DNA polymerase.

Synthesis of DNA proceeds in the 5' → 3' direction through the attachment of the 5'-phosphate of an incoming dNTP to the existing 3'-OH in the elongating DNA strands with the concomitant release of pyrophosphate. Initiation of synthesis, at origins of replication, occurs simultaneously on both strands of DNA. Synthesis then proceeds bidirectionally, with one strand in each direction being copied continuously and one strand in each direction being copied discontinuously. During the process of DNA polymerases incorporating dNTPs into DNA in the 5' → 3' direction they are moving in the 3' → 5' direction with respect to the template strand. In order for DNA synthesis to occur simultaneously on both template strands as well as bidirectionally one strand appears to be synthesized in the 3' → 5' direction. In actuality one strand of newly synthesized DNA is produced discontinuously.

The strand of DNA synthesized continuously is termed the **leading strand** and the discontinuous strand is termed the **lagging strand**. The lagging strand of DNA is composed of short stretches of RNA primer plus newly synthesized DNA approximately 100-200 bases long (the approximate distance between adjacent nucleosomes). The lagging strands of DNA are also called **Okazaki fragments**. The concept of continuous strand synthesis is somewhat of a misnomer since DNA polymerases do not remain associated with a template strand indefinitely. The ability of a particular polymerase to remain associated with the template strand is termed its **processivity**. The longer it associates the higher the processivity of the enzyme. DNA polymerase processivity is enhanced by additional protein activities of the replisome identified as **processivity accessory proteins**.

How is it that DNA polymerase can copy both strands of DNA in the 5' → 3' direction simultaneously? A model has been proposed where DNA polymerases exist as dimers associated with the other necessary proteins at the replication fork and identified as the **replisome**. The template for the lagging strand is temporarily looped through the replisome such that the DNA polymerases are moving along both strands in the 3' → 5' direction simultaneously for short distances, the distance of an Okazaki fragment. As the replication forks progress along the template strands the newly synthesized daughter strands and parental template strands reform a DNA double helix. This means that only a small stretch of the template duplex is single-stranded at any given time.

The progression of the replication fork requires that the DNA ahead of the fork be continuously unwound. Due to the fact that eukaryotic chromosomal DNA is attached to a protein scaffold the progressive movement of the replication fork introduces severe torsional stress into the duplex ahead of the fork. This torsional stress is relieved by DNA topoisomerases. Topoisomerases relieve torsional stresses in duplexes of DNA by introducing either double- (topoisomerases II) or single-stranded (topoisomerases I) breaks into the backbone of the DNA. These breaks allow unwinding of the duplex and removal of the replication-induced torsional strain. The nicks are then resealed by the topoisomerases.

The RNA primers of the leading strands and Okazaki fragments are removed by the repair DNA polymerases simultaneously replacing the ribonucleotides with deoxyribonucleotides. The gaps that exist between the 3'-OH of one leading strand and the 5'-



phosphate of another as well as between one Okazaki fragment and another are repaired by DNA ligases thereby, completing the process of replication.

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## Additional DNA Polymerase Activities

The main enzymatic activity of DNA polymerases is the 5' ----> 3' synthetic activity. However, DNA polymerases possess two additional activities of importance for both replication and repair. These additional activities include a 5' ----> 3' exonuclease function and a 3' -----> 5' exonuclease function. The 5' -----> 3' exonuclease activity allows the removal of ribonucleotides of the RNA primer, utilized to initiate DNA synthesis, along with their simultaneous replacement with deoxyribonucleotides by the 5' -----> 3' polymerase activity. The 5' -----> 3' exonuclease activity is also utilized during the repair of damaged DNA. The 3' -----> 5' exonuclease function is utilized during replication to allow DNA polymerase to remove mismatched bases. It is possible (but rare) for DNA polymerases to incorporate an incorrect base during replication. These mismatched bases are recognized by the polymerase immediately due to the lack of Watson-Crick base-pairing. The mismatched base is then removed by the 3' -----> 5' exonuclease activity and the correct base inserted prior to progression of replication.

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## Post-Replicative Modification of DNA, Methylation

One of the major post-replicative reactions that modifies the DNA is **methylation**. The sites of natural methylation (i.e. not chemically induced) of eukaryotic DNA is always on cytosine residues that are present in CpG dinucleotides. However, it should be noted that not all CpG dinucleotides are methylated at the C residue. The cytidine is methylated at the 5 position of the pyrimidine ring generating 5-methylcytidine.

Methylation of DNA in prokaryotic cells also occurs. The function of this methylation is to prevent degradation of host DNA in the presence of enzymatic activities synthesized by bacteria called **restriction endonucleases**. These enzymes recognize specific nucleotide sequences of DNA. The role of this system in prokaryotic cells (called the restriction-modification system) is to degrade invading viral DNAs. Since the viral DNAs are not modified by methylation they are degraded by the host restriction enzymes. The methylated host genome is resistant to the action of these enzymes.

The precise role of methylation in eukaryotic DNA is unclear. It was originally thought that methylated DNA would be less transcriptionally active. Indeed, experiments have been carried out to demonstrate that this is true for certain genes. For example, under-methylation of the MyoD gene (a master control gene regulating the differentiation of muscle cells through the control of the expression of muscle-specific genes) results in the conversion of fibroblasts to myoblasts. The experiments were carried out by allowing replicating fibroblasts to incorporate 5-azacytidine into their newly synthesized DNA. This analog of cytidine prevents methylation.



The net result is that the maternal pattern of methylation is lost and numerous genes become under methylated. However, lack of methylation nor the presence of methylation is a clear indicator of whether a gene will be transcriptionally active or silent.

The pattern of methylation is copied post-replicatively by the **maintenance methylase system**. This activity recognizes the pattern of methylated C residues in the maternal DNA strand following replication and methylates the C residue present in the corresponding CpG dinucleotide of the daughter strand.

The phenomenon of **genomic imprinting** refers to the fact that the expression of some genes depends on whether or not they are inherited maternally or paternally. **Insulin-like growth factor-2 (Igf2)** is a gene whose expression is required for normal fetal development and growth. Expression of Igf2 occurs exclusively from the paternal copy of the gene. Imprinted genes are "marked" by their state of methylation. In the case of Igf2 an element in the paternal locus, called an **insulator element**, is methylated blocking its function. The function of the un-methylated insulator is to bind a protein that when bound blocks activation of Igf2 expression. When methylated the protein cannot bind the insulator thus allowing a distant enhancer element to drive expression of the Igf2 gene. In the maternal genome, the insulator is not methylated, thus protein binds to it blocking the action of the distant enhancer element.

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## DNA Recombination

DNA recombination refers to the phenomenon whereby two parental strands of DNA are spliced together resulting in an exchange of portions of their respective strands. This process leads to new molecules of DNA that contain a mix of genetic information from each parental strand. There are 3 main forms of genetic recombination. These are **homologous recombination**, **site-specific recombination** and **transposition**.

Homologous recombination is the process of genetic exchange that occurs between any two molecules of DNA that share a region (or regions) of homologous DNA sequences. This form of recombination occurs frequently while sister chromatids are paired during meiosis. Indeed, it is the process of homologous recombination between the maternal and paternal chromosomes that imparts genetic diversity to an organism. Homologous recombination generally involves exchange of large regions of the chromosomes.

Site-specific recombination involves exchange between much smaller regions of DNA sequence (approximately 20 - 200 base pairs) and requires the recognition of specific sequences by the proteins involved in the recombination process. Site-specific recombination events occur primarily as a mechanism to alter the program of genes expressed at specific stages of development. The most significant site-specific recombinational events in humans are the somatic cell gene rearrangements that take place in the immunoglobulin genes during B-cell differentiation in response to antigen presentation. These gene rearrangements in the immunoglobulin genes result in an extremely diverse potential for antibody production. A typical antibody molecule is composed of both heavy and light chains. The genes for both these peptide chains undergo somatic cell rearrangement yielding the potential for

approximately 3000 different light chain combinations and approximately 5000 heavy chain combinations. Then because any given heavy chain can combine with any given light chain the potential diversity exceeds 10,000,000 possible different antibody molecules.

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## DNA Transposition

Transposition is a unique form of recombination where **mobile genetic elements** can virtually move from one region to another within one chromosome or to another chromosome entirely. There is no requirement for sequence homology for a transpositional event to occur. Because the potential exists for the disruption of a vitally important gene by a transposition event this process must be tightly regulated. The exact nature of how transpositional events are controlled is unclear.

Transposition occurs with a higher frequency in bacteria and yeasts than it does in humans. The identification of the occurrence of transposition in the human genome resulted when it was found that certain processed genes were present in the genome. These processed genes are nearly identical to the mRNA encoded by the normal gene. The processed genes contain the poly(A) tail that would have been present in the RNA and they lack the introns of the normal gene. These particular forms of genes must have arisen through a reverse transcription event, similar to the life cycle of retroviral genomes, and then been incorporated into the genome by a transpositional event. Since most of the processed genes that have been identified are non-functional they have been termed pseudogenes.

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## Repair of Damaged DNA

Cancer, in most non-viral induced cases, is the severe medically relevant consequence of the inability to repair damaged DNA. It is clear that multiple somatic cell mutations in DNA can lead to the genesis of the transformed phenotype. Therefore, it should be obvious that complete understanding of DNA repair mechanisms would be invaluable in the design of potential therapeutic agents in the treatment of cancer.

DNA damage can occur as the result of exposure to environmental stimuli such as alkylating chemicals or ultraviolet or radioactive irradiation and free radicals generated spontaneously in the oxidizing environment of the cell. These phenomena can, and do, lead to the introduction of mutations in the coding capacity of the DNA. Mutations in DNA can also, but rarely, arise from the spontaneous tautomerization of the bases.

Modification of the DNA bases by alkylation (predominately the incorporation of -CH<sub>3</sub> groups) predominately occurs on purine residues. Methylation of G residues allows them to base pair with T instead of C. A unique activity called O<sup>6</sup>-alkylguanine transferase removes the alkyl group from G residues. The protein itself becomes alkylated and is no longer active, thus, a single protein molecule can remove only one alkyl group.

Mutations in DNA are of two types. **Transition mutations** result from the exchange of



one purine, or pyrimidine, for another purine, or pyrimidine. **Transversion mutations** result from the exchange of a purine for a pyrimidine or visa versa.

The prominent by-product from *uv*irradiation of DNA is the formation of thymine dimers. These form from two adjacent T residues in the DNA. Repair of thymine dimers is most understood from consideration of the mechanisms used in *E. coli*. However, several mechanism are common to both prokaryotes and eukaryotes.

Thymine dimers are removed by several mechanisms. Specific glycohydrolases recognize the dimer as abnormal and cleave the N-glycosidic bond of the bases in the dimer. This results in the base leaving and generates an apyrimidinic site in the DNA. This is repaired by DNA polymerase and ligase. Glycohydrolases are also responsible for the removal of other abnormal bases, not just thymine dimers.

Another, widely distributed activity, is DNA photolyase or photoreactivating enzyme. This protein binds to thymine dimers in the dark. In response to visible light stimulation the enzyme cleaves the pyrimidine rings. The chromophore associated with this enzyme that allows visible light activation is FADH<sub>2</sub>

Humans defective in DNA repair, (in particular the repair of *uv*-induced thymine dimers), due to autosomal recessive genetic defects suffer from the disease **Xeroderma pigmentosum** (XP). There are at least nine distinct genetic defects associated with this disease. One of these is due to a defect in the gene coding for the glycohydrolase that cleaves the N-glycosidic bond of the thymine dimers. There are two major clinical forms of XP, one which leads to progressive degenerative changes in the eyes and skin and the other which also includes progressive neurological degeneration.

Another inherited disorder affecting DNA repair in which patients suffer from sun sensitivity, short stature and progressive neurological degeneration without an increased incidence of skin cancer is **Cockayne Syndrome**.

**Ataxia telangiectasia** (AT) is an autosomal recessive disorder resulting in neurological disability and suppressed immune function. Patients develop a disabling cerebellar ataxia early in life and have recurrent infections. Patients suffering from AT have an increased sensitivity to x-irradiation suggesting a role for the AT gene in DNA repair.

Several diseases associated with defective repair of damaged DNA can be found in the **Inborn Errors** page.

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## Chemotherapies Targeting Replication

The class of compounds that have been used the longest as anticancer drugs are the **alkylating agents**. The major alkylating agents are derived from nitrogenous mustards that were originally developed for use by the military. Commonly used alkylating agents include **cyclophosphamide** (Cytosan, Neosar), **ifosfamide**, **decarbazine**, **chlorambucil** (Leukeran) and **procarbazine** (Matulane, Natulan).

Alkylating agents function by reacting with and disrupting the structure of DNA. Some agents react with alkyl groups in DNA resulting in fragmentation of the DNA as a consequence



of the action of DNA repair enzymes. Some agents catalyze the cross-linking of bases in the DNA which prevents the separation of the two strands during DNA replication. Some agents induce mispairing of nucleotides resulting in permanent mutations in the DNA. Alkylating agents act upon DNA at all stages of the **cell cycle**, thus they are potent anticancer drugs. However, because of their potency, prolonged use of alkylating agents can lead to secondary cancers, particularly leukemias.

Several classes of anticancer drugs function through interference with the actions of the topoisomerases. Two of these classes are the **anthracyclines** and the **camptothecins**.

The anthracyclins were originally isolated from the fungus, *Streptomyces*. **Doxorubicin** (Adriamycin, Doxil, Rubex) and **daunorubicin** (Cerubidine, DaunoXome, Daunomycin, Rubidomycin) have similar modes of action, although doxorubicin is the more potent of the two and is used in the treatment breast cancers, lymphomas and sarcomas. The anthracyclins inhibit the actions of topoisomerase II whose function is to introduce double-strand breaks in DNA during the process of replication as a means to relieve torsional stresses. Anthracyclines also function by inducing the formation of oxygen free radicals that cause DNA strand breaks resulting in inhibition of replication.

Another plant-derived anticancer compound that functions through inhibition of topoisomerase II is **etoposide** (VP-16, Vepesid, Etophos, Eposin). Etoposide is isolated from the mandrake plant and is in a class of compounds referred to as epipodophyllotoxins.

The camptothecins were originally found in the bark of the *Camptotheca accuminata* tree and include **irinotecan** (Campto, Camptosar) and **topotecan** (Hycamtin). Camptothecins inhibit the action of topoisomerase I, an enzyme that induces single-strand breaks in DNA during replication.

Anticancer compounds that are extracted from the periwinkle plant, *Vinca rosea*, are called the vinca alkaloids. These compounds bind to tubulin monomers leading to the disruption of the microtubules of the mitotic spindle fibers that are necessary for cell division during mitosis. There are four major vinca alkaloids that are currently used as chemotherapeutics, **vincristine** (Oncovin, Vincrex), **vinblastine** (Velbe, Velban, Velsar), **vinorelbin** (VIN, Navelbine), and **vindesine** (Eldisine).

The **taxanes** are another class of plant-derived compounds that act via interference with microtubule function. These compounds are isolated from the Pacific yew tree, *Taxus brevifolia* and include **paclitaxel** (Taxol) and **docetaxel** (Taxotere). The taxanes function by hyperstabilizing microtubules which prevents cell division (cytokinesis). These compounds are used to treat a wide range of cancers including head and neck, lung, ovarian, breast, bladder, and prostate cancers. Taxol has proven highly effective in the treatment of certain forms of breast cancer.

In order for DNA replication to proceed, proliferating cells require a pool of nucleotides. The class of anticancer drugs that has been developed to interfere with aspects of nucleotide metabolism is known as the **antimetabolites**. There are two major types of antimetabolites used in the treatment of a broad range of cancers: compounds that inhibit thymidylate synthase and compounds that inhibit dihydrofolate reductase (DHFR). Both of these enzymes are involved in **thymidine nucleotide biosynthesis**. Drugs that inhibit thymidylate synthase include **5-fluorouracil** (5-FU, Adrucil, Efudex) and **5-fluorodeoxyuridine**. Those that inhibit DHFR are

analogs of the vitamin **folic acid** and include **methotrexate** (Trexall, Rheumatrex) and **trimethoprim** (Proloprim, Trimplex).

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## Chemical Nature of the Amino Acids

All peptides and polypeptides are polymers of alpha-amino acids. There are 20  $\alpha$ -amino acids that are relevant to the make-up of mammalian proteins (see below). Several other amino acids are found in the body free or in combined states (i.e. not associated with peptides or proteins). These non-protein associated amino acids perform specialized functions. Several of the amino acids found in proteins also serve functions distinct from the formation of peptides and proteins, e.g., tyrosine in the formation of thyroid hormones or glutamate acting as a neurotransmitter.

The  $\alpha$ -amino acids in peptides and proteins (excluding proline) consist of a carboxylic acid (**-COOH**) and an amino (**-NH<sub>2</sub>**) functional group attached to the same tetrahedral carbon atom. This carbon is the  $\alpha$ -carbon. Distinct R-groups, that distinguish one amino acid from another, also are attached to the alpha-carbon (except in the case of glycine where the R-group is hydrogen). The fourth substitution on the tetrahedral  $\alpha$ -carbon of amino acids is hydrogen.

## Table of $\alpha$ -Amino Acids Found in Proteins

| Amino Acid                          | Symbol  | Structure*                                                                                                                             | pK <sub>1</sub><br>(COOH) | pK <sub>2</sub><br>(NH <sub>2</sub> ) | pK R<br>Group |
|-------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------|---------------|
| Amino Acids with Aliphatic R-Groups |         |                                                                                                                                        |                           |                                       |               |
| Glycine                             | Gly - G | $\begin{array}{c} \text{H}-\text{CH}-\text{COOH} \\   \\ \text{NH}_2 \end{array}$                                                      | 2.4                       | 9.8                                   |               |
| Alanine                             | Ala - A | $\begin{array}{c} \text{CH}_3-\text{CH}-\text{COOH} \\   \\ \text{NH}_2 \end{array}$                                                   | 2.4                       | 9.9                                   |               |
| Valine                              | Val - V | $\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{COOH} \\   \quad   \\ \text{H}_3\text{C} \quad \text{NH}_2 \end{array}$ | 2.2                       | 9.7                                   |               |

|                                                        |         |                                                                                                                                                                          |     |      |     |
|--------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|-----|
| Leucine                                                | Leu - L | $  \begin{array}{c}  \text{H}_3\text{C}-\text{CH}-\text{CH}_2-\text{CH}-\text{COOH} \\    \quad \quad   \\  \text{H}_3\text{C} \quad \quad \text{NH}_2  \end{array}  $   | 2.3 | 9.7  |     |
| Isoleucine                                             | Ile - I | $  \begin{array}{c}  \text{H}_3\text{C}-\text{CH}_2-\text{CH}-\text{CH}-\text{COOH} \\    \quad \quad   \\  \text{H}_3\text{C} \quad \quad \text{NH}_2  \end{array}  $   | 2.3 | 9.8  |     |
| <b>Non-Aromatic Amino Acids with Hydroxyl R-Groups</b> |         |                                                                                                                                                                          |     |      |     |
| Serine                                                 | Ser - S | $  \begin{array}{c}  \text{HO}-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  \end{array}  $                                                                   | 2.2 | 9.2  | ~13 |
| Threonine                                              | Thr - T | $  \begin{array}{c}  \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{COOH} \\    \quad   \\  \text{HO} \quad \text{NH}_2  \end{array}  $                                    | 2.1 | 9.1  | ~13 |
| <b>Amino Acids with Sulfur-Containing R-Groups</b>     |         |                                                                                                                                                                          |     |      |     |
| Cysteine                                               | Cys - C | $  \begin{array}{c}  \text{HS}-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  \end{array}  $                                                                   | 1.9 | 10.8 | 8.3 |
| Methionine                                             | Met-M   | $  \begin{array}{c}  \text{H}_3\text{C}-\text{S}-(\text{CH}_2)_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  \end{array}  $                                             | 2.1 | 9.3  |     |
| <b>Acidic Amino Acids and their Amides</b>             |         |                                                                                                                                                                          |     |      |     |
| Aspartic Acid                                          | Asp - D | $  \begin{array}{c}  \text{HOOC}-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  \end{array}  $                                                                 | 2.0 | 9.9  | 3.9 |
| Asparagine                                             | Asn - N | $  \begin{array}{c}  \text{H}_2\text{N}-\text{C}-\text{CH}_2-\text{CH}-\text{COOH} \\     \quad \quad   \\  \text{O} \quad \quad \text{NH}_2  \end{array}  $             | 2.1 | 8.8  |     |
| Glutamic Acid                                          | Glu - E | $  \begin{array}{c}  \text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  \end{array}  $                                                     | 2.1 | 9.5  | 4.1 |
| Glutamine                                              | Gln - Q | $  \begin{array}{c}  \text{H}_2\text{N}-\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}-\text{COOH} \\     \quad \quad   \\  \text{O} \quad \quad \text{NH}_2  \end{array}  $ | 2.2 | 9.1  |     |
| <b>Basic Amino Acids</b>                               |         |                                                                                                                                                                          |     |      |     |

|                                        |         |                                                                                                                                                                                                              |     |      |      |
|----------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|
| Arginine                               | Arg - R | $  \begin{array}{c}  \text{HN}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}-\text{COOH} \\    \qquad \qquad \qquad   \\  \text{C}=\text{NH} \qquad \qquad \text{NH}_2 \\    \\  \text{NH}_2  \end{array}  $ | 1.8 | 9.0  | 12.5 |
| Lysine                                 | Lys - K | $  \text{H}_2\text{N}-(\text{CH}_2)_4-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  $                                                                                                                         | 2.2 | 9.2  | 10.8 |
| Histidine                              | His - H | $  \begin{array}{c}  \text{CH}_2-\text{CH}-\text{COOH} \\    \qquad \qquad   \\  \text{NH} \quad \text{N:} \qquad \text{NH}_2  \end{array}  $                                                                | 1.8 | 9.2  | 6.0  |
| <b>Amino Acids with Aromatic Rings</b> |         |                                                                                                                                                                                                              |     |      |      |
| Phenylalanine                          | Phe - F | $  \text{C}_6\text{H}_5-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  $                                                                                                                           | 2.2 | 9.2  |      |
| Tyrosine                               | Tyr - Y | $  \text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-\text{CH}-\text{COOH} \\    \\  \text{NH}_2  $                                                                                                                 | 2.2 | 9.1  | 10.1 |
| Tryptophan                             | Trp-W   | $  \begin{array}{c}  \text{CH}_2-\text{CH}-\text{COOH} \\    \qquad \qquad   \\  \text{N} \qquad \qquad \text{NH}_2  \end{array}  $                                                                          | 2.4 | 9.4  |      |
| <b>Imino Acids</b>                     |         |                                                                                                                                                                                                              |     |      |      |
| Proline                                | Pro - P | $  \begin{array}{c}  \text{COOH} \\    \\  \text{N}^+ \\    \quad   \\  \text{H} \quad \text{H}  \end{array}  $                                                                                              | 2.0 | 10.6 |      |

\*Backbone of the amino acids is red, R-groups are black

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## Amino Acid Classifications

Each of the 20  $\alpha$ -amino acids found in proteins can be distinguished by the R-group substitution on the  $\alpha$ -carbon atom. There are two broad classes of amino acids based upon whether the R-group is hydrophobic or hydrophilic.

The hydrophobic amino acids tend to repel the aqueous environment and, therefore, reside

predominantly in the interior of proteins. This class of amino acids does not ionize nor participate in the formation of H-bonds. The hydrophilic amino acids tend to interact with the aqueous environment, are often involved in the formation of H-bonds and are predominantly found on the exterior surfaces proteins or in the reactive centers of enzymes.

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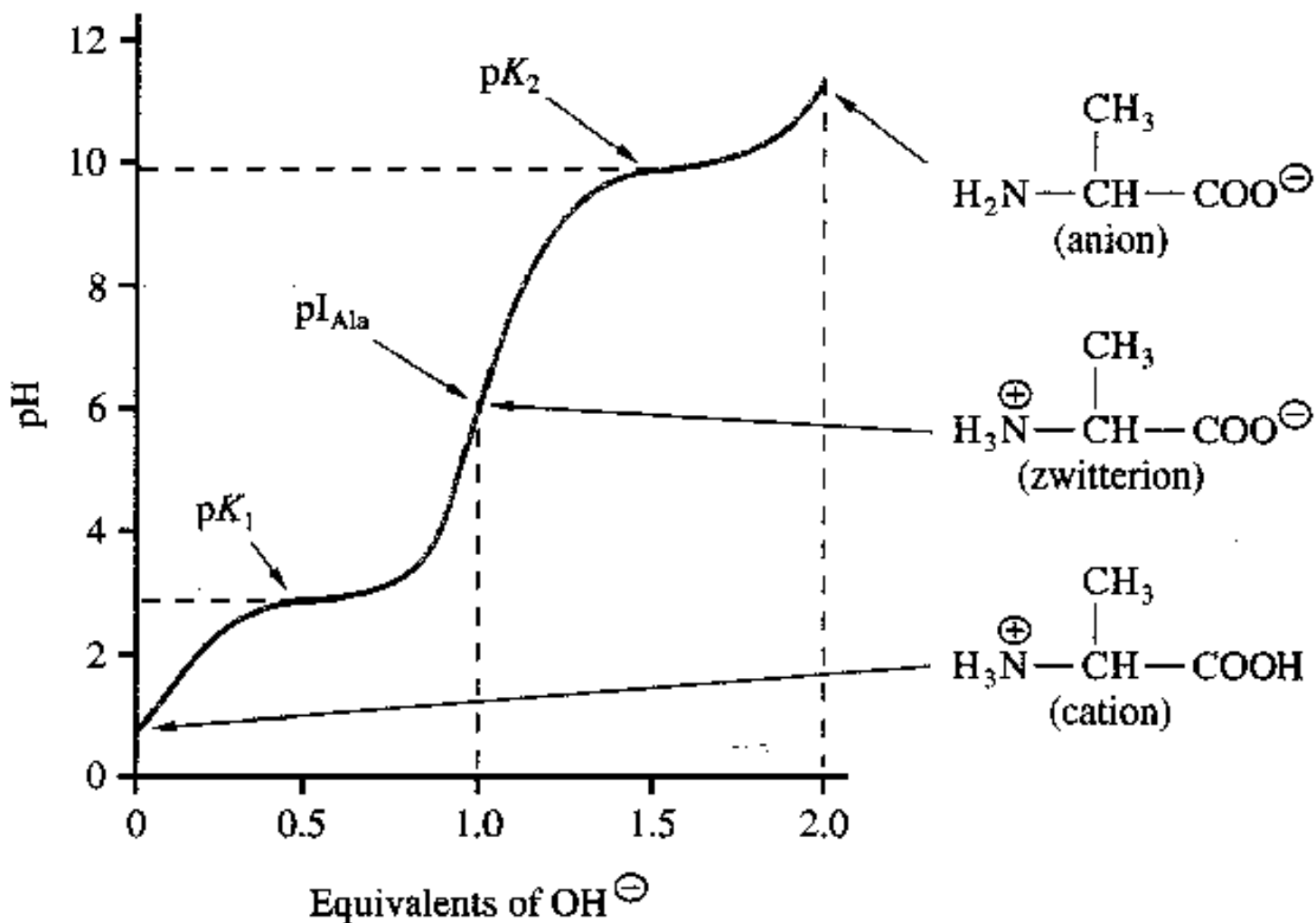
## Acid-Base Properties of the Amino Acids

The  $\alpha$ -COOH and  $\alpha$ -NH<sub>2</sub> groups in amino acids are capable of ionizing (as are the acidic and basic R-groups of the amino acids). As a result of their ionizability the following ionic equilibrium reactions may be written:



The equilibrium reactions, as written, demonstrate that amino acids contain at least two weakly acidic groups. However, the carboxyl group is a far stronger acid than the amino group. At physiological pH (around 7.4) the carboxyl group will be unprotonated and the amino group will be protonated. An amino acid with no ionizable R-group would be electrically neutral at this pH. This species is termed a **zwitterion**.

Like typical organic acids, the acidic strength of the carboxyl, amino and ionizable R-groups in amino acids can be defined by the association constant,  $K_a$  or more commonly the negative logarithm of  $K_a$ , the  $pK_a$ . The **net charge** (the algebraic sum of all the charged groups present) of any amino acid, peptide or protein, will depend upon the pH of the surrounding aqueous environment. As the pH of a solution of an amino acid or protein changes so too does the net charge. This phenomenon can be observed during the titration of any amino acid or protein. When the net charge of an amino acid or protein is zero the pH will be equivalent to the **isoelectric point: pI**.



Titration curve for Alanine

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## Functional Significance of Amino Acid R-Groups

In solution it is the nature of the amino acid R-groups that dictate structure-function relationships of peptides and proteins. The hydrophobic amino acids will generally be encountered in the interior of proteins shielded from direct contact with water. Conversely, the hydrophilic amino acids are generally found on the exterior of proteins as well as in the active centers of enzymatically active proteins. Indeed, it is the very nature of certain amino acid R-groups that allow enzyme reactions to occur.

The imidazole ring of histidine allows it to act as either a proton donor or acceptor at physiological pH. Hence, it is frequently found in the reactive center of enzymes. Equally important is the ability of histidines in hemoglobin to buffer the H<sup>+</sup> ions from carbonic acid ionization in red blood cells. It is this property of hemoglobin that allows it to exchange O<sub>2</sub> and CO<sub>2</sub> at the tissues or lungs, respectively.

The primary alcohol of serine and threonine as well as the thiol (-SH) of cysteine allow these amino acids to act as nucleophiles during enzymatic catalysis. Additionally, the thiol of cysteine is able to form a disulfide bond with other cysteines:

## Cysteine-SH + HS-Cysteine <-----> Cysteine-S-S-Cysteine

This simple disulfide is identified as cystine. The formation of disulfide bonds between cysteines present within proteins is important to the formation of active structural domains in a large number of proteins. Disulfide bonding between cysteines in different polypeptide chains of oligomeric proteins plays a crucial role in ordering the structure of complex proteins, e.g. the insulin receptor.

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## Optical Properties of the Amino Acids

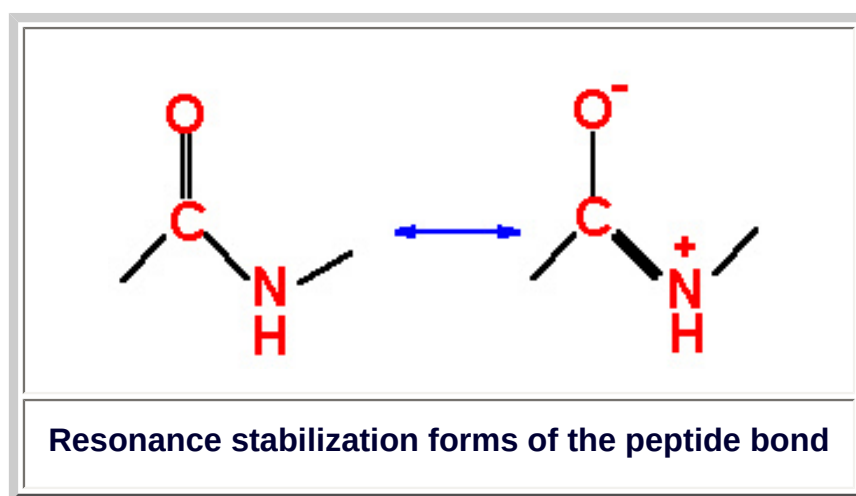
A tetrahedral carbon atom with 4 distinct constituents is said to be **chiral**. The one amino acid not exhibiting chirality is glycine since its "R-group" is a hydrogen atom. Chirality describes the handedness of a molecule that is observable by the ability of a molecule to rotate the plane of polarized light either to the right (**dextrorotatory**) or to the left (**levorotatory**). All of the amino acids in proteins exhibit the same absolute steric configuration as **L-glyceraldehyde**. Therefore, they are all L- $\alpha$ -amino acids. D-amino acids are never found in proteins, although they exist in nature. D-amino acids are often found in polypeptide antibiotics.

The aromatic R-groups in amino acids absorb ultraviolet light with an absorbance maximum in the range of 280nm. The ability of proteins to absorb ultraviolet light is predominantly due to the presence of the tryptophan which strongly absorbs ultraviolet light.

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## The Peptide Bond

Peptide bond formation is a condensation reaction leading to the polymerization of amino acids into peptides and proteins. Peptides are small consisting of few amino acids. A number of hormones and neurotransmitters are peptides. Additionally, several antibiotics and antitumor agents are peptides. Proteins are polypeptides of greatly divergent length. The simplest peptide, a **dipeptide**, contains a single peptide bond formed by the condensation of the carboxyl group of one amino acid with the amino group of the second with the concomitant elimination of water. The presence of the carbonyl group in a peptide bond allows electron resonance stabilization to occur such that the peptide bond exhibits rigidity not unlike the typical -C=C- double bond. The peptide bond is, therefore, said to have **partial double-bond character**.



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Last modified: **Tuesday, 12-Aug-2003 20:00:34 EST**

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- [Carbohydrate Nomenclature](#)
- [Monosaccharides](#)
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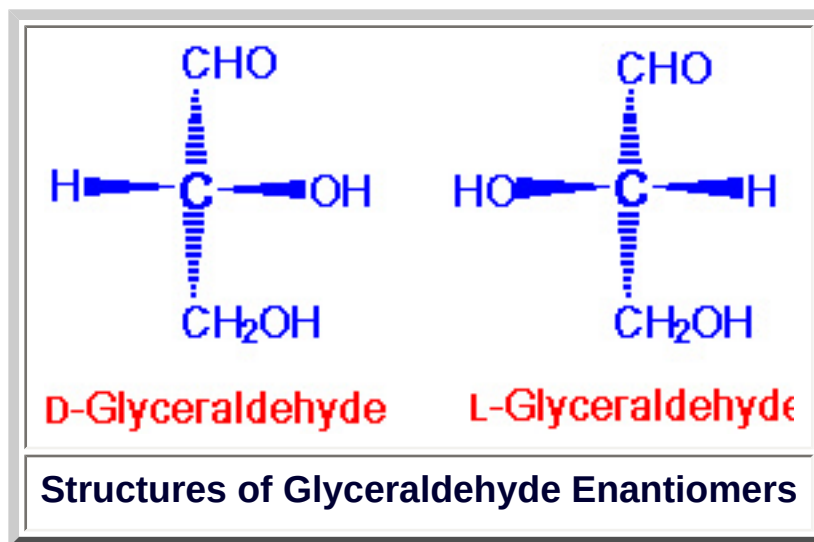
## Introduction

Carbohydrates are carbon compounds that contain large quantities of hydroxyl groups. The simplest carbohydrates also contain either an aldehyde moiety (these are termed **polyhydroxyaldehydes**) or a ketone moiety (**polyhydroxyketones**). All carbohydrates can be classified as either **monosaccharides**, **oligosaccharides** or **polysaccharides**. Anywhere from two to ten monosaccharide units, linked by glycosidic bonds, make up an oligosaccharide. Polysaccharides are much larger, containing hundreds of monosaccharide units. The presence of the hydroxyl groups allows carbohydrates to interact with the aqueous environment and to participate in hydrogen bonding, both within and between chains. Derivatives of the carbohydrates can contain nitrogens, phosphates and sulfur compounds. Carbohydrates also can combine with lipid to form **glycolipids** or with protein to form **glycoproteins**.

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## Carbohydrate Nomenclature

The predominant carbohydrates encountered in the body are structurally related to the aldotriose **glyceraldehyde** and to the ketotriose **dihydroxyacetone**. All carbohydrates contain at least one asymmetrical (chiral) carbon and are, therefore, optically active. In addition, carbohydrates can exist in either of two conformations, as determined by the orientation of the hydroxyl group about the asymmetric carbon farthest from the carbonyl. With a few exceptions, those carbohydrates that are of physiological significance exist in the **D-conformation**. The mirror-image conformations, called **enantiomers**, are in the **L-conformation**.



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## Monosaccharides

The monosaccharides commonly found in humans are classified according to the number of carbons they contain in their backbone structures. The major monosaccharides contain four to six carbon atoms.

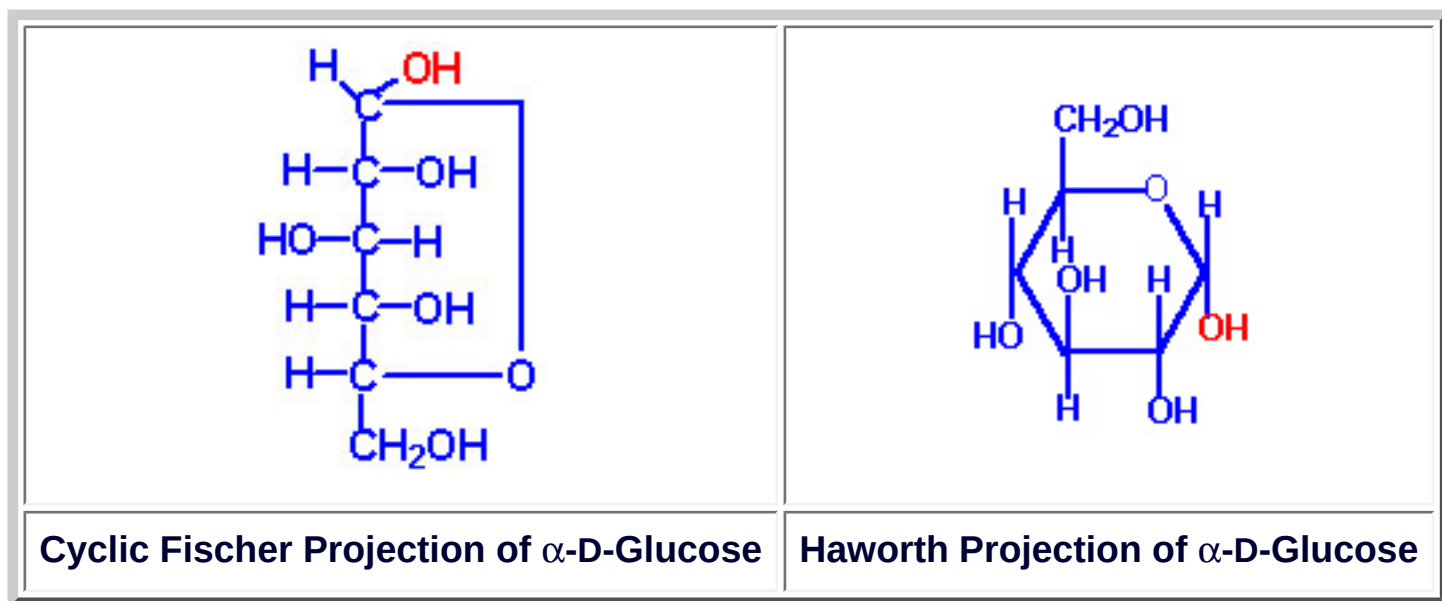
### Carbohydrate Classifications

| # Carbons | Category Name | Relevant examples                     |
|-----------|---------------|---------------------------------------|
| 3         | Triose        | Glyceraldehyde, Dihydroxyacetone      |
| 4         | Tetrose       | Erythrose                             |
| 5         | Pentose       | Ribose, Ribulose, Xylulose            |
| 6         | Hexose        | Glucose, Galactose, Mannose, Fructose |
| 7         | Heptose       | Sedoheptulose                         |

|   |        |                                            |
|---|--------|--------------------------------------------|
| 9 | Nonose | Neuraminic acid<br>also called sialic acid |
|---|--------|--------------------------------------------|

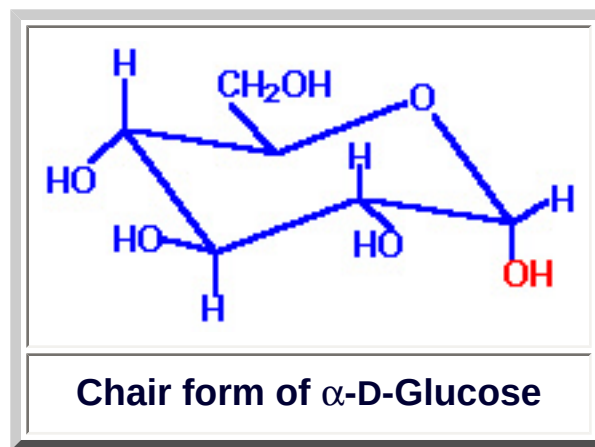
The aldehyde and ketone moieties of the carbohydrates with five and six carbons will spontaneously react with alcohol groups present in neighboring carbons to produce intramolecular **hemiacetals** or **hemiketals**, respectively. This results in the formation of five- or six-membered rings. Because the five-membered ring structure resembles the organic molecule **furan**, derivatives with this structure are termed **furanoses**. Those with six-membered rings resemble the organic molecule **pyran** and are termed **pyranoses**.

Such structures can be depicted by either **Fischer** or **Haworth** style diagrams. The numbering of the carbons in carbohydrates proceeds from the carbonyl carbon, for aldoses, or the carbon nearest the carbonyl, for ketoses.



The rings can open and re-close, allowing rotation to occur about the carbon bearing the reactive carbonyl yielding two distinct configurations ( $\alpha$  and  $\beta$ ) of the hemiacetals and hemiketals. The carbon about which this rotation occurs is the **anomeric carbon** and the two forms are termed anomers. Carbohydrates can change spontaneously between the  $\alpha$  and  $\beta$  configurations-- a process known as **mutarotation**. When drawn in the Fischer projection, the  $\alpha$  configuration places the hydroxyl attached to the anomeric carbon to the right, towards the ring. When drawn in the Haworth projection, the  $\alpha$  configuration places the hydroxyl downward.

The spatial relationships of the atoms of the furanose and pyranose ring structures are more correctly described by the two conformations identified as the **chair form** and the **boat form**. The chair form is the more stable of the two. Constituents of the ring that project above or below the plane of the ring are **axial** and those that project parallel to the plane are **equatorial**. In the chair conformation, the orientation of the hydroxyl group about the anomeric carbon of  $\alpha$ -D-glucose is axial and equatorial in  $\beta$ -D-glucose.

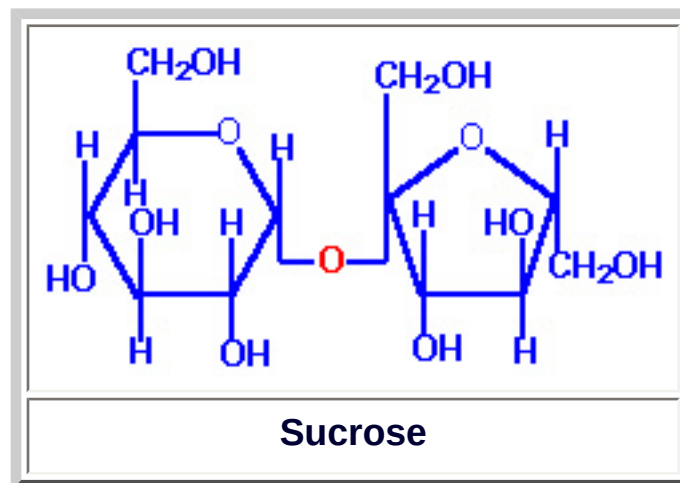


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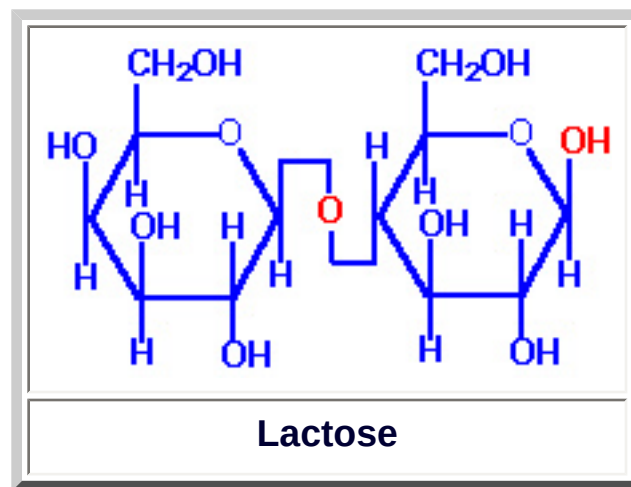
## Disaccharides

Covalent bonds between the anomeric hydroxyl of a cyclic sugar and the hydroxyl of a second sugar (or another alcohol containing compound) are termed **glycosidic bonds**, and the resultant molecules are **glycosides**. The linkage of two monosaccharides to form disaccharides involves a glycosidic bond. Several physiologically important disaccharides are sucrose, lactose and maltose.

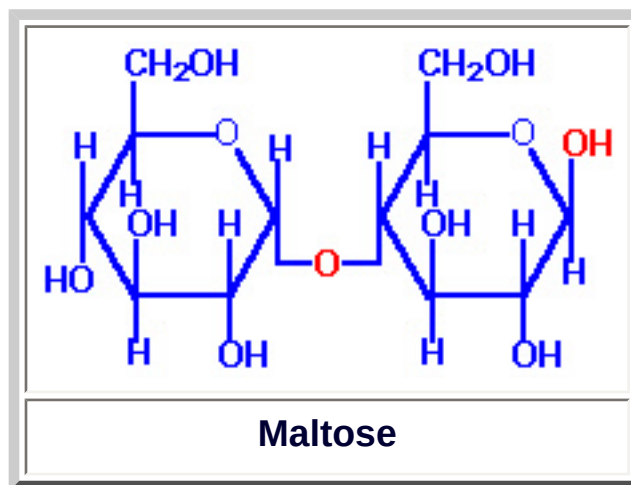
- **Sucrose**: prevalent in sugar cane and sugar beets, is composed of glucose and fructose through an  $\alpha$ -(1,2) $\beta$ -glycosidic bond.



- **Lactose**: is found exclusively in the milk of mammals and consists of galactose and glucose in a  $\beta$ -(1,4) glycosidic bond.



- **Maltose**: the major degradation product of starch, is composed of 2 glucose monomers in an  $\alpha$ -(1,4) glycosidic bond.



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## Polysaccharides

Most of the carbohydrates found in nature occur in the form of high molecular weight polymers called **polysaccharides**. The monomeric building blocks used to generate polysaccharides can be varied; in all cases, however, the predominant monosaccharide found in polysaccharides is D-glucose. When polysaccharides are composed of a single monosaccharide building block, they are termed **homopolysaccharides**. Polysaccharides composed of more than one type of monosaccharide are termed **heteropolysaccharides**.

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## Glycogen

Glycogen is the major form of stored carbohydrate in animals. This crucial molecule is a homopolymer of glucose in  $\alpha$ -(1,4) linkage; it is also highly branched, with  $\alpha$ -(1,6) branch



linkages occurring every 8-10 residues. Glycogen is a very compact structure that results from the coiling of the polymer chains. This compactness allows large amounts of carbon energy to be stored in a small volume, with little effect on cellular osmolarity.

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## Starch

Starch is the major form of stored carbohydrate in plant cells. Its structure is identical to glycogen, except for a much lower degree of branching (about every 20-30 residues). Unbranched starch is called **amylose**; branched starch is called **amylopectin**.

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Last modified: **Wednesday, 06-Oct-2004 14:58:12 EST**

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- [Basic Biochemistry of Fatty Acids](#)
- [Physiologically Relevant Fatty Acids](#)
- [Basic Structure of Complex Lipids](#)
- [Triacylglycerides](#)
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## Major Roles of Biological of Lipids

Biological molecules that are insoluble in aqueous solutions and soluble in organic solvents are classified as lipids. The lipids of physiological importance for humans have four major functions:

- **1.** They serve as structural components of biological membranes.
- **2.** They provide energy reserves, predominantly in the form of triacylglycerols.
- **3.** Both lipids and lipid derivatives serve as vitamins and hormones.
- **4.** Lipophilic bile acids aid in lipid solubilization.

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## Fatty Acids

Fatty acids fill two major roles in the body:

- **1.** as the components of more complex membrane lipids.
- **2.** as the major components of stored fat in the form of triacylglycerols.

Fatty acids are long-chain hydrocarbon molecules containing a carboxylic acid moiety at one end. The numbering of carbons in fatty acids begins with the carbon of the carboxylate group. At physiological pH, the carboxyl group is readily ionized, rendering a negative charge onto fatty acids in bodily fluids.

Fatty acids that contain no carbon-carbon double bonds are termed **saturated fatty acids**; those that contain double bonds are **unsaturated fatty acids**. The numeric designations used for fatty acids come from the number of carbon atoms, followed by the number of sites of unsaturation (eg, palmitic acid is a 16-carbon fatty acid with no unsaturation and is designated by 16:0). The site of unsaturation in a fatty acid is indicated by the symbol  $\Delta$  and the number of the first carbon of the double bond (e.g. palmitoleic acid is a 16-carbon fatty acid with one site of unsaturation between carbons 9 and 10, and is designated by 16:1 <sup>$\Delta$</sup> <sub>9</sub>).

Saturated fatty acids of less than eight carbon atoms are liquid at physiological temperature, whereas those containing more than ten are solid. The presence of double bonds in fatty acids significantly lowers the melting point relative to a saturated fatty acid.

The majority of body fatty acids are acquired in the diet. However, the lipid biosynthetic capacity of the body (fatty acid synthase and other fatty acid modifying enzymes) can supply the body with all the various fatty acid structures needed. Two key exceptions to this are the highly unsaturated fatty acids known as **linoleic acid** and **linolenic acid**, containing unsaturation sites beyond carbons 9 and 10. These two fatty acids cannot be synthesized from precursors in the body, and are thus considered the **essential fatty acids**; essential in the sense that they must be provided in the diet. Since plants are capable of synthesizing linoleic and linolenic acid humans can acquire these fats by consuming a variety of plants or else by eating the meat of animals that have consumed these plant fats.

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## Physiologically Relevant Fatty Acids

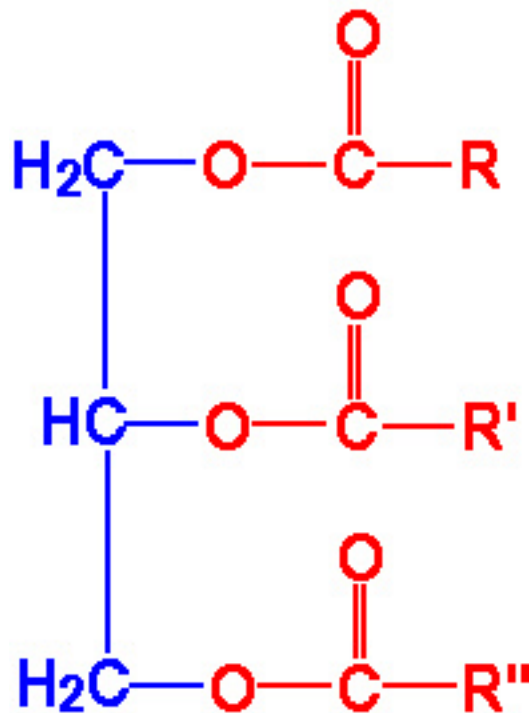
| Numerical Symbol | Common Name | Structure | Comments |
|------------------|-------------|-----------|----------|
|                  |             |           |          |

|                                        |                  |                                                                                                               |                                                                                        |
|----------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 14:0                                   | Myristic acid    | $\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$                                                                    | Often found attached to the N-term. of plasma membrane-associated cytoplasmic proteins |
| 16:0                                   | Palmitic acid    | $\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$                                                                    | End product of mammalian fatty acid synthesis                                          |
| 16:1 <sup>Δ</sup> <sub>9</sub>         | Palmitoleic acid | $\text{CH}_3(\text{CH}_2)_5\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$                                       |                                                                                        |
| 18:0                                   | Stearic acid     | $\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$                                                                    |                                                                                        |
| 18:1 <sup>Δ</sup> <sub>9</sub>         | Oleic acid       | $\text{CH}_3(\text{CH}_2)_7\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$                                       |                                                                                        |
| 18:2 <sup>Δ</sup> <sub>9,12</sub>      | Linoleic acid    | $\text{CH}_3(\text{CH}_2)_4\text{C}=\text{CCH}_2\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$                  | <b>Essential fatty acid</b>                                                            |
| 18:3 <sup>Δ</sup> <sub>9,12,15</sub>   | Linolenic acid   | $\text{CH}_3\text{CH}_2\text{C}=\text{CCH}_2\text{C}=\text{CCH}_2\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$ | <b>Essential fatty acid</b>                                                            |
| 20:4 <sup>Δ</sup> <sub>5,8,11,14</sub> | Arachidonic acid | $\text{CH}_3(\text{CH}_2)_3(\text{CH}_2\text{C}=\text{C})_4(\text{CH}_2)_3\text{COOH}$                        | Precursor for <b>eicosanoid synthesis</b>                                              |

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## Basic Structure of Triacylglycerides

Triacylglycerides are composed of a glycerol backbone to which 3 fatty acids are esterified.

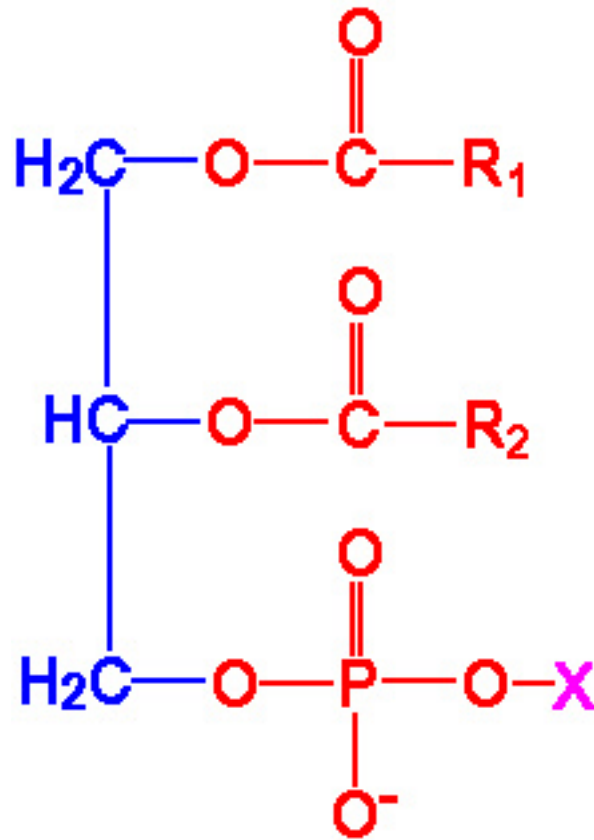


Basic composition of a triacylglyceride. The glycerol backbone is in blue.

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## Basic Structure of Phospholipids

The basic structure of phospholipids is very similar to that of the triacylglycerides except that C-3 (*sn*3) of the glycerol backbone is esterified to phosphoric acid. The building block of the phospholipids is **phosphatidic acid** which results when the X substitution in the basic structure shown in the Figure below is a hydrogen atom. Substitutions include ethanolamine (**phosphatidylethanolamine**), choline (**phosphatidylcholine**, also called **lecithins**), serine (**phosphatidylserine**), glycerol (**phosphatidylglycerol**), *myo*-inositol (**phosphatidylinositol**), these compounds can have a variety in the numbers of inositol alcohols that are phosphorylated generating **polyphosphatidylinositols**, and phosphatidylglycerol (**diphosphatidylglycerol** more commonly known as **cardiolipins**).



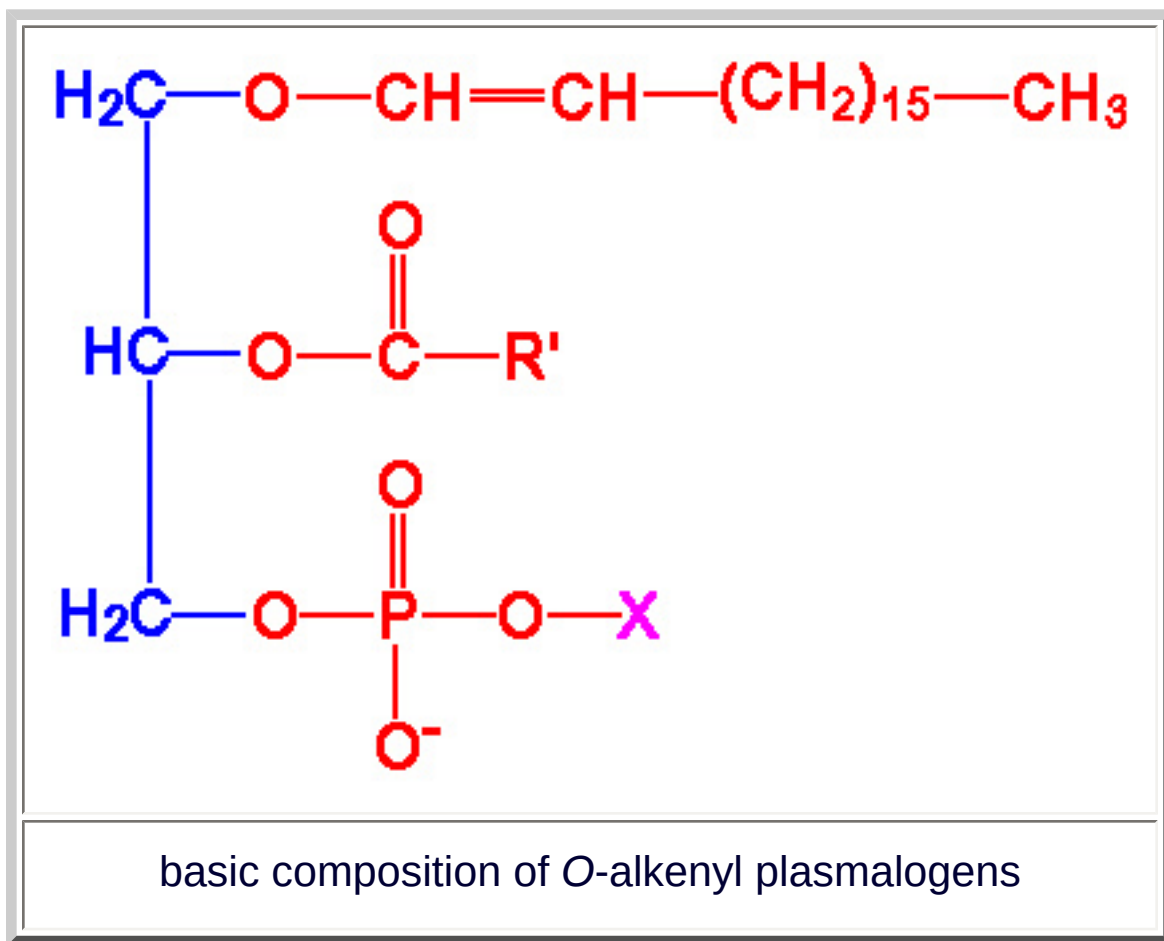
**Basic composition of a phospholipid. X can be a number of different substituents.**

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## Basic Structure of Plasmalogens

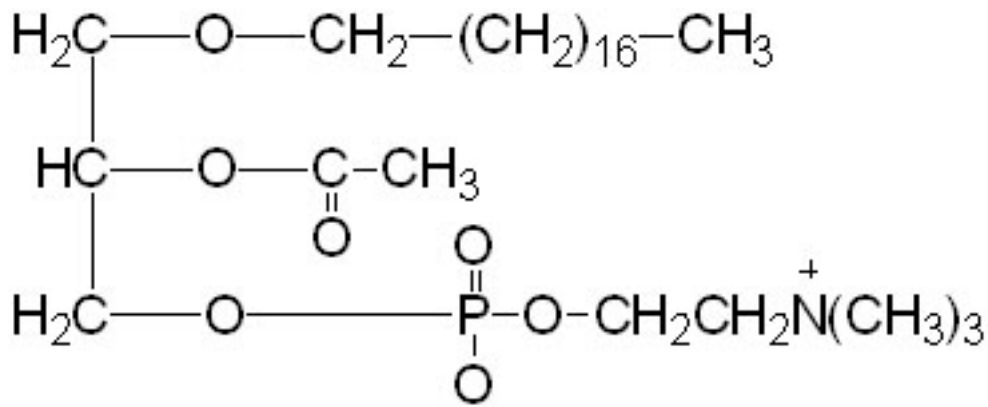
Plasmalogens are complex membrane lipids that resemble phospholipids, principally phosphatidylcholine. The major difference is that the fatty acid at C-1 (*sn*1) of glycerol contains either an *O*-alkyl (-O-CH<sub>2</sub>-) or *O*-alkenyl ether (-O-CH=CH-) species. A basic *O*-alkenyl ether species is shown in the Figure below.





One of the most potent alkyl ether plasmalogens is **platelet activating factor** (PAF: 1-O-1'-enyl-2-acetyl-*sn*-glycero-3-phosphocholine) which is a choline plasmalogen in which the C-2 (*sn*2) position of glycerol is esterified with an acetyl group instead of a long chain fatty acid.

PAF functions as a mediator of hypersensitivity, acute inflammatory reactions and anaphylactic shock. PAF is synthesized in response to the formation of antigen-IgE complexes on the surfaces of basophils, neutrophils, eosinophils, macrophages and monocytes. The synthesis and release of PAF from cells leads to platelet aggregation and the release of serotonin from platelets. PAF also produces responses in liver, heart, smooth muscle, and uterine and lung tissues.



structure of PAF

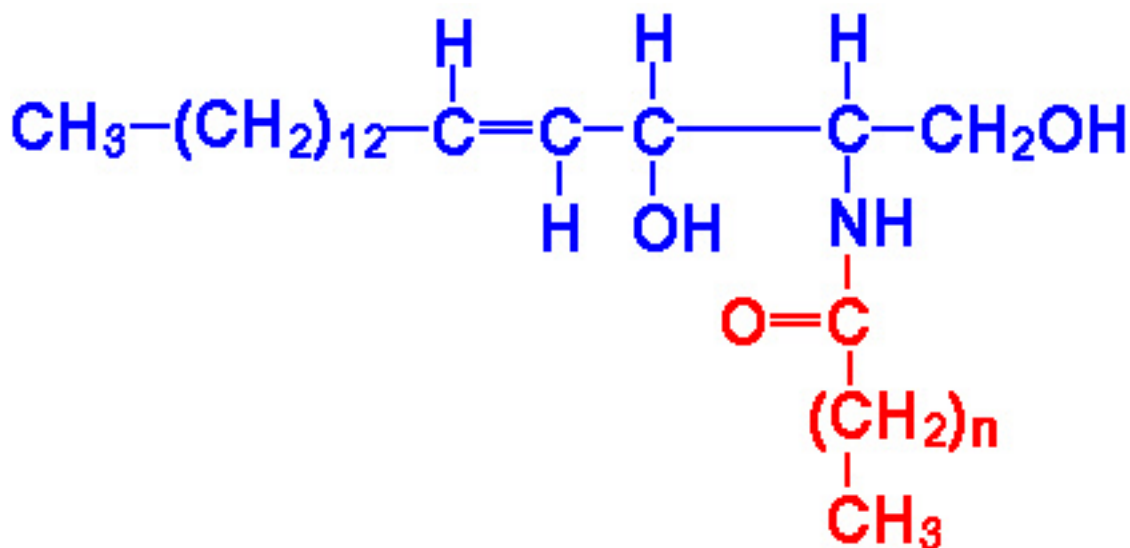
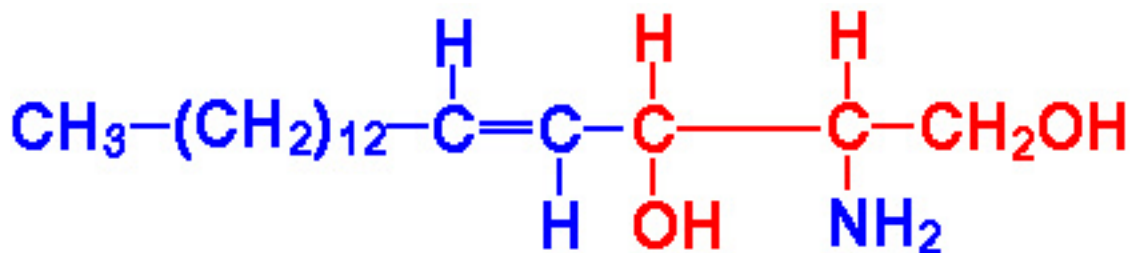
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## Basic Structure of Sphingolipids

Sphingolipids are composed of a backbone of sphingosine which is derived itself from glycerol. Sphingosine is N-acetylated by a variety of fatty acids generating a family of molecules referred to as **ceramides**. Sphingolipids predominate in the myelin sheath of nerve fibers. **Sphingomyelin** is an abundant sphingolipid generated by transfer of the phosphocholine moiety of phosphatidylcholine to a ceramide, thus sphingomyelin is a unique form of a phospholipid.

The other major class of sphingolipids (besides the sphingomyelins) are the **glycosphingolipids** generated by substitution of carbohydrates to the *sn1* carbon of the glycerol backbone of a ceramide. There are 4 major classes of glycosphingolipids:

- **Cerebrosides:** contain a single moiety, principally galactose.
- **Sulfatides:** sulfuric acid esters of galactocerebrosides.
- **Globosides:** contain 2 or more sugars.
- **Gangliosides:** similar to globosides except also contain sialic acid.



Top: Sphingosine  
the atoms in red are derived from glycerol.

Bottom: Basic composition of a ceramide  
n indicates any fatty acid may be N-acetylated at this position.

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Last modified: Thursday, 05-May-2005 10:33:09 EST

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## Introduction

AMP-activated protein kinase (AMPK) was first discovered as an activity that inhibited preparations of [acetyl-CoA carboxylase](#) (ACC) and [3-hydroxy-3-methylglutaryl-CoA reductase](#) (HMG-CoA reductase, HMGR) and was induced by AMP. AMPK induces a cascade of events within cells in response to the ever changing energy charge of the cell. The role of AMPK in regulating cellular energy charge places this enzyme at a central control point in maintaining energy homeostasis. More recent evidence has shown that AMPK activity can also be regulated by physiological stimuli, independent of the energy charge of the cell, including hormones and nutrients.

Once activated, AMPK-mediated phosphorylation events switch cells from active ATP consumption (e.g. [fatty acid](#) and [cholesterol biosynthesis](#)) to active ATP production (e.g. [fatty acid](#) and [glucose oxidation](#)). These events are rapidly initiated and are referred to as short-term regulatory processes. The activation of AMPK also exerts long-term effects at the level of both gene expression and protein synthesis. Other important activities attributable to AMPK are regulation of insulin synthesis and secretion in pancreatic islet  $\beta$ -cells and modulation of hypothalamic functions involved in the regulation of satiety. How these latter two functions impact obesity and [diabetes](#) will be discussed below.

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## Structure of AMPK

The mammalian AMPK is a trimeric enzyme composed of a catalytic  $\alpha$  subunit and the non-catalytic  $\beta$  and  $\gamma$  subunits. There are two genes encoding isoforms of both the  $\alpha$  and  $\beta$  subunits ( $\alpha 1$ ,  $\alpha 2$ ,  $\beta 1$  and  $\beta 2$ ) and three genes encoding isoforms of the  $\gamma$  subunit ( $\gamma 1$ - $\gamma 3$ ). The  $\alpha 2$  isoform is the subunit of AMPK found predominantly within skeletal and cardiac muscle, whereas, approximately equal distribution of both the  $\alpha 1$  and  $\alpha 2$  isoforms are present in hepatic AMPK. Within pancreatic islet  $\beta$ -cells the  $\alpha 1$  isoform predominates.

The N-terminal half of the  $\alpha$  subunits contains a typical serine/threonine kinase catalytic domain. Interaction with the  $\beta$  and  $\gamma$  subunits occurs via the C-terminal half of the  $\alpha$  subunits. The yeast AMPK  $\beta$  subunits are lipid modified with myristic acid. Myristoylation may account for the membrane association mammalian AMPK. The core of the  $\beta$  subunits have a glycogen-binding domain (GBD). This domain is closely related to the isoamylase N domain subfamily and weakly related to domains in the glycogen-targeting phosphatase subunits and several starch-binding proteins. The close proximity of AMPK to cellular glycogen stores allows it to rapidly affect changes in [glycogen metabolism](#) in response to changes in metabolic demands. The  $\gamma$  subunits of AMPK have been shown to contain nucleotide binding sites with similarity to cystathionine  $\beta$ -synthase (CBS) domains. Indeed, direct AMP-binding studies have shown that AMP is bound to the  $\gamma$  subunits by a pair of CBS domains. Of clinical significance is the observation that mutations in the CBS domains of the  $\gamma 2$  subunit are associated with [Wolff-Parkinson-White syndrome](#) and familial hypertrophic cardiomyopathy.

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## Regulation of AMPK

AMPK is activated by phosphorylation by one or more upstream AMPK kinases (AMPKKs). In the absence of phosphorylation there is no detectable activity of AMPK towards any substrates. Phosphorylation of AMPK occurs in the  $\alpha$  subunit at threonine 172 (Thr-172) which lies in the activation loop. However, as the name implies, AMPK is also regulated by AMP. The effects of AMP are two-fold: a direct allosteric activation and making AMPK a poorer substrate for dephosphorylation. Because AMP affects both the rate of AMPK phosphorylation in the positive direction and dephosphorylation in the negative direction, the cascade is ultrasensitive. This means that a very small rise in AMP levels can induce a dramatic increase in the activity of AMPK. The activity of adenylate kinase, catalyzing the reaction shown below, ensure that AMPK is highly sensitive to small changes in the intracellular [ATP]/[ADP] ratio.



Negative allosteric regulation of AMPK also occurs and this effect is exerted by phosphocreatine. As indicated above, the  $\beta$  subunits of AMPK have a glycogen-binding domain, GBD. In muscle, a high glycogen content represses AMPK activity and this is likely the result of interaction between the GBD and glycogen, although this has not been shown directly. As suggested above, the GBD of AMPK allows association of the enzyme with the regulation of glycogen metabolism by placing AMPK in close proximity to one of its substrates glycogen synthase.

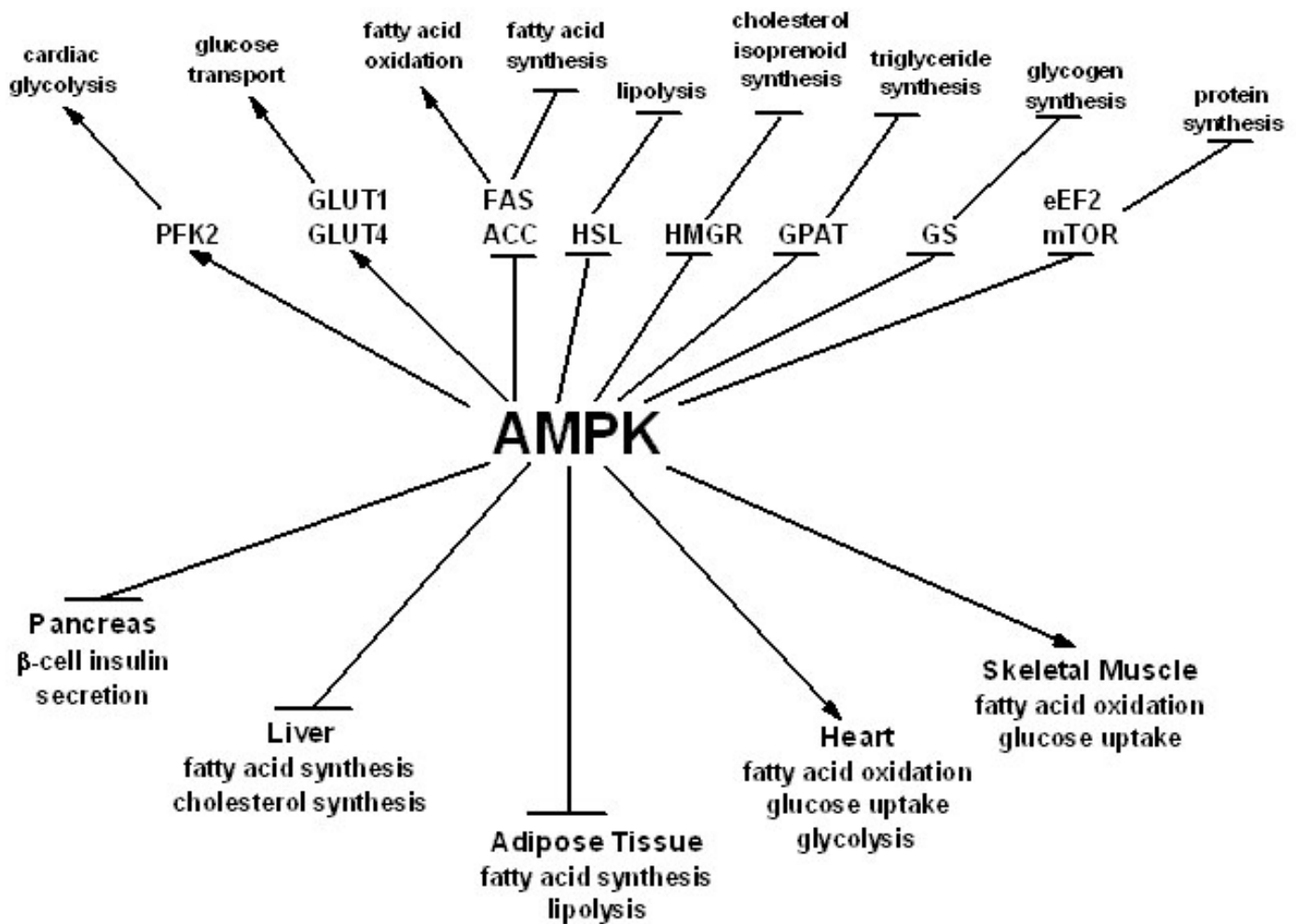
AMPK has also been shown to be activated by receptors that are coupled to phospholipase C- $\gamma$  (PLC- $\gamma$ ) and by cytokines secreted by adipose tissue (adipocytokines) such as leptin and adiponectin.

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## Targets of AMPK

The signaling cascades initiated by the activation of AMPK exert effects on glucose and lipid metabolism, gene expression and protein synthesis. These effects are most important for regulating metabolic events in the liver, skeletal muscle, heart, adipose tissue, and pancreas.



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Demonstration of the central role of AMPK in the regulation of metabolism in response to events such as nutrient- or exercise-induced stress. Several of the known physiologic targets for AMPK are included as well as several pathways whose flux is affected by AMPK activation. Arrows indicate positive effects of AMPK, whereas, T-lines indicate inhibitory effects. See text for definition of enzyme abbreviations.

The uptake, by skeletal muscle, accounts for >70% of the glucose removal from the serum in humans. Therefore, it should be obvious that this event is extremely important for overall glucose homeostasis, keeping in mind, of course, that glucose uptake by cardiac muscle and adipocytes cannot be excluded from consideration. An important fact related to skeletal muscle glucose uptake is that this process is markedly impaired in individuals with type 2 diabetes. The uptake of glucose increases dramatically in response to stress (such as ischemia) and exercise and is stimulated by insulin-induced recruitment of glucose transporters to the plasma membrane, primarily GLUT4. Insulin-independent recruitment of glucose transporters also occurs in skeletal muscle in response to contraction (exercise). The activation of AMPK plays an important, albeit not an exclusive, role in the induction of GLUT4 recruitment to the plasma membrane. In addition, there is some demonstration that AMPK may regulate glucose transport through GLUT1. Increased glucose uptake will result in an increase in glycolysis and ATP production.

Under ischemic conditions in the heart the activation of AMPK leads to the phosphorylation and activation of phosphofructokinase-2, PFK-2 (6-phosphofructo-2-kinase). PFK-2 is one of the most potent regulators of the rate of flux through [glycolysis](#) and [gluconeogenesis](#). It is important to note that like many enzymes, there are multiple



isoforms of PFK-2 and neither the liver nor the skeletal muscle isoforms contain the AMPK phosphorylation sites of the inducible form of PFK-2. Of particular significance is the fact that the inducible form of PFK-2 is commonly found in tumor cells and this may allow AMPK to play an important role in protecting tumor cells from hypoxic stress. Indeed, techniques for depleting AMPK in tumor cells have shown that these cells become sensitized to nutritional stress upon loss of AMPK activity.

Whereas, stress and exercise are powerful inducers of AMPK activity in skeletal muscle, additional regulators of its activity have been identified. [Insulin-sensitizing drugs](#) of the thiazolidinedione family as well as the hypoglycemia drug **metformin** may exert a portion of their effects through regulation of the activity of AMPK. Also, the adipocytokine, **leptin** has been shown to activate AMPK.

Within skeletal muscle and heart activation of AMPK leads to the phosphorylation and inhibition of acetyl-CoA carboxylase (ACC). This inhibition results in a drop in the level of malonyl-CoA which itself is an inhibitor of carnitine palmitoyltransferase I (CPT I). With a drop in the inhibition of CPT I a concomitant increase in  [\$\beta\$ -oxidation of fatty acid](#) will occur within the mitochondria. An increase in fatty acid oxidation, like increases in glycolysis, will lead to increases in ATP production. In addition to ACC, AMPK has been shown to phosphorylate and thus regulate the activities of HMG-CoA reductase, (HMGR); hormone-sensitive lipase, (HSL); glycerophosphate acyltransferase, (GPAT); malonyl-CoA decarboxylase, (MCD); glycogen synthase, (GS) and creatine kinase, (CK). Therefore, the effects of AMPK activation are exerted on not only glucose homeostasis and fatty acid metabolism but overall energy homeostasis including glycogen metabolism, cholesterol metabolism and phosphocreatine metabolism.

Not only does activation of AMPK exerts effects on enzyme activity through phosphorylation, there are marked effects on the expression of a number of glycolytic and lipogenic enzymes in the liver and adipose tissue. Included in this list are the genes for the liver isoform of pyruvate kinase (L-PK), fatty acid synthase (FAS), and ACC. Activation of AMPK leads to a reduction in the level of [SREBP](#) a transcription factor that is a key regulator of the expression of numerous lipogenic enzymes. Another transcription factor reduced in response to AMPK activation is **hepatocyte nuclear factor 4 $\alpha$** , **HNF4 $\alpha$**  which is a member of the steroid/thyroid hormone superfamily. HNF4 $\alpha$  is known to regulate the expression of several liver and pancreatic  $\beta$ -cell genes such as GLUT2, L-PK and preproinsulin. Of clinical significance is that mutations in HNF4 $\alpha$  are responsible for **maturity-onset diabetes of the young, MODY-1**. Recent evidence indicates that the newly discovered **carbohydrate-response-element-binding protein (ChREBP)** is a target for AMPK in the liver. ChREBP is implicated in the transcriptional regulation of L-PK.

The target of the thiazolidinedione class of drugs used to treat [type 2 diabetes](#) is the **peroxisome proliferator-activated receptor  $\gamma$** , **PPAR $\gamma$**  which itself may be a target for the action of AMPK. The transcription co-activator, p300, is phosphorylated by AMPK which inhibits interaction of p300 with not only PPAR $\gamma$  but also the retinoic acid receptor, retinoid X receptor, and thyroid receptor. Another transcription factor target of AMPK is the forkhead protein, **FKHR** (now referred to as **FoxO1a**). FoxO1a is involved in the activation of glucose-6-phosphatase expression and, therefore, loss of FoxO1a activity in response to AMPK activation will lead to reduced hepatic output of glucose.

AMPK activation in response to hypoxia exerts effects on rates of protein synthesis. Hepatic translation elongation factor, eEF2 is a target for phosphorylation in response to AMPK activation. AMPK phosphorylates and activates the kinase that phosphorylates eEF2 (eEF2K) leading to inhibition of protein synthesis. Another indirect substrate for AMPK that plays a role in protein synthesis is the **mammalian target of rapamycin, mTOR**. A detailed description of the role of mTOR in regulating protein synthesis can be found on the [Protein Synthesis](#) page.

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## Relevance of AMPK to Type 2 Diabetes

An impairment in fuel metabolism occurs in obesity and this impairment is a leading pathogenic factor in [type 2 diabetes](#). The insulin resistance associated with type 2 diabetes is most profound at the level of skeletal muscle as this is the primary site of glucose and fatty acid utilization. Therefore, an understanding of how to activate AMPK in skeletal muscle would offer significant pharmacologic benefits in the treatment of type 2 diabetes. As indicated above, it has already been shown that metformin and the thiazolidinedione drugs exert some of their effects via activation of AMPK. In the non-pharmacologic context, activation of AMPK occurs in response to exercise, an activity known to have significant benefit for type 2 diabetics. Future targets for type 2 diabetes treatments will likely be those that can effect beneficial changes in the activity of AMPK.

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Last modified: **Thursday, 07-Oct-2004 13:16:39 EST**

# Disorders of Glycosphingolipid Metabolism

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- GM<sub>1</sub> Gangliosidoses
  - Generalized GM<sub>1</sub> Type II - Juvenile Form
  - Generalized GM<sub>1</sub> Type III - Adult Form
- GM<sub>2</sub> Gangliosidosis
  - Sandhoff-Jatzkewitz Disease
- GM<sub>3</sub> Gangliosidosis
- Tay-Sachs Disease
- Tay-Sachs AB Variant
- Gaucher Disease
- Niemann-Pick Disease, Type A and B
  - Types A and B are due to **acid sphingomyelinase** deficiencies
- Niemann-Pick Disease, Type C1
- Niemann-Pick Disease, Type C2
- Niemann-Pick Disease, Type D
  - Types C and D are not due to **acid sphingomyelinase** deficiencies
- Schindler Disease
- Fabry Disease
- Lactosylceramidosis
- Farber Disease
- Krabbe Disease
- Multiple Sulfatase Deficiency - Austin Disease
- Metachromatic Leukodystrophy - Sulfatide Lipodosis
- Fucosidosis

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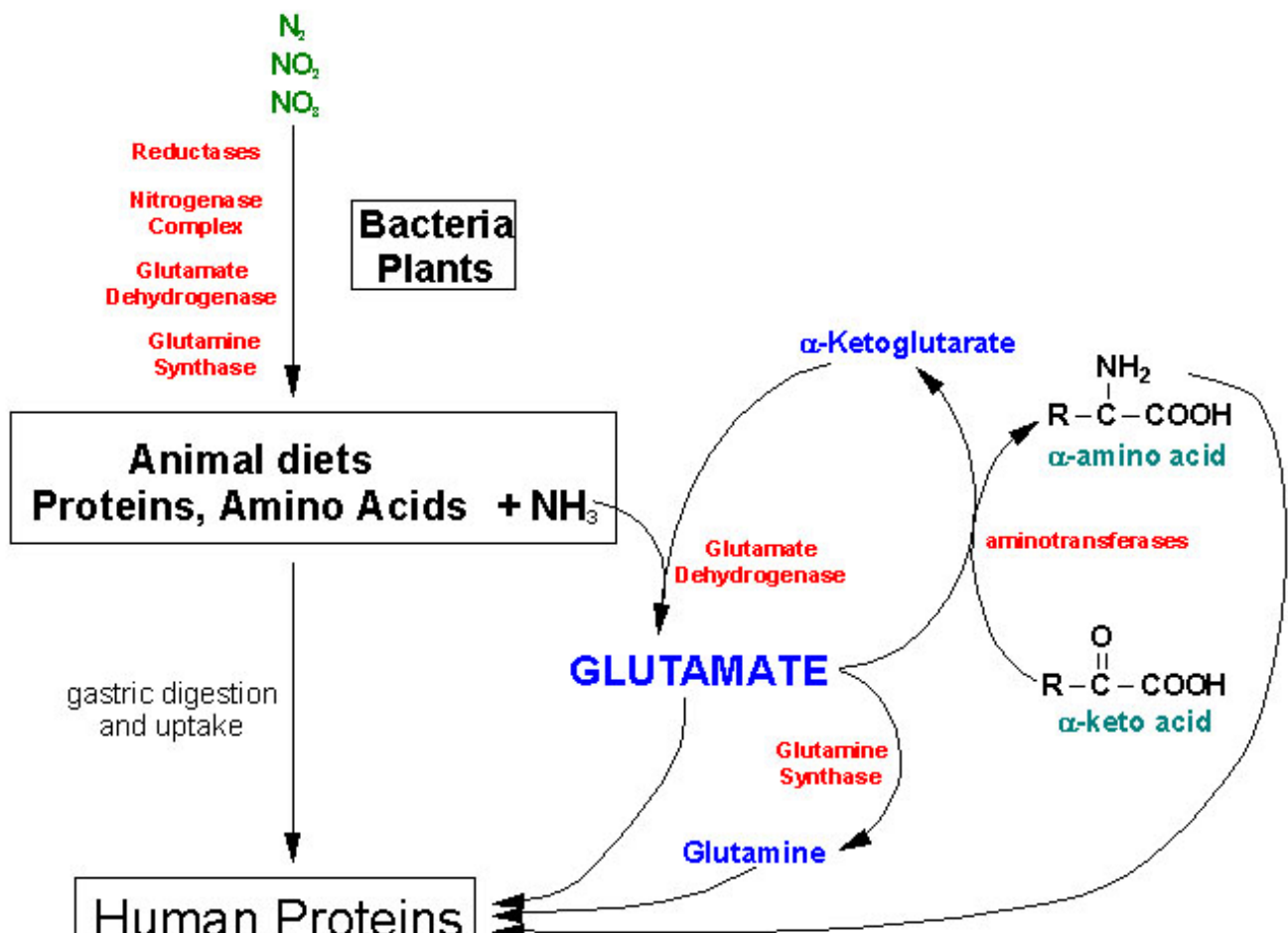
Last modified: **Thursday, 24-Jun-2004 15:46:04 EST**

- Introduction to Nitrogen Metabolism
- Glutamate Dehydrogenase
- Glutamine Synthase
- Digestive Tract Nitrogen
- Essential vs. Nonessential Amino Acids
- Removal of Nitrogen from Amino Acids
- The Urea Cycle
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- Urea Cycle Defects (UCDs)
- Table of Urea Cycle Defects
- Neurotoxicity of Ammonia

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## Introduction

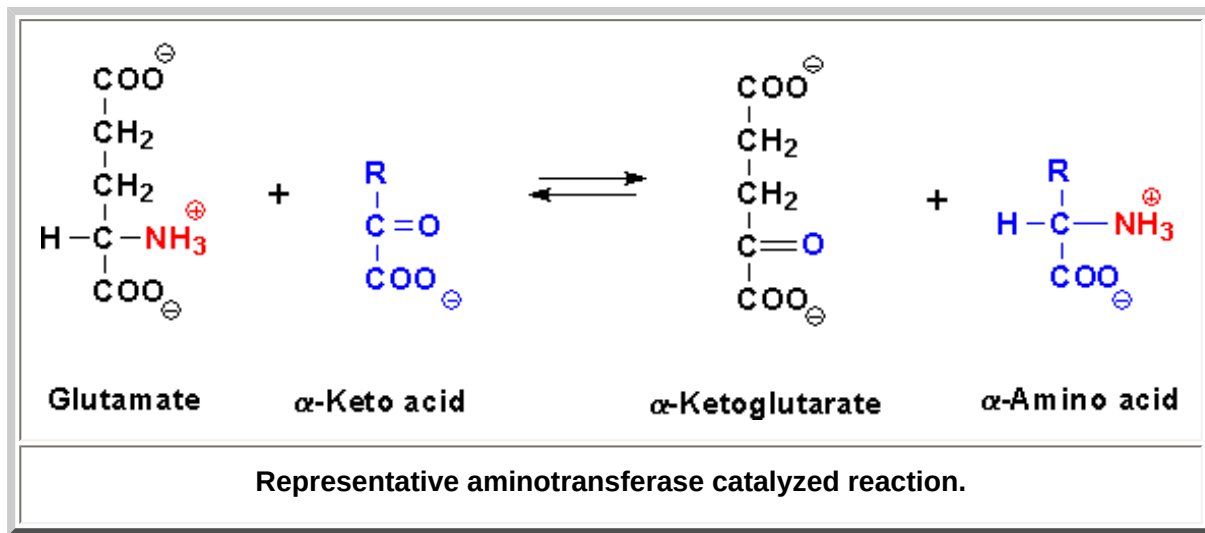
Humans are totally dependent on other organisms for converting atmospheric nitrogen into forms available to the body. Nitrogen fixation is carried out by bacterial nitrogensases forming reduced nitrogen,  $\text{NH}_4^+$  which can then be used by all organisms to form amino acids.



# Human Proteins

Overview of the flow of nitrogen in the biosphere. Nitrogen, nitrites and nitrates are acted upon by bacteria (nitrogen fixation) and plants and we assimilate these compounds as protein in our diets. Ammonia incorporation in animals occurs through the actions of glutamate dehydrogenase and glutamine synthase. Glutamate plays the central role in mammalian nitrogen flow, serving as both a nitrogen donor and nitrogen acceptor.

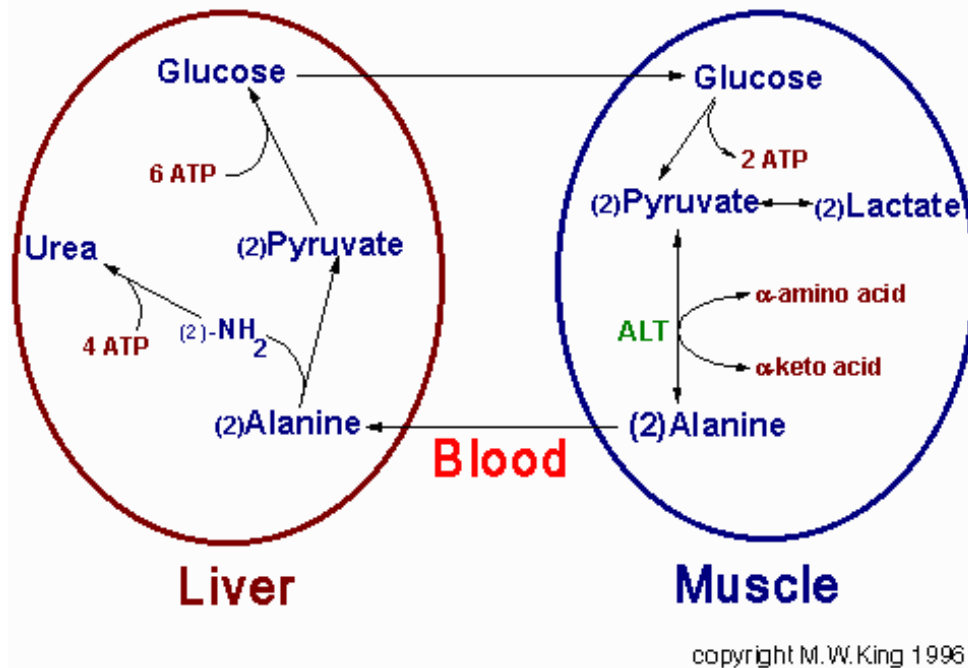
Reduced nitrogen enters the human body as dietary free amino acids, protein, and the ammonia produced by intestinal tract bacteria. A pair of principal enzymes, glutamate dehydrogenase and glutamine synthetase, are found in all organisms and effect the conversion of ammonia into the amino acids **glutamate** and **glutamine**, respectively. Amino and amide groups from these 2 substances are freely transferred to other carbon skeletons by **transamination** and **transamidation** reactions.



Aminotransferases exist for all amino acids except threonine and lysine. The most common compounds involved as a donor/acceptor pair in transamination reactions are **glutamate** and **α-ketoglutarate** (α-KG), which participate in reactions with many different aminotransferases. Serum aminotransferases such as serum glutamate-oxaloacetate-aminotransferase (SGOT) (also called aspartate aminotransferase, AST) and serum glutamate-pyruvate aminotransferase (SGPT) (also called alanine transaminase, ALT) have been used as clinical markers of tissue damage, with increasing serum levels indicating an increased extent of damage. Alanine transaminase has an important function in the delivery of skeletal muscle carbon and nitrogen (in the form of alanine) to the liver. In skeletal muscle, pyruvate is transaminated to alanine, thus affording an additional route of nitrogen transport from muscle to liver. In the liver alanine transaminase transfers the ammonia to α-KG and regenerates pyruvate. The pyruvate can then be diverted into gluconeogenesis. This process is referred to as the **glucose-alanine cycle**.



## Glucose/Alanine Cycle

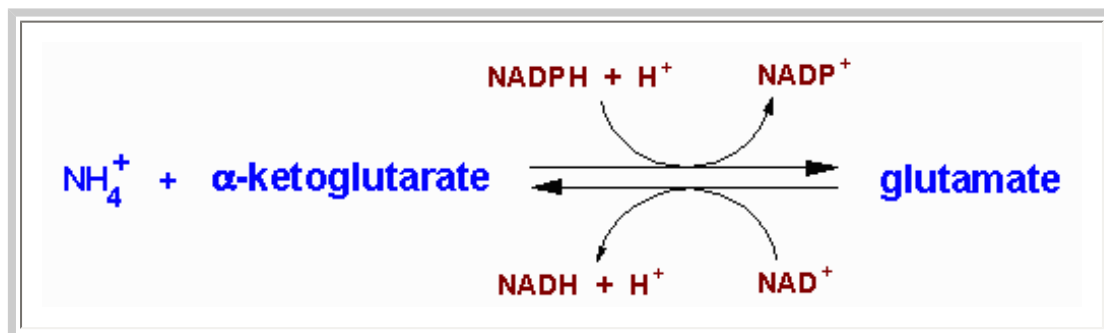


The glucose-alanine cycle is used primarily as a mechanism for skeletal muscle to eliminate nitrogen while replenishing its energy supply. Glucose oxidation produces pyruvate which can undergo transamination to alanine. This reaction is catalyzed by alanine transaminase, ALT. Additionally, during periods of fasting, skeletal muscle protein is degraded for the energy value of the amino acid carbons and alanine is a major amino acid in protein. The alanine then enters the blood stream and is transported to the liver. Within the liver alanine is converted back to pyruvate which is then a source of carbon atoms for gluconeogenesis. The newly formed glucose can then enter the blood for delivery back to the muscle. The amino group transported from the muscle to the liver in the form of alanine is converted to urea in the urea cycle and excreted.

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## The Glutamate Dehydrogenase Reaction

The reaction catalyzed by glutamate dehydrogenase is:



The glutamate dehydrogenase utilizes both nicotinamide nucleotide cofactors; NAD<sup>+</sup> in the direction of nitrogen liberation and NADP<sup>+</sup> for nitrogen incorporation. In the forward reaction as shown above glutamate dehydrogenase is important in converting free ammonia and α-ketoglutarate (α-KG) to glutamate, forming one of the 20 amino acids required for

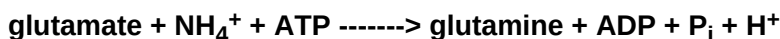
protein synthesis. However, it should be recognized that the reverse reaction is a key anapleurotic process linking amino acid metabolism with [TCA cycle](#) activity. In the reverse reaction, glutamate dehydrogenase provides an oxidizable carbon source used for the production of energy as well as a reduced electron carrier, NADH. As expected for a branch point enzyme with an important link to energy metabolism, glutamate dehydrogenase is regulated by the cell energy charge. ATP and GTP are positive allosteric effectors of the formation of glutamate, whereas ADP and GDP are positive allosteric effectors of the reverse reaction. Thus, when the level of ATP is high, conversion of glutamate to  $\alpha$ -KG and other TCA cycle intermediates is limited; when the cellular energy charge is low, glutamate is converted to ammonia and oxidizable TCA cycle intermediates. Glutamate is also a principal amino donor to other amino acids in subsequent transamination reactions. The multiple roles of glutamate in nitrogen balance make it a gateway between free ammonia and the amino groups of most amino acids.

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## The Glutamine Synthase Reaction

The reaction catalyzed by glutamine synthetase is:



The glutamine synthetase reaction is also important in several respects. First it produces glutamine, one of the 20 major amino acids. Second, in animals, glutamine is the major amino acid found in the circulatory system. Its role there is to carry ammonia to and from various tissues but principally from peripheral tissues to the kidney, where the amide nitrogen is hydrolyzed by the enzyme glutaminase (reaction below); this process regenerates glutamate and free ammonium ion, which is excreted in the urine.



Note that, in this function, ammonia arising in peripheral tissue is carried in a nonionizable form which has none of the neurotoxic or alkalosis-generating properties of free ammonia.

Liver contains both glutamine synthetase and glutaminase but the enzymes are localized in different cellular segments. This ensures that the liver is neither a net producer nor consumer of glutamine. The differences in cellular location of these two enzymes allows the liver to scavenge ammonia that has not been incorporated into urea. The enzymes of the urea cycle are located in the same cells as those that contain glutaminase. The result of the differential distribution of these two hepatic enzymes makes it possible to control ammonia incorporation into either urea or glutamine, the latter leads to excretion of ammonia by the kidney.

When acidosis occurs the body will divert more glutamine from the liver to the kidney. This allows for the conservation of bicarbonate ion since the incorporation of ammonia into urea requires bicarbonate (see below). When glutamine enters the kidney, glutaminase releases one mole of ammonia generating glutamate and then glutamate dehydrogenase releases another mole of ammonia generating  $\alpha$ -ketoglutarate. The ammonia will ionize to ammonium ion ( $\text{NH}_4^+$ ) which is excreted. The net effect is a reduction in the pH (see also [Kidneys and Acid-Base Balance](#)).

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## Digestive Tract Nitrogen

While glutamine, glutamate, and the remaining nonessential amino acids can be made by animals, the majority of the amino acids found in human tissues necessarily come from dietary sources (about 400g of protein per day). Protein digestion begins in the stomach, where a proenzyme called pepsinogen is secreted, autocatalytically converted to Pepsin A, and used for the first step of proteolysis. However, most proteolysis takes place in the duodenum as a consequence of enzyme activities secreted by the pancreas. All of the serine proteases and the zinc peptidases of pancreatic secretions are produced in the form of their respective proenzymes. These proteases are both endopeptidase and exopeptidase, and their combined action in the intestine leads to the production of amino acids, dipeptides, and tripeptides, all of which are taken up by enterocytes of the mucosal wall.

A circuitous regulatory pathway leading to the secretion of proenzymes into the intestine is triggered by the appearance of food in the intestinal lumen. Special mucosal endocrine cells secrete the peptide hormones **cholecystokinin (CCK)** and **secretin** into the circulatory system. Together, CCK and secretin cause contraction of the gall bladder and the exocrine secretion of a bicarbonate-rich, alkaline fluid, containing protease proenzymes from the pancreas into the intestine. A second,

paracrine role of CCK is to stimulate adjacent intestinal cells to secrete enteropeptidase, a protease that cleaves trypsinogen to produce trypsin. Trypsin also activates trypsinogen as well as all the other proenzymes in the pancreatic secretion, producing the active proteases and peptidases that hydrolyze dietary polypeptides.

Subsequent to luminal hydrolysis, small peptides and amino acids are transferred through enterocytes to the portal circulation by diffusion, facilitated diffusion, or active transport. A number of  $\text{Na}^+$ -dependent amino acid transport systems with overlapping amino acid specificity have been described. In these transport systems,  $\text{Na}^+$  and amino acids at high luminal concentrations are co-transported down their concentration gradient to the interior of the cell. The ATP-dependent  $\text{Na}^+/\text{K}^+$  pump exchanges the accumulated  $\text{Na}^+$  for extracellular  $\text{K}^+$ , reducing intracellular  $\text{Na}^+$  levels and maintaining the high extracellular  $\text{Na}^+$  concentration (high in the intestinal lumen, low in enterocytes) required to drive this transport process.

Transport mechanisms of this nature are ubiquitous in the body. Small peptides are accumulated by a proton ( $\text{H}^+$ ) driven transport process and hydrolyzed by intracellular peptidases. Amino acids in the circulatory system and in extracellular fluids are transported into cells of the body by at least 7 different ATP-requiring active transport systems with overlapping amino acid specificities.

**Hartnup disorder** is an autosomal recessive impairment of neutral amino acid transport affecting the kidney tubules and small intestine. It is believed that the defect lies in a specific system responsible for neutral amino acid transport across the brush-border membrane of renal and intestinal epithelium. The exact defect has not yet been characterized. The characteristic diagnostic feature of Hartnup disorder is a dramatic neutral hyperaminoaciduria. Additionally, individuals excrete indolic compounds that originate from the bacterial degradation of unabsorbed tryptophan. The reduced intestinal absorption and increased renal loss of tryptophan lead to a reduced availability of tryptophan for niacin and nicotinamide nucleotide biosynthesis. As a consequence affected individuals frequently exhibit pellegra-like rashes.

Many other nitrogenous compounds are found in the intestine. Most are bacterial products of protein degradation. Some have powerful pharmacological (vasopressor) effects.

| Products of Intestinal Bacterial Activity |                    |                    |
|-------------------------------------------|--------------------|--------------------|
| Substrates                                | Products           |                    |
|                                           | Vasopressor Amines | Other              |
| Lysine                                    | Cadaverene         |                    |
| Arginine                                  | Agmatine           |                    |
| Tyrosine                                  | Tyramine           |                    |
| Ornithine                                 | Putrescine         |                    |
| Histidine                                 | Histamine          |                    |
| Tryptophan                                |                    | Indole and skatole |
| All amino acids                           |                    | $\text{NH}_4^+$    |

Prokaryotes such as *E. coli* can make the carbon skeletons of all 20 amino acids and transaminate those carbon skeletons with nitrogen from glutamine or glutamate to complete the amino acid structures. Humans cannot synthesize the branched carbon chains found in branched chain amino acids or the ring systems found in phenylalanine and the aromatic amino acids; nor can we incorporate sulfur into covalently bonded structures. Therefore, the 10 so-called **essential amino acids** (see Table below) must be supplied from the diet. Nevertheless, it should be recognized that, depending on the composition of the diet and physiological state of an individual, one or another of the non-essential amino acids may also become a required dietary component. For example, arginine is only normally considered to be essential amino acid during

early childhood development because enough for adult needs is made by the urea cycle.

To take a different type of example, cysteine and tyrosine are considered non-essential but are formed from the essential amino acids methionine and phenylalanine, respectively. If sufficient cysteine and tyrosine are present in the diet, the requirements for methionine and phenylalanine are markedly reduced; conversely, if methionine and phenylalanine are present in only limited quantities, cysteine and tyrosine can become essential dietary components. Finally, it should be recognized that if the  $\alpha$ -keto acids corresponding to the carbon skeletons of the essential amino acids are supplied in the diet, aminotransferases in the body will convert the keto acids to their respective amino acids, largely supplying the basic needs.

Unlike fats and carbohydrates, nitrogen has no designated storage depots in the body. Since the half-life of many proteins is short (on the order of hours), insufficient dietary quantities of even one amino acid can quickly limit the synthesis and lower the body levels of many essential proteins. The result of limited synthesis and normal rates of protein degradation is that the balance of nitrogen intake and nitrogen excretion is rapidly and significantly altered. Normal, healthy adults are generally in **nitrogen balance**, with intake and excretion being very well matched. Young growing children, adults recovering from major illness, and pregnant women are often in **positive nitrogen balance**. Their intake of nitrogen exceeds their loss as net protein synthesis proceeds. When more nitrogen is excreted than is incorporated into the body, an individual is in **negative nitrogen balance**. Insufficient quantities of even one essential amino acid is adequate to turn an otherwise normal individual into one with a negative nitrogen balance.

The biological value of dietary proteins is related to the extent to which they provide all the necessary amino acids. Proteins of animal origin generally have a high biological value; plant proteins have a wide range of values from almost none to quite high. In general, plant proteins are deficient in lysine, methionine, and tryptophan and are much less concentrated and less digestible than animal proteins. The absence of lysine in low-grade cereal proteins, used as a dietary mainstay in many underdeveloped countries, leads to an inability to synthesize protein (because of missing essential amino acids) and ultimately to a syndrome known as **kwashiorkor**, common among children in these countries.

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## Essential vs. Nonessential Amino Acids

| Nonessential | Essential      |
|--------------|----------------|
| Alanine      | Arginine*      |
| Asparagine   | Histidine      |
| Aspartate    | Isoleucine     |
| Cysteine     | Leucine        |
| Glutamate    | Lysine         |
| Glutamine    | Methionine*    |
| Glycine      | Phenylalanine* |
| Proline      | Threonine      |
| Serine       | Tyrtptophan    |
| Tyrosine     | Valine         |

\*The amino acids **arginine**, **methionine** and **phenylalanine** are considered essential for reasons not directly related to lack of synthesis. Arginine is synthesized by mammalian cells but at a rate that is insufficient to meet the growth needs of the body and the majority that is synthesized is cleaved to form urea. Methionine is required in large amounts to produce cysteine if the latter amino acid is not adequately supplied in the diet. Similarly, phenylalanine is needed in large amounts to form tyrosine if the latter is not adequately supplied in the diet.

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## Removal of Nitrogen from Amino Acids

The dominant reactions involved in removing amino acid nitrogen from the body are known as **transaminations**. This class of reactions funnels nitrogen from all free amino acids into a small number of compounds; then, either they are oxidatively deaminated, producing ammonia, or their amine groups are converted to urea by the urea cycle. Transaminations involve moving an  $\alpha$ -amino group from a donor  $\alpha$ -amino acid to the keto carbon of an acceptor  $\alpha$ -keto acid. These reversible reactions are catalyzed by a group of intracellular enzymes known as **aminotransferases**, which generally employ covalently bound pyridoxal phosphate as a cofactor. However, some aminotransferases employ pyruvate as a cofactor.

Aminotransferases exist for all amino acids except threonine and lysine. The most common compounds involved as a donor/acceptor pair in transamination reactions are glutamate and  $\alpha$ -ketoglutarate, which participate in reactions with many different aminotransferases. Serum aminotransferases such as serum glutamate-oxaloacetate-aminotransferase (SGOT) (also called aspartate aminotransferase, AST) and serum glutamate-pyruvate aminotransferase (SGPT) (also called alanine transaminase, ALT) have been used as clinical markers of tissue damage, with increasing serum levels indicating an increased extent of damage. Alanine transaminase has an important function in the delivery of skeletal muscle carbon and nitrogen (in the form of alanine) to the liver. In skeletal muscle, pyruvate is transaminated to alanine, thus affording an additional route of nitrogen transport from muscle to liver. In the liver alanine transaminase transfers the ammonia to  $\alpha$ -KG and regenerates pyruvate. The pyruvate can then be diverted into gluconeogenesis. This process is referred to as the **glucose-alanine cycle**.

Because of the participation of  $\alpha$ -ketoglutarate in numerous transaminations, glutamate is a prominent intermediate in nitrogen elimination as well as in anabolic pathways. Glutamate formed in the course of nitrogen elimination is either oxidatively deaminated by liver glutamate dehydrogenase, forming ammonia, or converted to glutamine by glutamine synthase and transported to kidney tubule cells. There the glutamine is sequentially deamidated by glutaminase and deaminated by kidney glutamate dehydrogenase.

The ammonia produced in the latter two reactions is excreted as  $\text{NH}_4^+$  in the urine, where it helps maintain urine pH in the normal range of pH 4 to pH 8. The extensive production of ammonia by peripheral tissue or hepatic glutamate dehydrogenase is not feasible because of the highly toxic effects of circulating ammonia. Normal serum ammonium concentrations are in the range of 20 - 40 mM, and an increase in circulating ammonia to about 400 mM causes alkalosis and neurotoxicity.

A final, therapeutically useful amino acid-related reaction is the amidation of aspartic acid to produce asparagine. The enzyme asparagine synthase catalyzes the ATP-requiring transamidation reaction shown below:



Most cells perform this reaction well enough to produce all the asparagine they need. However, some leukemia cells require exogenous asparagine, which they obtain from the plasma. Chemotherapy using the enzyme asparaginase takes advantage of this property of leukemic cells by hydrolyzing serum asparagine to ammonia and aspartic acid, thus depriving the neoplastic cells of the asparagine that is essential for their characteristic rapid growth.

In the peroxisomes of mammalian tissues, especially liver, there exist a minor enzymatic pathway for the removal of amino groups from amino acids. L-amino acid oxidase is FMN-linked and has broad specificity for the L-amino acids.

A number of substances, including oxygen, can act as electron acceptors from the flavoproteins. If oxygen is the acceptor the product is hydrogen peroxide, which is then rapidly degraded by the catalases found in liver and other tissues.

Missing or defective biogenesis of peroxisomes or L-amino acid oxidase causes generalized hyperaminoacidemia and hyperaminoaciduria, generally leading to neurotoxicity and early death.

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## The Urea Cycle

Earlier it was noted that kidney glutaminase was responsible for converting excess glutamine from the liver to urine ammonium. However, about 80% of the excreted nitrogen is in the form of urea which is also largely made in the liver, in a series of reactions that are distributed between the mitochondrial matrix and the cytosol. The series of reactions that form urea is known as the **Urea Cycle** or the **Krebs-Henseleit Cycle**.

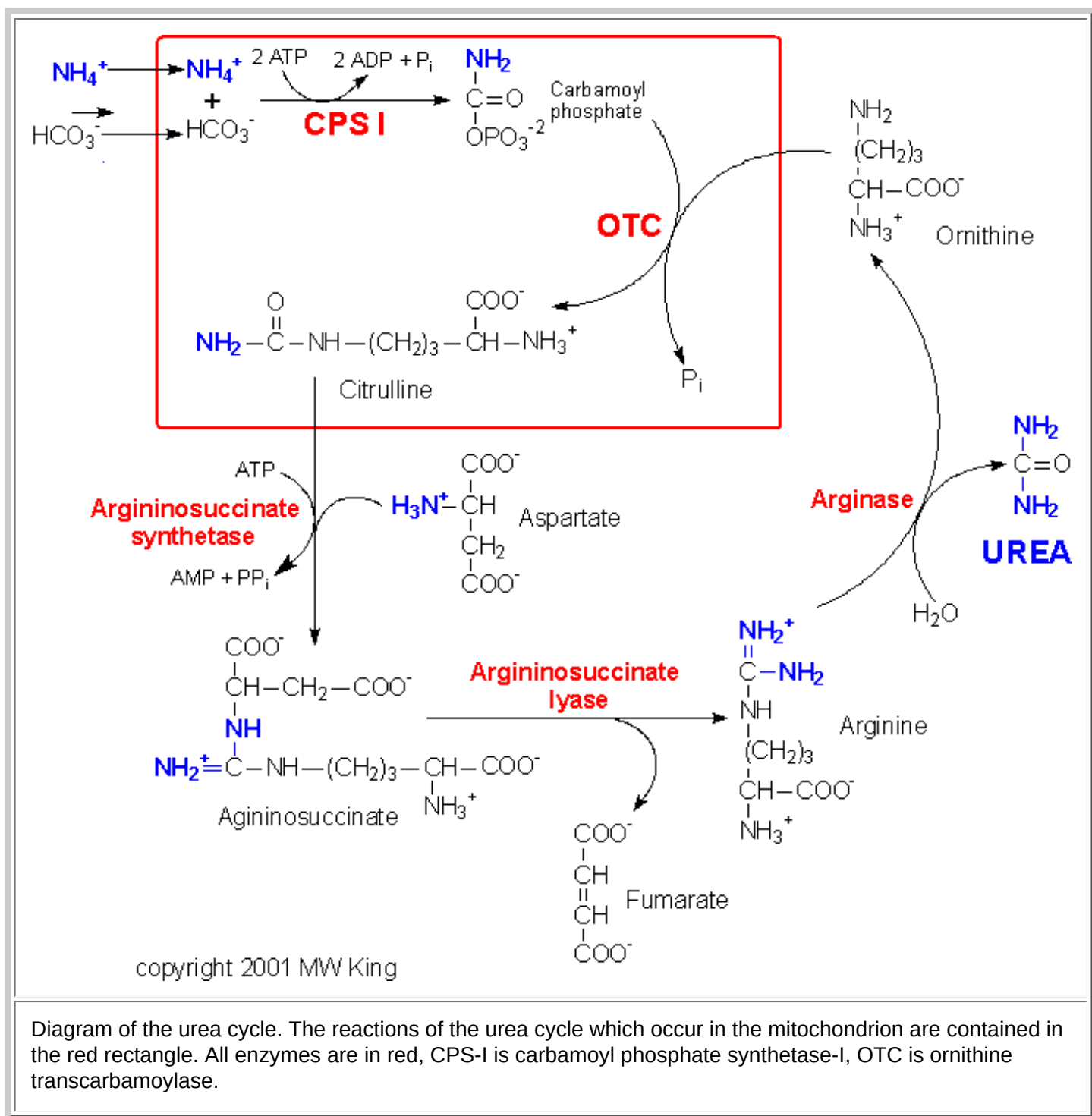


Diagram of the urea cycle. The reactions of the urea cycle which occur in the mitochondrion are contained in the red rectangle. All enzymes are in red, CPS-I is carbamoyl phosphate synthetase-I, OTC is ornithine transcarbamoylase.

The essential features of the urea cycle reactions and their metabolic regulation are as follows: Arginine from the diet or from protein breakdown is cleaved by the cytosolic enzyme arginase, generating urea and ornithine. In subsequent reactions of the urea cycle a new urea residue is built on the ornithine, regenerating arginine and perpetuating the cycle.

Ornithine arising in the cytosol is transported to the mitochondrial matrix, where ornithine transcarbamoylase catalyzes the condensation of ornithine with carbamoyl phosphate, producing citrulline. The energy for the reaction is provided by the high-energy anhydride of carbamoyl phosphate. The product, citrulline, is then transported to the cytosol, where the remaining reactions of the cycle take place.

The synthesis of citrulline requires a prior activation of carbon and nitrogen as carbamoyl phosphate (CP). The activation step requires 2 equivalents of ATP and the mitochondrial matrix enzyme carbamoyl phosphate synthetase-I (CPS-I). There are two CP synthetases: a mitochondrial enzyme, CPS-I, which forms CP destined for inclusion in the urea cycle, and a cytosolic CP synthetase (CPS-II), which is involved in **pyrimidine nucleotide biosynthesis**. CPS-I is positively regulated by the allosteric effector **N-acetylglutamate**, while the cytosolic enzyme is acetylglutamate independent.

In a 2-step reaction, catalyzed by cytosolic argininosuccinate synthetase, citrulline and aspartate are condensed to form



argininosuccinate. The reaction involves the addition of AMP (from ATP) to the amido carbonyl of citrulline, forming an activated intermediate on the enzyme surface (AMP-citrulline), and the subsequent addition of aspartate to form argininosuccinate.

Arginine and fumarate are produced from argininosuccinate by the cytosolic enzyme argininosuccinate lyase (also called argininosuccinase). In the final step of the cycle arginase cleaves urea from aspartate, regenerating cytosolic ornithine, which can be transported to the mitochondrial matrix for another round of urea synthesis. The fumarate, generated via the action of argininosuccinate lyase, is reconverted to aspartate for use in the argininosuccinate synthetase reaction. This occurs through the actions of cytosolic versions of the [TCA cycle](#) enzymes, fumarase (which yields malate) and malate dehydrogenase (which yields oxaloacetate). The oxaloacetate is then transaminated to aspartate by AST.

Beginning and ending with ornithine, the reactions of the cycle consumes 3 equivalents of ATP and a total of 4 high-energy nucleotide phosphates. Urea is the only new compound generated by the cycle; all other intermediates and reactants are recycled. The energy consumed in the production of urea is more than recovered by the release of energy formed during the synthesis of the urea cycle intermediates. Ammonia released during the glutamate dehydrogenase reaction is coupled to the formation of NADH. In addition, when fumarate is converted back to aspartate, the malate dehydrogenase reaction used to convert malate to oxaloacetate generates a mole of NADH. These two moles of NADH, thus, are oxidized in the mitochondria yielding 6 moles of ATP.

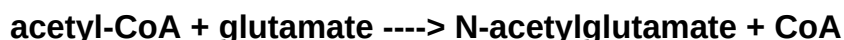
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## Regulation of the Urea Cycle

The urea cycle operates only to eliminate excess nitrogen. On high-protein diets the carbon skeletons of the amino acids are oxidized for energy or stored as fat and glycogen, but the amino nitrogen must be excreted. To facilitate this process, enzymes of the urea cycle are controlled at the gene level. With long-term changes in the quantity of dietary protein, changes of 20-fold or greater in the concentration of cycle enzymes are observed. When dietary proteins increase significantly, enzyme concentrations rise. On return to a balanced diet, enzyme levels decline. Under conditions of starvation, enzyme levels rise as proteins are degraded and amino acid carbon skeletons are used to provide energy, thus increasing the quantity of nitrogen that must be excreted.

Short-term regulation of the cycle occurs principally at CPS-I, which is relatively inactive in the absence of its allosteric activator **N-acetylglutamate**. The steady-state concentration of N-acetylglutamate is set by the concentration of its components acetyl-CoA and glutamate and by arginine, which is a positive allosteric effector of N-acetylglutamate synthetase.



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## Urea Cycle Defects (UCDs)

A complete lack of any one of the enzymes of the urea cycle will result in death shortly after birth. However, deficiencies in each of the enzymes of the urea cycle, including N-acetylglutamate synthase, have been identified. These disorders are referred to as **urea cycle disorders or UCDs**. More information on the individual UCDs can be found in the [Inborn Errors in Metabolism](#) pages. A common thread to most UCDs is **hyperammonemia** leading to ammonia intoxication with the consequences described [below](#). Deficiencies in arginase do not lead to symptomatic hyperammonemia as severe or as commonly as in the other UCDs.

Clinical symptoms are most severe when the UCD is at the level of carbamoyl phosphate synthetase I. Symptoms of UCDs usually arise at birth and encompass, ataxia, convulsions, lethargy, poor feeding and eventually coma and death if not recognized and treated properly. In fact, the mortality rate is 100% for UCDs that are left undiagnosed. Several UCDs manifest with late-onset such as in adulthood. In these cases the symptoms are hyperactivity, hepatomegaly and an avoidance of high protein foods.

In general, the treatment of UCDs has as common elements the reduction of protein in the diet, removal of excess ammonia and replacement of intermediates missing from the urea cycle. Administration of **levulose** reduces ammonia through its action of acidifying the colon. Bacteria metabolize levulose to acidic byproducts which then promotes excretion of ammonia in the feces as ammonium ions,  $\text{NH}_4^+$ . Antibiotics can be administered to kill intestinal ammonia producing bacteria. Sodium benzoate and sodium phenylbutyrate can be administered to covalently bind glycine (forming hippurate) and

glutamine (forming phenylacetylglutamine), respectively. These latter compounds, which contain the ammonia nitrogen, are excreted in the feces. Dietary supplementation with arginine or citrulline can increase the rate of urea production in certain UCDs.

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**Table of UCDs**

| UCD                                     | Enzyme Deficiency                           | Symptoms/Comments                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type I Hyperammonemia                   | Carbamoylphosphate synthetase I             | with 24h - 72h after birth infant becomes lethargic, needs stimulation to feed, vomiting, increasing lethargy, hypothermia and hyperventilation; without measurement of serum ammonia levels and appropriate intervention infant will die: treatment with arginine which activates N-acetylglutamate synthetase                                                                                                                     |
| N-acetylglutamate synthetase Deficiency | N-acetylglutamate synthetase                | severe hyperammonemia, mild hyperammonemia associated with deep coma, acidosis, recurrent diarrhea, ataxia, hypoglycemia, hyperornithinemia: treatment includes administration of carbamoyl glutamate to activate CPS I                                                                                                                                                                                                             |
| Type 2 Hyperammonemia                   | Ornithine transcarbamoylase                 | most commonly occurring UCD, only X-linked UCD, ammonia and amino acids elevated in serum, increased serum orotic acid due to mitochondrial carbamoylphosphate entering cytosol and being incorporated into pyrimidine nucleotides which leads to excess production and consequently excess catabolic products: treat with high carbohydrate, low protein diet, ammonia detoxification with sodium phenylacetate or sodium benzoate |
| Classic Citrullinemia                   | Argininosuccinate synthetase                | episodic hyperammonemia, vomiting, lethargy, ataxia, seizures, eventual coma: treat with arginine administration to enhance citrulline excretion, also with sodium benzoate for ammonia detoxification                                                                                                                                                                                                                              |
| Argininosuccinic aciduria               | Argininosuccinate lyase (argininosuccinase) | episodic symptoms similar to classic citrullinemia, elevated plasma and cerebral spinal fluid argininosuccinate: treat with arginine and sodium benzoate                                                                                                                                                                                                                                                                            |

## Hyperargininemia

## Arginase

rare UCD, progressive spastic quadriplegia and mental retardation, ammonia and arginine high in cerebral spinal fluid and serum, arginine, lysine and ornithine high in urine: treatment includes diet of essential amino acids excluding arginine, low protein diet

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## Neurotoxicity Associated with Ammonia

Earlier it was noted that ammonia was neurotoxic. Marked brain damage is seen in cases of failure to make urea via the urea cycle or to eliminate urea through the kidneys. The result of either of these events is a buildup of circulating levels of ammonium ion. Aside from its effect on blood pH, ammonia readily traverses the brain blood barrier and in the brain is converted to glutamate via glutamate dehydrogenase, depleting the brain of  $\alpha$ -ketoglutarate. As the  $\alpha$ -ketoglutarate is depleted, oxaloacetate falls correspondingly, and ultimately TCA cycle activity comes to a halt. In the absence of aerobic oxidative phosphorylation and TCA cycle activity, irreparable cell damage and neural cell death ensue.

In addition, the increased glutamate leads to glutamine formation. This depletes glutamate stores which are needed in neural tissue since glutamate is both a neurotransmitter and a precursor for the synthesis of  $\gamma$ -aminobutyrate, GABA, another neurotransmitter. Therefore, reductions in brain glutamate affect energy production as well as neurotransmission.

Additional untoward consequences are the result of elevations in neural glutamine concentration. Glial cell (astrocytes) volume is controlled by intracellular organic osmolyte metabolism. The organic osmolyte is glutamine. As glutamine levels rise in the brain the volume of fluid within glial cells increases resulting in the cerebral edema seen in infants with hyperammonemia caused by urea cycle defects.

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# Urea Cycle Defects

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- Hyperammonemias

- Carbamoyl Phosphate Synthetase I (CPS-I) Deficiency
- Ornithine Transcarbamylase (OTC) Deficiency
- N-Acetylglutamate Synthetase Deficiency

- Argininosuccinic Aciduria - Argininosuccinate Lyase Deficiency
- Hyperargininemia - Arginase Deficiency
- Citrullinemia - Argininosuccinate Synthetase Deficiency
- Ornithine Aminotransferase Deficiency

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- [Essential versus Non-Essential Amino Acids](#)
- [OMIM Links for Inborn Errors in Amino Acid Metabolism](#)

- **Nonessential Amino Acid Biosynthesis**

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## Introduction

All tissues have some capability for synthesis of the [non-essential amino acids](#), amino acid remodeling, and conversion of non-amino acid carbon skeletons into amino acids and other derivatives that contain nitrogen. However, the liver is the major site of [nitrogen metabolism](#) in the body. In times of dietary surplus, the potentially toxic nitrogen of amino acids is eliminated via transaminations, deamination, and urea formation; the carbon skeletons are generally conserved as carbohydrate, via [gluconeogenesis](#), or as fatty acid via [fatty acid synthesis](#) pathways. In this respect amino acids fall into three categories: **glucogenic**, **ketogenic**, or **glucogenic and ketogenic**. Glucogenic amino acids are those that give rise to a net production of pyruvate or [TCA cycle](#) intermediates, such as  $\alpha$ -ketoglutarate or oxaloacetate, all of which are precursors to glucose via gluconeogenesis. All amino acids except lysine and leucine are at least partly glucogenic. Lysine and leucine are the only amino acids that are solely ketogenic, giving rise only to acetylCoA or acetoacetylCoA, neither of which can bring about net glucose production.

A small group of amino acids comprised of isoleucine, phenylalanine, threonine, tryptophan, and tyrosine give rise to both glucose and fatty acid precursors and are thus characterized as being glucogenic and ketogenic. Finally, it should be recognized that amino acids have a third possible fate. During times of starvation the reduced carbon skeleton is used for energy production, with the result that it is oxidized to CO<sub>2</sub> and H<sub>2</sub>O.

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## Essential vs. Nonessential Amino Acids

| Nonessential | Essential      |
|--------------|----------------|
| Alanine      | Arginine*      |
| Asparagine   | Histidine      |
| Aspartate    | Isoleucine     |
| Cysteine     | Leucine        |
| Glutamate    | Lysine         |
| Glutamine    | Methionine*    |
| Glycine      | Phenylalanine* |
| Proline      | Threonine      |
| Serine       | Tryptophan     |
| Tyrosine     | Valine         |

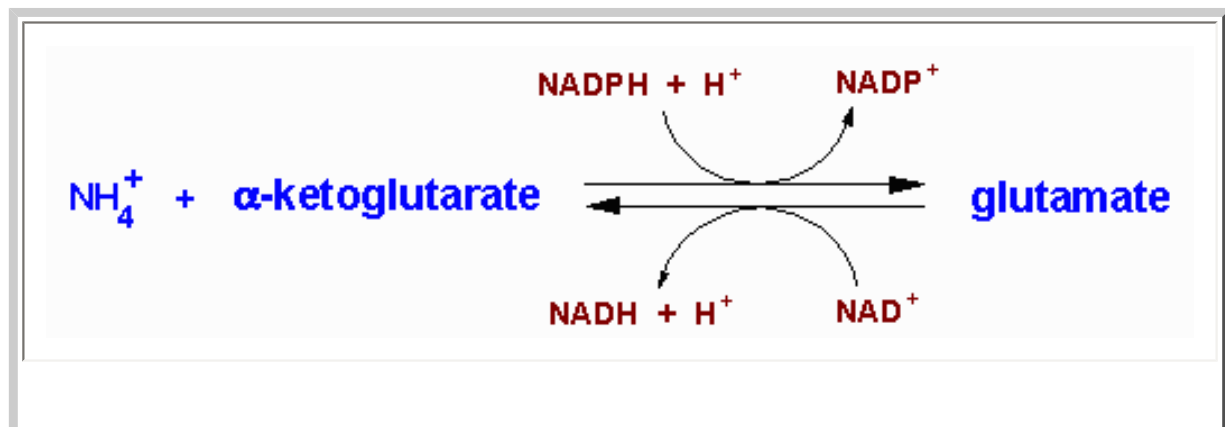
\*The amino acids **arginine**, **methionine** and **phenylalanine** are considered essential for reasons not directly related to lack of synthesis. Arginine is synthesized by mammalian cells but at a rate that is insufficient to meet the growth needs of the body and the majority that is synthesized is cleaved to form urea. Methionine is required in large amounts to produce cysteine if the latter amino acid is not adequately supplied in the diet. Similarly, phenylalanine is needed in large amounts to form tyrosine if the latter is not adequately supplied in the diet.

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## Amino Acid Biosynthesis

### Glutamate and Aspartate

Glutamate and aspartate are synthesized from their widely distributed  $\alpha$ -keto acid precursors by simple 1-step transamination reactions. The former catalyzed by glutamate dehydrogenase and the latter by aspartate aminotransferase, AST.





## Reactions of *glutamate dehydrogenase*

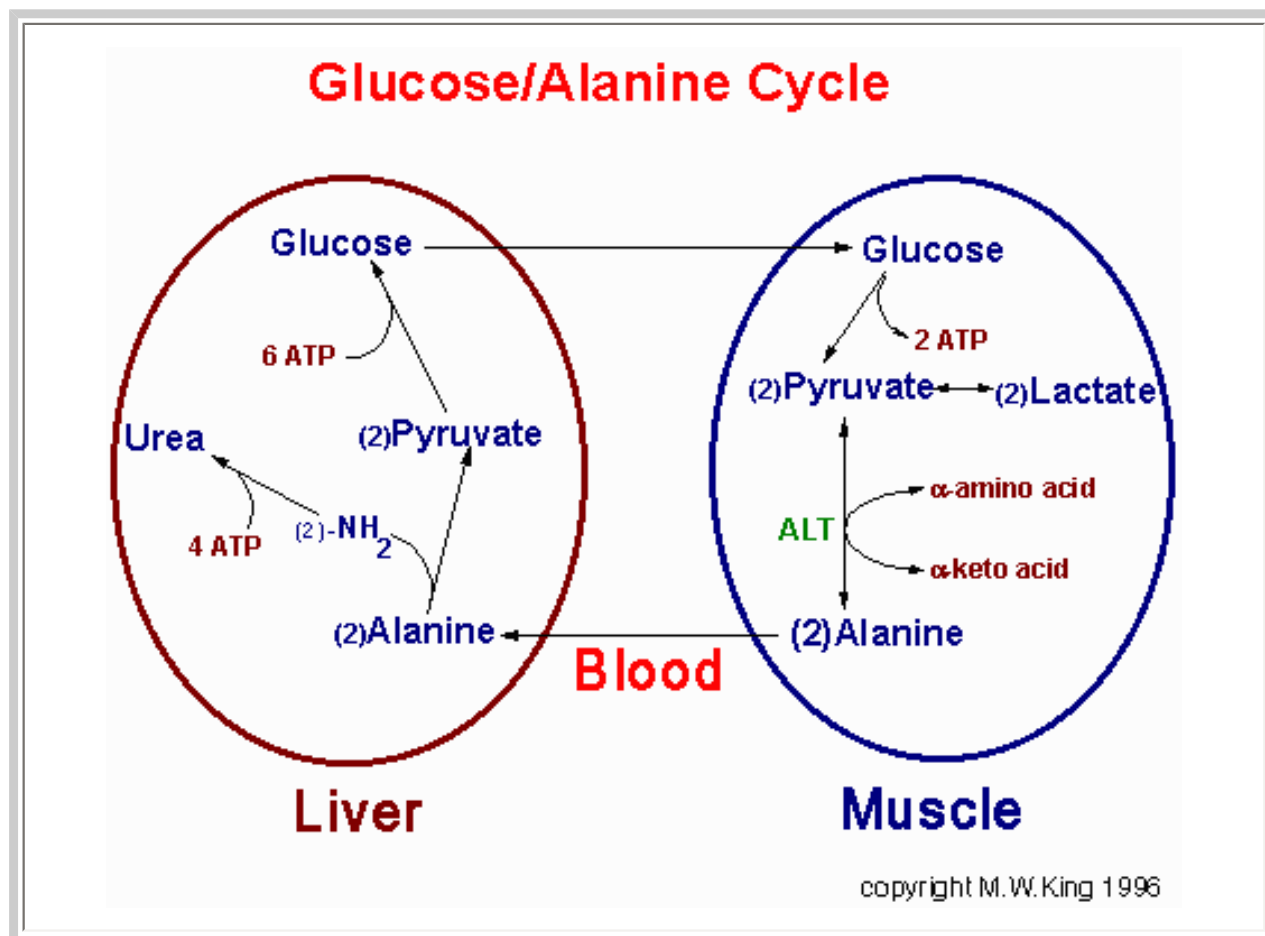
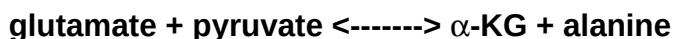
Aspartate is also derived from asparagine through the action of asparaginase. The importance of glutamate as a common intracellular amino donor for transamination reactions and of aspartate as a precursor of ornithine for the [urea cycle](#) is described in the [Nitrogen Metabolism page](#).

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## Alanine and the Glucose-Alanine Cycle

Aside from its role in protein synthesis, alanine is second only to glutamine in prominence as a circulating amino acid. In this capacity it serves a unique role in the transfer of nitrogen from peripheral tissue to the liver. Alanine is transferred to the circulation by many tissues, but mainly by muscle, in which alanine is formed from pyruvate at a rate proportional to intracellular pyruvate levels. Liver accumulates plasma alanine, reverses the transamination that occurs in muscle, and proportionately increases urea production. The pyruvate is either oxidized or converted to glucose via [gluconeogenesis](#). When alanine transfer from muscle to liver is coupled with glucose transport from liver back to muscle, the process is known as the **glucose-alanine cycle**. The key feature of the cycle is that in 1 molecule, alanine, peripheral tissue exports pyruvate and ammonia (which are potentially rate-limiting for metabolism) to the liver, where the carbon skeleton is recycled and most nitrogen eliminated.

There are 2 main pathways to production of muscle alanine: directly from protein degradation, and via the transamination of pyruvate by alanine transaminase, ALT (also referred to as serum glutamate-pyruvate transaminase, SGPT).

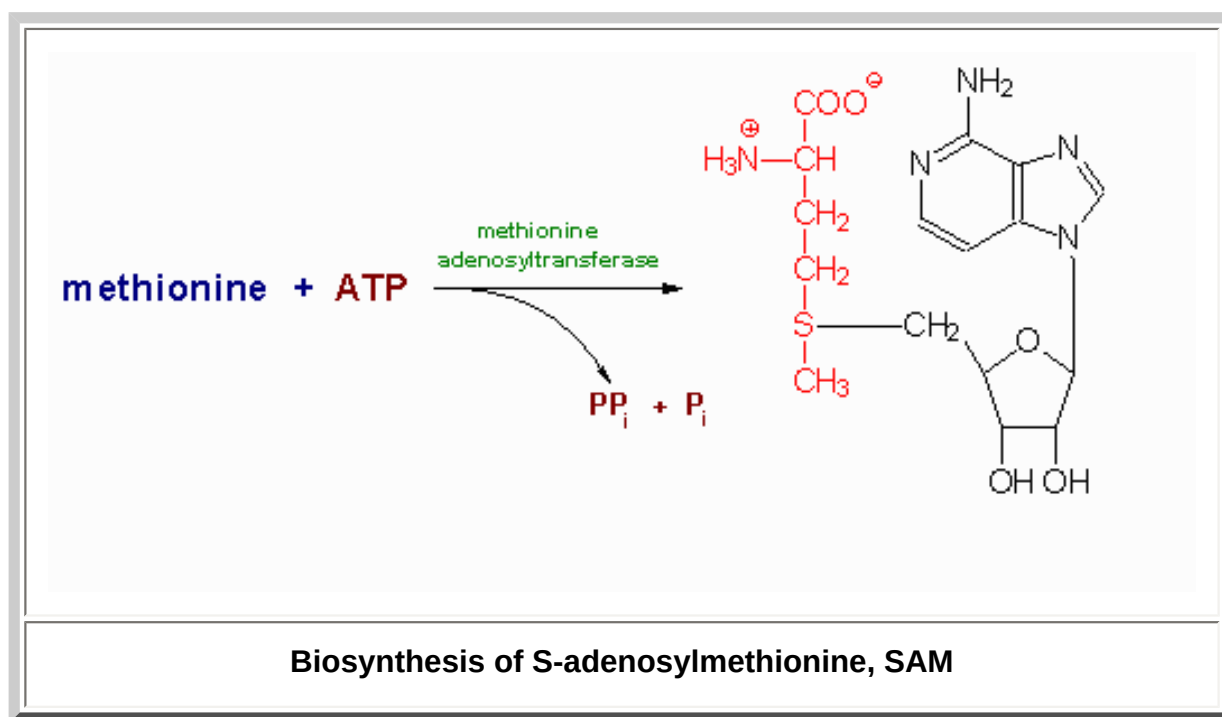


The glucose-alanine cycle is used primarily as a mechanism for skeletal muscle to eliminate nitrogen while replenishing its energy supply. Glucose oxidation produces pyruvate which can undergo transamination to alanine. This reaction is catalyzed by alanine transaminase, ALT (ALT used to be called serum glutamate-pyruvate transaminase, SGPT). Additionally, during periods of fasting, skeletal muscle protein is degraded for the energy value of the amino acid carbons and alanine is a major amino acid in protein. The alanine then enters the blood stream and is transported to the liver. Within the liver alanine is converted back to pyruvate which is then a source of carbon atoms for gluconeogenesis. The newly formed glucose can then enter the blood for delivery back to the muscle. The amino group transported from the muscle to the liver in the form of alanine is converted to urea in the urea cycle and excreted.

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## Cysteine Biosynthesis

The sulfur for cysteine synthesis comes from the essential amino acid methionine. A condensation of ATP and methionine catalyzed by methionine adenosyltransferase yields **S-adenosylmethionine (SAM or AdoMet)**.

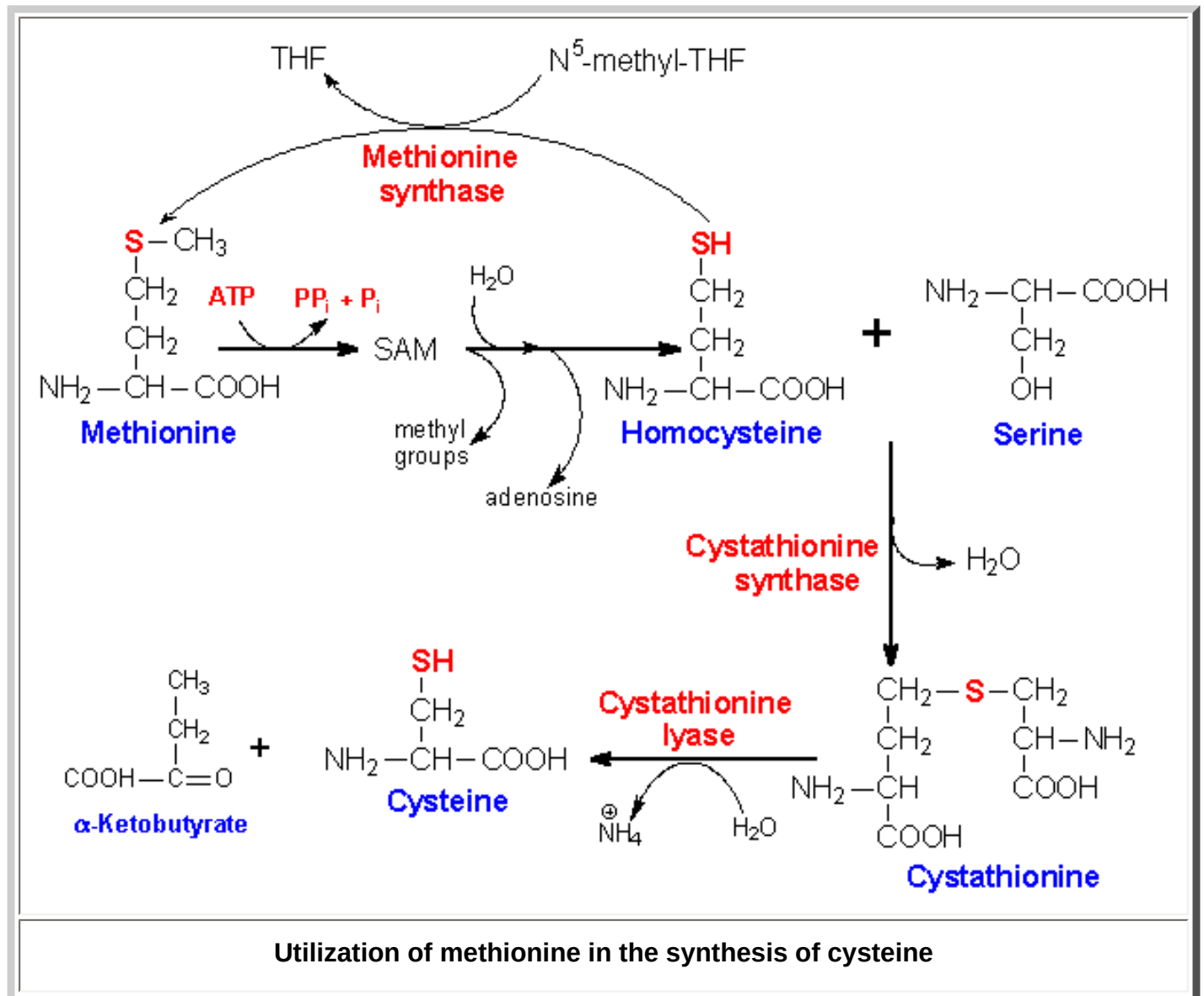


SAM serves as a precursor for numerous methyl transfer reactions (e.g. the conversion of norepinephrine to epinephrine, see [Specialized Products of Amino Acids](#)). The result of methyl transfer is the conversion of SAM to S-adenosylhomocysteine. S-adenosylhomocysteine is then cleaved by adenosylhomocysteinease to yield homocysteine and adenosine. Homocysteine can be converted back to methionine by methionine synthase, a reaction that occurs under methionine-sparing conditions and requires *N*<sup>5</sup>-methyl-tetrahydrofolate as methyl donor. This reaction was discussed in the context of vitamin B<sub>12</sub>-requiring enzymes in the [Vitamins](#) page.

Transmethylation reactions employing SAM are extremely important, but in this case the role of S-adenosylmethionine in transmethylation is secondary to the production of homocysteine (essentially a by-product of transmethylation activity). In the production of SAM all phosphates of an ATP are lost: one as P<sub>i</sub> and two as PP<sub>i</sub>. It is adenosine which is transferred to methionine and not AMP.

In cysteine synthesis, homocysteine condenses with serine to produce cystathionine, which is subsequently cleaved by cystathionase to produce cysteine and  $\alpha$ -ketobutyrate. The sum of the latter two reactions is known as **trans-sulfuration**.

Cysteine is used for protein synthesis and other body needs, while the  $\alpha$ -ketobutyrate is decarboxylated and converted to propionyl-CoA. While cysteine readily oxidizes in air to form the disulfide cystine, cells contain little if any free cystine because the ubiquitous reducing agent, **glutathione** effectively reverses the formation of cystine by a non-enzymatic reduction reaction.



The 2 key enzymes of this pathway, cystathionine synthase and cystathionase (cystathionine lyase), both use pyridoxal phosphate as a cofactor, and both are under regulatory control. Cystathionase is under negative allosteric control by cysteine, as well, cysteine inhibits the expression of the cystathionine synthase gene.

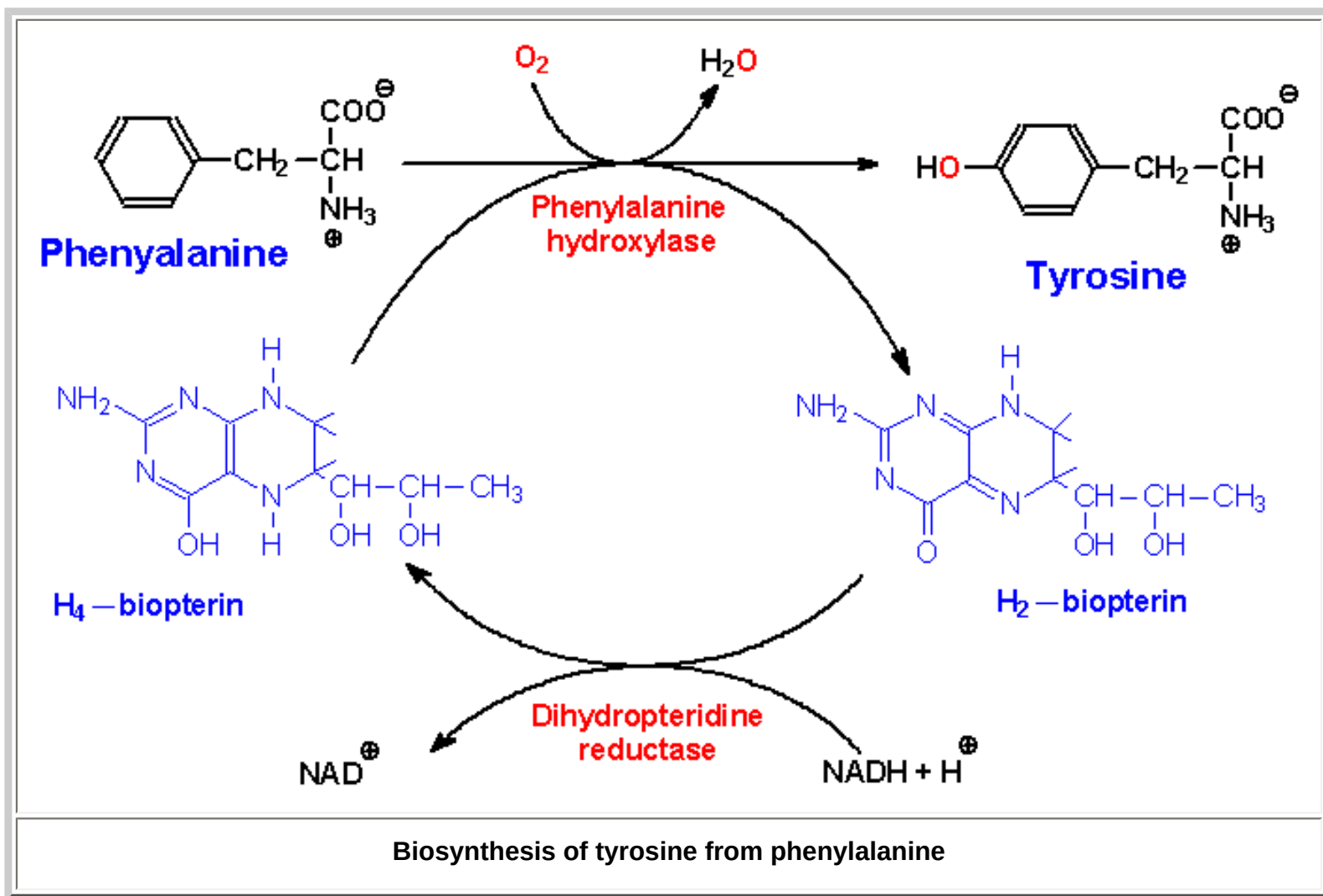
Genetic defects are known for both the synthase and the lyase. Missing or impaired cystathionine synthase leads to **homocystinuria** and is often associated with mental retardation, although the complete syndrome is multifaceted and many individuals with this disease are mentally normal. Some instances of genetic homocystinuria respond favorably to pyridoxine therapy, suggesting that in these cases the defect in cystathionine synthase is a decreased affinity for the cofactor. Missing or impaired cystathionase leads to excretion of cystathionine in the urine but does not have any other untoward effects. Rare cases are known in which cystathionase is defective and operates at a low level. This genetic disease leads to methioninuria with no other consequences.

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## Tyrosine Biosynthesis

Tyrosine is produced in cells by hydroxylating the essential amino acid phenylalanine. This relationship is much like that between cysteine and methionine. Half of the phenylalanine required goes into the production of tyrosine; if the diet is rich in tyrosine itself, the requirements for phenylalanine are reduced by about 50%.

Phenylalanine hydroxylase is a mixed-function oxygenase: one atom of oxygen is incorporated into water and the other into the hydroxyl of tyrosine. The reductant is the tetrahydrofolate-related cofactor **tetrahydrobiopterin**, which is maintained in the reduced state by the NADH-dependent enzyme dihydropteridine reductase (DHPR).



Missing or deficient phenylalanine hydroxylase results in **hyperphenylalaninemia**. Hyperphenylalaninemia is defined as a plasma phenylalanine concentration greater than 2mg/dL (120 $\mu$ M). The most widely recognized hyperphenylalaninemia (and most severe) is the genetic disease known as **phenylketonuria (PKU)**. Patients suffering from PKU have plasma phenylalanine levels >1000 $\mu$ M, whereas the non-PKU hyperphenylalaninemias exhibit levels of plasma phenylalanine <1000 $\mu$ M. Untreated PKU leads to severe mental retardation. The mental retardation is caused by the accumulation of phenylalanine, which becomes a major donor of amino groups in aminotransferase activity and depletes neural tissue of  $\alpha$ -ketoglutarate. This absence of  $\alpha$ -ketoglutarate in the brain shuts down the **TCA cycle** and the associated production of aerobic energy, which is essential to normal brain development.

The product of phenylalanine transamination, phenylpyruvic acid, is reduced to phenylacetate and phenyllactate, and all 3 compounds appear in the urine. The presence of phenylacetate in the urine imparts a "mousy" odor. If the problem is diagnosed early, the addition of tyrosine and restriction of phenylalanine from the diet can minimize the extent of mental retardation.

Because of the requirement for tetrahydrobiopterin in the function of phenylalanine hydroxylase, deficiencies in DHPR can manifest with hyperphenylalaninemia. However, since tetrahydrobiopterin is a cofactor in several other enzyme catalyzed reactions (e.g. see the synthesis of the tyrosine- and tryptophan-derived neurotransmitters as well as nitric oxide in [Specialized Products of Amino Acids](#)), the effects of missing or defective DHPR cause even more severe neurological difficulties than those usually associated with PKU caused by deficient phenylalanine hydroxylase activity.

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## Ornithine and Proline Biosynthesis

Glutamate is the precursor of both proline and ornithine, with glutamate semialdehyde being a branch point intermediate leading to one or the other of these 2 products. While ornithine is not one of the 20 amino acids used in protein synthesis, it plays a significant role as the acceptor of carbamoyl phosphate in the [urea cycle](#). Ornithine serves an additional important role as the precursor for the synthesis of the [polyamines](#). The production of ornithine from glutamate is important when dietary arginine, the other principal source of ornithine, is limited.

The fate of glutamate semialdehyde depends on prevailing cellular conditions. Ornithine production occurs from the semialdehyde via a simple glutamate-dependent transamination, producing ornithine. When arginine concentrations become elevated, the ornithine contributed from the [urea cycle](#) plus that from glutamate semialdehyde inhibit the aminotransferase reaction, with accumulation of the semialdehyde as a result. The semialdehyde cyclizes spontaneously to  $\Delta^1$ pyrroline-5-carboxylate which is then reduced to proline by an NADPH-dependent reductase.

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## Serine Biosynthesis

The main pathway to serine starts with the glycolytic intermediate 3-phosphoglycerate. An NADH-linked dehydrogenase converts 3-phosphoglycerate into a keto acid, 3-phosphopyruvate, suitable for subsequent transamination. Aminotransferase activity with glutamate as a donor produces 3-phosphoserine, which is converted to serine by phosphoserine phosphatase.

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## Glycine Biosynthesis

The main pathway to glycine is a 1-step reaction catalyzed by serine hydroxymethyltransferase. This reaction involves the transfer of the hydroxymethyl group from serine to the cofactor tetrahydrofolate (THF), producing glycine and  $N^5,N^{10}$ -methylene-THF. Glycine produced from serine or from the diet can also be oxidized by glycine cleavage complex, GCC, to yield a second equivalent of  $N^5,N^{10}$ -methylene-tetrahydrofolate as well as ammonia and  $\text{CO}_2$ .

Glycine is involved in many anabolic reactions other than protein synthesis including the synthesis of [purine nucleotides](#), [heme](#), [glutathione](#), [creatine](#) and serine.

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## Aspartate/Asparagine and Glutamate/Glutamine Biosynthesis

Glutamate is synthesized by the reductive amination of  $\alpha$ -ketoglutarate catalyzed by glutamate dehydrogenase; it is thus a nitrogen-fixing reaction. In addition, glutamate arises by aminotransferase reactions, with the amino nitrogen being donated by a number of different amino acids. Thus, glutamate is a general collector of amino nitrogen.

Aspartate is formed in a transamination reaction catalyzed by aspartate transaminase, AST. This reaction uses the aspartate  $\alpha$ -keto acid analog, oxaloacetate, and glutamate as the amino donor. Aspartate can also be formed by deamination of asparagine catalyzed by asparaginase.

Asparagine synthetase and glutamine synthetase, catalyze the production of asparagine and glutamine from their respective  $\alpha$ -amino acids. Glutamine is produced from glutamate by the direct incorporation of ammonia; and this can be considered another nitrogen fixing reaction. Asparagine, however, is formed by an amidotransferase reaction.

Aminotransferase reactions are readily reversible. The direction of any individual transamination depends principally on the concentration ratio of reactants and products. By contrast, transamidation reactions, which are dependent on ATP, are considered irreversible. As a consequence, the degradation of asparagine and glutamine take place by a hydrolytic pathway rather than by a reversal of the pathway by which they were formed. As indicated above, asparagine can be degraded to aspartate.

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## Amino Acid Catabolism

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### Glutamine/Glutamate and Asparagine/Aspartate Catabolism

Glutaminase is an important kidney tubule enzyme involved in converting glutamine (from liver and from other tissue) to glutamate and  $\text{NH}_3^+$ , with the  $\text{NH}_3^+$  being excreted in the urine. Glutaminase activity is present in many other tissues as well, although its activity is not nearly as prominent as in the kidney. The glutamate produced from glutamine is converted to  $\alpha$ -ketoglutarate, making glutamine a glucogenic amino acid.

Asparaginase is also widely distributed within the body, where it converts asparagine into ammonia and aspartate. Aspartate transaminates to oxaloacetate, which follows the gluconeogenic pathway to glucose.

Glutamate and aspartate are important in collecting and eliminating amino nitrogen via glutamine synthetase and the [urea cycle](#), respectively. The catabolic path of the carbon skeletons involves simple 1-step aminotransferase reactions that directly produce net quantities of a [TCA cycle](#) intermediate. The glutamate dehydrogenase reaction operating in the direction of  $\alpha$ -ketoglutarate production provides a second avenue leading from glutamate to gluconeogenesis.

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### Alanine Catabolism

Alanine is also important in intertissue nitrogen transport as part of the [glucose-alanine cycle](#). Alanine's catabolic pathway involves a simple aminotransferase reaction that directly produces pyruvate. Generally pyruvate produced by this pathway will result in the formation of oxaloacetate, although when the energy charge of a cell is low the pyruvate will be oxidized to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  via the PDH complex and the [TCA cycle](#). This makes alanine a glucogenic amino acid.

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### Arginine, Ornithine and Proline Catabolism

The catabolism of arginine begins within the context of the [urea cycle](#). It is hydrolyzed to urea and ornithine



by arginase.

Ornithine, in excess of [urea cycle](#) needs, is transaminated to form glutamate semialdehyde. Glutamate semialdehyde can serve as the precursor for proline biosynthesis as described above or it can be converted to glutamate.

Proline catabolism is a reversal of its synthesis process.

The glutamate semialdehyde generated from ornithine and proline catabolism is oxidized to glutamate by an ATP-independent glutamate semialdehyde dehydrogenase. The glutamate can then be converted to  $\alpha$ -ketoglutarate in a transamination reaction. Thus arginine, ornithine and proline, are glucogenic.

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## Serine Catabolism

The conversion of serine to glycine and then glycine oxidation to  $\text{CO}_2$  and  $\text{NH}_3$ , with the production of two equivalents of  $N^5,N^{10}$ -methyleneTHF, was described above. Serine can be catabolized back to the glycolytic intermediate, 3-phosphoglycerate, by a pathway that is essentially a reversal of serine biosynthesis. However, the enzymes are different. Serine can also be converted to pyruvate through a deamination reaction catalyzed by serine/threonine dehydratase.

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## Threonine Catabolism

There are at least 3 pathways for threonine catabolism. One involves a pathway initiated by threonine dehydrogenase yielding  $\alpha$ -amino- $\beta$ -ketobutyrate. The  $\alpha$ -amino- $\beta$ -ketobutyrate is either converted to acetyl-CoA and glycine or spontaneously degrades to aminoacetone which is converted to pyruvate. The second pathway involves serine/threonine dehydratase yielding  $\alpha$ -ketobutyrate which is further catabolized to propionyl-CoA and finally the [TCA cycle](#) intermediate, succinyl-CoA. The third pathway utilizes threonine aldolase. The products of this reaction are both ketogenic (acetyl-CoA) and glucogenic (pyruvate).

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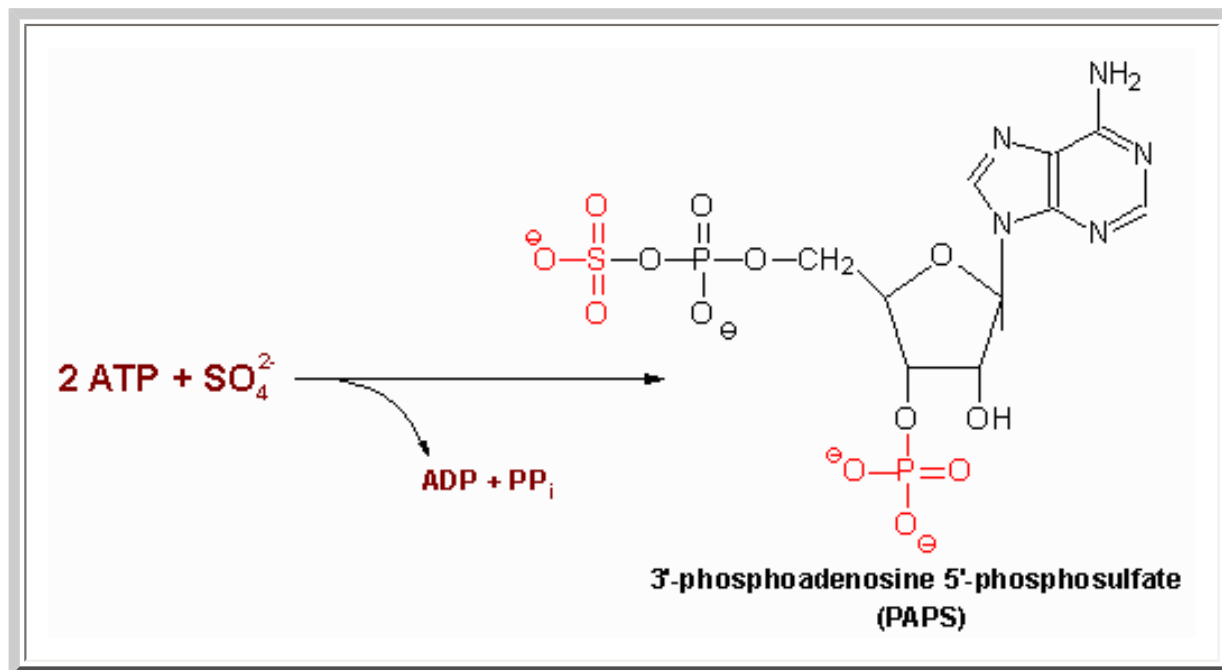
## Glycine Catabolism

Glycine is classified as a glucogenic amino acid, since it can be converted to serine by serine hydroxymethyltransferase, and serine can be converted back to the glycolytic intermediate, 3-phosphoglycerate or to pyruvate by serine/threonine dehydratase. Nevertheless, the main glycine catabolic pathway leads to the production of  $\text{CO}_2$ , ammonia, and one equivalent of  $N^5,N^{10}$ -methyleneTHF by the mitochondrial glycine cleavage complex.

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## Cysteine Catabolism

There are several pathways for cysteine catabolism. The simplest, but least important pathway is catalyzed by a liver desulfurase and produces hydrogen sulfide, ( $\text{H}_2\text{S}$ ) and pyruvate. The major catabolic pathway in animals is via cysteine dioxygenase that oxidizes the cysteine sulfhydryl to sulfinic acid, producing the intermediate **cysteinesulfinic acid**. Cysteinesulfinic acid can serve as a biosynthetic intermediate undergoing decarboxylation and oxidation to produce **taurine**. Catabolism of cysteinesulfinic acid proceeds through transamination to  $\beta$ -sulfinylpyruvate which then undergoes desulfuration yielding bisulfite, ( $\text{HSO}_3^-$ ) and the glucogenic product, pyruvate. The enzyme sulfite oxidase uses  $\text{O}_2$  and  $\text{H}_2\text{O}$  to convert  $\text{HSO}_3^-$  to sulfate, ( $\text{SO}_4^{2-}$ ) and  $\text{H}_2\text{O}_2$ . The resultant sulfate is used as a precursor for the formation of **3'-phosphoadenosine-5'-phosphosulfate, PAPS**.



PAPS is used for the transfer of sulfate to biological molecules such as the sugars of the [glycosphingolipids](#).

Other than protein, the most important product of cysteine metabolism is the bile salt precursor **taurine**, which is used to form the bile acid conjugates **taurocholate** and **taurochenodeoxycholate**.

The enzyme cystathionase can also transfer the sulfur from one cysteine to another generating thiocysteine and pyruvate. Transamination of cysteine yields  $\beta$ -mercaptopyruvate which then reacts with sulfite, ( $\text{SO}_3^{2-}$ ), to produce thiosulfate, ( $\text{S}_2\text{O}_3^{2-}$ ) and pyruvate. Both thiocysteine and thiosulfate can be used by the enzyme rhodanese to incorporate sulfur into cyanide, ( $\text{CN}^-$ ), thereby detoxifying the cyanide to thiocyanate.

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## Methionine Catabolism

The principal fates of the essential amino acid methionine are incorporation into polypeptide chains, and use in the production of  $\alpha$ -ketobutyrate and cysteine via SAM as described above. The transulfuration reactions that produce cysteine from homocysteine and serine also produce  $\alpha$ -ketobutyrate, the latter being converted to succinyl-CoA.

Regulation of the methionine metabolic pathway is based on the availability of methionine and cysteine. If both amino acids are present in adequate quantities, SAM accumulates and is a positive effector on cystathionine synthase, encouraging the production of cysteine and  $\alpha$ -ketobutyrate (both of which are glucogenic). However, if methionine is scarce, SAM will form only in small quantities, thus limiting cystathionine synthase activity. Under these conditions accumulated homocysteine is remethylated to methionine, using  $N^5$ -methylTHF and other compounds as methyl donors.

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## Valine, Leucine and Isoleucine Catabolism

This group of essential amino acids are identified as the **branched-chain amino acids, BCAAs**. Because this arrangement of carbon atoms cannot be made by humans, these amino acids are an essential element in the diet. The catabolism of all three compounds initiates in muscle and yields NADH and  $\text{FADH}_2$  which can be utilized

for ATP generation. The catabolism of all three of these amino acids uses the same enzymes in the first two steps. The first step in each case is a transamination using a single BCAA aminotransferase, with  $\alpha$ -ketoglutarate as amine acceptor. As a result, three different  $\alpha$ -keto acids are produced and are oxidized using a common branched-chain  $\alpha$ -keto acid dehydrogenase, yielding the three different CoA derivatives. Subsequently the metabolic pathways diverge, producing many intermediates.

The principal product from valine is propionylCoA, the glucogenic precursor of succinyl-CoA. Isoleucine catabolism terminates with production of acetylCoA and propionylCoA; thus isoleucine is both glucogenic and ketogenic. Leucine gives rise to acetylCoA and acetoacetylCoA, and is thus classified as strictly ketogenic.

There are a number of genetic diseases associated with faulty catabolism of the BCAAs. The most common defect is in the branched-chain  $\alpha$ -keto acid dehydrogenase. Since there is only one dehydrogenase enzyme for all three amino acids, all three  $\alpha$ -keto acids accumulate and are excreted in the urine. The disease is known as [Maple syrup urine disease](#) because of the characteristic odor of the urine in afflicted individuals. Mental retardation in these cases is extensive. Unfortunately, since these are essential amino acids, they cannot be heavily restricted in the diet; ultimately, the life of afflicted individuals is short and development is abnormal. The main neurological problems are due to poor formation of myelin in the CNS.

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## Phenylalanine and Tyrosine Catabolism

Phenylalanine normally has only two fates: incorporation into polypeptide chains, and production of tyrosine via the tetrahydrobiopterin-requiring phenylalanine hydroxylase. Thus, phenylalanine catabolism always follows the pathway of tyrosine catabolism. The main pathway for tyrosine degradation involves conversion to fumarate and acetoacetate, allowing phenylalanine and tyrosine to be classified as both glucogenic and ketogenic.

Tyrosine is equally important for protein biosynthesis as well as an intermediate in the biosynthesis of several physiologically important metabolites e.g. dopamine, norepinephrine and epinephrine (see [Specialized Products of Amino Acids](#)).

As in phenylketonuria (deficiency of phenylalanine hydroxylase), deficiency of tyrosine aminotransferase (TAT) leads to hypertyrosinemia and the urinary excretion of tyrosine and the catabolic intermediates between phenylalanine and tyrosine. The adverse neurological symptoms are similar for phenylalanine hydroxylase and TAT deficiencies. In addition, hypertyrosinemia leads to painful corneal eruptions and photophobia.

The first genetic disease ever recognized, [alcaptonuria](#), was demonstrated to be the result of a defect in phenylalanine and tyrosine catabolism. Alcaptonuria is caused by defective homogentisic acid oxidase. Homogentisic acid accumulation is relatively innocuous, causing urine to darken on exposure to air, but no life-threatening effects accompany the disease.

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## Lysine Catabolism

Lysine catabolism is unusual in the way that the  $\epsilon$ -amino group is transferred to  $\alpha$ -ketoglutarate and into the general nitrogen pool. The reaction is a transamination in which the  $\epsilon$ -amino group is transferred to the  $\alpha$ -keto carbon of  $\alpha$ -ketoglutarate forming the metabolite, **saccharopine**. Unlike the majority of transamination reactions, this one does not employ pyridoxal phosphate as a cofactor. Saccharopine is immediately hydrolyzed by the enzyme  $\alpha$ -aminoadipic semialdehyde synthase in such a way that the amino nitrogen remains with the  $\alpha$ -carbon of  $\alpha$ -ketoglutarate, producing glutamate and  $\alpha$ -aminoadipic semialdehyde. Because this transamination reaction is not reversible, lysine is an essential amino acid. The ultimate end-product of lysine catabolism is acetoacetyl-CoA.

Genetic deficiencies in the enzyme  $\alpha$ -aminoadipic semialdehyde synthase have been observed in individuals who excrete large quantities of urinary lysine and some saccharopine. The lysinemia and associated lysinuria are benign. Other serious disorders associated with lysine metabolism are due to failure of the transport

system for lysine and the other dibasic amino acids across the intestinal wall. Lysine is essential for protein synthesis; a deficiencies of its transport into the body can cause seriously diminished levels of protein synthesis. Probably more significant however, is the fact that arginine is transported on the same dibasic amino acid carrier, and resulting arginine deficiencies limit the quantity of ornithine available for the [urea cycle](#). The result is severe hyperammonemia after a meal rich in protein. The addition of citrulline to the diet prevents the hyperammonemia.

Lysine is also important as a precursor for the synthesis of **carnitine**, required for the transport of fatty acids into the mitochondria for oxidation. Free lysine does not serve as the precursor for this reaction, rather the modified lysine found in certain proteins. Some proteins modify lysine to **trimethyllysine** using SAM as the methyl donor to transfer methyl groups to the  $\epsilon$ -amino of the lysine side chain. Hydrolysis of proteins containing trimethyllysine provide the substrate for the subsequent conversion to carnitine

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## Histidine Catabolism

Histidine catabolism begins with release of the  $\alpha$ -amino group catalyzed by histidase, introducing a double bond into the molecule. As a result, the deaminated product, urocanate, is not the usual  $\alpha$ -keto acid associated with loss of  $\alpha$ -amino nitrogens. The end product of histidine catabolism is glutamate, making histidine one of the glucogenic amino acids.

Another key feature of histidine catabolism is that it serves as a source of ring nitrogen to combine with tetrahydrofolate (THF), producing the 1-carbon THF intermediate known as  $N^5$ -formiminoTHF. The latter reaction is one of two routes to  $N^5$ -formiminoTHF.

The principal genetic deficiency associated with histidine metabolism is absence or deficiency of the first enzyme of the pathway, histidase. The resultant histidinemia is relatively benign. The disease, which is of relatively high incidence (1 in 10,000), is most easily detected by the absence of urocanate from skin and sweat, where it is normally found in relative abundance.

Decarboxylation of histidine in the intestine by bacteria gives rise to **histamine**. Similarly, histamine arises in many tissues by the decarboxylation of histidine, which in excess causes constriction or dilation of various blood vessels. The general symptoms are those of asthma and various allergic reactions.

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## Tryptophan Catabolism

A number of important side reactions occur during the catabolism of tryptophan on the pathway to acetoacetate. The first enzyme of the catabolic pathway is an iron porphyrin oxygenase that opens the indole ring. The latter enzyme is highly inducible, its concentration rising almost 10-fold on a diet high in tryptophan.

**Kynurenine** is the first key branch point intermediate in the pathway. Kynurenine undergoes deamination in a standard transamination reaction yielding **kynurenic acid**. Kynurenic acid and metabolites have been shown to act as antiexcitotoxics and anticonvulsives.

A second side branch reaction produces anthranilic acid plus alanine. Another equivalent of alanine is produced further along the main catabolic pathway, and it is the production of these alanine residues that allows tryptophan to be classified among the glucogenic and ketogenic amino acids.

The second important branch point converts kynurenine into 2-amino-3-carboxymuconic semialdehyde, which has two fates. The main flow of carbon elements from this intermediate is to glutarate. An important side reaction in liver is a transamination and several rearrangements to produce limited amounts of **nicotinic acid**, which leads to production of a small amount of  $NAD^+$  and  $NADP^+$

Aside from its role as an amino acid in protein biosynthesis, tryptophan also serves as a precursor for the synthesis of **serotonin** and **melatonin**. These products are discussed in [Specialized Products of Amino Acids](#)

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# Defects in Amino Acid Metabolism

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- [Phenylketonuria](#)
- [Type I Tyrosinemia](#) - Tyrosinosis
- [Type II Tyrosinemia](#) - Richner-Hanhart Syndrome
- [Type III Tyrosinemia](#)
- [Alcaptonuria](#)
- [Homocystinuria](#)
- [Histidinemia](#)
- [Maple Syrup Urine Disease, MSUD](#)
  - [MSUD Type Ib](#)
  - [MSUD Type II](#)
- [Methylmalonic Aciduria](#)
- [Non-ketonic Hyperglycinemia Type I \(NKHI\)](#)
- [Hyperlysinemia](#)

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# Defects in Amino Acid Transport

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- [Cystinuria Type I](#)
  - [Cystinuria Type III](#)
  - [Hartnup Disease](#)
  - [Hyperammonemia-hyperornithinemia-homocitrullinuria \(HHH\) Syndrome](#)
- 

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# The Porphyrrias

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- Congenital Erythropoietic Porphyrria (CEP)
- Erythropoietic Protoporphyrria (EPP)
- ALA Dehydratase Deficiency Porphyrria (ADP)
- Acute Intermittent Porphyrria (AIP)
- Hereditary Coproporphyrria (HCP)
- Variegate Porphyrria (VP)
- Porphyrria Cutanea Tarda (PCT)
- Hepatoerythropoietic Porphyrria (HEP)

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# Disorders in Bilirubin Metabolism

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- [Gilbert Syndrome](#)
- [Crigler-Najjar Syndrome Type I](#)
- [Crigler-Najjar Syndrome Type II](#)
- [Dubin-Johnson Syndrome](#)
- [Rotor Syndrome](#)

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## Disorders in Mitochondrial $\beta$ -Oxidation

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- [Very-Long-Chain Acyl-CoA Dehydrogenase Deficiency](#) - VLCAD
  - [Long-Chain Acyl-CoA Dehydrogenase Deficiency](#) - LCAD
  - [Medium-Chain Acyl-CoA Dehydrogenase Deficiency](#) - MCAD
  - [Short-Chain Acyl-CoA Dehydrogenase Deficiency](#) - SCAD
  
  - [Carnitine Translocase Deficiency](#)
  - [Carnitine Palmitoyltransferase I \(CPT I\) Deficiency](#)
  - [Carnitine Palmitoyltransferase II \(CPT II\) Deficiency](#)
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# Defects in Purine Nucleotide Metabolism

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- [Lesch-Nyhan Syndrome](#)
  - [Severe Combined Immunodeficiency Disease \(SCID\)](#) - due to adenosine deaminase (ADA) deficiency
  - [Gout](#)
  - [Renal Lithiasis](#) - due to adenine phosphoribosyltransferase (APRT) deficiency
  - [Xanthinuria](#) - due to xanthine oxidase deficiency
- 

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# Defects in Pyrimidine Nucleotide Metabolism

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- [Type I Orotic Aciduria](#)
  - [Type II Orotic Aciduria](#)
  - [Ornithine Transcarbamoylase Deficiency](#)
- 

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# Disorders in Metal Metabolism and Transport

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- [Wilson Disease](#)
- [Menkes Disease](#)
- [Occipital Horn Syndrome](#)
- [Hemochromatosis](#)

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# Disorders in Peroxisomes

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- [Zellweger Syndrome](#)
  - [X-Linked Adreoleukodystrophy](#)
  - [Neonatal Adrenoleukodystrophy \(NALD\)](#)
  - [Rhizomelic Chondrodysplasia Punctata \(RCDP\)](#)
  - [Infantile Refsum's Disease \(IRD\)](#)
- 

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# Disorders Due to Defective DNA Repair

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- [Ataxia telangiectasia \(AT\)](#)
- [Xeroderma pigmentosum \(XP\)](#)
- [Cockayne Syndrome](#)
- [Bloom Syndrome](#)
- [Fanconi anemia](#)

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# Medical Biochemistry Course Guide

## Indiana University School of Medicine

### Terre Haute Center for Medical Education

## Fall 2005 Course Information

**COURSE DIRECTOR:** Michael W. King, Ph.D.

**Recommended Text:**

**Biochemistry: with Clinical  
Correlations**

by Devlin 5th ed. Wiley-Liss

A good companion text for the Cell and Molecular Biology material encountered in Medical Biochemistry, Histology and Physiology, is: **Molecular Cell Biology**, Lodish et al., 5th ed. Freeman, 2004

## Grading Methods

|              |           |
|--------------|-----------|
| H = Honors   | 93 - 100% |
| HP = Hi Pass | 83 -- 92% |
| P = Pass     | 72 -- 82% |
| F = Fail     | Below 72% |

|                                      |                            |
|--------------------------------------|----------------------------|
| 4 Hourly Exams                       | Value = 55% of Final Grade |
| PBL and Class Participation          | Value = 10% of Final Grade |
| PBL Exam Questions (on hourly tests) | Value = 10% of Final Grade |

Comprehensive Final Exam

Value = 25% of  
Final Grade

% of Final Grade Values Remains Constant

Total = 100%

## Lecture and Exam Schedule

## Core Subjects for Medical Biochemistry



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# Medical Biochemistry Lecture Schedule, 2005

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| Day - Date    | Time and Subject Content                                                                                       | Devlin Readings required           | Alberts Readings optional | Instructor   |
|---------------|----------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------|--------------|
| <b>WEEK 1</b> |                                                                                                                |                                    |                           |              |
| Mon Aug 15    | Registration - Orientation                                                                                     |                                    |                           |              |
| Tues Aug 16   | <b>No Biochemistry</b>                                                                                         |                                    |                           |              |
| Wed Aug 17    | 1:00 <a href="#">Protein Structure</a> and Disease                                                             | Chapter 9                          | Chapter 3                 | King         |
| Thurs Aug 18  | <b>1:00 – 4:00 PBL #1</b>                                                                                      |                                    |                           |              |
| Fri Aug 19    | 1:00 <a href="#">Hemoglobin and Myoglobin</a>                                                                  | Chapter 9                          | Chapter 3 and 7           | King         |
| <b>WEEK 2</b> |                                                                                                                |                                    |                           |              |
| Mon Aug 22    | 1:00 <a href="#">Enzyme Kinetics</a><br>2:00 <a href="#">Enzyme Kinetics</a>                                   |                                    | Chapter 2<br>Chapter 2    | King<br>King |
| Tues Aug 23   | <b>No Biochemistry</b>                                                                                         |                                    |                           |              |
| Wed Aug 24    | 1:00 <a href="#">DNA Synthesis</a><br>2:00 <a href="#">DNA Modifications</a>                                   | Chapter 2 and 4<br>Chapter 2 and 4 | Chapter 2<br>Chapter 5    | King<br>King |
| Thurs Aug 25  | <b>1:00 – 4:00 PBL #1</b>                                                                                      |                                    |                           |              |
| Fri Aug 26    | 1:00 <a href="#">RNA Structure and Synthesis</a>                                                               | Chapter 2 and 5                    | Chapter 6                 | King         |
| <b>WEEK 3</b> |                                                                                                                |                                    |                           |              |
| Mon Aug 29    | 1:00 <a href="#">Protein Synthesis</a>                                                                         | Chapter 6                          | Chapter 6                 | King         |
| Tues Aug 30   | <b>No Biochemistry</b>                                                                                         |                                    |                           |              |
| Wed Aug 31    | 1:00 <a href="#">Protein Synthesis</a><br>2:00 Review for Exam 1                                               | Chapter 6                          | Chapter 6                 | King<br>King |
| Thurs Sept 1  | <b>1:00 EXAM 1</b> (lectures through Sept 1)                                                                   |                                    |                           |              |
| Fri Sept 2    | 1:00 <a href="#">Protein Processing/Modifications</a><br>2:00 <a href="#">Protein Processing/Modifications</a> | Chapter 6<br>Chapter 6             | Chapter 6<br>Chapter 6    | King<br>King |
| <b>WEEK 4</b> |                                                                                                                |                                    |                           |              |
| Mon Sept 5    | <b>LABOR DAY NO CLASSES</b>                                                                                    |                                    |                           |              |
| Tues Sept 6   | <b>No Biochemistry</b>                                                                                         |                                    |                           |              |
| Wed Sept 7    | 1:00 <a href="#">Protein Processing/Modifications</a><br>2:00 Exam #1 Discussion                               | Chapter 6                          | Chapter 6                 | King         |
| Thurs Sept 8  | <b>1:00 – 4:00 PBL #2</b>                                                                                      |                                    |                           |              |
| Fri Sept 9    | 1:00 Cell Cycle and Control                                                                                    | Chapter 4                          | Chapter 17 and 18         | King         |
| <b>WEEK 5</b> |                                                                                                                |                                    |                           |              |
| Mon Sept 12   | 1:00 <a href="#">Control of Gene Expression</a><br>2:00 <a href="#">Control of Gene Expression</a>             | Chapter 8<br>Chapter 8             | Chapter 7<br>Chapter 7    | King<br>King |
| Tues Sept 13  | <b>No Biochemistry</b>                                                                                         |                                    |                           |              |



|               |                                                                   |                          |                        |              |
|---------------|-------------------------------------------------------------------|--------------------------|------------------------|--------------|
| Wed Sept 14   | 1:00 Glycolysis<br>2:00 Glycolysis                                | Chapter 14<br>Chapter 14 | Chapter 2<br>Chapter 2 | King<br>King |
| Thurs Sept 15 | 1:00 – 4:00 PBL #2                                                |                          |                        |              |
| Fri Sept 16   | 1:00 Gluconeogenesis<br>2:00 Pentose Phosphate Pathway            | Chapter 14<br>Chapter 15 | Chapter 2<br>Chapter 2 | King<br>King |
| WEEK 6        |                                                                   |                          |                        |              |
| Mon Sept 19   | 1:00 TCA Cycle                                                    | Chapter 13               | Chapter 2              | King         |
| Tues Sept 20  | No Biochemistry                                                   |                          |                        |              |
| Wed Sept 21   | 1:00 Oxidative Phosphorylation<br>2:30 Review for Exam #2         | Chapter 13               | Chapter 14             | King<br>King |
| Thurs Sept 22 | 1:00 EXAM 2 (lectures through Sept 22)                            |                          |                        |              |
| Fri Sept 23   | 1:00 Glycogen Metabolism<br>2:00 Glycogen Metabolism              | Chapter 14<br>Chapter 14 | Chapter 2<br>Chapter 2 | King<br>King |
| WEEK 7        |                                                                   |                          |                        |              |
| Mon Sept 26   | 1:00 Diabetes                                                     | Chapter 20               |                        | King         |
| Tues Sept 27  | No Biochemistry                                                   |                          |                        |              |
| Wed Sept 28   | 1:00 Cholesterol and Bile<br>2:00 Exam #2 Discussion              | Chapter 17               | Chapter 2              | King         |
| Thurs Sept 29 | 1:00 – 4:00 PBL #3                                                |                          |                        |              |
| Fri Sept 30   | 1:00 Fatty Acid Synthesis<br>2:00 Complex Lipid Synthesis         | Chapter 16               | Chapter 2<br>Chapter 2 | King<br>King |
| WEEK 8        |                                                                   |                          |                        |              |
| Mon Oct 3     | 1:00 Fatty Acid Oxidation                                         | Chapter 16               | Chapter 2              | King         |
| Tues Oct 5    | No Biochemistry                                                   |                          |                        |              |
| Wed Oct 5     | 1:00 Lipoproteins and Disease                                     | Chapter 17               |                        | King         |
| Thurs Oct 6   | 1:00 – 4:00 PBL #3                                                |                          |                        |              |
| Fri Oct 7     | 1:00 Amino Acid Metabolism<br>2:00 Special Amino Acid Derivatives | Chapter 18               | Chapter 2              | King         |
| WEEK 9        |                                                                   |                          |                        |              |
| Mon Oct 10    | Fall Break no classes                                             |                          |                        |              |
| Tues Oct 11   | Fall Break no classes                                             |                          |                        |              |
| Wed Oct 12    | Fall Break no classes                                             |                          |                        |              |
| Thurs Oct 13  | Fall Break no classes                                             |                          |                        |              |
| Fri Oct 14    | Fall Break no classes                                             |                          |                        |              |
| WEEK 10       |                                                                   |                          |                        |              |
| Mon Oct 17    | 1:00 Nitrogen Metabolism                                          | Chapter 18 and 19        | Chapter 2              | King         |
| Tues Oct 18   | No Biochemistry                                                   |                          |                        |              |
| Wed Oct 19    | 1:00 Nitrogen Metabolism<br>2:00 Review for Exam #3               | Chapter 18 and 19        | Chapter 2              | King         |
| Thurs Oct 20  | 1:00 – 4:00 PBL #4                                                |                          |                        |              |
| Fri Oct 21    | 1:00 Exam #3 (lectures through Oct 20)                            |                          |                        |              |
| WEEK 11       |                                                                   |                          |                        |              |
| Mon Oct 24    | 1:00 Urea Cycle and Diseases                                      | Chapter 19               |                        | King         |

|              |                                                              |                          |                          |              |
|--------------|--------------------------------------------------------------|--------------------------|--------------------------|--------------|
| Tues Oct 25  | No Biochemistry                                              |                          |                          |              |
| Wed Oct 26   | 1:00 Nucleotide Metabolism                                   | Chapter 19               | Chapter 2                | King         |
| Thurs Oct 27 | 1:00 – 4:00 PBL #4                                           |                          |                          |              |
| Fri Oct 28   | 1:00 Nucleotide Metabolism<br>2:00 Exam #3 Discussion        | Chapter 19               | Chapter 2                | King         |
| WEEK 12      |                                                              |                          |                          |              |
| Mon Oct 31   | 1:00 Porphyrins and Heme                                     | Chapter 24               |                          | King         |
| Tues Nov 1   | No Biochemistry                                              |                          |                          |              |
| Wed Nov 2    | 1:00 Vitamins and Co-enzymes<br>2:00 Vitamins and Co-enzymes | Chapter 27<br>Chapter 27 |                          | King<br>King |
| Thurs Nov 3  | 1:00 – 4:00 PBL #5                                           |                          |                          |              |
| Fri Nov 4    | 1:00 Blood Coagulation                                       | Chapter 23               |                          | King         |
| WEEK 13      |                                                              |                          |                          |              |
| Mon Nov 7    | 1:00 Genes and Cancer<br>2:00 Tumor Suppressors and Cancer   | Chapter 21<br>Chapter 21 | Chapter 23<br>Chapter 23 | King<br>King |
| Tues Nov 8   | No Biochemistry                                              |                          |                          |              |
| Wed Nov 9    | 1:00 Molecular Medicine<br>2:00 Molecular Medicine           | Chapter 7<br>Chapter 7   | Chapter 8<br>Chapter 8   | King<br>King |
| Thurs Nov 10 | 1:00 – 4:00 PBL #5                                           |                          |                          |              |
| Fri Nov 11   | 1:00 Molecular Medicine                                      | Chapter 7                | Chapter 8                | King         |
| WEEK 14      |                                                              |                          |                          |              |
| Mon Nov 14   | 1:00 Peptide Hormones and Receptors                          | Chapter 21               |                          | King         |
| Tues Nov 15  | No Biochemistry                                              |                          |                          |              |
| Wed Nov 16   | 1:00 Steroid Hormones and Receptors                          | Chapter 21               |                          | King         |
| Thurs Nov 17 | 1:00 – 4:00 PBL #6                                           |                          |                          |              |
| Fri Nov 18   | 1:00 Signal Transduction<br>1:00 Signal Transduction         |                          | Chapter 15<br>Chapter 15 | King<br>King |
| WEEK 15      |                                                              |                          |                          |              |
| Mon Nov 21   | 1:00 – 4:00 PBL #6                                           |                          |                          |              |
| Tues Nov 22  | No Biochemistry                                              |                          |                          |              |
| Wed Nov 23   | THANKSGIVING HOLIDAY                                         |                          |                          |              |
| Thurs Nov 24 | THANKSGIVING HOLIDAY                                         |                          |                          |              |
| Fri Nov 25   | THANKSGIVING HOLIDAY                                         |                          |                          |              |
| WEEK 16      |                                                              |                          |                          |              |
| Mon Nov 28   | 1:00 Inborn Errors in Metabolism                             |                          |                          | King         |
| Tues Nov 29  | No Biochemistry                                              |                          |                          |              |
| Wed Nov 30   | 1:00 Inborn Errors in Metabolism                             |                          |                          | King         |
| Thurs Dec 1  | 1:00 Review for Exam #4                                      |                          |                          |              |
| Fri Dec 2    | 1:00 Exam #4 (lectures through Dec 1)                        |                          |                          |              |
| WEEK 17      |                                                              |                          |                          |              |
| Mon Dec 5    | Reading Week                                                 |                          |                          |              |

|            |                            |
|------------|----------------------------|
| Tues Dec 6 | Reading Week               |
| Wed Dec 7  | Reading Week               |
| Thur Dec 8 | Reading Week               |
| Fri Dec 9  | Reading Week               |
| WEEK 18    |                            |
| Mon Dec 12 | BIOCHEMISTRY FINAL 9:00 AM |



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- [Introduction to Enzymes](#)
- [Enzyme Classifications](#)
- [Role of Coenzymes](#)
- [Enzyme Activity Relative to Substrate Type](#)
- [Enzyme-Substrate Interactions](#)
- [Chemical Reactions and Rates](#)
- [Chemical Reaction Order](#)
- [Enzymes as Biological Catalysts](#)
- [Michaelis-Menton Kinetics](#)
- [Inhibition of Enzyme Catalyzed Reactions](#)
- [Regulation of Enzyme Activity](#)
- [Allosteric Enzymes](#)
- [Enzymes in the Diagnosis of Pathology](#)

[Enzyme Kinetics](#) by Dr. Peter Birch, University of Paisley

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## Introduction to Enzymes

Enzymes are **biological catalysts** responsible for supporting almost all of the chemical reactions that maintain animal homeostasis. Because of their role in maintaining life processes, the assay and pharmacological regulation of enzymes have become key elements in clinical diagnosis and therapeutics. The macromolecular components of almost all enzymes are composed of protein, except for a class of RNA modifying catalysts known as **ribozymes**. Ribozymes are molecules of ribonucleic acid that catalyze reactions on the phosphodiester bond of other RNAs.

Enzymes are found in all tissues and fluids of the body. Intracellular enzymes catalyze the reactions of metabolic pathways. Plasma membrane enzymes regulate catalysis within cells in response to extracellular signals, and enzymes of the circulatory system are responsible for regulating the [clotting of blood](#). Almost every significant life process is dependent on enzyme activity.

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## Enzyme Classifications

Traditionally, enzymes were simply assigned names by the investigator who discovered the enzyme. As knowledge expanded, systems of enzyme classification became more comprehensive and complex. Currently enzymes are grouped into six functional classes by the International Union of Biochemists (I.U.B.).

| Number | Classification | Biochemical Properties |
|--------|----------------|------------------------|
|--------|----------------|------------------------|

|    |                 |                                                                                                                                                                                       |
|----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Oxidoreductases | Act on many chemical groupings to add or remove hydrogen atoms.                                                                                                                       |
| 2. | Transferases    | Transfer functional groups between donor and acceptor molecules. Kinases are specialized transferases that regulate metabolism by transferring phosphate from ATP to other molecules. |
| 3. | Hydrolases      | Add water across a bond, hydrolyzing it.                                                                                                                                              |
| 4. | Lyases          | Add water, ammonia or carbon dioxide across double bonds, or remove these elements to produce double bonds.                                                                           |
| 5. | Isomerases      | Carry out many kinds of isomerization: L to D isomerizations, mutase reactions (shifts of chemical groups) and others.                                                                |
| 6. | Ligases         | Catalyze reactions in which two chemical groups are joined (or <b>ligated</b> ) with the use of energy from ATP.                                                                      |

These rules give each enzyme a unique number. The I.U.B. system also specifies a textual name for each enzyme. The enzyme's name is comprised of the names of the substrate(s), the product(s) and the enzyme's functional class. Because many enzymes, such as alcohol dehydrogenase, are widely known in the scientific community by their common names, the change to I.U.B.-approved nomenclature has been slow. In everyday usage, most enzymes are still called by their common name.

Enzymes are also classified on the basis of their composition. Enzymes composed wholly of protein are known as **simple enzymes** in contrast to **complex enzymes**, which are composed of protein plus a relatively small organic molecule. Complex enzymes are also known as **holoenzymes**. In this terminology the protein component is known as the **apoenzyme**, while the non-protein component is known as the **coenzyme** or **prosthetic group** where prosthetic group describes a complex in which the small organic molecule is bound to the apoenzyme by covalent bonds; when the binding between the apoenzyme and non-protein components is non-covalent, the small organic molecule is called a **coenzyme**. Many prosthetic groups and coenzymes are water-soluble derivatives of **vitamins**. It should be noted that the main clinical symptoms of dietary vitamin insufficiency generally arise from the malfunction of enzymes, which lack sufficient cofactors derived from vitamins to maintain homeostasis.

The non-protein component of an enzyme may be as simple as a metal ion or as complex as a small non-protein organic molecule. Enzymes that require a metal in their composition are known as **metalloenzymes** if they bind and retain their metal atom(s) under all conditions, that is with very high affinity. Those which have a lower affinity for metal ion, but still require the metal ion for activity, are known as metal-activated enzymes.

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## Role of Coenzymes

The functional role of coenzymes is to act as transporters of chemical groups from one reactant to another. The chemical groups carried can be as simple as the hydride ion ( $\text{H}^+ + 2\text{e}^-$ ) carried by NAD or the mole of hydrogen carried by FAD; or they can be even more complex than the amine ( $-\text{NH}_2$ ) carried by pyridoxal phosphate.

Since coenzymes are chemically changed as a consequence of enzyme action, it is often useful to consider coenzymes to be a special class of substrates, or **second substrates**, which are common to many different holoenzymes. In all cases, the coenzymes donate the carried chemical grouping to an acceptor molecule and are thus regenerated to their original form. This regeneration of coenzyme and holoenzyme fulfills the definition of an enzyme as a chemical catalyst, since (unlike the usual substrates, which are used up during the course of a reaction) coenzymes are generally regenerated.

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## Enzyme Relative to Substrate Type

Although enzymes are highly specific for the kind of reaction they catalyze, the same is not always true of substrates they attack. For example, while succinic dehydrogenase (SDH) always catalyzes an oxidation-reduction reaction and its substrate is invariably succinic acid, alcohol dehydrogenase (ADH) always catalyzes oxidation-reduction reactions but attacks a number of different alcohols, ranging from methanol to butanol. Generally, enzymes having broad substrate specificity are most active against one particular substrate. In the case of ADH, ethanol is the preferred substrate.

Enzymes also are generally specific for a particular steric configuration (optical isomer) of a substrate. Enzymes that attack D sugars will not attack the corresponding L isomer. Enzymes that act on L amino acids will not employ the corresponding D optical isomer as a substrate. The enzymes known as racemases provide a striking exception to these generalities; in fact, the role of racemases is to convert D isomers to L isomers and vice versa. Thus racemases attack both D and L forms of their substrate.

As enzymes have a more or less broad range of substrate specificity, it follows that a given substrate may be acted on by a number of different enzymes, each of which uses the same substrate(s) and produces the same product(s). The individual members of a set of enzymes sharing such characteristics are known as **isozymes**. These are the products of genes that vary only slightly; often, various isozymes of a group are expressed in different tissues of the body. The best studied set of isozymes is the lactate dehydrogenase (LDH) system. LDH is a tetrameric enzyme composed of all possible arrangements of two different protein subunits; the subunits are known as H (for heart) and M (for skeletal muscle). These subunits combine in various combinations leading to 5 distinct isozymes. The all H isozyme is characteristic of that from heart tissue, and the all M isozyme is typically found in skeletal muscle and liver. These isozymes all catalyze the same chemical reaction, but they exhibit differing degrees of efficiency. The detection of specific LDH isozymes in the blood is highly diagnostic of tissue damage such as occurs during cardiac infarct.

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## Enzyme-Substrate Interactions

The favored model of enzyme substrate interaction is known as the **induced fit model**. This model proposes that the initial interaction between enzyme and substrate is relatively weak, but that these weak interactions rapidly induce conformational changes in the enzyme that strengthen binding and bring catalytic sites close to substrate bonds to be altered. After binding takes place, one or more mechanisms of catalysis generates transition- state complexes and reaction products. The possible mechanisms of catalysis are four in number:



**1. Catalysis by Bond Strain:** In this form of catalysis, the induced structural rearrangements that take place with the binding of substrate and enzyme ultimately produce strained substrate bonds, which more easily attain the transition state. The new conformation often forces substrate atoms and bulky catalytic groups, such as aspartate and glutamate, into conformations that strain existing substrate bonds.

**2. Catalysis by Proximity and Orientation:** Enzyme-substrate interactions orient reactive groups and bring them into proximity with one another. In addition to inducing strain, groups such as aspartate are frequently chemically reactive as well, and their proximity and orientation toward the substrate thus favors their participation in catalysis.

**3. Catalysis Involving Proton Donors (Acids) and Acceptors (Bases):** Other mechanisms also contribute significantly to the completion of catalytic events initiated by a strain mechanism, for example, the use of glutamate as a general acid catalyst (proton donor).

**4. Covalent Catalysis:** In catalysis that takes place by covalent mechanisms, the substrate is oriented to active sites on the enzymes in such a way that a covalent intermediate forms between the enzyme or coenzyme and the substrate. One of the best-known examples of this mechanism is that involving proteolysis by **serine proteases**, which include both digestive enzymes (trypsin, chymotrypsin, and elastase) and several enzymes of the **blood clotting cascade**. These proteases contain an active site serine whose R group hydroxyl forms a covalent bond with a carbonyl carbon of a peptide bond, thereby causing hydrolysis of the peptide bond.

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## Chemical Reactions and Rates

According to the conventions of biochemistry, the rate of a chemical reaction is described by the number of molecules of reactant(s) that are converted into product(s) in a specified time period. Reaction rate is always dependent on the concentration of the chemicals involved in the process and on rate constants that are characteristic of the reaction. For example, the reaction in which A is converted to B is written as follows:



The rate of this reaction is expressed algebraically as either a decrease in the concentration of reactant A:

$$-[\text{A}] = k[\text{B}]$$

or an increase in the concentration of product B:

$$[\text{B}] = k[\text{A}]$$

In the second equation (of the 3 above) the negative sign signifies a decrease in concentration of A as the reaction progresses, brackets define concentration in molarity and the  $k$  is known as a rate constant. Rate constants are simply proportionality constants that provide a quantitative connection between chemical concentrations and reaction rates. Each chemical reaction has characteristic values for its rate constants; these in turn directly relate to the **equilibrium constant** for that reaction. Thus, reaction can be rewritten as

an equilibrium expression in order to show the relationship between reaction rates, rate constants and the equilibrium constant for this simple case. The rate constant for the forward reaction is defined as  $k_{+1}$  and the reverse as  $k_{-1}$ .

At equilibrium the rate ( $v$ ) of the forward reaction ( $A \rightleftharpoons B$ ) is--- by definition--- equal to that of the reverse or back reaction ( $B \rightleftharpoons A$ ), a relationship which is algebraically symbolized as:

$$v_{\text{forward}} = v_{\text{reverse}}$$

where, for the forward reaction:

$$v_{\text{forward}} = k_{+1}[A]$$

and for the reverse reaction:

$$v_{\text{reverse}} = k_{-1}[B]$$

In the above equations,  $k_{+1}$  and  $k_{-1}$  represent rate constants for the forward and reverse reactions, respectively. The negative subscript refers only to a reverse reaction, not to an actual negative value for the constant. To put the relationships of the two equations into words, we state that the rate of the forward reaction [ $v_{\text{forward}}$ ] is equal to the product of the forward rate constant  $k_{+1}$  and the molar concentration of A. The rate of the reverse reaction is equal to the product of the reverse rate constant  $k_{-1}$  and the molar concentration of B.

At equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction leading to the **equilibrium constant** of the reaction and is expressed by:

$$[B]/[A] = k_{+1}/k_{-1} = K_{\text{eq}}$$

This equation demonstrates that the equilibrium constant for a chemical reaction is not only equal to the equilibrium ratio of product and reactant concentrations, but is also equal to the ratio of the characteristic rate constants of the reaction.

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## Chemical Reaction Order

Reaction order refers to the number of molecules involved in forming a reaction complex that is competent to proceed to product(s). Empirically, order is easily determined by summing the exponents of each concentration term in the rate equation for a reaction. A reaction characterized by the conversion of one molecule of A to one molecule of B with no influence from any other reactant or solvent is a **first-order** reaction. The exponent on the substrate concentration in the rate equation for this type of reaction is 1. A reaction with two substrates forming two products would be a **second-order** reaction. However, the reactants in second- and higher- order reactions need not be different chemical species. An example of a second order reaction is the formation of ATP through the condensation of ADP with orthophosphate:



For this reaction the forward reaction rate would be written as:

$$v_{\text{forward}} = k_1[\text{ADP}][\text{H}_2\text{PO}_4]$$

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## Enzymes as Biological Catalysts

In cells and organisms most reactions are catalyzed by enzymes, which are regenerated during the course of a reaction. These **biological catalysts** are physiologically important because they speed up the rates of reactions that would otherwise be too slow to support life. Enzymes increase reaction rates---sometimes by as much as one millionfold, but more typically by about one thousand fold. Catalysts speed up the forward and reverse reactions proportionately so that, although the magnitude of the rate constants of the forward and reverse reactions is increased, the ratio of the rate constants remains the same in the presence or absence of enzyme. Since the equilibrium constant is equal to a ratio of rate constants, it is apparent that enzymes and other catalysts have no effect on the equilibrium constant of the reactions they catalyze.

Enzymes increase reaction rates by decreasing the amount of energy required to form a complex of reactants that is competent to produce reaction products. This complex is known as the activated state or **transition state complex** for the reaction. Enzymes and other catalysts accelerate reactions by lowering the energy of the transition state. The free energy required to form an activated complex is much lower in the catalyzed reaction. The amount of energy required to achieve the transition state is lowered; consequently, at any instant a greater proportion of the molecules in the population can achieve the transition state. The result is that the reaction rate is increased.

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## Michaelis-Menton Kinetics

In typical enzyme-catalyzed reactions, reactant and product concentrations are usually hundreds or thousands of times greater than the enzyme concentration. Consequently, each enzyme molecule catalyzes the conversion to product of many reactant molecules. In biochemical reactions, reactants are commonly known as substrates. The catalytic event that converts substrate to product involves the formation of a transition state, and it occurs most easily at a specific binding site on the enzyme. This site, called the **catalytic site** of the enzyme, has been evolutionarily structured to provide specific, high-affinity binding of substrate(s) and to provide an environment that favors the catalytic events. The complex that forms when substrate(s) and enzyme combine is called the **enzyme substrate (ES) complex**. Reaction products arise when the ES complex breaks down releasing free enzyme.

Between the binding of substrate to enzyme, and the reappearance of free enzyme and product, a series of complex events must take place. At a minimum an ES complex must be formed; this complex must pass to the transition state ( $\text{ES}^*$ ); and the transition state complex must advance to an enzyme product complex (EP). The latter is finally competent to dissociate to product and free enzyme. The series of events can be shown thus:



The kinetics of simple reactions like that above were first characterized by biochemists Michaelis and Menten. The concepts underlying their analysis of enzyme kinetics continue to provide the cornerstone for understanding metabolism today, and for the development and clinical use of drugs aimed at selectively altering rate constants and interfering with the progress of disease states. The **Michaelis-Menten equation**:

$$V_1 = \frac{V_{\max}[S]}{\{K_m + [S]\}}$$

is a quantitative description of the relationship among the rate of an enzyme-catalyzed reaction [ $v_1$ ], the concentration of substrate  $[S]$  and two constants,  $V_{\max}$  and  $K_m$  (which are set by the particular equation). The symbols used in the Michaelis-Menten equation refer to the reaction rate [ $v_1$ ], maximum reaction rate ( $V_{\max}$ ), substrate concentration  $[S]$  and the Michaelis-Menten constant ( $K_m$ ).

The Michaelis-Menten equation can be used to demonstrate that at the substrate concentration that produces exactly half of the maximum reaction rate, i.e.,  $1/2 V_{\max}$ , the substrate concentration is numerically equal to  $K_m$ . This fact provides a simple yet powerful bioanalytical tool that has been used to characterize both normal and altered enzymes, such as those that produce the symptoms of genetic diseases. Rearranging the Michaelis-Menten equation leads to:

$$K_m = [S] \left\{ \left[ \frac{V_{\max}}{V_1} \right] - 1 \right\}$$

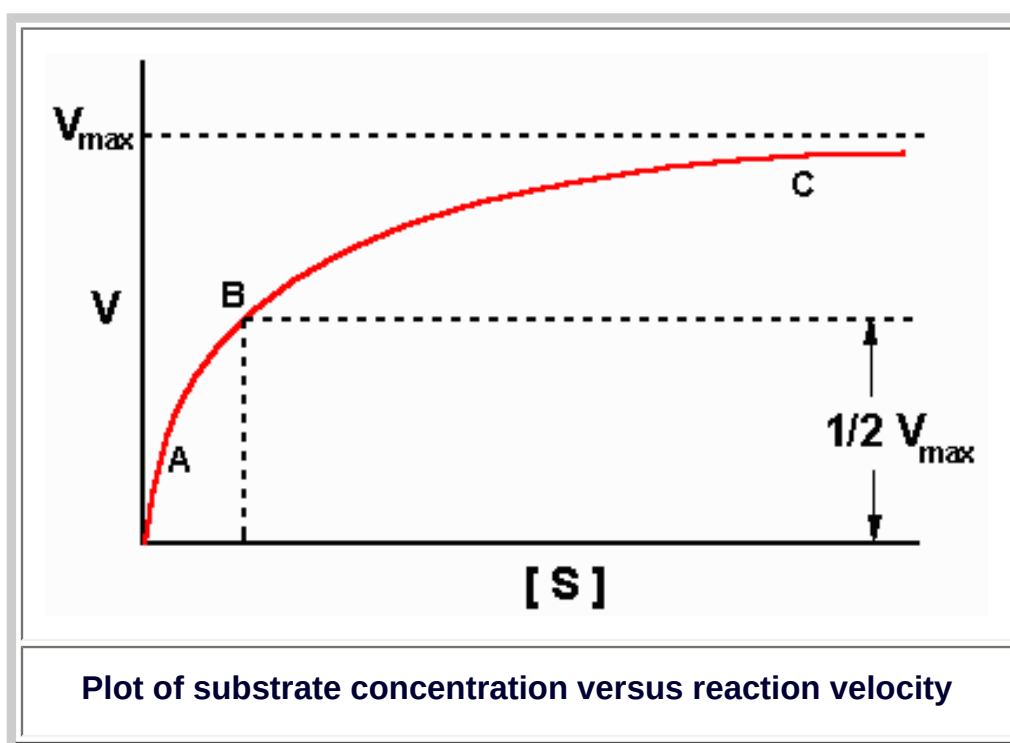
From this equation it should be apparent that when the substrate concentration is half that required to support the maximum rate of reaction, the observed rate,  $v_1$ , will be equal to  $V_{\max}$  divided by 2; in other words,  $v_1 = [V_{\max}/2]$ . At this substrate concentration  $V_{\max}/v_1$  will be exactly equal to 2, with the result that

$$[S](1) = K_m$$

The latter is an algebraic statement of the fact that, for enzymes of the Michaelis-Menten type, when the observed reaction rate is half of the maximum possible reaction rate, the substrate concentration is numerically equal to the Michaelis-Menten constant. In this derivation, the units of  $K_m$  are those used to specify the concentration of  $S$ , usually Molarity.

The Michaelis-Menten equation has the same form as the equation for a rectangular hyperbola;

graphical analysis of reaction rate ( $v$ ) versus substrate concentration  $[S]$  produces a hyperbolic rate plot.



The key features of the plot are marked by points A, B and C. At high substrate concentrations the rate represented by point C the rate of the reaction is almost equal to  $V_{\max}$ , and the difference in rate at nearby concentrations of substrate is almost negligible. If the Michaelis-Menten plot is extrapolated to infinitely high substrate concentrations, the extrapolated rate is equal to  $V_{\max}$ . When the reaction rate becomes independent of substrate concentration, or nearly so, the rate is said to be zero order. (Note that the reaction is zero order only with respect to this substrate. If the reaction has two substrates, it may or may not be zero order with respect to the second substrate). The very small differences in reaction velocity at substrate concentrations around point C (near  $V_{\max}$ ) reflect the fact that at these concentrations almost all of the enzyme molecules are bound to substrate and the rate is virtually independent of substrate, hence zero order. At lower substrate concentrations, such as at points A and B, the lower reaction velocities indicate that at any moment only a portion of the enzyme molecules are bound to the substrate. In fact, at the substrate concentration denoted by point B, exactly half the enzyme molecules are in an ES complex at any instant and the rate is exactly one half of  $V_{\max}$ . At substrate concentrations near point A the rate appears to be directly proportional to substrate concentration, and the reaction rate is said to be first order.

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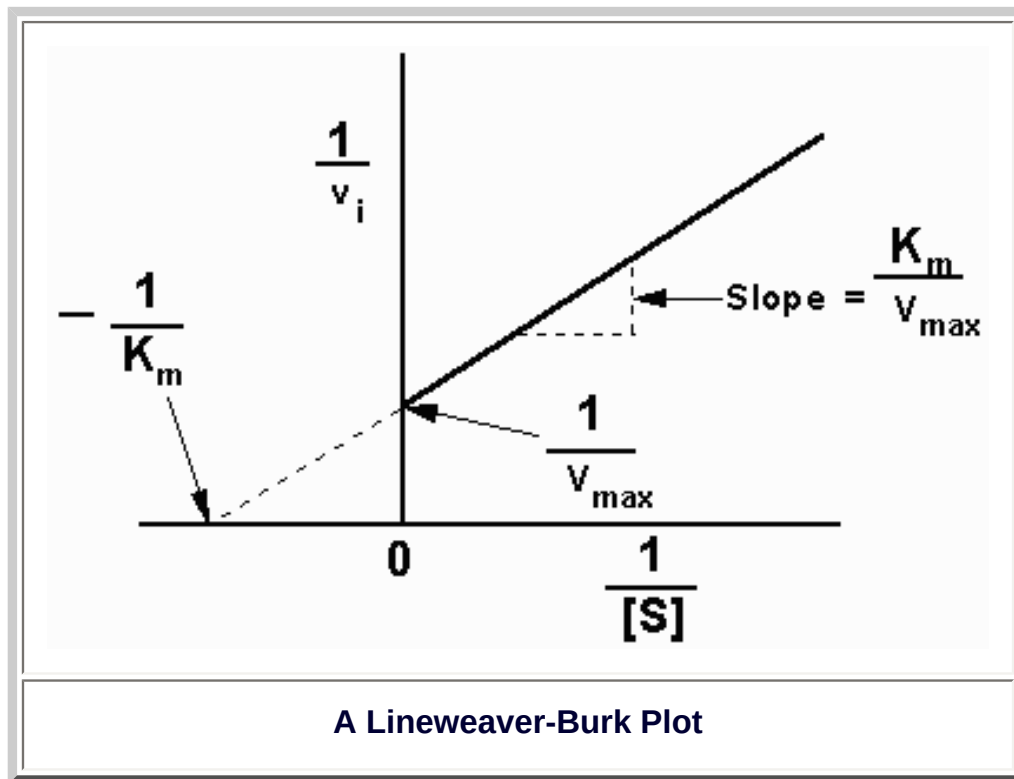
## Inhibition of Enzyme Catalyzed Reactions

To avoid dealing with curvilinear plots of enzyme catalyzed reactions, biochemists Lineweaver and Burk introduced an analysis of enzyme kinetics based on the following rearrangement of the Michaelis-Menten equation:

$$[1/v] = [K_m (1)/ V_{\max}[S] + (1)/V_{\max}]$$

Plots of  $1/v$  versus  $1/[S]$  yield straight lines having a slope of  $K_m/V_{\max}$  and an intercept on the

ordinate at  $1/V_{\max}$ .



An alternative linear transformation of the Michaelis-Menten equation is the Eadie-Hofstee transformation:

$$v/[S] = -v [1/K_m] + [V_{\max}/K_m]$$

and when  $v/[S]$  is plotted on the y-axis versus  $v$  on the x-axis, the result is a linear plot with a slope of  $-1/K_m$  and the value  $V_{\max}/K_m$  as the intercept on the y-axis and  $V_{\max}$  as the intercept on the x-axis.

Both the Lineweaver-Burk and Eadie-Hofstee transformation of the Michaelis-Menten equation are useful in the analysis of enzyme inhibition. Since most clinical drug therapy is based on inhibiting the activity of enzymes, analysis of enzyme reactions using the tools described above has been fundamental to the modern design of pharmaceuticals. Well-known examples of such therapy include the use of methotrexate in cancer chemotherapy to semi-selectively inhibit DNA synthesis of malignant cells, the use of aspirin to inhibit the synthesis of prostaglandins which are at least partly responsible for the aches and pains of arthritis, and the use of sulfa drugs to inhibit the folic acid synthesis that is essential for the metabolism and growth of disease-causing bacteria. In addition, many poisons--- such as cyanide, carbon monoxide and polychlorinated biphenols (PCBs)--- produce their life-threatening effects by means of enzyme inhibition.

Enzyme inhibitors fall into two broad classes: those causing irreversible inactivation of enzymes and those whose inhibitory effects can be reversed. Inhibitors of the first class usually cause an inactivating, covalent modification of enzyme structure. Cyanide is a classic example of an irreversible enzyme inhibitor: by covalently binding mitochondrial cytochrome oxidase, it inhibits all the reactions associated with electron transport. The kinetic effect of irreversible inhibitors is to decrease the concentration of active enzyme, thus decreasing the maximum possible concentration of ES complex. Since the limiting enzyme reaction rate is often  $k_2[ES]$ , it is clear that under these circumstances the reduction of enzyme concentration will lead to decreased reaction rates. Note that when enzymes in cells are only partially inhibited by irreversible inhibitors, the remaining unmodified enzyme molecules are not distinguishable from those in untreated cells;



in particular, they have the same turnover number and the same  $K_m$ . **Turnover number**, related to  $V_{max}$ , is defined as the maximum number of moles of substrate that can be converted to product per mole of catalytic site per second. Irreversible inhibitors are usually considered to be poisons and are generally unsuitable for therapeutic purposes.

Reversible inhibitors can be divided into two main categories--- **competitive inhibitors** and **noncompetitive inhibitors**---with a third category, **uncompetitive inhibitors**, rarely encountered.

| Inhibitor Type           | Binding Site on Enzyme                                                                                                                                                                                    | Kinetic effect                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Competitive Inhibitor    | Specifically at the catalytic site, where it competes with substrate for binding in a dynamic equilibrium- like process. Inhibition is reversible by substrate.                                           | $V_{max}$ is unchanged; $K_m$ , as defined by [S] required for 1/2 maximal activity, is increased.       |
| Noncompetitive Inhibitor | Binds E or ES complex other than at the catalytic site. Substrate binding unaltered, but ESI complex cannot form products. Inhibition cannot be reversed by substrate.                                    | $K_m$ appears unaltered; $V_{max}$ is decreased proportionately to inhibitor concentration.              |
| Uncompetitive Inhibitor  | Binds only to ES complexes at locations other than the catalytic site. Substrate binding modifies enzyme structure, making inhibitor- binding site available. Inhibition cannot be reversed by substrate. | Apparent $V_{max}$ decreased; $K_m$ , as defined by [S] required for 1/2 maximal activity, is decreased. |

The hallmark of all the reversible inhibitors is that when the inhibitor concentration drops, enzyme activity is regenerated. Usually these inhibitors bind to enzymes by non-covalent forces and the inhibitor

maintains a reversible equilibrium with the enzyme. The equilibrium constant for the dissociation of enzyme inhibitor complexes is known as  $K_i$ :

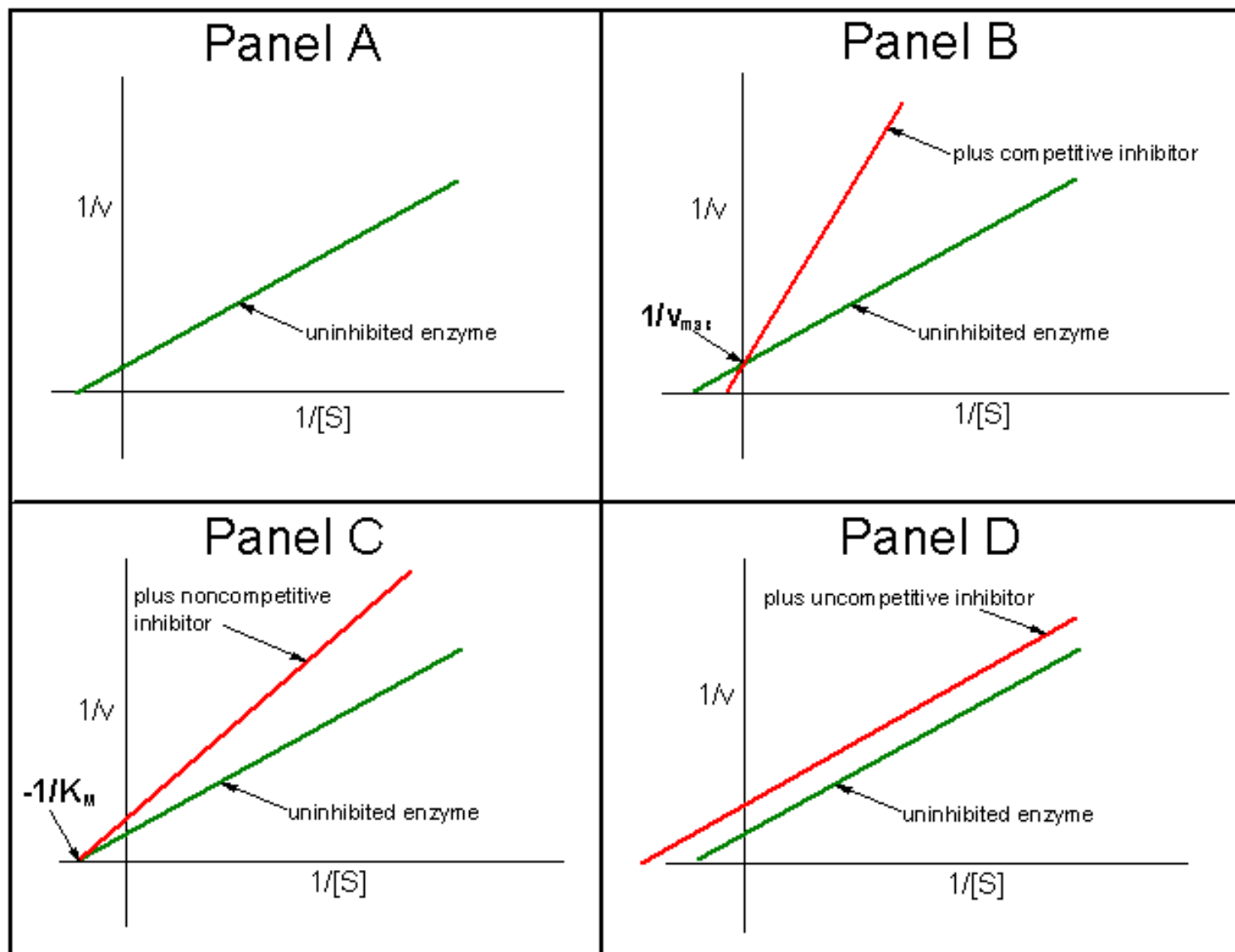
$$K_i = \frac{[E][I]}{[E-I\text{-complex}]}$$

The importance of  $K_i$  is that in all enzyme reactions where substrate, inhibitor and enzyme interact, the normal  $K_m$  and or  $V_{max}$  for substrate enzyme interaction appear to be altered. These changes are a consequence of the influence of  $K_i$  on the overall rate equation for the reaction. The effects of  $K_i$  are best observed in Lineweaver-Burk plots.

Probably the best known reversible inhibitors are competitive inhibitors, which always bind at the catalytic or active site of the enzyme. Most drugs that alter enzyme activity are of this type. Competitive inhibitors are especially attractive as clinical modulators of enzyme activity because they offer two routes for the reversal of enzyme inhibition, while other reversible inhibitors offer only one. First, as with all kinds of reversible inhibitors, a decreasing concentration of the inhibitor reverses the equilibrium, regenerating active free enzyme. Second, since substrate and competitive inhibitors both bind at the same site, they compete with one another for binding

Raising the concentration of substrate (S), while holding the concentration of inhibitor constant, provides the second route for reversal of competitive inhibition. The greater the proportion of substrate, the greater the proportion of enzyme present in competent ES complexes.

As noted earlier, high concentrations of substrate can displace virtually all competitive inhibitor bound to active sites. Thus, it is apparent that  $V_{max}$  should be unchanged by competitive inhibitors. This characteristic of competitive inhibitors is reflected in the identical vertical-axis intercepts of Lineweaver-Burk plots, with and without inhibitor.



**Lineweaver-Burk Plots of Inhibited Enzymes**

Since attaining  $V_{\max}$  requires appreciably higher substrate concentrations in the presence of competitive inhibitor,  $K_m$  (the substrate concentration at half maximal velocity) is also higher, as demonstrated by the differing negative intercepts on the horizontal axis in panel B.

Analogously, panel C illustrates that noncompetitive inhibitors appear to have no effect on the intercept at the x-axis implying that noncompetitive inhibitors have no effect on the  $K_m$  of the enzymes they inhibit. Since noncompetitive inhibitors do not interfere in the equilibration of enzyme, substrate and ES complexes, the  $K_m$ 's of Michaelis-Menten type enzymes are not expected to be affected by noncompetitive inhibitors, as demonstrated by x-axis intercepts in panel C. However, because complexes that contain inhibitor (ESI) are incapable of progressing to reaction products, the effect of a noncompetitive inhibitor is to reduce the concentration of ES complexes that can advance to product. Since  $V_{\max} = k_2[E_{\text{total}}]$ , and the concentration of competent  $E_{\text{total}}$  is diminished by the amount of ESI formed, noncompetitive inhibitors are expected to decrease  $V_{\max}$ , as illustrated by the y-axis intercepts in panel C.

A corresponding analysis of uncompetitive inhibition leads to the expectation that these inhibitors should change the apparent values of  $K_m$  as well as  $V_{\max}$ . Changing both constants leads to double

reciprocal plots, in which intercepts on the x and y axes are proportionately changed; this leads to the production of parallel lines in inhibited and uninhibited reactions.

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## Regulation of Enzyme Activity

While it is clear that enzymes are responsible for the catalysis of almost all biochemical reactions, it is important to also recognize that rarely, if ever, do enzymatic reactions proceed in isolation. The most common scenario is that enzymes catalyze individual steps of multi-step metabolic pathways, as is the case with [glycolysis](#), [gluconeogenesis](#) or the [synthesis of fatty acids](#). As a consequence of these lock- step sequences of reactions, any given enzyme is dependent on the activity of preceding reaction steps for its substrate.

In humans, substrate concentration is dependent on food supply and is not usually a physiologically important mechanism for the routine regulation of enzyme activity. Enzyme concentration, by contrast, is continually modulated in response to physiological needs. Three principal mechanisms are known to regulate the concentration of active enzyme in tissues:

- **1.** Regulation of gene expression controls the quantity and rate of enzyme synthesis.
- **2.** Proteolytic enzyme activity determines the rate of enzyme degradation.
- **3.** Covalent modification of preexisting pools of inactive proenzymes produces active enzymes.

Enzyme synthesis and proteolytic degradation are comparatively slow mechanisms for regulating enzyme concentration, with response times of hours, days or even weeks. Proenzyme activation is a more rapid method of increasing enzyme activity but, as a regulatory mechanism, it has the disadvantage of not being a reversible process. Proenzymes are generally synthesized in abundance, stored in secretory granules and covalently activated upon release from their storage sites. Examples of important proenzymes include pepsinogen, trypsinogen and chymotrypsinogen, which give rise to the proteolytic digestive enzymes. Likewise, many of the proteins involved in the cascade of chemical reactions responsible for blood clotting are synthesized as proenzymes. Other important proteins, such as peptide hormones and collagen, are also derived by covalent modification of precursors.

Another mechanism of regulating enzyme activity is to sequester enzymes in compartments where access to their substrates is limited. For example, the proteolysis of cell proteins and glycolipids by enzymes responsible for their degradation is controlled by sequestering these enzymes within the lysosome.

In contrast to regulatory mechanisms that alter enzyme concentration, there is an important group of regulatory mechanisms that do not affect enzyme concentration, are reversible and rapid in action, and actually carry out most of the moment- to- moment physiological regulation of enzyme activity. These mechanisms include allosteric regulation, regulation by reversible covalent modification and regulation by control proteins such as calmodulin. Reversible covalent modification is a major mechanism for the rapid and transient regulation of enzyme activity. The best examples, again, come from studies on the regulation of glycogen metabolism where phosphorylation of glycogen synthase and glycogen phosphorylase kinase results in the stimulation of glycogen degradation while glycogen synthesis is coordinately inhibited. Numerous other enzymes of intermediary metabolism are affected by phosphorylation, either positively or negatively. These covalent phosphorylations can be reversed by a separate sub-subclass of enzymes known as phosphatases. Recent research has indicated that the aberrant phosphorylation of growth factor and hormone receptors, as well as of proteins that regulate cell division, often leads to unregulated cell growth or cancer. The usual sites for phosphate addition to proteins are the serine, threonine and tyrosine R group hydroxyl residues.

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## Allosteric Enzymes

In addition to simple enzymes that interact only with substrates and inhibitors, there is a class of enzymes that bind small, physiologically important molecules and modulate activity in ways other than those described above. These are known as **allosteric enzymes**; the small regulatory molecules to which they bind are known as **effectors**. Allosteric effectors bring about catalytic modification by binding to the enzyme at distinct allosteric sites, well removed from the catalytic site, and causing conformational changes that are transmitted through the bulk of the protein to the catalytically active site(s).

The hallmark of effectors is that when they bind to enzymes, they alter the catalytic properties of an enzyme's active site. Those that increase catalytic activity are known as positive effectors. Effectors that reduce or inhibit catalytic activity are negative effectors.

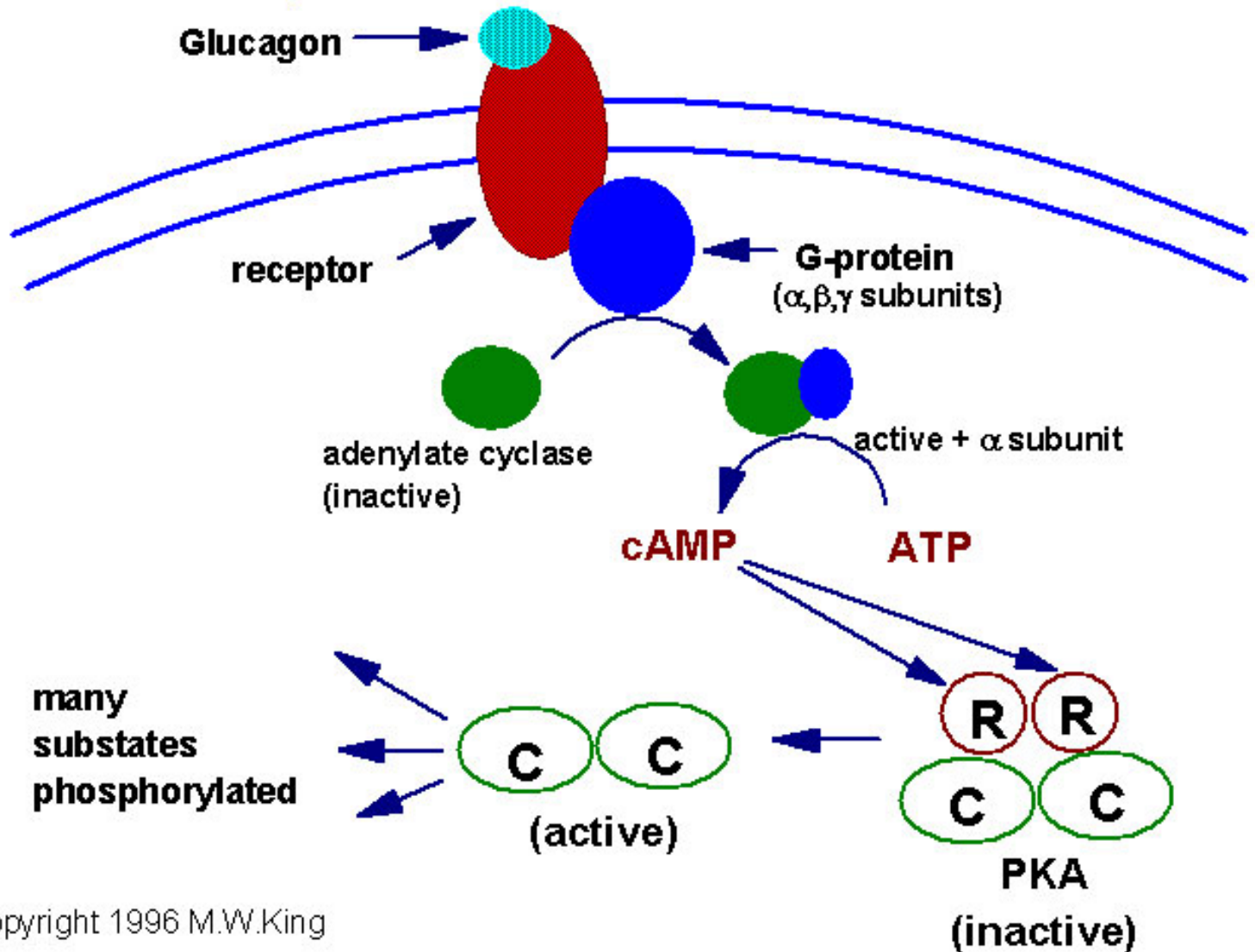
Most allosteric enzymes are oligomeric (consisting of multiple subunits); generally they are located at or near branch points in metabolic pathways, where they are influential in directing substrates along one or another of the available metabolic paths. The effectors that modulate the activity of these allosteric enzymes are of two types. Those activating and inhibiting effectors that bind at allosteric sites are called **heterotropic effectors**. (Thus there exist both positive and negative heterotropic effectors.) These effectors can assume a vast diversity of chemical forms, ranging from simple inorganic molecules to complex nucleotides such as cyclic adenosine monophosphate (cAMP). Their single defining feature is that they are not identical to the substrate.

In many cases the substrate itself induces distant allosteric effects when it binds to the catalytic site. Substrates acting as effectors are said to be **homotropic effectors**. When the substrate is the effector, it can act as such, either by binding to the substrate-binding site, or to an allosteric effector site. When the substrate binds to the catalytic site it transmits an activity-modulating effect to other subunits of the molecule. Often used as the model of a homotropic effector is hemoglobin, although it is not a branch-point enzyme and thus does not fit the definition on all counts.

There are two ways that enzymatic activity can be altered by effectors: the  $V_{\max}$  can be increased or decreased, or the  $K_m$  can be raised or lowered. Enzymes whose  $K_m$  is altered by effectors are said to be **K-type enzymes** and the effector a K-type effector. If  $V_{\max}$  is altered, the enzyme and effector are said to be V-type. Many allosteric enzymes respond to multiple effectors with V-type and K-type behavior. Here again, hemoglobin is often used as a model to study allosteric interactions, although it is not strictly an enzyme.

In the preceding discussion we assumed that allosteric sites and catalytic sites were homogeneously present on every subunit of an allosteric enzyme. While this is often the case, there is another class of allosteric enzymes that are comprised of separate catalytic and regulatory subunits. The archetype of this class of enzymes is cAMP-dependent protein kinase (PKA), whose mechanism of activation is illustrated in the Figure below. The enzyme is tetrameric, containing two catalytic subunits and two regulatory subunits, and enzymatically inactive. When intracellular cAMP levels rise, one molecule of cAMP binds to each regulatory subunit, causing the tetramer to dissociate into one regulatory dimer and two catalytic monomers. In the dissociated form, the catalytic subunits are fully active; they catalyze the phosphorylation of a number of other enzymes, such as those involved in regulating glycogen metabolism. The regulatory subunits have no catalytic activity.

## Receptor-Mediated Activation of PKA



Representative pathway for the activation of cAMP-dependent protein kinase (PKA). In this example glucagon binds to its' cell-surface receptor, thereby activating the receptor. Activation of the receptor is coupled to the activation of a receptor-coupled G-protein (GTP-binding and hydrolyzing protein) composed of 3 subunits. Upon activation the  $\alpha$ -subunit dissociates and binds to and activates adenylate cyclase. Adenylate cyclase then converts ATP to cyclic-AMP (cAMP). The cAMP thus produced then binds to the regulatory subunits of PKA leading to dissociation of the associated catalytic subunits. The catalytic subunits are inactive until dissociated from the regulatory subunits. Once released the catalytic subunits of PKA phosphorylate numerous substrate using ATP as the phosphate donor.

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## Enzymes in the Diagnosis of Pathology



The measurement of the serum levels of numerous enzymes has been shown to be of diagnostic significance. This is because the presence of these enzymes in the serum indicates that tissue or cellular damage has occurred resulting in the release of intracellular components into the blood. Hence, when a physician indicates that he/she is going to assay for liver enzymes, the purpose is to ascertain the potential for liver cell damage. Commonly assayed enzymes are the amino transferases: alanine transaminase, ALT (sometimes still referred to as serum glutamate-pyruvate aminotransferase, SGPT) and aspartate aminotransferase, AST (also referred to as serum glutamate-oxaloacetate aminotransferase, SGOT); lactate dehydrogenase, LDH; creatine kinase, CK (also called creatine phosphokinase, CPK); gamma-glutamyl transpeptidase, GGT. Other enzymes are assayed under a variety of different clinical situations but they will not be covered here.

The typical "liver enzymes" measured are AST and ALT. ALT is particularly diagnostic of liver involvement as this enzyme is found predominantly in hepatocytes. When assaying for both ALT and AST the ratio of the level of these two enzymes can also be diagnostic. Normally in liver disease or damage that is not of viral origin the ratio of ALT/AST is less than 1. However, with viral hepatitis the ALT/AST ratio will be greater than 1. Measurement of AST is useful not only for liver involvement but also for heart disease or damage. The level of AST elevation in the serum is directly proportional to the number of cells involved as well as on the time following injury that the AST assay was performed. Following injury, levels of AST rise within 8 hours and peak 24-36 hours later. Within 3-7 days the level of AST should return to pre-injury levels, provided a continuous insult is not present or further injury occurs. Although measurement of AST is not, in and of itself, diagnostic for myocardial infarction, taken together with LDH and CK measurements (see below) the level of AST is useful for timing of the infarct.

The measurement of LDH is especially diagnostic for myocardial infarction because this enzyme exists in 5 closely related, but slightly different forms (isozymes). The 5 types and their normal distribution and levels in non-disease/injury are listed below.

LDH 1 - Found in heart and red-blood cells and is 17% - 27% of the normal serum total.

LDH 2 - Found in heart and red-blood cells and is 27% - 37% of the normal serum total.

LDH 3 - Found in a variety of organs and is 18% - 25% of the normal serum total.

LDH 4 - Found in a variety of organs and is 3% - 8% of the normal serum total.

LDH 5 - Found in liver and skeletal muscle and is 0% - 5% of the normal serum total.

Following a myocardial infarct the serum levels of LDH rise within 24-48 hours reaching a peak by 2-3 days and return to normal in 5-10 days. Especially diagnostic is a comparison of the LDH-1/LDH-2 ratio. Normally, this ratio is less than 1. A reversal of this ratio is referred to as a "flipped LDH.". Following an acute myocardial infarct the flipped LDH ratio will appear in 12-24 hours and is definitely present by 48 hours in over 80% of patients. Also important is the fact that persons suffering chest pain due to angina only will not likely have altered LDH levels.

CPK is found primarily in heart and skeletal muscle as well as the brain. Therefore, measurement of serum CPK levels is a good diagnostic for injury to these tissues. The levels of CPK will rise within 6 hours of injury and peak by around 18 hours. If the injury is not persistent the level of CK returns to normal within 2-3 days. Like LDH, there are tissue-specific isozymes of CPK and their designations are described below.

CPK3 (CPK-MM) is the predominant isozyme in muscle and is 100% of the normal serum total.

CPK2 (CPK-MB) accounts for about 35% of the CPK activity in cardiac muscle, but less than 5% in skeletal

muscle and is 0% of the normal serum total.

CPK1 (CPK-BB) is the characteristic isozyme in brain and is in significant amounts in smooth muscle and is 0% of the normal serum total.

Since most of the released CPK after a myocardial infarction is CPK-MB, an increased ratio of CPK-MB to total CPK may help in diagnosis of an acute infarction, but an increase of total CPK in itself may not. CPK-MB levels rise 3-6 hours after a myocardial infarct and peak 12-24 hours later if no further damage occurs and returns to normal 12-48 hours after the infarct.

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# Medical Biochemistry Core

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## 1. General Principles

### 1.1 Biochemistry and molecular biology

#### 1.1.1 Gene expression: DNA structure, replication, and exchange

1.1.1.1 DNA structure: single- and double-stranded DNA, stabilizing forces, supercoiling;

1.1.1.2 analysis of DNA: sequencing, restriction analysis, PCR amplification, hybridization

1.1.1.3 DNA replication, mutation, repair (e.g., xeroderma pigmentosa) and degradation

1.1.1.4 gene structure and organization; chromosomes in prokaryotes and eukaryotes

1.1.1.5 recombination, insertion sequences, transposons

1.1.1.7 plasmids and bacteriophages

#### 1.1.2 gene expression: transcription

1.1.2.1 transcription of DNA into RNA; enzymatic reactions, RNA; RNA degradation

1.1.2.2 regulation: cis-regulatory elements, transcription factors, enhancers, promoters

1.1.2.3 defects in transcription and RNA processing (e.g., thalassemias)

#### 1.1.3 gene expression: translation

1.1.3.1 the genetic code

1.1.3.2 structure and function of tRNA

1.1.3.3 structure and function of ribosomes

1.1.3.4 protein synthesis

1.1.3.5  
regulation of  
translation

1.1.3.6 post-translational modifications

1.1.3.7 protein degradation

1.1.3.8 defects in translation and protein structure (e.g., hemoglobinopathies, cystic fibrosis)

#### 1.1.4 structure and function of proteins

1.1.4.1 principles of protein structure and folding

### 1.1.4.2 enzymes: kinetics, thermodynamics, reaction mechanisms

### 1.1.4.3 structural and regulatory proteins; ligand binding, self-assembly

1.1.4.4 mutations that alter proteins (e.g., hemoglobinopathies, familial hypercholesterolemia)

### 1.1.5 energy metabolism; metabolic sequences and regulation

1.1.5.1 generation of energy from carbohydrates, fatty acids, and nonessential amino acids; [glycolysis](#), [glycogenolysis](#), [pentose phosphate pathway](#), [tricarboxylic acid cycle](#), [electron transport and oxidative phosphorylation](#)

1.1.5.2 storage of energy: [gluconeogenesis](#), [glycogenesis](#), [fatty acid and triglyceride synthesis](#)

1.1.5.3 [thermodynamics](#): free energy, chemical equilibria and group transfer potential; the energetics of ATP and other high-energy compounds

1.1.5.4 altered energy metabolism (e.g., cyanide poisoning, mitochondrial myopathies, diabetic ketoacidosis)

### 1.1.6 metabolic pathways of small molecules and associated diseases

1.1.6.1 [biosynthesis and degradation of amino acids](#) (e.g., Phenylketonuria, maple syrup urine disease)

1.1.6.2 [biosynthesis and degradation of purine and pyrimidine nucleotides](#) (e.g., gout, Lesch-Nyhan syndrome)

1.1.6.3 biosynthesis and [degradation of lipids](#) and [cholesterol](#), [steroid hormones](#), prostaglandins, and thromboxanes (e.g., adrenogenital syndrome)

1.1.6.4 biosynthesis and degradation of [porphyrins](#) and [bile acids](#) (e.g., porphyrias)

## 1.2 Biology of cells

1.2.1 structure and composition of cell membranes; ion channels and pumps; endocytosis

1.2.2 second messenger systems (e.g., G proteins)

1.2.3 organelles, protein sorting and secretion

1.2.7 cell cycle

1.2.11 cell-to-cell communication, [signal transduction](#), cell junction

## 1.3 Human development and genetics; Biochemical and molecular aspects

1.3.5 disease-producing mutations

1.3.5.1 chromosomal abnormalities: translocations, deletions, duplications, inversions (e.g., Down's syndrome, Philadelphia chromosome, fragile X syndrome)

1.3.5.3 single gene defects e.g. sickle cell anemia

1.3.5.3.1 autosomal dominant

(e.g., Huntington's disease, osteogenesis imperfecta)

1.3.5.3.2 autosomal recessive (e.g., cystic fibrosis, phenylketonuria, hemoglobinopathies, thalassemias)

1.3.5.3.3 X-linked (e.g., Duchenne's muscular dystrophy, hemophilia)

1.3.6 multifactorial diseases (e.g., diabetes mellitus, atherosclerotic heart disease,)

1.3.7 principles of therapy

1.3.7. diagnostic methods; predictive testing, screening, prenatal diagnosis

1.3.7.2 potentials for gene therapy: ( application for cancer, adenosine deaminase deficiency, cystic fibrosis)

1.3.8. mouse model (systems, knouts, transgenic, and embryonic stem cells.

Implications and disease

## 1.4 Biology of tissues and their responses to disease

1.4.2.3 lysosomal enzymes, reactive oxygen metabolites

1.4.4 neoplasia

1.4.4.2 cell biology, biochemistry, and molecular biology of neoplastic cells: cell kinetics, doubling time; control points, transformation, retroviruses, oncogenes, tumor-suppressor genes, apoptosis

1.4.4.3 polypeptide growth factors; signal transduction

1.4.4.4 hereditary neoplastic disorders; chromosomal abnormalities; translocations, karyotypic abnormalities, predisposing Mendelian conditions

1.4.4.5 environmental carcinogenesis

## 1.6 Multisystem processes

1.6.1 nutrition

1.6.1.1 generation, expenditure, and storage of energy at the whole-body level

1.6.1.2 digestion and absorption of nutrients

1.6.1.3 functions of essential nutrients (e.g., vitamins, minerals, amino acids, fatty acids)

1.6.1.4 caloric and nitrogen balance

1.6.1.5 protein-calorie malnutrition (e.g., marasmus, kwashiorkor)

1.6.1.6 vitamin deficiencies (e.g., folate/anemia, vitamin C/scurvy,) and toxicities (e.g., vitamin D/hypercalcemia)

1.6.1.7 mineral deficiencies/toxicities: iron

1.6.1.8 obesity, anorexia, bulimia, food fads

## **1.7 Pharmacodynamic and pharmacokinetic processes**

### **1.7.2 Biochemical aspects of autotoxins**

#### **1.7.2.1 [prostaglandins, leukotrienes, thromboxanes](#), and their inhibitors**

##### **1.7.2.2 peptides and analogs (e.g., endorphins, substance P, erythropoietin)**

##### **1.7.2.3 biogenic amines (e.g., [adrenergic, cholinergic, dopaminergic, serotonergic, histaminergic, GABAergic](#))**

##### **1.7.2.4 smooth muscle/endothelial autotoxins; nitric oxide, platelet-activating factor, atrial natriuretic peptide, endothelin**

#### **1.7.2.5 [cytokines](#) (e.g., interleukins, tumor necrosis factor)**

## **2. Hematopoietic and lymphoreticular systems**

### **2.2 Normal Processes**

#### **2.2.1 metabolic, physiologic, and regulatory processes**

##### **2.2.1.1 erythropoiesis, [hemoglobin](#), normal erythrocyte function, O<sub>2</sub> and CO<sub>2</sub> transport, transport proteins, hemoglobin structure, function, and development**

##### **2.2.1.2 function of platelets, [coagulation](#), homeostasis**

### **2.3 Abnormal processes**

#### **2.3.1 genetic/congenital disorders (e.g., hemoglobinopathies, thalassemias, enzyme deficiency)**

## **3. Central and peripheral nervous systems**

### **3.2 Normal processes**

#### **3.2.1.7 [metabolism of the nervous system](#); energy metabolism**

## **7. Cardiovascular system**

### **7.2 Normal processes**

#### **7.2.1 metabolic, physiologic, and regulatory processes**

##### **7.2.1.1 heart muscle, metabolism, biochemistry, and secretory function (e.g., atrial natriuretic peptide)**

### **7.3 Abnormal processes**

#### **7.3.1 genetic/congenital disorders (e.g., glycogen storage disease)**

### **7.4 Principles of therapeutics**

#### **7.4.1 therapeutic mechanisms of action and uses of drugs for treatment of**



disorders of the cardiovascular system

7.4.1.5 [drugs affecting cholesterol and lipid metabolism](#)

7.4.1.6 [drugs that affect blood coagulation; thrombolytic agents](#)

## 8. Gastrointestinal system

### 8.2 Normal processes

8.2.1 metabolic, and regulatory processes

8.2.1.3 salivary, gastrointestinal, pancreatic, hepatic secretory products and processes

8.2.1.4 [digestion and absorption](#)

8.2.1.5 synthetic and metabolic functions of the liver; functions of gallbladder, bile ducts

8.2.1.6 [enterohepatic circulation](#)

### 8.3 Abnormal processes

8.3.4 disorders involving metabolic, physiologic, or regulatory processes

8.3.4.2 malabsorption (e.g., pancreatic insufficiency, sprue)

8.3.4.3 hepatic failure, jaundice, encephalopathy; cirrhosis

8.3.4.4 cholelithiasis, cholestasis

8.3.5 neoplastic disorders (e.g., pancreas, colon)

## 9. Renal/urinary system

### 9.2 Normal processes

9.2.1.6 renal metabolism

## 11. Endocrine system

### 11.2 Normal processes

11.2.1 hormone production, transport, and metabolism; hormone and intracellular actions, whole-body effects

11.2.1.4 [adrenal cortex; adrenal medulla](#)

11.2.1.5 [endocrine pancreas](#)

11.2.1.6 ovary and testis

11.2.1.8 [steroids and molecular actions](#)

### 11.3 Abnormal processes

11.3.1 genetic/congenital disorders; [inborn errors of metabolism](#) (e.g. adrenal hyperplasia)

11.3.3 disorders involving metabolic, or regulatory processes

11.3.3.4 endocrine pancreatic disorders (e.g., [diabetes mellitus, types I and II](#)), diabetic ketoacidosis, hyperosmolar coma, hyperinsulinism

- First Law of Thermodynamics
  - Enthalpy
  - Reversible and Irreversible Reactions
- Second Law of Thermodynamics and Entropy
- Standard State Conditions for Biological Reactions
- Coupled Reactions

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## First Law of Thermodynamics

Stated simply; The total energy of the universe does not change. This does not mean that the form of the energy cannot change. Indeed, chemical energies of a molecule can be converted to thermal, electrical or mechanical energies.

The internal energy of a system can change only by work or heat exchanges. From this the change in the free energy of a system can be shown by the following equation:

$$\Delta E = q - w \text{ Eqn. 1}$$

When  $q$  is negative heat has flowed from the system and when  $q$  is positive heat has been absorbed by the system. Conversely when  $w$  is negative work has been done on the system by the surrounding and when positive, work has been done by the system on the surroundings.

In a reaction carried out at constant volume no work will be done on or by the system, only heat will be transferred from the system to the surroundings. The end result is that:

$$\Delta E = q \text{ Eqn. 2}$$

When the same reaction is performed at constant pressure the reaction vessel will do work on the surroundings. In this case:

$$\Delta E = q - w \text{ Eqn. 3}$$

$$\text{where } w = P\Delta V \text{ Eqn. 4}$$

When the initial and final temperatures are essentially equal (e.g. in the case of biological systems):

$$\Delta V = \Delta n[RT/P] \text{ Eqn. 5}$$

$$\text{therefore, } w = \Delta nRT \text{ Eqn. 6}$$

by rearrangement of equation 3 and incorporation of the statements in equations 4-6, one can calculate the amount of heat released under constant pressure:

$$q = \Delta E + w = \Delta E + P\Delta V = \Delta E + \Delta nRT \text{ Eqn. 7}$$

In equation 7  $\Delta n$  is the change in moles of gas per mole of substance oxidized (or reacted),  $R$  is the gas constant and  $T$  is absolute temperature.

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## Enthalpy

Since all biological reactions take place at constant pressure and temperature the state function of reactions defined to account for the heat evolved (or absorbed) by a system is **enthalpy** given the symbol, **H**.

The changes in enthalpy are related to changes in free energy by the following equation:

$$\Delta H = \Delta E + P\Delta V \text{ Eqn. 8}$$

Equation 8 is in this form because we are addressing the constant pressure situation. In the biological setting most all reaction occur in a large excess of fluid, therefore, essentially no gases are formed during the course of the reaction. This means that the value  $\Delta V$ , is extremely small and thus the product  $P\Delta V$  is very small as well. The values  $\Delta E$  and  $\Delta H$  are very nearly equivalent in biological reactions

Stated above was the fact that state functions, like  $\Delta H$  and  $\Delta E$ , do not depend on the path taken during a reaction. These functions pertain only to the differences between the initial and final states of a reaction. However, heat ( $q$ ) and work ( $w$ ) are not state functions and their values are affected by the pathway taken.

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## Reversible and Irreversible Reactions

In an idealized irreversible reaction such as one done by expanding an ideal gas against

zero pressure, no work will be done by or on the system so the:

$$w = 0 \text{ Eqn. 9}$$

In the case of an ideal gas (whose molecules do not interact) there will be no change in internal energy either so:

$$\Delta E = 0 \text{ Eqn. 10}$$

since  $\Delta E = q - w$ , in this irreversible reaction  $q = 0$  also.

In a reversible reaction involving an ideal gas,  $\Delta E$  still will equal zero, however, the pressure will be changing continuously and work ( $w$ ) is a function of  $P$ , work done must be determined over the entire course of the reaction. This results in the following mathematical reduction:

$$w = RT \ln[V_2/V_1] \text{ Eqn. 11}$$

Since in this situation  $\Delta E = 0$ ,  $q = w$ . This demonstrates that some of the heat of the surroundings has to be absorbed by the system in order to perform the work of changing the system volume.

Reversible reactions differ from irreversible in that the former always proceeds infinitely slowly through a series of intermediate steps in which the system is always in the equilibrium state. Whereas, in the irreversible reaction no equilibrium states are encountered. Irreversible reactions are also spontaneous or favorable processes. Thermodynamic calculations do not give information as to the rates of reaction only whether they are favorable or not.

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## Second Law of Thermodynamics: Entropy

The second law of thermodynamics states that the universe (i.e. all systems) tend to the greatest degree of randomization. This concept is defined by the term **entropy, S**.

$$S = k \ln W \text{ Eqn. 12}$$

where  $k$  = Boltzmann constant (the gas constant,  $R$ , divided by Avogadro's number) and  $W$  = the number of substrates. For an isothermal reversible reaction the change in entropy can be reduced to the term:

$$\Delta S = \Delta H/T \text{ Eqn. 13}$$

Whereas, enthalpy is a term whose value is largely dependent upon electronic internal energies, entropy values are dependent upon translational, vibrational and rotational internal energies. Entropy also differs from enthalpy in that the values of enthalpy that indicate favored reactions are negative and the values of entropy are positive. Together the terms enthalpy and entropy demonstrate that a system tends toward the highest entropy and the lowest enthalpy.

In order to effectively evaluate the course (spontaneity or lack thereof) of a reaction and taking into account both the first and second laws of thermodynamics, Josiah Gibbs defined the term, free energy. Free energy:

$$\Delta G = \Delta H - T\Delta S \text{ Eqn. 14}$$

Free energy is a valuable concept because it allows one to determine whether a reaction will proceed and allows one to calculate the equilibrium constant of the reaction which defines the extent to which a reaction can proceed. The discussion above indicated that a decrease in energy, a negative  $\Delta H$ , and an increase in entropy, a positive  $\Delta S$ , are indicative of favorable reactions. These terms would, therefore, make  $\Delta G$  a negative value. Reactions with negative  $\Delta G$  values are termed exergonic and those with positive  $\Delta G$  values endergonic. However, when a system is at equilibrium:

$$\Delta G = 0 \text{ Eqn. 15}$$

Gibb's free energy calculations allow one to determine whether a given reaction will be thermodynamically favorable. The sign of  $\Delta G$  states that a reaction as written or its reverse process is the favorable step. If  $\Delta G$  is negative then the forward reaction is favored and visa versa for  $\Delta G$  values that are calculated to be positive.

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## Standard State Conditions in Biological Reactions

To effectively interpret the course of a reaction in the presence of a mixture of components, such as in the cell, one needs to account for the free energies of the contributing components. This is accomplished by calculating total free energy which is comprised of the individual free energies. In order to carry out these calculations one needs to have a reference state from which to calculate free energies. This reference state, termed the **Standard State**, is chosen to be the condition where each component in a reaction is at 1M. Standard state free energies are given the symbol:

$$G^0$$

The partial molar free energy of any component of the reaction is related to the standard free energy by the following:



$$G = G^0 + RT\ln[X] \text{ Eqn. 16}$$

From equation 16 one can see that when the component X, or any other component, is at 1M the  $\ln[1]$  term will become zero and:

$$G = G^0 \text{ Eqn. 17}$$

The utility of free energy calculations can be demonstrated in a consideration of the diffusion of a substance across a membrane. The calculation needs to take into account the changes in the concentration of the substance on either side of the membrane. This means that there will be a  $\Delta G$  term for both chambers and, therefore, the total free energy change is the sum of the  $\Delta G$  values for each chamber:

$$\Delta G = \Delta G_1 + \Delta G_2 = RT\ln\{[A]_2/[A]_1\} \text{ Eqn. 18}$$

Equation 18 tells one that if  $[A]_2$  is less than  $[A]_1$  the value of  $\Delta G$  will be negative and transfer from region 1 to 2 is favored. Conversely if  $[A]_2$  is greater than  $[A]_1$   $\Delta G$  will be positive and transfer from region 1 to 2 is not favorable, the reverse direction is.

One can expand upon this theme when dealing with chemical reactions. It is apparent from the derivation of  $\Delta G$  values for a given reaction that one can utilize this value to determine the equilibrium constant,  $K_{eq}$ . As for the example above dealing with transport across a membrane, calculation of the total free energy of a reaction includes the free energies of the reactants and products:

$$\Delta G = G_{(products)} - G_{(reactants)} \text{ Eqn. 19}$$

Since this calculation involves partial molar free energies the  $\Delta G^0$  terms of all the reactants and products are included. The end result of the reduction of all the terms in the equation is:

$$\Delta G = \Delta G^0 + RT\ln\{[C][D]/[A][B]\} \text{ Eqn. 20}$$

When equation 20 is used for a reaction that is at equilibrium the concentration values of A, B, C and D will all be equilibrium concentrations and, therefore, will be equal to  $K_{eq}$ . Also, when at equilibrium  $\Delta G = 0$ . Therefore:

$$0 = \Delta G^0 + RT\ln K_{eq} \text{ Eqn. 21}$$

$$K_{eq} = e^{-\{\Delta G^0/RT\}} \text{ Eqn. 22}$$

This demonstrates the relationship between the free energy values and the equilibrium constants for any reaction.

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## Coupled Reactions

Two or more reactions in a cell sometimes can be coupled so that thermodynamically unfavorable reactions and favorable reactions are combined to drive the overall process in the favorable direction. In this circumstance the overall free energy is the sum of individual free energies of each reaction. This process of coupling reactions is carried out at all levels within cells. The predominant form of coupling is the use of compounds with high energy to drive unfavorable reactions.

The predominant form of high energy compounds in the cell are those which contain phosphate. Hydrolysis of the phosphate group can yield free energies in the range of -10 to -62 kJ/mol. These molecules contain energy in the phosphate bonds due to:

- 1. Resonance stabilization of the phosphate products
- 2. Increased hydration of the products
- 3. Electrostatic repulsion of the products
- 4. Resonance stabilization of products
- 5. Proton release in buffered solutions

The latter phenomenon indicates that the pH of the solution a reaction is performed in will influence the equilibrium of the reaction. To account for the fact that all cellular reactions take place in an aqueous environment and that the  $[H_2O]$  and  $[H^+]$  are essentially constant these terms in the free energy calculation have been incorporated into a free energy term identified as:

$$\Delta G^{0'} = \Delta G^0 + RT \ln\{[H^+]/[H_2O]\} \text{ Eqn. 23}$$

Incorporation of equation 23 into a free energy calculation for any reaction in the cell yields:

$$\Delta G = \Delta G^{0'} + RT \ln\{[products]/[reactants]\} \text{ Eqn. 24}$$

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